

# **Processor Programming Reference (PPR) for AMD Family 19h Model 11h, Revision B2 Processors Volume 7 of 9**

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## 14 Fusion Controller Hub (FCH)

### 14.1 FCH Overview

#### 14.1.1 Acronyms

Table 239: List of Acronyms used in FCH

| Acronym             | Definition                                           |
|---------------------|------------------------------------------------------|
| ASF                 | Alert Standard Format.                               |
| ASL                 | ACPI Source Language. See <a href="#">docACPI</a> .  |
| DASH                | Desktop and mobile Architecture for System Hardware. |
| SCH                 | Server Controller Hub.                               |
| <a href="#">SSC</a> | Spread Spectrum Clocking.                            |
| TPM                 | Trusted Platform Module.                             |

#### 14.1.2 Functional

This section describes the integrated [FCH](#), also known as the SCH. The SCH, which replaces the South Bridge Gasket (SBG), integrates the FCH with an interface to a standard Scalable Data Fabric Port (SDP). It also adapts additional DFX features such as the Remote System Management Unit (RSMU) and Remote Design-For-Test (RDFT) structures.

In terms of IO functions, the [I2C](#), [I3C](#), UART, and eSPI are available in the Family 19h Models 10h-1Fh Processors. In addition, ACPI Power management, basic low level system functions, boot, and clock generation (ClkGen) logic are provided by the processor.

The following is the list of IP blocks and functions:

AB/SB-SDP(SB Scalable Data Port): SDP is designed to meet Family 19h Models 10h-1Fh architecture and is the interface to Scalable Data Fabric (SDF). See 14.3.2 [AB Configuration Register] for register descriptions.

SPI/eSPI – Serial peripheral interface: this is the bridge logic to the BIOS/firmware flash and SPI TPM. eSPI is multiplexed on the SPI bus to support an eSPI device such as embedded controller (ec).

Family 19h Models 10h-1Fh Processors have the following new features:

1. We have 2 sets of eSPI and SPI muxed pads. eSPI is the default function.
2. A second eSPI controller is added

UART – Universal Asynchronous Receiver/Transmitter: a low speed communication bus that is typically used for interfacing to serial ports. There are four UART ports and the speed can be upgraded to 4Mbps depending on availability of required reference clock. At least one UART is required to support legacy IO addresses (0x3F8:0x3FF). The UART doesn't support FIFO mode in legacy IO mode.

I2C - Inter-Integrated Circuit: a low level serial bus that is popular for connecting to external devices. Two I2Cs can be configured to be in secured mode where they can only be accessible by the [PSP](#) through the RSMU interface. In secure mode, the two I2C blocks become inaccessible by the host CPU.

I3C - The I3C interface is intended to improve upon the features of the I2C interface, preserving backward compatibility.



It provides more power efficiency and speed improvement over I2C.

**Clkgen – Clock Generation:** integrated clock generation function for the entire system. This block adds a non-spread display and more [PCIe®](#) GPP clock outputs. It also adds more internal reference clocks for various PHYs. In a multi-die configuration, Clkgen is configured to a different internal clock mode where the secondary dies use a 100MHz reference (coming from primary die) instead of a 48MHz crystal source. See 14.3.14.1 [Miscellaneous (MISC) Registers].

**SMBus – System Management Bus:** a low level serial bus used for communicating with on board devices. The SMBus Host Controller provides two independent control spaces, one for the SMBus and one for the Alert Standard Format (ASF) bus. For register definitions of these control spaces, see 14.3.12 [SMBUS Host Controller] and 14.3.12.2 [ASF (Alert Standard Format) Registers] for register descriptions.

**GPIO – General Purpose IO:** used as GPIO or interrupt inputs. See 14.3.15 [GPIO Pin control registers] for register descriptions.

**ACPI – Advanced Configuration and Power Interface:** power management and reset functions. This block also includes other system functions such as IOAPIC and various system timers. See 14.3.4 [I/O Advanced Programmable Interrupt Control] for register descriptions.

**Power Management (PM) Supervisory –** custom logic to manage iSCH clock and power gating to support AOAC. See 14.3.14.3 [Power Management (PM) Registers and Standard ACPI Registers] for register description.

**Remote System Management Unit (RSMU).** The RSMU is included in the iSCH (integrated server controller hub). Its functions include efuse distribution (both functional and memory repair), as well as secured and non-secured access from the PSP to specific functional blocks inside the iSCH. This includes ACPI, I2C, I3C and SPI. To support secured I2C, two GPIO interrupt lines are to be routed to RSMU allowing line status to be read by the PSP.

### 14.1.3 MMIO Programming for Legacy Devices

The legacy devices, IOAPIC, TPM, ACPI, and Watchdog Timer, require the base address of the Memory Mapped IO registers to be assigned before these registers can be accessed. The Memory Mapped IO register base address and its entire range should be mapped to a non-posted memory region by programming the CPU register.

#### 14.1.3.1 Description for FCH::IO::PCIInterrupt Map

FCH::IO::PCI\_INTR\_INDEX bit [7] means PciIntrApic, set as 0 means IRQ routing to PIC, set as 1 means IRQ routing to IOAPIC.

FCH::IO::PCI\_INTR\_INDEX bit[6:0] are PCI interrupt index. Select which PCI interrupt to map. Following are detail description.

*Table 240: PCI interrupt index list of FCH::IO::PCI\_INTR\_INDEX bit[6:0]*

| Bit[6:0] | Description               |
|----------|---------------------------|
| 04h-00h  | INT[E:A]#.                |
| 05h      | INTF#/GENINT2.            |
| 06h      | INTG#.                    |
| 07h      | INTH#.                    |
| 08h      | Misc.                     |
| 0Bh-09h  | Misc[2:0].                |
| 0Fh-0Ch  | INT[D:A] from serial IRQ. |
| 10h      | SCI.                      |
| 11h      | SMBUS0.                   |

|         |                               |
|---------|-------------------------------|
| 12h     | ASF.                          |
| 15h-13h | Reserved.                     |
| 16h     | PerMon.                       |
| 18h-17h | Reserved.                     |
| 1Ah     | SDIO.                         |
| 43h-1Bh | Reserved.                     |
| 53h-50h | GPPInt3/2/1/0.                |
| 60h     | SCI Interrupt                 |
| 61h     | <a href="#">SMI</a> Interrupt |
| 62h     | GPIO controller interrupt.    |
| 70h     | I2C0/I3C0.                    |
| 71h     | I2C1/I3C1.                    |
| 72h     | I2C2/I3C2.                    |
| 73h     | I2C3/I3C3.                    |
| 75h-74h | UART1/UART0.                  |
| 77h-76h | I2C5/I2C4.                    |
| 79h-78h | UART3/UART2.                  |
| 7Fh-7Ah | Reserved.                     |

#### 14.1.4 ASF controller

The ASF controller may be used in a slave mode to support DASH clients. When the platform is configured for DASH the BIOS should set the following registers to ensure the ASF is configured in slave mode and the control commands are defined properly in the ASF tables:

When operating in Master mode,

BIOS programs FCH::PM::DECODEEN[asfsmmasteren] = 1

BIOS programs FCH::PM::DECODEEN[asfsmmasteren] back to 0 after the master operations are complete.

When operating in slave mode,

BIOS programs FCH::PM::DECODEEN[asfsmmasteren] = 0

In addition to supporting remote DASH operations, BIOS should ensure that the ASF tables define the correct command codes and must be programmed with the same value reported in the ASF control data table.

The ASF controller can behave like a remote control device to accept commands from the client side to do the reset down and up power cycles. BIOS reports those in the ASF control data table. The command and command data are defined the below ASF Remote Control Commands table:

1. If Remote Control is rest, then control command is 50h, control data is 00h.
2. If Remote Control is Power Up, then control command is 51h, control data is 00h.
3. If Remote Control is Power Down, then control command is 52h, control data is 00h.
4. If Remote Control is Power Cycle, then control command is 53h, control data is 00h.

#### 14.1.5 A-Link Bridge

The A-link bridge (AB) sits behind the [FCH](#) gasket and acts as the bridge between the root complex and A-Link, B-Link bus.

Program the following sequence to detect upstream interrupts:

BIOS enables AB to detect upstream interrupts for the purpose of system management.

- Program FCH::AB::MsiCtl[MsiAddr\_39\_20] with CPU interrupt delivery address [39:20].

- Program FCH::AB::MsiCtl[MsiAddrEn] = 1.

Program the following sequence to allow AB 32/64 Byte DMA writes:

1. Program FCH::AB::MiscCtl1[UpWr16BMode] = 0.
2. FCH::AB::MiscCtl1[UpSWrByteCntSbgMode] = 1.
3. Software ensures that SDP Upstream Control[Dma16ByteMode] == 0.

### 14.1.6 FCH Gasket

The FCH\_SDP replaces the FCH\_SBG and used to connect between the [FCH](#) and IO Hub.

The FCH\_SDP is responsible for converting between the protocols used by the SDP interface and the A-link bridge interface. BIOS programs FCH::PM::ABREGBAR to set up base address before accessing the AB configuration registers. The SDP memory power saving list follows:

The following sequence enables SBG memory power saving:

1. Program FCH\_MISC\_MEMPWRSAPVCNTRL[sd\_bypass\_mem\_dsd] = 0 to enable SDP memory power savings.
2. Program SDP Misc Control[HostMemShutDownEn] = 1 to enable SDP host memory shutdown.
3. Program SDP Misc Control[DmaMemShutdownEn] = 1 to enable SDP DMA memory shutdown.

SBG internal clock gating list as below:

The following sequence enables the SBG internal clock gating:

1. Program SDP Misc Control[MedBCLKGateEnHost] = 1 to enable medium grain BCLK gating.
2. Program SDP Misc Control[DBG\_CLK\_GATE\_EN] = 1 to disable debug BCLK.

### 14.1.7 AOAC Function

#### 14.1.7.1 Wake Enable

The D3 cold and S3/S5 wake is triggered by GPIO. In order to enable the wake on specified GPIO pins, the corresponding GPIO registers need to be programmed to enable wake capability. See 14.1.14.1.2 [Wakeup GPIO] for GPIO wake up function.

For the S3 or S4 state, the wake goes through Pm1a and Gevent or the GPIO controller.

#### 14.1.7.2 WinBlue power button

Windows® 8.1 has redefined the power button. The power button connects to GPIO controller and can be programmed by the OS to trigger on both edges. In addition, the 4 second shutdown is changed to a 10 second shutdown. To support the 10 second shutdown, set FCH\_GPIO\_GPIO\_WAKE\_INTERRUPT\_MASTER\_SWITCH[enwinbluebtn] = 1.

### 14.1.8 ACPI

#### 14.1.8.1 Enable legacy IO access to DMA controller 1

To enable legacy IO access to FCH::IO::DMA\_CH\_0, program FCH\_PM\_LegacySS[LegacySS] to an unoccupied port address.

FCH\_PM\_LegacySS[LegacySS] used to define the IO base address of a function which is no longer used. It is defaulted to be 0x0000 that conflicts with the legacy DMA BaseAddress register. Therefore software needs to program FCH\_PM\_LegacySS[LegacySS] to an unoccupied IO port address other than 0x0000.

### 14.1.9 SPI Bus Interface

This section describes the SPI Bus related programming requirements.

RomProtect applies to ROM, those ROM controls does not apply to eSPI bus as eSPI controller is another independent IP.

#### 14.1.9.1 Serial Peripheral Interface (SPI) ROM

##### 14.1.9.1.1 Introduction

Security is important to AMD. A feature called "ROM locking" enables platform developers to add a layer of security to their platforms. Regions of SPI ROM can be flagged as "Read" or "No-Read or Write" or "No-Write" ("Write-only" is not supported). Up to 64 MB of ROM can be managed through the ROM Protection regions.

The SPI ROM can be accessed by two methods or "modes" of operation: "Direct" and "Indexed".

- "Direct" is a program read or write to a memory location (MMIO in the [Physical address](#) range) which gets translated by HW to an SPI bus command behind the scene. This is the mode used for most ROM reads during boot.
- "Indexed" mode uses SPI bus commands to access the SPI device. This requires setting up the SPI command/control registers and issuing the command on the SPI bus. This mode is used for special operations like erasing a block and programming a block in the flash part. It can be used for reads; however, they are much slower. AMD recommends Index Mode not be used for reads.

When accessing the ROM in the various modes, it is important to keep in mind the addressing mechanisms in use. There are multiple places in the physical memory map where a ROM can be located.

- ROM1 – Traditionally a smaller ROM based at 1 MB.
- ROM2 – Typically located at the 4GB address boundary, supporting 16 MB ROMs, although larger ROMs may be used and paged into the 16 MB aperture.
- ROM3 – AMD supports a special address space which can support larger ROMs (up to 64 MB).

Also, when using Indexed mode, the address become a "relative" address. 0X0000 being the first storage in the device running up to ROM\_Size-1 being the last storage byte in the device. Multiple SPI devices can be present on the bus selected by the Alt\_SPI\_CS register as described in the PPR. Protection is applied identically to multiple ROMs.

AMD's architecture includes support for protection in all three ROM address spaces with the described ROM protection mechanisms.

##### 14.1.9.1.2 Normal Flash ROM on SPI Bus

This level of protection is characterized by having no added HW protection beyond that provided by the SPI ROM device or that is built into the platform.

This is the level most common for client PCs. Manufacturers use various schemes to prevent accidental erasure, and to ensure recovery from an update gone wrong.

SPI memory can be protected from being written to or read from by using the following sequence. This is optional and used as required.

##### 14.1.9.1.2.1 Steps to Prepare

Set the address(es) for the ROM(s) in the system. ROM Address\_Range registers are 16 bits => address [31:12]. These start and end registers contain the upper 20 bits of the start and end address of a ROM range. The lower 12 bits of a start address are 000h. The lower 12 bits of an end address are FFFh. These registers apply to SPI bus.

**14.1.9.1.3 Standard HW Protection**

This level is characterized by invoking hardware provided features for locking access to certain SPI registers. The PPR describes registers where the platform BIOS engineer can:

- Define Protection regions
- Set restricted commands (14.1.9.1.3.1.5 [Set the Block List])
- Invoke a hardware lock of protection registers

**14.1.9.1.3.1 Steps to Invoke HW Protection****14.1.9.1.3.1.1 Prepare**

Set the address(es) for the ROM(s) in the system. ROM Address\_Range registers are 16 bits => address [31:12]. These start and end registers contain the upper 20 bits of the start and end address of a ROM range. The lower 12 bits of a start address are 000h. The lower 12 bits of an end address are FFFh. These registers apply to SPI bus.

**14.1.9.1.3.1.2 Set the Protect Regions**

There are four ROM\_PROTECT registers that allow you define up to 16 [MB](#) each of protected address space. Protection can be read-only, no-read-nor-write, or unprotected (write-only is not supported). It is considered an error to define a write-only area and SPI ROM Locking FW does not allow it. A summary of points of interest for these registers are described in the PPR:

- The 4 regions must not overlap.
- "Range" is 0-based (#blocks=N+1); max field value is 255, representing 256 RangeUnits.
- "RangeUnit" can be 4k (bit=0) or 64k (bit=1) in size.
- When programmed, all 4 protection registers must contain valid protection bits. If both protection bits are 0, the HW will not look at the range in a particular register.

**14.1.9.1.3.1.3 ROM2 Address Override**

The developer must also ensure that the correct value for the ROM2\_ADDR\_OVERRIDE register is set in the same way as the ROM\_PROTECTx registers. Note that the HW only looks at unified addresses.

**14.1.9.1.3.1.4 SPI Bank Mapping**

The PPR defines ROM2\_ADD\_OVERRIDE register that can be used to map banks of a large (up to 64M) ROM into a 16M window. This mapping is applied the ROM [Physical address](#) has been determined. This affects the protection setting chosen by the platform BIOS engineer. The protection regions must be aligned to the pre-paged physical addresses. When used in ROM2 space, the SPI Controller can address from the beginning to the end of the ROM subject to the ROM Protect registers. However, since the aperture is only 16MB, only the lower 16MB can be protected by HW mechanisms (Rom Protect registers) alone.

**14.1.9.1.3.1.5 Set the Block List**

There are 2 SPI\_RestrictedCmd registers that allow for 8 commands to be in block list. List the hex command code(s) to be blocked. There are 5 commands (RestrictedCmdx) that restrict access when the OpCode matches and the address is above 512k. The 3 RestrictCmdWoAddrx entries restrict access with or without address parameters.

Note: For general protection from any/all SPI writes or erasures – include the WE (Write Enable) command in block list. The SPI protocol requires the WE command to be issued prior to any write or erase command.

**14.1.9.1.4 Set SPI ROM Protection**

SPI memory can be protected from being written to or read from by using the following sequence. This is optional and used as required.

**14.1.9.1.4.1 Enable SPI ROM Protection**

1. Program FCH::LPCPCICFG::ROM\_PROTECT\_0 to enable protection for Read or Write memory accesses to SPI flash memory space. Up to eight ranges of memory can be protected.
2. Program FCH::LPCHOSTSPIREG::SPI\_RESTRICTEDCMD\_REGISTER or FCH::LPCHOSTSPIREG::SPI\_RESTRICTEDCMD2\_REGISTER with SPI commands that are required before doing write accesses to the SPI ROM. Use commands such as WR\_EN, WR\_Status and Erase when programming these restricted command registers.
3. Program FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER[spiaccessmacromen] = 0 or FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER[spihostaccessromen] = 0 to protect the registers above from being reprogrammed.

Accesses are now blocked by the SPI host preventing write accesses to the SPI ROM.

Notes:

Step 1 enables protection for direct memory access to SPI (Direct memory access is when Host access SPI memory using normal Read / Write commands these access are converted to SPI Read and Write commands by the SPI controller).

Step 2 enables protection for indirect memory access (a.k.a. Indexed mode) to the SPI controller. Indirect memory accesses (a.k.a. Indexed mode) are when the FCH::LPCHOSTSPIREG::CMDCODE is used to send SPI commands to the SPI ROM.

Step 3 By programming the FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER[spiaccessmacromen] = 0 or FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER[spihostaccessromen] = 0, all SPI commands (defined in register FCH::LPCHOSTSPIREG::SPI\_CMDVALUE1\_REGISTER and FCH::LPCHOSTSPIREG::SPI\_CMDVALUE2\_REGISTER) and Restricted commands (defined in FCH::LPCHOSTSPIREG::SPI\_RESTRICTEDCMD\_REGISTER and FCH::LPCHOSTSPIREG::SPI\_RESTRICTEDCMD2\_REGISTER) are protected.

**14.1.9.1.4.2 Disable SPI ROM Protection**

1. Generate [SMI](#).
2. Program FCH::LPCPCICFG::ROM\_PROTECT\_0 [read\_protect][write\_protect] = 0 to unprotect Read or Write memory access to the SPI flash memory space.
3. Program FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER[spiaccessmacromen] = 1 and FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER[spihostaccessromen] = 1 to unprotect the registers above from being reprogrammed.
4. Exit SMI mode.

**14.1.10 eSPI**

eSPI is configured to run at 16.67 MHz, 33 MHz, and 66 MHz. See 14.3.13.4 [eSPI Registers] about how to program the speed.

**14.1.11 UART**

The UART is controlled by driver software. There are no programming requirements for the BIOS, except for pin configuration and enablement.

If Program UART0\_1p843MCLK\_EN=1, enable UART0 1.843MHz clock.

If Program UART1\_1p843MCLK\_EN=1, enable UART1 1.843MHz clock.

If Program UART2\_1p843MCLK\_EN=1, enable UART2 1.843MHz clock.

If Program UART3\_1p843MCLK\_EN=1, enable UART3 1.843MHz clock.

**14.1.12 I2C Master**

For I2C0~I2C3, the [I2C](#) pins interface pins can be either at 1.1V or 1.8V. I2C interface pins for I2C4 and I2C5 supports 1.8V. The following table lists the I2C PAD control registers.

*Table 241: I2C PAD Control Configuration*

| I2C modules | Description          | Register Location       |
|-------------|----------------------|-------------------------|
| I2C_0       | Pad control Register | FCH::MISC::I2C0_PADCTRL |
| I2C_1       | Pad control Register | FCH::MISC::I2C1_PADCTRL |
| I2C_2       | Pad control Register | FCH::MISC::I2C2_PADCTRL |
| I2C_3       | Pad control Register | FCH::MISC::I2C3_PADCTRL |
| I2C_4       | Pad control Register | FCH::MISC::I2C4_PADCTRL |
| I2C_5       | Pad control Register | FCH::MISC::I2C5_PADCTRL |

**14.1.13 I3C Master**

The [I3C](#) interface is improved from [I2C](#) interface and preserves the backward compatibility with I2C.

Following are the supporting features:

1. Two wire serial interface up to 12.5MHz using Push-Pull
2. Legacy I2C Device co-existence on the same Bus (with some limitations)
3. Dynamic Addressing while supporting Static Addressing for Legacy I2C Devices
4. Broadcast and Direct Common Command Code ([CCC](#)) message
5. Legacy I2C messaging
6. Support Single Data Rate messaging ([SDR](#))
7. In-Band Interrupt support
8. Hot-Join support
9. Master mode only

For I3C0~I3C3, the I3C pins interface pins can be either at 1.1V or 1.8V

*Table 242: I3C PAD Control Configuration*

| I3C modules | Description          | Register Location       |
|-------------|----------------------|-------------------------|
| I3C_0       | Pad control Register | FCH::MISC::I2C0_PADCTRL |
| I3C_1       | Pad control Register | FCH::MISC::I2C1_PADCTRL |
| I3C_2       | Pad control Register | FCH::MISC::I2C2_PADCTRL |
| I3C_3       | Pad control Register | FCH::MISC::I2C3_PADCTRL |



**14.1.14 GPIO**

There are three types of pads in [FCH](#): GPIO, [I2C](#) and [I3C](#).

- For the GPIO pads, most of their functional connections are controlled by GPIO registers. When IOMUX is set to select the GPIO function, the pad is controlled by GPIO register, including the output enable, output value, pull-up and pull-down and driving strength functions.
- For the I2C pad, only the output enable comes from a GPIO register, as the pad output is an open drain output. Open drain outputs require an external (on-board) pull-up resistor. After programming the I2C pad to be GPIO (through the IOMUX register), drive the output enable high if you want the output PAD driven low. Conversely, drive the output enable low in order to allow the external pull-up resistor to pull the output PAD high.
- For the I3C pads, additional control ports are required. An additional 32-bit control register is available to control these GPIOs. After programming each I3C pad to be GPIO (through the IOMUX register), the register should be used to drive the GPIO accordingly.
  - I3C have more control ports than I2C. Besides one DWORD ACPI register, there is more GPIO register bits in use, the following is the details:
    - I2C0~3\_PadCtrl of MISC register
      - Bit31: SpikeRcSel1
      - Bit30: SpikeRcSel0
      - Bit29: Mode\_1p8\_1p1\_1
      - Bit28: Mode\_1p8\_1p1\_0
      - Bit27: Mode\_I3c\_I2c1
      - Bit26: Mode\_I3c\_I2c0
      - Bit25: Slewp1
      - Bit24: Slewp0
      - Bit23: ResBiasEn1
      - Bit22: ResBiasEn0
      - Bit21: CompSel1
      - Bit20: CompSel0
      - Bit19: Spare1
      - Bit18: Spare0
      - Bit17: BiasCrtEn1
      - Bit16: BiasCrtEn0
      - Bit15: RSel1p1
      - Bit14: RSel0p9
      - Bit13: CSel1p1
      - Bit12: CSel0p9
      - Bit11: SpikeRcEn1
      - Bit10: SpikeRcEn0
      - Bit9: FallSlewSel1
      - Bit8: FallSlewSel0
      - Bit7: Slewn1
      - Bit6: Slewn0
      - Bit5: RxSel1
      - Bit4: RxSel0
      - Bit3: OD\_Rp\_Sw3
      - Bit2: OD\_Rp\_Sw2
      - Bit1: OD\_Rp\_Sw1\_pre
      - Bit0: OD\_Rp\_Sw0\_pre
    - GPIO register bits
      - NG0\_SCL\_drv\_sel\_bit0



- NG1 SCL\_drv\_sel\_bit1
- NG2 SDA\_drv\_sel\_bit0
- NG3 SDA\_drv\_sel\_bit1
- DIN0 SCL\_IN\_DIN
- DIN1 SDA\_IN\_DIN
- OE0 SCL\_IN\_OE
- OE1 SDA\_IN\_OE
- PU0 SCL\_IN\_PU
- PU1 SDA\_IN\_PU
- PD0 SCL\_IN\_PD
- PD1 SDA\_IN\_PD

Table 243: GPIO capable pads

| Bump Pin Name                          | Pad Macro                  |
|----------------------------------------|----------------------------|
| Power                                  |                            |
| BP_PWR_BTN_L/AGPIO0                    | GPIO Pad                   |
| BP_SYS_RESET_L/AGPIO1                  | GPIO Pad                   |
| BP_WAKE_L/AGPIO2                       | GPIO Pad                   |
| BP_RSMRST_L                            | GPIO Pad                   |
| BP_SLP_S3_L                            | GPIO Pad                   |
| BP_SLP_S5_L                            | GPIO Pad                   |
| BP_PWR_GOOD                            | GPIO Pad                   |
| Clock                                  |                            |
| BP_RTCCLK                              | GPIO Pad                   |
| GPIO S5                                |                            |
| BP_AGPIO3/SPD_HOST_CTRL_L              | GPIO Pad                   |
| BP_AGPIO4/SATA_ACT_L                   | GPIO Pad                   |
| BP_AGPIO5/DEVSLP0/SATA_ZP0_L           | GPIO Pad                   |
| BP_AGPIO6/DEVSLP1/SATA_ZP1_L           | GPIO Pad                   |
| BP_AGPIO7                              | GPIO Pad                   |
| BP_AGPIO21                             | GPIO Pad                   |
| BP_AGPIO22                             | GPIO Pad                   |
| BP_SMERR_L/AGPIO24                     | GPIO Pad                   |
| BP_AGPIO256/SGPIO0_CLK                 | GPIO Pad                   |
| BP_AGPIO257/SGPIO1_CLK/CLK_REQ01_L     | GPIO Pad                   |
| BP_AGPIO258/SGPIO2_CLK/CLK_REQ02_L     | GPIO Pad                   |
| BP_AGPIO259/SGPIO3_CLK                 | GPIO Pad                   |
| BP_SGPIO_DATAOUT/AGPIO260              | GPIO Pad                   |
| BP_SGPIO_LOAD/AGPIO261                 | GPIO Pad                   |
| BP_PWRGD_OUT/AGPIO12                   | GPIO Pad                   |
| BP_USB10_OC_L/AGPIO16                  | GPIO Pad                   |
| BP_USB11_OC_L/AGPIO17                  | GPIO Pad                   |
| BP_USB00_OC_L/AGPIO264                 | GPIO Pad                   |
| BP_USB01_OC_L/AGPIO265                 | GPIO Pad                   |
| I2C                                    |                            |
| BP_I2C5_SCL/BMC_SCL/SMBUS1_SCL/AGPIO19 | I2C Pad with SMBUS support |
| BP_I2C5_SDA/BMC_SDA/SMBUS1_SDA/AGPIO20 | I2C Pad with SMBUS support |
| BP_I2C4_SCL/HP_SCL/UBM_SCL/AGPIO13     | I2C Pad                    |
| BP_I2C4_SDA/HP_SDA/UBM_SDA/AGPIO14     | I2C Pad                    |

| Legacy                             |          |
|------------------------------------|----------|
| BP_PCIE_RST1_L/AGPIO26             | GPIO Pad |
| BP_PCIE_RST0_L/AGPIO266            | GPIO Pad |
| BP_SOCKET_ID[0]                    | GPIO Pad |
| BP_SOCKET_ID[1]                    | GPIO Pad |
| GPIO S0                            |          |
| BP_AGPIO87                         | GPIO Pad |
| BP_AGPIO88                         | GPIO Pad |
| BP_GENINT_L/PM_INTR_L/AGPIO89      | GPIO Pad |
| BP_AGPIO104                        | GPIO Pad |
| BP_AGPIO105                        | GPIO Pad |
| BP_AGPIO106                        | GPIO Pad |
| BP_AGPIO107                        | GPIO Pad |
| BP_AGPIO109                        | GPIO Pad |
| BP_AGPIO115/CLK_REQ11_L            | GPIO Pad |
| BP_AGPIO116/CLK_REQ12_L            | GPIO Pad |
| BP_NMI_SYNC_FLOOD_L                | GPIO Pad |
| Thermal                            |          |
| BP_THERMTRIP_L                     | GPIO Pad |
| SPI/ESPI                           |          |
| BP_ESPI_CLK0/SPI_CLK0/AGPIO117     | GPIO Pad |
| BP_ESPI0_ALERT_L_ESPI_IO1/AGPIO108 | GPIO Pad |
| BP_ESPI_CLK1/SPI_CLK1/AGPIO75      | GPIO Pad |
| BP_ESPI_CLK2/AGPIO74               | GPIO Pad |
| BP_SPI_CS0_L/AGPIO118              | GPIO Pad |
| BP_ESPI_CS0_L/AGPIO124             | GPIO Pad |
| BP_ESPI_CS1_L/AGPIO125             | GPIO Pad |
| BP_SPI_CS2_L/AGPIO126              | GPIO Pad |
| BP_SPI_CS1_L/AGPIO119              | GPIO Pad |
| BP_ESPI0_DAT0/SPI0_DAT0/AGPIO120   | GPIO Pad |
| BP_ESPI0_DAT1/SPI0_DAT1/AGPIO121   | GPIO Pad |
| BP_ESPI0_DAT2/SPI0_DAT2/AGPIO122   | GPIO Pad |
| BP_ESPI0_DAT3/SPI0_DAT3/AGPIO123   | GPIO Pad |
| BP_AGPIO76/SPI_TPM_CS_L            | GPIO Pad |
| BP_ESPI_RSTIN_L/KBRST_L/AGPIO129   | GPIO Pad |
| BP_ESPI1_DAT0/SPI1_DAT0/AGPIO131   | GPIO Pad |
| BP_ESPI1_DAT1/SPI1_DAT1/AGPIO132   | GPIO Pad |
| BP_ESPI1_DAT2/SPI1_DAT2/AGPIO133   | GPIO Pad |
| BP_ESPI1_DAT3/SPI1_DAT3/AGPIO134   | GPIO Pad |
| BP_ESPI1_ALERT_L/AGPIO110          | GPIO Pad |
| BP_ESPI_RSTOUT_L/AGPIO23           | GPIO Pad |
| UART                               |          |
| BP_UART0_CTS_L/UART2_TXD/AGPIO135  | GPIO Pad |
| BP_UART0_RXD/AGPIO136              | GPIO Pad |
| BP_UART0_RTS_L/UART2_RXD/AGPIO137  | GPIO Pad |
| BP_UART0_TXD/AGPIO138              | GPIO Pad |
| BP_UART0_INTR/AGPIO139             | GPIO Pad |
| BP_UART1_RXD/AGPIO141              | GPIO Pad |

|                                                   |                            |
|---------------------------------------------------|----------------------------|
| BP_UART1_TXD/AGPIO142                             | GPIO Pad                   |
| I3C                                               |                            |
| BP_I3C0_SCL/I2C0_SCL/SPD0_SCL/SMBUS0_SCL/AGPIO145 | I3C Pad with SMBUS support |
| BP_I3C0_SDA/I2C0_SDA/SPD0_SDA/SMBUS0_SDA/AGPIO146 | I3C Pad with SMBUS support |
| BP_I3C1_SCL/I2C1_SCL/SPD1_SCL/AGPIO147            | I3C Pad                    |
| BP_I3C1_SDA/I2C1_SDA/SPD1_SDA/AGPIO148            | I3C Pad                    |
| BP_I3C2_SCL/I2C2_SCL/SPD2_SCL/AGPIO149            | I3C Pad                    |
| BP_I3C2_SDA/I2C2_SDA/SPD2_SDA/AGPIO150            | I3C Pad                    |
| BP_I3C3_SCL/I2C3_SCL/SPD3_SCL/AGPIO151            | I3C Pad                    |
| BP_I3C3_SDA/I2C3_SDA/SPD3_SDA/AGPIO152            | I3C Pad                    |

#### 14.1.14.1 Groups of GPIO

There are three groups of GPIO: GPIOBank0, GPIOBank1 and GPIOBank2, please refer to 14.3.15.2 [GPIO Registers]. To enable the GPIO controller, follow the programming steps below:

##### 14.1.14.1.1 Interrupt GPIO

All the configuration bits (Bit[x]) used in the following steps come from GPIOBank0, GPIOBank1 and GPIOBank2. If the pin is used as an interrupt, see 14.3.15.2 [GPIO Registers]:

1. The driver programs the DebounceTmrOut/DebounceTmrOutUnit/DebounceCntrl bits according to the data passed by BIOS through the ASL code.
2. Configure bit[10:8] according to the data passed by BIOS through ASL code.
3. Set bit[11] = 1 and bit[12] = 1 to enable interrupt delivery and interrupt status. Interrupt status is at bit[28] when the GPIO input is asserted.
4. Configure the GPIO controller interrupt. The GPIO controller interrupt is an active low level interrupt. The driver clears the interrupt status, then sets FCH::GPIO::GPIO\_WAKE\_INTERRUPT\_MASTER\_SWITCH[eoi] = 1 to deassert the interrupt request after acknowledging the interrupt.

##### 14.1.14.1.2 Wakeup GPIO

All the configuration bits (Bit[x]) used in the following steps come from GPIOBank0, GPIOBank1 and GPIOBank2. If the pin is used as wake, see 14.3.15.2 [GPIO Registers]:

1. BIOS programs bit[15:13] to enable wake in S3/S5 by BIOS. The wake status is bit[29]. BIOS/SMU reads the wake status to determine the wake source, then clears them after wake.
2. The driver programs DebounceTmrOut/DebounceTmrOutUnit/DebounceCntrl bits according the data from BIOS through the ASL code.
3. The driver sets bit[10:8] to configure assertion condition according to the data from BIOS through the ASL code.
4. Set bit[11] = 1 and bit[12] = 1 to enable interrupt delivery and interrupt status then its interrupt status is found at bit[28] when the GPIO input is asserted.
5. Configure the GPIO controller interrupt. The GPIO controller interrupt is an active low level interrupt. BIOS configures it and sets it to 0s. The driver clears the interrupt status then sets FCH::GPIO::GPIO\_WAKE\_INTERRUPT\_MASTER\_SWITCH[eoi] = 1 after the interrupt is acknowledged.

##### 14.1.14.1.3 Pure GPIO

All the configuration bits (Bit[x]) used in the following steps come from GPIOBank0, GPIOBank1 and GPIOBank2. If the pin is used as a GPIO, see 14.3.15.2 [GPIO Registers]:

1. Bit[23:17] provides the state of the control on pull-up, pull-down, drive strength, output enable, output value, and pin value through bit[16].

**14.1.15 Sticky/non-sticky attribute of PMIO registers**

The table below list the sticky/non-sticky attribute of all PMIO registers. Sticky/non-sticky attribute is directly related to the register's reset condition:

A "Sticky" register is reset in the following conditions:

- Resume Reset: This reset is asserted in G3 state, and deasserted during G3 to S5 transition.
- System Reset: From system reset button

A "Non-sticky" register is reset in the following conditions:

- Resume Reset: This reset is asserted in G3 state, and deasserted during G3 to S5 transition.
- System Reset: From system reset button.
- S0 Reset events: Some events that happen in S0 state, such as CF9 and Keyboard Reset.
- Traditional sleep: S3 and S5 states.

See 14.3.14.3 [Power Management (PM) Registers and Standard ACPI Registers] for PMIO register descriptions.

*Table 244: PMIO Register Sticky/Non-sticky Attributes*

| Dword# | Offset    | Sticky/Non-sticky                                                                                                                                |
|--------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | 00h ~ 03h | Sticky                                                                                                                                           |
| 1      | 04h ~ 07h | Sticky                                                                                                                                           |
| 2      | 08h ~ 0Bh | Sticky                                                                                                                                           |
| 3      | 0Ch ~ 0Fh | Sticky                                                                                                                                           |
| 4      | 10h ~ 13h | Sticky                                                                                                                                           |
| 5      | 14h ~ 17h | Sticky                                                                                                                                           |
| 6      | 18h ~ 1Bh | Sticky                                                                                                                                           |
| 7      | 1Ch ~ 1Fh | Sticky                                                                                                                                           |
| 8      | 20h ~ 23h | Sticky                                                                                                                                           |
| 9      | 24h ~ 27h | Sticky                                                                                                                                           |
| 10     | 28h ~ 2Bh | Sticky                                                                                                                                           |
| 11     | 2Ch ~ 2Fh | Sticky                                                                                                                                           |
| 12     | 30h ~ 33h | Sticky                                                                                                                                           |
| 13     | 34h ~ 37h | Sticky                                                                                                                                           |
| 14     | 38h ~ 3Bh | Sticky                                                                                                                                           |
| 15     | 3Ch ~ 3Fh | Sticky                                                                                                                                           |
| 16     | 40h ~ 43h | Sticky                                                                                                                                           |
| 17     | 44h ~ 47h | [31] non-sticky<br>[30] sticky<br>[29] non-sticky<br>[28:25] sticky<br>[24:0] read-only                                                          |
| 18     | 48h ~ 4Bh | Sticky                                                                                                                                           |
| 19     | 4Ch ~ 4Fh | Sticky                                                                                                                                           |
| 20     | 50h ~ 53h | Non-sticky                                                                                                                                       |
| 21     | 54h ~ 57h | Sticky                                                                                                                                           |
| 22     | 58h ~ 5Bh | [31:28] no reset (these bits are actually in RTC)<br>[27:23] sticky<br>[22] read-only<br>[21] write-only, read always return 0<br>[20:19] sticky |

|    |           |                                                                                                                                                       |
|----|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
|    |           | [18] read-only<br>[17] sticky<br>[16:0] sticky                                                                                                        |
| 23 | 5Ch ~ 5Fh | [31:24] no reset (these bits are actually in RTC)<br>[23:11] sticky<br>[10] reserved, read always return 1<br>[9:0] sticky                            |
| 24 | 60h ~ 63h | Sticky                                                                                                                                                |
| 25 | 64h ~ 67h | Sticky                                                                                                                                                |
| 26 | 68h ~ 6Bh | Sticky                                                                                                                                                |
| 27 | 6Ch ~ 6Fh | Sticky                                                                                                                                                |
| 28 | 70h ~ 73h | Sticky                                                                                                                                                |
| 29 | 74h ~ 77h | [31:8] sticky<br>[7] write-only, read always return 0<br>[6:0] sticky                                                                                 |
| 30 | 78h ~ 7Bh | Non-sticky                                                                                                                                            |
| 31 | 7Ch ~ 7Fh | Non-sticky                                                                                                                                            |
| 32 | 80h ~ 83h | Sticky                                                                                                                                                |
| 33 | 84h ~ 87h | Non-sticky                                                                                                                                            |
| 34 | 88h ~ 8Bh | Non-sticky                                                                                                                                            |
| 35 | 8Ch ~ 8Fh | Sticky                                                                                                                                                |
| 36 | 90h ~ 93h | Sticky                                                                                                                                                |
| 37 | 94h ~ 97h | Sticky                                                                                                                                                |
| 38 | 98h ~ 9Bh | Sticky                                                                                                                                                |
| 39 | 9Ch ~ 9Fh | Read-only                                                                                                                                             |
| 40 | A0h ~ A3h | [31:16] non-sticky<br>[15:0] is read-only if 0xD0[23] is 1, otherwise it is non-sticky                                                                |
| 41 | A4h ~ A7h | [31] non-sticky<br>[30:29] reserved, read always return 0<br>[28:0] is read-only if [31] is 1, otherwise it is non-sticky                             |
| 42 | A8h ~ Abh | [31] non-sticky<br>[30:0] sticky                                                                                                                      |
| 43 | Ach ~ Afh | Reserved, read always return 0                                                                                                                        |
| 44 | B0h ~ B3h | [31:28] non-sticky<br>[27:23] read-only<br>[22:0] non-sticky                                                                                          |
| 45 | B4h ~ B7h | Sticky                                                                                                                                                |
| 46 | B8h ~ BBh | [31:21] sticky<br>[20] read-only<br>[19:0] sticky                                                                                                     |
| 47 | BCh ~ BFh | Sticky                                                                                                                                                |
| 48 | C0h ~ C3h | [31:30] sticky<br>[29:0] is sticky if 0xC4[2] is 1, otherwise it is read-only                                                                         |
| 49 | C4h ~ C7h | [31:11] sticky<br>[10] write-only, read always return 0 [9] sticky [8] reserved, read always return 0 [7:1] sticky [0] reserved, read always return 0 |
| 50 | C8h ~ CBh | Sticky                                                                                                                                                |
| 51 | CCh ~ CFh | Sticky                                                                                                                                                |
| 52 | D0h ~ D3h | [31:16] sticky<br>[15:8] sticky, but is indirect access to four 8-bit registers [7:0]                                                                 |

|    |           |                                                                                                           |
|----|-----------|-----------------------------------------------------------------------------------------------------------|
|    |           | sticky                                                                                                    |
| 53 | D4h ~ D7h | Sticky                                                                                                    |
| 54 | D8h ~ DBh | [31:16] sticky<br>[15:0] is indirect access to a stack of registers, which are reset by Resume Reset only |
| 55 | DCh ~ DFh | Sticky                                                                                                    |
| 56 | E0h ~ E3h | Sticky                                                                                                    |
| 57 | E4h ~ E7h | Sticky                                                                                                    |
| 58 | E8h ~ Ebh | Sticky                                                                                                    |
| 59 | Ech ~ Efh | [31:20] sticky<br>[19] read-only<br>[18:0] sticky                                                         |
| 60 | F0h ~ F3h | Sticky                                                                                                    |
| 61 | F4h ~ F7h | Sticky                                                                                                    |
| 62 | F8h ~ FBh | Sticky                                                                                                    |
| 63 | FCh ~ FFh | Sticky                                                                                                    |

## 14.1.16 CLK IP Overview

### 14.1.16.1 iCLK/eCLK mode

There are two clocking modes in which the processor can be brought up based on the boot strap pin SPI\_CLK: iCLK and eCLK clocking modes. iCLK clocking mode means that chip takes an internal crystal generated clocks as reference clock while eCLK clocking mode means that chip takes an external clock as reference clock.

BIOS should read FCH::MISC::STRAPSTATUS[clkgenstrap] to determine the clocking mode.

### 14.1.16.2 CGPLL SSC Enable/Disable

The CGPLL [SSC](#) is disabled by default.

In eCLK mode, Spread Spectrum Clocking (SSC) should remain disabled.

In iCLK mode, SSC can be enabled on the [BSP](#) (socket 0) CG1PLL only. The CGPLL default spread amount is -0.375%. Alternatively, the spread amount can be set to -0.3%.

SSC enable sequence only need be executed during initial boot-up sequence or reset sequence triggered by pushing reset button, that means, for example, this sequence should not be executed after system exit Sleep State.

Note: When spread spectrum is turned on, the effective clock frequency will be slowed down by the same amount. This is applicable to functions that derive clocks from this PLL. For example, the effective CPU core frequency will be slowed down approximately ½ of the spread amount.

#### 14.1.16.2.1 CG1PLL -0.375% SSC Programming

Software can enable the CG1PLL default -0.375% [SSC](#) setting by the following programming:

1. FCH::MISC::CGPLLCONFIG1[cg1\_spread\_spectrum\_enable] = 1
2. FCH::MISC::CGPLLCONFIG3[cg1pll\_fracn\_en\_override] = 1
3. FCH::MISC::MISCCLKCNTRL0[cg1\_cfg\_update\_req] = 1

Software can disable the CG1PLL default -0.375% SSC setting by the following programming:

1. FCH::MISC::CGPLLCONFIG1[cg1\_spread\_spectrum\_enable] = 0
2. FCH::MISC::CGPLLCONFIG3[cg1pll\_fracn\_en\_override] = 0
3. FCH::MISC::MISCCLKCNTRL0[cg1\_cfg\_update\_req] = 1

#### 14.1.16.2.2 CG1PLL -0.3% SSC Programming

Software can enable the CG1PLL -0.3% [SSC](#) setting by the following programming:

1. FCH::MISC::CGPLLCONFIG4[cg1pll\_fcw1\_frac\_override] = 0x1333
2. FCH::MISC::CGPLLCONFIG5[cg1pll\_fcw\_slew\_frac\_override] = 0x3A
3. FCH::MISC::CGPLLCONFIG1[cg1\_spread\_spectrum\_enable] = 1
4. FCH::MISC::CGPLLCONFIG3[cg1pll\_fracn\_en\_override] = 1
5. FCH::MISC::MISCCLKCNTRL0[cg1\_fbdiv\_loaden] = 1
6. FCH::MISC::MISCCLKCNTRL0[cg1\_cfg\_update\_req] = 1

Software can disable the CG1PLL -0.3% SSC setting by the following programming:

1. Program FCH::MISC::CGPLLCONFIG4[cg1pll\_fcw1\_frac\_override] = 0
2. Program FCH::MISC::CGPLLCONFIG5[cg1pll\_fcw\_slew\_frac\_override] = 0
3. Program FCH::MISC::CGPLLCONFIG1[cg1\_spread\_spectrum\_enable] = 0
4. Program FCH::MISC::CGPLLCONFIG3[cg1pll\_fracn\_en\_override] = 0
5. Program FCH::MISC::MISCCLKCNTRL0[cg1\_fbdiv\_loaden] = 0
6. Program FCH::MISC::MISCCLKCNTRL0[cg1\_cfg\_update\_req] = 1

#### 14.1.16.3 Global A-Link/B-Link Clock Gating

A-Link and B-Link clocks are 2 global clocks used inside [FCH](#) for most of sub-IPs.

Software enables the global A-Link clock gate off function by programming:

1. FCH::MISC::CLKGATEDCNTL[alinkclk\_gateoffen] = 1
2. FCH::PM::ISACONTROL[abclkgateen] = 1

Software enables the global B-Link clock gate off function by programming:

1. FCH::MISC::CLKGATEDCNTL[blinkclk\_gateoffen] = 1
2. FCH::PM::ISACONTROL[abclkgateen] = 1
3. FCH::SDP::SDP\_DISCONNECT\_CTL[SDP\_ONESIDE\_DISCONNECT\_EN\_HOST] = 1
4. FCH::SDP::SDP\_DISCONNECT\_CTL[SDP\_ONESIDE\_DISCONNECT\_EN\_DMA] = 1

FCH::MISC::CLKGATEDCNTL[alinkclk\_gateoffen] and FCH::MISC::CLKGATEDCNTL[blinkclk\_gateoffen] are sticky bits.

FCH::PM::ISACONTROL[abclkgateen] is a non-sticky bit that should be re-programmed to 1 after resets defined in 14.1.15 [Sticky/non-sticky attribute of PMIO registers] if global A-Link/B-Link gating functions been enabled.

Note: FCH::SDP::SDP\_DISCONNECT\_CTL[SDP\_ONESIDE\_DISCONNECT\_EN\_HOST] = 1 and FCH::SDP::SDP\_DISCONNECT\_CTL[SDP\_ONESIDE\_DISCONNECT\_EN\_DMA] = 1 to enable FCH\_SDP disconnect to allow LCLK deep sleep.

**14.1.16.4 GPP CLKREQB Mapping Table***Table 245: GPP ClkREQB Mapping*

| CLKREQB Map Register                                           | Package Pin Name                           |
|----------------------------------------------------------------|--------------------------------------------|
| FCH::MISC::GPPCLKCNTROL[gpp_clk11_clock_request_mapping]       | BP_AGPI0115/<br>CLK_REQ11_L                |
| FCH::MISC::GPPCLKCNTROL[gpp_clk12_clock_request_mapping]       | BP_AGPI0116/<br>CLK_REQ12_L                |
| FCH::MISC2::RMT_CLKCNTL_0_REG[gpp_clk01_clock_request_mapping] | BP_AGPI0257/<br>SGPIO1_CLK/<br>CLK_REQ01_L |
| FCH::MISC2::RMT_CLKCNTL_0_REG[gpp_clk02_clock_request_mapping] | BP_AGPI0258/<br>SGPIO2_CLK/<br>CLK_REQ02_L |

**14.1.17 Reset Overview**

Below is definition of all kind of reset types:

Type 0 reset (S5 Reset): RsmRst and UsrRstB

Type 1 reset (reset initiated by SW or system): CF9, KBRst, Sync\_flood, ASF\_remote\_reset, Fail\_boot, Watchdog timer reset, toggling of PwrGood (SLP\_S3#/SLP\_S5# remain deasserted at high), SHUTDOWN command, INIT/PORT92

Type 2 reset (Sleep Reset): S3/S4/S5 reset

Type 3 reset (Fatal\_error\_reset or reset caused by hw exception): 4s-shutdown, thermal trip, ASF\_remotePowerDown

Type 4 reset (any reset from above): Type 0 or Type 1 or Type 2 or Type 3

*Table 246: Reset Type*

| Register block                                                                 | Power domain             | Reset source                                                               |
|--------------------------------------------------------------------------------|--------------------------|----------------------------------------------------------------------------|
| PCI Configure                                                                  | S0                       | Type 4                                                                     |
| <a href="#">SMI</a> (14.3.5 [SMI Registers])                                   | S5                       | Some register: Type 0 or Type 3<br>Some register: Type 4                   |
| PMIO (14.3.14.3 [Power Management (PM) Registers and Standard ACPI Registers]) | S5                       | Some register: Type 0<br>Some register: Type 4                             |
| ACPI (14.3.14.3 [Power Management (PM) Registers and Standard ACPI Registers]) | S5                       | Type 0 or Type 3                                                           |
| ASF (14.3.12.2 [ASF (Alert Standard Format) Registers])                        | S5                       | Type 0                                                                     |
| SMBUS (14.3.12.3 [SMBUS Register])                                             | S5                       | Type 4                                                                     |
| WatchDog (14.3.9 [Watchdog Timer (WDT) Registers])                             | S5                       | Type 0                                                                     |
| HPET (14.3.6 [High Precision Event Timer (HPET) Registers])                    | S0                       | Type 4                                                                     |
| IOMUX (14.3.15.1 [IOMUX Registers])                                            | S5 (0-42)<br>S0 (67-148) | S5-IoMux: Type 0<br>S0-IoMux: Type 4                                       |
| MISC (14.3.14.1 [Miscellaneous (MISC) Registers])                              | S5                       | Most registers are Type 0<br>Some CLK register are Type 4, 0x4C, 0x94, ... |
| Serial Debug                                                                   | S5                       | Type 0                                                                     |
| Shadow System Counter                                                          | S5                       | Type 0 or Type 1                                                           |
| GPIO-0 (14.3.15.2 [GPIO Registers])                                            | S5                       | Type 0 or Type 1                                                           |



|                                                                             |    |                  |
|-----------------------------------------------------------------------------|----|------------------|
| GPIO-1 (14.3.15.2 [GPIO Registers])                                         | S0 | Type 4           |
| GPIO-2 (14.3.15.2 [GPIO Registers])                                         | S0 | Type 4           |
| GPIO-3 (14.3.15.2 [GPIO Registers])                                         | S0 | Type 4           |
| Wake Alarm (ACDC timer) (14.3.10 [Wake Alarm Device (AcDcTimer) Registers]) | S5 | Type 0           |
| AOAC (14.3.11 [Always On Always Connected (AOAC) Registers])                | S5 | Type 0 or Type 1 |
| Remote-GPIO+IoMuX (14.3.15.3 [REMOTE GPIO and IOMUX Registers])             | S5 | Type 0           |

Below is a simplified top level reset paths.

When a reset is generated from acpi\_s5, all S0 logic within [FCH](#) are reset by it. Other IPs within SOC are not by it directly. Instead, they are being reset by [MP1](#). CpuRst (FCH\_SOC\_RESETh) is connected to MP1 and was treated as an interrupt. Upon assertion of this signal, MP1 will proceed to do its "house cleaning" activities first; then it will issue resets to all other IPs that are not inside FCH. CupPwrGood (FCH\_SOC\_PWROK) is the actual reset to MP1. When this signal is deasserted (to low), MP1 will propagate it reset to all IPs within SOC. We will define a warm reset as an assertion of CupRstB (low) and cold reset of toggling of both CpuRstB (low) and CpuPwrGood (low).

Table 247: Reset Behavior Overview

| S5 reset                                    | Behavior                                                                                                                                             |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| RSMRST_L                                    | Reset all logic (S0 and S5). In addition, SOC will go back to default power state (always-off, always-on, previous state).                           |
| SYS_RST_L                                   | Reset all logic (S0 and S5).                                                                                                                         |
| Sleep induced reset                         | Behavior                                                                                                                                             |
| SLEEP3/4/5                                  | Sleep entry to S3/4/5 will assert cold reset to all S0 logic prior to assertion of SLP_S3_L/SLP_S5_L.                                                |
| Pin included reset                          | Behavior                                                                                                                                             |
| PWR_BTN_L                                   | 4 second override – unconditional shutdown (Sleep S5) → generates cold reset and will reset all S0 logic..                                           |
| THERMTRIP_L                                 | Over temperature indicator from CPU – unconditional shutdown (Sleep S5) → generates cold reset and will reset all S0 logic.                          |
| KBRST_L                                     | Keyboard reset → reset to S0 logic; configurable to warm or cold reset.                                                                              |
| PWR_GOOD                                    | Deassertion of PWRGOOD will always generate cold reset to S0 logic.                                                                                  |
| SW induced reset                            | Behavior                                                                                                                                             |
| CF9                                         | Write to CF9 → reset S0 logic; configurable to warm or cold reset; or momentary S0 → S5 → S0 transition.                                             |
| SW write to PMIO_Reg 0xC4[6], ResetAllAcpi  | Generates reset to S0 and S5 logic.                                                                                                                  |
| HW based reset                              | Behavior                                                                                                                                             |
| watchdog timer                              | watchdog timer → reset S0 logic; configurable to warm or cold reset.                                                                                 |
| AMD (boot_timer) watchdog timer             | AMD defined watchdog timer → reset S0 logic; configurable to warm or cold reset.                                                                     |
| Remote Command reset                        | Behavior                                                                                                                                             |
| ASF – remote reset                          | Reset S0 logic; configurable to warm or cold reset.                                                                                                  |
| ASF – remote sleep                          | Remote sleep S5 command – logic will sequence to S5 and assert cold reset to all S0 logic.                                                           |
| Internal events                             | Behavior                                                                                                                                             |
| <a href="#">Shutdown</a> (message from CPU) | Triple faults in CPU will cause an internal SHUTDOWN message broadcasted. FCH will generate a reset to S0 logic; configurable to warm or cold reset. |

|                      |                                                                                                                                                                                          |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SYNC_FLOOD (message) | Internal data fabric logic detects an error (eg. parity error) and broadcasts an internal SYNC_FLOOD message. FCH will generate a reset to S0 logic; configurable to warm or cold reset. |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 14.1.18 ROM Address Mapping Support in FCH implementation

[FCH](#) needs to implement the following register bits.

1. spirom\_addr\_32 (FCH::LPCHOSTSPIREG::ADDR32CTRL0): 0: 24-bit SPI ROM address, default. 1: 32-bit SPI ROM address.
2. spirom\_page[31:24] (FCH::LPCHOSTSPIREG::ADDR32CTRL3[spirom\_page\_31\_24]): these bits will be used to control the address mapping, "XOR" will be used for spirom\_page[31:24] and HostMemAddr[31:24] to produce SPIROMAddr[31:24].

Following table shows the address mapping.

Table 248: ROM Address Mapping Support in FCH implementation

|                                                                          | ROM1 (1MB)                                    | ROM2 (16MB)                                    | ROM3 (64MB)                                    |
|--------------------------------------------------------------------------|-----------------------------------------------|------------------------------------------------|------------------------------------------------|
| Default address space                                                    | 0x0000_0000 - 0x000F_FFFF                     | 0xFF00_0000 - 0xFFFF_FFFF                      | 0xFD_0000_0000 - 0xFD_03FF_FFFF                |
| SPI ROM, 24bit address                                                   | 0xF0_0000 - 0xFF_FFFF<br>(top 1MB of 16MB)    | 0x00_0000 – 0xFF_FFFF<br>(16MB)                | 0x00_0000 – 0xFF_FFFF<br>(16MB)                |
| SPI ROM, 32bit address:<br>Bit[31:24] are before XOR<br>with SPIROM_page | 0xFFF0_0000 – 0xFFFF_FFFF<br>(top 1MB of 4GB) | 0xFF00_0000 – 0xFFFF_FFFF<br>(top 16MB of 4GB) | 0xFC00_0000 – 0xFFFF_FFFF<br>(top 64MB of 4GB) |

### 14.1.19 RTC

#### 14.1.19.1 RTC Register

The value in Time/Alarm or CMOS registers or CMOS RAM is undefined/indeterministic when power up first time. It is highly recommended to add a checksum or CRC over CMOS RAM so BIOS can detect whether CMOS has been corrupted or erased.

You can see RTC registers at 14.3.1 [Legacy Block Configuration Registers (IO)]

#### 14.1.19.2 RTC Extended Register

The value in these RTC Extended registers is undefined/indeterministic when power up first time.

You can see RTC Extended registers at 14.3.14.4 [RTC External Registers]

### 14.1.20 Strap Definition

Table 249: Strap Pins/Function

| Ball Name | Strap Name | Type | Default Value | Description                                                                                                                                                 |
|-----------|------------|------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ESPI_CLK0 | CLKGEN     | II   | NA            | Defines clock generator<br>"0" – Use 100MHz PCIE clock as reference clock and generate internal clocks only. Note a 48MHz crystal is still required in this |

|                        |                       |    |    |                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------|-----------------------|----|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                        |                       |    |    | configuration<br>"1" – Use 48MHz crystal clock and generate both internal and external clocks External pull-up/Pull-down is required on the pin. (This strap does have a weak internal pull-down resistor (50K).)                                                                                                                                                                                         |
| UART1_TXD,<br>LFRAME_L | ROMTYPE[1:0]          | II | NA | The strap pin should be configured to the corresponding state that matches the hardware ROM type installed. External pull-up/Pull-down is required on the pin.<br>00: Reserved<br>01: SPI ROM<br>10: Reserved<br>11: eSPI without SAFS support                                                                                                                                                            |
| ESPI_CLK1              | AlternateImage        | II | NA | Note this strap does not have any real hardware function. It only serves as a software bit for <a href="#">PSP</a> to read.<br>This strap does have a weak internal pull-down resistor (50K). This strap has no meaning if the ROM is 16MB or less<br>1: PSP should modify SPI page register bits [25:24] to remap physical ROM to upper image<br>0: PSP should not modify SPI page register bits [25:24] |
| ESPI_CLK2              | WAFL_TRAINING_DISABLE | II | 0  | Note this strap does not have any real hardware function. It only serves as a software bit to be read by PSP/SMU. This strap does have a weak internal pull-down resistor (50K).<br><ul style="list-style-type: none"> <li>WAFL_TRAINING_DISABLE = 0 (default) means WAFL link will be trained.</li> <li>WAFL_TRAINING_DISABLE = 1 means WAFL link will not be trained.</li> </ul>                        |

Even though some of the pins may have integrated pull-up enabled, external pull-up/down are required on all strap pins except for the two pins explicitly mentioned above.

Type I strap: Use BP\_RSMRST\_L rising edge to latch the strap value. The strap pad can only locate in S5.

Type II strap: Use BP\_PWR\_GOOD as source and after S0 power up, count another counter (16 osc clock cycles), then latch the strap done. All straps on S0 pads use this type of strap.

*Table 250: Address Space Mapping under APB BUS*

| Function name | Address Mapping |
|---------------|-----------------|
| I2C_0         | 0xFEDC_2xxx     |
| I2C_1         | 0xFEDC_3xxx     |
| I2C_2         | 0xFEDC_4xxx     |
| I2C_3         | 0xFEDC_5xxx     |

|         |                                   |
|---------|-----------------------------------|
| I2C_4   | 0xFEDC_6xxx                       |
| DMA_0   | 0xFEDC_7xxx                       |
| DMA_1   | 0xFEDC_8xxx                       |
| UART_0  | 0xFEDC_9xxx                       |
| UART_1  | 0xFEDC_Axxx                       |
| I2C_5   | 0xFEDC_Bxxx                       |
| DMA_2   | 0xFEDC_Cxxx                       |
| DMA_3   | 0xFEDC_Dxxx                       |
| UART_2  | 0xFEDC_Exxx                       |
| UART_3  | 0xFEDC_Fxxx                       |
| I3C_0   | 0xFEDD_2xxx                       |
| I3C_1   | 0xFEDD_3xxx                       |
| I3C_2   | 0xFEDD_4xxx                       |
| I3C_3   | 0xFEDD_6xxx                       |
| ACPI    | 0xFED8_0000 8KB memory Space      |
| ACPI    | 0xFD_F95F_FF00 256B memory Space  |
| SPI     | 0xFD_F95F_FE00 256B memory Space  |
| SPI ROM | Top of 4G & 1M; 256B memory Space |

## 14.2 FCH BTS Check Table

Table 251: FCH BTS Common Table

| RegisterField                               | Type  | Content |
|---------------------------------------------|-------|---------|
| FCH::MISC::ECOBIT2[xtalpciclk_notalwaysen]  | Check | 1b      |
| FCH::MISC::ECOBIT2[sel_intr_disable_method] | Check | 1b      |
| FCH::MISC::CLKGATEDCNTL[blinkclk_gateoffen] | Check | 1b      |
| FCH::MISC::CLKGATEDCNTL[alinkclk_gateoffen] | Check | 1b      |

## 14.3 Registers

### 14.3.1 Legacy Block Configuration Registers (IO)

#### 14.3.1.1 Registers

| IOx00C01_x00 [PCI INT[H#,G#,F#,E#,D#,C#,B#,A#] Map] (FCH::IO::PciIntMap)     |                                                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Read-write. Reset: 1Fh.                                                      |                                                                                                                                                                                                                                                                            |
| _aliasIO; IOx00C01_x00; IO=0000_0000h; DataPortWrite=FCH::IO::PCI_INTR_INDEX |                                                                                                                                                                                                                                                                            |
| Bits                                                                         | Description                                                                                                                                                                                                                                                                |
| 7:5                                                                          | Reserved.                                                                                                                                                                                                                                                                  |
| 4:0                                                                          | <b>Pci2IntrMap.</b> Read-write. Reset: 1Fh. If (FCH::IO::PCI_INTR_INDEX[7] == 1) then Pci2IntrMap specifies mapping of INT[H#:A#] to APIC interrupt number. If (FCH::IO::PCI_INTR_INDEX[7] == 0) then Pci2IntrMap specifies mapping of INT[H#:A#] to PIC interrupt number. |

**IOx00C01\_x08 [Intr\_Misc\_Map] (FCH::IO::IntrMiscMap)**

Read-write. Reset: 00h.

\_aliasIO; IOx00C01\_x08; IO=0000\_0000h; DataPortWrite=FCH::IO::PCI\_INTR\_INDEX

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                         |       |             |    |                             |    |                           |    |                        |    |                                              |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|----|-----------------------------|----|---------------------------|----|------------------------|----|----------------------------------------------|
| 7:6   | <b>Pci2Intr15Map.</b> Read-write. Reset: 0h.<br><b>ValidValues:</b> <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>IRQ15 mapped to legacy IDE.</td></tr> <tr> <td>1h</td><td>IRQ15 mapped to SATA IDE.</td></tr> <tr> <td>2h</td><td>IRQ15 mapped to SATA2.</td></tr> <tr> <td>3h</td><td>IRQ15 come from Serial IRQ or PCI interrupt.</td></tr> </table> | Value | Description | 0h | IRQ15 mapped to legacy IDE. | 1h | IRQ15 mapped to SATA IDE. | 2h | IRQ15 mapped to SATA2. | 3h | IRQ15 come from Serial IRQ or PCI interrupt. |
| Value | Description                                                                                                                                                                                                                                                                                                                                                                         |       |             |    |                             |    |                           |    |                        |    |                                              |
| 0h    | IRQ15 mapped to legacy IDE.                                                                                                                                                                                                                                                                                                                                                         |       |             |    |                             |    |                           |    |                        |    |                                              |
| 1h    | IRQ15 mapped to SATA IDE.                                                                                                                                                                                                                                                                                                                                                           |       |             |    |                             |    |                           |    |                        |    |                                              |
| 2h    | IRQ15 mapped to SATA2.                                                                                                                                                                                                                                                                                                                                                              |       |             |    |                             |    |                           |    |                        |    |                                              |
| 3h    | IRQ15 come from Serial IRQ or PCI interrupt.                                                                                                                                                                                                                                                                                                                                        |       |             |    |                             |    |                           |    |                        |    |                                              |
| 5:4   | <b>Pci2Intr14Map.</b> Read-write. Reset: 0h.<br><b>ValidValues:</b> <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>IRQ14 mapped to legacy IDE.</td></tr> <tr> <td>1h</td><td>IRQ14 mapped to SATA IDE.</td></tr> <tr> <td>2h</td><td>IRQ14 mapped to SATA2.</td></tr> <tr> <td>3h</td><td>IRQ14 mapped to Serial IRQ or PCI interrupt.</td></tr> </table> | Value | Description | 0h | IRQ14 mapped to legacy IDE. | 1h | IRQ14 mapped to SATA IDE. | 2h | IRQ14 mapped to SATA2. | 3h | IRQ14 mapped to Serial IRQ or PCI interrupt. |
| Value | Description                                                                                                                                                                                                                                                                                                                                                                         |       |             |    |                             |    |                           |    |                        |    |                                              |
| 0h    | IRQ14 mapped to legacy IDE.                                                                                                                                                                                                                                                                                                                                                         |       |             |    |                             |    |                           |    |                        |    |                                              |
| 1h    | IRQ14 mapped to SATA IDE.                                                                                                                                                                                                                                                                                                                                                           |       |             |    |                             |    |                           |    |                        |    |                                              |
| 2h    | IRQ14 mapped to SATA2.                                                                                                                                                                                                                                                                                                                                                              |       |             |    |                             |    |                           |    |                        |    |                                              |
| 3h    | IRQ14 mapped to Serial IRQ or PCI interrupt.                                                                                                                                                                                                                                                                                                                                        |       |             |    |                             |    |                           |    |                        |    |                                              |
| 3     | <b>PciIntrIrq12.</b> Read-write. Reset: 0. 0=IMC as IRQ12 input source. 1=Serial IRQ or PCI devices as IRQ12 input source.                                                                                                                                                                                                                                                          |       |             |    |                             |    |                           |    |                        |    |                                              |
| 2     | <b>PciIntrIrq8.</b> Read-write. Reset: 0. 0=RTC is IRQ8 input source. 1=Serial IRQ or PCI devices as IRQ8 input source.                                                                                                                                                                                                                                                             |       |             |    |                             |    |                           |    |                        |    |                                              |
| 1     | <b>PciIntrIrq1.</b> Read-write. Reset: 0. 0=IMC as IRQ1 input source. 1=Serial IRQ or PCI devices as IRQ1 input source.                                                                                                                                                                                                                                                             |       |             |    |                             |    |                           |    |                        |    |                                              |
| 0     | <b>PciIntrIrq0.</b> Read-write. Reset: 0. 0=8254 timer as IRQ0 input source. 1=Serial IRQ or PCI devices as IRQ0 input source.                                                                                                                                                                                                                                                      |       |             |    |                             |    |                           |    |                        |    |                                              |

**IOx00C01\_x09 [Intr\_Misc0Map] (FCH::IO::IntrMisc0Map)**

Read-write. Reset: E7h.

\_aliasIO; IOx00C01\_x09; IO=0000\_0000h; DataPortWrite=FCH::IO::PCI\_INTR\_INDEX

| Bits | Description                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>IntrDelay.</b> Read-write. Reset: 1. INTR 600 ns delay.                                                                                                                                           |
| 6    | <b>IRQ12FilterEnable.</b> Read-write. Reset: 1. IRQ12 filter enable.                                                                                                                                 |
| 5    | <b>IRQ1FilterEnable.</b> Read-write. Reset: 1. IRQ1 filter enable.                                                                                                                                   |
| 4    | <b>IrqInputEn.</b> Read-write. Reset: 0. 0=Mask off IRQ input. 1=Enable IRQ input.                                                                                                                   |
| 3    | <b>MaskIrq1Irq12.</b> Read-write. Reset: 0. 0=Enable IRQ1 and IRQ12. 1=Mask off IRQ1 and IRQ12.                                                                                                      |
| 2    | <b>Merge_Ec_irq12.</b> Read-write. Reset: 1. 0=Route serial IRQ12 to USB IRQ12 input. 1=Route IMC IRQ12 to USB IRQ12 input.                                                                          |
| 1    | <b>Merge_Ec_irq1.</b> Read-write. Reset: 1. 0=Route serial IRQ1 to USB IRQ1 input. 1=Route IMC IRQ1 to USB IRQ1 input.                                                                               |
| 0    | <b>IntMap.</b> Read-write. Reset: 1. 0=INT0 in IOAPIC comes from IRQ0 in PIC, INT2 in IOAPIC comes from INTR in PIC. 1=INT2 in IOAPIC comes from IRQ0 in PIC, INT0 in IOAPIC comes from INTR in PIC. |

**IOx00C01\_x0A [Intr\_Misc1Map] (FCH::IO::IntrMisc1Map)**

Read-write. Reset: 00h.

\_aliasIO; IOx00C01\_x0A; IO=0000\_0000h; DataPortWrite=FCH::IO::PCI\_INTR\_INDEX

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>HPET</b> . Read-write. Reset: 00h. Writes to this register update the bits in FCH::TMR::HPET::TMR0_CONF_CAP_H [7:0], FCH::TMR::HPET::TMR1_CONF_CAP_H [7:0] and FCH::TMR::HPET::TMR2_CONF_CAP_H [7:0]; All 3 registers, HPETx104[7:0], HPETx124[7:0], and HPETx144[7:0], are updated at the same time. FCH::IO::IntrMisc1Map updates the lower 8 bits. FCH::IO::IntrMisc2Map updates the upper 8 bits. |

**IOx00C01\_x0B [Intr\_Misc2Map] (FCH::IO::IntrMisc2Map)**

Read-write. Reset: 00h.

\_aliasIO; IOx00C01\_x0B; IO=0000\_0000h; DataPortWrite=FCH::IO::PCI\_INTR\_INDEX

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>HPET</b> . Read-write. Reset: 00h. Writes to this register update this bits in FCH::TMR::HPET::TMR0_CONF_CAP_H [15:8], FCH::TMR::HPET::TMR1_CONF_CAP_H [15:8] and FCH::TMR::HPET::TMR2_CONF_CAP_H [15:8]; All 3 registers, HPETx104[15:8], HPETx124[15:8], and HPETx144[15:8], are updated at the same time. FCH::IO::IntrMisc1Map updates the lower 8 bits. FCH::IO::IntrMisc2Map updates the upper 8 bits. |

**IOx00C01\_x0C [PCIInterruptMap] (FCH::IO::PCIInterruptMap)**

Read-write. Reset: 1Fh.

\_aliasIO; IOx00C01\_x0C; IO=0000\_0000h; DataPortWrite=FCH::IO::PCI\_INTR\_INDEX

| Bits | Description                                                                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:5  | Reserved.                                                                                                                                                                                                                                                                                                                     |
| 4:0  | <b>Pci2IntrMap</b> . Read-write. Reset: 1Fh. If (FCH::IO::PCI_INTR_INDEX[7] == 1), then Pci2IntrMap specifies the APIC interrupt number that the corresponding PCI interrupt maps to. If (FCH::IO::PCI_INTR_INDEX[7] == 0), then Pci2IntrMap specifies the PIC interrupt number that the corresponding PCI interrupt maps to. |

**LEGACYIOx00000000 (FCH::IO::DMA\_CH\_0)**

Read-write. Reset: 0000h.

Dma\_Ch 0 register

\_aliasHOSTLEGACY; LEGACYIOx00000000; LEGACYIO=FF00\_0000h

| Bits | Description                                                                   |
|------|-------------------------------------------------------------------------------|
| 15:0 | <b>dma_ch_0</b> . Read-write. Reset: 0000h. DMA1 Ch0 Base and Current Address |

**LEGACYIOx00000002 (FCH::IO::DMA\_CH\_1)**

Read-write. Reset: 0000h.

Dma\_Ch 1 register

\_aliasHOSTLEGACY; LEGACYIOx00000002; LEGACYIO=FF00\_0000h

| Bits | Description                                                                   |
|------|-------------------------------------------------------------------------------|
| 15:0 | <b>dma_ch_1</b> . Read-write. Reset: 0000h. DMA1 Ch1 Base and Current Address |

**LEGACYIOx00000004 (FCH::IO::DMA\_CH\_2)**

Read-write. Reset: 0000h.

Dma\_Ch 2 register

\_aliasHOSTLEGACY; LEGACYIOx00000004; LEGACYIO=FF00\_0000h

| Bits | Description                                                                   |
|------|-------------------------------------------------------------------------------|
| 15:0 | <b>dma_ch_2</b> . Read-write. Reset: 0000h. DMA2 Ch2 Base and Current Address |

**LEGACYIOx00000006 (FCH::IO::DMA\_CH\_3)**

Read-write. Reset: 0000h.

Dma\_Ch 3 register

\_aliasHOSTLEGACY; LEGACYIOx00000006; LEGACYIO=FF00\_0000h

**Bits Description**15:0 **dma\_ch\_3.** Read-write. Reset: 0000h. DMA1 Ch3 Base and Current Address**LEGACYIOx00000008 (FCH::IO::DMA\_STATUS)**

Read-write. Reset: 00h.

Dma\_Status register

\_aliasHOSTLEGACY; LEGACYIOx00000008; LEGACYIO=FF00\_0000h

**Bits Description**7:0 **dma\_status.** Read-write. Reset: 00h. Returns status when read command for write**LEGACYIOx00000009 (FCH::IO::DMA\_WRITEREQUEST)**

Read-write. Reset: 00h.

Dma\_WriteRequest register

\_aliasHOSTLEGACY; LEGACYIOx00000009; LEGACYIO=FF00\_0000h

**Bits Description**7:0 **dma\_writerequest.** Read-write. Reset: 00h. Request register.**LEGACYIOx0000000A (FCH::IO::DMA\_WRITEMASK)**

Read-write. Reset: 00h.

Dma\_WriteMask register

\_aliasHOSTLEGACY; LEGACYIOx0000000A; LEGACYIO=FF00\_0000h

**Bits Description**7:0 **dma\_writemask.** Read-write. Reset: 00h. Channel mask register.**LEGACYIOx0000000B (FCH::IO::DMA\_WRITEMODE)**

Read-write. Reset: 00h.

Dma\_WriteMode register

\_aliasHOSTLEGACY; LEGACYIOx0000000B; LEGACYIO=FF00\_0000h

**Bits Description**7:0 **dma\_writemode.** Read-write. Reset: 00h. Mode register.**LEGACYIOx0000000C (FCH::IO::DMA\_CLEAR)**

Read-write. Reset: 00h.

Dma\_Clear register

\_aliasHOSTLEGACY; LEGACYIOx0000000C; LEGACYIO=FF00\_0000h

**Bits Description**7:0 **dma\_clear.** Read-write. Reset: 00h. Channel 0-3 DMA clear byte pointer**LEGACYIOx0000000D (FCH::IO::DMA\_MASTERCLR)**

Read-write. Reset: 00h.

Dma\_MasterClr register

\_aliasHOSTLEGACY; LEGACYIOx0000000D; LEGACYIO=FF00\_0000h

**Bits Description**7:0 **dma\_masterclr.** Read-write. Reset: 00h. Intermediate register.

**LEGACYIOx0000000E (FCH::IO::DMA\_CLRMASK)**

Read-write. Reset: 00h.

Dma\_ClrMask register

\_aliasHOSTLEGACY; LEGACYIOx0000000E; LEGACYIO=FF00\_0000h

| Bits | Description                                                            |
|------|------------------------------------------------------------------------|
| 7:0  | <b>dma_clrmask.</b> Read-write. Reset: 00h. Channel 0-3 DMA Clear Mask |

**LEGACYIOx0000000F (FCH::IO::DMA\_ALLMASK)**

Read-write. Reset: 00h.

Dma\_AllMask register

\_aliasHOSTLEGACY; LEGACYIOx0000000F; LEGACYIO=FF00\_0000h

| Bits | Description                                                |
|------|------------------------------------------------------------|
| 7:0  | <b>dma_allmask.</b> Read-write. Reset: 00h. Mask register. |

**LEGACYIOx00000020 (FCH::IO::INTRCNTRL1REG1)**

Read-write. Reset: 00h.

IntrCntrl1Reg1 register

\_aliasHOSTLEGACY; LEGACYIOx00000020; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>intrcntrl1reg1.</b> Read-write. Reset: 00h.<br><b>Description:</b> IRQ0 IRQ7:<br>Read IRR, ISR<br>Write ICW1, OCW2, OCW3 |

**LEGACYIOx00000021 (FCH::IO::INTRCNTRL1REG2)**

Read-write. Reset: 00h.

IntrCntrl1Reg2 register

\_aliasHOSTLEGACY; LEGACYIOx00000021; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>intrcntrl1reg2.</b> Read-write. Reset: 00h.<br><b>Description:</b> IRQ0 IRQ7:<br>Read IMR<br>Write ICW2, ICW3, ICW4, OCW1 |

**LEGACYIOx00000022 (FCH::IO::IMCR\_INDEX)**

Read-write. Reset: 00h.

IMCR\_Index register

\_aliasHOSTLEGACY; LEGACYIOx00000022; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>imcr_index.</b> Read-write. Reset: 00h. The IMCR is supported by two read/writeable IO ports 22/23h which are used as index and data port respectively. The actual IMCR register is located at index 70h. |

**LEGACYIOx00000023 (FCH::IO::IMCR\_DATA)**

Read-write. Reset: 00h.

IMCR\_Data register

\_aliasHOSTLEGACY; LEGACYIOx00000023; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>imcr_data.</b> Read-write. Reset: 00h. The IMCR is supported by two read/writeable IO ports 22/23h which are used as index and data port respectively. The actual IMCR register is located at index 70h and it is at bit 0. The actual IMCR bit can only be accessed when bit port 22 is set to 70h. Default value of IMCR is 0. |



**LEGACYIOx00000040 (FCH::IO::TIMERCH0)**

Read-write. Reset: 00h.

TimerCh0 register

\_aliasHOSTLEGACY; LEGACYIOx00000040; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>timerch0.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                                 |
|      | <b>Description:</b> 8254 Timer 1: Counter 0 Data Port<br>This timer is known as the System Clock timer and it is always on. It is clocked internally by OSC/12 (1.19318MHz), and asserts IRQ0 every time the timer rolls over. This timer is used for time-of-day, diskette time-out, and other system timing functions. |

**LEGACYIOx00000041 (FCH::IO::TIMERCH1)**

Read-write. Reset: 00h.

TimerCh1 register

\_aliasHOSTLEGACY; LEGACYIOx00000041; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>timerch1.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                          |
|      | <b>Description:</b> 8254 Timer 1: Counter 1 Data Port<br>This timer is normally used for ISA refresh cycles and is also clocked by OSC/12 (1.19818MHz). Since this refresh function is no longer needed (we don't have an external ISA bus), it can be used as a general purpose timing function. |

**LEGACYIOx00000042 (FCH::IO::TIMERCH2)**

Read-write. Reset: 00h.

TimerCh2 register

\_aliasHOSTLEGACY; LEGACYIOx00000042; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>timerch2.</b> Read-write. Reset: 00h.                                                                                                                                                                                    |
|      | <b>Description:</b> 8254 Timer 1: Counter 2 Data Port<br>This is the speaker tone generator and is enabled by IO port 61H. It is clocked by OSC/12 (1.19318MHz) and directly drives the output SPKR that goes to a speaker. |

**LEGACYIOx00000043 (FCH::IO::TMR1CNTRLWORD)**

Write-only. Reset: 00h.

Tmr1CntrlWord register: This is the control word to access the 8254 timer 1. It is used to select which counter will be accessed and how it will be accessed. This register specifies the counter, the operating mode, the order and size of the count value, and whether it counts down in a 16 bit or [BCD](#) format.

If a counter is programmed to read or write two-byte counts, the following precaution applies: A program must not transfer control between writing the first and second byte to another routine which also writes into that same counter otherwise, the counter will be loaded with an incorrect value. The count must always be completely loaded with both bytes.

Tmr1CntrlWord is write-only. Read always returns FF.

\_aliasHOSTLEGACY; LEGACYIOx00000043; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:6  | <b>counterselect.</b> Write-only. Reset: 0h.<br><b>Description:</b> 00: Select counter 0<br>01: Select counter 1<br>10: Select counter 2<br>11: Read back command                                                                                                                            |
| 5:4  | <b>cmmandselect.</b> Write-only. Reset: 0h.<br><b>Description:</b> 00: Counter latch command<br>01: Read/write least significant byte<br>10: Read/write most significant byte<br>11: Read/write least, and then most significant byte                                                        |
| 3:1  | <b>modeselect.</b> Write-only. Reset: 0h.<br><b>Description:</b> 000: Asserts OUT signal at end of count<br>001: Hardware re-triggerable one-shot<br>010: Rate generator<br>011: Square wave output<br>100: Software triggered strobe<br>101: Hardware triggered strobe<br>110 111: Not used |
| 0    | <b>cntdownselect.</b> Write-only. Reset: 0.<br><b>Description:</b> 0: Binary countdown<br>1: <a href="#">BCD</a> countdown                                                                                                                                                                   |

**LEGACYIOx00000060 (FCH::IO::IO\_PORT\_60)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; LEGACYIOx00000060; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>io_port_60.</b> Read-write. Reset: 00h. This is for Legacy USB emulation, please refer to 4.25 USB Legacy register for detail |

**LEGACYIOx00000061 (FCH::IO::NMI\_STATUS)**

Read-write.

Nmi\_Status register: Independent read and write registers will be accessed at this port. When writing to port 61H, bits[3:0] allow software to enable/disable parity error NMI's and control the speaker timer. When reading port 61H, status on parity errors, speaker count, speaker control and refresh cycles is returned.

\_aliasHOSTLEGACY; LEGACYIOx00000061; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>parerr_nmi.</b> Read-write. Reset: X. NMI is caused by parity error (either PERR# or SERR#). [Read-only]                                                                                    |
| 6    | <b>iochk_nmi.</b> Read-write. Reset: X. NMI is triggered by serial IOCHK. [Read-only]                                                                                                          |
| 5    | <b>spkrclk.</b> Read-write. Reset: X. The output of the counter 2. [Read-only]                                                                                                                 |
| 4    | <b>refclk.</b> Read-write. Reset: X. The output of the counter 1 (8254). [Read-only]                                                                                                           |
| 3    | <b>iochk_nmi_en.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Enable IoChk to NMI generation<br>1: Disable IoChk to NMI generation                                                      |
| 2    | <b>parity_nmi_en.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Enable Parity Error to NMI generation (from SERR# or PERR#)<br>1: Disable Parity Error to NMI generation and clear bit 7 |
| 1    | <b>spkrtmrenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Speaker timer off<br>1: Speaker timer on                                                                                 |
| 0    | <b>spkrenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable counter 2<br>1: Enable counter 2                                                                                    |

**LEGACYIOx00000064 (FCH::IO::IO\_PORT\_64)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; LEGACYIOx00000064; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>io_port_64.</b> Read-write. Reset: 00h. This is for Legacy USB emulation, please refer to 4.25 USB Legacy register for detail |

**LEGACYIOx00000070 (FCH::IO::NMI\_ENABLE)**

Read-write. Reset: 80h.

Nmi\_Enable register

\_aliasHOSTLEGACY; LEGACYIOx00000070; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                            |
|------|--------------------------------------------------------------------------------------------------------|
| 7    | <b>nmienableb.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: NMI enable<br>1: NMI disable        |
| 6:0  | <b>rtc_address_port.</b> Read-write. Reset: 00h. This is used with either internal RTC or external RTC |

**LEGACYIOx00000071 (FCH::IO::RTCDATAPORT)**

Read-write. Reset: 00h.

RtcDataPort

\_aliasHOSTLEGACY; LEGACYIOx00000071; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7:0  | <b>rtc_data_port.</b> Read-write. Reset: 00h. This is used with either internal RTC or external RTC |

**LEGACYIOx00000072 (FCH::IO::ALTERNATRTCADDRPORT)**

Read-write. Reset: 00h.

AlternatRtcAddrPort

\_aliasHOSTLEGACY; LEGACYIOx00000072; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>alternatrtcaddrport.</b> Read-write. Reset: 00h. This is used with internal RTC. This port allows user to specify the full 8 bit address (instead of bank0/bank1 indexing) to access the 256 byte RTC RAM |

**LEGACYIOx00000073 (FCH::IO::ALTERNATRTCdataport)**

Read-write. Reset: 00h.

AlternatRtcDataPort

\_aliasHOSTLEGACY; LEGACYIOx00000073; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>alternatrtc_data_port.</b> Read-write. Reset: 00h. This is used with internal RTC in conjunction with port h72 |

**LEGACYIOx00000080 (FCH::IO::IO\_PORT\_80)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; LEGACYIOx00000080; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>io_port_80.</b> Read-write. Reset: 00h. Write to this port will send data to be displayed on "POST Code display module" |

**LEGACYIOx00000081 (FCH::IO::DMA\_PAGECH2)**

Read-write. Reset: 00h.

Dma\_PageCh2 register

\_aliasHOSTLEGACY; LEGACYIOx00000081; LEGACYIO=FF00\_0000h

| Bits | Description                                                         |
|------|---------------------------------------------------------------------|
| 7:0  | <b>dma_pagech2.</b> Read-write. Reset: 00h. DMA2 ch 2 page register |

**LEGACYIOx00000082 (FCH::IO::DMA\_PAGECH3)**

Read-write. Reset: 00h.

Dma\_PageCh3 register

\_aliasHOSTLEGACY; LEGACYIOx00000082; LEGACYIO=FF00\_0000h

| Bits | Description                                                         |
|------|---------------------------------------------------------------------|
| 7:0  | <b>dma_pagech3.</b> Read-write. Reset: 00h. DMA2 ch 3 page register |

**LEGACYIOx00000083 (FCH::IO::DMA\_PAGECH1)**

Read-write. Reset: 00h.

Dma\_PageCh1 register

\_aliasHOSTLEGACY; LEGACYIOx00000083; LEGACYIO=FF00\_0000h

| Bits | Description                                                         |
|------|---------------------------------------------------------------------|
| 7:0  | <b>dma_pagech1.</b> Read-write. Reset: 00h. DMA2 ch 1 page register |

**LEGACYIOx00000084 (FCH::IO::DMA\_PAGE\_RESERVED1)**

Read-write. Reset: 00h.

Dma\_Page\_Reserved1 register

\_aliasHOSTLEGACY; LEGACYIOx00000084; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma_page_reserved1.</b> Read-write. Reset: 00h. DMA Page Reserved1 register |

**LEGACYIOx00000085 (FCH::IO::DMA\_PAGE\_RESERVED2)**

Read-write. Reset: 00h.

Dma\_Page\_Reserved2 register

\_aliasHOSTLEGACY; LEGACYIOx00000085; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma_page_reserved2.</b> Read-write. Reset: 00h. DMA Page Reserved2 register |

**LEGACYIOx00000086 (FCH::IO::DMA\_PAGE\_RESERVED3)**

Read-write. Reset: 00h.

Dma\_Page\_Reserved3 register

\_aliasHOSTLEGACY; LEGACYIOx00000086; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma_page_reserved3.</b> Read-write. Reset: 00h. DMA Page Reserved3 register |

**LEGACYIOx00000087 (FCH::IO::DMA\_PAGECH0)**

Read-write. Reset: 00h.

Dma\_PageCh0 register

\_aliasHOSTLEGACY; LEGACYIOx00000087; LEGACYIO=FF00\_0000h

| Bits | Description                                                         |
|------|---------------------------------------------------------------------|
| 7:0  | <b>dma_pagech0.</b> Read-write. Reset: 00h. DMA2 ch 0 page register |

**LEGACYIOx00000088 (FCH::IO::DMA\_PAGE\_RESERVED4)**

Read-write. Reset: 00h.

Dma\_Page\_Reserved4 register

\_aliasHOSTLEGACY; LEGACYIOx00000088; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma_page_reserved4.</b> Read-write. Reset: 00h. Dma Page Reserved4 register |

**LEGACYIOx00000089 (FCH::IO::DMA\_PAGECH6)**

Read-write. Reset: 00h.

Dma\_PageCh6 register

\_aliasHOSTLEGACY; LEGACYIOx00000089; LEGACYIO=FF00\_0000h

| Bits | Description                                                         |
|------|---------------------------------------------------------------------|
| 7:0  | <b>dma_pagech6.</b> Read-write. Reset: 00h. DMA2 ch 6 page register |

**LEGACYIOx0000008A (FCH::IO::DMA\_PAGECH7)**

Read-write. Reset: 00h.

Dma\_PageCh7 register

\_aliasHOSTLEGACY; LEGACYIOx0000008A; LEGACYIO=FF00\_0000h

| Bits | Description                                                         |
|------|---------------------------------------------------------------------|
| 7:0  | <b>dma_pagech7.</b> Read-write. Reset: 00h. DMA2 ch 7 page register |

**LEGACYIOx0000008B (FCH::IO::DMA\_PAGECH5)**

Read-write. Reset: 00h.

Dma\_PageCh5 register

\_aliasHOSTLEGACY; LEGACYIOx0000008B; LEGACYIO=FF00\_0000h

| Bits | Description                                                         |
|------|---------------------------------------------------------------------|
| 7:0  | <b>dma_pagech5.</b> Read-write. Reset: 00h. DMA2 ch 5 page register |

**LEGACYIOx0000008C (FCH::IO::DMA\_PAGE\_RESERVED5)**

Read-write. Reset: 00h.

Dma\_Page\_Reserved5 register

\_aliasHOSTLEGACY; LEGACYIOx0000008C; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma_page_reserved5.</b> Read-write. Reset: 00h. Dma Page Reserved5 register |

**LEGACYIOx0000008D (FCH::IO::DMA\_PAGE\_RESERVED6)**

Read-write. Reset: 00h.

Dma\_Page\_Reserved6 register

\_aliasHOSTLEGACY; LEGACYIOx0000008D; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma_page_reserved6.</b> Read-write. Reset: 00h. Dma Page Reserved6 register |

**LEGACYIOx0000008E (FCH::IO::DMA\_PAGE\_RESERVED7)**

Read-write. Reset: 00h.

Dma\_Page\_Reserved7 register

\_aliasHOSTLEGACY; LEGACYIOx0000008E; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma_page_reserved7.</b> Read-write. Reset: 00h. Dma Page Reserved7 register |

**LEGACYIOx0000008F (FCH::IO::DMA\_REFRESH)**

Read-write. Reset: 00h.

Dma\_Refresh register

\_aliasHOSTLEGACY; LEGACYIOx0000008F; LEGACYIO=FF00\_0000h

| Bits | Description                                                         |
|------|---------------------------------------------------------------------|
| 7:0  | <b>dma_refresh.</b> Read-write. Reset: 00h. DMA2 ch4 page register. |

**LEGACYIOx00000092 (FCH::IO::FASTINIT)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; LEGACYIOx00000092; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                                                                                                                                                                 |
| 1    | <b>a20enb.</b> Read-write. Reset: 0.<br><b>Description:</b> A20Enable Bar bit if set to 1 A20M# function is disabled.                                                                                                                                                                                                                     |
| 0    | <b>fastinit.</b> Read-write. Reset: 0.<br><b>Description:</b> FAST_INIT. This read/write bit provides a fast software executed processor reset function. Writing a 1 to this bit will cause the INIT assertion for approximately 4ms. Before another INIT pulse can be generated via this register, this bit must be written back to a 0. |

**LEGACYIOx000000A0 (FCH::IO::INTRCNTRL2REG1)**

Read-write. Reset: 00h.

IntrCntrl2Reg1 register

\_aliasHOSTLEGACY; LEGACYIOx000000A0; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>intrcntrl2reg1.</b> Read-write. Reset: 00h.<br><b>Description:</b> IRQ8 IRQ15:<br>Read IRR, ISR<br>Write ICW1, OCW2, OCW3 |

**LEGACYIOx000000A1 (FCH::IO::INTRCNTRL2REG2)**

Read-write. Reset: 00h.

IntrCntrl2Reg2 register

\_aliasHOSTLEGACY; LEGACYIOx000000A1; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>intrcntrl2reg2.</b> Read-write. Reset: 00h.<br><b>Description:</b> IRQ8 IRQ15:<br>Read IMR<br>Write ICW2, ICW3, ICW4, OCW1 |

**LEGACYIOx000000C0 (FCH::IO::DMA2\_CH4ADDR)**

Read-write. Reset: 00h.

Dma2\_Ch4Addr register

\_aliasHOSTLEGACY; LEGACYIOx000000C0; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma2_ch4addr.</b> Read-write. Reset: 00h. DMA2 Ch4 Base and Current Address |

**LEGACYIOx000000C2 (FCH::IO::DMA2\_CH4CNT)**

Read-write. Reset: 00h.

Dma2\_Ch4Cnt register

\_aliasHOSTLEGACY; LEGACYIOx000000C2; LEGACYIO=FF00\_0000h

| Bits | Description                                                                 |
|------|-----------------------------------------------------------------------------|
| 7:0  | <b>dma2_ch4cnt.</b> Read-write. Reset: 00h. DMA2 Ch4 Base and Current Count |

**LEGACYIOx000000C4 (FCH::IO::DMA2\_CH5ADDR)**

Read-write. Reset: 00h.

Dma2\_Ch5Addr register

\_aliasHOSTLEGACY; LEGACYIOx000000C4; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma2_ch5addr.</b> Read-write. Reset: 00h. DMA2 Ch5 Base and Current Address |

**LEGACYIOx000000C6 (FCH::IO::DMA2\_CH5CNT)**

Read-write. Reset: 00h.

Dma2\_Ch5Cnt register

\_aliasHOSTLEGACY; LEGACYIOx000000C6; LEGACYIO=FF00\_0000h

| Bits | Description                                                                 |
|------|-----------------------------------------------------------------------------|
| 7:0  | <b>dma2_ch5cnt.</b> Read-write. Reset: 00h. DMA2 Ch4 Base and Current Count |

**LEGACYIOx000000C8 (FCH::IO::DMA2\_CH6ADDR)**

Read-write. Reset: 00h.

Dma2\_Ch6Addr register

\_aliasHOSTLEGACY; LEGACYIOx000000C8; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma2_ch6addr.</b> Read-write. Reset: 00h. DMA2 Ch6 Base and Current Address |

**LEGACYIOx000000CA (FCH::IO::DMA2\_CH6CNT)**

Read-write. Reset: 00h.

Dma2\_Ch6Cnt register

\_aliasHOSTLEGACY; LEGACYIOx000000CA; LEGACYIO=FF00\_0000h

| Bits | Description                                                                 |
|------|-----------------------------------------------------------------------------|
| 7:0  | <b>dma2_ch6cnt.</b> Read-write. Reset: 00h. DMA2 Ch6 Base and Current Count |

**LEGACYIOx000000CC (FCH::IO::DMA2\_CH7ADDR)**

Read-write. Reset: 00h.

Dma2\_Ch7Addr register

\_aliasHOSTLEGACY; LEGACYIOx000000CC; LEGACYIO=FF00\_0000h

| Bits | Description                                                                    |
|------|--------------------------------------------------------------------------------|
| 7:0  | <b>dma2_ch7addr.</b> Read-write. Reset: 00h. DMA2 Ch5 Base and Current Address |

**LEGACYIOx000000CE (FCH::IO::DMA2\_CH7CNT)**

Read-write. Reset: 00h.

Dma2\_Ch7Cnt register

\_aliasHOSTLEGACY; LEGACYIOx000000CE; LEGACYIO=FF00\_0000h

| Bits | Description                                                                      |
|------|----------------------------------------------------------------------------------|
| 7:0  | <b>dma2_ch7cnt.</b> Read-write. Reset: 00h. Channel 7 DMA base and current count |

**LEGACYIOx000000D0 (FCH::IO::DMA2\_STATUS)**

Read-write. Reset: 00h.

Dma\_Status register

\_aliasHOSTLEGACY; LEGACYIOx000000D0; LEGACYIO=FF00\_0000h

| Bits | Description                                                     |
|------|-----------------------------------------------------------------|
| 7:0  | <b>dma_status.</b> Read-write. Reset: 00h. DMA2 status register |

**LEGACYIOx000000D2 (FCH::IO::DMA2\_WRITEREQUEST)**

Read-write. Reset: 00h.

Dma\_WriteRequest register

\_aliasHOSTLEGACY; LEGACYIOx000000D2; LEGACYIO=FF00\_0000h

| Bits | Description                                                            |
|------|------------------------------------------------------------------------|
| 7:0  | <b>dma_writerequest.</b> Read-write. Reset: 00h. DMA2 request register |

**LEGACYIOx000000D4 (FCH::IO::DMA2\_WRITEMASK)**

Read-write. Reset: 00h.

Dma\_WriteMask register

\_aliasHOSTLEGACY; LEGACYIOx000000D4; LEGACYIO=FF00\_0000h

| Bits | Description                                                              |
|------|--------------------------------------------------------------------------|
| 7:0  | <b>dma_writemask.</b> Read-write. Reset: 00h. DMA2 channel mask register |

**LEGACYIOx000000D6 (FCH::IO::DMA2\_WRITEMODE)**

Read-write. Reset: 00h.

Dma\_WriteMode register

\_aliasHOSTLEGACY; LEGACYIOx000000D6; LEGACYIO=FF00\_0000h

| Bits | Description                                                      |
|------|------------------------------------------------------------------|
| 7:0  | <b>dma_writemode.</b> Read-write. Reset: 00h. DMA2 mode register |

**LEGACYIOx000000D8 (FCH::IO::DMA2\_CLEAR)**

Read-write. Reset: 00h.

Dma\_Clear register

\_aliasHOSTLEGACY; LEGACYIOx000000D8; LEGACYIO=FF00\_0000h

| Bits | Description                                                              |
|------|--------------------------------------------------------------------------|
| 7:0  | <b>dma_clear.</b> Read-write. Reset: 00h. Channel 4-7 clear byte pointer |



**LEGACYIOx000000DA (FCH::IO::DMA2\_MASTER\_CLEAR)**

Read-write. Reset: 00h.

Dma\_Clear register

\_aliasHOSTLEGACY; LEGACYIOx000000DA; LEGACYIO=FF00\_0000h

**Bits Description**7:0 **dma\_clear.** Read-write. Reset: 00h. Channel 4-7 DMA master clear**LEGACYIOx000000DC (FCH::IO::DMA2\_CLRMASK)**

Read-write. Reset: 00h.

Dma\_ClrMask register

\_aliasHOSTLEGACY; LEGACYIOx000000DC; LEGACYIO=FF00\_0000h

**Bits Description**7:0 **dma\_clrmask.** Read-write. Reset: 00h. Channel 4-7 DMA Clear Mask**LEGACYIOx000000DE (FCH::IO::DMA2\_ALLMASK)**

Read-write. Reset: 00h.

Dma\_AllMask register

\_aliasHOSTLEGACY; LEGACYIOx000000DE; LEGACYIO=FF00\_0000h

**Bits Description**7:0 **dma\_allmask.** Read-write. Reset: 00h. DMA2 mask register**LEGACYIOx000000F0 (FCH::IO::NCP\_ERROR)**

Read-write. Reset: 00h.

NCP\_Error register: In addition to the WarmBoot function, writing to this port will assert IGNNE# if FERR# is true. If FERR# is false, then write to this port will not assert IGNNE#.

\_aliasHOSTLEGACY; LEGACYIOx000000F0; LEGACYIO=FF00\_0000h

**Bits Description**

7 **warmboot.** Read-write. Reset: 0.  
**Description:** Warm or cold boot indicator  
0: Cold  
1: Warm, this bit is set when any value is written to this register

6:0 Reserved.

**LEGACYIOx000004D0 (FCH::IO::INTREDGECONTROL)**

Read-write. Reset: 0000h.

IntrEdgeControl register: This register programs each interrupt to be either edge or level sensitive.

\_aliasHOSTLEGACY; LEGACYIOx000004D0; LEGACYIO=FF00\_0000h

| Bits | Description                                                                           |
|------|---------------------------------------------------------------------------------------|
| 15   | <b>irq15control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge |
| 14   | <b>irq14control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge |
| 13   | Reserved.                                                                             |
| 12   | <b>irq12control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge |
| 11   | <b>irq11control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge |
| 10   | <b>irq10control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge |
| 9    | <b>irq9control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge  |
| 8    | <b>irq8control.</b> Read-write. Reset: 0. (Read Only) Always Edge                     |
| 7    | <b>irq7control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge  |
| 6    | <b>irq6control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge  |
| 5    | <b>irq5control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge  |
| 4    | <b>irq4control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge  |
| 3    | <b>irq3control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge  |
| 2    | Reserved.                                                                             |
| 1    | <b>irq1control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge  |
| 0    | <b>irq0control.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Level<br>0: Edge  |

**LEGACYIOx00000C00 (FCH::IO::PCI\_INTR\_INDEX)**

Read-write. Reset: 00h.

Pci\_Intr\_Index register

\_aliasHOSTLEGACY; LEGACYIOx00000C00; LEGACYIO=FF00\_0000h

| Bits | Description                                                        |
|------|--------------------------------------------------------------------|
| 7:0  | <b>pci_intr_index.</b> Read-write. Reset: 00h. PCI interrupt index |

**LEGACYIOx00000C01 (FCH::IO::PCI\_INTR\_DATA)**

Read-write. Reset: 1Fh.

Pci\_Intr\_Data register

\_aliasHOSTLEGACY; LEGACYIOx00000C01; LEGACYIO=FF00\_0000h

| Bits | Description                                                            |
|------|------------------------------------------------------------------------|
| 7:0  | <b>pci_intr_data.</b> Read-write. Reset: 1Fh. PCI redirection register |

**LEGACYIOx00000C14 (FCH::IO::PCI\_ERROR)**

Read-write.

Pci\_Error register. This register is enabled/disabled by PM\_Reg:00[20].

\_aliasHOSTLEGACY; LEGACYIOx00000C14; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | Reserved.                                                                                                                                                                                                  |
| 3    | <b>perr_nmi.</b> Read-write. Reset: 1.<br><b>Description:</b> Enable NMI generation from PERR#<br>0: Enable<br>1: Disable                                                                                  |
| 2    | <b>serr_nmi.</b> Read-write. Reset: 1.<br><b>Description:</b> Enable NMI generation from SERR#<br>0: Enable<br>1: Disable                                                                                  |
| 1    | <b>perr_nmi_status.</b> Read-write. Reset: X. Set to 1 when NMI generation is enabled and PERR# has been asserted due to a PCI data parity error. Cleared by writing a one to port 61h, bit 2. [Read-only] |
| 0    | <b>serr_nmi_status.</b> Read-write. Reset: X. Set to 1 when NMI generation is enabled and SERR# has been asserted due to a PCI error. Cleared by writing a one to port 61h, bit 2. [Read-only]             |

**LEGACYIOx00000CF9 (FCH::IO::IO\_PORT\_CF9)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; LEGACYIOx00000CF9; LEGACYIO=FF00\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                            |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | Reserved.                                                                                                                                                                                                                                                                                              |
| 3:1  | <b>io_port_cf9.</b> Read-write. Reset: 0h.<br><b>Description:</b> 010: do init, send INIT HT message<br>011: do reset, generate <a href="#">Warm reset</a><br>111: do full reset, generate <a href="#">Cold reset</a> and place system in S5 state for 3 to 5 seconds<br>All other values are reserved |
| 0    | Reserved.                                                                                                                                                                                                                                                                                              |

**RTCHOSTx00000000 (FCH::IO::SECONDS)**

Read-write.

Seconds register

\_aliasHOSTLEGACY; RTCHOSTx00000000; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>seconds.</b> Read-write. Reset: XXXXXXXXb.                                                                                                                                                                                                                                                                                             |
|      | <b>Description:</b> Binary-Code-Decimal format. Range:00 59<br>This register can be set by software (set bit of Register B = 1) or can be automatically updated by hardware every second. When set by software, hardware updating is disabled.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000001 (FCH::IO::SECONDS\_ALARM)**

Read-write.

Seconds Alarm register

\_aliasHOSTLEGACY; RTCHOSTx00000001; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>seconds_alarm.</b> Read-write. Reset: XXXXXXXXb.                                                                                                                                                                                                                                                                             |
|      | <b>Description:</b> Binary-Code-Decimal format.<br>If set bit = 1, the Seconds Alarm Register will never match with Seconds Register, else<br>If bits [7:6] = [11], the Seconds Alarm Register always matches with Seconds Register.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000002 (FCH::IO::MINUTES)**

Read-write.

Minutes register

\_aliasHOSTLEGACY; RTCHOSTx00000002; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>minutes.</b> Read-write. Reset: XXXXXXXXb.                                                                                                                                                                                                                                                                                             |
|      | <b>Description:</b> Binary-Code-Decimal format. Range:00 59<br>This register can be set by software (set bit of Register B = 1) or can be automatically updated by hardware every minute. When set by software, hardware updating is disabled.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000003 (FCH::IO::MINUTES\_ALARM)**

Read-write.

Minutes Alarm register

\_aliasHOSTLEGACY; RTCHOSTx00000003; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>minutes_alarm.</b> Read-write. Reset: XXXXXXXXb.                                                                                                                                                                                                                                                                             |
|      | <b>Description:</b> Binary-Code-Decimal format.<br>If set bit = 1, the Minutes Alarm Register will never match with Minutes Register, else<br>If bits [7:6] = [11], the Minutes Alarm Register always matches with Minutes Register.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000004 (FCH::IO::HOURS)**

Read-write.

Hours register

\_aliasHOSTLEGACY; RTCHOSTx00000004; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                             |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>hours.</b> Read-write. Reset: XXXXXXXXb.                                                                                                                                                                                                                                                                                             |
|      | <b>Description:</b> Binary-Code-Decimal format. Range:00 23<br>This register can be set by software (set bit of Register B = 1) or can be automatically updated by hardware every hour. When set by software, hardware updating is disabled.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000005 (FCH::IO::HOURS\_ALARM)**

Read-write.

Hours Alarm register

\_aliasHOSTLEGACY; RTCHOSTx00000005; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>hours_alarm.</b> Read-write. Reset: XXXXXXXXb.                                                                                                                                                                                                                                                                       |
|      | <b>Description:</b> Binary-Code-Decimal format.<br>If set bit = 1, the Hours Alarm Register will never match with Hours Register, else<br>If bits [7:6] = [11], the Hours Alarm Register always matches with Hours Register.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000006 (FCH::IO::DAY\_OF\_WEEK)**

Read-write.

Day of Week register

\_aliasHOSTLEGACY; RTCHOSTx00000006; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>day_of_week.</b> Read-write. Reset: XXXXXXXXb.                                                                                                                                                                                                                                                                                                                                                                                           |
|      | <b>Description:</b> Binary-Code-Decimal format. Range: 01 07 (Sunday = 1). No leap year correction capability. Leap year correction has to be done by software.<br>This register can be set by a software (set bit of Register B = 1) or can be automatically updated by hardware everyday. When set by software, hardware updating is disabled.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000007 (FCH::IO::DATE\_OF\_MONTH)**

Read-write.

Date of Month register

\_aliasHOSTLEGACY; RTCHOSTx00000007; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>date_of_month.</b> Read-write. Reset: XXXXXXXXb.                                                                                                                                                                                                                                                                                                                                                                               |
|      | <b>Description:</b> Binary-Code-Decimal format. Range: 01 28 for February and no leap year capability. Leap year correction has to be done by software.<br>This register can be set by software (set bit of Register B = 1) or can be automatically updated by hardware everyday. When set by software, hardware updating is disabled.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000008 (FCH::IO::MONTH)**

Read-write.

Month register

\_aliasHOSTLEGACY; RTCHOSTx00000008; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>month.</b> Read-write. Reset: XXXXXXXXb.<br><b>Description:</b> Binary-Code-Decimal format. Range: 01 12. No leap year correction capability. Leap year correction has to be done by software.<br>This register can be set by software (set bit of Register B = 1) or can be automatically updated by hardware every month. When set by software, hardware updating is disabled.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000009 (FCH::IO::YEAR)**

Read-write.

Year register

\_aliasHOSTLEGACY; RTCHOSTx00000009; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>year.</b> Read-write. Reset: XXXXXXXXb.<br><b>Description:</b> Binary-Code-Decimal format. Range: 00 99. No leap year correction capability. Leap year correction has to be done by software.<br>This register can be set by software (set bit of Register B = 1) or can be automatically updated by hardware every year. When set by software, hardware updating is disabled.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx0000000A (FCH::IO::REGISTER\_A)**

Read-write.

Register A: Control register

\_aliasHOSTLEGACY; RTCHOSTx0000000A; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>update_in_progress_uip.</b> Read-write. Reset: X.<br><b>Description:</b> If set bit = 1, UIP is cleared.<br>If UIP = 1, the update transfer will soon occur.<br>If UIP = 0, the update transfer will not occur for at least 244us. [Read-only]<br>Note: the value of this register is undefined/indeterministic when power up first time.                                                                                          |
| 6:5  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4    | <b>bank_selection_dv0.</b> Read-write. Reset: X.<br><b>Description:</b> DV0 = 0 selects Bank 0 DV0 = 1 selects Bank 1.<br>The SB800 has an alternate way to access the RAM without the use of bank select bit. Port 72/73 can be used as the index to access the full 256 bytes of RAM directly.<br>Note: the value of this register is undefined/indeterministic when power up first time.                                           |
| 3    | <b>rate_selection_rs3.</b> Read-write. Reset: X.                                                                                                                                                                                                                                                                                                                                                                                      |
| 2    | <b>rate_selection_rs2.</b> Read-write. Reset: X.                                                                                                                                                                                                                                                                                                                                                                                      |
| 1    | <b>rate_selection_rs1.</b> Read-write. Reset: X.                                                                                                                                                                                                                                                                                                                                                                                      |
| 0    | <b>rate_selection_rs0.</b> Read-write. Reset: X.<br><b>Description:</b> These four rate-selection bits select one of the 13 taps on the 15-stage frequency divider or disable the divider output (flat output signal). The tap selected can be used to generate a periodic interrupt. See the following table for the frequency selection.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx0000000B (FCH::IO::REGISTER\_B)**

Read-write.

Register B: Control register

\_aliasHOSTLEGACY; RTCHOSTx0000000B; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>set_new_time_set.</b> Read-write. Reset: X.<br><b>Description:</b> If set bit = 1, no internal updating for Time Registers is allowed. If set bit = 0, the Time Registers are updated every second.<br>Note: the value of this register is undefined/indeterministic when power up first time. |
| 6    | <b>periodic_interrupt_enable_pie.</b> Read-write. Reset: X.<br><b>Description:</b> PIE enables the Periodic Interrupt Flag (PF) bit in Register C to assert IRQ.<br>Note: the value of this register is undefined/indeterministic when power up first time.                                       |
| 5    | <b>alarm_interrupt_enable_aie.</b> Read-write. Reset: X.<br><b>Description:</b> AIE enables the Alarm Flag (AF) bit in Register C to assert IRQ.<br>Note: the value of this register is undefined/indeterministic when power up first time.                                                       |
| 4    | <b>update_ended_interrupt_enable_uie.</b> Read-write. Reset: X.<br><b>Description:</b> UIE enables the Update End Flag (UF) bit in Register C to assert IRQ. If set bit = 1, UIE is cleared.<br>Note: the value of this register is undefined/indeterministic when power up first time.           |
| 3:2  | Reserved.                                                                                                                                                                                                                                                                                         |
| 1    | <b>hourmode.</b> Read-write. Reset: X.<br><b>Description:</b> Hour mode<br>0: 12 hour mode<br>1: 24 hour mode<br>Note: the value of this register is undefined/indeterministic when power up first time.                                                                                          |
| 0    | <b>daylight_saving_enable.</b> Read-write. Reset: X.<br><b>Description:</b> Both this bit and RtcExt_Reg: 00h bit[0] need to be set to 1 to enable RTC daylight saving feature.<br>Note: the value of this register is undefined/indeterministic when power up first time.                        |

**RTCHOSTx0000000C (FCH::IO::REGISTER\_C)**

Read-only.

Register C: Control register

\_aliasHOSTLEGACY; RTCHOSTx0000000C; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>interrupt_request_flag_irqf.</b> Read-only. Reset: X.<br><b>Description:</b> Logically, IRQF = (PF*PIE)+(AF*AIE)+(UF*UIE)+(WF*WIE) where WF and WIE are defined in Extended Control Register 4A and 4B. Reading Register C clears IRQF bit. Any time the IRQF bit is set to one, the #IRQ pin is driven low.<br>Note: the value of this register is undefined/indeterministic when power up first time. |
| 6    | <b>xperiodic_interrupt_flag_pf.</b> Read-only. Reset: X.<br><b>Description:</b> This bit is set to one when an edge is detected on the selected tap (through RS3 to RS0) of the frequency divider. Reading Register C clears PF bit.<br>Note: the value of this register is undefined/indeterministic when power up first time.                                                                            |
| 5    | <b>alarm_interrupt_flag_af.</b> Read-only. Reset: X.<br><b>Description:</b> This bit is set to one if second, minute and hour time has matched the second, minute and hour alarm time. Reading Register C clears AF bit.<br>Note: the value of this register is undefined/indeterministic when power up first time.                                                                                        |
| 4    | <b>update_ended_interrupt_flag_uf.</b> Read-only. Reset: X.<br><b>Description:</b> This bit is set to one after each update cycle. Reading Register C clears UF.<br>Note: the value of this register is undefined/indeterministic when power up first time.                                                                                                                                                |
| 3:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                  |

**RTCHOSTx0000000D (FCH::IO::DATEALARM)**

Read-write.

Date Alarm Register

\_aliasHOSTLEGACY; RTCHOSTx0000000D; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>vrt.</b> Read-write. Reset: X.<br><b>Description:</b> Valid RAM and Time refer to VRT_T1 and VRT_T2 registers (PMIO 3E/3F)<br>Note: the value of this register is undefined/indeterministic when power up first time.                                                                                                               |
| 6    | <b>scratchbit.</b> Read-write. Reset: X.                                                                                                                                                                                                                                                                                               |
| 5:0  | <b>datealarm.</b> Read-write. Reset: XXXXXXb.<br><b>Description:</b> DateAlarm in <a href="#">BCD</a> format and is considered when it is set to non-zero value. If this value is set to 0, then date is not compared for alarm generation.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000032 (FCH::IO::ALTCENTURY)**

Read-write.

AltCentury Register

\_aliasHOSTLEGACY; RTCHOSTx00000032; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>altcentury.</b> Read-write. Reset: XXXXXXXXb.<br><b>Description:</b> (This register is accessed only when DV0=0 and PM_Reg 56h Bit12=1.) Binary-Code-Decimal format. Leap year correction is done through hardware. This register can be set by software (set bit of Register B = 1) or can be automatically updated by hardware every century. When set by software, hardware updating is disabled.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000048 (FCH::IO::CENTURY)**

Read-write.

Century Register

\_aliasHOSTLEGACY; RTCHOSTx00000048; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>century.</b> Read-write. Reset: XXXXXXXXb.<br><b>Description:</b> (This register is accessed only when DV0=1) Binary-Code-Decimal format. Leap year correction is done through hardware. This register can be set by software (set bit of Register B = 1) or can be automatically updated by hardware every century. When set by software, hardware updating is disabled.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx00000050 (FCH::IO::EXTENDED\_RAM\_ADDRESS\_PORT)**

Read-write.

Extended RAM Address Port register: The address port to access Extended RAM.

\_aliasHOSTLEGACY; RTCHOSTx00000050; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6:0  | <b>extendedramaddr.</b> Read-write. Reset: XXXXXXXb.<br><b>Description:</b> Because only 7 address bits are used in port x70, only lower 128 bytes are accessible through port x71. The Extended RAM (upper 128 bytes) are physically located at address 80H to FFH. In order to access these address, an address offset should be programmed into this register and access them through Extended RAMDataPort. (An offset of x80H will automatically add to this 7-bit address).<br>Note: the value of this register is undefined/indeterministic when power up first time. |



**RTCHOSTx00000053 (FCH::IO::EXTENDED\_RAM\_DATA\_PORT)**

Read-write.

Extended RAM Data Port register.

\_aliasHOSTLEGACY; RTCHOSTx00000053; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>extended_ram_data_port.</b> Read-write. Reset: XXXXXXXXb. There is no physical register corresponding to this data port but the data port address is used for decoding to generate appropriate internal control signals. |

**RTCHOSTx0000007E (FCH::IO::RTC\_TIME\_CLEAR)**

Read-write.

RTC Time Clear register.

\_aliasHOSTLEGACY; RTCHOSTx0000007E; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | Reserved.                                                                                                                                                                                                                                                                                   |
| 0    | <b>rtctimeclear.</b> Read-write. Reset: X.<br><b>Description:</b> Setting this bit '1' will clear the RTC second and RTC time will stop. When PSP_regxFC[9]=1, this bit can only be written '0'.<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**RTCHOSTx0000007F (FCH::IO::RTC\_RAM\_ENABLE)**

Read-write.

RTC RAM Enable register.

\_aliasHOSTLEGACY; RTCHOSTx0000007F; RTCHOST=FED8\_0700h

| Bits | Description                                                                                                                                                                                                     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | Reserved.                                                                                                                                                                                                       |
| 0    | <b>rtcramenable.</b> Read-write. Reset: X.<br><b>Description:</b> Setting this bit will enable access to the RTC RAM<br>Note: the value of this register is undefined/indeterministic when power up first time. |

**14.3.2 AB Configuration Register****14.3.2.1 Registers**

AB Configuration registers are accessed indirectly through FCH::AB::ABIndex and FCH::AB::ABData. Before accessing FCH::AB::ABIndex and FCH::AB::ABData software needs to program FCH::PM::ABREGBAR[abregbar] = CD8h.

**ABx00000 [AB Index Register] (FCH::AB::ABIndex)**

Read-write. Reset: 0000\_0000h.

\_aliasIO; ABx00000; AB=0000\_0CD8h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |       |           |    |                                                        |    |           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-------|-----------|----|--------------------------------------------------------|----|-----------|
| 31:29 | <b>RegSpace.</b> Read-write. Reset: 0h. Once a valid register block address is written to this field, only valid register block addresses can be written to this field.<br><b>ValidValues:</b> <table border="1"> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>5h-0h</td><td>Reserved.</td></tr> <tr> <td>6h</td><td>ABCFG, A-Link Bridge Configuration; SDP Configuration.</td></tr> <tr> <td>7h</td><td>Reserved.</td></tr> </table> | Value | Description | 5h-0h | Reserved. | 6h | ABCFG, A-Link Bridge Configuration; SDP Configuration. | 7h | Reserved. |
| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |       |           |    |                                                        |    |           |
| 5h-0h | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |       |           |    |                                                        |    |           |
| 6h    | ABCFG, A-Link Bridge Configuration; SDP Configuration.                                                                                                                                                                                                                                                                                                                                                                                          |       |             |       |           |    |                                                        |    |           |
| 7h    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |       |           |    |                                                        |    |           |
| 28:17 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |       |           |    |                                                        |    |           |
| 16:0  | <b>RegAddr: Register Address.</b> Read-write. Reset: 0_0000h.                                                                                                                                                                                                                                                                                                                                                                                   |       |             |       |           |    |                                                        |    |           |

**ABx00004 [AB Data Register] (FCH::AB::ABData)**

Read-write. Reset: 0000\_0000h.

\_aliasIO; ABx00004; AB=0000\_0CD8h

| Bits | Description                                                   |
|------|---------------------------------------------------------------|
| 31:0 | <b>Data: Register Address.</b> Read-write. Reset: 0000_0000h. |

**ABx0CDC\_xC000\_0054 [Misc Control 1] (FCH::AB::MiscCtl1)**

Read-write. Reset: 0010\_0001h.

\_aliasIO; ABx0CDC\_xC0000054; AB=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:25 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                              |
| 24    | <b>BIClkGateEn.</b> Read-write. Reset: 0. 1=Enable gating of B-Link clocks when idle is detected.                                                                                                                                                                                                                                                                                                      |
| 23:16 | <b>BIClkGateDelay.</b> Read-write. Reset: 10h. Specifies the number of B-Link clocks to delay before gating B-Link clocks after idle condition is detected.                                                                                                                                                                                                                                            |
| 15:5  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                              |
| 4     | <b>DbgClkGateEn.</b> Read-write. Reset: 0. 1=Enable gating of B-Link debug clocks; hence no debug clock running.                                                                                                                                                                                                                                                                                       |
| 3     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                              |
| 2     | <b>UpSWrByteCntSbgMode.</b> Read-write. Reset: 0. 1=Upstream 32/64 byte MST_BIF write requests are issued only when all byte enables in the request are 1. This field is mutually exclusive with UpWr16BMode.                                                                                                                                                                                          |
| 1     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                              |
| 0     | <b>UpWr16BMode.</b> Read-write. Reset: 1. 0=Disable 16 Byte mode; AB can send DMA write request with length 16, 32, or 64 bytes. 1=Enable 16 Byte mode; AB only sends DMA write request with length 16 bytes. Enables upstream writes to be limited to a maximum of 16-bytes per transaction. This bit should be mutually exclusive with UpSWrByteCntSbgMode. See FCH::SDP::SDP_US_CTL[Dma16ByteMode]. |

**ABx0CDC\_xC000\_0058 [B-Link RAB Control] (FCH::AB::BLinkRABCtl)**

Read-write. Reset: 0000\_0000h.

\_aliasIO; ABx0CDC\_xC0000058; AB=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits | Description                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>AlMemSDEn.</b> Read-write. Reset: 0. 1=Enable AB A-Link Memory Shut Down Feature. See FCH::AB::AL_Arb_Ctl[27:26] for idle counter settings.  |
| 30   | <b>AlMemDSEn.</b> Read-write. Reset: 0. 1=Enable AB A-Link Memory Deep Sleep Feature. See FCH::AB::AL_Arb_Ctl[27:26] for idle counter settings. |
| 29   | <b>BIMemSDEn.</b> Read-write. Reset: 0. 1=Enable AB B-Link Memory Shut Down Feature. See FCH::AB::AL_Arb_Ctl[27:26] for idle counter settings.  |
| 28   | <b>BIMemDSEn.</b> Read-write. Reset: 0. 1=Enable AB B-Link Memory Deep Sleep Feature. See FCH::AB::AL_Arb_Ctl[27:26] for idle counter settings. |
| 27:0 | Reserved.                                                                                                                                       |

**ABx0CDC\_xC000\_0090 [BIF Control 0] (FCH::AB::BIFCtl0)**

Read-write.

\_aliasIO; ABx0CDC\_xC0000090; AB=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits  | Description                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:22 | Reserved.                                                                                                                                                   |
| 21    | <b>PpNpRfwCplCtrl.</b> Read-write. Reset: 0. Init: BIOS,1. RFW completion control for posted pass non-posted. Required by "Posted pass Non-posted" feature. |
| 20:0  | Reserved.                                                                                                                                                   |

**ABx0CDC\_xC000\_0094 [MSI Control] (FCH::AB::MsiCtl)**

Read-write. Reset: 0000\_0FEEh.

\_aliasIO; ABx0CDC\_xC0000094; AB=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits  | Description                                                                                                                                                    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:21 | Reserved.                                                                                                                                                      |
| 20    | <b>MsiAddrEn.</b> Read-write. Reset: 0. Init: BIOS,1. 1=Enable AB to detect MSI sent upstream from <a href="#">FCH</a> and inform power-management controller. |
| 19:0  | <b>MsiAddr_39_20.</b> Read-write. Reset: 0_0FEEh. Specifies MSI Address[39:20].                                                                                |

**ABx0CDC\_xC001\_0050 [Misc Control 2] (FCH::AB::MiscCtl2)**

Read-write. Reset: 0000\_0004h.

\_aliasIO; ABx0CDC\_xC0010050; AB=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits | Description                                                                                                                                                                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:3 | Reserved.                                                                                                                                                                                     |
| 2    | <b>SyncfloodEn.</b> Read-write. Reset: 1. 1=Enable AB to forward downstream sync-flood message to system controller. Enable AB to forward downstream sync-flood message to system controller. |
| 1:0  | Reserved.                                                                                                                                                                                     |

**ABx0CDC\_xC001\_0054 [A-Link Arbitration Control and Clock Control] (FCH::AB::AL\_Arb\_Ctl)**

Read-write. Reset: 0010\_0000h.

\_aliasIO; ABx0CDC\_xC0010054; AB=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |    |            |    |            |    |            |    |              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|----|------------|----|------------|----|------------|----|--------------|
| 31:28 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |             |    |            |    |            |    |            |    |              |
| 27:26 | <b>DsdSslvIdleCounter.</b> Read-write. Reset: 0h. This counter is for sslv idle cycle counts before allowing AB memory entering DSD mode. When there is rapid frequent traffic then holds off entering on short idle conditions. Unit is number of A-Link cycles. See FCH::AB::BLinkRABCtl [31:28] for AB memory DSD enable bits.<br><b>ValidValues:</b> <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>0h</td><td>64 cycles.</td></tr> <tr> <td>1h</td><td>32 cycles.</td></tr> <tr> <td>2h</td><td>16 cycles.</td></tr> <tr> <td>3h</td><td>Immediately.</td></tr> </table> | Value | Description | 0h | 64 cycles. | 1h | 32 cycles. | 2h | 16 cycles. | 3h | Immediately. |
| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |    |            |    |            |    |            |    |              |
| 0h    | 64 cycles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |             |    |            |    |            |    |            |    |              |
| 1h    | 32 cycles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |             |    |            |    |            |    |            |    |              |
| 2h    | 16 cycles.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |             |    |            |    |            |    |            |    |              |
| 3h    | Immediately.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |             |    |            |    |            |    |            |    |              |
| 25    | <b>DbgClkGateEn.</b> Read-write. Reset: 0. 1=Enable gating of A-Link debug clocks; hence debug clock is disabled. DbgClkGateEn.                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |             |    |            |    |            |    |            |    |              |
| 24    | <b>AlClkGateEn.</b> Read-write. Reset: 0. 1=Enable gating of A-Link clocks when idle is detected. AlClkGateEn.                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |             |    |            |    |            |    |            |    |              |
| 23:16 | <b>AlClkGateDelay.</b> Read-write. Reset: 10h. Specifies the number of cycles to delay before gating A-Link clocks after idle condition is detected.                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |             |    |            |    |            |    |            |    |              |
| 15:12 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |             |    |            |    |            |    |            |    |              |
| 11:0  | <b>ArbCtl.</b> Read-write. Reset: 000h. Init: BIOS,7FFh. A-Link Arbiter Control. Enable the A-Link int_arbiter enhancement to allow the A-Link bandwidth to be used more efficiently.                                                                                                                                                                                                                                                                                                                                                                                                        |       |             |    |            |    |            |    |            |    |              |

**ABx0CDC\_xC001\_0060 [A-Link DMA Prefetch Enable] (FCH::AB::ALinkDmaPrefetchEn)**

Read-write. Reset: 0000\_0000h.

\_aliasIO; ABx0CDC\_xC0010060; AB=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|-----|------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------|
| 31    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| 30:16 | <b>DmaExpPrefEn0.</b> Read-write. Reset: 0000h. Init: BIOS,0200h. Explicit prefetch enable. Each bit inside [30:16] has dedicated meaning for each function, For example, Bit[25] is used for SD. Each bit[x] has below meaning.<br><b>ValidValues:</b> <table> <tr> <th>Bit</th><th>Description</th></tr> <tr> <td>[0]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[1]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[2]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[3]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[4]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[5]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[6]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[7]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[8]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[9]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[10]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[11]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[12]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[13]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> <tr> <td>[14]</td><td>0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.</td></tr> </table> | Bit | Description | [0] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [1] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [2] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [3] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [4] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [5] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [6] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [7] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [8] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [9] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [10] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [11] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [12] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [13] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. | [14] | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location. |
| Bit   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [0]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [1]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [2]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [3]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [4]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [5]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [6]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [7]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [8]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [9]   | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [10]  | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [11]  | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [12]  | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [13]  | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| [14]  | 0=Disable implicit prefetch device in this bit location. 1=Enable implicit prefetch device in this bit location.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |
| 15:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |             |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |     |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |      |                                                                                                                  |

**ABx0CDC\_xC001\_0090 [Misc Control 3] (FCH::AB::MiscCtl3)**

Read-write. Reset: 0001\_0000h.

\_aliasIO; ABx0CDC\_xC0010090; AB=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits  | Description                                                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                  |
| 16    | <b>IOTrapDelayEnable.</b> Read-write. Reset: 1. Init: BIOS,1. 1=Enable the IO trap delay logic for the <a href="#">SMI</a> message to ensure that the SMI messages are sent to the CPU in the right order. |
| 15:0  | Reserved.                                                                                                                                                                                                  |

### 14.3.3 Scalable Data Port Configuration Register

#### 14.3.3.1 Registers

FCH\_SDP Configuration registers are accessed indirectly through FCH::AB::ABIndex and FCH::AB::ABData.

| FCHSDPx0CDC_xC000_0204 [SDP Upstream Control] (FCH::SDP::SDP_US_CTL)               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Read-write. Reset: 0000_0000h.                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| _aliasIO; FCHSDPx0CDC_xC0000204; FCHSDP=0000_0000h; DataPortWrite=FCH::AB::ABIndex |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Bits                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 31:1                                                                               | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0                                                                                  | <b>Dma16ByteMode.</b> Read-write. Reset: 0. 0=All the DMA write requests has the size that AB passes up. 1=Split the DMA write requests to be 16 bytes requests. To always split DMA write request to be 16 bytes request. Overrides FCH::AB::MiscCtl1[UpWr16BMode].                                                                                                                                                                                                                                  |
| FCHSDPx0CDC_xC000_0208 [SDP Misc Control] (FCH::SDP::SDP_MISC_CTL)                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Read-write. Reset: 0000_1000h.                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| _aliasIO; FCHSDPx0CDC_xC0000208; FCHSDP=0000_0000h; DataPortWrite=FCH::AB::ABIndex |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Bits                                                                               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 31:23                                                                              | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 22                                                                                 | <b>DBG_CLK_GATE_EN.</b> Read-write. Reset: 0. 1=statically gate off BCLK for debug features.                                                                                                                                                                                                                                                                                                                                                                                                          |
| 21:18                                                                              | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 17                                                                                 | <b>MedBCLKGateEnXrsmu.</b> Read-write. Reset: 0. 1=Enable XRSMU path medium grain BCLK gating. MedBCLKGateEnXrsmu                                                                                                                                                                                                                                                                                                                                                                                     |
| 16                                                                                 | <b>MedBCLKGateEnDma.</b> Read-write. Reset: 0.<br><b>Description:</b> MedBCLKGateEnDma. This bit must be used in conjunction with FCH::SDP::SDP_DISCONNECT_CTL[SDP_ONESIDE_DISCONNECT_FIX_DMA]. Allowed value of { FCH::SDP::SDP_DISCONNECT_CTL[SDP_ONESIDE_DISCONNECT_FIX_DMA], FCH::SDP::SDP_MISC_CTL[MedBCLKGateEnDma]}:<br>2'b00: Disable SDP DMA path medium grain BCLK gating.<br>2'b10: Disable SDP DMA path medium grain BCLK gating.<br>2'b11: Enable SDP DMA path medium grain BCLK gating. |
| 15                                                                                 | <b>MedBCLKGateEnHost.</b> Read-write. Reset: 0. 1=Enable HOST path medium grain BCLK gating. MedBCLKGateEnHost                                                                                                                                                                                                                                                                                                                                                                                        |
| 14:10                                                                              | <b>MedBCLKGateDly.</b> Read-write. Reset: 04h. Specifies the number of cycles to delay before asserting OkToStopBCLK or gate off internal BCLK after idle condition is detected. Delay counter resets whenever SDP is not idle.                                                                                                                                                                                                                                                                       |
| 9                                                                                  | <b>DmaMemShutdownEn.</b> Read-write. Reset: 0. 1=Enable shut-down power saving in DMA memories. DmaMemShutdownEn.                                                                                                                                                                                                                                                                                                                                                                                     |
| 8                                                                                  | <b>DmaMemDeepSleepEn.</b> Read-write. Reset: 0. 1=Enable deep sleep power saving in DMA memories.                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7                                                                                  | <b>HostMemShutDownEn.</b> Read-write. Reset: 0. 1=Enable shut-down power saving in host memories. HostMemShutDownEn.                                                                                                                                                                                                                                                                                                                                                                                  |
| 6                                                                                  | <b>HostMemDeepSleepEn.</b> Read-write. Reset: 0. 1=Enable deep-sleep power saving in host memories.                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5                                                                                  | <b>SdpMemPSDis.</b> Read-write. Reset: 0. 1=Disable SDP internal memory power saving. Does not apply to deep-sleep and shut-down.                                                                                                                                                                                                                                                                                                                                                                     |
| 4:0                                                                                | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**FCHSDPx0CDC\_xC000\_0224 [SDP Disconnect Control] (FCH::SDP::SDP\_DISCONNECT\_CTL0)**

Read-write. Reset: 0000\_FFFFh.

\_aliasIO; FCHSDPx0CDC\_xC0000224; FCHSDP=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits | Description                                                                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>PORT_WAIT_IBC_DISCON_TIMER_CTL.</b> Read-write. Reset: 0000_FFFFh. Number of cycles to delay before completing the handshake for full disconnected initiated by HOST/DMA OBC when HOST/DMA IBC has not acknowledged full-disconnect yet. |

**FCHSDPx0CDC\_xC000\_0228 [SDP Disconnect Control] (FCH::SDP::SDP\_DISCONNECT\_CTL)**

Read-write. Reset: 0000\_0000h.

\_aliasIO; FCHSDPx0CDC\_xC0000228; FCHSDP=0000\_0000h; DataPortWrite=FCH::AB::ABIndex

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7    | <b>SDP_ONESIDE_DISCONNECT_FIX_DMA.</b> Read-write. Reset: 0.<br><b>Description:</b> This bit must be used in conjunction with FCH::SDP::SDP_MISC_CTL[MedBCLKGateEnDma]. Allowed value of { FCH::SDP::SDP_DISCONNECT_CTL[SDP_ONESIDE_DISCONNECT_FIX_DMA], FCH::SDP::SDP_MISC_CTL[MedBCLKGateEnDma] }:<br>2'b00: Disable SDP DMA path medium grain BCLK gating.<br>2'b10: Disable SDP DMA path medium grain BCLK gating.<br>2'b11: Enable SDP DMA path medium grain BCLK gating. |
| 6    | <b>SDP_FULL_DISCONNECT_WAKEUP_DMA.</b> Read-write. Reset: 0. 1= Let DMA OBC connect from full-disconnected state to connected state. To enable DMA out-band-channel initiate one-side disconnect.                                                                                                                                                                                                                                                                              |
| 5    | <b>SDP_FULL_DISCONNECT_WAKEUP_HOST.</b> Read-write. Reset: 0. 1= Let HOST OBC connect from full-disconnected state to connected state. To enable DMA out-band-channel initiate one-side disconnect.                                                                                                                                                                                                                                                                            |
| 4    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3    | <b>SDP_ONESIDE_DISCONNECT_EN_DMA.</b> Read-write. Reset: 0. 0= Disable DMA OBC initiate one-side disconnect, and host OBC can connect from one-side disconnected state to connected state by configuring it to 1'b0. 1= Enable DMA OBC initiate one-side disconnect. To enable DMA out-band-channel initiate one-side disconnect.                                                                                                                                              |
| 2    | <b>SDP_FULL_DISCONNECT_EN_DMA.</b> Read-write. Reset: 0. 0= Disable DMA OBC initiate full-disconnect. 1= Enable DMA OBC initiate full-disconnect. To enable DMA out-band-channel initiate full-disconnect.                                                                                                                                                                                                                                                                     |
| 1    | <b>SDP_ONESIDE_DISCONNECT_EN_HOST.</b> Read-write. Reset: 0. 0= Disable HOST OBC initiate one-side disconnect, and host OBC can connect from one-side disconnected state to connected state by configuring it to 1'b0. 1= Enable HOST OBC initiate one-side disconnect. To enable HOST out-band-channel initiate one-side disconnect.                                                                                                                                          |
| 0    | <b>SDP_FULL_DISCONNECT_EN_HOST.</b> Read-write. Reset: 0. 0= Disable HOST OBC initiate full-disconnect. 1= Enable HOST OBC initiate full-disconnect. To enable HOST out-band-channel initiate full-disconnect.                                                                                                                                                                                                                                                                 |

## 14.3.4 I/O Advanced Programmable Interrupt Control

### 14.3.4.1 IOAPIC Registers

#### IOAPICx00000000 (FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER)

Read-write.

Used to determine which register is manipulated during an IO Window Register read/write operation.

\_aliasHOSTLEGACY; IOAPICx00000000; IOAPIC=FEC0\_0000h

| Bits | Description                                                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                                                                                                                                  |
| 7:0  | <b>indirect_address_offset</b> . Read-write. Reset: 00h. Indirect Address Offset to IO Window Register, used to determine which register is manipulated during an IO Window Register read/write operation. |

#### IOAPICx00000010 (FCH::IOAPIC::IO\_WINDOW\_REGISTER)

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010; IOAPIC=FEC0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>io_window</b> . Read-write. Reset: 0000_0000h.<br><b>Description:</b> Mapped by the value in the IO Register Select Register to the designated indirect access register. Technically a RW register however, the read/write capability is determined by the indirect access register referenced by the IO Register Select Register. |

#### IOAPICx00000010\_indirectaddressoffset00 (FCH::IOAPIC::IOAPIC\_ID\_REGISTER)

Read-write. Reset: 0000\_0000h.

Not used in XAPIC PCI bus delivery mode.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset00; IOAPIC=FEC0\_0000h;  
DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                            |
|-------|----------------------------------------------------------------------------------------|
| 31:24 | <b>id</b> . Read-write. Reset: 00h. IOAPIC device ID for APIC serial bus delivery mode |
| 23:0  | Reserved.                                                                              |

#### IOAPICx00000010\_indirectaddressoffset01 (FCH::IOAPIC::IOXAPIC\_VERSION\_REGISTER)

Read-only. Reset: 0017\_8021h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset01; IOAPIC=FEC0\_0000h;  
DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                               |
|-------|---------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                 |
| 23:16 | <b>max_redirection_entries</b> . Read-only. Reset: 17h. 24 entries [23:0] |
| 15    | <b>prq</b> . Read-only. Reset: 1. IRQ pin assertion supported             |
| 14:8  | Reserved.                                                                 |
| 7:0   | <b>version</b> . Read-only. Reset: 21h. PCI 2.2 compliant                 |

#### IOAPICx00000010\_indirectaddressoffset02 (FCH::IOAPIC::IOAPIC\_ARBITRATION\_REGISTER)

Read-only. Reset: 0000\_0000h.

Not used in XAPIC PCI bus delivery mode.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset02; IOAPIC=FEC0\_0000h;  
DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                    |
|-------|------------------------------------------------------------------------------------------------|
| 31:28 | Reserved.                                                                                      |
| 27:24 | <b>arbitration_id</b> . Read-only. Reset: 0h. Arbitration ID for APIC serial bus delivery mode |
| 23:0  | Reserved.                                                                                      |

**IOAPICx00000010\_indirectaddressoffset10**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_0\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset10; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset11**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_0\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset11; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |



**IOAPICx00000010\_indirectaddressoffset12**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_1\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset12; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset13**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_1\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset13; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset14**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_2\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset14; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset15**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_2\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset15; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset16**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_3\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset16; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                              |
| 16    | <b>mask</b> . Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode</b> . Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector</b> . Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset17**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_3\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset17; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id</b> . Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                  |

**IOAPICx00000010\_indirectaddressoffset18**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_4\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset18; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset19**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_4\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset19; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset1A**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_5\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset1A; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                              |
| 16    | <b>mask</b> . Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode</b> . Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector</b> . Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset1B**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_5\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset1B; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id</b> . Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                  |

**IOAPICx00000010\_indirectaddressoffset1C**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_6\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset1C; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset1D**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_6\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset1D; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset1E  
(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_7\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset1E; IOAPIC=FEC0\_0000h;  
DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset1F  
(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_7\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset1F; IOAPIC=FEC0\_0000h;  
DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset20**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_8\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset20; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                              |
| 16    | <b>mask</b> . Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode</b> . Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector</b> . Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset21**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_8\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset21; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id</b> . Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                  |



**IOAPICx00000010\_indirectaddressoffset22**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_9\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset22; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset23**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_9\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset23; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset24**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_10\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset24; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset25**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_10\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset25; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset26**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_11\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset26; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset27**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_11\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset27; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset28**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_12\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset28; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset29**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_12\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset29; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset2A**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_13\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

`_alias`HOSTLEGACY; IOAPICx00000010\_indirectaddressoffset2A; IOAPIC=FEC0\_0000h;

DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                              |
| 16    | <b>mask</b> . Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode</b> . Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector</b> . Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset2B**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_13\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

`_alias`HOSTLEGACY; IOAPICx00000010\_indirectaddressoffset2B; IOAPIC=FEC0\_0000h;

DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id</b> . Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                  |

**IOAPICx00000010\_indirectaddressoffset2C**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_14\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset2C; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                              |
| 16    | <b>mask</b> . Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status</b> . Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode</b> . Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector</b> . Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset2D**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_14\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset2D; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id</b> . Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                  |

**IOAPICx00000010\_indirectaddressoffset2E**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_15\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset2E; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset2F**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_15\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset2F; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset30**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_16\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset30; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset31**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_16\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset31; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |



**IOAPICx00000010\_indirectaddressoffset32**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_17\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset32; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset33**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_17\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset33; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset34**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_18\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset34; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset35**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_18\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset35; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset36**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_19\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset36; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset37**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_19\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset37; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset38**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_20\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset38; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset39**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_20\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

 \_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset39; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset3A**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_21\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset3A; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset3B**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_21\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset3B; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset3C**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_22\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset3C; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset3D**  
**(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_22\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset3D; IOAPIC=FEC0\_0000h;  
 DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000010\_indirectaddressoffset3E  
(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_23\_LOW\_32BIT)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset3E; IOAPIC=FEC0\_0000h;  
DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                             |
| 16    | <b>mask.</b> Read-write. Reset: 1.<br><b>Description:</b> Masks the interrupt injection at the input of this device.<br>Write 0 to unmask                                                                                             |
| 15    | <b>trigger_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge<br>1: Level                                                                                                                                                 |
| 14    | <b>remote_irr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only.<br>Used for level triggered interrupts only.<br>Set when interrupt message delivered.<br>Cleared by EOI special cycle transaction or write to EOI register |
| 13    | <b>interrupt_pin_polarity.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: High<br>1: Low                                                                                                                                         |
| 12    | <b>delivery_status.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only<br>0: Idle<br>1: Send Pending                                                                                                                          |
| 11    | <b>destination_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Physical<br>1: Logical                                                                                                                                       |
| 10:8  | <b>delivery_mode.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Fixed<br>001: Lowest Priority<br>010: <a href="#">SMI</a> /PMI<br>011: Reserved<br>100: NMI<br>101: INIT<br>110: Reserved<br>111: ExtINT                     |
| 7:0   | <b>vector.</b> Read-write. Reset: 00h. Interrupt vector associated with this interrupt input                                                                                                                                          |

**IOAPICx00000010\_indirectaddressoffset3F  
(FCH::IOAPIC::REDIRECTION\_TABLE\_ENTRY\_23\_HIGH\_32BIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; IOAPICx00000010\_indirectaddressoffset3F; IOAPIC=FEC0\_0000h;  
DataPortWrite=FCH::IOAPIC::IO\_REGISTER\_SELECT\_REGISTER[indirect\_address\_offset]

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 31:24 | <b>destination_id.</b> Read-write. Reset: 00h. Bits [19:12] of the address field of the interrupt message |
| 23:0  | Reserved.                                                                                                 |

**IOAPICx00000020 (FCH::IOAPIC::IRQ\_PIN\_ASSERTION\_REGISTER)**

Read-write. Reset: 0000\_0000h.

Write to this register will trigger an interrupt associated with the redirection table entry referenced by the IRQ number. Currently the redirection table has 24 entries. Write with IRQ number greater than 17H has no effect.

\_aliasHOSTLEGACY; IOAPICx00000020; IOAPIC=FEC0\_0000h

| Bits | Description                                                                       |
|------|-----------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                         |
| 7:0  | <b>input_irq</b> . Read-write. Reset: 00h. IRQ number for the requested interrupt |

**IOAPICx00000040 (FCH::IOAPIC::EOI\_REGISTER)**

Write-only. Reset: 0000\_0000h.

Write to this register will clear the remote IRR bit in the redirection table entry found matching the interrupt vector. This provides an alternate mechanism other than PCI special cycle for EOI to reach IOXAPIC.

\_aliasHOSTLEGACY; IOAPICx00000040; IOAPIC=FEC0\_0000h

| Bits | Description                                              |
|------|----------------------------------------------------------|
| 31:8 | Reserved.                                                |
| 7:0  | <b>vector</b> . Write-only. Reset: 00h. Interrupt vector |

**14.3.5 SMI Registers**

[SMI](#) register space is accessed through the AcpiMmio region. The SMI registers range from FED8\_0000h+200h to FED8\_0000h+2FFh. See FCH::PM::ISACONTROL[mmioen].

**SMIx00000000 (FCH::SMI::EVENT\_STATUS)**

Read-write.

\_aliasHOSTLEGACY; SMIx00000000; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>eventstatus</b> . Read-write. Reset: XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXb.<br><b>Description:</b> This is a mirror register of the standard ACPI EVENT_STATUS register. Writing 1 to each bit clears the corresponding status bit.<br>Each Event status is set when the selected event input equals to the corresponding value in SciTrig. |

**SMIx00000004 (FCH::SMI::EVENT\_ENABLE)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000004; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>eventenable</b> . Read-write. Reset: 0000_0000h. This is the mirror register of the standard ACPI EVENT_ENABLE register. Each bit controls whether ACP should generate wakeup and Sci interrupt. |



**SMIx00000008 (FCH::SMI::SCITRIG)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; SMIx00000008; SMI=FED8\_0200h

| Bits | Description                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>scitrig31.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 31<br>0: Active low<br>1: Active high |
| 30   | <b>scitrig30.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 30<br>0: Active low<br>1: Active high |
| 29   | <b>scitrig29.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 29<br>0: Active low<br>1: Active high |
| 28   | <b>scitrig28.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 28<br>0: Active low<br>1: Active high |
| 27   | <b>scitrig27.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 27<br>0: Active low<br>1: Active high |
| 26   | <b>scitrig26.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 26<br>0: Active low<br>1: Active high |
| 25   | <b>scitrig25.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 25<br>0: Active low<br>1: Active high |
| 24   | <b>scitrig24.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 24<br>0: Active low<br>1: Active high |
| 23   | <b>scitrig23.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 23<br>0: Active low<br>1: Active high |
| 22   | <b>scitrig22.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 22<br>0: Active low<br>1: Active high |
| 21   | <b>scitrig21.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 21<br>0: Active low<br>1: Active high |
| 20   | <b>scitrig20.</b> Read-write. Reset: 1.                                                                                                               |

|    |                                                                                                                                                         |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Description:</b> The bit controls the way to set Event_Status bit 20<br>0: Active low<br>1: Active high                                              |
| 19 | <b>scitrig19.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 19<br>0: Active low<br>1: Active high   |
| 18 | <b>scitrig18.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 18<br>0: Active low<br>1: Active high   |
| 17 | <b>scitrig17.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 17<br>0: Active low<br>1: Active high   |
| 16 | <b>scitrig16.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 16<br>0: Active low<br>1: Active high   |
| 15 | <b>scitrig15.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 15<br>0: Active low<br>1: Active high   |
| 14 | <b>scitrig14.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 14<br>0: Active low<br>1: Active high   |
| 13 | <b>scitrig13.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 13<br>0: Active low<br>1: Active high   |
| 12 | <b>scitrig12.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 12<br>0: Active low<br>1: Active high   |
| 11 | <b>scitrig11.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 11<br>0: Falling edge<br>1: Active high |
| 10 | <b>scitrig10.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 10<br>0: Active low<br>1: Active high   |
| 9  | <b>scitrig9.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 9<br>0: Active low<br>1: Active high     |
| 8  | <b>scitrig8.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 8<br>0: Active low<br>1: Active high     |

|   |                                                                                                                                                     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 7 | <b>scitrig7.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 7<br>0: Active low<br>1: Active high |
| 6 | <b>scitrig6.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 6<br>0: Active low<br>1: Active high |
| 5 | <b>scitrig5.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 5<br>0: Active low<br>1: Active high |
| 4 | <b>scitrig4.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 4<br>0: Active low<br>1: Active high |
| 3 | <b>scitrig3.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 3<br>0: Active low<br>1: Active high |
| 2 | <b>scitrig2.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 2<br>0: Active low<br>1: Active high |
| 1 | <b>scitrig1.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 1<br>0: Active low<br>1: Active high |
| 0 | <b>scitrig0.</b> Read-write. Reset: 1.<br><b>Description:</b> The bit controls the way to set Event_Status bit 0<br>0: Active low<br>1: Active high |

**SMIx0000000C (FCH::SMI::SCILEVL)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx0000000C; SMI=FED8\_0200h

| Bits | Description                                                                                        |
|------|----------------------------------------------------------------------------------------------------|
| 31   | <b>scilevl31.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 30   | <b>scilevl30.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 29   | <b>scilevl29.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 28   | <b>scilevl28.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 27   | <b>scilevl27.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 26   | <b>scilevl26.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 25   | <b>scilevl25.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 24   | <b>scilevl24.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 23   | <b>scilevl23.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 22   | <b>scilevl22.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 21   | <b>scilevl21.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 20   | <b>scilevl20.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 19   | <b>scilevl19.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 18   | <b>scilevl18.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 17   | <b>scilevl17.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 16   | <b>scilevl16.</b> Read-write. Reset: 0.                                                            |

|    |                                                         |
|----|---------------------------------------------------------|
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 15 | <b>scilevl15.</b> Read-write. Reset: 0.                 |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 14 | <b>scilevl14.</b> Read-write. Reset: 0.                 |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 13 | <b>scilevl13.</b> Read-write. Reset: 0.                 |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 12 | <b>scilevl12.</b> Read-write. Reset: 0.                 |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 11 | <b>scilevl11.</b> Read-write. Reset: 0.                 |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 10 | <b>scilevl10.</b> Read-write. Reset: 0.                 |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 9  | <b>scilevl9.</b> Read-write. Reset: 0.                  |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 8  | <b>scilevl8.</b> Read-write. Reset: 0.                  |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 7  | <b>scilevl7.</b> Read-write. Reset: 0.                  |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 6  | <b>scilevl6.</b> Read-write. Reset: 0.                  |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 5  | <b>scilevl5.</b> Read-write. Reset: 0.                  |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 4  | <b>scilevl4.</b> Read-write. Reset: 0.                  |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 3  | <b>scilevl3.</b> Read-write. Reset: 0.                  |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 2  | <b>scilevl2.</b> Read-write. Reset: 0.                  |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 1  | <b>scilevl1.</b> Read-write. Reset: 0.                  |
|    | <b>Description:</b> 0: Edge trigger<br>1: Level trigger |
| 0  | <b>scilevl0.</b> Read-write. Reset: 0.                  |

|  |                                                                                                                                 |
|--|---------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Description:</b> This register defines the trigger mode for each of the Event_Status:<br>0: Edge trigger<br>1: Level trigger |
|--|---------------------------------------------------------------------------------------------------------------------------------|

**SMIx00000010 (FCH::SMI::SMISCISTATUS)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000010; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>smiscistatus.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> Each bit indicates the corresponding SmiSci status. The input of each bit is controlled by the corresponding SciTrig bit.<br>Each status bit can be cleared to 0 by writing 1. Note this function can be considered as a superset of Event_Status. When one of this bit is set (and its SmiSciEn is also set), it will trigger a <a href="#">SMI</a> to call the BIOS. After the BIOS has serviced the <a href="#">SMM</a> and clears its status, the internal logic will automatically set the corresponding Event_Status bit and thereby triggering a SCI. |

**SMIx00000014 (FCH::SMI::SMISCIEEN)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000014; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>smiscien.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> Each bit controls if <a href="#">SMI</a> message will be generated when the corresponding SmiSciStatus bit is set to 1.<br>0: Not to send SMI message when the corresponding SmiSciStatus bit is set<br>1: Send SMI message when the corresponding SmiSciStatus bit is set |

**SMIx00000018 (FCH::SMI::SOFTWARESCIEN)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000018; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                                                                                                                               |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>softwarescien.</b> Read-write. Reset: 0000_0000h. When set, software can write to softwareSciData and set the corresponding Event_Status bit (note the setting of this bit will need to match with SciTrig and SciLevl in order to set the status bit). This register is meant as a software mechanism to trigger SCI. |

**SMIx0000001C (FCH::SMI::SOFTWARESCIDATA)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx0000001C; SMI=FED8\_0200h

| Bits | Description                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>softwarescidata.</b> Read-write. Reset: 0000_0000h. This is the software data path to set the corresponding Event_Status when softwareSciEn is set |

**SMIx00000020 (FCH::SMI::SCISLEEPDISABLE)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000020; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>scisleepdisable.</b> Read-write. Reset: 0000_0000h. When set, the corresponding Event_Status bit will be masked off whenever the system goes to S3 or higher sleep state. This is meant for ignoring EVENT pins that are powered in the main power domain (instead of aux. power domain). |

**SMIx00000030 (FCH::SMI::CAPTUREDDATA)**

Read-only.

\_aliasHOSTLEGACY; SMIx00000030; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>captureddata.</b> Read-only. Reset: XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXb. This is the buffer to capture write data for the last transaction that caused an <a href="#">SMI</a> #. Note: this buffer has no meaning for a read trap |

**SMIx00000034 (FCH::SMI::CAPTUREDVALID)**

Read-only.

\_aliasHOSTLEGACY; SMIx00000034; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                                             |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | Reserved.                                                                                                                                                                                                                               |
| 3:0  | <b>capturedvalid.</b> Read-only. Reset: XXXXb. This is the byte valid buffer to signal which byte is captured for the last transaction that caused the <a href="#">SMI</a> . Bit 0 for byte 0, 1 for byte 1, 2 for byte 2, 3 for byte 3 |

**SMIx00000038 (FCH::SMI::EPBIF\_AER\_STRAPS)**

Read-write. Reset: 07FE\_FFEh.

\_aliasHOSTLEGACY; SMIx00000038; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 31:28 | Reserved.                                                                                                                                   |
| 27    | <b>strap_bif_internal_err_en_sb.</b> Read-write. Reset: 0. Internal error enable.                                                           |
| 26    | <b>strap_bif_poisoned_advisory_nonfatal_a_sb.</b> Read-write. Reset: 1. Poisoned TLP as advisory nonfatal.                                  |
| 25    | <b>strap_bif_acs_direct_translated_p2p_sb.</b> Read-write. Reset: 1. ACS direct translated <a href="#">P2P</a> enable.                      |
| 24    | <b>strap_bif_acs_upstream_forwarding_sb.</b> Read-write. Reset: 1. ACS upstream forwarding enable.                                          |
| 23    | <b>strap_bif_acs_p2p_completion_redirect_sb.</b> Read-write. Reset: 1. ACS <a href="#">P2P</a> completion redirect enable.                  |
| 22    | <b>strap_bif_acs_p2p_request_redirect_sb.</b> Read-write. Reset: 1. ACS <a href="#">P2P</a> request redirect enable.                        |
| 21    | <b>strap_bif_acs_translation_blocking_sb.</b> Read-write. Reset: 1. ACS translation blocking enable.                                        |
| 20    | <b>strap_bif_acs_source_validation_sb.</b> Read-write. Reset: 1. ACS source validation enable.                                              |
| 19    | <b>strap_bif_acs_en_sb.</b> Read-write. Reset: 1. ACS enable.                                                                               |
| 18    | <b>strap_bif_first_rcvd_err_log_sb.</b> Read-write. Reset: 1. First received error log.                                                     |
| 17    | <b>strap_bif_ecrc_check_en_sb.</b> Read-write. Reset: 1. ECRC check enable.                                                                 |
| 16    | <b>strap_bif_ecrc_gen_en_sb.</b> Read-write. Reset: 0. ECRC generate enable.                                                                |
| 15    | <b>strap_bif_cpl_abort_err_en_sb.</b> Read-write. Reset: 1. Completer abort error enable.                                                   |
| 14    | <b>strap_bif_rx_ignore_vend0_ur_sb.</b> Read-write. Reset: 1. Ignore Vendor 0 error.                                                        |
| 13    | <b>strap_bif_rx_ignore_tc_err_sb.</b> Read-write. Reset: 1. Ignore traffic class error.                                                     |
| 12    | <b>strap_bif_rx_ignore_msg_err_sb.</b> Read-write. Reset: 1. Ignore message error.                                                          |
| 11    | <b>strap_bif_rx_ignore_max_payload_err_sb.</b> Read-write. Reset: 1. Ignore maximum payload error.                                          |
| 10    | <b>strap_bif_rx_ignore_len_mismatch_err_sb.</b> Read-write. Reset: 1. Ignore length mismatch error.                                         |
| 9     | <b>strap_bif_rx_ignore_io_ur_err_sb.</b> Read-write. Reset: 1. Ignore IO UR error.                                                          |
| 8     | <b>strap_bif_rx_ignore_io_err_sb.</b> Read-write. Reset: 1. Ignore IO error.                                                                |
| 7     | <b>strap_bif_rx_ignore_ep_err_sb.</b> Read-write. Reset: 1. Ignore poisoned TLP error.                                                      |
| 6     | <b>strap_bif_rx_ignore_cpl_err_sb.</b> Read-write. Reset: 1. Ignore completion error.                                                       |
| 5     | <b>strap_bif_rx_ignore_cfg_ur_sb.</b> Read-write. Reset: 1. Ignore config. UR error.                                                        |
| 4     | <b>strap_bif_rx_ignore_cfg_err_sb.</b> Read-write. Reset: 1. Ignore configuration error.                                                    |
| 3     | <b>strap_bif_rx_ignore_be_err_sb.</b> Read-write. Reset: 1. Ignore byte enable error.                                                       |
| 2     | <b>strap_bif_err_reporting_dis_sb.</b> Read-write. Reset: 1. Error reporting disable.                                                       |
| 1     | <b>strap_bif_aer_en_sb.</b> Read-write. Reset: 1. AER enable.                                                                               |
| 0     | <b>strap_bif_sticky_override_s5.</b> Read-write. Reset: 0. When set to 1, values in this register would override straps loaded from EEPROM. |

**SMIx0000003C (FCH::SMI::DATAERRORSTATUS)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx0000003C; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                                                                                               |
| 7    | <b>sirq_iochk</b> . Read-write. Reset: 0. Serial Iochk error write 1 to clear it to 0                                                                                   |
| 6    | <b>sataperr</b> . Read-write. Reset: 0. SATA controller internal parity error status write 1 to clear it to 0.                                                          |
| 5    | <b>umi_uncorrectable_err</b> . Read-write. Reset: 0. <a href="#">UMI</a> uncorrectable error status write 1 to clear it to 0.                                           |
| 4    | <b>umi_correctable_err</b> . Read-write. Reset: 0. <a href="#">UMI</a> correctable error status write 1 to clear it to 0.                                               |
| 3    | <b>abumigppperr</b> . Read-write. Reset: 0. AB/ <a href="#">UMI</a> /GPP parity error status write 1 to clear it to 0.                                                  |
| 2    | <b>internalgppserr</b> . Read-write. Reset: 0. Internal error status: <a href="#">FCH</a> has detected an internal error from upstream bridge write 1 to clear it to 0. |
| 1    | <b>internalperr</b> . Read-write. Reset: 0. Internal devices Perr error status write 1 to clear it to 0.                                                                |
| 0    | <b>internalserr</b> . Read-write. Reset: 0. Internal devices serr error status write 1 to clear it to 0.                                                                |

**SMIx00000040 (FCH::SMI::SCIMAP0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000040; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                                        |
| 28:24 | <b>scimap_3</b> . Read-write. Reset: 00h.<br><b>Description:</b> Mapping of AGPIO22 to one of 32 Event_Status.<br>00000: map event source 3 to the input of Event_Status bit 0<br>00001: map event source 3 to the input of Event_Status bit 1<br>11111: map event source 3 to the input of Event_Status bit 31  |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                                                        |
| 20:16 | <b>scimap_2</b> . Read-write. Reset: 00h.<br><b>Description:</b> Mapping of AGPIO3 to one of 32 Event_Status.<br>00000: map event source 2 to the input of Event_Status bit 0<br>00001: map event source 2 to the input of Event_Status bit 1<br>11111: map event source 2 to the input of Event_Status bit 31   |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                                        |
| 12:8  | <b>scimap_1</b> . Read-write. Reset: 00h.<br><b>Description:</b> Mapping of AGPIO115 to one of 32 Event_Status.<br>00000: map event source 1 to the input of Event_Status bit 0<br>00001: map event source 1 to the input of Event_Status bit 1<br>11111: map event source 1 to the input of Event_Status bit 31 |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                                        |
| 4:0   | <b>scimap_0</b> . Read-write. Reset: 00h.<br><b>Description:</b> Mapping of GENINT_L to one of 32 Event_Status.<br>00000: map event source 0 to the input of Event_Status bit 0<br>00001: map event source 0 to the input of Event_Status bit 1<br>11111: map input event0 to the input of Event_Status bit 31   |



**SMIx00000044 (FCH::SMI::SCIMAP1)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000044; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                           |
| 28:24 | <b>scimap_7.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                            |
|       | <b>Description:</b> Mapping of AGPIO5 to one of 32 Event_Status.<br>00000: map event source 7 to the input of Event_Status bit 0<br>00001: map event source 7 to the input of Event_Status bit 1<br>11111: map event source 7 to the input of Event_Status bit 31   |
| 23:21 | Reserved.                                                                                                                                                                                                                                                           |
| 20:16 | <b>scimap_6.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                            |
|       | <b>Description:</b> Mapping of AGPIO116 to one of 32 Event_Status.<br>00000: map event source 6 to the input of Event_Status bit 0<br>00001: map event source 6 to the input of Event_Status bit 1<br>11111: map event source 6 to the input of Event_Status bit 31 |
| 15:13 | Reserved.                                                                                                                                                                                                                                                           |
| 12:8  | <b>scimap_5.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                            |
|       | <b>Description:</b> Mapping of AGPIO21 to one of 32 Event_Status.<br>00000: map event source 5 to the input of Event_Status bit 0<br>00001: map event source 5 to the input of Event_Status bit 1<br>11111: map event source 5 to the input of Event_Status bit 31  |
| 7:5   | Reserved.                                                                                                                                                                                                                                                           |
| 4:0   | <b>scimap_4.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                            |
|       | <b>Description:</b> Mapping of AGPIO4 to one of 32 Event_Status.<br>00000: map event source 4 to the input of Event_Status bit 0<br>00001: map event source 4 to the input of Event_Status bit 1<br>11111: map event source 4 to the input of Event_Status bit 31   |

**SMIx00000048 (FCH::SMI::SCIMAP2)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000048; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                   |
| 28:24 | <b>scimap_11.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                   |
|       | <b>Description:</b> Mapping of AGPIO76 to one of 32 Event_Status.<br>00000: map event source 11 to the input of Event_Status bit 0<br>00001: map event source 11 to the input of Event_Status bit 1<br>11111: map event source 11 to the input of Event_Status bit 31       |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                   |
| 20:16 | <b>scimap_10.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                   |
|       | <b>Description:</b> Mapping of AGPIO6 to one of 32 Event_Status.<br>00000: map event source 10 to the input of Event_Status bit 0<br>00001: map event source 10 to the input of Event_Status bit 1<br>11111: map event source 10 to the input of Event_Status bit 31        |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                   |
| 12:8  | <b>scimap_9.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                    |
|       | <b>Description:</b> Mapping of NMI_SYNC_FLOOD_L to one of 32 Event_Status.<br>00000: map event source 9 to the input of Event_Status bit 0<br>00001: map event source 9 to the input of Event_Status bit 1<br>11111: map event source 9 to the input of Event_Status bit 31 |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                   |
| 4:0   | <b>scimap_8.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                    |
|       | <b>Description:</b> Mapping of WAKE_L to one of 32 Event_Status.<br>00000: map event source 8 to the input of Event_Status bit 0<br>00001: map event source 8 to the input of Event_Status bit 1<br>11111: map event source 8 to the input of Event_Status bit 31           |

**SMIx0000004C (FCH::SMI::SCIMAP3)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx0000004C; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                 |
| 28:24 | <b>scimap_15.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of PCIE_RST1_L to one of 32 Event_Status.<br>00000: map event source 15 to the input of Event_Status bit 0<br>00001: map event source 15 to the input of Event_Status bit 1<br>11111: map event source 15 to the input of Event_Status bit 31 |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                 |
| 20:16 | <b>scimap_14.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of SMERR_L to one of 32 Event_Status.<br>00000: map event source 14 to the input of Event_Status bit 0<br>00001: map event source 14 to the input of Event_Status bit 1<br>11111: map event source 14 to the input of Event_Status bit 31     |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                 |
| 12:8  | <b>scimap_13.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of USB11_OC_L to one of 32 Event_Status.<br>00000: map event source 13 to the input of Event_Status bit 0<br>00001: map event source 13 to the input of Event_Status bit 1<br>11111: map event source 13 to the input of Event_Status bit 31  |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                 |
| 4:0   | <b>scimap_12.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of USB10_OC_L to one of 32 Event_Status.<br>00000: map event source 12 to the input of Event_Status bit 0<br>00001: map event source 12 to the input of Event_Status bit 1<br>11111: map event source 12 to the input of Event_Status bit 31  |

**SMIx00000050 (FCH::SMI::SCIMAP4)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000050; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                   |
| 28:24 | <b>scimap_19.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                   |
|       | <b>Description:</b> Mapping of SYS_RESET_L to one of 32 Event_Status.<br>00000: map event source 19 to the input of Event_Status bit 0<br>00001: map event source 19 to the input of Event_Status bit 1<br>11111: map event source 19 to the input of Event_Status bit 31   |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                   |
| 20:16 | <b>scimap_18.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                   |
|       | <b>Description:</b> Mapping of X48M_OUT to one of 32 Event_Status.<br>00000: map event source 18 to the input of Event_Status bit 0<br>00001: map event source 18 to the input of Event_Status bit 1<br>11111: map event source 18 to the input of Event_Status bit 31      |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                   |
| 12:8  | <b>scimap_17.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                   |
|       | <b>Description:</b> Mapping of ESPI_RSTIN_L to one of 32 Event_Status.<br>00000: map event source 17 to the input of Event_Status bit 0<br>00001: map event source 17 to the input of Event_Status bit 1<br>11111: map event source 17 to the input of Event_Status bit 31  |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                   |
| 4:0   | <b>scimap_16.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                   |
|       | <b>Description:</b> Mapping of ESPI_RSTOUT_L to one of 32 Event_Status.<br>00000: map event source 16 to the input of Event_Status bit 0<br>00001: map event source 16 to the input of Event_Status bit 1<br>11111: map event source 16 to the input of Event_Status bit 31 |

**SMIx00000054 (FCH::SMI::SCIMAP5)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000054; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                               |
| 28:24 | <b>scimap_23.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                               |
|       | <b>Description:</b> Mapping of AGPIO106 to one of 32 Event_Status.<br>00000: map event source 23 to the input of Event_Status bit 0<br>00001: map event source 23 to the input of Event_Status bit 1<br>11111: map event source 23 to the input of Event_Status bit 31  |
| 23:21 | Reserved.                                                                                                                                                                                                                                                               |
| 20:16 | <b>scimap_22.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                               |
|       | <b>Description:</b> Mapping of AGPIO105 to one of 32 Event_Status.<br>00000: map event source 22 to the input of Event_Status bit 0<br>00001: map event source 22 to the input of Event_Status bit 1<br>11111: map event source 22 to the input of Event_Status bit 31  |
| 15:13 | Reserved.                                                                                                                                                                                                                                                               |
| 12:8  | <b>scimap_21.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                               |
|       | <b>Description:</b> Mapping of PWR_BTN_L to one of 32 Event_Status.<br>00000: map event source 21 to the input of Event_Status bit 0<br>00001: map event source 21 to the input of Event_Status bit 1<br>11111: map event source 21 to the input of Event_Status bit 31 |
| 7:5   | Reserved.                                                                                                                                                                                                                                                               |
| 4:0   | <b>scimap_20.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                               |
|       | <b>Description:</b> Mapping of AGPIO104 to one of 32 Event_Status.<br>00000: map event source 20 to the input of Event_Status bit 0<br>00001: map event source 20 to the input of Event_Status bit 1<br>11111: map event source 20 to the input of Event_Status bit 31  |

**SMIx00000058 (FCH::SMI::SCIMAP6)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000058; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                                                    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                                      |
| 28:24 | <b>scimap_27.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                      |
|       | <b>Description:</b> Mapping of eSPI_WAKE_PME (~eSPI_WAKE_PME_B, active high) to one of 32 Event_Status.<br>00000: map event source 27 to the input of Event_Status bit 0<br>00001: map event source 27 to the input of Event_Status bit 1<br>11111: map event source 27 to the input of Event_Status bit 31    |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                                                      |
| 20:16 | <b>scimap_26.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                      |
|       | <b>Description:</b> Mapping of eSPI system event (~eSPI_SYS_EVT_B, Active high) to one of 32 Event_Status.<br>00000: map event source 26 to the input of Event_Status bit 0<br>00001: map event source 26 to the input of Event_Status bit 1<br>11111: map event source 26 to the input of Event_Status bit 31 |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                                      |
| 12:8  | <b>scimap_25.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                      |
|       | <b>Description:</b> Mapping of MP2 GPIO[0] to one of 32 Event_Status.<br>00000: map event source 25 to the input of Event_Status bit 0<br>00001: map event source 25 to the input of Event_Status bit 1<br>11111: map event source 25 to the input of Event_Status bit 31                                      |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                                      |
| 4:0   | <b>scimap_24.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                      |
|       | <b>Description:</b> Mapping of MP2 wakeup event to one of 32 Event_Status.<br>00000: map event source 24 to the input of Event_Status bit 0<br>00001: map event source 24 to the input of Event_Status bit 1<br>11111: map event source 24 to the input of Event_Status bit 31                                 |

**SMIx0000005C (FCH::SMI::SCIMAP7)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx0000005C; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:21 | Reserved.                                                                                                                                                                                                                                                                     |
| 20:16 | <b>scimap_30.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                     |
|       | <b>Description:</b> Mapping of NB GPP Hot Plug to one of 32 Event_Status.<br>00000: map event source 30 to the input of Event_Status bit 0<br>00001: map event source 30 to the input of Event_Status bit 1<br>11111: map event source 30 to the input of Event_Status bit 31 |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                     |
| 12:8  | <b>scimap_29.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                     |
|       | <b>Description:</b> Mapping of NB GPP_PME to one of 32 Event_Status.<br>00000: map event source 29 to the input of Event_Status bit 0<br>00001: map event source 29 to the input of Event_Status bit 1<br>11111: map event source 29 to the input of Event_Status bit 31      |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                     |
| 4:0   | <b>scimap_28.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                     |
|       | <b>Description:</b> Mapping of MP2 GPIO[1] to one of 32 Event_Status.<br>00000: map event source 28 to the input of Event_Status bit 0<br>00001: map event source 28 to the input of Event_Status bit 1<br>11111: map event source 28 to the input of Event_Status bit 31     |

**SMIx00000060 (FCH::SMI::SCIMAP8)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000060; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                 |
| 28:24 | <b>scimap_35.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of PM FakeSts2 to one of 32 Event_Status.<br>00000: map event source 35 to the input of Event_Status bit 0<br>00001: map event source 35 to the input of Event_Status bit 1<br>11111: map event source 35 to the input of Event_Status bit 31 |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                 |
| 20:16 | <b>scimap_34.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of FakeSts1 to one of 32 Event_Status.<br>00000: map event source 34 to the input of Event_Status bit 0<br>00001: map event source 34 to the input of Event_Status bit 1<br>11111: map event source 34 to the input of Event_Status bit 31    |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                 |
| 12:8  | <b>scimap_33.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of FakeSts0 to one of 32 Event_Status.<br>00000: map event source 33 to the input of Event_Status bit 0<br>00001: map event source 33 to the input of Event_Status bit 1<br>11111: map event source 33 to the input of Event_Status bit 31    |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                 |
| 4:0   | <b>scimap_32.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of WAKE_L to one of 32 Event_Status.<br>00000: map event source 32 to the input of Event_Status bit 0<br>00001: map event source 32 to the input of Event_Status bit 1<br>11111: map event source 32 to the input of Event_Status bit 31      |

**SMIx00000064 (FCH::SMI::SCIMAP9)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000064; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                                        |
| 28:24 | <b>scimap_39.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                        |
|       | <b>Description:</b> Mapping of Azalia PME to one of 32 Event_Status.<br>00000: map event source 39 to the input of Event_Status bit 0<br>00001: map event source 39 to the input of Event_Status bit 1<br>11111: map event source 39 to the input of Event_Status bit 31                                         |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                                                        |
| 20:16 | <b>scimap_38.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                        |
|       | <b>Description:</b> Mapping of eSPI1_WAKE_PME (~eSPI1_WAKE_PME_B, active high) to one of 32 Event_Status.<br>00000: map event source 38 to the input of Event_Status bit 0<br>00001: map event source 38 to the input of Event_Status bit 1<br>11111: map event source 38 to the input of Event_Status bit 31    |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                                        |
| 12:8  | <b>scimap_37.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                        |
|       | <b>Description:</b> Mapping of eSPI1 system event (~eSPI1_SYS_EVT_B, Active high) to one of 32 Event_Status.<br>00000: map event source 37 to the input of Event_Status bit 0<br>00001: map event source 37 to the input of Event_Status bit 1<br>11111: map event source 37 to the input of Event_Status bit 31 |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                                        |
| 4:0   | <b>scimap_36.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                        |
|       | <b>Description:</b> Mapping of ESPI SCI (~eSPI_SCI_B, active high) to one of 32 Event_Status.<br>00000: map event source 36 to the input of Event_Status bit 0<br>00001: map event source 36 to the input of Event_Status bit 1<br>11111: map event source 36 to the input of Event_Status bit 31                |



**SMIx00000068 (FCH::SMI::SCIMAP10)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000068; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                                        |
| 28:24 | <b>scimap_43.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                        |
|       | <b>Description:</b> Mapping of AltHPET timer event to one of 32 Event_Status.<br>00000: map event source 43 to the input of Event_Status bit 0<br>00001: map event source 43 to the input of Event_Status bit 1<br>11111: map event source 43 to the input of Event_Status bit 31                                |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                                                        |
| 20:16 | <b>scimap_42.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                        |
|       | <b>Description:</b> Mapping of ESPI1 SCI (~eSPI1_SCI_B, active high) to one of 32 Event_Status.<br>00000: map event source 42 to the input of Event_Status bit 0<br>00001: map event source 42 to the input of Event_Status bit 1<br>11111: map event source 42 to the input of Event_Status bit 31              |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                                        |
| 12:8  | <b>scimap_41.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                        |
|       | <b>Description:</b> Mapping of Gpio Interrupt to one of 32 Event_Status.<br>00000: map event source 41 to the input of Event_Status bit 0<br>00001: map event source 41 to the input of Event_Status bit 1<br>11111: map event source 41 to the input of Event_Status bit 31                                     |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                                        |
| 4:0   | <b>scimap_40.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                                        |
|       | <b>Description:</b> Mapping of USB PD I2C4 INTR (~USB_PD_I2C4_intrB, active high) to one of 32 Event_Status.<br>00000: map event source 41 to the input of Event_Status bit 0<br>00001: map event source 41 to the input of Event_Status bit 1<br>11111: map event source 41 to the input of Event_Status bit 31 |

**SMIx0000006C (FCH::SMI::SCIMAP11)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx0000006C; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                                        |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                          |
| 28:24 | <b>scimap_47.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                          |
|       | <b>Description:</b> Mapping of SMBUS0 Interrupt event to one of 32 Event_Status.<br>00000: map event source 47 to the input of Event_Status bit 0<br>00001: map event source 47 to the input of Event_Status bit 1<br>11111: map event source 47 to the input of Event_Status bit 31               |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                                          |
| 20:16 | <b>scimap_46.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                          |
|       | <b>Description:</b> Mapping of I2S wake event to one of 32 Event_Status.<br>00000: map event source 46 to the input of Event_Status bit 0<br>00001: map event source 46 to the input of Event_Status bit 1<br>11111: map event source 46 to the input of Event_Status bit 31                       |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                          |
| 12:8  | <b>scimap_45.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                          |
|       | <b>Description:</b> Mapping of ASF Master and Slave Interrupt event to one of 32 Event_Status.<br>00000: map event source 45 to the input of Event_Status bit 0<br>00001: map event source 45 to the input of Event_Status bit 1<br>11111: map event source 45 to the input of Event_Status bit 31 |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                          |
| 4:0   | <b>scimap_44.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                          |
|       | <b>Description:</b> Mapping of internal FAN/THERMAL event to one of 32 Event_Status.<br>00000: map event source 44 to the input of Event_Status bit 0<br>00001: map event source 44 to the input of Event_Status bit 1<br>11111: map event source 44 to the input of Event_Status bit 31           |

**SMIx00000070 (FCH::SMI::SCIMAP12)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000070; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                               |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                 |
| 28:24 | <b>scimap_51.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of power button event to one of the 32 Event_Status bits.<br>00000: map event source 50 to the input of Event_Status bit 0<br>00001: map event source 50 to the input of Event_Status bit 1<br>11111: map event source 50 to the input of Event_Status bit 31 |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                                 |
| 20:16 | <b>scimap_50.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of LLB# to one of the 32 Event_Status bits.<br>00000: map event source 50 to the input of Event_Status bit 0<br>00001: map event source 50 to the input of Event_Status bit 1<br>11111: map event source 50 to the input of Event_Status bit 31               |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                 |
| 12:8  | <b>scimap_49.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of internal traffic monitor to one of 32 Event_Status.<br>00000: map event source 49 to the input of Event_Status bit 0<br>00001: map event source 49 to the input of Event_Status bit 1<br>11111: map event source 49 to the input of Event_Status bit 31    |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                 |
| 4:0   | <b>scimap_48.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                 |
|       | <b>Description:</b> Mapping of TWARN pin to one of 32 Event_Status.<br>00000: map event source 48 to the input of Event_Status bit 0<br>00001: map event source 48 to the input of Event_Status bit 1<br>11111: map event source 48 to the input of Event_Status bit 31                   |

**SMIx00000074 (FCH::SMI::SCIMAP13)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000074; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                               |
| 28:24 | <b>scimap_55.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                               |
|       | <b>Description:</b> Mapping of RAS_event status to one of the 32 Event_Status bits.<br>00000: map event source 55 to the input of Event_Status bit 0<br>00001: map event source 55 to the input of Event_Status bit 1<br>11111: map event source 55 to the input of Event_Status bit 31                 |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                                               |
| 20:16 | <b>scimap_54.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                               |
|       | <b>Description:</b> Mapping of "SCI assertion message from APU" to one of the 32 Event_Status bits.<br>00000: map event source 54 to the input of Event_Status bit 0<br>00001: map event source 54 to the input of Event_Status bit 1<br>11111: map event source 54 to the input of Event_Status bit 31 |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                               |
| 12:8  | <b>scimap_53.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                               |
|       | <b>Description:</b> Mapping of "HW assertion message from APU" to one of the 32 Event_Status bits.<br>00000: map event source 53 to the input of Event_Status bit 0<br>00001: map event source 53 to the input of Event_Status bit 1<br>11111: map event source 53 to the input of Event_Status bit 31  |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                               |
| 4:0   | <b>scimap_52.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                               |
|       | <b>Description:</b> Mapping of PROHOT# pin to one of the 32 Event_Status bits.<br>00000: map event source 52 to the input of Event_Status bit 0<br>00001: map event source 52 to the input of Event_Status bit 1<br>11111: map event source 52 to the input of Event_Status bit 31                      |

**SMIx00000078 (FCH::SMI::SCIMAP14)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000078; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                         |
| 28:24 | <b>scimap_59.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                         |
|       | <b>Description:</b> Mapping of DSM Cross Trigger[0] event to one of the 32 Event_Status bits.<br>00000: map event source 59 to the input of Event_Status bit 0<br>00001: map event source 59 to the input of Event_Status bit 1<br>11111: map event source 59 to the input of Event_Status bit 31 |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                                         |
| 20:16 | <b>scimap_58.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                         |
|       | <b>Description:</b> Mapping of AC/DC timer event to one of the 32 Event_Status bits.<br>00000: map event source 58 to the input of Event_Status bit 0<br>00001: map event source 58 to the input of Event_Status bit 1<br>11111: map event source 58 to the input of Event_Status bit 31          |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                         |
| 12:8  | <b>scimap_57.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                         |
|       | <b>Description:</b> Mapping of XHC-1 PME event to one of the 32 Event_Status bits.<br>00000: map event source 57 to the input of Event_Status bit 0<br>00001: map event source 57 to the input of Event_Status bit 1<br>11111: map event source 57 to the input of Event_Status bit 31            |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                         |
| 4:0   | <b>scimap_56.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                         |
|       | <b>Description:</b> Mapping of XHC-0 PME event to one of the 32 Event_Status bits.<br>00000: map event source 56to the input of Event_Status bit 0<br>00001: map event source 56to the input of Event_Status bit 1<br>11111: map event source 56 to the input of Event_Status bit 31              |

**SMIx0000007C (FCH::SMI::SCIMAP15)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx0000007C; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                       |
| 28:24 | <b>scimap_63.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                       |
|       | <b>Description:</b> Mapping of current temp status to one of the 32 Event_Status bits.<br>00000: map event source 63 to the input of Event_Status bit 0<br>00001: map event source 63 to the input of Event_Status bit 1<br>11111: map event source 63 to the input of Event_Status bit 31      |
| 23:21 | Reserved.                                                                                                                                                                                                                                                                                       |
| 20:16 | <b>scimap_62.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                       |
|       | <b>Description:</b> Mapping of DSM Cross Trigger[3] pin to one of the 32 Event_Status bits.<br>00000: map event source 62 to the input of Event_Status bit 0<br>00001: map event source 62 to the input of Event_Status bit 1<br>11111: map event source 62 to the input of Event_Status bit 31 |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                       |
| 12:8  | <b>scimap_61.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                       |
|       | <b>Description:</b> Mapping of DSM Cross Trigger[2] pin to one of the 32 Event_Status bits.<br>00000: map event source 61 to the input of Event_Status bit 0<br>00001: map event source 61 to the input of Event_Status bit 1<br>11111: map event source 61 to the input of Event_Status bit 31 |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                       |
| 4:0   | <b>scimap_60.</b> Read-write. Reset: 00h.                                                                                                                                                                                                                                                       |
|       | <b>Description:</b> Mapping of DSM Cross Trigger[1] pin to one of the 32 Event_Status bits.<br>00000: map event source 60 to the input of Event_Status bit 0<br>00001: map event source 60 to the input of Event_Status bit 1<br>11111: map event source 60 to the input of Event_Status bit 31 |

**SMIx00000080 (FCH::SMI::SMISTATUS0)**

Read,Write-1-to-clear. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000080; SMI=FED8\_0200h

| Bits | Description                                                                                   |
|------|-----------------------------------------------------------------------------------------------|
| 31   | Reserved.                                                                                     |
| 30   | <b>nbgpphp_event30</b> . Read,Write-1-to-clear. Reset: 0. Status of NB Hot Plug event         |
| 29   | <b>nbgpppme_event29</b> . Read,Write-1-to-clear. Reset: 0. Status of NB GPP PME               |
| 28   | <b>smi_event28</b> . Read,Write-1-to-clear. Reset: 0. Status of MP2 GPIO[1]                   |
| 27   | <b>espi_wake_pme_event27</b> . Read,Write-1-to-clear. Reset: 0. Status of ESPI WAKE/PME event |
| 26   | <b>smi_event26</b> . Read,Write-1-to-clear. Reset: 0. Status of ~eSPI_SYS_EVT_B               |
| 25   | <b>smi_event25</b> . Read,Write-1-to-clear. Reset: 0. Status of MP2 GPIO[0]                   |
| 24   | <b>smi_event24</b> . Read,Write-1-to-clear. Reset: 0. Status of MP2 wakeup event              |
| 23   | <b>gevent23status_event23</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO106           |
| 22   | <b>gevent22status_event22</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO105           |
| 21   | <b>gevent21status_event21</b> . Read,Write-1-to-clear. Reset: 0. Status of PWR_BTN_L          |
| 20   | <b>gevent20status_event20</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO104           |
| 19   | <b>gevent19status_event19</b> . Read,Write-1-to-clear. Reset: 0. Status of SYS_RESET_L        |
| 18   | <b>gevent18status_event18</b> . Read,Write-1-to-clear. Reset: 0. Status of X48M_OUT           |
| 17   | <b>gevent17status_event17</b> . Read,Write-1-to-clear. Reset: 0. Status of ESPI_RSTIN_L       |
| 16   | <b>gevent16status_event16</b> . Read,Write-1-to-clear. Reset: 0. Status of ESPI_RSTOUT_L      |
| 15   | <b>gevent15status_event15</b> . Read,Write-1-to-clear. Reset: 0. Status of PCIE_RST1_L        |
| 14   | <b>gevent14status_event14</b> . Read,Write-1-to-clear. Reset: 0. Status of SMERR_L            |
| 13   | <b>gevent13status_event13</b> . Read,Write-1-to-clear. Reset: 0. Status of USB_OC1_L          |
| 12   | <b>gevent12status_event12</b> . Read,Write-1-to-clear. Reset: 0. Status of USB_OC0_L          |
| 11   | <b>gevent11status_event11</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO76            |
| 10   | <b>gevent10status_event10</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO6             |
| 9    | <b>gevent9status_event9</b> . Read,Write-1-to-clear. Reset: 0. Status of NMI_SYNC_FLOOD       |
| 8    | <b>gevent8status_event8</b> . Read,Write-1-to-clear. Reset: 0. Status of WAKE_L               |
| 7    | <b>gevent7status_event7</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO5               |
| 6    | <b>gevent6status_event6</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO116             |
| 5    | <b>gevent5status_event5</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO21              |
| 4    | <b>gevent4status_event4</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO4               |
| 3    | <b>gevent3status_event3</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO22              |
| 2    | <b>gevent2status_event2</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO3               |
| 1    | <b>gevent1status_event1</b> . Read,Write-1-to-clear. Reset: 0. Status of AGPIO115             |
| 0    | <b>gevent0status_event0</b> . Read,Write-1-to-clear. Reset: 0. Status of GENINT1_L            |

**SMIx00000084 (FCH::SMI::SMISTATUS1)**

Read,Write-1-to-clear. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000084; SMI=FED8\_0200h

| Bits  | Description                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------|
| 31:27 | Reserved.                                                                                                                   |
| 26    | <b>acdctimerevent_event58.</b> Read,Write-1-to-clear. Reset: 0. Status of AcDcTimer wake up event (Wake Alarm Device)       |
| 25    | <b>xhc1pme_event57.</b> Read,Write-1-to-clear. Reset: 0. Status of XHC1 PME                                                 |
| 24    | <b>xhc0pme_event56.</b> Read,Write-1-to-clear. Reset: 0. Status of XHC0 PME                                                 |
| 23    | <b>ras_event55.</b> Read,Write-1-to-clear. Reset: 0. Internal devices SERR error status                                     |
| 22    | <b>apusciasrtion_event54.</b> Read,Write-1-to-clear. Reset: 0. Status of APU SCI request                                    |
| 21    | <b>apuhwassersion_event53.</b> Read,Write-1-to-clear. Reset: 0. Status of APU Hw assertion                                  |
| 20    | <b>prochot_event52.</b> Read,Write-1-to-clear. Reset: 0. Status of ProcHot event                                            |
| 19    | <b>pwrbutton_event51.</b> Read,Write-1-to-clear. Reset: 0. Status of PwrButton (rising edge) writing 1 to clear it to 0.    |
| 18    | <b>illb_event50.</b> Read,Write-1-to-clear. Reset: 0. Status of iLLB# assertion                                             |
| 17    | <b>trafficmonitorintr_event49.</b> Read,Write-1-to-clear. Reset: 0. Status of <a href="#">FCH</a> Traffic Monitor Interrupt |
| 16    | <b>twarn_event48.</b> Read,Write-1-to-clear. Reset: 0. Status of <a href="#">FCH</a> TWARN                                  |
| 15    | <b>smbus0_event47.</b> Read,Write-1-to-clear. Reset: 0. Status of <a href="#">FCH</a> SMBUS0 Master interrupt               |
| 14    | <b>i2swake_event46.</b> Read,Write-1-to-clear. Reset: 0. Status of I2S wake event                                           |
| 13    | <b>asfrintr_event45.</b> Read,Write-1-to-clear. Reset: 0. Status of <a href="#">FCH</a> ASF Master and Slave interrupt      |
| 12    | <b>fanthermalgevent_event44.</b> Read,Write-1-to-clear. Reset: 0. Status of <a href="#">FCH</a> FanThermal                  |
| 11    | <b>altmmtimersts_event43.</b> Read,Write-1-to-clear. Reset: 0. Status of AltMmTimer Alarm                                   |
| 10    | <b>smi_event42.</b> Read,Write-1-to-clear. Reset: 0. Status of ~eSPI1_SCI_B                                                 |
| 9     | <b>gpointr_event41.</b> Read,Write-1-to-clear. Reset: 0. Status of GPIO interrupt                                           |
| 8     | <b>smi_event40.</b> Read,Write-1-to-clear. Reset: 0. Status of ~USB_PD_I2C4_intrB                                           |
| 7     | <b>azpme_event39.</b> Read,Write-1-to-clear. Reset: 0. Status of Azaila wake event                                          |
| 6     | <b>espi1_wake_pme_event38.</b> Read,Write-1-to-clear. Reset: 0. Status of ESPI1 WAKE/PME event                              |
| 5     | <b>smi_event37.</b> Read,Write-1-to-clear. Reset: 0. Status of ~eSPI1_SYS_EVT_B                                             |
| 4     | <b>smi_event36.</b> Read,Write-1-to-clear. Reset: 0. Status of ~eSPI_SCI_B                                                  |
| 3     | <b>fakests2_event35.</b> Read,Write-1-to-clear. Reset: 0. Status of Fake2                                                   |
| 2     | <b>fakests1_event34.</b> Read,Write-1-to-clear. Reset: 0. Status of Fake1                                                   |
| 1     | <b>fakests0_event33.</b> Read,Write-1-to-clear. Reset: 0. Status of Fake0                                                   |
| 0     | <b>smi_event32.</b> Read,Write-1-to-clear. Reset: 0. Status of WAKE_L                                                       |



**SMIx00000088 (FCH::SMI::SMISTATUS2)**

Read,Write-1-to-clear. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000088; SMI=FED8\_0200h

| Bits  | Description                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------|
| 31:27 | Reserved.                                                                                                              |
| 26    | <b>emulate64_event90.</b> Read,Write-1-to-clear. Reset: 0. Status of Emulation Io Port 60/64h                          |
| 25:21 | Reserved.                                                                                                              |
| 20    | <b>pciserr_event84.</b> Read,Write-1-to-clear. Reset: 0. Status of Serr assertion on Pci bus                           |
| 19    | <b>prothot_event83.</b> Read,Write-1-to-clear. Reset: 0. Status of ProtHot event                                       |
| 18    | <b>vbatlow_event82.</b> Read,Write-1-to-clear. Reset: 0. Status of VBAT low                                            |
| 17    | <b>sim_event81.</b> Read,Write-1-to-clear. Reset: 0. Status of Intruder Alert Status                                   |
| 16:15 | Reserved.                                                                                                              |
| 14    | <b>smbus0intr_event78.</b> Read,Write-1-to-clear. Reset: 0. Status of SMBUS0 interrupt request                         |
| 13    | <b>serialirqsmi_event77.</b> Read,Write-1-to-clear. Reset: 0. Status of Smi request from Serial IRQ                    |
| 12    | <b>usbsmi_event76.</b> Read,Write-1-to-clear. Reset: 0. Status of Usb Smi request                                      |
| 11    | <b>smicmdport_event75.</b> Read,Write-1-to-clear. Reset: 0. Status of Writing Smi Command Port                         |
| 10    | <b>pwrbtn_event74.</b> Read,Write-1-to-clear. Reset: 0. Status of Power Button being pressed                           |
| 9     | <b>bios_rls_event73.</b> Read,Write-1-to-clear. Reset: 0. Status of BIOS_RLS                                           |
| 8     | <b>gbl_rls_event72.</b> Read,Write-1-to-clear. Reset: 0. Status of GBL event                                           |
| 7:3   | Reserved.                                                                                                              |
| 2     | <b>ial2h_acpi_assertion_event66.</b> Read,Write-1-to-clear. Reset: 0. Status of iAL2H_ACPI_Assertion                   |
| 1     | <b>slp_type_event65.</b> Read,Write-1-to-clear. Reset: 0. Status is set when ACPI SLP_TYP register bit 2 is programmed |
| 0     | <b>smi_event64.</b> Read,Write-1-to-clear. Reset: 0. Status of KBRst                                                   |

**SMIx00000090 (FCH::SMI::SMISTATUS4)**

Read,Write-1-to-clear. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx00000090; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                  |
| 28    | <b>cfgtrapping0_event156.</b> Read,Write-1-to-clear. Reset: 0. Status of Pci configuration cycle Trapping0 Smi request                                     |
| 27:25 | Reserved.                                                                                                                                                  |
| 24    | <b>memtrapping0_event152.</b> Read,Write-1-to-clear. Reset: 0. Status of memory Trapping0 Smi request                                                      |
| 23    | <b>iotrapping3_event151.</b> Read,Write-1-to-clear. Reset: 0. Status of Io Trapping3 Smi request                                                           |
| 22    | <b>iotrapping2_event150.</b> Read,Write-1-to-clear. Reset: 0. Status of Io Trapping2 Smi request                                                           |
| 21    | <b>iotrapping1_event149.</b> Read,Write-1-to-clear. Reset: 0. Status of Io Trapping1 Smi request                                                           |
| 20    | <b>iotrapping0_event148.</b> Read,Write-1-to-clear. Reset: 0. Status of Io Trapping0 Smi request                                                           |
| 19    | <b>espi1_smi_event147.</b> Read,Write-1-to-clear. Reset: 0. Status of eSPI1 <a href="#">SMI</a> event                                                      |
| 18    | <b>espi_smi_event146.</b> Read,Write-1-to-clear. Reset: 0. Status of eSPI <a href="#">SMI</a> event                                                        |
| 17    | Reserved.                                                                                                                                                  |
| 16    | <b>absmitrap_event144.</b> Read,Write-1-to-clear. Reset: 0. Status of AB Smi trapping request                                                              |
| 15    | <b>longtimer_event143.</b> Read,Write-1-to-clear. Reset: 0. Status of Long Timer Smi request                                                               |
| 14    | <b>shorttimer_event142.</b> Read,Write-1-to-clear. Reset: 0. Status of Short Timer Smi request                                                             |
| 13    | <b>cf9write_event141.</b> Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 1: Cf9 bit 2 has been written to 1.<br>0: Cf9 bit 2 is not written to 1. |
| 12:6  | Reserved.                                                                                                                                                  |
| 5     | <b>fanin0sts_event133.</b> Read,Write-1-to-clear. Reset: 0. Status of FanIn0 event                                                                         |
| 4:0   | Reserved.                                                                                                                                                  |

**SMIx00000094 (FCH::SMI::SMIPointer)**

Read-only.

This register is meant as a faster mechanism to locate the [SMI](#) source. BIOS can examine this register to find out the SMI source instead of reading SmiStatus0 through SmiStatus4 individually

\_aliasHOSTLEGACY; SMIx00000094; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:6 | Reserved.                                                                                                                                                             |
| 5    | <b>smistatussource4</b> . Read-only. Reset: 0. Indicate whether the <a href="#">SMI</a> source is from SmiStatus4[31:0] if the corresponding SMI enable is selected.  |
| 4    | <b>smistatussource3</b> . Read-only. Reset: 0. Indicate whether the <a href="#">SMI</a> source is from SmiStatus3[31:0] if the corresponding SMI enable is selected.  |
| 3    | <b>smistatussource2</b> . Read-only. Reset: 0. Indicates whether the <a href="#">SMI</a> source is from SmiStatus2[31:0] if the corresponding SMI enable is selected. |
| 2    | <b>smistatussource1</b> . Read-only. Reset: 0. Indicates whether the <a href="#">SMI</a> source is from SmiStatus1[31:0] if the corresponding SMI enable is selected. |
| 1    | <b>smistatussource0</b> . Read-only. Reset: 0. Indicates whether the <a href="#">SMI</a> source is from SmiStatus0[31:0] if the corresponding SMI enable is selected. |
| 0    | <b>smiscisource</b> . Read-only. Reset: 0. Indicates whether the <a href="#">SMI</a> source is from SMISCI.                                                           |

**SMIx00000096 (FCH::SMI::SMI\_SHORT\_LONG\_TIMER)**

Read-write. Reset: 0000h.

\*Note: This register 96h can be either "SmiShortTimer" or "SmiLongTimer," depending on the select bit "SmiTimerSel" in SMI\_Reg 98[29]. The default setting (SmiTimerSel=0) selects this register as "SmiShortTimer" software needs to set the "SmiTimerSel=1" to select this register as "SmiLongTimer".

\_aliasHOSTLEGACY; SMIx00000096; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | <b>timeren</b> . Read-write. Reset: 0.<br><b>Description:</b> Enable the <a href="#">SMI</a> short Timer or long timer, which is selected by SmiTimerEn (PMIO_98[29]).<br>0 = Disable<br>1 = Enable |
| 14:0 | <b>smitimercount</b> . Read-write. Reset: 0000h.<br><b>Description:</b> Actual timer duration = (TimerTime + 1) * 2us (Short Timer)<br>Actual timer duration = (TimerTime + 1) * 1ms (Long Timer)   |

**SMIx00000098 (FCH::SMI::SMITRIG0)**

Read-write. Reset: 8FFF\_FFFFh.

\_aliasHOSTLEGACY; SMIx00000098; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>smienb.</b> Read-write. Reset: 1.<br><b>Description:</b> Enable <a href="#">SMI</a> function.<br>0: Enable<br>1: Disable                                                                                                       |
| 30   | Reserved.                                                                                                                                                                                                                         |
| 29   | <b>smitimersel.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Selects the SMI_Reg 96h to be SMIShortTimer register.<br>1: Selects the SMI_Reg 96h to be SMILongTimer register.                                              |
| 28   | <b>eos.</b> Read-write. Reset: 0. This bit is set to 1 by SW to enable <a href="#">SMI</a> generation. It is cleared by hardware after a SMI event has occurred. When Eos is clear, subsequent pending SMI event will be blocked. |
| 27   | <b>fakests2.</b> Read-write. Reset: 1. Program the value to emulate an <a href="#">SMI</a> input event.                                                                                                                           |
| 26   | <b>fakests1.</b> Read-write. Reset: 1. Program the value to emulate an <a href="#">SMI</a> input event.                                                                                                                           |
| 25   | <b>fakests0.</b> Read-write. Reset: 1. Program the value to emulate an <a href="#">SMI</a> input event.                                                                                                                           |
| 24   | <b>trappingirqonpic.</b> Read-write. Reset: 1.<br><b>Description:</b> <a href="#">SMI</a> will be generated when<br>0: Trapping Irq0 ~ 15 of IoAPIC<br>1: Trapping Irq0 ~ 15 of PIC                                               |
| 23   | <b>smitrig23.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                                                                    |
| 22   | <b>smitrig22.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                                                                    |
| 21   | <b>smitrig21.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                                                                    |
| 20   | <b>smitrig20.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                                                                    |
| 19   | <b>smitrig19.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                                                                    |
| 18   | <b>smitrig18.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                                                                    |
| 17   | <b>smitrig17.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                                                                    |
| 16   | <b>smitrig16.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                                                                    |
| 15   | <b>smitrig15.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                                                                    |
| 14   | <b>smitrig14.</b> Read-write. Reset: 1.                                                                                                                                                                                           |

|    |                                                                                                                                                                                            |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Description:</b> 1: Active high<br>0: Active low                                                                                                                                        |
| 13 | <b>smitrig13.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                             |
| 12 | <b>smitrig12.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                             |
| 11 | <b>smitrig11.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                             |
| 10 | <b>smitrig10.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                             |
| 9  | <b>smitrig9.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                              |
| 8  | <b>smitrig8.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                              |
| 7  | <b>smitrig7.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                              |
| 6  | <b>smitrig6.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                              |
| 5  | <b>smitrig5.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                              |
| 4  | <b>smitrig4.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                              |
| 3  | <b>smitrig3.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                              |
| 2  | <b>smitrig2.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                              |
| 1  | <b>smitrig1.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                                                              |
| 0  | <b>smitrig0.</b> Read-write. Reset: 1.<br><b>Description:</b> This defines the trigger mode for SmiStatus0[23:0]. Note these are different from SciTrig<br>0: Active low<br>1: Active high |

**SMIx0000009C (FCH::SMI::SMITRIG1)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx0000009C; SMI=FED8\_0200h

| Bits | Description                                                                                       |
|------|---------------------------------------------------------------------------------------------------|
| 31   | <b>smiirq31trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 30   | <b>smiirq30trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 29   | <b>smiirq29trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 28   | <b>smiirq28trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 27   | <b>smiirq27trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 26   | <b>smiirq26trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 25   | <b>smiirq25trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 24   | <b>smiirq24trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 23   | <b>smiirq23trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 22   | <b>smiirq22trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 21   | <b>smiirq21trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 20   | <b>smiirq20trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 19   | <b>smiirq19trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 18   | <b>smiirq18trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 17   | <b>smiirq17trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 16   | <b>smiirq16trig.</b> Read-write. Reset: 0.                                                        |

|    |                                                                                                   |
|----|---------------------------------------------------------------------------------------------------|
|    | <b>Description:</b> 0: Active low<br>1: Active high                                               |
| 15 | <b>smiirq15trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 14 | <b>smiirq14trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 13 | <b>smiirq13trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 12 | <b>smiirq12trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 11 | <b>smiirq11trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 10 | <b>smiirq10trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 9  | <b>smiirq9trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |
| 8  | <b>smiirq8trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |
| 7  | <b>smiirq7trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |
| 6  | <b>smiirq6trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |
| 5  | <b>smiirq5trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |
| 4  | <b>smiirq4trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |
| 3  | <b>smiirq3trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |
| 2  | <b>smiirq2trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |
| 1  | <b>smiirq1trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |
| 0  | <b>smiirq0trig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Active low<br>1: Active high  |

**SMIx000000A0 (FCH::SMI::SMICONTROL0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000A0; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>smicontrol_15.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT15<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 29:28 | <b>smicontrol_14.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT14<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 27:26 | <b>smicontrol_13.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT13<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 25:24 | <b>smicontrol_12.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT12<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 23:22 | <b>smicontrol_11.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT11<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 21:20 | <b>smicontrol_10.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT10<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 19:18 | <b>smicontrol_9.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT9<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13   |
| 17:16 | <b>smicontrol_8.</b> Read-write. Reset: 0h.                                                                                                                |

|       |                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <b>Description:</b> Control for GEVENT8<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                |
| 15:14 | <b>smicontrol_7.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT7<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 13:12 | <b>smicontrol_6.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT6<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 11:10 | <b>smicontrol_5.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT5<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 9:8   | <b>smicontrol_4.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT4<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 7:6   | <b>smicontrol_3.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT3<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 5:4   | <b>smicontrol_2.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT2<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 3:2   | <b>smicontrol_1.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT1<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 1:0   | <b>smicontrol_0.</b> Read-write. Reset: 0h.                                                                                                              |



|  |                                                                                                           |
|--|-----------------------------------------------------------------------------------------------------------|
|  | <b>Description:</b> Control for GEVENT0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
|--|-----------------------------------------------------------------------------------------------------------|

**SMIx000000A4 (FCH::SMI::SMICONTROL1)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000A4; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>smicontrol_31.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GPP_PME (dev 21, func 3)<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13   |
| 29:28 | <b>smicontrol_30.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GPP_PME (dev 21, func 2)<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13   |
| 27:26 | <b>smicontrol_29.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GPP_PME (dev 21, func 1)<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13   |
| 25:24 | <b>smicontrol_28.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GPP_PME (dev 21, func 0~3)<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 23:22 | <b>smicontrol_27.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for eSPI_WAKE_PME_B<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13            |
| 21:20 | <b>smicontrol_26.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for eSPI_SYS_EVT_B<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13             |
| 19:18 | Reserved.                                                                                                                                                                    |
| 17:16 | <b>smicontrol_24.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for USB_PME (dev 18)<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13           |
| 15:14 | <b>smicontrol_23.</b> Read-write. Reset: 0h.                                                                                                                                 |

|       |                                                                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <b>Description:</b> Control for GEVENT23<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                 |
| 13:12 | <b>smicontrol_22.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT22<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 11:10 | <b>smicontrol_21.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT21<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 9:8   | <b>smicontrol_20.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT20<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 7:6   | <b>smicontrol_19.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT19<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 5:4   | <b>smicontrol_18.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT18<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 3:2   | <b>smicontrol_17.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT17<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 1:0   | <b>smicontrol_16.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GEVENT16<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |

**SMIx000000A8 (FCH::SMI::SMICONTROL2)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000A8; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>smicontrol_47.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for SMBUS0 interrupt<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13     |
| 29:28 | <b>smicontrol_46.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for ASF Slave interrupt<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13  |
| 27:26 | <b>smicontrol_45.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for ASF Master interrupt<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 25:24 | <b>smicontrol_44.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for FanThermal Gevent<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13    |
| 23:22 | <b>smicontrol_43.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for ALTHPET_TimerSts<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13     |
| 21:20 | <b>smicontrol_42.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for eSPI1_SCI_B<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13          |
| 19:18 | <b>smicontrol_41.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GPIO interrupt<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13       |
| 17:16 | <b>smicontrol_40.</b> Read-write. Reset: 0h.                                                                                                                           |

|       |                                                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <b>Description:</b> Control for Ec Gevent0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                                     |
| 15:14 | Reserved.                                                                                                                                                                        |
| 13:12 | <b>smicontrol_38.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for eSPI1_WAKE_PME_B<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13               |
| 11:10 | <b>smicontrol_37.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for eSPI1_SYS_EVT_B<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                |
| 9:8   | <b>smicontrol_36.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for eSPI_SCI_B<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                     |
| 7:6   | <b>smicontrol_35.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for FakeSts2<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                       |
| 5:4   | <b>smicontrol_34.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for FakeSts1<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                       |
| 3:2   | <b>smicontrol_33.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for FakeSts0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                       |
| 1:0   | <b>smicontrol_32.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for GPP_HotPlug (dev 21, func 0~3)<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |

**SMIx000000AC (FCH::SMI::SMICONTROL3)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000AC; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                        |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>smicontrol_63.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for TempTsi event<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                    |
| 29:28 | <b>smicontrol_62.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for DSM Cross Trigger event 3<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                        |
| 27:26 | <b>smicontrol_61.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for DSM Cross Trigger event 2<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                        |
| 25:24 | <b>smicontrol_60.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for DSM Cross Trigger event 1<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                        |
| 23:22 | <b>smicontrol_59.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for DSM Cross Trigger event 0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                        |
| 21:20 | <b>smicontrol_58.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for AcDcTimer wake up event (Wake Device in ACPI4.0)<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 19:18 | <b>smicontrol_57.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for XHC1 (dev 16, func 1) PME<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                        |
| 17:16 | <b>smicontrol_56.</b> Read-write. Reset: 0h.                                                                                                                                                       |

|       |                                                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <b>Description:</b> Control for XHC0 (dev 16, func 0) PME<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                          |
| 15:14 | <b>smicontrol_55.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Internal devices SERR error status<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 13:12 | <b>smicontrol_54.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for APU SCI request<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                    |
| 11:10 | <b>smicontrol_53.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for APU Hw assertion<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                   |
| 9:8   | <b>smicontrol_52.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Prochot event<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                      |
| 7:6   | <b>smicontrol_51.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Power button event<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                 |
| 5:4   | <b>smicontrol_50.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for iLLB#<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                              |
| 3:2   | <b>smicontrol_49.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for internal Traffic monitor interrupt<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 1:0   | <b>smicontrol_48.</b> Read-write. Reset: 0h.                                                                                                                                         |

|  |                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------|
|  | <b>Description:</b> Control for TWARN#<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
|--|----------------------------------------------------------------------------------------------------------|



**SMIx000000B0 (FCH::SMI::SMICONTROL4)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000B0; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>smicontrol_79.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Ec Smi request0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13             |
| 29:28 | <b>smicontrol_78.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for SMBUS0 interrupt<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13            |
| 27:26 | <b>smicontrol_77.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Smi request form serial Irq<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 25:24 | <b>smicontrol_76.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Usb Smi request<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13             |
| 23:22 | <b>smicontrol_75.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for writing Smi command port<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13    |
| 21:20 | <b>smicontrol_74.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for power button being pressed<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13  |
| 19:18 | <b>smicontrol_73.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for writing BIOS_RLS<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13            |
| 17:16 | <b>smicontrol_72.</b> Read-write. Reset: 0h.                                                                                                                                  |

|      |                                                                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <b>Description:</b> Control for writing GBL_RLS<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                                               |
| 15:8 | Reserved.                                                                                                                                                                                       |
| 7:6  | <b>smicontrol_67.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Sata AHCI event<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                               |
| 5:4  | <b>smicontrol_66.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for iAL2H_ACPI_Assertion<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                          |
| 3:2  | <b>smicontrol_65.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for writing SLP_TYP to put the system in S state.<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 1:0  | <b>smicontrol_64.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Keyboard Reset event.<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                         |

**SMIx000000B4 (FCH::SMI::SMICONTROL5)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000B4; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:28 | Reserved.                                                                                                                                                                |
| 27:26 | <b>smicontrol_93.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for HD audio FLR<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13           |
| 25:24 | <b>smicontrol_92.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Sata FLR<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13               |
| 23:22 | <b>smicontrol_91.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Usb FLR<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                |
| 21:20 | <b>smicontrol_90.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Emulation64<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13            |
| 19:18 | <b>smicontrol_89.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for ThermalTrip# assertion<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 17:12 | Reserved.                                                                                                                                                                |
| 11:10 | <b>smicontrol_85.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for SB GPP Serr#(dev 21, func 0~3)<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13        |
| 9:8   | <b>smicontrol_84.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for SERR#<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                  |
| 7:6   | <b>smicontrol_83.</b> Read-write. Reset: 0h.                                                                                                                             |

|     |                                                                                                                                                                   |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> Control for ProcHot<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                         |
| 5:4 | <b>smicontrol_82.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for VBAT low<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13        |
| 3:2 | <b>smicontrol_81.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Intruder event.<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 1:0 | <b>smicontrol_80.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Ec Smi request1<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |

**SMIx000000B8 (FCH::SMI::SMICONTROL6)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000B8; SMI=FED8\_0200h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**SMIx000000BC (FCH::SMI::SMICONTROL7)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000BC; SMI=FED8\_0200h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**SMIx000000C0 (FCH::SMI::SMICONTROL8)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000C0; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>smicontrol_143.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Long timer<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                                                                          |
| 29:28 | <b>smicontrol_142.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Short timer<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                                                                         |
| 27:26 | <b>smicontrol_141.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Cf9 Io write<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                                                                        |
| 25:24 | <b>smicontrol_140.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for FakeSts2<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13<br>Note: FakeSts2 defined in PMIO can be programmed to generate SMI/NMI/IRQ13 specified in those two bits. |
| 23:22 | <b>smicontrol_139.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for FakeSts1<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13<br>Note: FakeSts1 defined in PMIO can be programmed to generate SMI/NMI/IRQ13 specified in those two bits. |
| 21:20 | <b>smicontrol_138.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for FakeSts0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13<br>Note: FakeSts0 defined in PMIO can be programmed to generate SMI/NMI/IRQ13 specified in those two bits. |
| 19:12 | Reserved.                                                                                                                                                                                                                                                              |
| 11:10 | <b>smicontrol_133.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Fan Tach 0 too slow event<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                                                           |
| 9:0   | Reserved.                                                                                                                                                                                                                                                              |

**SMIx000000C4 (FCH::SMI::SMICONTROL9)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000C4; SMI=FED8\_0200h

| Bits  | Description                                                                                                                                                                       |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:26 | Reserved.                                                                                                                                                                         |
| 25:24 | <b>smicontrol_156.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for configuration cycle trapping 0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |
| 23:18 | Reserved.                                                                                                                                                                         |
| 17:16 | <b>smicontrol_152.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for memory trapping 0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13              |
| 15:14 | <b>smicontrol_151.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Io trapping 3<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                  |
| 13:12 | <b>smicontrol_150.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Io trapping 2<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                  |
| 11:10 | <b>smicontrol_149.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Io trapping 1<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                  |
| 9:8   | <b>smicontrol_148.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for Io trapping 0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                  |
| 7:6   | <b>smicontrol_147.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for eSPI1_SMI_B<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                    |
| 5:4   | <b>smicontrol_146.</b> Read-write. Reset: 0h.                                                                                                                                     |

|     |                                                                                                                                                                            |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> Control for eSPI_SMI_B<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13                                                               |
| 3:2 | <b>smicontrol_145.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for P state message 0<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13       |
| 1:0 | <b>smicontrol_144.</b> Read-write. Reset: 0h.<br><b>Description:</b> Control for AB Smi trapping request<br>00: Disable<br>01: <a href="#">SMI</a><br>10: NMI<br>11: IRQ13 |

**SMIx000000C8 (FCH::SMI::SMILEVEL0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000C8; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>smilevel31.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 30   | <b>smilevel30.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 29   | <b>smilevel29.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 28   | <b>smilevel28.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 27   | <b>smilevel27.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 26   | <b>smilevel26.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 25   | <b>smilevel25.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 24   | <b>smilevel24.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 23   | <b>smilevel23.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 22   | <b>smilevel22.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 21   | <b>smilevel21.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 20   | <b>smilevel20.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 19   | <b>smilevel19.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 18   | <b>smilevel18.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 17   | <b>smilevel17.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 16   | <b>smilevel16.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 15   | <b>smilevel15.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 14   | <b>smilevel14.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 13   | <b>smilevel13.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 12   | <b>smilevel12.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 11   | <b>smilevel11.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 10   | <b>smilevel10.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                          |
| 9    | <b>smilevel9.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                           |
| 8    | <b>smilevel8.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                           |
| 7    | <b>smilevel7.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                           |
| 6    | <b>smilevel6.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                           |
| 5    | <b>smilevel5.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                           |
| 4    | <b>smilevel4.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                           |
| 3    | <b>smilevel3.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                           |
| 2    | <b>smilevel2.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                           |
| 1    | <b>smilevel1.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                           |
| 0    | <b>smilevel0.</b> Read-write. Reset: 0.<br><b>Description:</b> This defines the edge/level trigger mode for SmiLevel[31:0]. Note these are different from SciLevl 0: Edge trigger mode, 1: Level trigger mode |



**SMIx000000CC (FCH::SMI::SMILEVELTRIG0)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; SMIx000000CC; SMI=FED8\_0200h

| Bits | Description                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>smitrig31.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                      |
| 30   | <b>smitrig30.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                      |
| 29   | <b>smitrig29.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                      |
| 28   | <b>smitrig28.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                      |
| 27   | <b>smitrig27.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                      |
| 26   | <b>smitrig26.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                      |
| 25   | <b>smitrig25.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high                                                      |
| 24   | <b>smitrig24.</b> Read-write. Reset: 1.<br><b>Description:</b> This defines the trigger mode for SmiTrig[31:24].<br>1: Active high<br>0: Active low |
| 23:0 | Reserved.                                                                                                                                           |

**SMIx000000D0 (FCH::SMI::SMILEVEL1)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000D0; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>smilevel63.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 30   | <b>smilevel62.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 29   | <b>smilevel61.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 28   | <b>smilevel60.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 27   | <b>smilevel59.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 26   | <b>smilevel58.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 25   | <b>smilevel57.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 24   | <b>smilevel56.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 23   | <b>smilevel55.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 22   | <b>smilevel54.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 21   | <b>smilevel53.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 20   | <b>smilevel52.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 19   | <b>smilevel51.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 18   | <b>smilevel50.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 17   | <b>smilevel49.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 16   | <b>smilevel48.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 15   | <b>smilevel47.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 14   | <b>smilevel46.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 13   | <b>smilevel45.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 12   | <b>smilevel44.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 11   | <b>smilevel43.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 10   | <b>smilevel42.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 9    | <b>smilevel41.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 8    | <b>smilevel40.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 7    | <b>smilevel39.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 6    | <b>smilevel38.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 5    | <b>smilevel37.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 4    | <b>smilevel36.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 3    | <b>smilevel35.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 2    | <b>smilevel34.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 1    | <b>smilevel33.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 0    | <b>smilevel32.</b> Read-write. Reset: 0.<br><b>Description:</b> This defines the edge/level trigger mode for SmiLevel[63:32]. Note these are different from SciLevl 0: Edge trigger mode, 1: Level trigger mode |

**SMIx000000D4 (FCH::SMI::SMILEVEL2)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000D4; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>smilevel95.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 30   | <b>smilevel94.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 29   | <b>smilevel93.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 28   | <b>smilevel92.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 27   | <b>smilevel91.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 26   | <b>smilevel90.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 25   | <b>smilevel89.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 24   | <b>smilevel88.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 23   | <b>smilevel87.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 22   | <b>smilevel86.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 21   | <b>smilevel85.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 20   | <b>smilevel84.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 19   | <b>smilevel83.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 18   | <b>smilevel82.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 17   | <b>smilevel81.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 16   | <b>smilevel80.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 15   | <b>smilevel79.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 14   | <b>smilevel78.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 13   | <b>smilevel77.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 12   | <b>smilevel76.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 11   | <b>smilevel75.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 10   | <b>smilevel74.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 9    | <b>smilevel73.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 8    | <b>smilevel72.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 7    | <b>smilevel71.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 6    | <b>smilevel70.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 5    | <b>smilevel69.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 4    | <b>smilevel68.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 3    | <b>smilevel67.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 2    | <b>smilevel66.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 1    | <b>smilevel65.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                            |
| 0    | <b>smilevel64.</b> Read-write. Reset: 0.<br><b>Description:</b> This defines the edge/level trigger mode for SmiLevel[95:64]. Note these are different from SciLevl 0: Edge trigger mode, 1: Level trigger mode |

**SMIx000000D8 (FCH::SMI::SMILEVEL3)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000D8; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>smilevel127.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 30   | <b>smilevel126.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 29   | <b>smilevel125.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 28   | <b>smilevel124.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 27   | <b>smilevel123.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 26   | <b>smilevel122.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 25   | <b>smilevel121.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 24   | <b>smilevel120.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 23   | <b>smilevel119.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 22   | <b>smilevel118.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 21   | <b>smilevel117.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 20   | <b>smilevel116.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 19   | <b>smilevel115.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 18   | <b>smilevel114.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 17   | <b>smilevel113.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 16   | <b>smilevel112.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 15   | <b>smilevel111.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 14   | <b>smilevel110.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 13   | <b>smilevel109.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 12   | <b>smilevel108.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 11   | <b>smilevel107.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 10   | <b>smilevel106.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 9    | <b>smilevel105.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 8    | <b>smilevel104.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 7    | <b>smilevel103.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 6    | <b>smilevel102.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 5    | <b>smilevel101.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 4    | <b>smilevel100.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                |
| 3    | <b>smilevel99.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                 |
| 2    | <b>smilevel98.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                 |
| 1    | <b>smilevel97.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                 |
| 0    | <b>smilevel96.</b> Read-write. Reset: 0.<br><b>Description:</b> This defines the edge/level trigger mode for SmiLevel[127:96]. Note these are different from SciLevel<br>0: Edge trigger mode, 1: Level trigger mode |

**SMIx000000DC (FCH::SMI::SMILEVEL4)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMIx000000DC; SMI=FED8\_0200h

| Bits | Description                                                                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>smilevel159.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 30   | <b>smilevel158.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 29   | <b>smilevel157.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 28   | <b>smilevel156.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 27   | <b>smilevel155.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 26   | <b>smilevel154.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 25   | <b>smilevel153.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 24   | <b>smilevel152.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 23   | <b>smilevel151.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 22   | <b>smilevel150.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 21   | <b>smilevel149.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 20   | <b>smilevel148.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 19   | <b>smilevel147.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 18   | <b>smilevel146.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 17   | <b>smilevel145.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 16   | <b>smilevel144.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 15   | <b>smilevel143.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 14   | <b>smilevel142.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 13   | <b>smilevel141.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 12   | <b>smilevel140.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 11   | <b>smilevel139.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 10   | <b>smilevel138.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 9    | <b>smilevel137.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 8    | <b>smilevel136.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 7    | <b>smilevel135.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 6    | <b>smilevel134.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 5    | <b>smilevel133.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 4    | <b>smilevel132.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 3    | <b>smilevel131.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 2    | <b>smilevel130.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 1    | <b>smilevel129.</b> Read-write. Reset: 0. 0: Edge trigger mode, 1: Level trigger mode                                                                                                                                  |
| 0    | <b>smilevel128.</b> Read-write. Reset: 0.<br><b>Description:</b> This defines the edge/level trigger mode for SmiLevel[159:128]. Note these are different from SciLevel<br>0: Edge trigger mode, 1: Level trigger mode |

**SMIx000000E0 (FCH::SMI::SMITRIG2)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; SMIx000000E0; SMI=FED8\_0200h

| Bits | Description                                                                                    |
|------|------------------------------------------------------------------------------------------------|
| 31   | <b>smitrig63.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 30   | <b>smitrig62.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 29   | <b>smitrig61.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 28   | <b>smitrig60.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 27   | <b>smitrig59.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 26   | <b>smitrig58.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 25   | <b>smitrig57.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 24   | <b>smitrig56.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Active high<br>0: Active low |
| 23   | <b>smitrig55.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 22   | <b>smitrig54.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 21   | <b>smitrig53.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 20   | <b>smitrig52.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 19   | <b>smitrig51.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 18   | <b>smitrig50.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 17   | <b>smitrig49.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 16   | <b>smitrig48.</b> Read-write. Reset: 1.                                                        |

|    |                                                                                                |
|----|------------------------------------------------------------------------------------------------|
|    | <b>Description:</b> 0: Active low<br>1: Active high                                            |
| 15 | <b>smitrig47.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 14 | <b>smitrig46.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Active high<br>0: Active low |
| 13 | <b>smitrig45.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 12 | <b>smitrig44.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 11 | <b>smitrig43.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 10 | <b>smitrig42.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 9  | <b>smitrig41.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 8  | <b>smitrig40.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 7  | <b>smitrig39.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 6  | <b>smitrig38.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 5  | <b>smitrig37.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 4  | <b>smitrig36.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 3  | <b>smitrig35.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 2  | <b>smitrig34.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 1  | <b>smitrig33.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 0  | <b>smitrig32.</b> Read-write. Reset: 1.                                                        |

|  |                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------|
|  | <b>Description:</b> This defines the trigger mode for SmiTrig[63:32].<br>0: Active low<br>1: Active high |
|--|----------------------------------------------------------------------------------------------------------|



**SMIx000000E4 (FCH::SMI::SMITRIG3)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; SMIx000000E4; SMI=FED8\_0200h

| Bits | Description                                                                                    |
|------|------------------------------------------------------------------------------------------------|
| 31   | <b>smitrig95.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 30   | <b>smitrig94.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 29   | <b>smitrig93.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 28   | <b>smitrig92.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 27   | <b>smitrig91.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 26   | <b>smitrig90.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 25   | <b>smitrig89.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 24   | <b>smitrig88.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Active high<br>0: Active low |
| 23   | <b>smitrig87.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 22   | <b>smitrig86.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 21   | <b>smitrig85.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 20   | <b>smitrig84.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 19   | <b>smitrig83.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 18   | <b>smitrig82.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 17   | <b>smitrig81.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 16   | <b>smitrig80.</b> Read-write. Reset: 1.                                                        |

|    |                                                                                                |
|----|------------------------------------------------------------------------------------------------|
|    | <b>Description:</b> 0: Active low<br>1: Active high                                            |
| 15 | <b>smitrig79.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 14 | <b>smitrig78.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Active high<br>0: Active low |
| 13 | <b>smitrig77.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 12 | <b>smitrig76.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 11 | <b>smitrig75.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 10 | <b>smitrig74.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 9  | <b>smitrig73.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 8  | <b>smitrig72.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 7  | <b>smitrig71.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 6  | <b>smitrig70.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 5  | <b>smitrig69.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 4  | <b>smitrig68.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 3  | <b>smitrig67.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 2  | <b>smitrig66.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 1  | <b>smitrig65.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 0  | <b>smitrig64.</b> Read-write. Reset: 1.                                                        |

|  |                                                                                                          |
|--|----------------------------------------------------------------------------------------------------------|
|  | <b>Description:</b> This defines the trigger mode for SmiTrig[95:64].<br>0: Active low<br>1: Active high |
|--|----------------------------------------------------------------------------------------------------------|

**SMIx000000E8 (FCH::SMI::SMITRIG4)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; SMIx000000E8; SMI=FED8\_0200h

| Bits | Description                                                                                     |
|------|-------------------------------------------------------------------------------------------------|
| 31   | <b>smitrig159.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 30   | <b>smitrig158.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 29   | <b>smitrig157.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 28   | <b>smitrig156.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 27   | <b>smitrig155.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 26   | <b>smitrig154.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 25   | <b>smitrig153.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 24   | <b>smitrig152.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Active high<br>0: Active low |
| 23   | <b>smitrig151.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 22   | <b>smitrig150.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 21   | <b>smitrig149.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 20   | <b>smitrig148.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 19   | <b>smitrig147.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 18   | <b>smitrig146.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 17   | <b>smitrig145.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 16   | <b>smitrig144.</b> Read-write. Reset: 1.                                                        |

|    |                                                                                                 |
|----|-------------------------------------------------------------------------------------------------|
|    | <b>Description:</b> 0: Active low<br>1: Active high                                             |
| 15 | <b>smitrig143.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 14 | <b>smitrig142.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Active high<br>0: Active low |
| 13 | <b>smitrig141.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 12 | <b>smitrig140.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 11 | <b>smitrig139.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 10 | <b>smitrig138.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 9  | <b>smitrig137.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 8  | <b>smitrig136.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 7  | <b>smitrig135.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 6  | <b>smitrig134.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 5  | <b>smitrig133.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 4  | <b>smitrig132.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 3  | <b>smitrig131.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 2  | <b>smitrig130.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 1  | <b>smitrig129.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Active low<br>1: Active high |
| 0  | <b>smitrig128.</b> Read-write. Reset: 1.                                                        |

|  |                                                                                                            |
|--|------------------------------------------------------------------------------------------------------------|
|  | <b>Description:</b> This defines the trigger mode for SmiTrig[159:128].<br>0: Active low<br>1: Active high |
|--|------------------------------------------------------------------------------------------------------------|

### 14.3.6 High Precision Event Timer (HPET) Registers

HPET registers are accessed through two methods:

- Memory access to HPET memory address range from FED0\_0000h to FED0\_01FFh. Program PMx00[HpetEn]=1 to enable HPET decoding.
- Memory mapped access through the AcpiMmio region. The HPET registers range from FED8\_0000h+C00h to FED8\_0000h+CFFh. See PMx04[MmioEn].

All registers in this block are reset by PciRstB, which will be asserted in the following conditions:

- Resume Reset: This reset is asserted in G3 state and deasserted during G3 to S5 transition.
- System Reset: From system reset button.
- S0 Reset events: some events that happen in S0 state, such as CF9 and Keyboard Reset.
- Sleep states: S3, S5 and Power saving mode states.

If the HPET Timer comparator value to be programmed is very close to the current value of HPET Timer, by the time the new comparator value is loaded to the register, the value of HPET Timer could have increased to a value greater than the revised comparator value.

| HPETx00000000 (FCH::TMR::HPET::ID)               |                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Read-only. Reset: 1022_8201h.                    |                                                                                                                                                                                                                                                                                                                                    |
| _aliasHOSTLEGACY; HPETx00000000; HPET=FED0_0000h |                                                                                                                                                                                                                                                                                                                                    |
| Bits                                             | Description                                                                                                                                                                                                                                                                                                                        |
| 31:16                                            | <b>vendorid.</b> Read-only. Reset: 1022h. AMD vendor ID.                                                                                                                                                                                                                                                                           |
| 15                                               | <b>legacy_cap.</b> Read-only. Reset: 1. Legacy replacement interrupt is supported.                                                                                                                                                                                                                                                 |
| 14                                               | Reserved.                                                                                                                                                                                                                                                                                                                          |
| 13                                               | <b>counter_size_cap.</b> Read-only. Reset: 0.<br><b>Description:</b> Main counter is 32-bits wide or 64-bit mode. The read only register value depends on the HPET_width_sel register bit:<br>When HPET_width_sel is 0, Counter_Size_Cap=0, means 32-bits wide<br>When HPET_width_sel is 1, Counter_Size_Cap=1, means 64-bits wide |
| 12:8                                             | <b>num_tmr_cap.</b> Read-only. Reset: 02h. Three timers are supported.                                                                                                                                                                                                                                                             |
| 7:0                                              | <b>revid.</b> Read-only. Reset: 01h. Revision ID.                                                                                                                                                                                                                                                                                  |

| HPETx00000004 (FCH::TMR::HPET::CLKPERIOD)        |                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Read-only. Reset: 0429_B17Eh.                    |                                                                                                                                                                                                                                                                                  |
| _aliasHOSTLEGACY; HPETx00000004; HPET=FED0_0000h |                                                                                                                                                                                                                                                                                  |
| Bits                                             | Description                                                                                                                                                                                                                                                                      |
| 31:0                                             | <b>counter_clk_period.</b> Read-only. Reset: 0429_B17Eh. Specifies the clock period of each HPET timer tick. HPET main counter runs at 14.31818/48 MHz. The unit is femptoseconds ( $10^{-15}$ seconds). Note: The value of this register can be modified through MISC_Reg: 34h. |

**HPETx00000010 (FCH::TMR::HPET::HPETCONFIG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; HPETx00000010; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:9 | Reserved.                                                                                                                                                                                                                                                                                                                         |
| 8    | <b>rst_sync_pingpong.</b> Read-write. Reset: 0. There is sync ping-pong logic to read timer value in Host clock domain, if this sync ping-pong logic stopped, timer value can no longer be read. Writing '1' to this bit will reset the logic to start sync ping-pong again, writing '0' will do nothing. This bit is write only. |
| 7:2  | Reserved.                                                                                                                                                                                                                                                                                                                         |
| 1    | <b>legacyen.</b> Read-write. Reset: 0.<br><b>Description:</b> If LegacyEn is set to 1b then:<br>Timer0 interrupt goes to IRQ0 of PIC controller, INT2 of IoAPIC<br>Timer1 interrupt goes to IRQ8 of PIC controller, INT8 of IoAPIC.                                                                                               |
| 0    | <b>tmren.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pause main counter and disable all timer interrupts.<br>1: Allow main counter to run and allow timer interrupts if enabled.                                                                                                                                         |

**HPETx00000020 (FCH::TMR::HPET::INTERRUPT\_STATUS)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; HPETx00000020; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:3 | Reserved.                                                                                                                                                                                                                                                                                                |
| 2    | <b>tmr2intrsts.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Timer2 interrupt is not active.<br>1: Timer2 interrupt is active.<br>Write 1 to clear if timer2 is set to level-triggered mode.<br>When set to edge-triggered mode, software should ignore this bit and always write 0b to this bit. |
| 1    | <b>tmr1intrsts.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Timer1 interrupt is not active.<br>1: Timer1 interrupt is active.<br>Write 1 to clear if timer1 is set to level-triggered mode.<br>When set to edge-triggered mode, software should ignore this bit and always write 0b to this bit. |
| 0    | <b>tmr0intrsts.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Timer0 interrupt is not active.<br>1: Timer0 interrupt is active.<br>Write 1 to clear if timer0 is set to level-triggered mode.<br>When set to edge-triggered mode, software should ignore this bit and always write 0b to this bit. |

**HPETx000000F0 (FCH::TMR::HPET::MAIN\_COUNTER\_L)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; HPETx000000F0; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>maincounter_l.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> HPET main counter, increment by 1 on every clock.<br>Counter should be written to only when halted.<br>The width of main counter depends on HPET_width_sel register bit:<br>0: HPET main counter is 32-bit. MainCounter_L is the valid counter bits. MainCounter_H is not valid.<br>1: HPET main counter is 64-bit. Both MainCounter_L and MainCounter_H are valid. |

**HPETx000000F4 (FCH::TMR::HPET::MAIN\_COUNTER\_H)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; HPETx000000F4; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>maincounter_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> HPET main counter, increment by 1 on every clock.<br>Counter should be written to only when halted.<br>The width of main counter depends on HPET_width_sel register bit:<br>0: HPET main counter is 32-bit. MainCounter_L is the valid counter bits. MainCounter_H is not valid.<br>1: HPET main counter is 64-bit. Both MainCounter_L and MainCounter_H are valid. |

**HPETx00000100 (FCH::TMR::HPET::TMR0\_CONF\_CAP\_L)**

Read-write. Reset: 0000\_8010h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000100; HPET=FED0\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 15    | <b>tmrfsbcap.</b> Read-write. Reset: 1. FSB delivery is supported. Read only.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 14    | <b>tmrfsben.</b> Read-write. Reset: 0. Set to 1 to enable FSB (Front Side Bus) delivery of interrupt.                                                                                                                                                                                                                                                                                                                                                                                                       |
| 13:9  | <b>tmrintroute.</b> Read-write. Reset: 00h. These 5 bits specify which INT entry of IoAPIC the timer is routed to when LegacyEn is not set.                                                                                                                                                                                                                                                                                                                                                                 |
| 8     | <b>tmr32modeen.</b> Read-write. Reset: 0. Timer n 32-bit Mode: (where n is the timer number: 0 to 2). Software can set this read/write bit to force a 64-bit timer to behave as a 32-bit timer. This is typically needed if the software is not willing to halt the main counter to read or write a particular timer, and the software is not capable of doing an atomic 64-bit read to the timer. If the timer is not 64 bits wide, then this bit will always be read as 0 and writes will have no effect. |
| 7     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6     | <b>tmrsetper.</b> Read-write. Reset: 0.<br><b>Description:</b> Set to 1 to allow software to set the timer's accumulator if the timer is set to periodic mode. The bit is automatically cleared when 'Comparator' is written by software.                                                                                                                                                                                                                                                                   |
| 5     | <b>tmrsizecap.</b> Read-write. Reset: 0.<br><b>Description:</b> The timer is 32-bits wide/or 64bit wide. This read only register value depend on the HPET_width_sel register bit:<br>When HPET_width_sel is 0, TmrSizeCap =0, means 32-bits wide<br>When HPET_width_sel is 1, TmrSizeCap =1, means 64-bits wide                                                                                                                                                                                             |
| 4     | <b>tmrtypcap.</b> Read-write. Reset: 1. The timer supports periodic interrupt delivery mode. Read only.                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3     | <b>tmrtyp.</b> Read-write. Reset: 0.<br><b>Description:</b> Select the timer interrupt type:<br>0: Non-periodic<br>1: Periodic                                                                                                                                                                                                                                                                                                                                                                              |
| 2     | <b>tmrinten.</b> Read-write. Reset: 0. Set to 1 to enable timer interrupt.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1     | <b>tmrinttyp.</b> Read-write. Reset: 0.<br><b>Description:</b> Control timer interrupt polarity:<br>0: Edge triggered<br>1: Level triggered                                                                                                                                                                                                                                                                                                                                                                 |
| 0     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



**HPETx00000104 (FCH::TMR::HPET::TMR0\_CONF\_CAP\_H)**

Read-only. Reset: 00C0\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000104; HPET=FED0\_0000h

| Bits | Description                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmrintroutecap.</b> Read-only. Reset: 00C0_0000h. Indicates which INT entry of IoAPIC can be assigned to the timer interrupt. Read only. |

**HPETx00000108 (FCH::TMR::HPET::TMR0\_COMP\_L)**

Read-write. Reset: FFFF\_FFFFh.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000108; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>comparator_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> The timer comparator.<br>In non-periodic mode: 'Comparator' is writeable.<br>In periodic mode: 'Comparator' can be modified after TmrSetPer is set to 1. 'Comparator' is periodically incremented by the value last written to this register. By default, value is incremented by 0xFFFFFFFF for 32-bit mode or 0xFFFFFFFFFFFFFFFF for 64-bit mode.<br>The width of Comparator depends on HPET_width_sel register bit:<br>0: The Comparator is 32-bit. Comparator_L is the valid bits. Comparator_H is not valid.<br>1: The Comparator is 64-bit. Both Comparator_L and Comparator_H are valid. |

**HPETx0000010C (FCH::TMR::HPET::TMR0\_COMP\_H)**

Read-write. Reset: 0000\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

Note1: The HPET spec requires different default values for 32-bit and 64-bit HPET. For 32-bit HPET, Comparator\_H default should be 00000000h. For 64-bit HPET, Comparator\_H default should be FFFFFFFFh. Our HPET can be configured as 32-bit or 64-bit by HPET\_width\_sel register bit. By default, HPET is 32-bit, so the default of Comparator\_H is 00000000h.

\_aliasHOSTLEGACY; HPETx0000010C; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>comparator_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> The timer comparator.<br>In non-periodic mode: 'Comparator' is writeable.<br>In periodic mode: 'Comparator' can be modified after TmrSetPer is set to 1. 'Comparator' is periodically incremented by the value last written to this register. By default, value is incremented by 0xFFFFFFFF for 32-bit mode or 0xFFFFFFFFFFFFFFFF for 64-bit mode.<br>The width of Comparator depends on HPET_width_sel register bit:<br>0: The Comparator is 32-bit. Comparator_L is the valid bits. Comparator_H is not valid.<br>1: The Comparator is 64-bit. Both Comparator_L and Comparator_H are valid. |

**HPETx00000110 (FCH::TMR::HPET::TMR0\_FSBINTVAL)**

Read-write. Reset: 0000\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000110; HPET=FED0\_0000h

| Bits | Description                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tn_fsb_int_val.</b> Read-write. Reset: 0000_0000h. Software sets this 32-bit field to specify the write data of FSB Interrupt Message. |

**HPETx00000114 (FCH::TMR::HPET::TMR0\_FSBINTADDR)**

Read-write. Reset: 0000\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000114; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tn_fsb_int_addr.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> Software sets this 32-bit field to specify the address of FSB interrupt Message. |

**HPETx00000120 (FCH::TMR::HPET::TMR1\_CONF\_CAP\_L)**

Read-write. Reset: 0000\_8010h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000120; HPET=FED0\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 15    | <b>tmrfsbcap.</b> Read-write. Reset: 1. FSB delivery is supported. Read only.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 14    | <b>tmrfsben.</b> Read-write. Reset: 0. Set to 1 to enable FSB (Front Side Bus) delivery of interrupt.                                                                                                                                                                                                                                                                                                                                                                                                       |
| 13:9  | <b>tmrintroute.</b> Read-write. Reset: 00h. These 5 bits specify which INT entry of IoAPIC the timer is routed to when LegacyEn is not set.                                                                                                                                                                                                                                                                                                                                                                 |
| 8     | <b>tmr32modeen.</b> Read-write. Reset: 0. Timer n 32-bit Mode: (where n is the timer number: 0 to 2). Software can set this read/write bit to force a 64-bit timer to behave as a 32-bit timer. This is typically needed if the software is not willing to halt the main counter to read or write a particular timer, and the software is not capable of doing an atomic 64-bit read to the timer. If the timer is not 64 bits wide, then this bit will always be read as 0 and writes will have no effect. |
| 7     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6     | <b>tmrsetper.</b> Read-write. Reset: 0.<br><b>Description:</b> Set to 1 to allow software to set the timer's accumulator if the timer is set to periodic mode. The bit is automatically cleared when 'Comparator' is written by software.                                                                                                                                                                                                                                                                   |
| 5     | <b>tmrsizecap.</b> Read-write. Reset: 0.<br><b>Description:</b> The timer is 32-bits wide/or 64bit wide. This read only register value depend on the HPET_width_sel register bit:<br>When HPET_width_sel is 0, TmrSizeCap =0, means 32-bits wide<br>When HPET_width_sel is 1, TmrSizeCap =1, means 64-bits wide                                                                                                                                                                                             |
| 4     | <b>tmrtypcap.</b> Read-write. Reset: 1. The timer supports periodic interrupt delivery mode. Read only.                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3     | <b>tmrtyp.</b> Read-write. Reset: 0.<br><b>Description:</b> Select the timer interrupt type:<br>0: Non-periodic<br>1: Periodic                                                                                                                                                                                                                                                                                                                                                                              |
| 2     | <b>tmrinten.</b> Read-write. Reset: 0. Set to 1 to enable timer interrupt.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1     | <b>tmrinttyp.</b> Read-write. Reset: 0.<br><b>Description:</b> Control timer interrupt polarity:<br>0: Edge triggered<br>1: Level triggered                                                                                                                                                                                                                                                                                                                                                                 |
| 0     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**HPETx00000124 (FCH::TMR::HPET::TMR1\_CONF\_CAP\_H)**

Read-only. Reset: 00C0\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000124; HPET=FED0\_0000h

| Bits | Description                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmrintroutecap.</b> Read-only. Reset: 00C0_0000h. Indicates which INT entry of IoAPIC can be assigned to the timer interrupt. Read only. |

**HPETx00000128 (FCH::TMR::HPET::TMR1\_COMP\_L)**

Read-write. Reset: FFFF\_FFFFh.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000128; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>comparator_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> The timer comparator.<br>In non-periodic mode: 'Comparator' is writeable.<br>In periodic mode: 'Comparator' can be modified after TmrSetPer is set to 1. 'Comparator' is periodically incremented by the value last written to this register. By default, value is incremented by 0xFFFFFFFF for 32-bit mode or 0xFFFFFFFFFFFFFFFF for 64-bit mode.<br>The width of Comparator depends on HPET_width_sel register bit:<br>0: The Comparator is 32-bit. Comparator_L is the valid bits. Comparator_H is not valid.<br>1: The Comparator is 64-bit. Both Comparator_L and Comparator_H are valid. |

**HPETx0000012C (FCH::TMR::HPET::TMR1\_COMP\_H)**

Read-write. Reset: 0000\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

Note1: The HPET spec requires different default values for 32-bit and 64-bit HPET. For 32-bit HPET, Comparator\_H default should be 00000000h. For 64-bit HPET, Comparator\_H default should be FFFFFFFFh. Our HPET can be configured as 32-bit or 64-bit by HPET\_width\_sel register bit. By default, HPET is 32-bit, so the default of Comparator\_H is 00000000h.

\_aliasHOSTLEGACY; HPETx0000012C; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>comparator_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> The timer comparator.<br>In non-periodic mode: 'Comparator' is writeable.<br>In periodic mode: 'Comparator' can be modified after TmrSetPer is set to 1. 'Comparator' is periodically incremented by the value last written to this register. By default, value is incremented by 0xFFFFFFFF for 32-bit mode or 0xFFFFFFFFFFFFFFFF for 64-bit mode.<br>The width of Comparator depends on HPET_width_sel register bit:<br>0: The Comparator is 32-bit. Comparator_L is the valid bits. Comparator_H is not valid.<br>1: The Comparator is 64-bit. Both Comparator_L and Comparator_H are valid. |

**HPETx00000130 (FCH::TMR::HPET::TMR1\_FSBINTVAL)**

Read-write. Reset: 0000\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000130; HPET=FED0\_0000h

| Bits | Description                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tn_fsb_int_val.</b> Read-write. Reset: 0000_0000h. Software sets this 32-bit field to specify the write data of FSB Interrupt Message. |

**HPETx00000134 (FCH::TMR::HPET::TMR1\_FSBINTADDR)**

Read-write. Reset: 0000\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000134; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tn_fsb_int_addr.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> Software sets this 32-bit field to specify the address of FSB interrupt Message. |

**HPETx00000140 (FCH::TMR::HPET::TMR2\_CONF\_CAP\_L)**

Read-write. Reset: 0000\_8010h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000140; HPET=FED0\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 15    | <b>tmrfsbcap.</b> Read-write. Reset: 1. FSB delivery is supported. Read only.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 14    | <b>tmrfsben.</b> Read-write. Reset: 0. Set to 1 to enable FSB (Front Side Bus) delivery of interrupt.                                                                                                                                                                                                                                                                                                                                                                                                       |
| 13:9  | <b>tmrintroute.</b> Read-write. Reset: 00h. These 5 bits specify which INT entry of IoAPIC the timer is routed to when LegacyEn is not set.                                                                                                                                                                                                                                                                                                                                                                 |
| 8     | <b>tmr32modeen.</b> Read-write. Reset: 0. Timer n 32-bit Mode: (where n is the timer number: 0 to 2). Software can set this read/write bit to force a 64-bit timer to behave as a 32-bit timer. This is typically needed if the software is not willing to halt the main counter to read or write a particular timer, and the software is not capable of doing an atomic 64-bit read to the timer. If the timer is not 64 bits wide, then this bit will always be read as 0 and writes will have no effect. |
| 7     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 6     | <b>tmrsetper.</b> Read-write. Reset: 0.<br><b>Description:</b> Set to 1 to allow software to set the timer's accumulator if the timer is set to periodic mode. The bit is automatically cleared when 'Comparator' is written by software.                                                                                                                                                                                                                                                                   |
| 5     | <b>tmrsizecap.</b> Read-write. Reset: 0.<br><b>Description:</b> The timer is 32-bits wide/or 64bit wide. This read only register value depend on the HPET_width_sel register bit:<br>When HPET_width_sel is 0, TmrSizeCap =0, means 32-bits wide<br>When HPET_width_sel is 1, TmrSizeCap =1, means 64-bits wide                                                                                                                                                                                             |
| 4     | <b>tmrtypcap.</b> Read-write. Reset: 1. The timer supports periodic interrupt delivery mode. Read only.                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3     | <b>tmrtyp.</b> Read-write. Reset: 0.<br><b>Description:</b> Select the timer interrupt type:<br>0: Non-periodic<br>1: Periodic                                                                                                                                                                                                                                                                                                                                                                              |
| 2     | <b>tmrinten.</b> Read-write. Reset: 0. Set to 1 to enable timer interrupt.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1     | <b>tmrinttyp.</b> Read-write. Reset: 0.<br><b>Description:</b> Control timer interrupt polarity:<br>0: Edge triggered<br>1: Level triggered                                                                                                                                                                                                                                                                                                                                                                 |
| 0     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**HPETx00000144 (FCH::TMR::HPET::TMR2\_CONF\_CAP\_H)**

Read-only. Reset: 00C0\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000144; HPET=FED0\_0000h

| Bits | Description                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmrintroutecap.</b> Read-only. Reset: 00C0_0000h. Indicates which INT entry of IoAPIC can be assigned to the timer interrupt. Read only. |

**HPETx00000148 (FCH::TMR::HPET::TMR2\_COMP\_L)**

Read-write. Reset: FFFF\_FFFFh.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000148; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>comparator_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> The timer comparator.<br>In non-periodic mode: 'Comparator' is writeable.<br>In periodic mode: 'Comparator' can be modified after TmrSetPer is set to 1. 'Comparator' is periodically incremented by the value last written to this register. By default, value is incremented by 0xFFFFFFFF for 32-bit mode or 0xFFFFFFFFFFFFFFFF for 64-bit mode.<br>The width of Comparator depends on HPET_width_sel register bit:<br>0: The Comparator is 32-bit. Comparator_L is the valid bits. Comparator_H is not valid.<br>1: The Comparator is 64-bit. Both Comparator_L and Comparator_H are valid. |

**HPETx0000014C (FCH::TMR::HPET::TMR2\_COMP\_H)**

Read-write. Reset: 0000\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

Note1: The HPET spec requires different default values for 32-bit and 64-bit HPET. For 32-bit HPET, Comparator\_H default should be 00000000h. For 64-bit HPET, Comparator\_H default should be FFFFFFFFh. Our HPET can be configured as 32-bit or 64-bit by HPET\_width\_sel register bit. By default, HPET is 32-bit, so the default of Comparator\_H is 00000000h.

\_aliasHOSTLEGACY; HPETx0000014C; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>comparator_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> The timer comparator.<br>In non-periodic mode: 'Comparator' is writeable.<br>In periodic mode: 'Comparator' can be modified after TmrSetPer is set to 1. 'Comparator' is periodically incremented by the value last written to this register. By default, value is incremented by 0xFFFFFFFF for 32-bit mode or 0xFFFFFFFFFFFFFFFF for 64-bit mode.<br>The width of Comparator depends on HPET_width_sel register bit:<br>0: The Comparator is 32-bit. Comparator_L is the valid bits. Comparator_H is not valid.<br>1: The Comparator is 64-bit. Both Comparator_L and Comparator_H are valid. |

**HPETx00000150 (FCH::TMR::HPET::TMR2\_FSBINTVAL)**

Read-write. Reset: 0000\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000150; HPET=FED0\_0000h

| Bits | Description                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tn_fsb_int_val.</b> Read-write. Reset: 0000_0000h. Software sets this 32-bit field to specify the write data of FSB Interrupt Message. |

**HPETx00000154 (FCH::TMR::HPET::TMR2\_FSBINTADDR)**

Read-write. Reset: 0000\_0000h.

Hardware supports 3 timers, &lt;N&gt; is 0, 1 or 2.

\_aliasHOSTLEGACY; HPETx00000154; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tn_fsb_int_addr.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> Software sets this 32-bit field to specify the address of FSB interrupt Message. |

**HPETx000001B0 (FCH::TMR::HPET::TMR0\_COMP\_BASE\_SHADOW\_L)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; HPETx000001B0; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr0_comp_base_shadow_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> This is shadow of the base value of Tmr0 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 0 Comparator. Writing the register will change the base value of Timer 0 Comparator. |

**HPETx000001B4 (FCH::TMR::HPET::TMR0\_COMP\_BASE\_SHADOW\_H)**

Read-write. Reset: 0000\_0000h.

Note1: The HPET spec requires different default values for 32-bit and 64-bit HPET. For 32-bit HPET, Comparator\_H default should be 00000000h. For 64-bit HPET, Comparator\_H default should be FFFFFFFFh. Our HPET can be configured as 32-bit or 64-bit by HPET\_width\_sel register bit. By default, HPET is 32-bit, so the default of Comparator\_H is 00000000h.

\_aliasHOSTLEGACY; HPETx000001B4; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr0_comp_base_shadow_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> This is shadow of the base value of Tmr0 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 0 Comparator. Writing the register will change the base value of Timer 0 Comparator. |

**HPETx000001B8 (FCH::TMR::HPET::TMR0\_COMP\_SHADOW\_L)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; HPETx000001B8; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr0_comp_shadow_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> This is shadow of the current value of Tmr0 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 0 Comparator. Writing the register will change the base value of Timer 0 Comparator. |

**HPETx000001BC (FCH::TMR::HPET::TMR0\_COMP\_SHADOW\_H)**

Read-write. Reset: 0000\_0000h.

Note1: The HPET spec requires different default values for 32-bit and 64-bit HPET. For 32-bit HPET, Comparator\_H default should be 00000000h. For 64-bit HPET, Comparator\_H default should be FFFFFFFFh. Our HPET can be configured as 32-bit or 64-bit by HPET\_width\_sel register bit. By default, HPET is 32-bit, so the default of Comparator\_H is 00000000h.

\_aliasHOSTLEGACY; HPETx000001BC; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr0_comp_shadow_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> This is shadow of the current value of Tmr0 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 0 Comparator. Writing the register will change the base value of Timer 0 Comparator. |

**HPETx000001C0 (FCH::TMR::HPET::TMR1\_COMP\_BASE\_SHADOW\_L)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; HPETx000001C0; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr1_comp_base_shadow_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> This is shadow of the base value of Tmr1 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 1 Comparator. Writing the register will change the base value of Timer 1 Comparator. |

**HPETx000001C4 (FCH::TMR::HPET::TMR1\_COMP\_BASE\_SHADOW\_H)**

Read-write. Reset: 0000\_0000h.

Note1: The HPET spec requires different default values for 32-bit and 64-bit HPET. For 32-bit HPET, Comparator\_H default should be 00000000h. For 64-bit HPET, Comparator\_H default should be FFFFFFFFh. Our HPET can be configured as 32-bit or 64-bit by HPET\_width\_sel register bit. By default, HPET is 32-bit, so the default of Comparator\_H is 00000000h.

\_aliasHOSTLEGACY; HPETx000001C4; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr1_comp_base_shadow_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> This is shadow of the base value of Tmr1 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 1 Comparator. Writing the register will change the base value of Timer 1 Comparator. |

**HPETx000001C8 (FCH::TMR::HPET::TMR1\_COMP\_SHADOW\_L)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; HPETx000001C8; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr1_comp_shadow_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> This is shadow of the current value of Tmr1 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 1 Comparator. Writing the register will change the base value of Timer 1 Comparator. |

**HPETx000001CC (FCH::TMR::HPET::TMR1\_COMP\_SHADOW\_H)**

Read-write. Reset: 0000\_0000h.

Note1: The HPET spec requires different default values for 32-bit and 64-bit HPET. For 32-bit HPET, Comparator\_H default should be 00000000h. For 64-bit HPET, Comparator\_H default should be FFFFFFFFh. Our HPET can be configured as 32-bit or 64-bit by HPET\_width\_sel register bit. By default, HPET is 32-bit, so the default of Comparator\_H is 00000000h.

\_aliasHOSTLEGACY; HPETx000001CC; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr1_comp_shadow_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> This is shadow of the current value of Tmr1 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 1 Comparator. Writing the register will change the base value of Timer 1 Comparator. |

**HPETx000001D0 (FCH::TMR::HPET::TMR2\_COMP\_BASE\_SHADOW\_L)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; HPETx000001D0; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr2_comp_base_shadow_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> This is shadow of the base value of Tmr2 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 2 Comparator. Writing the register will change the base value of Timer 2 Comparator. |

**HPETx000001D4 (FCH::TMR::HPET::TMR2\_COMP\_BASE\_SHADOW\_H)**

Read-write. Reset: 0000\_0000h.

Note1: The HPET spec requires different default values for 32-bit and 64-bit HPET. For 32-bit HPET, Comparator\_H default should be 00000000h. For 64-bit HPET, Comparator\_H default should be FFFFFFFFh. Our HPET can be configured as 32-bit or 64-bit by HPET\_width\_sel register bit. By default, HPET is 32-bit, so the default of Comparator\_H is 00000000h.

\_aliasHOSTLEGACY; HPETx000001D4; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr2_comp_base_shadow_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> This is shadow of the base value of Tmr2 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 2 Comparator. Writing the register will change the base value of Timer 2 Comparator. |

**HPETx000001D8 (FCH::TMR::HPET::TMR2\_COMP\_SHADOW\_L)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; HPETx000001D8; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr2_comp_shadow_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> This is shadow of the current value of Tmr2 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 2 Comparator. Writing the register will change the base value of Timer 2 Comparator. |

**HPETx000001DC (FCH::TMR::HPET::TMR2\_COMP\_SHADOW\_H)**

Read-write. Reset: 0000\_0000h.

Note1: The HPET spec requires different default values for 32-bit and 64-bit HPET. For 32-bit HPET, Comparator\_H default should be 00000000h. For 64-bit HPET, Comparator\_H default should be FFFFFFFFh. Our HPET can be configured as 32-bit or 64-bit by HPET\_width\_sel register bit. By default, HPET is 32-bit, so the default of Comparator\_H is 00000000h.

\_aliasHOSTLEGACY; HPETx000001DC; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmr2_comp_shadow_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> This is shadow of the current value of Tmr2 Comparator (HPET_Reg:108h). Reading this register returns the base value of Timer 2 Comparator. Writing the register will change the base value of Timer 2 Comparator. |

**HPETx000001E0 (FCH::TMR::HPET::MAIN\_COUNTER\_RTC\_L)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; HPETx000001E0; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>main_counter_rtc_l.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> This is a shadow of Main_Counter register. It samples the value of Main_Counter at every falling edge of Rtc 32kHz clock for SW to read. When SW writes this register, Main_Counter (offset F0) will be updated with the same value written to this register, and then enabled counting at the next RtcClk falling edge. The purpose of this register is for the convenience of SW save/restore HPET.<br>For 32-bit HPET mode, only Main_Counter_rtc_L is valid.<br>For 64-bit HPET mode, both Main_Counter_rtc_L and Main_Counter_rtc_H are valid. |



**HPETx000001E4 (FCH::TMR::HPET::MAIN\_COUNTER\_RTC\_H)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; HPETx000001E4; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>main_counter_rtc_h.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> This is a shadow of Main_Counter register. It samples the value of Main_Counter at every falling edge of Rtc 32kHz clock for SW to read. When SW writes this register, Main_Counter (offset F0) will be updated with the same value written to this register, and then enabled counting at the next RtcClk falling edge. The purpose of this register is for the convenience of SW save/restore HPET.<br>For 32-bit HPET mode, only Main_Counter_rtc_L is valid.<br>For 64-bit HPET mode, both Main_Counter_rtc_L and Main_Counter_rtc_H are valid. |

**HPETx000001E8 (FCH::TMR::HPET::NXTTMRREMAIN\_L)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; HPETx000001E8; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>nxttmrremain_l.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> Read-only. This register tells SW how many timer ticks remain before the next enabled comparator interrupt.<br>For 32-bit HPET mode, only NxtTmrRemain_L is valid.<br>For 64-bit HPET mode, both NxtTmrRemain_L and NxtTmrRemain_H are valid. |

**HPETx000001EC (FCH::TMR::HPET::NXTTMRREMAIN\_H)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; HPETx000001EC; HPET=FED0\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>nxttmrremain_h.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> Read-only. This register tells SW how many timer ticks remain before the next enabled comparator interrupt.<br>For 32-bit HPET mode, only NxtTmrRemain_L is valid.<br>For 64-bit HPET mode, both NxtTmrRemain_L and NxtTmrRemain_H are valid. |

**14.3.7 I2C Registers**See 14.1.12 [I2C Master] for [I2C](#) pad selection to choose 1.1V or 1.8V or 3.3V.

**FCHI2Cx FEDC[2...B]000 [I2C Control Register] (FCH::I2C::IC\_CON)**

Reset: 0000\_007Fh.

**I2C** Control Register.

This register can be written only when the I2C\_IP\_CORE is disabled, which corresponds to the IC\_ENABLE[0] register being set to 0. Writes at other times have no effect.

Read/Write Access:

- If configuration parameter I2C\_DYNAMIC\_TAR\_UPDATE=1, bit 4 is read only.
- If configuration parameter IC\_RX\_FULL\_HLD\_BUS\_EN=0, bit 9 is read only.
- If configuration parameter IC\_STOP\_DET\_IF\_MASTER\_ACTIVE=0, bit 10 is read only.
- If configuration parameter IC\_BUS\_CLEAR\_FEATURE=0, bit 11 is read only
- If configuration parameter IC\_OPTIONAL\_SAR=0, bit 16 is read only
- If configuration parameter IC\_SMBUS=0, bit 17 is read only
- If configuration parameter IC\_SMBUS\_ARP=0, bits 18 and 19 are read only.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB000,FEDC\_{6:2}000]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:20 | <b>RSVD_IC_CON_2.</b> Read-only,Volatile. Reset: 000h. IC_CON_2 Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 19    | <b>RSVD_SMBUS_PERSISTENT_SLV_ADDR_EN.</b> Read-only,Volatile. Reset: 0.<br>SMBUS_PERSISTENT_SLV_ADDR_EN Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 18    | <b>RSVD_SMBUS_ARP_EN.</b> Read-only,Volatile. Reset: 0. SMBUS_ARP_EN Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 17    | <b>RSVD_SMBUS_SLAVE_QUICK_EN.</b> Read-only,Volatile. Reset: 0. SMBUS_SLAVE_QUICK_EN Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 16    | <b>RSVD_OPTIONAL_SAR_CTRL.</b> Read-only,Volatile. Reset: 0. OPTIONAL_SAR_CTRL Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 15:12 | <b>RSVD_IC_CON_1.</b> Read-only,Volatile. Reset: 0h. IC_CON_1 Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 11    | <b>BUS_CLEAR_FEATURE_CTRL.</b> Read-write,Volatile. Reset: 0.<br><b>Description:</b> In Master mode:<br>- 1'b1: Bus Clear Feature is enabled.<br>- 1'b0: Bus Clear Feature is Disabled.<br>In Slave mode, this register bit is not applicable.                                                                                                                                                                                                                                                                                                                     |
| 10    | <b>STOP_DET_IF_MASTER_ACTIVE.</b> Read-only,Volatile. Reset: 0.<br><b>Description:</b> In Master mode:<br>- 1'b1: issues the STOP_DET interrupt only when master is active.<br>- 1'b0: issues the STOP_DET irrespective of whether master is active or not.                                                                                                                                                                                                                                                                                                        |
| 9     | <b>RX_FIFO_FULL_HLD_CTRL.</b> Read-write,Volatile. Reset: 0.<br><b>Description:</b> This bit controls whether I2C_IP_CORE should hold the bus when the Rx FIFO is physically full to its RX_BUFFER_DEPTH, as described in the IC_RX_FULL_HLD_BUS_EN parameter.                                                                                                                                                                                                                                                                                                     |
| 8     | <b>TX_EMPTY_CTRL.</b> Read-write,Volatile. Reset: 0.<br><b>Description:</b> This bit controls the generation of the TX_EMPTY interrupt, as described in the IC_RAW_INTR_STAT register.                                                                                                                                                                                                                                                                                                                                                                             |
| 7     | <b>STOP_DET_IFADDRESSED.</b> Read-write,Volatile. Reset: 0.<br><b>Description:</b> In slave mode:<br>- 1'b1: issues the STOP_DET interrupt only when it is addressed.<br>- 0'b0: issues the STOP_DET irrespective of whether it's addressed or not.<br><br>NOTE: During a general call address, this slave does not issue the STOP_DET interrupt if STOP_DET_IF_ADDRESSED = 1'b1, even if the slave responds to the general call address by generating ACK. The STOP_DET interrupt is generated only when the transmitted address matches the slave address (SAR). |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6   | <p><b>IC_SLAVE_DISABLE.</b> Read-write,Volatile. Reset: 1.</p> <p><b>Description:</b> This bit controls whether <a href="#">I2C</a> has its slave disabled, which means once the preseln signal is applied, then this bit takes on the value of the configuration parameter IC_SLAVE_DISABLE. You have the choice of having the slave enabled or disabled after reset is applied, which means software does not have to configure the slave. By default, the slave is always enabled (in reset state as well). If you need to disable it after reset, set this bit to 1.</p> <p>If this bit is set (slave is disabled), I2C_IP_CORE functions only as a master and does not perform any action that requires a slave.</p> <p>NOTE: Software should ensure that if this bit is written with 0, then bit 0 should also be written with a 0.</p>                                                                                                                                          |
| 5   | <p><b>IC_RESTART_EN.</b> Read-write,Volatile. Reset: 1.</p> <p><b>Description:</b> Determines whether RESTART conditions may be sent when acting as a master. Some older slaves do not support handling RESTART conditions; however, RESTART conditions are used in several I2C_IP_CORE operations. When RESTART is disabled, the master is prohibited from performing the following functions:</p> <ul style="list-style-type: none"> <li>- Sending a START BYTE</li> <li>- Performing any high-speed mode operation</li> <li>- High-speed mode operation</li> <li>- Performing direction changes in combined format mode</li> <li>- Performing a read operation with a 10-bit address</li> </ul> <p>By replacing RESTART condition followed by a STOP and a subsequent START condition, split operations are broken down into multiple I2C_IP_CORE transfers. If the above operations are performed, it will result in setting bit 6 (TX_ABRT) of the IC_RAW_INTR_STAT register.</p> |
| 4   | <p><b>IC_10BITADDR_MASTER.</b> Read-write,Volatile. Reset: 1.</p> <p><b>Description:</b> If the I2C_DYNAMIC_TAR_UPDATE configuration parameter is set to 'No' (0), this bit is named IC_10BITADDR_MASTER and controls whether the I2C_IP_CORE starts its transfers in 7- or 10-bit addressing mode when acting as a master.</p> <p>If I2C_DYNAMIC_TAR_UPDATE is set to 'Yes' (1), the function of this bit is handled by bit 12 of IC_TAR register, and becomes a read-only copy called IC_10BITADDR_MASTER_rd_only.</p> <ul style="list-style-type: none"> <li>- 0: 7-bit addressing</li> <li>- 1: 10-bit addressing</li> </ul>                                                                                                                                                                                                                                                                                                                                                       |
| 3   | <p><b>IC_10BITADDR_SLAVE.</b> Read-write,Volatile. Reset: 1.</p> <p><b>Description:</b> When acting as a slave, this bit controls whether the I2C_IP_CORE responds to 7- or 10-bit addresses.</p> <ul style="list-style-type: none"> <li>- 0: 7-bit addressing. The I2C_IP_CORE ignores transactions that involve 10-bit addressing; for 7-bit addressing, only the lower 7 bits of the IC_SAR register are compared.</li> <li>- 1: 10-bit addressing. The I2C_IP_CORE responds to only 10-bit addressing transfers that match the full 10 bits of the IC_SAR register.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2:1 | <p><b>SPEED.</b> Read-write,Volatile. Reset: 3h.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|   |                                                                                                                                                                                                                                                                                                                                                                                                     |             |                                     |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------------------------------|
|   | <b>Description:</b> This register should be programmed only with a value in the range of 1 to IC_MAX_SPEED_MODE; otherwise, hardware updates this register with the value of IC_MAX_SPEED_MODE.<br>1: standard mode (100 kbit/s)<br>2: fast mode (<=400 kbit/s) or fast mode plus (<=1000Kbit/s)<br>3: high speed mode (3.4 Mbit/s)<br>Note: This field is not applicable when IC_ULTRA_FAST_MODE=1 |             |                                     |
|   | <b>ValidValues:</b>                                                                                                                                                                                                                                                                                                                                                                                 |             |                                     |
|   | <b>Value</b>                                                                                                                                                                                                                                                                                                                                                                                        | <b>Name</b> | <b>Description</b>                  |
|   | 0h                                                                                                                                                                                                                                                                                                                                                                                                  |             | Reserved.                           |
|   | 1h                                                                                                                                                                                                                                                                                                                                                                                                  | STANDARD    | Standard Speed mode of operation    |
| 0 | 2h                                                                                                                                                                                                                                                                                                                                                                                                  | FAST        | Fast or Fast Plus mode of operation |
|   | 3h                                                                                                                                                                                                                                                                                                                                                                                                  | HIGH        | High Speed mode of operation        |
|   | <b>MASTER_MODE.</b> Read-write,Volatile. Reset: 1.<br><b>Description:</b> This bit controls whether the I2C_IP_CORE master is enabled.<br><br>NOTE: Software should ensure that if this bit is written with '1' then bit 6 should also be written with a '1'.                                                                                                                                       |             |                                     |

**FCH I2CxFEDC[2...B]004 [I2C Target Address Register] (FCH::I2C::IC\_TAR)**

Reset: 0000\_0055h.

**I2C** Target Address Register

If the configuration parameter I2C\_DYNAMIC\_TAR\_UPDATE is set to 'No' (0), this register is 12 bits wide, and bits 31:12 are reserved. This register can be written to only when IC\_ENABLE[0] is set to 0.

However, if I2C\_DYNAMIC\_TAR\_UPDATE = 1, then the register becomes 13 bits wide.

In this case, writes to IC\_TAR succeed when one of the following conditions are true:

- I2C\_IP\_CORE is NOT enabled (IC\_ENABLE[0] is set to 0);
- I2C\_IP\_CORE is enabled (IC\_ENABLE[0]=1); AND I2C\_IP\_CORE is NOT engaged in any Master (tx, rx) operation (IC\_STATUS[5]=0); AND I2C\_IP\_CORE is enabled to operate in Master mode (IC\_CON[0]=1); AND there are NO entries in the TX FIFO (IC\_STATUS[2]=1)

You can change the TAR address dynamically without losing the bus, only if the following conditions are met.

- I2C\_IP\_CORE is enabled (IC\_ENABLE[0]=1); AND IC\_EMPTYFIFO\_HOLD\_MASTER\_EN configuration parameter is set to 1; AND I2C\_IP\_CORE is enabled to operate in Master mode (IC\_CON[0]=1); AND there are NO entries in the Tx FIFO and the master is in HOLD state (IC\_INTR\_STAT[13]=1).

Note: If the software or application is aware that the I2C\_IP\_CORE is not using the TAR address for the pending commands in the Tx FIFO, then it is possible to update the TAR address even while the Tx FIFO has entries (IC\_STATUS[2]= 0).

- It is not necessary to perform any write to this register if I2C\_IP\_CORE is enabled as an I2C slave only.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB004,FEDC\_[6:2]004]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | <b>RSVD_IC_TAR_2.</b> Read-only,Volatile. Reset: 0000h. IC_TAR_2 Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 16    | <b>RSVD_SMBUS_QUICK_CMD.</b> Read-only,Volatile. Reset: 0. SMBUS_QUICK_CMD Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 15:14 | <b>RSVD_IC_TAR_1.</b> Read-only,Volatile. Reset: 0h. IC_TAR_1 Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 13    | <b>RSVD_DEVICE_ID.</b> Read-only,Volatile. Reset: 0. DEVICE_ID Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 12    | <b>RSVD_IC_10BITADDR_MASTER.</b> Read-only,Volatile. Reset: 0. IC_10BITADDR_MASTER Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 11    | <b>SPECIAL.</b> Read-write,Volatile. Reset: 0.<br><b>Description:</b> This bit indicates whether software performs a Device-ID or General Call or START BYTE command.<br>- 0: ignore bit 10 GC_OR_START and use IC_TAR normally<br>- 1: perform special <b>I2C</b> command as specified in Device_ID or GC_OR_START bit                                                                                                                                                                                                                                                      |
| 10    | <b>GC_OR_START.</b> Read-write,Volatile. Reset: 0.<br><b>Description:</b> If bit 11 (SPECIAL) is set to 1 and bit 13(Device-ID) is set to 0, then this bit indicates whether a General Call or START byte command is to be performed by the I2C_IP_CORE.<br>- 0: General Call Address - after issuing a General Call, only writes may be performed. Attempting to issue a read command results in setting bit 6 (TX_ABRT) of the IC_RAW_INTR_STAT register. The I2C_IP_CORE remains in General Call mode until the SPECIAL bit value (bit 11) is cleared.<br>- 1: START BYTE |
| 9:0   | <b>IC_TAR.</b> Read-write,Volatile. Reset: 055h.<br><b>Description:</b> This is the target address for any master transaction. When transmitting a General Call, these bits are ignored. To generate a START BYTE, the CPU needs to write only once into these bits.<br><br>If the IC_TAR and IC_SAR are the same, loopback exists but the FIFOs are shared between master and slave, so full loopback is not feasible. Only one direction loopback mode is supported (simplex), not duplex. A master cannot transmit to itself; it can transmit to only a slave.            |

**FCHI2Cx FEDC[2...B]008 [I2C Slave Address Register] (FCH::I2C::IC\_SAR)**

Reset: 0000\_0055h.

**I2C** Slave Address Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB008,FEDC\_[6:2]008]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:10 | <b>RSVD_IC_SAR.</b> Read-only,Volatile. Reset: 00_0000h. IC_SAR Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 9:0   | <p><b>IC_SAR.</b> Read-write,Volatile. Reset: 055h.</p> <p><b>Description:</b> The IC_SAR holds the slave address when the <b>I2C</b> is operating as a slave. For 7-bit addressing, only IC_SAR[6:0] is used.</p> <p>This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect.</p> <p>Note:</p> <p>The default values cannot be any of the reserved address locations: that is, 0x00 to 0x07, or 0x78 to 0x7f. The correct operation of the device is not ensured if you program the IC_SAR or IC_TAR to a reserved value.</p> |

**FCHI2Cx FEDC[2...B]00C [I2C High Speed Master Mode Code Address Register] (FCH::I2C::IC\_HS\_MADDR)**

Reset: 0000\_0001h.

**I2C** High Speed Master Mode Code Address Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB00C,FEDC\_[6:2]00C]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:3 | <b>RSVD_IC_HS_MAR.</b> Read-only,Volatile. Reset: 0000_0000h. IC_HS_MAR Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2:0  | <p><b>IC_HS_MAR.</b> Read-write,Volatile. Reset: 1h.</p> <p><b>Description:</b> This bit field holds the value of the <b>I2C</b> HS mode master code. HS-mode master codes are reserved 8-bit codes (00001xxx) that are not used for slave addressing or other purposes. Each master has its unique master code; up to eight high-speed mode masters can be present on the same I2C bus system.</p> <p>Valid values are from 0 to 7. This register goes away and becomes read-only returning 0's if the IC_MAX_SPEED_MODE configuration parameter is set to either Standard (1) or Fast (2).</p> <p>This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect.</p> |

**FCHI2Cx FEDC[2...B]010 [I2C Rx/Tx Data Buffer and Command Register] (FCH::I2C::IC\_DATA\_CMD)**

Reset: 0000\_0000h.

**I2C** Rx/Tx Data Buffer and Command Register; this is the register the CPU writes to when filling the TX FIFO and the CPU reads from when retrieving bytes from RX FIFO.

The size of the register changes as follows:

Write:

- 11 bits when IC\_EMPTYFIFO\_HOLD\_MASTER\_EN=1
- 9 bits when IC\_EMPTYFIFO\_HOLD\_MASTER\_EN=0

Read:

- 12 bits when IC\_FIRST\_DATA\_BYTE\_STATUS = 1
- 8 bits when IC\_FIRST\_DATA\_BYTE\_STATUS = 0

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB010,FEDC\_[6:2]010]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | <b>RSVD_IC_DATA_CMD</b> . Read-only, Volatile. Reset: 0_0000h. IC_DATA_CMD Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 11    | <p><b>FIRST_DATA_BYTE</b>. Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> Indicates the first data byte received after the address phase for receive transfer in Master receiver or Slave receiver mode.</p> <p>NOTE: In case of APB_DATA_WIDTH=8,</p> <ol style="list-style-type: none"> <li>1. The user has to perform two APB Reads to IC_DATA_CMD in order to get status on 11 bit.</li> <li>2. In order to read the 11 bit, the user has to perform the first data byte read [7:0] (offset 0x10) and then perform the second read [15:8] (offset 0x11) in order to know the status of 11 bit (whether the data received in previous read is a first data byte or not).</li> <li>3. The 11th bit is an optional read field, user can ignore 2nd byte read [15:8] (offset 0x11) if not interested in FIRST_DATA_BYTE status.</li> </ol>                                                       |
| 10    | <p><b>RESTART</b>. Write-only, Volatile. Reset: 0.</p> <p><b>Description:</b> This bit controls whether a RESTART is issued before the byte is sent or received. This bit is available only if IC_EMPTYFIFO_HOLD_MASTER_EN is configured to 1.</p> <p>1 - If IC_RESTART_EN is 1, a RESTART is issued before the data is sent/received (according to the value of CMD), regardless of whether or not the transfer direction is changing from the previous command; if IC_RESTART_EN is 0, a STOP followed by a START is issued instead.</p> <p>0 - If IC_RESTART_EN is 1, a RESTART is issued only if the transfer direction is changing from the previous command; if IC_RESTART_EN is 0, a STOP followed by a START is issued instead.</p>                                                                                                                                                                 |
| 9     | <p><b>STOP</b>. Write-only, Volatile. Reset: 0.</p> <p><b>Description:</b> This bit controls whether a STOP is issued after the byte is sent or received. This bit is available only if IC_EMPTYFIFO_HOLD_MASTER_EN is configured to 1.</p> <ul style="list-style-type: none"> <li>- 1 - STOP is issued after this byte, regardless of whether or not the Tx FIFO is empty. If the Tx FIFO is not empty, the master immediately tries to start a new transfer by issuing a START and arbitrating for the bus.</li> <li>- 0 - STOP is not issued after this byte, regardless of whether or not the Tx FIFO is empty. If the Tx FIFO is not empty, the master continues the current transfer by sending/receiving data bytes according to the value of the CMD bit. If the Tx FIFO is empty, the master holds the SCL line low and stalls the bus until a new command is available in the Tx FIFO.</li> </ul> |
| 8     | <b>CMD</b> . Write-only, Volatile. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p><b>Description:</b> This bit controls whether a read or a write is performed. This bit does not control the direction when the I2C_IP_CORE acts as a slave. It controls only the direction when it acts as a master.</p> <p>When a command is entered in the TX FIFO, this bit distinguishes the write and read commands. In slave-receiver mode, this bit is a "don't care" because writes to this register are not required. In slave-transmitter mode, a "0" indicates that the data in IC_DATA_CMD is to be transmitted.</p> <p>When programming this bit, you should remember the following: attempting to perform a read operation after a General Call command has been sent results in a TX_ABRT interrupt (bit 6 of the IC_RAW_INTR_STAT register), unless bit 11 (SPECIAL) in the IC_TAR register has been cleared. If a "1" is written to this bit after receiving a RD_REQ interrupt, then a TX_ABRT interrupt occurs.</p> |
| 7:0 | <p><b>DAT.</b> Read-write, Volatile. Reset: 00h.</p> <p><b>Description:</b> This register contains the data to be transmitted or received on the <a href="#">I2C</a> bus. If you are writing to this register and want to perform a read, bits 7:0 (DAT) are ignored by the I2C_IP_CORE. However, when you read this register, these bits return the value of data received on the I2C_IP_CORE interface.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

#### FCHI2Cx FEDC[2...B]014 [Standard Speed I2C Clock SCL High Count Register] (FCH::I2C::IC\_SS\_SCL\_HCNT)

Reset: 0000\_0285h.

Standard Speed [I2C](#) Clock SCL High Count Register

[\\_link\[5:0\]](#) alias HOSTLEGACY; FCHI2Cx[FEDCB014, FEDC\_[6:2]014]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <p><b>RSVD_IC_SS_SCL_HIGH_COUNT.</b> Read-only, Volatile. Reset: 0000h. IC_SS_SCL_HCNT Reserved bits - Read Only</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 15:0  | <p><b>IC_SS_SCL_HCNT.</b> Read-write, Volatile. Reset: 0285h.</p> <p><b>Description:</b> This register must be set before any <a href="#">I2C</a> bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high-period count for standard speed. This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect. The minimum valid value is 6; hardware prevents values less than this being written, and if attempted results in 6 being set. When the configuration parameter IC_HC_COUNT_VALUES is set to 1, this register is read only. NOTE: This register must not be programmed to a value higher than 65525.</p> |



**FCHI2Cx FEDC[2...B]018 [Standard Speed I2C Clock SCL Low Count Register]  
(FCH::I2C::IC\_SS\_SCL\_LCNT)**

Reset: 0000\_0357h.

Standard Speed [I2C](#) Clock SCL Low Count Register[\\_link](#)[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB018,FEDC\_[6:2]018]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>RSVD_IC_SS_SCL_LOW_COUNT.</b> Read-only, Volatile. Reset: 0000h. RSVD_IC_SS_SCL_LOW_COUNT Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 15:0  | <b>IC_SS_SCL_LCNT.</b> Read-write, Volatile. Reset: 0357h.<br><b>Description:</b> This register must be set before any <a href="#">I2C</a> bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock low period count for standard speed. This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect. The minimum valid value is 8; hardware prevents values less than this being written, and if attempted, results in 8 being set. When the configuration parameter IC_HC_COUNT_VALUES is set to 1, this register is read only. |

**FCHI2Cx FEDC[2...B]01C [Fast Mode or Fast Mode Plus I2C Clock SCL High Count Register]  
(FCH::I2C::IC\_FS\_SCL\_HCNT)**

Reset: 0000\_0087h.

Fast Mode or Fast Mode Plus [I2C](#) Clock SCL High Count Register[\\_link](#)[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB01C,FEDC\_[6:2]01C]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>RSVD_IC_FS_SCL_HCNT.</b> Read-only, Volatile. Reset: 0000h. IC_FS_SCL_HCNT Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 15:0  | <b>IC_FS_SCL_HCNT.</b> Read-write, Volatile. Reset: 0087h.<br><b>Description:</b> This register must be set before any <a href="#">I2C</a> bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high-period count for fast mode or fast mode plus. It is used in high-speed mode to send the Master Code and START BYTE or General CALL. This register goes away and becomes read-only returning 0s if IC_MAX_SPEED_MODE = standard. This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect. The minimum valid value is 6; hardware prevents values less than this being written, and if attempted results in 6 being set. |

**FCHI2Cx FEDC[2...B]020 [Fast Mode or Fast Mode Plus I2C Clock SCL Low Count Register]  
(FCH::I2C::IC\_FS\_SCL\_LCNT)**

Reset: 0000\_00F0h.

Fast Mode or Fast Mode Plus [I2C](#) Clock SCL Low Count Register[\\_link](#)[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB020,FEDC\_[6:2]020]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>RSVD_IC_FS_SCL_LCNT.</b> Read-only, Volatile. Reset: 0000h. IC_FS_SCL_LCNT Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 15:0  | <b>IC_FS_SCL_LCNT.</b> Read-write, Volatile. Reset: 00F0h.<br><b>Description:</b> This register must be set before any <a href="#">I2C</a> bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock low period count for fast speed. It is used in high-speed mode to send the Master Code and START BYTE or General CALL. This register goes away and becomes read-only returning 0s if IC_MAX_SPEED_MODE = standard. This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect. The minimum valid value is 8; hardware prevents values less than this being written, and if attempted results in 8 being set. When the configuration parameter IC_HC_COUNT_VALUES is set to 1, this register is read only. |

**FCHI2Cx FEDC[2...B]024 [High Speed I2C Clock SCL High Count Register] (FCH::I2C::IC\_HS\_SCL\_HCNT)**

Reset: 0000\_0006h.

High Speed [I2C](#) Clock SCL High Count Register[\\_link\[5:0\]](#)\_aliasHOSTLEGACY; FCHI2Cx[FEDCB024,FEDC\_[6:2]024]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>RSVD_IC_HS_SCL_HCNT.</b> Read-only,Volatile. Reset: 0000h. IC_HS_SCL_HCNT Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 15:0  | <p><b>IC_HS_SCL_HCNT.</b> Read-write,Volatile. Reset: 0006h.</p> <p><b>Description:</b> This register must be set before any <a href="#">I2C</a> bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock high period count for high speed. The SCL High time depends on the loading of the bus. For 100pF loading, the SCL High time is 60ns; for 400pF loading, the SCL High time is 120ns. This register goes away and becomes read-only returning 0s if IC_MAX_SPEED_MODE != high. This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect. The minimum valid value is 6; hardware prevents values less than this being written, and if attempted results in 6 being set. The lower byte must be programmed first. Then the upper byte is programmed.</p> |

**FCHI2Cx FEDC[2...B]028 [High Speed I2C Clock SCL Low Count Register] (FCH::I2C::IC\_HS\_SCL\_LCNT)**

Reset: 0000\_0010h.

High Speed [I2C](#) Clock SCL Low Count Register[\\_link\[5:0\]](#)\_aliasHOSTLEGACY; FCHI2Cx[FEDCB028,FEDC\_[6:2]028]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>RSVD_IC_HS_SCL_LOW_CNT.</b> Read-only,Volatile. Reset: 0000h. IC_HS_SCL_LCNT Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 15:0  | <p><b>IC_HS_SCL_LCNT.</b> Read-write,Volatile. Reset: 0010h.</p> <p><b>Description:</b> This register must be set before any <a href="#">I2C</a> bus transaction can take place to ensure proper I/O timing. This register sets the SCL clock low period count for high speed. The SCL low time depends on the loading of the bus. For 100pF loading, the SCL low time is 160ns; for 400pF loading, the SCL low time is 320ns. This register goes away and becomes read-only returning 0s if IC_MAX_SPEED_MODE != high. This register can be written only when the I2C interface is disabled, which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect. The minimum valid value is 8; hardware prevents values less than this being written, and if attempted results in 8 being set. The lower byte must be programmed first. Then the upper byte is programmed. If the value is less than 8 then the count value gets changed to 8.</p> |

**FCHI2Cx FEDC[2...B]02C [I2C Interrupt Status Register] (FCH::I2C::IC\_INTR\_STAT)**

Read-only, Volatile. Reset: 0000\_0000h.

**I2C** Interrupt Status Register

Each bit in this register has a corresponding mask bit in the IC\_INTR\_MASK register. These bits are cleared by reading the matching interrupt clear register. The unmasked raw versions of these bits are available in the IC\_RAW\_INTR\_STAT register.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB02C,FEDC\_[6:2]02C]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                          |
|-------|--------------------------------------------------------------------------------------------------------------------------------------|
| 31:15 | <b>RSVD_IC_INTR_STAT.</b> Read-only, Volatile. Reset: 0_0000h. IC_INTR_STAT Reserved bits - Read Only                                |
| 14    | <b>R_SCL_STUCK_AT_LOW.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_SCL_STUCK_AT_LOW bit. |
| 13    | <b>R_MASTER_ON_HOLD.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_MASTER_ON_HOLD bit.     |
| 12    | <b>R_RESTART_DET.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_RESTART_DET bit.           |
| 11    | <b>R_GEN_CALL.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_GEN_CALL bit.                 |
| 10    | <b>R_START_DET.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_START_DET bit.               |
| 9     | <b>R_STOP_DET.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_STOP_DET bit.                 |
| 8     | <b>R_ACTIVITY.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_ACTIVITY bit.                 |
| 7     | <b>R_RX_DONE.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_RX_DONE bit.                   |
| 6     | <b>R_TX_ABRT.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_TX_ABRT bit.                   |
| 5     | <b>R_RD_REQ.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_RD_REQ bit.                     |
| 4     | <b>R_TX_EMPTY.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_TX_EMPTY bit.                 |
| 3     | <b>R_TX_OVER.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_TX_OVER bit.                   |
| 2     | <b>R_RX_FULL.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_RX_FULL bit.                   |
| 1     | <b>R_RX_OVER.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_RX_OVER bit.                   |
| 0     | <b>R_RX_UNDER.</b> Read-only, Volatile. Reset: 0. See IC_RAW_INTR_STAT for a detailed description of R_RX_UNDER bit.                 |

**FCHI2Cx FEDC[2...B]030 [I2C Interrupt Mask Register] (FCH::I2C::IC\_INTR\_MASK)**

Reset: 0000\_48FFh.

[I2C](#) Interrupt Mask Register.

These bits mask their corresponding interrupt status bits. This register is active low; a value of 0 masks the interrupt, whereas a value of 1 unmasks the interrupt.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB030,FEDC\_[6:2]030]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:15 | <b>RSVD_IC_INTR_STAT.</b> Read-only, Volatile. Reset: 0_0000h. IC_INTR_STAT Reserved bits - Read Only                                                                |
| 14    | <b>M_SCL_STUCK_AT_LOW.</b> Read-write, Volatile. Reset: 1. This bit masks the R_SCL_STUCK_AT_LOW interrupt in IC_INTR_STAT register.                                 |
| 13    | <b>M_MASTER_ON_HOLD_read_only.</b> Read-only, Volatile. Reset: 0. This M_MASTER_ON_HOLD_read_only bit masks the R_MASTER_ON_HOLD interrupt in IC_INTR_STAT register. |
| 12    | <b>M_RESTART_DET_read_only.</b> Read-only, Volatile. Reset: 0. This M_RESTART_DET_read_only bit masks the R_RESTART_DET interrupt in IC_INTR_STAT register.          |
| 11    | <b>M_GEN_CALL.</b> Read-write, Volatile. Reset: 1. This bit masks the R_GEN_CALL interrupt in IC_INTR_STAT register.                                                 |
| 10    | <b>M_START_DET.</b> Read-write, Volatile. Reset: 0. This bit masks the R_START_DET interrupt in IC_INTR_STAT register.                                               |
| 9     | <b>M_STOP_DET.</b> Read-write, Volatile. Reset: 0. This bit masks the R_STOP_DET interrupt in IC_INTR_STAT register.                                                 |
| 8     | <b>M_ACTIVITY.</b> Read-write, Volatile. Reset: 0. This bit masks the R_ACTIVITY interrupt in IC_INTR_STAT register.                                                 |
| 7     | <b>M_RX_DONE.</b> Read-write, Volatile. Reset: 1. This bit masks the R_RX_DONE interrupt in IC_INTR_STAT register.                                                   |
| 6     | <b>M_TX_ABRT.</b> Read-write, Volatile. Reset: 1. This bit masks the R_TX_ABRT interrupt in IC_INTR_STAT register.                                                   |
| 5     | <b>M_RD_REQ.</b> Read-write, Volatile. Reset: 1. This bit masks the R_RD_REQ interrupt in IC_INTR_STAT register.                                                     |
| 4     | <b>M_TX_EMPTY.</b> Read-write, Volatile. Reset: 1. This bit masks the R_TX_EMPTY interrupt in IC_INTR_STAT register.                                                 |
| 3     | <b>M_TX_OVER.</b> Read-write, Volatile. Reset: 1. This bit masks the R_TX_OVER interrupt in IC_INTR_STAT register.                                                   |
| 2     | <b>M_RX_FULL.</b> Read-write, Volatile. Reset: 1. This bit masks the R_RX_FULL interrupt in IC_INTR_STAT register.                                                   |
| 1     | <b>M_RX_OVER.</b> Read-write, Volatile. Reset: 1. This bit masks the R_RX_OVER interrupt in IC_INTR_STAT register.                                                   |
| 0     | <b>M_RX_UNDER.</b> Read-write, Volatile. Reset: 1. This bit masks the R_RX_UNDER interrupt in IC_INTR_STAT register.                                                 |

**FCHI2Cx FEDC[2...B]034 [I2C Raw Interrupt Status Register] (FCH::I2C::IC\_RAW\_INTR\_STAT)**

Read-only, Volatile. Reset: 0000\_0000h.

[I2C](#) Raw Interrupt Status Register.[\\_link\[5:0\]\\_alias](#)HOSTLEGACY; FCHI2Cx[FEDCB034, FEDC\_[6:2]034]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:15 | <b>RSVD_IC_RAW_INTR_STAT.</b> Read-only, Volatile. Reset: 0_0000h. IC_RAW_INTR_STAT Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 14    | <b>SCL_STUCK_AT_LOW.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Indicates whether the SCL Line is stuck at low for the IC_SCL_STUCK_LOW_TIMEOUT number of ic_clk periods.<br>Enabled only when IC_BUS_CLEAR_FEATURE=1 and IC_ULTRA_FAST_MODE=0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 13    | <b>MASTER_ON_HOLD.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Indicates whether master is holding the bus and TX FIFO is empty.<br>Enabled only when I2C_DYNAMIC_TAR_UPDATE=1 and IC_EMPTYFIFO_HOLD_MASTER_EN=1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 12    | <b>RESTART_DET.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Indicates whether a RESTART condition has occurred on the <a href="#">I2C</a> interface when the Slave mode is being addressed.<br>Enabled only when IC_SLV_RESTART_DET_EN=1.<br>Note: In high-speed mode or during a START BYTE transfer, the RESTART comes before the address field as per the I2C protocol. In this case, the slave is not the addressed slave when the RESTART is issued                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 11    | <b>GEN_CALL.</b> Read-only, Volatile. Reset: 0. Set only when a General Call address is received and it is acknowledged.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 10    | <b>START_DET.</b> Read-only, Volatile. Reset: 0. Indicates whether a START or RESTART condition has occurred on the <a href="#">I2C</a> interface regardless of it's in Slave mode or Master Mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 9     | <b>STOP_DET.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Indicates whether a STOP condition has occurred on the <a href="#">I2C</a> interface regardless of it's in Slave mode or Master mode.<br>In Slave Mode:<br>- If IC_CON[7]=1'b1 (STOP_DET_IFADDRESSED), the STOP_DET interrupt will be issued only if slave is addressed.<br>Note: During a general call address, this slave does not issue a STOP_DET interrupt if STOP_DET_IF_ADDRESSED=1'b1, even if the slave responds to the general call address by generating ACK. The STOP_DET interrupt is generated only when the transmitted address matches the slave address (SAR).<br>- If IC_CON[7]=1'b0 (STOP_DET_IFADDRESSED), the STOP_DET interrupt is issued irrespective of whether it is being addressed.<br>In Master Mode:<br>- If IC_CON[10]=1'b1 (STOP_DET_IF_MASTER_ACTIVE), the STOP_DET interrupt will be issued only if Master is active.<br>- If IC_CON[10]=1'b0 (STOP_DET_IFADDRESSED), the STOP_DET interrupt will be issued irrespective of whether master is active or not. |
| 8     | <b>ACTIVITY.</b> Read-only, Volatile. Reset: 0. Once this bit is set, it remains set unless it is cleared, which indicates that there was activity on the bus.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7     | <b>RX_DONE.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> In slave-transmitter mode, this bit is set to 1 if the master does not acknowledge a transmitted byte. This occurs on the last byte of the transmission, indicating that the transmission is done.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 6     | <b>TX_ABRT.</b> Read-only, Volatile. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p><b>Description:</b> When this bit is set to 1, the IC_TX_ABRT_SOURCE register indicates the reason why the transmit abort takes places.</p> <p><b>Note:</b><br/>The Tx FIFO remains in this flushed state until the register IC_CLR_TX_ABRT is read. Once this read is performed, the Tx FIFO is then ready to accept more data bytes from the APB interface. RX FIFO flush because of TX_ABRT is controlled by the IP parameter IC_AVOID_RX_FIFO_FLUSH_ON_TX_ABRT.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 5 | <p><b>RD_REQ.</b> Read-only, Volatile. Reset: 0. This bit is set to 0 just after the processor reads the IC_CLR_RD_REQ register.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 4 | <p><b>TX_EMPTY.</b> Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> The behavior of the TX_EMPTY interrupt status differs based on the TX_EMPTY_CTRL selection in the IC_CON register.</p> <ul style="list-style-type: none"> <li>- When TX_EMPTY_CTRL = 0:<br/>This bit is set to 1 when the transmit buffer is at or below the threshold value set in the IC_TX_TL register.</li> <li>- When TX_EMPTY_CTRL = 1:<br/>This bit is set to 1 when the transmit buffer is at or below the threshold value set in the IC_TX_TL register and the transmission of the address/data from the internal shift register for the most recently popped command is completed.</li> </ul> <p>It is automatically cleared by hardware when the buffer level goes above the threshold. When IC_ENABLE[0] is set to 0, the TX FIFO is flushed and held in reset. There the TX FIFO looks like it has no data within it, so this bit is set to 1, provided there is activity in the master or slave state machines. When there is no longer any activity, then with ic_en=0, this bit is set to 0.</p> |
| 3 | <p><b>TX_OVER.</b> Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> Set during transmit if the transmit buffer is filled to IC_TX_BUFFER_DEPTH and the processor attempts to issue another <a href="#">I2C</a> command by writing to the IC_DATA_CMD register. When the module is disabled, this bit keeps its level until the master or slave state machines go into idle, and when ic_en goes to 0, this interrupt is cleared.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2 | <p><b>RX_FULL.</b> Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> Set when the receive buffer reaches or goes above the RX_TL threshold in the IC_RX_TL register. It is automatically cleared by hardware when buffer level goes below the threshold.</p> <p>If the module is disabled (IC_ENABLE[0]=0), the RX FIFO is flushed and held in reset; therefore the RX FIFO is not full. So this bit is cleared once the IC_ENABLE bit 0 is programmed with a 0, regardless of the activity that continues.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 1 | <p><b>RX_OVER.</b> Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> Set if the receive buffer is completely filled to IC_RX_BUFFER_DEPTH and an additional byte is received from an external <a href="#">I2C</a> device.</p> <p>If the module is disabled (IC_ENABLE[0]=0), this bit keeps its level until the master or slave state machines go into idle, and when ic_en goes to 0, this interrupt is cleared.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0 | <p><b>RX_UNDER.</b> Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> Set if the processor attempts to read the receive buffer when it is empty by reading from the IC_DATA_CMD register.</p> <p>If the module is disabled (IC_ENABLE[0]=0), this bit keeps its level until the master or slave state machines go into idle, and when ic_en goes to 0, this interrupt is cleared.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |



**FCHI2Cx FEDC[2...B]038 [I2C Receive FIFO Threshold Register] (FCH::I2C::IC\_RX\_TL)**

Reset: 0000\_0000h.

**I2C** Receive FIFO Threshold Register[\\_link\[5:0\]](#) alias HOSTLEGACY; FCHI2Cx[FEDCB038,FEDC\_[6:2]038]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | <b>RSVD_IC_RX_TL.</b> Read-only, Volatile. Reset: 00_0000h. IC_RX_TL Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 7:0  | <b>RX_TL.</b> Read-write, Volatile. Reset: 00h.<br><b>Description:</b> Receive FIFO Threshold Level.<br><br>Controls the level of entries (or above) that triggers the RX_FULL interrupt (bit 2 in IC_RAW_INTR_STAT register). The valid range is 0-255, with the additional restriction that hardware does not allow this value to be set to a value larger than the depth of the buffer. If an attempt is made to do that, the actual value set will be the maximum depth of the buffer. A value of 0 sets the threshold for 1 entry, and a value of 255 sets the threshold for 256 entries. |

**FCHI2Cx FEDC[2...B]03C [I2C Transmit FIFO Threshold Register] (FCH::I2C::IC\_TX\_TL)**

Reset: 0000\_0000h.

**I2C** Transmit FIFO Threshold Register[\\_link\[5:0\]](#) alias HOSTLEGACY; FCHI2Cx[FEDCB03C,FEDC\_[6:2]03C]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | <b>RSVD_IC_TX_TL.</b> Read-only, Volatile. Reset: 00_0000h. IC_TX_TL Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7:0  | <b>TX_TL.</b> Read-write, Volatile. Reset: 00h.<br><b>Description:</b> Transmit FIFO Threshold Level.<br><br>Controls the level of entries (or below) that trigger the TX_EMPTY interrupt (bit 4 in IC_RAW_INTR_STAT register). The valid range is 0-255, with the additional restriction that it may not be set to value larger than the depth of the buffer. If an attempt is made to do that, the actual value set will be the maximum depth of the buffer. A value of 0 sets the threshold for 0 entries, and a value of 255 sets the threshold for 255 entries. |

**FCHI2Cx FEDC[2...B]040 [Clear Combined and Individual Interrupt Register] (FCH::I2C::IC\_CLR\_INTR)**

Read-only, Volatile. Reset: 0000\_0000h.

## Clear Combined and Individual Interrupt Register

[\\_link\[5:0\]](#) alias HOSTLEGACY; FCHI2Cx[FEDCB040,FEDC\_[6:2]040]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_INTR.</b> Read-only, Volatile. Reset: 0000_0000h. CLR_INTR Reserved bits - Read Only                                                                                                                                                                                                                                                                                 |
| 0    | <b>CLR_INTR.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the combined interrupt, all individual interrupts, and the IC_TX_ABRT_SOURCE register. This bit does not clear hardware clearable interrupts but software clearable interrupts. Refer to Bit 9 of the IC_TX_ABRT_SOURCE register for an exception to clearing IC_TX_ABRT_SOURCE. |

**FCHI2Cx FEDC[2...B]044 [Clear RX\_UNDER Interrupt Register] (FCH::I2C::IC\_CLR\_RX\_UNDER)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear RX\_UNDER Interrupt Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB044, FEDC\_[6:2]044]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_RX_UNDER.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_RX_UNDER Reserved bits - Read Only                                                          |
| 0    | <b>CLR_RX_UNDER.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the RX_UNDER interrupt (bit 0) of the IC_RAW_INTR_STAT register. |

**FCHI2Cx FEDC[2...B]048 [Clear RX\_OVER Interrupt Register] (FCH::I2C::IC\_CLR\_RX\_OVER)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear RX\_OVER Interrupt Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB048, FEDC\_[6:2]048]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_RX_OVER.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_RX_OVER Reserved bits - Read Only                                                          |
| 0    | <b>CLR_RX_OVER.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the RX_OVER interrupt (bit 1) of the IC_RAW_INTR_STAT register. |

**FCHI2Cx FEDC[2...B]04C [Clear TX\_OVER Interrupt Register] (FCH::I2C::IC\_CLR\_TX\_OVER)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear TX\_OVER Interrupt Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB04C, FEDC\_[6:2]04C]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_TX_OVER.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_TX_OVER Reserved bits - Read Only                                                          |
| 0    | <b>CLR_TX_OVER.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the TX_OVER interrupt (bit 3) of the IC_RAW_INTR_STAT register. |

**FCHI2Cx FEDC[2...B]050 [Clear RD\_REQ Interrupt Register] (FCH::I2C::IC\_CLR\_RD\_REQ)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear RD\_REQ Interrupt Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB050, FEDC\_[6:2]050]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_RD_REQ.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_RD_REQ Reserved bits - Read Only                                                          |
| 0    | <b>CLR_RD_REQ.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the RD_REQ interrupt (bit 5) of the IC_RAW_INTR_STAT register. |



**FCHI2CxFEDC[2...B]054 [Clear TX\_ABRT Interrupt Register] (FCH::I2C::IC\_CLR\_TX\_ABRT)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear TX\_ABRT Interrupt Register

[\\_link\[5:0\]](#) [\\_aliasHOSTLEGACY](#); FCHI2Cx[FEDCB054, FEDC\_ [6:2]054]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_TX_ABRT.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_TX_ABRT Reserved bits - Read Only                                                                                                                                                                                                                                                                                                 |
| 0    | <b>CLR_TX_ABRT.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the TX_ABRT interrupt (bit 6) of the IC_RAW_INTR_STAT register, and the IC_TX_ABRT_SOURCE register. This also releases the TX FIFO from the flushed/reset state, allowing more writes to the TX FIFO. Refer to Bit 9 of the IC_TX_ABRT_SOURCE register for an exception to clearing IC_TX_ABRT_SOURCE. |

**FCHI2CxFEDC[2...B]058 [Clear RX\_DONE Interrupt Register] (FCH::I2C::IC\_CLR\_RX\_DONE)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear RX\_DONE Interrupt Register

[\\_link\[5:0\]](#) [\\_aliasHOSTLEGACY](#); FCHI2Cx[FEDCB058, FEDC\_ [6:2]058]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_RX_DONE.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_RX_DONE Reserved bits - Read Only                                                          |
| 0    | <b>CLR_RX_DONE.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the RX_DONE interrupt (bit 7) of the IC_RAW_INTR_STAT register. |

**FCHI2CxFEDC[2...B]05C [Clear ACTIVITY Interrupt Register] (FCH::I2C::IC\_CLR\_ACTIVITY)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear ACTIVITY Interrupt Register

[\\_link\[5:0\]](#) [\\_aliasHOSTLEGACY](#); FCHI2Cx[FEDCB05C, FEDC\_ [6:2]05C]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_ACTIVITY.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_ACTIVITY Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0    | <b>CLR_ACTIVITY.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Reading this register clears the ACTIVITY interrupt if the <a href="#">I2C</a> is not active anymore. If the I2C module is still active on the bus, the ACTIVITY interrupt bit continues to be set. It is automatically cleared by hardware if the module is disabled and if there is no further activity on the bus. The value read from this register to get status of the ACTIVITY interrupt (bit 8) of the IC_RAW_INTR_STAT register. |

**FCHI2Cx FEDC[2...B]060 [Clear STOP\_DET Interrupt Register] (FCH::I2C::IC\_CLR\_STOP\_DET)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear STOP\_DET Interrupt Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB060, FEDC\_[6:2]060]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_STOP_DET.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_STOP_DET Reserved bits - Read Only                                                          |
| 0    | <b>CLR_STOP_DET.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the STOP_DET interrupt (bit 9) of the IC_RAW_INTR_STAT register. |

**FCHI2Cx FEDC[2...B]064 [Clear START\_DET Interrupt Register] (FCH::I2C::IC\_CLR\_START\_DET)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear START\_DET Interrupt Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB064, FEDC\_[6:2]064]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_START_DET.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_START_DET Reserved bits - Read Only                                                           |
| 0    | <b>CLR_START_DET.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the START_DET interrupt (bit 10) of the IC_RAW_INTR_STAT register. |

**FCHI2Cx FEDC[2...B]068 [Clear GEN\_CALL Interrupt Register] (FCH::I2C::IC\_CLR\_GEN\_CALL)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear GEN\_CALL Interrupt Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB068, FEDC\_[6:2]068]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_CLR_GEN_CALL.</b> Read-only, Volatile. Reset: 0000_0000h. IC_CLR_GEN_CALL Reserved bits - Read Only                                                       |
| 0    | <b>CLR_GEN_CALL.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Read this register to clear the GEN_CALL interrupt (bit 11) of IC_RAW_INTR_STAT register. |

**FCHI2Cx FEDC[2...B]06C [I2C ENABLE Register] (FCH::I2C::IC\_ENABLE)**

Reset: 0000\_0000h.

[I2C](#) Enable Register

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB06C,FEDC\_[6:2]06C]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:19 | <b>RSVD_IC_ENABLE_2.</b> Read-only, Volatile. Reset: 0000h. IC_ENABLE Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 18    | <b>RSVD_SMBUS_ALERT_EN.</b> Read-only, Volatile. Reset: 0. SMBUS_ALERT_EN Register field Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 17    | <b>RSVD_SMBUS_SUSPEND_EN.</b> Read-only, Volatile. Reset: 0. SMBUS_SUSPEND_EN Register field Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 16    | <b>RSVD_SMBUS_CLK_RESET.</b> Read-only, Volatile. Reset: 0. SMBUS_CLK_RESET Register field Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 15:4  | <b>RSVD_IC_ENABLE_1.</b> Read-only, Volatile. Reset: 000h. RSVD_IC_ENABLE_1 Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3     | <b>SDA_STUCK_RECOVERY_ENABLE.</b> Read-write, Volatile. Reset: 0.<br><b>Description:</b> If SDA is stuck at low indicated through the TX_ABORT interrupt (IC_TX_ABRT_SOURCE[17]), then this bit is used as a control knob to initiate the SDA Recovery Mechanism (that is, send at most 9 SCL clocks and STOP to release the SDA line) and then this bit gets auto clear                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2     | <b>TX_CMD_BLOCK.</b> Read-write, Volatile. Reset: 0.<br><b>Description:</b> In Master mode:<br>- 1'b1: Blocks the transmission of data on <a href="#">I2C</a> bus even if Tx FIFO has data to transmit.<br>- 1'b0: The transmission of data starts on I2C bus automatically, as soon as the first data is available in the Tx FIFO.<br>Note: To block the execution of Master commands, set the TX_CMD_BLOCK bit only when Tx FIFO is empty (IC_STATUS[2]==1) and Master is in Idle state (IC_STATUS[5] == 0).<br>Any further commands put in the Tx FIFO are not executed until TX_CMD_BLOCK bit is unset.                                                                                                                                                                                                                                |
| 1     | <b>ABORT.</b> Read-write, Volatile. Reset: 0.<br><b>Description:</b> When set, the controller initiates the transfer abort.<br>- 0: ABORT not initiated or ABORT done<br>- 1: ABORT operation in progress<br>The software can abort the <a href="#">I2C</a> transfer in master mode by setting this bit. The software can set this bit only when ENABLE is already set; otherwise, the controller ignores any write to ABORT bit. The software cannot clear the ABORT bit once set. In response to an ABORT, the controller issues a STOP and flushes the Tx FIFO after completing the current transfer, then sets the TX_ABORT interrupt after the abort operation. The ABORT bit is cleared automatically after the abort operation.<br><br>For a detailed description on how to abort I2C transfers, refer to "Aborting I2C Transfers". |
| 0     | <b>ENABLE.</b> Read-write, Volatile. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**Description:** Controls whether the I2C\_IP\_CORE is enabled.

- 0: Disables I2C\_IP\_CORE (TX and RX FIFOs are held in an erased state)

- 1: Enables I2C\_IP\_CORE

Software can disable I2C\_IP\_CORE while it is active.

When I2C\_IP\_CORE is disabled, the following occurs:

- The TX FIFO and RX FIFO get flushed.

- Status bits in the IC\_INTR\_STAT register are still active until DW\_apb\_i2c goes into IDLE state.

If the module is transmitting, it stops as well as deletes the contents of the transmit buffer after the current transfer is complete.

If the module is receiving, the I2C\_IP\_CORE stops the current transfer at the end of the current byte and does not acknowledge the transfer.

In systems with asynchronous pclk and ic\_clk when IC\_CLK\_TYPE parameter set to asynchronous (1), there is a two ic\_clk delay when enabling or disabling the I2C\_IP\_CORE.

**FCHI2Cx FEDC[2...B]070 [I2C STATUS Register] (FCH::I2C::IC\_STATUS)**

Read-only, Volatile. Reset: 0000\_0006h.

**I2C** Status Register. This is a read-only register used to indicate the current transfer status and FIFO status. The status register may be read at any time. None of the bits in this register request an interrupt.

When the I2C is disabled by writing 0 in bit 0 of the IC\_ENABLE register:

- Bits 1 and 2 are set to 1
- Bits 3 and 10 are set to 0

When the master or slave state machines goes to idle and ic\_en=0:

- Bits 5 and 6 are set to 0

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB070,FEDC\_[6:2]070]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:21 | <b>RSVD_IC_STATUS_2.</b> Read-only, Volatile. Reset: 000h. IC_STATUS Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 20    | <b>RSVD_SMBUS_ALERT_STATUS.</b> Read-only, Volatile. Reset: 0. SMBUS_ALERT_STATUS Register field Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                     |
| 19    | <b>RSVD_SMBUS_SUSPEND_STATUS.</b> Read-only, Volatile. Reset: 0. SMBUS_SUSPEND_STATUS Register field Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                 |
| 18    | <b>RSVD_SMBUS_SLAVE_ADDR_RESOLVED.</b> Read-only, Volatile. Reset: 0. SMBUS_SLAVE_ADDR_RESOLVED Register field Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                       |
| 17    | <b>RSVD_SMBUS_SLAVE_ADDR_VALID.</b> Read-only, Volatile. Reset: 0. SMBUS_SLAVE_ADDR_VALID Register field Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                             |
| 16    | <b>RSVD_SMBUS_QUICK_CMD_BIT.</b> Read-only, Volatile. Reset: 0. SMBUS_QUICK_CMD_BIT Register field Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                   |
| 15:12 | <b>RSVD_IC_STATUS_1.</b> Read-only, Volatile. Reset: 0h. RSVD_IC_STATUS_1 Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 11    | <b>SDA_STUCK_NOT_RECOVERED.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> This bit indicates that SDA stuck at low is not recovered after the recovery mechanism. In Slave mode, this register bit is not applicable.                                                                                                                                                                                                                                                                                              |
| 10    | <b>SLV_HOLD_RX_FIFO_FULL.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> This bit indicates the BUS Hold in Slave mode due to Rx FIFO is Full and an additional byte has been received (This kind of Bus hold is applicable if IC_RX_FULL_HLD_BUS_EN is set to 1).                                                                                                                                                                                                                                                  |
| 9     | <b>SLV_HOLD_TX_FIFO_EMPTY.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> This bit indicates the BUS Hold in Slave mode for the Read request when the Tx FIFO is empty. The Bus is in hold until the Tx FIFO has data to Transmit for the read request.                                                                                                                                                                                                                                                             |
| 8     | <b>MST_HOLD_RX_FIFO_FULL.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> This bit indicates the BUS Hold in Master mode due to Rx FIFO is Full and additional byte has been received (This kind of Bus hold is applicable if IC_RX_FULL_HLD_BUS_EN is set to 1).                                                                                                                                                                                                                                                    |
| 7     | <b>MST_HOLD_TX_FIFO_EMPTY.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> If the IC_EMPTYFIFO_HOLD_MASTER_EN parameter is set to 1, I2C_IP_CORE stalls the write transfer when Tx FIFO is empty, and the the last byte does not have the Stop bit set. This bit indicates the BUS hold when the master holds the bus because of the Tx FIFO being empty, and the the previous transferred command does not have the Stop bit set. (This kind of Bus hold is applicable if IC_EMPTYFIFO_HOLD_MASTER_EN is set to 1). |
| 6     | <b>SLV_ACTIVITY.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Slave FSM Activity Status. When the Slave Finite State Machine (FSM) is not in the IDLE state, this bit is set.<br>- 0: Slave FSM is in IDLE state so the Slave part I2C_IP_CORE is not Active.<br>- 1: Slave FSM is not in IDLE state so the Slave part of I2C_IP_CORE is Active.                                                                                                                                                                  |
| 5     | <b>MST_ACTIVITY.</b> Read-only, Volatile. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <b>Description:</b> Master FSM Activity Status. When the Master Finite State Machine (FSM) is not in the IDLE state, this bit is set.<br>- 0: Master FSM is in IDLE state so the Master part of I2C_IP_CORE is not Active.<br>- 1: Master FSM is not in IDLE state so the Master part of I2C_IP_CORE is Active.<br>Note:<br>IC_STATUS[0]-that is, ACTIVITY bit-is the OR of SLV_ACTIVITY and MST_ACTIVITY bits. |
| 4 | <b>RFF.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Receive FIFO Completely Full.<br>When the receive FIFO is completely full, this bit is set. When the receive FIFO contains one or more empty location, this bit is cleared.<br>- 0: Receive FIFO is not full<br>- 1: Receive FIFO is full                                                                                                     |
| 3 | <b>RFNE.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> Receive FIFO Not Empty.<br>This bit is set when the receive FIFO contains one or more entries; it is cleared when the receive FIFO is empty.<br>- 0: Receive FIFO is empty<br>- 1: Receive FIFO is not empty                                                                                                                                 |
| 2 | <b>TFE.</b> Read-only, Volatile. Reset: 1.<br><b>Description:</b> Transmit FIFO Completely Empty.<br>When the transmit FIFO is completely empty, this bit is set. When it contains one or more valid entries, this bit is cleared. This bit field does not request an interrupt.<br>- 0: Transmit FIFO is not empty<br>- 1: Transmit FIFO is empty                                                              |
| 1 | <b>TFNF.</b> Read-only, Volatile. Reset: 1.<br><b>Description:</b> Transmit FIFO Not Full.<br>Set when the transmit FIFO contains one or more empty locations, and is cleared when the FIFO is full.<br>- 0: Transmit FIFO is full<br>- 1: Transmit FIFO is not full                                                                                                                                            |
| 0 | <b>ACTIVITY.</b> Read-only, Volatile. Reset: 0. <a href="#">I2C</a> Activity Status.                                                                                                                                                                                                                                                                                                                            |

**FCHI2Cx FEDC[2...B]074 [I2C Transmit FIFO Level Register] (FCH::I2C::IC\_TXFLR)**

Read-only, Volatile. Reset: 0000\_0000h.

[I2C](#) Transmit FIFO Level Register

This register contains the number of valid data entries in the transmit FIFO buffer. It is cleared whenever:

- The I2C is disabled
- There is a transmit abort - that is, TX\_ABRT bit is set in the IC\_RAW\_INTR\_STAT register
- The slave bulk transmit mode is aborted

The register increments whenever data is placed into the transmit FIFO and decrements when data is taken from the transmit FIFO.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB074, FEDC\_[6:2]074]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:9 | <b>RSVD_TXFLR.</b> Read-only, Volatile. Reset: 00_0000h. TXFLR Register field Reserved bits - Read Only                                                        |
| 8:0  | <b>TXFLR.</b> Read-only, Volatile. Reset: 000h.<br><b>Description:</b> Transmit FIFO Level.<br>Contains the number of valid data entries in the transmit FIFO. |

**FCHI2CxFEDC[2...B]078 [I2C Receive FIFO Level Register] (FCH::I2C::IC\_RXFLR)**

Read-only, Volatile. Reset: 0000\_0000h.

[I2C](#) Receive FIFO Level Register

This register contains the number of valid data entries in the receive FIFO buffer. It is cleared whenever:

- The I2C is disabled
- Whenever there is a transmit abort caused by any of the events tracked in IC\_TX\_ABRT\_SOURCE

The register increments whenever data is placed into the receive FIFO and decrements when data is taken from the receive FIFO.

[\\_link\[5:0\]](#) [aliasHOSTLEGACY](#); FCHI2Cx[FEDCB078, FEDC\_[6:2]078]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:9 | <b>RSVD_RXFLR.</b> Read-only, Volatile. Reset: 00_0000h. RXFLR Reserved bits - Read Only                                                                     |
| 8:0  | <b>RXFLR.</b> Read-only, Volatile. Reset: 000h.<br><b>Description:</b> Receive FIFO Level.<br>Contains the number of valid data entries in the receive FIFO. |

**FCHI2CxFEDC[2...B]07C [I2C SDA Hold Time Length Register] (FCH::I2C::IC\_SDA\_HOLD)**

Reset: 0000\_0001h.

[I2C](#) SDA Hold Time Length Register

The bits [15:0] of this register are used to control the hold time of SDA during transmit in both slave and master mode (after SCL goes from HIGH to LOW).

The bits [23:16] of this register are used to extend the SDA transition (if any) whenever SCL is HIGH in the receiver in either master or slave mode.

Writes to this register succeed only when IC\_ENABLE[0]=0.

The values in this register are in units of ic\_clk period. The value programmed in IC\_SDA\_TX\_HOLD must be greater than the minimum hold time in each mode one cycle in master mode, seven cycles in slave mode for the value to be implemented.

The programmed SDA hold time during transmit (IC\_SDA\_TX\_HOLD) cannot exceed at any time the duration of the low part of scl. Therefore the programmed value cannot be larger than N\_SCL\_LOW-2, where N\_SCL\_LOW is the duration of the low part of the scl period measured in ic\_clk cycles.

[\\_link\[5:0\]](#) [aliasHOSTLEGACY](#); FCHI2Cx[FEDCB07C, FEDC\_[6:2]07C]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>RSVD_IC_SDA_HOLD.</b> Read-only, Volatile. Reset: 00h. IC_SDA_HOLD Reserved bits - Read Only                                                               |
| 23:16 | <b>IC_SDA_RX_HOLD.</b> Read-write, Volatile. Reset: 00h. Sets the required SDA hold time in units of ic_clk period, when I2C_IP_CORE acts as a receiver.      |
| 15:0  | <b>IC_SDA_TX_HOLD.</b> Read-write, Volatile. Reset: 0001h. Sets the required SDA hold time in units of ic_clk period, when I2C_IP_CORE acts as a transmitter. |

**FCHI2Cx FEDC[2...B]080 [I2C Transmit Abort Source Register] (FCH::I2C::IC\_TX\_ABRT\_SOURCE)**

Read-only, Volatile. Reset: 0000\_0000h.

**I2C** Transmit Abort Source Register

This register has 32 bits that indicate the source of the TX\_ABRT bit. Except for Bit 9, this register is cleared whenever the IC\_CLR\_TX\_ABRT register or the IC\_CLR\_INTR register is read. To clear Bit 9, the source of the ABRT\_SBYTE\_NORSTRT must be fixed first; RESTART must be enabled (IC\_CON[5]=1), the SPECIAL bit must be cleared (IC\_TAR[11]), or the GC\_OR\_START bit must be cleared (IC\_TAR[10]).

Once the source of the ABRT\_SBYTE\_NORSTRT is fixed, then this bit can be cleared in the same manner as other bits in this register. If the source of the ABRT\_SBYTE\_NORSTRT is not fixed before attempting to clear this bit, Bit 9 clears for one cycle and is then re-asserted.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB080,FEDC\_[6:2]080]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:23 | <b>TX_FLUSH_CNT.</b> Read-only, Volatile. Reset: 000h.<br><b>Description:</b> This field indicates the number of Tx FIFO Data Commands which are flushed due to TX_ABRT interrupt. It is cleared whenever <b>I2C</b> is disabled.                                                                                                                                                                                                                                                                                                              |
| 22:21 | <b>RSVD_IC_TX_ABRT_SOURCE.</b> Read-only, Volatile. Reset: 0h. IC_TX_ABRT_SOURCE Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 20:18 | <b>RSVD_ABRT_DEVICE_WRITE.</b> Read-only, Volatile. Reset: 0h. ABRT_DEVICE_WRITE Register field Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 17    | <b>ABRT_SDA_STUCK_AT_LOW.</b> Read-only, Volatile. Reset: 0. This is a master-mode-only bit. Master detects the SDA Stuck at low for the IC_SDA_STUCK_AT_LOW_TIMEOUT value of ic_clks.                                                                                                                                                                                                                                                                                                                                                         |
| 16    | <b>ABRT_USER_ABRT.</b> Read-only, Volatile. Reset: 0. This is a master-mode-only bit. Master has detected the transfer abort.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 15    | <b>ABRT_SLVRD_INTX.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> 1: When the processor side responds to a slave mode request for data to be transmitted to a remote master and user writes a 1 in CMD (bit 8) of IC_DATA_CMD register.                                                                                                                                                                                                                                                                                            |
| 14    | <b>ABRT_SLV_ARBLOST.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> This field indicates that a Slave has lost the bus while transmitting data to a remote master. IC_TX_ABRT_SOURCE[12] is set at the same time.<br>Note: Even though the slave never 'owns' the bus, something could go wrong on the bus. This is a fail safe check. For instance, during a data transmission at the low-to-high transition of SCL, if what is on the data bus is not what is supposed to be transmitted, then I2C_IP_CORE no longer own the bus. |
| 13    | <b>ABRT_SLVFLUSH_TXFIFO.</b> Read-only, Volatile. Reset: 0. This field specifies that the Slave has received a read command and some data exists in the TX FIFO, so the slave issues a TX_ABRT interrupt to flush old data in TX FIFO.                                                                                                                                                                                                                                                                                                         |
| 12    | <b>ARB_LOST.</b> Read-only, Volatile. Reset: 0. This field specifies that the Master has lost arbitration, or if IC_TX_ABRT_SOURCE[14] is also set, then the slave transmitter has lost arbitration.                                                                                                                                                                                                                                                                                                                                           |
| 11    | <b>ABRT_MASTER_DIS.</b> Read-only, Volatile. Reset: 0. This field indicates that the User tries to initiate a Master operation with the Master mode disabled.                                                                                                                                                                                                                                                                                                                                                                                  |
| 10    | <b>ABRT_10B_RD_NORSTRT.</b> Read-only, Volatile. Reset: 0. This field indicates that the restart is disabled (IC_RESTART_EN bit (IC_CON[5]) =0) and the master sends a read command in 10-bit addressing mode.                                                                                                                                                                                                                                                                                                                                 |
| 9     | <b>ABRT_SBYTE_NORSTRT.</b> Read-only, Volatile. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p><b>Description:</b> To clear Bit 9, the source of the ABRT_SBYTE_NORSTRT must be fixed first; restart must be enabled (IC_CON[5]=1), the SPECIAL bit must be cleared (IC_TAR[11]), or the GC_OR_START bit must be cleared (IC_TAR[10]).</p> <p>Once the source of the ABRT_SBYTE_NORSTRT is fixed, then this bit can be cleared in the samemanner as other bits in this register.</p> <p>If the source of the ABRT_SBYTE_NORSTRT is not fixed before attempting to clear this bit, bit 9 clears for one cycle and then gets reasserted.</p> <p>When this field is set to 1, the restart is disabled (IC_RESTART_EN bit (IC_CON[5]) =0) and the user is trying to send a START Byte.</p> |
| 8 | <b>ABRT_HS_NORSTRT.</b> Read-only, Volatile. Reset: 0. This field indicates that the restart is disabled (IC_RESTART_EN bit (IC_CON[5]) =0) and the user is trying to use the master to transfer data in High Speed mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 7 | <b>ABRT_SBYTE_ACKDET.</b> Read-only, Volatile. Reset: 0. This field indicates that the Master has sent a START Byte and the START Byte was acknowledged (wrong behavior).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 6 | <b>ABRT_HS_ACKDET.</b> Read-only, Volatile. Reset: 0. This field indicates that the Master is in High Speed mode and the High Speed Master code was acknowledged (wrong behavior).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 5 | <b>ABRT_GCALL_READ.</b> Read-only, Volatile. Reset: 0. This field indicates that I2C_IP_CORE in the master mode has sent a General Call but the user programmed the byte following the General Call to be a read from the bus (IC_DATA_CMD[9] is set to 1).                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4 | <b>ABRT_GCALL_NOACK.</b> Read-only, Volatile. Reset: 0. This field indicates that I2C_IP_CORE in master mode has sent a General Call and no slave on the bus acknowledged the General Call.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3 | <b>ABRT_TXDATA_NOACK.</b> Read-only, Volatile. Reset: 0. This field indicates the master-mode only bit. When the master receives an acknowledgement for the address, but when it sends data byte(s) following the address, it did not receive an acknowledge from the remote slave(s).                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2 | <b>ABRT_10ADDR2_NOACK.</b> Read-only, Volatile. Reset: 0. This field indicates that the Master is in 10-bit address mode and that the second address byte of the 10-bit address was not acknowledged by any slave.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1 | <b>ABRT_10ADDR1_NOACK.</b> Read-only, Volatile. Reset: 0. This field indicates that the Master is in 10-bit address mode and the first 10-bit address byte was not acknowledged by any slave. Reset value: 0x0                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0 | <b>ABRT_7B_ADDR_NOACK.</b> Read-only, Volatile. Reset: 0. This field indicates that the Master is in 7-bit addressing mode and the address sent was not acknowledged by any slave.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**FCHI2CxFEDC[2...B]084 [Generate Slave Data NACK Register] (FCH::I2C::IC\_SLV\_DATA\_NACK\_ONLY)**

Reset: 0000\_0000h.

**Generate Slave Data NACK Register**

The register is used to generate a NACK for the data part of a transfer when I2C\_IP\_CORE is acting as a slave-receiver. This register only exists when the IC\_SLV\_DATA\_NACK\_ONLY parameter is set to 1. When this parameter disabled, this register does not exist and writing to the register's address has no effect.

A write can occur on this register if both of the following conditions are met:

- I2C\_IP\_CORE is disabled (IC\_ENABLE[0] = 0)
- Slave part is inactive (IC\_STATUS[6] = 0)

Note: The IC\_STATUS[6] is a register read-back location for the internal slv\_activity signal; the user should poll this before writing the ic\_slv\_data\_nack\_only bit.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB084,FEDC\_[6:2]084]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_IC_SLV_DATA_NACK_ONLY.</b> Read-only,Volatile. Reset: 0000_0000h.<br>IC_SLV_DATA_NACK_ONLY Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0    | <b>NACK.</b> Read-write,Volatile. Reset: 0.<br><b>Description:</b> Generate NACK. This NACK generation only occurs when I2C_IP_CORE is a slave-receiver. If this register is set to a value of 1, it can only generate a NACK after a data byte is received; hence, the data transfer is aborted and the data received is not pushed to the receive buffer.<br>When the register is set to a value of 0, it generates NACK/ACK, depending on normal criteria.<br>- 1: generate NACK after data byte received<br>- 0: generate NACK/ACK normally |

**FCHI2CxFEDC[2...B]094 [I2C SDA Setup Register] (FCH::I2C::IC\_SDA\_SETUP)**

Reset: 0000\_0064h.

**I2C SDA Setup Register.**

This register controls the amount of time delay (in terms of number of ic\_clk clock periods) introduced in the rising edge of SCL - relative to SDA changing - when I2C\_IP\_CORE services a read request in a slave-transmitter operation. This register must be programmed with a value equal to or greater than 2.

Writes to this register succeed only when IC\_ENABLE[0] = 0.

Note: The length of setup time is calculated using  $[(IC\_SDA\_SETUP - 1) * (ic\_clk\_period)]$ , so if the user requires 10 ic\_clk periods of setup time, they should program a value of 11.

The IC\_SDA\_SETUP register is only used by the I2C\_IP\_CORE when operating as a slave transmitter.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB094,FEDC\_[6:2]094]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                            |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | <b>RSVD_IC_SDA_SETUP.</b> Read-only,Volatile. Reset: 00_0000h. IC_SDA_SETUP Reserved bits - Read Only                                                                                                                                                                                                  |
| 7:0  | <b>SDA_SETUP.</b> Read-write,Volatile. Reset: 64h.<br><b>Description:</b> SDA Setup.<br>It is recommended that if the required delay is 1000ns, then for an ic_clk frequency of 10 MHz, IC_SDA_SETUP should be programmed to a value of 11. IC_SDA_SETUP must be programmed with a minimum value of 2. |

**FCHI2Cx FEDC[2...B]09C [I2C Enable Status Register] (FCH::I2C::IC\_ENABLE\_STATUS)**

Read-only, Volatile. Reset: 0000\_0000h.

**I2C** Enable Status Register.

The register is used to report the I2C\_IP\_CORE hardware status when the IC\_ENABLE[0] register is set from 1 to 0; that is, when I2C\_IP\_CORE is disabled.

If IC\_ENABLE[0] has been set to 1, bits 2:1 are forced to 0, and bit 0 is forced to 1.

If IC\_ENABLE[0] has been set to 0, bits 2:1 is only be valid as soon as bit 0 is read as '0'.

**Note:**

When IC\_ENABLE[0] has been set to 0, a delay occurs for bit 0 to be read as 0 because disabling the I2C\_IP\_CORE depends on I2C bus activities.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB09C, FEDC\_[6:2]09C]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:3 | <b>RSVD_IC_ENABLE_STATUS.</b> Read-only, Volatile. Reset: 0000_0000h. IC_ENABLE_STATUS Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 2    | <p><b>SLV_RX_DATA_LOST.</b> Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> Slave Received Data Lost.</p> <p>This bit indicates if a Slave-Receiver operation has been aborted with at least one data byte received from an <b>I2C</b> transfer due to the setting bit 0 of IC_ENABLE from 1 to 0.</p> <p>When read as 1, I2C_IP_CORE is deemed to have been actively engaged in an aborted I2C transfer (with matching address) and the data phase of the I2C transfer has been entered, even though a data byte has been responded with a NACK.</p> <p>Note: If the remote I2C master terminates the transfer with a STOP condition before the I2C_IP_CORE has a chance to NACK a transfer, and IC_ENABLE[0] has been set to 0, then this bit is also set to 1.</p> <p>When read as 0, I2C_IP_CORE is deemed to have been disabled without being actively involved in the data phase of a Slave-Receiver transfer.</p> <p>Note: The CPU can safely read this bit when IC_EN (bit 0) is read as 0.</p>                                                                                                                                                                                                                                                                                                                            |
| 1    | <p><b>SLV_DISABLED_WHILE_BUSY.</b> Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> Slave Disabled While Busy (Transmit, Receive).</p> <p>This bit indicates if a potential or active Slave operation has been aborted due to the setting bit 0 of the IC_ENABLE register from 1 to 0.</p> <p>This bit is set when the CPU writes a 0 to the IC_ENABLE register while:</p> <p>(a) I2C_IP_CORE is receiving the address byte of the Slave-Transmitter operation from a remote master; OR,</p> <p>(b) address and data bytes of the Slave-Receiver operation from a remote master.</p> <p>When read as 1, I2C_IP_CORE is deemed to have forced a NACK during any part of an <b>I2C</b> transfer, irrespective of whether the I2C address matches the slave address set in I2C_IP_CORE (IC_SAR register) OR if the transfer is completed before IC_ENABLE is set to 0 but has not taken effect.</p> <p>Note: If the remote I2C master terminates the transfer with a STOP condition before the I2C_IP_CORE has a chance to NACK a transfer, and IC_ENABLE[0] has been set to 0, then this bit will also be set to 1.</p> <p>When read as 0, I2C_IP_CORE is deemed to have been disabled when there is master activity, or when the I2C bus is idle.</p> <p>Note: The CPU can safely read this bit when IC_EN (bit 0) is read as 0.</p> |
| 0    | <p><b>IC_EN.</b> Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> ic_en Status.</p> <p>This bit always reflects the value driven on the output port ic_en.</p> <ul style="list-style-type: none"> <li>- When read as 1, I2C_IP_CORE is deemed to be in an enabled state.</li> <li>- When read as 0, I2C_IP_CORE is deemed completely inactive.</li> </ul> <p>Note: The CPU can safely read this bit anytime. When this bit is read as 0, the CPU can safely read SLV_RX_DATA_LOST (bit 2) and SLV_DISABLED_WHILE_BUSY (bit 1).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**FCHI2Cx FEDC[2...B]0A0 [I2C SS, FS or FM+ spike suppression limit] (FCH::I2C::IC\_FS\_SPKLEN)**

Reset: 0000\_0005h.

**I2C** SS, FS or FM+ spike suppression limit

This register is used to store the duration, measured in ic\_clk cycles, of the longest spike that is filtered out by the spike suppression logic when the component is operating in SS, FS or FM+ modes.

The relevant I2C requirement is tSP (table 4) as detailed in the I2C Bus Specification. This register must be programmed with a minimum value of 1.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB0A0,FEDC\_[6:2]0A0]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | <b>RSVD_IC_FS_SPKLEN.</b> Read-only, Volatile. Reset: 00_0000h. IC_FS_SPKLEN Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7:0  | <p><b>IC_FS_SPKLEN.</b> Read-write, Volatile. Reset: 05h.</p> <p><b>Description:</b> This register must be set before any <a href="#">I2C</a> bus transaction can take place to ensure stable operation. This register sets the duration, measured in ic_clk cycles, of the longest spike in the SCL or SDA lines that will be filtered out by the spike suppression logic.</p> <p>This register can be written only when the I2C interface is disabled which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect.</p> <p>The minimum valid value is 1; hardware prevents values less than this being written, and if attempted results in 1 being set. or more information, refer to "Spike Suppression".</p> |

**FCHI2Cx FEDC[2...B]0A4 [I2C HS spike suppression limit register] (FCH::I2C::IC\_HS\_SPKLEN)**

Reset: 0000\_0001h.

**I2C** HS spike suppression limit register

This register is used to store the duration, measured in ic\_clk cycles, of the longest spike that is filtered out by the spike suppression logic when the component is operating in HS modes.

The relevant I2C requirement is tSP (table 6) as detailed in the I2C Bus Specification. This register must be programmed with a minimum value of 1 and is implemented only if the component is configured to support HS mode; that is, if the IC\_MAX\_SPEED\_MODE parameter is set to 3.

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB0A4,FEDC\_[6:2]0A4]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | <b>RSVD_IC_HS_SPKLEN.</b> Read-only, Volatile. Reset: 00_0000h. IC_HS_SPKLEN Reserved bits - Read Only                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 7:0  | <p><b>IC_HS_SPKLEN.</b> Read-write, Volatile. Reset: 01h.</p> <p><b>Description:</b> This register must be set before any <a href="#">I2C</a> bus transaction can take place to ensure stable operation. This register sets the duration, measured in ic_clk cycles, of the longest spike in the SCL or SDA lines that will be filtered out by the spike suppression logic; for more information, refer to "Spike Suppression"</p> <p>This register can be written only when the I2C interface is disabled which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect.</p> <p>The minimum valid value is 1; hardware prevents values less than this being written, and if attempted results in 1 being set.</p> |

**FCHI2Cx FEDC[2...B]0AC [I2C SCL Stuck at Low Timeout register]  
(FCH::I2C::IC\_SCL\_STUCK\_AT\_LOW\_TIMEOUT)**

Read-write, Volatile. Reset: FFFF\_FFFFh.

[I2C](#) SCL Stuck at Low Timeout

This register is used to store the duration, measured in ic\_clk cycles, used to Generate an Interrupt (SCL\_STUCK\_AT\_LOW) if SCL is held low for the IC\_SCL\_STUCK\_LOW\_TIMEOUT duration.

[\\_link\[5:0\]](#) alias HOSTLEGACY; FCHI2Cx[FEDCB0AC, FEDC\_[6:2]0AC]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>IC_SCL_STUCK_LOW_TIMEOUT.</b> Read-write, Volatile. Reset: FFFF_FFFFh.<br><b>Description:</b> I2C_IP_CORE generate the interrupt to indicate SCL stuck at low (SCL_STUCK_AT_LOW) if it detects the SCL stuck at low for the IC_SCL_STUCK_LOW_TIMEOUT in units of ic_clk period. This register can be written only when the <a href="#">I2C</a> interface is disabled which corresponds to the IC_ENABLE[0] register being set to 0. Writes at other times have no effect. |

**FCHI2Cx FEDC[2...B]0B0 [I2C SDA Stuck at Low Timeout register]  
(FCH::I2C::IC\_SDA\_STUCK\_AT\_LOW\_TIMEOUT)**

Read-write, Volatile. Reset: FFFF\_FFFFh.

[I2C](#) SDA Stuck at Low Timeout

This register is used to store the duration, measured in ic\_clk cycles, used to Recover the Data (SDA) line through sending SCL pulses if SDA is held low for the mentioned duration.

[\\_link\[5:0\]](#) alias HOSTLEGACY; FCHI2Cx[FEDCB0B0, FEDC\_[6:2]0B0]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>IC_SDA_STUCK_LOW_TIMEOUT.</b> Read-write, Volatile. Reset: FFFF_FFFFh. I2C_IP_CORE initiates the recovery of SDA line through enabling the SDA_STUCK_RECOVERY_EN (IC_ENABLE[3]) register bit, if it detects the SDA stuck at low for the IC_SDA_STUCK_LOW_TIMEOUT in units of ic_clk period. |

**FCHI2Cx FEDC[2...B]0B4 [Clear SCL Stuck at Low Detect interrupt Register]  
(FCH::I2C::IC\_CLR\_SCL\_STUCK\_DET)**

Read-only, Volatile. Reset: 0000\_0000h.

Clear SCL Stuck at Low Detect Interrupt Register

[\\_link\[5:0\]](#) alias HOSTLEGACY; FCHI2Cx[FEDCB0B4, FEDC\_[6:2]0B4]; FCHI2C=0000\_0000h

| Bits | Description                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | <b>RSVD_CLR_SCL_STUCK_DET.</b> Read-only, Volatile. Reset: 0000_0000h. CLR_SCL_STUCK_DET Reserved bits - Read Only                                             |
| 0    | <b>CLR_SCL_STUCK_DET.</b> Read-only, Volatile. Reset: 0. Read this register to clear the SCL_STUCK_AT_LOW interrupt (bit 15) of the IC_RAW_INTR_STAT register. |

**FCHI2Cx FEDC[2...B]0F4 [Component Parameter Register 1] (FCH::I2C::IC\_COMP\_PARAM\_1)**

Read-only, Volatile. Reset: 00FF\_FFAEh.

Component Parameter Register 1

Note

This is a constant read-only register that contains encoded information about the component's parameter settings.

The reset value depends on IP parameter(s).

\_link[5:0]\_aliasHOSTLEGACY; FCHI2Cx[FEDCB0F4, FEDC\_[6:2]0F4]; FCHI2C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>RSVD_IC_COMP_PARAM_1.</b> Read-only, Volatile. Reset: 00h. IC_COMP_PARAM_1 Reserved bits - Read Only                                                                                                                                                                                                                                                                                                        |
| 23:16 | <b>TX_BUFFER_DEPTH.</b> Read-only, Volatile. Reset: FFh.<br><b>Description:</b> The value of this register is derived from the IC_TX_BUFFER_DEPTH IP parameter.<br>- 0x00 = Reserved<br>- 0x01 = 2<br>- 0x02 = 3<br>- ...<br>- 0xFF = 256                                                                                                                                                                      |
| 15:8  | <b>RX_BUFFER_DEPTH.</b> Read-only, Volatile. Reset: FFh.<br><b>Description:</b> The value of this register is derived from the IC_RX_BUFFER_DEPTH IP parameter.<br>- 0x00: Reserved<br>- 0x01: 2<br>- 0x02: 3<br>- ...<br>- 0xFF: 256                                                                                                                                                                          |
| 7     | <b>ADD_ENCODED_PARAMS.</b> Read-only, Volatile. Reset: 1.<br><b>Description:</b> The value of this register is derived from the IC_ADD_ENCODED_PARAMS IP parameter.<br>Reading 1 in this bit means that the capability of reading these encoded parameters via software has been included. Otherwise, the entire register is 0 regardless of the setting of any other parameters that are encoded in the bits. |
| 6     | <b>HAS_DMA.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> The value of this register is derived from the IC_HAS_DMA IP parameter.                                                                                                                                                                                                                                                                  |
| 5     | <b>INTR_IO.</b> Read-only, Volatile. Reset: 1.<br><b>Description:</b> The value of this register is derived from the IC_INTR_IO IP parameter.                                                                                                                                                                                                                                                                  |
| 4     | <b>HC_COUNT_VALUES.</b> Read-only, Volatile. Reset: 0.<br><b>Description:</b> The value of this register is derived from the IC_HC_COUNT_VALUES IP parameter.                                                                                                                                                                                                                                                  |
| 3:2   | <b>MAX_SPEED_MODE.</b> Read-only, Volatile. Reset: 3h.                                                                                                                                                                                                                                                                                                                                                         |

|     |                                                                                                                                                                        |             |                                               |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------|
|     | <b>Description:</b> The value of this register is derived from the IC_MAX_SPEED_MODE IP parameter.<br>- 0x0: Reserved<br>- 0x1: Standard<br>- 0x2: Fast<br>- 0x3: High |             |                                               |
|     | <b>ValidValues:</b>                                                                                                                                                    |             |                                               |
|     | <b>Value</b>                                                                                                                                                           | <b>Name</b> | <b>Description</b>                            |
|     | 0h                                                                                                                                                                     |             | Reserved.                                     |
|     | 1h                                                                                                                                                                     | STANDARD    | IC <a href="#">MAX</a> SPEED is STANDARD MODE |
| 1:0 | 2h                                                                                                                                                                     | FAST        | IC MAX SPEED is FAST MODE                     |
|     | 3h                                                                                                                                                                     | HIGH        | IC MAX SPEED is HIGH MODE                     |
|     | <b>APB_DATA_WIDTH.</b> Read-only,Volatile. Reset: 2h.                                                                                                                  |             |                                               |
|     | <b>Description:</b> The value of this register is derived from the APB_DATA_WIDTH IP parameter.                                                                        |             |                                               |
|     | <b>ValidValues:</b>                                                                                                                                                    |             |                                               |
|     | <b>Value</b>                                                                                                                                                           | <b>Name</b> | <b>Description</b>                            |
|     | 0h                                                                                                                                                                     | APB_08BITS  | APB data bus width is 08 bits                 |
|     | 1h                                                                                                                                                                     | APB_16BITS  | APB data bus width is 16 bits                 |
|     | 2h                                                                                                                                                                     | APB_32BITS  | APB data bus width is 32 bits                 |
|     | 3h                                                                                                                                                                     | RESERVED    | Reserved bits                                 |

### **14.3.8 I3C Registers**

#### **14.3.8.1 MIPI I3C® Registers**



**FCHI3Cx02DE[2...5]000 [Device Control Register] (FCH::I3C::DEVICE\_CTRL)**

Reset: 0000\_0001h.

mipi\_i3c control Register

This Register controls the transfer properties and disposition of controller's capabilities.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]000; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6000,FEDD\_[4:2]000]; FCHI3C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <p><b>ENABLE.</b> Read-write, Volatile. Reset: 0.</p> <p><b>Description:</b> Controls whether or not mipi_i3c is enabled.</p> <ul style="list-style-type: none"> <li>- 1: Enables the mipi_i3c controller.</li> <li>- 0: Disables the mipi_i3c controller.</li> </ul> <p>In Controller mode of operation, software can disable mipi_i3c while it is active. However, the controller may not get disabled immediately and is 'Disabled' after commands in the Command queue (if any) are executed leading to a STOP condition on the bus and Controller FSM is in IDLE state (as indicated by PRESENT_STATE Register).</p>                                                       |
| 30   | <p><b>RESUME.</b> Read-write, Volatile. Reset: 0.</p> <p><b>Description:</b> mipi_i3c Resume</p> <p>This bit is used to resume the controller after it goes to the halt state.</p> <p>In the Controller mode of operation, the controller goes to the halt state (as indicated in PRESENT_STATE Register) due to any type of error in the transfer (the type of error is indicated by ERR_STATUS field in the RESPONSE_QUEUE_PORT register).</p> <p>After the controller goes to the halt state, the application has to write 1'b1 to this bit to resume the controller. This bit is auto-cleared once the controller resumes the transfers by initiating the next command.</p> |
| 29   | <p><b>ABORT.</b> Read-write, Volatile. Reset: 0.</p> <p><b>Description:</b> mipi_i3c Abort</p> <p>This bit is used in Controller mode of operation.</p> <p>This bit allows the controller to relinquish the mipi_i3c bus before completing the issued transfer.</p> <p>In response to an ABORT request, the controller issues the STOP condition after the complete data byte is transferred or received.</p> <p>This bit is auto-cleared once the transfer is aborted and the controller issues a 'Transfer Abort' interrupt.</p>                                                                                                                                              |
| 28:9 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8    | <p><b>HOT_JOIN_CTRL.</b> Read-write. Reset: 0.</p> <p><b>Description:</b> Hot-Join ACK/NACK Control</p> <p>This bit is used in Controller mode of operation.</p> <p>This bit acts as a global control to ACK/NACK the Hot-Join request from the devices. The mipi_i3c Controller ACK/NACKs the Hot-Join request from other devices connected on the mipi_i3c bus, based on programming of this bit.</p> <ul style="list-style-type: none"> <li>- 0: ACK the Hot-join request.</li> <li>- 1: NACK and send broadcast <a href="#">CCC</a> to disable Hot-Join.</li> </ul>                                                                                                         |
| 7    | <b>I2C_Target_PRESENT.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> <a href="#">I2C</a> Target Present<br><br>This bit is used in Controller mode of operation.<br><br>This bit indicates whether any Legacy I2C devices are present in the system.<br><br>In HDR mode, this field is used to select TSL over TSP in a mixed bus configuration.                                                                                                                                                                       |
| 6:1 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0   | <b>IBA_INCLUDE.</b> Read-write. Reset: 1.<br><b>Description:</b> <a href="#">I3C</a> Broadcast Address include<br><br>This bit is used in Controller mode of operation.<br><br>This bit is used to include mipi_i3c broadcast address (0x7E) for private transfer.<br><br>Note: If mipi_i3c broadcast address is not included for the private transfers, In-band Interrupts (IBI) driven from Targets might not win address arbitration. Hence, the IBIs get delayed. |

**FCHI3Cx02DE[2...5]004 [Device Address Register] (FCH::I3C::DEVICE\_ADDR)**

Read-write, Volatile. Reset: 8000\_0000h.

In the Controller mode of operation this Register is used to program the Device Dynamic Addresses and its respective valid bit.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]004; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6004,FEDD\_[4:2]004]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DYNAMIC_ADDR_VALID.</b> Read-write, Volatile. Reset: 1.<br><b>Description:</b> Dynamic Address Valid<br><br>This bit is used to control whether the DYNAMIC_ADDR is valid or not.<br><br><ul style="list-style-type: none"> <li>- In <a href="#">I3C</a> Main Controller mode, the user sets this bit to 1 as it self-assigns its dynamic address.</li> <li>- In all other operation modes, the Controller sets this bit to 1 when Main Controller assigns the Dynamic address during <a href="#">ENTDAA</a> or <a href="#">SETDASA</a> mechanism.</li> </ul>                                        |
| 30:23 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 22:16 | <b>DYNAMIC_ADDR.</b> Read-write, Volatile. Reset: 00h.<br><b>Description:</b> Device Dynamic Address.<br><br>This field is used to program the Device Dynamic Address. The Controller uses this address for <a href="#">I3C</a> transfers.<br><br><ul style="list-style-type: none"> <li>- In Main Controller mode, the user/application has to program the Dynamic Address through the Target interface as it self-assigns its Dynamic Address.</li> <li>- In all other modes, the Main Controller assigns this address during <a href="#">ENTDAA</a> or <a href="#">SETDASA</a> mechanism.</li> </ul> |
| 15:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**FCHI3Cx02DE[2...5]008 [Hardware Capability register] (FCH::I3C::HW\_CAPABILITY)**

Read-only. Reset: 0001\_40A1h.

Hardware Capability register

This register reflects the configured capabilities of mipi\_i3c.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]008; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6008,FEDD\_[4:2]008]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:20 | Reserved.                                                                                                                                                                                                                                                                                                           |
| 19    | <b>SLV_IBI_CAP.</b> Read-only. Reset: 0.<br><b>Description:</b> Reflects the IC_SLV_IBI <a href="#">Configurable</a> Parameter.<br>Specifies Target's capability to initiate Target interrupt requests.                                                                                                             |
| 18    | <b>SLV_HJ_CAP.</b> Read-only. Reset: 0.<br><b>Description:</b> Reflects the IC_SLV_HJ <a href="#">Configurable</a> Parameter.<br>Specifies Target's capability to initiate Hot-join request.                                                                                                                        |
| 17    | <b>DMA_EN.</b> Read-only. Reset: 0.<br><b>Description:</b> Reflects the IC_HAS_DMA <a href="#">Configurable</a> Parameter.<br>Specifies whether controller is configured to have DMA handshaking interface.                                                                                                         |
| 16:11 | <b>HDR_TX_CLOCK_PERIOD.</b> Read-only. Reset: 28h. Reflects the IC_HDR_TX_CLK_PERIOD <a href="#">Configurable</a> Parameter.                                                                                                                                                                                        |
| 10:5  | <b>CLOCK_PERIOD.</b> Read-only. Reset: 05h. Reflects the IC_CLK_PERIOD <a href="#">Configurable</a> Parameter                                                                                                                                                                                                       |
| 4     | <b>HDR_TS_EN.</b> Read-only. Reset: 0.<br><b>Description:</b> Reflects the IC_SPEED_HDR_TS <a href="#">Configurable</a> Parameter.<br>Specifies the Controllers capability to perform HDR-TS transfers.<br>- 0: HDR-TS not supported<br>- 1: HDR-TS supported                                                       |
| 3     | <b>HDR_DDR_EN.</b> Read-only. Reset: 0.<br><b>Description:</b> Reflects the IC_SPEED_HDR_DDR <a href="#">Configurable</a> Parameter.<br>Specifies the Controllers capability to perform HDR-DDR transfers.<br>- 0: HDR-DDR not supported<br>- 1: HDR-DDR supported                                                  |
| 2:0   | <b>DEVICE_ROLE_CONFIG.</b> Read-only. Reset: 1h.<br><b>Description:</b> Reflects the IC_DEVICE_ROLE <a href="#">Configurable</a> Parameter.<br>Specifies the configured role of mipi_i3c controller<br>- 1: Controller Only<br>- 2: Programmable Controller-Target<br>- 3: Secondary Controller<br>- 4: Target Only |

**FCHI3Cx02DE[2...5]00C [COMMAND\_QUEUE\_PORT] (FCH::I3C::COMMAND\_QUEUE\_PORT)**

Write-only. Reset: 0000\_0000h.

COMMAND\_QUEUE\_PORT.

In Controller mode of operation:

Command Descriptor structure is used to schedule the transfers to devices on [I3C](#) bus.

There are four types of commands defined

- Transfer Command
- Transfer Argument
- Short Data Argument
- Address Assignment Command

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]00C; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD600C,FEDD\_[4:2]00C]; FCHI3C=0000\_0000h

| Bits | Description                                                   |
|------|---------------------------------------------------------------|
| 31:0 | <b>COMMAND.</b> Write-only. Reset: 0000_0000h. 32 bit command |

**FCHI3Cx02DE[2...5]010 [RESPONSE\_QUEUE\_PORT] (FCH::I3C::RESPONSE\_QUEUE\_PORT)**

Read-only. Reset: 0000\_0000h.

In Controller mode of operation:

The response status for each Command is written into the Response Queue by the controller if ROC (Response On Completion) bit is set or

if transfer error occurs. The Response Queue can be read through this register.

It is expected that this register

is be read whenever RESP\_READY\_STAT\_INTR bit is set in INTR\_STATUS register. Not doing so might result in execution of new commands getting stalled. A new command is executed only if there is space available in Response Queue to push the corresponding response.

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]010; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD6010,FEDD\_[4:2]010]; FCHI3C=0000\_0000h

| Bits | Description                                                    |
|------|----------------------------------------------------------------|
| 31:0 | <b>RESPONSE.</b> Read-only. Reset: 0000_0000h. 32 bit Response |

**FCHI3Cx02DE[2...5]014 [Receive Data Port Register] (FCH::I3C::RX\_DATA\_PORT)**

Read-only. Reset: 0000\_0000h.

Receive Data Port Register

This register when read from, reads data from the RX Buffer. This has the same offset as TX\_DATA\_PORT to provide a single bi-directional data port for transmitting or receiving the data from the DWC\_mipi\_i3c.

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]014; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD6014,FEDD\_[4:2]014]; FCHI3C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>RX_DATA_PORT.</b> Read-only. Reset: 0000_0000h.<br><b>Description:</b> Receive Data Port.<br><br>The Receive data port is mapped to the Rx-Data Buffer.<br><br>The Receive data is always packed in 4-byte aligned data words and stored in the Rx-Data Buffer.<br>If the command length is not aligned to the 4-bytes, then the additional data bytes have to be ignored. |

**FCHI3Cx02DE[2...5]014 [Transmit Data Port Register] (FCH::I3C::TX\_DATA\_PORT)**

Write-only. Reset: 0000\_0000h.

**Transmit Data Port Register**

This register when written into, writes data to the TX Buffer. This has the same offset as RX\_DATA\_PORT to provide a single bi-directional data port for transmitting or receiving the data from the mipi\_i3c.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]014; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6014,FEDD\_[4:2]014]; FCHI3C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>TX_DATA_PORT.</b> Write-only. Reset: 0000_0000h.<br><b>Description:</b> Transmit Data Port<br><br>The Transmit Data port is mapped to the Tx-Data Buffer.<br><br>The transmit data should always be packed as 4-byte aligned data words and written to the Transmit Data Port register.<br>If the Command length is not aligned to 4-bytes, then the additional bytes are ignored. |

**FCHI3Cx02DE[2...5]018 [In-Band Interrupt Queue Status Register] (FCH::I3C::IBI\_QUEUE\_STATUS)**

Read-only. Reset: 0000\_0000h.

**In-Band Interrupt Queue Status Register**

This register is used in Controller mode of operation. It is expected that this register is read whenever INTR\_STATUS.IBI\_THLD\_STS is set.

This register when read from, returns the data from the IBI Queue and indicates how the controller responded to incoming IBI (SIR, [MR](#) and HJ).

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]018; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6018,FEDD\_[4:2]018]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:28 | <b>IBI_STS.</b> Read-only. Reset: 0h.<br><b>Description:</b> IBI Received Status.<br><br>Defines the Controller response for IBI received.<br><br>- 4'b0xxx: Responded with ACK<br>- 4'b1xxx: Responded with NACK<br>- Others: RESERVED                                                                |
| 27:16 | Reserved.                                                                                                                                                                                                                                                                                              |
| 15:8  | <b>IBI_ID.</b> Read-only. Reset: 00h.<br><b>Description:</b> IBI Identifier.<br><br>The byte received after START which includes the address and the R/W bit.<br>- Device address and R/W bit in case of Target Interrupt or Controller Request.<br>- Hot-Join ID and R/W bit in case of Hot-Join IBI. |
| 7:0   | <b>DATA_LENGTH.</b> Read-only. Reset: 00h.<br><b>Description:</b> In-Band Interrupt data length.<br><br>This field represents the length of data received along with the IBI, in bytes.                                                                                                                |

**FCHI3Cx02DE[2...5]01C [Queue Threshold Control Register] (FCH::I3C::QUEUE\_THLD\_CTRL)**

Read-write. Reset: 0100\_0001h.

Queue Threshold Control Register

This register is used to program the threshold settings for the Queues in mipi\_i3c.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]01C; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD601C,FEDD\_[4:2]01C]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>IBI_STATUS_THLD.</b> Read-write. Reset: 01h.<br><b>Description:</b> In-Band Interrupt Status Threshold Value.<br><br>Every In Band Interrupt received (with or without Payload) by <a href="#">I3C</a> controller generates an IBI status. This field controls the number of IBI status entries (or greater) in the IBI queue that trigger the IBI_THLD_STAT interrupt.<br><br>The valid range is 0 to IC_IBI_BUF_DEPTH-1. The software programs only valid values. A value of 0 sets the threshold for 1 entry, and a value of N sets the threshold for N+1 entries.<br><br>NOTE: The valid value is only 0 if IBI with payload is selected in the configuration.<br><br>Each IBI status entry can represent the complete (IBI payload byte size <= 4*IBI_DATA_THLD) IBI payload or a segment (IBI payload byte size > 4*IBI_DATA_THLD) of the IBI payload. |
| 23:16 | <b>IBI_DATA_THLD.</b> Read-write. Reset: 00h.<br><b>Description:</b> IBI Data Threshold Value<br><br>This field represents the IBI data segment size in Dwords (4 bytes). The minimum supported segment size is 1 (4 bytes) and the maximum supported size is IC_IBI_DATA_BUF_DEPTH-1. The IBI_DATA_THLD field enables the slicing of the incoming IBI data and generate individual status and thereby promotes the cut-through operation in reading out the IBI data.                                                                                                                                                                                                                                                                                                                                                                                          |
| 15:8  | <b>RESP_BUF_THLD.</b> Read-write. Reset: 00h.<br><b>Description:</b> Response Buffer Threshold Value.<br><br>Controls the number of entries (or greater) in the Response Queue that trigger the RESP_READY_STAT_INTR interrupt.<br><br>The valid range is 0 to IC_RESP_BUF_DEPTH-1. The software programs only valid values. A value of 0 sets the threshold for 1 entry, and a value of N sets the threshold for N+1 entries.                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 7:0   | <b>CMD_EMPTY_BUF_THLD.</b> Read-write. Reset: 01h.<br><b>Description:</b> Command Buffer Empty Threshold Value.<br>Controls the number of empty locations (or greater) in the Command Queue that trigger CMD_QUEUE_READY_STAT interrupt.<br><br>The valid range is 0 to IC_CMD_BUF_DEPTH-1. The software programs only valid values. Value of N ranging from 1 to IC_CMD_BUF_DEPTH-1 sets the threshold to N empty locations and a value of 0 sets the threshold to indicate that the queue is completely empty.                                                                                                                                                                                                                                                                                                                                                |

**FCHI3Cx02DE[2...5]020 [Data Buffer Threshold Control Register] (FCH::I3C::DATA\_BUFFER\_THLD\_CTRL)**

Read-write. Reset: 0404\_0404h.

Data Buffer Threshold Control Register

This register is used to program the threshold settings for the Data Buffers in mipi\_i3c.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]020; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6020,FEDD\_[4:2]020]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:27 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 26:24 | <b>RX_START_THLD.</b> Read-write. Reset: 4h.<br><b>Description:</b> Receive Start Threshold Value<br><br>In Controller mode of operation when the controller is set up to initiate a read transfer, it waits until either one of the conditions are met before it initiates the read transfer on the <a href="#">I3C</a> Interface. <ul style="list-style-type: none"> <li>- Data length (as specified in the command) number of locations are empty in the Receive FIFO.</li> <li>- Threshold number of locations (or more) are empty in the Receive FIFO.</li> </ul> <p>The supported values for RX_START_THLD are:</p> <ul style="list-style-type: none"> <li>- 000 - 1</li> <li>- 001 - 4</li> <li>- 010 - 8</li> <li>- 011 - 16</li> <li>- 100 - 32</li> <li>- 101 - 64</li> </ul>           |
| 23:19 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 18:16 | <b>TX_START_THLD.</b> Read-write. Reset: 4h.<br><b>Description:</b> Transfer Start Threshold Value<br><br>In Controller mode of operation when the controller is set up to initiate a write transfer, it waits until either one of the following conditions are met before it initiates the write transfer on the <a href="#">I3C</a> Interface. <ul style="list-style-type: none"> <li>- Data length (as specified in the command) number of locations are filled in the Transmit FIFO</li> <li>- Threshold number of entries (or more) are available in the Transmit FIFO</li> </ul> <p>The supported values for TX_START_THLD are:</p> <ul style="list-style-type: none"> <li>- 000: 1</li> <li>- 001: 4</li> <li>- 010: 8</li> <li>- 011: 16</li> <li>- 100: 32</li> <li>- 101: 64</li> </ul> |
| 15:11 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 10:8  | <b>RX_BUF_THLD.</b> Read-write. Reset: 4h.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p><b>Description:</b> Receive Buffer Threshold Value</p> <p>This field controls the number of entries (or above) in the Receive FIFO that trigger the RX_THLD_STAT interrupt.</p> <p>If the programmed value is greater than the buffer depth, then threshold is set to IC_RX_BUF_DEPTH.</p> <p>The supported values for RX_BUF_THLD are</p> <ul style="list-style-type: none"> <li>- 000: 1</li> <li>- 001: 4</li> <li>- 010: 8</li> <li>- 011: 16</li> <li>- 100: 32</li> <li>- 101: 64</li> </ul>                                                                   |
| 7:3 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2:0 | <p><b>TX_EMPTY_BUF_THLD.</b> Read-write. Reset: 4h.</p> <p><b>Description:</b> Transmit Buffer Threshold Value</p> <p>This field controls the number of empty locations (or above) in the Transmit FIFO that trigger the TX_THLD_STAT interrupt.</p> <p>If the programmed value is greater than the buffer depth, then threshold is set to IC_TX_BUF_DEPTH.</p> <p>The supported values for TX_BUF_THLD are</p> <ul style="list-style-type: none"> <li>- 000: 1</li> <li>- 001: 4</li> <li>- 010: 8</li> <li>- 011: 16</li> <li>- 100: 32</li> <li>- 101: 64</li> </ul> |



**FCHI3Cx02DE[2...5]024 [IBI Queue Control Register] (FCH::I3C::IBI\_QUEUE\_CTRL)**

Read-write. Reset: 0000\_0000h.

This Register is used to control whether or not to intimate the application if an IBI request is rejected (Nacked).

This register is only used in Controller mode of operation

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]024; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6024,FEDD\_[4:2]024]; FCHI3C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:4 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3    | <p><b>NOTIFY_SIR_REJECTED.</b> Read-write. Reset: 0.</p> <p><b>Description:</b> Notify Rejected Target Interrupt Request Control.</p> <p>This bit is used to suppress reporting to the application about SIR request rejected.</p> <ul style="list-style-type: none"> <li>- 0: Suppress passing the IBI Status to the IBI FIFO (hence not notifying the application) when a Target Interrupt Request is NACKed and auto-disabled based on the IBI_SIR_REQ_REJECT Register.</li> <li>- 1: Writes IBI Status to the IBI FIFO (hence notifying the application) when a Target Interrupt Request is NACKed and auto-disabled based on the IBI_SIR_REQ_REJECT Register.</li> </ul> |
| 2:1  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 0    | <p><b>NOTIFY_HJ_REJECTED.</b> Read-write. Reset: 0.</p> <p><b>Description:</b> Notify Rejected Hot-Join Control.</p> <p>This bit is used to suppress reporting to the application about Hot-Join request rejected (NACK and Auto Disable).</p> <ul style="list-style-type: none"> <li>- 0: Suppress passing the IBI Status to the IBI FIFO (hence not notifying the application) when a HJ Request is NACKed and auto-disabled based on the DEVICE_CTRL.HOT_JOIN_CTRL.</li> <li>- 1: Writes IBI Status to the IBI FIFO (hence notifying the application) when a HJ Request is NACKed and auto-disabled based on the DEVICE_CTRL.HOT_JOIN_CTRL.</li> </ul>                     |

**FCHI3Cx02DE[2...5]034 [Reset Control Register] (FCH::I3C::RESET\_CTRL)**

Read-write, Volatile. Reset: 0000\_0000h.

This Register is used for general software reset and for individual buffer reset.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]034; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6034,FEDD\_[4:2]034]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>BUS_RESET.</b> Read-write, Volatile. Reset: 0.<br><b>Description:</b> Bus Reset.<br><br>This bit is only used in Controller mode of operation.<br><br>Write 1'b1 to this bit to exercise Bus Reset Reset Pattern Generation based on Bus Reset Type selection.<br>This bit is cleared automatically once the Bus Reset Pattern Generation is completed. |
| 30:29 | <b>BUS_RESET_TYPE.</b> Read-write, Volatile. Reset: 0h.<br><b>Description:</b> Bus Reset type<br><br>Type of bus reset triggered by BUS_RESET field.<br><br>Values:<br><br>2'b00: EXIT: Exit Pattern<br><br>2'b11: SCL_LOW_RESET Pattern<br><br>Others: Reserved                                                                                           |
| 28:6  | Reserved.                                                                                                                                                                                                                                                                                                                                                  |
| 5     | <b>IBI_QUEUE_RST.</b> Read-write, Volatile. Reset: 0.<br><b>Description:</b> IBI Queue Software Reset.<br><br>This bit is only used in Controller mode of operation.<br><br>Write 1'b1 to this bit to exercise IBI Queue reset<br>This bit is cleared automatically once the IBI Queue reset is completed.                                                 |
| 4     | <b>RX_FIFO_RST.</b> Read-write, Volatile. Reset: 0.<br><b>Description:</b> Receive Buffer Software Reset.<br><br>Write 1'b1 to this bit to exercise Receive Buffer reset.<br>This bit is cleared automatically once the Receive buffer reset is completed.                                                                                                 |
| 3     | <b>TX_FIFO_RST.</b> Read-write, Volatile. Reset: 0.<br><b>Description:</b> Transmit Buffer Software Reset<br><br>Write 1'b1 to this bit to exercise Transmit Buffer reset.<br>This bit is cleared automatically once the Transmit Buffer reset is completed.                                                                                               |
| 2     | <b>RESP_QUEUE_RST.</b> Read-write, Volatile. Reset: 0.<br><b>Description:</b> Response Queue Software Reset<br><br>Write 1'b1 to this bit to exercise Response Queue reset.<br>This bit is cleared automatically once the Response Queue reset is complete.                                                                                                |
| 1     | <b>CMD_QUEUE_RST.</b> Read-write, Volatile. Reset: 0.                                                                                                                                                                                                                                                                                                      |

|   |                                                                                                                                                                                                                                                                                                    |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <b>Description:</b> Command Queue Software Reset<br><br>Write 1'b1 to this bit to exercise Command Queue reset.<br>This bit is cleared automatically once the Command Queue reset is complete.                                                                                                     |
| 0 | <b>SOFT_RST.</b> Read-write, Volatile. Reset: 0.<br><b>Description:</b> Core Software Reset.<br><br>Write 1'b1 to this bit to exercise software reset.<br>This resets all Buffers - Receive, Transmit, Command, and Response<br>This bit is cleared automatically once the core reset is complete. |

**FCH13Cx02DE[2...5]038 [Target Event Status Register] (FCH::I3C::SLV\_EVENT\_STATUS)**

Reset: 0000\_0000h.

This register indicates the status/values of some events/controls that are relevant to Target mode of operation. These values are set by Controller initiated CCCs.

\_link[3:0]\_aliasSMN; FCH13Cx02DE\_[5:2]038; FCH13C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCH13Cx[FEDD6038,FEDD\_[4:2]038]; FCH13C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                                                                                                                                                                                                                                                                                              |
| 7    | <b>MWL_UPDATED.</b> Read-write, Read, Write-1-to-clear. Reset: 0.<br><b>Description:</b> MWL Updated Status.<br><br>This bit indicates a SETMWL <a href="#">CCC</a> is received by the Target. The updated MWL value can be read from SLV_MAX_LEN register.<br>This status can be cleared by writing 1'b1 to this field after reading the updated MWL.                 |
| 6    | <b>MRL_UPDATED.</b> Read-write, Read, Write-1-to-clear. Reset: 0.<br><b>Description:</b> <a href="#">MRL</a> Updated Status.<br><br>This bit indicates a SETMRL <a href="#">CCC</a> is received by the Target. The updated MRL value can be read from SLV_MAX_LEN register.<br>This status can be cleared by writing 1'b1 to this field after reading the updated MRL. |
| 5:4  | <b>ACTIVITY_STATE.</b> Read-only. Reset: 0h.<br><b>Description:</b> Activity State Status.<br><br>- ENTAS0 - 00<br><br>- ENTAS1 - 01<br><br>- ENTAS2 - 10<br><br>- ENTAS3 - 11<br><br>This bit reflects the Activity State of Target set by the Controller.                                                                                                            |
| 3:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                              |

**FCHI3Cx02DE[2...5]03C [Interrupt Status Register] (FCH::I3C::INTR\_STATUS)**

Reset: 0000\_0000h.

**Interrupt Status Register**

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]03C; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD603C,FEDD\_[4:2]03C]; FCHI3C=0000\_0000h

| <b>Bits</b> | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16       | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 15          | <b>BUS_RESET_DONE_STS.</b> Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.<br><b>Description:</b> Bus Reset Pattern Generation Done Status.<br><br>This field is used only in Controller mode of operation.<br><br>This interrupt is generated when the SCL Low Timeout Bus Reset Pattern Generation is completed. This bit can be cleared by writing 1'b1.                                                                                                        |
| 14:10       | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 9           | <b>TRANSFER_ERR_STS.</b> Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.<br><b>Description:</b> Transfer Error Status.<br><br>This interrupt is generated if any error occurs during transfer. The error type is specified in the response packet associated with the command (in ERR_STATUS field of RESPONSE_QUEUE_PORT register).<br>This bit can be cleared by writing 1'b1.                                                                                   |
| 8:6         | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5           | <b>TRANSFER_ABORT_STS.</b> Read-write,Read,Write-1-to-clear,Volatile. Reset: 0.<br><b>Description:</b> Transfer Abort Status.<br><br>This field is used only in Controller mode of operation.<br><br>This interrupt is generated if transfer is aborted. This interrupt can be cleared by writing 1'b1.                                                                                                                                                                 |
| 4           | <b>RESP_READY_STS.</b> Read-only,Volatile. Reset: 0.<br><b>Description:</b> Response Queue Ready Status.<br><br>This interrupt is generated when number of entries in response queue is greater than or equal to threshold value specified by RESP_BUF_THLD field in QUEUE_THLD_CTRL register. This interrupt is cleared automatically when number of entries in response buffer is less than threshold value specified.                                                |
| 3           | <b>CMD_QUEUE_READY_STS.</b> Read-only,Volatile. Reset: 0.<br><b>Description:</b> Command Queue Ready.<br><br>This interrupt is generated when number of empty locations in command queue is greater than or equal to threshold value specified by CMD_EMPTY_BUF_THLD field in QUEUE_THLD_CTRL register. This interrupt is cleared automatically when number of empty locations in command buffer is less than threshold value specified.                                |
| 2           | <b>IBI_THLD_STS.</b> Read-only,Volatile. Reset: 0.<br><b>Description:</b> IBI Buffer Threshold Status.<br><br>This field is only used in Controller mode of operation<br>This interrupt is generated when number of entries in IBI buffer is greater than or equal to threshold value specified by IBI_BUF_THLD field in QUEUE_THLD_CTRL register. This interrupt is cleared automatically when number of entries in IBI buffer is less than threshold value specified. |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>RX_THLD_STS.</b> Read-only,Volatile. Reset: 0.<br><b>Description:</b> Receive Buffer Threshold Status.<br><br>This interrupt is generated when number of entries in receive buffer is greater than or equal to threshold value specified by RX_BUF_THLD field in DATA_BUFFER_THLD_CTRL register. This interrupt is cleared automatically when number of entries in receive buffer is less than threshold value specified.                             |
| 0 | <b>TX_THLD_STS.</b> Read-only,Volatile. Reset: 0.<br><b>Description:</b> Transmit Buffer Threshold Status<br><br>This interrupt is generated when the number of empty locations in transmit buffer is greater than or equal to threshold value specified by TX_EMPTY_BUF_THLD field in DATA_BUFFER_THLD_CTRL register. This interrupt is cleared automatically when number of empty locations in transmit buffer is less than threshold value specified. |

**FCH13Cx02DE[2...5]040 [Interrupt Status Enable Register] (FCH::I3C::INTR\_STATUS\_EN)**

Read-write. Reset: 0000\_0000h.

Interrupt Status Enable Register.

The interrupt status is updated in INTR\_STATUS register only if corresponding Status Enable bit is set.

\_link[3:0]\_aliasSMN; FCH13Cx02DE\_[5:2]040; FCH13C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCH13Cx[FEDD6040,FEDD\_[4:2]040]; FCH13C=0000\_0000h

| Bits  | Description                                                                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                   |
| 15    | <b>BUS_RESET_DONE_STS_EN.</b> Read-write. Reset: 0.<br><b>Description:</b> Bus Reset Pattern Generation Done Status Enable.<br><br>This field is used only in Controller mode of operation. |
| 14:10 | Reserved.                                                                                                                                                                                   |
| 9     | <b>TRANSFER_ERR_STS_EN.</b> Read-write. Reset: 0. Transfer Error Status Enable                                                                                                              |
| 8:6   | Reserved.                                                                                                                                                                                   |
| 5     | <b>TRANSFER_ABORT_STS_EN.</b> Read-write. Reset: 0.<br><b>Description:</b> Transfer Abort Status Enable.<br><br>This field is used only in Controller mode of operation.                    |
| 4     | <b>RESP_READY_STS_EN.</b> Read-write. Reset: 0. Response Queue Ready Status Enable                                                                                                          |
| 3     | <b>CMD_QUEUE_READY_STS_EN.</b> Read-write. Reset: 0. Command Queue Ready Status Enable                                                                                                      |
| 2     | <b>IBI_THLD_STS_EN.</b> Read-write. Reset: 0.<br><b>Description:</b> IBI Buffer Threshold Status Enable.<br><br>This field is used only in Controller mode of operation.                    |
| 1     | <b>RX_THLD_STS_EN.</b> Read-write. Reset: 0. Receive Buffer Threshold Status Enable                                                                                                         |
| 0     | <b>TX_THLD_STS_EN.</b> Read-write. Reset: 0. Transmit Buffer Threshold Status Enable.                                                                                                       |

**FCHI3Cx02DE[2...5]044 [Interrupt Signal Enable Register] (FCH::I3C::INTR\_SIGNAL\_EN)**

Read-write. Reset: 0000\_0000h.

Interrupt Signal Enable Register

The interrupt pin is triggered based on INTR\_STATUS only if corresponding Signal Enable bit is set.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]044; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6044,FEDD\_[4:2]044]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                    |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                      |
| 15    | <b>BUS_RESET_DONE_SIGNAL_EN.</b> Read-write. Reset: 0.<br><b>Description:</b> Bus Reset Pattern Generation Done Signal Enable.<br><br>This field is used only in Controller mode of operation. |
| 14:10 | Reserved.                                                                                                                                                                                      |
| 9     | <b>TRANSFER_ERR_SIGNAL_EN.</b> Read-write. Reset: 0. Transfer Error Signal Enable                                                                                                              |
| 8:6   | Reserved.                                                                                                                                                                                      |
| 5     | <b>TRANSFER_ABORT_SIGNAL_EN.</b> Read-write. Reset: 0.<br><b>Description:</b> Transfer Abort Signal Enable<br>This field is used in Controller mode of operation.                              |
| 4     | <b>RESP_READY_SIGNAL_EN.</b> Read-write. Reset: 0. Response Queue Ready Signal Enable                                                                                                          |
| 3     | <b>CMD_QUEUE_READY_SIGNAL_EN.</b> Read-write. Reset: 0. Command Queue Ready Signal Enable                                                                                                      |
| 2     | <b>IBI_THLD_SIGNAL_EN.</b> Read-write. Reset: 0.<br><b>Description:</b> IBI Buffer Threshold Signal Enable<br>This field is used in Controller mode of operation.                              |
| 1     | <b>RX_THLD_SIGNAL_EN.</b> Read-write. Reset: 0. Receive Buffer Threshold Signal Enable                                                                                                         |
| 0     | <b>TX_THLD_SIGNAL_EN.</b> Read-write. Reset: 0. Transmit Buffer Threshold Signal Enable                                                                                                        |

**FCHI3Cx02DE[2...5]048 [Interrupt Force Enable Register] (FCH::I3C::INTR\_FORCE)**

Write-only. Reset: 0000\_0000h.

Interrupt Force Enable Register

Individual interrupts can be forcefully triggered if corresponding Force Enable bit is set, provided the corresponding bit in the INTR\_STATUS\_EN register is set.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]048; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6048,FEDD\_[4:2]048]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                    |
| 15    | <b>BUS_RESET_DONE_FORCE_EN.</b> Write-only. Reset: 0.<br><b>Description:</b> Bus Reset Pattern Generation Done Force Enable.<br><br>This field is used only in Controller mode of operation. |
| 14:10 | Reserved.                                                                                                                                                                                    |
| 9     | <b>TRANSFER_ERR_FORCE_EN.</b> Write-only. Reset: 0. Transfer Error Force Enable                                                                                                              |
| 8:6   | Reserved.                                                                                                                                                                                    |
| 5     | <b>TRANSFER_ABORT_FORCE_EN.</b> Write-only. Reset: 0.<br><b>Description:</b> Transfer Abort Force Enable<br>This field is used in Controller mode of operation.                              |
| 4     | <b>RESP_READY_FORCE_EN.</b> Write-only. Reset: 0. Response Queue Ready Force Enable                                                                                                          |
| 3     | <b>CMD_QUEUE_READY_FORCE_EN.</b> Write-only. Reset: 0. Command Queue Ready Force Enable                                                                                                      |
| 2     | <b>IBI_THLD_FORCE_EN.</b> Write-only. Reset: 0.<br><b>Description:</b> IBI Buffer Threshold Force Enable<br>This field is used in Controller mode of operation.                              |
| 1     | <b>RX_THLD_FORCE_EN.</b> Write-only. Reset: 0. Receive Buffer Threshold Force Enable                                                                                                         |
| 0     | <b>TX_THLD_FORCE_EN.</b> Write-only. Reset: 0. Transmit Buffer Threshold Force Enable                                                                                                        |

**FCHI3Cx02DE[2...5]04C [Queue Status Level Register] (FCH::I3C::QUEUE\_STATUS\_LEVEL)**

Read-only. Reset: 0000\_0000h.

Queue Status Level Register.

This register reflects the status level of the Queues in mipi\_i3c.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]04C; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD604C,FEDD\_[4:2]04C]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                                                                                    |
| 28:24 | <b>IBI_STS_CNT.</b> Read-only. Reset: 00h.<br><b>Description:</b> IBI Buffer Status Count.<br><br>When IC_HAS_IBI_DATA =0, this field is reserved and always returns 0.<br><br>When IC_HAS_IBI_DATA=1, Contains the number of valid IBI Status entries in the IBI Status Buffer.<br><br>This field is used in Controller mode of operation.                  |
| 23:16 | <b>IBI_BUF_BLR.</b> Read-only. Reset: 00h.<br><b>Description:</b> IBI Buffer Level Value.<br><br>When IC_HAS_IBI_DATA =0, Contains the number of Valid IBI Status entries in the IBI Buffer.<br><br>When IC_HAS_IBI_DATA=1, Contains the number of valid IBI Data entries in the IBI Data Buffer.<br><br>This field is used in Controller mode of operation. |
| 15:8  | <b>RESP_BUF_BLR.</b> Read-only. Reset: 00h.<br><b>Description:</b> Response Buffer Level Value.<br><br>Contains the number of valid data entries in the response Buffer.                                                                                                                                                                                     |
| 7:0   | <b>CMD_QUEUE_EMPTY_LOC.</b> Read-only. Reset: 00h.<br><b>Description:</b> Command Queue Empty Locations.<br><br>Contains the number of empty locations in the command Buffer.                                                                                                                                                                                |

**FCHI3Cx02DE[2...5]050 [Data Buffer Status Level Register] (FCH::I3C::DATA\_BUFFER\_STATUS\_LEVEL)**

Read-only. Reset: 0000\_0040h.

Data Buffer Status Level Register.

This register reflects the status level of the Buffers in mipi\_i3c.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]050; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6050,FEDD\_[4:2]050]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                       |
| 23:16 | <b>RX_BUF_BLR.</b> Read-only. Reset: 00h.<br><b>Description:</b> Receive Buffer Level Value.<br><br>Contains the number of valid data entries in the receive Buffer.            |
| 15:8  | Reserved.                                                                                                                                                                       |
| 7:0   | <b>TX_BUF_EMPTY_LOC.</b> Read-only. Reset: 40h.<br><b>Description:</b> Transmit Buffer Empty Level Value.<br><br>Contains the number of empty locations in the transmit Buffer. |



**FCHI3Cx02DE[2...5]054 [Present State Register] (FCH::I3C::PRESENT\_STATE)**

Reset: 1000\_0003h.

The user can get status of the mipi\_i3c Controller from this 32-bit read only register. This register is relevant in both Controller and Target mode of operation and is meant to be used to get debug information related to the controllers internal states.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]054; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6054,FEDD\_[4:2]054]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 28    | <b>Controller_IDLE.</b> Read-only, Volatile. Reset: 1. This field reflects whether the Controller Controller is in Idle state or not. This bit is set when all the Queues(Command , Response, IBI) and Buffers(Transmit and Receive) are empty along with the Controller State machine is in Idle state.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 27:24 | <b>CMD_TID.</b> Read-only. Reset: 0h. This field reflects the Transaction-ID of the current executing command.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 23:22 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 21:16 | <b>CM_TFR_ST_STS.</b> Read-only. Reset: 00h.<br><b>Description:</b> Current Controller Transfer State Status.<br><br>Indicates the state of current transfer currently executing by the mipi_i3c controller.<br>This is valid in Controller mode only.<br><br><ul style="list-style-type: none"> <li>- 6'h0: IDLE (Controller is Idle state, waiting for commands from application or Target initiated In-band Interrupt)</li> <li>- 6'h1: START Generation State.</li> <li>- 6'h2: RESTART Generation State.</li> <li>- 6'h3: STOP Generation State.</li> <li>- 6'h4: START Hold Generation for the Target Initiated START State.</li> <li>- 6'h5: Broadcast Write Address Header(7'h7E,W) Generation State.</li> <li>- 6'h6: Broadcast Read Address Header(7'h7E,R) Generation State.</li> <li>- 6'h7: Dynamic Address Assignment State.</li> <li>- 6'h8: Target Address Generation State.</li> <li>- 6'hB: <a href="#">CCC</a> Byte Generation State.</li> <li>- 6'hC: HDR Command Generation State.</li> <li>- 6'hD: Write Data Transfer State.</li> <li>- 6'hE: Read Data Transfer State.</li> <li>- 6'hF: In-Band Interrupt(SIR) Read Data State.</li> <li>- 6'h10: In-Band Interrupt Auto-Disable State</li> <li>- 6'h11: HDR-DDR CRC Data Generation/Receive State.</li> <li>- 6'h12: Clock Extension State.</li> <li>- 6'h13: Halt State.</li> </ul> |
| 15:14 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 13:8  | <b>CM_TFR_STS.</b> Read-only. Reset: 00h.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p><b>Description:</b> Transfer Type Status</p> <p>Indicates the type of transfer currently executing by the mipi_i3c controller.</p> <p>In Controller mode of operation:</p> <ul style="list-style-type: none"> <li>- 6'h0: IDLE (Controller is in Idle state, waiting for commands from application or Target initiated In-band Interrupt)</li> <li>- 6'h1: Broadcast <a href="#">CCC</a> Write Transfer.</li> <li>- 6'h2: Directed CCC Write Transfer.</li> <li>- 6'h3: Directed CCC Read Transfer.</li> <li>- 6'h4: <a href="#">ENTDAA</a> Address Assignment Transfer.</li> <li>- 6'h5: <a href="#">SETDASA</a> Address Assignment Transfer.</li> <li>- 6'h6: Private <a href="#">I3C</a> <a href="#">SDR</a> Write Transfer.</li> <li>- 6'h7: Private I3C SDR Read Transfer.</li> <li>- 6'h8: Private <a href="#">I2C</a> SDR Write Transfer.</li> <li>- 6'h9: Private I2C SDR Read Transfer.</li> <li>- 6'hA: Private HDR Ternary Symbol(TS) Write Transfer.</li> <li>- 6'hB: Private HDR Ternary Symbol(TS) Read Transfer.</li> <li>- 6'hC: Private HDR Double-Data Rate(DDR) Write Transfer.</li> <li>- 6'hD: Private HDR Double-Data Rate(DDR) Read Transfer.</li> <li>- 6'hE: Servicing In-Band Interrupt Transfer.</li> <li>- 6'hF: Halt state (Controller is in Halt State, waiting for the application to resume through DEVICE_CTRL Register)</li> </ul> |
| 7:3 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2   | <p><b>CURRENT_Controller.</b> Read-only, Volatile. Reset: 0.</p> <p><b>Description:</b> This Bit is used to check whether the Controller is Current Controller or not. The Current Controller is the Controller that owns the SCL line.</p> <p>If this bit is set to 0, the Controller is not Current Controller and requires to request and the ownership before initiating any transfer on the line.</p> <p>If this bit is set to 1, the Controller is the Current Controller and can initiate the transfers on the line.</p> <ul style="list-style-type: none"> <li>- 0: Controller is not Current Controller</li> <li>- 1: Controller is Current Controller</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 1   | <p><b>SDA_LINE_SIGNAL_LEVEL.</b> Read-only, Volatile. Reset: 1.</p> <p><b>Description:</b> This bit is used to check the SDA line level to recover from errors and for debugging. This bit reflects the value of synchronized sda_in_a signal. This is valid in Controller mode only.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 0   | <p><b>SCL_LINE_SIGNAL_LEVEL.</b> Read-only, Volatile. Reset: 1.</p> <p><b>Description:</b> This bit is used to check the SCL line level to recover from errors and for debugging. This bit reflects the value of synchronized scl_in_a signal. This is valid in Controller mode only</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

### FCHI3Cx02DE[2...5]05C [Pointer for Device Address Table Registers] (FCH::I3C::DEVICE\_ADDR\_TABLE\_POINTER)

Read-only. Reset: 000B\_02C0h.

Pointer for Device Address Table

This register is used in Controller mode of operation.

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]05C; FCHI3C=0000\_0000h

[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD605C,FEDD\_[4:2]05C]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------|
| 31:16 | <b>DEV_ADDR_TABLE_DEPTH.</b> Read-only. Reset: 000Bh. Depth of Device Address Table                 |
| 15:0  | <b>P_DEV_ADDR_TABLE_START_ADDR.</b> Read-only. Reset: 02C0h. Start Address of Device Address Table. |

### FCHI3Cx02DE[2...5]060 [Pointer for Device Characteristics Table] (FCH::I3C::DEV\_CHAR\_TABLE\_POINTER)

Reset: 0002\_C200h.

Pointer for Device Characteristics Table

This register is used in Controller mode of operation.

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]060; FCHI3C=0000\_0000h

[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD6060,FEDD\_[4:2]060]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:23 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 22:19 | <b>PRESENT_DEV_CHAR_TABLE_IDX.</b> Read-write. Reset: 0h.<br><b>Description:</b> Current index of Device Characteristics Table.<br><br>This field returns the current location of Device Characteristics Table index. Initially, this index points to 0.<br><br>Once the complete characteristics information of a Target device is written into Device Characteristics Table during <a href="#">ENTDAA</a> , this index increments by 1. The first winning device information is stored in Device Characteristics Table index 0, the second winning device information in index 1, and so on.<br>If required, this index can be used to override the location, where characteristic information of Target devices on the <a href="#">I3C</a> bus are written during ENTDAA. Hence, this field is useful only if the device is Current Controller. During DEFSLV <a href="#">CCC</a> , the index always starts from 0.<br>In Non-current Controller, this field is always read-only. |
| 18:12 | <b>DEV_CHAR_TABLE_DEPTH.</b> Read-only. Reset: 2Ch. Depth of Device Characteristics Table                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 11:0  | <b>P_DEV_CHAR_TABLE_START_ADDR.</b> Read-only. Reset: 200h. Start Address of Device Characteristics Table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

### FCHI3Cx02DE[2...5]06C [Pointer for Vendor specific Registers] (FCH::I3C::VENDOR\_SPECIFIC\_REG\_POINTER)

Read-only. Reset: 0000\_00B0h.

Pointer for Vendor Specific Registers.

This register is used in Controller mode of operation.

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]06C; FCHI3C=0000\_0000h

[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD606C,FEDD\_[4:2]06C]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                          |
|-------|------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                            |
| 15:0  | <b>P_VENDOR_REG_START_ADDR.</b> Read-only. Reset: 00B0h. Start Address of Vendor specific registers. |

**FCHI3Cx02DE[2...5]0B0 [Device Control Extended Register] (FCH::I3C::DEVICE\_CTRL\_EXTENDED)**

Read-write, Volatile. Reset: 0000\_0000h.

Device Control Extended register.

This register is relevant to both Controller and Target modes of operation and hosts functions related to Targets disposition to incoming GETACCMST [CCC](#) from current Controller. It also allows for the Device Role to be fixed through Software programming in some configurations.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]0B0; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD60B0, FEDD\_[4:2]0B0]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             |      |             |    |            |            |    |        |        |       |  |           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------|-------------|----|------------|------------|----|--------|--------|-------|--|-----------|
| 31:2  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |             |      |             |    |            |            |    |        |        |       |  |           |
| 1:0   | <b>DEV_OPERATION_MODE.</b> Read-write,Volatile. Reset: 0h.<br><b>Description:</b> This bit is used to select the Device Operation Mode before the controller is enabled.<br><br>This field is written only when the mipi_i3c is disabled.<br><br>- 0: Controller<br>- 1: Target<br>- 2: Reserved<br>- 3: Reserved<br>This field is automatically updated by the controller once the role change happens in secondary Controller mode..<br><b>ValidValues:</b> <table><tr><th>Value</th><th>Name</th><th>Description</th></tr><tr><td>0h</td><td>Controller</td><td>Controller</td></tr><tr><td>1h</td><td>Target</td><td>Target</td></tr><tr><td>3h-2h</td><td></td><td>Reserved.</td></tr></table> | Value       | Name | Description | 0h | Controller | Controller | 1h | Target | Target | 3h-2h |  | Reserved. |
| Value | Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Description |      |             |    |            |            |    |        |        |       |  |           |
| 0h    | Controller                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Controller  |      |             |    |            |            |    |        |        |       |  |           |
| 1h    | Target                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Target      |      |             |    |            |            |    |        |        |       |  |           |
| 3h-2h |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Reserved.   |      |             |    |            |            |    |        |        |       |  |           |

**FCHI3Cx02DE[2...5]0B4 [SCL I3C Open Drain Timing Register] (FCH::I3C::SCL\_I3C\_OD\_TIMING)**

Read-write. Reset: 0006\_002Ch.

SCL [I3C](#) Open Drain Timing Register

This register sets the SCL clock high period and low period count for I3C Open Drain transfers. The count value takes the number of core\_clks to maintain the I/O SCL High/Low Period timing.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]0B4; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD60B4, FEDD\_[4:2]0B4]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                                       |
| 23:16 | <b>I3C_OD_HCNT.</b> Read-write. Reset: 06h.<br><b>Description:</b> <a href="#">I3C</a> Open Drain High Count.<br><br>SCL open-drain High count (I3C) for I3C transfers targeted to I3C devices. |
| 15:8  | Reserved.                                                                                                                                                                                       |
| 7:0   | <b>I3C_OD_LCNT.</b> Read-write. Reset: 2Ch.<br><b>Description:</b> <a href="#">I3C</a> Open Drain Low Count.<br><br>SCL Open-drain low count for I3C transfers targeted to I3C devices.         |

**FCHI3Cx02DE[2...5]0B8 [SCL I3C Push Pull Timing Register] (FCH::I3C::SCL\_I3C\_PP\_TIMING)**

Read-write. Reset: 0007\_0009h.

SCL [I3C](#) Push Pull Timing Register

This register sets the SCL clock high period and low period count for I3C Push Pull transfers. The count value takes the number of core\_clks to maintain the I/O SCL High/Low Period timing.

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]0B8; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD60B8,FEDD\_[4:2]0B8]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                             |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                               |
| 23:16 | <b>I3C_PP_HCNT.</b> Read-write. Reset: 07h.<br><b>Description:</b> <a href="#">I3C</a> Push Pull High Count.<br><br>SCL push-pull High count for I3C transfers targeted to I3C devices. |
| 15:8  | Reserved.                                                                                                                                                                               |
| 7:0   | <b>I3C_PP_LCNT.</b> Read-write. Reset: 09h.<br><b>Description:</b> <a href="#">I3C</a> Push Pull Low Count.<br><br>SCL Push-pull low count for I3C transfers targeted to I3C devices.   |

**FCHI3Cx02DE[2...5]0BC [SCL I2C Fast Mode Timing Register] (FCH::I3C::SCL\_I2C\_FM\_TIMING)**

Read-write. Reset: 00B4\_0141h.

SCL [I2C](#) Fast Mode Timing Register

This register sets the SCL clock high period and low period count for I2C Fast Mode transfers. The count value takes the number of core\_clks to maintain the I/O SCL Low/High period timing.

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]0BC; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD60BC,FEDD\_[4:2]0BC]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>I2C_FM_HCNT.</b> Read-write. Reset: 00B4h.<br><b>Description:</b> <a href="#">I2C</a> Fast Mode High Count<br><br>The SCL open-drain high count timing for I2C fast mode transfers. |
| 15:0  | <b>I2C_FM_LCNT.</b> Read-write. Reset: 0141h.<br><b>Description:</b> <a href="#">I2C</a> Fast Mode Low Count<br><br>The SCL open-drain low count timing for I2C fast mode transfers.   |

**FCHI3Cx02DE[2...5]0C0 [SCL I2C Fast Mode Plus Timing Register] (FCH::I3C::SCL\_I2C\_FMP\_TIMING)**

Read-write. Reset: 004C\_007Ch.

SCL [I2C](#) Fast Mode Plus Timing Register

This register sets the SCL clock high period and low period count for I2C Fast Mode Plus transfers. The count value takes the number of core\_clks to maintain the I/O SCL Low/High period timing.

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]0C0; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD60C0,FEDD\_[4:2]0C0]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                                       |
| 23:16 | <b>I2C_FMP_HCNT.</b> Read-write. Reset: 4Ch.<br><b>Description:</b> <a href="#">I2C</a> Fast Mode Plus High Count<br><br>The SCL open-drain high count timing for I2C fast mode plus transfers. |
| 15:0  | <b>I2C_FMP_LCNT.</b> Read-write. Reset: 007Ch.<br><b>Description:</b> <a href="#">I2C</a> Fast Mode Plus Low Count<br><br>The SCL open-drain low count timing for I2C fast mode plus transfers. |

**FCHI3Cx02DE[2...5]0C8 [SCL Extended Low Count Timing Register] (FCH::I3C::SCL\_EXT\_LCNT\_TIMING)**

Read-write. Reset: 5D2B\_1A12h.

SCL Extended Low Count Timing Register.

This register sets the extended low periods for the [I3C](#) transfers to allow the low data rates of the Target devices as specified in GETMXDS [CCC](#). The Speed field of Transfer command (COMMAND\_QUEUE\_PORT\_TRANSFER\_COMMAND) decides the selection of extended low period to achieve the lower data rate for the transfers to Target devices.

- SDR1: Uses I3C\_EXT\_LCNT\_1 field for the data transfer.
- SDR2: Uses I3C\_EXT\_LCNT\_2 field for the data transfer.
- SDR3: Uses I3C\_EXT\_LCNT\_3 field for the data transfer.
- SDR4: Uses I3C\_EXT\_LCNT\_4 field for the data transfer.

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]0C8; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD60C8,FEDD\_[4:2]0C8]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>I3C_EXT_LCNT_4.</b> Read-write. Reset: 5Dh.<br><b>Description:</b> <a href="#">I3C</a> Extended Low Count Register 4<br><br>SDR4 uses this register field for data transfer. |
| 23:16 | <b>I3C_EXT_LCNT_3.</b> Read-write. Reset: 2Bh.<br><b>Description:</b> <a href="#">I3C</a> Extended Low Count Register 3<br><br>SDR3 uses this register field for data transfer. |
| 15:8  | <b>I3C_EXT_LCNT_2.</b> Read-write. Reset: 1Ah.<br><b>Description:</b> <a href="#">I3C</a> Extended Low Count Register 2<br><br>SDR2 uses this register field for data transfer. |
| 7:0   | <b>I3C_EXT_LCNT_1.</b> Read-write. Reset: 12h.<br><b>Description:</b> <a href="#">I3C</a> Extended Low Count Register 1<br><br>SDR1 uses this register field for data transfer. |

**FCHI3Cx02DE[2...5]0CC [SCL Termination Bit Low count Timing Register]  
(FCH::I3C::SCL\_EXT\_TERMN\_LCNT\_TIMING)**

Read-write. Reset: 0000\_0000h.

**SCL Termination Bit Low Count Timing Register**

This register is used to extend the SCL Low period for Read Termination Bit.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]0CC; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD60CC,FEDD\_[4:2]0CC]; FCHI3C=0000\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:4 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3:0  | <b>I3C_EXT_TERMN_LCNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> <a href="#">I3C</a> Read Termination Bit Low count.<br><br>Extended I3C Read Termination Bit low count for I3C Read transfers.<br>Effective Termination-Bit Low Period is derived based on the <a href="#">SDR</a> speed as shown below <ul style="list-style-type: none"> <li>- SDR0 speed: I3C_PP_LCNT + I3C_EXT_TERMN_LCNT</li> <li>- SDR1 speed: I3C_EXT_LCNT_1 + I3C_EXT_TERMN_LCNT</li> <li>- SDR2 speed: I3C_EXT_LCNT_2 + I3C_EXT_TERMN_LCNT</li> <li>- SDR3 speed: I3C_EXT_LCNT_3 + I3C_EXT_TERMN_LCNT</li> <li>- SDR4 speed: I3C_EXT_LCNT_4 + I3C_EXT_TERMN_LCNT</li> </ul> |

**FCHI3Cx02DE[2...5]0D0 [SDA Hold and Mode Switch Delay Timing Register]  
(FCH::I3C::SDA\_HOLD\_SWITCH\_DLY\_TIMING)**

Read-write. Reset: 0002\_0000h.

**SDA Hold and Mode Switch Delay Timing Register**

The Bits [2:0] of this register are used to shift the sda\_out with respect to sda\_oe while switching transfer from Open Drain timing to Push Pull timing.

The bits [10:8] of this register are used to shift the sda\_oe with respect to sda\_out while switching transfer from Push pull timing to Open Drain timing.

The bits [18:16] of this register are used to control the hold time of SDA during transmit mode in [SDR](#) & DDR transfers.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]0D0; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD60D0,FEDD\_[4:2]0D0]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:19 | Reserved.                                                                                                                                                                                                                                                                                                                                                                        |
| 18:16 | <b>SDA_TX_HOLD.</b> Read-write. Reset: 2h.<br><b>Description:</b> This field controls the hold time (in term of the core clock period) of the transmit data (SDA) with respect to the SCL edge in FM FM+ <a href="#">SDR</a> and DDR speed mode of operations.<br>This field is not applicable for the ternary speed modes.<br>The valid values are 1 to 7. Others are Reserved. |
| 15:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                        |

**FCHI3Cx02DE[2...5]0D4 [Bus Free Timing Register] (FCH::I3C::BUS\_FREE\_AVAIL\_TIMING)**

Read-write. Reset: 0000\_000Ah.

Bus Free and Available Timing Register

This register sets the Bus free time for initiating the transfer in Controller mode or generating IBI in non-current Controller mode.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]0D4; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD60D4,FEDD\_[4:2]0D4]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                |
| 15:0  | <p><b>BUS_FREE_TIME.</b> Read-write. Reset: 000Ah.</p> <p><b>Description:</b> This register field is used only in Controller mode of operation</p> <p><a href="#">I3C</a> Bus Free Count Value.</p> <p>In Pure Bus System, this field represents tCAS parameter. In Mixed Bus system, this field is expected to be programmed to tLOW of <a href="#">I2C</a> Timing.</p> |

**FCHI3Cx02DE[2...5]0DC [SCL\_LOW\_MST\_TIMEOUT] (FCH::I3C::SCL\_LOW\_MST\_EXT\_TIMEOUT)**

Read-write. Reset: 0035\_67E0h.

The SCL Low Controller Extended Timeout register is used to define the duration of the SCL Low Bus Reset Pattern.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]0DC; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD60DC,FEDD\_[4:2]0DC]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                          |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:26 | Reserved.                                                                                                                                                                            |
| 25:0  | <p><b>SCL_LOW_MST_TIMEOUT_COUNT.</b> Read-write. Reset: 035_67E0h. This count defines the number of core clock periods to count for generation of the SCL Low Bus Reset Pattern.</p> |



**FCHI3Cx02DE[2...5]0E8 [mipi\_i3c Queue Size Capability Register] (FCH::I3C::QUEUE\_SIZE\_CAPABILITY)**

Read-only. Reset: 0003\_2355h.

This register reflects the configured size of the Data Buffer and Queues in mipi\_i3c.

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]0E8; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD60E8,FEDD\_[4:2]0E8]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:20 | Reserved.                                                                                                                                                                                                                                                                                       |
| 19:16 | <b>IBI_BUF_SIZE.</b> Read-only. Reset: 3h.<br><b>Description:</b> IBI Queue Size<br><br>This field reflects the configured IBI Queue size (in DWORDS) in Encoded Values.<br><br>Values:<br><br>- 0x0: 2 DWORDS<br><br>- 0x1: 4 DWORDS<br><br>- 0x2: 8 DWORDS<br><br>- 0x3: 16 DWORDS            |
| 15:12 | <b>RESP_BUF_SIZE.</b> Read-only. Reset: 2h.<br><b>Description:</b> Response Queue Size<br><br>This field reflects the configured Response Queue size (in DWORDS) in Encoded Values.<br><br>Values:<br><br>- 0x0: 2 DWORDS<br><br>- 0x1: 4 DWORDS<br><br>- 0x2: 8 DWORDS<br><br>- 0x3: 16 DWORDS |
| 11:8  | <b>CMD_BUF_SIZE.</b> Read-only. Reset: 3h.<br><b>Description:</b> Command Queue Size<br><br>This field reflects the configured Command Queue size (in DWORDS) in Encoded Values.<br><br>Values:<br><br>- 0x0: 2 DWORDS<br><br>- 0x1: 4 DWORDS<br><br>- 0x2: 8 DWORDS<br><br>- 0x3: 16 DWORDS    |

|     |                                                                                                                                                                                                                                                                                                                                                      |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4 | <b>RX_BUF_SIZE.</b> Read-only. Reset: 5h.<br><b>Description:</b> Receive Data Buffer Size<br><br>This field reflects the configured Receive Buffer size (in DWORDS) in Encoded Values.<br><br>Values:<br><br>- 0x0: 2 DWORDS<br><br>- 0x1: 4 DWORDS<br><br>- 0x2: 8 DWORDS<br><br>- 0x3: 16 DWORDS<br><br>- 0x4: 32 DWORDS<br><br>- 0x5: 64 DWORDS   |
| 3:0 | <b>TX_BUF_SIZE.</b> Read-only. Reset: 5h.<br><b>Description:</b> Transmit Data Buffer Size<br><br>This field reflects the configured Transmit Buffer size (in DWORDS) in Encoded Values.<br><br>Values:<br><br>- 0x0: 2 DWORDS<br><br>- 0x1: 4 DWORDS<br><br>- 0x2: 8 DWORDS<br><br>- 0x3: 16 DWORDS<br><br>- 0x4: 32 DWORDS<br><br>- 0x5: 64 DWORDS |

**FCHI3Cx02DE[2...5]200 (FCH::I3C::DEV\_CHAR\_TABLE1\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device1

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_ [5:2]200; FCHI3C=0000\_0000h

\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6200,FEDD\_ [4:2]200]; FCHI3C=0000\_0000h

| Bits | Description                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------|
| 31:0 | <b>LSB_PROVISIONAL_ID.</b> Read-only, Volatile. Reset: 0000_0000h. The LSB 32-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]204 (FCH::I3C::DEV\_CHAR\_TABLE1\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device1

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]204; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6204,FEDD\_[4:2]204]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                             |
| 15:0  | <b>MSB_PROVISIONAL_ID</b> . Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]208 (FCH::I3C::DEV\_CHAR\_TABLE1\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device1

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]208; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6208,FEDD\_[4:2]208]; FCHI3C=0000\_0000h

| Bits  | Description                                                               |
|-------|---------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                 |
| 15:8  | <b>BCR</b> . Read-only, Volatile. Reset: 00h. Bus Characteristic Value    |
| 7:0   | <b>DCR</b> . Read-only, Volatile. Reset: 00h. Device Characteristic Value |

**FCHI3Cx02DE[2...5]20C (FCH::I3C::DEV\_CHAR\_TABLE1\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device1

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]20C; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD620C,FEDD\_[4:2]20C]; FCHI3C=0000\_0000h

| Bits | Description                                                                                 |
|------|---------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                   |
| 7:0  | <b>DEV_DYNAMIC_ADDR</b> . Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned. |

**FCHI3Cx02DE[2...5]210 (FCH::I3C::DEV\_CHAR\_TABLE2\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device2

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]210; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6210,FEDD\_[4:2]210]; FCHI3C=0000\_0000h

| Bits | Description                                                                                                |
|------|------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>LSB_PROVISIONAL_ID</b> . Read-only, Volatile. Reset: 0000_0000h. The LSB 32-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]214 (FCH::I3C::DEV\_CHAR\_TABLE2\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device2

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]214; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6214,FEDD\_[4:2]214]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                             |
| 15:0  | <b>MSB_PROVISIONAL_ID</b> . Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]218 (FCH::I3C::DEV\_CHAR\_TABLE2\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device2

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]218; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6218,FEDD\_[4:2]218]; FCHI3C=0000\_0000h

| Bits  | Description                                                               |
|-------|---------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                 |
| 15:8  | <b>BCR</b> . Read-only, Volatile. Reset: 00h. Bus Characteristic Value    |
| 7:0   | <b>DCR</b> . Read-only, Volatile. Reset: 00h. Device Characteristic Value |

**FCHI3Cx02DE[2...5]21C (FCH::I3C::DEV\_CHAR\_TABLE2\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device2

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]21C; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD621C,FEDD\_[4:2]21C]; FCHI3C=0000\_0000h

| Bits | Description                                                                                 |
|------|---------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                   |
| 7:0  | <b>DEV_DYNAMIC_ADDR</b> . Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned. |

**FCHI3Cx02DE[2...5]220 (FCH::I3C::DEV\_CHAR\_TABLE3\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device3

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]220; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6220,FEDD\_[4:2]220]; FCHI3C=0000\_0000h

| Bits | Description                                                                                                |
|------|------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>LSB_PROVISIONAL_ID</b> . Read-only, Volatile. Reset: 0000_0000h. The LSB 32-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]224 (FCH::I3C::DEV\_CHAR\_TABLE3\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device3

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]224; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6224,FEDD\_[4:2]224]; FCHI3C=0000\_0000h**Bits Description**

31:16 Reserved.

15:0 **MSB\_PROVISIONAL\_ID**. Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID**FCHI3Cx02DE[2...5]228 (FCH::I3C::DEV\_CHAR\_TABLE3\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device3

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]228; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6228,FEDD\_[4:2]228]; FCHI3C=0000\_0000h**Bits Description**

31:16 Reserved.

15:8 **BCR**. Read-only, Volatile. Reset: 00h. Bus Characteristic Value7:0 **DCR**. Read-only, Volatile. Reset: 00h. Device Characteristic Value**FCHI3Cx02DE[2...5]22C (FCH::I3C::DEV\_CHAR\_TABLE3\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device3

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]22C; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD622C,FEDD\_[4:2]22C]; FCHI3C=0000\_0000h**Bits Description**

31:8 Reserved.

7:0 **DEV\_DYNAMIC\_ADDR**. Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned.**FCHI3Cx02DE[2...5]230 (FCH::I3C::DEV\_CHAR\_TABLE4\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device4

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]230; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6230,FEDD\_[4:2]230]; FCHI3C=0000\_0000h**Bits Description**31:0 **LSB\_PROVISIONAL\_ID**. Read-only, Volatile. Reset: 0000\_0000h. The LSB 32-bit value of Provisional-ID

**FCHI3Cx02DE[2...5]234 (FCH::I3C::DEV\_CHAR\_TABLE4\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device4

This register is used in Controller mode of operation.

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]234; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6234,FEDD\\_\[4:2\]234\]; FCHI3C=0000\\_0000h](#)

| Bits  | Description                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                             |
| 15:0  | <b>MSB_PROVISIONAL_ID</b> . Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]238 (FCH::I3C::DEV\_CHAR\_TABLE4\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device4

This register is used in Controller mode of operation.

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]238; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6238,FEDD\\_\[4:2\]238\]; FCHI3C=0000\\_0000h](#)

| Bits  | Description                                                               |
|-------|---------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                 |
| 15:8  | <b>BCR</b> . Read-only, Volatile. Reset: 00h. Bus Characteristic Value    |
| 7:0   | <b>DCR</b> . Read-only, Volatile. Reset: 00h. Device Characteristic Value |

**FCHI3Cx02DE[2...5]23C (FCH::I3C::DEV\_CHAR\_TABLE4\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device4

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]23C; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD623C,FEDD\\_\[4:2\]23C\]; FCHI3C=0000\\_0000h](#)

| Bits | Description                                                                                 |
|------|---------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                   |
| 7:0  | <b>DEV_DYNAMIC_ADDR</b> . Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned. |

**FCHI3Cx02DE[2...5]240 (FCH::I3C::DEV\_CHAR\_TABLE5\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device5

This register is used in Controller mode of operation.

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]240; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6240,FEDD\\_\[4:2\]240\]; FCHI3C=0000\\_0000h](#)

| Bits | Description                                                                                                |
|------|------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>LSB_PROVISIONAL_ID</b> . Read-only, Volatile. Reset: 0000_0000h. The LSB 32-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]244 (FCH::I3C::DEV\_CHAR\_TABLE5\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device5

This register is used in Controller mode of operation.

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]244; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6244,FEDD\\_\[4:2\]244\]; FCHI3C=0000\\_0000h](#)

| Bits  | Description                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                             |
| 15:0  | <b>MSB_PROVISIONAL_ID</b> . Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]248 (FCH::I3C::DEV\_CHAR\_TABLE5\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device5

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]248; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6248,FEDD\_[4:2]248]; FCHI3C=0000\_0000h**Bits Description**

31:16 Reserved.

15:8 **BCR**. Read-only, Volatile. Reset: 00h. Bus Characteristic Value7:0 **DCR**. Read-only, Volatile. Reset: 00h. Device Characteristic Value**FCHI3Cx02DE[2...5]24C (FCH::I3C::DEV\_CHAR\_TABLE5\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device5

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]24C; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD624C,FEDD\_[4:2]24C]; FCHI3C=0000\_0000h**Bits Description**

31:8 Reserved.

7:0 **DEV\_DYNAMIC\_ADDR**. Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned.**FCHI3Cx02DE[2...5]250 (FCH::I3C::DEV\_CHAR\_TABLE6\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device6

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]250; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6250,FEDD\_[4:2]250]; FCHI3C=0000\_0000h**Bits Description**31:0 **LSB\_PROVISIONAL\_ID**. Read-only, Volatile. Reset: 0000\_0000h. The LSB 32-bit value of Provisional-ID**FCHI3Cx02DE[2...5]254 (FCH::I3C::DEV\_CHAR\_TABLE6\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device6

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]254; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6254,FEDD\_[4:2]254]; FCHI3C=0000\_0000h**Bits Description**

31:16 Reserved.

15:0 **MSB\_PROVISIONAL\_ID**. Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID

**FCHI3Cx02DE[2...5]258 (FCH::I3C::DEV\_CHAR\_TABLE6\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device6

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]258; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6258,FEDD\_[4:2]258]; FCHI3C=0000\_0000h**Bits Description**

31:16 Reserved.

15:8 **BCR**. Read-only, Volatile. Reset: 00h. Bus Characteristic Value7:0 **DCR**. Read-only, Volatile. Reset: 00h. Device Characteristic Value**FCHI3Cx02DE[2...5]25C (FCH::I3C::DEV\_CHAR\_TABLE6\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device6

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]25C; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD625C,FEDD\_[4:2]25C]; FCHI3C=0000\_0000h**Bits Description**

31:8 Reserved.

7:0 **DEV\_DYNAMIC\_ADDR**. Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned.**FCHI3Cx02DE[2...5]260 (FCH::I3C::DEV\_CHAR\_TABLE7\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device7

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]260; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6260,FEDD\_[4:2]260]; FCHI3C=0000\_0000h**Bits Description**31:0 **LSB\_PROVISIONAL\_ID**. Read-only, Volatile. Reset: 0000\_0000h. The LSB 32-bit value of Provisional-ID**FCHI3Cx02DE[2...5]264 (FCH::I3C::DEV\_CHAR\_TABLE7\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device7

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]264; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6264,FEDD\_[4:2]264]; FCHI3C=0000\_0000h**Bits Description**

31:16 Reserved.

15:0 **MSB\_PROVISIONAL\_ID**. Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID



**FCHI3Cx02DE[2...5]268 (FCH::I3C::DEV\_CHAR\_TABLE7\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device7

This register is used in Controller mode of operation.

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]268; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6268,FEDD\\_\[4:2\]268\]; FCHI3C=0000\\_0000h](#)**Bits Description**

31:16 Reserved.

15:8 **BCR**. Read-only, Volatile. Reset: 00h. Bus Characteristic Value7:0 **DCR**. Read-only, Volatile. Reset: 00h. Device Characteristic Value**FCHI3Cx02DE[2...5]26C (FCH::I3C::DEV\_CHAR\_TABLE7\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device7

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]26C; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD626C,FEDD\\_\[4:2\]26C\]; FCHI3C=0000\\_0000h](#)**Bits Description**

31:8 Reserved.

7:0 **DEV\_DYNAMIC\_ADDR**. Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned.**FCHI3Cx02DE[2...5]270 (FCH::I3C::DEV\_CHAR\_TABLE8\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device8

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]270; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6270,FEDD\\_\[4:2\]270\]; FCHI3C=0000\\_0000h](#)**Bits Description**31:0 **LSB\_PROVISIONAL\_ID**. Read-only, Volatile. Reset: 0000\_0000h. The LSB 32-bit value of Provisional-ID**FCHI3Cx02DE[2...5]274 (FCH::I3C::DEV\_CHAR\_TABLE8\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device8

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]274; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6274,FEDD\\_\[4:2\]274\]; FCHI3C=0000\\_0000h](#)**Bits Description**

31:16 Reserved.

15:0 **MSB\_PROVISIONAL\_ID**. Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID**FCHI3Cx02DE[2...5]278 (FCH::I3C::DEV\_CHAR\_TABLE8\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device8

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]278; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6278,FEDD\\_\[4:2\]278\]; FCHI3C=0000\\_0000h](#)**Bits Description**

31:16 Reserved.

15:8 **BCR**. Read-only, Volatile. Reset: 00h. Bus Characteristic Value7:0 **DCR**. Read-only, Volatile. Reset: 00h. Device Characteristic Value

**FCHI3Cx02DE[2...5]27C (FCH::I3C::DEV\_CHAR\_TABLE8\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device8

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]27C; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD627C,FEDD\\_\[4:2\]27C\]; FCHI3C=0000\\_0000h](#)**Bits Description**

31:8 Reserved.

7:0 **DEV\_DYNAMIC\_ADDR.** Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned.**FCHI3Cx02DE[2...5]280 (FCH::I3C::DEV\_CHAR\_TABLE9\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device9

This register is used in Controller mode of operation.

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]280; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6280,FEDD\\_\[4:2\]280\]; FCHI3C=0000\\_0000h](#)**Bits Description**31:0 **LSB\_PROVISIONAL\_ID.** Read-only, Volatile. Reset: 0000\_0000h. The LSB 32-bit value of Provisional-ID**FCHI3Cx02DE[2...5]284 (FCH::I3C::DEV\_CHAR\_TABLE9\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device9

This register is used in Controller mode of operation.

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]284; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6284,FEDD\\_\[4:2\]284\]; FCHI3C=0000\\_0000h](#)**Bits Description**

31:16 Reserved.

15:0 **MSB\_PROVISIONAL\_ID.** Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID**FCHI3Cx02DE[2...5]288 (FCH::I3C::DEV\_CHAR\_TABLE9\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device9

This register is used in Controller mode of operation.

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]288; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD6288,FEDD\\_\[4:2\]288\]; FCHI3C=0000\\_0000h](#)**Bits Description**

31:16 Reserved.

15:8 **BCR.** Read-only, Volatile. Reset: 00h. Bus Characteristic Value7:0 **DCR.** Read-only, Volatile. Reset: 00h. Device Characteristic Value**FCHI3Cx02DE[2...5]28C (FCH::I3C::DEV\_CHAR\_TABLE9\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device9

This register is used in Controller mode of operation.

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]28C; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD628C,FEDD\\_\[4:2\]28C\]; FCHI3C=0000\\_0000h](#)**Bits Description**

31:8 Reserved.

7:0 **DEV\_DYNAMIC\_ADDR.** Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned.

**FCHI3Cx02DE[2...5]290 (FCH::I3C::DEV\_CHAR\_TABLE10\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device10  
This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]290; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6290,FEDD\_[4:2]290]; FCHI3C=0000\_0000h

| Bits | Description                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------|
| 31:0 | <b>LSB_PROVISIONAL_ID.</b> Read-only, Volatile. Reset: 0000_0000h. The LSB 32-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]294 (FCH::I3C::DEV\_CHAR\_TABLE10\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device10  
This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]294; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6294,FEDD\_[4:2]294]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                          |
|-------|------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                            |
| 15:0  | <b>MSB_PROVISIONAL_ID.</b> Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]298 (FCH::I3C::DEV\_CHAR\_TABLE10\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device10  
This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]298; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD6298,FEDD\_[4:2]298]; FCHI3C=0000\_0000h

| Bits  | Description                                                              |
|-------|--------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                |
| 15:8  | <b>BCR.</b> Read-only, Volatile. Reset: 00h. Bus Characteristic Value    |
| 7:0   | <b>DCR.</b> Read-only, Volatile. Reset: 00h. Device Characteristic Value |

**FCHI3Cx02DE[2...5]29C (FCH::I3C::DEV\_CHAR\_TABLE10\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device10  
This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]29C; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD629C,FEDD\_[4:2]29C]; FCHI3C=0000\_0000h

| Bits | Description                                                                                |
|------|--------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                  |
| 7:0  | <b>DEV_DYNAMIC_ADDR.</b> Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned. |

**FCHI3Cx02DE[2...5]2A0 (FCH::I3C::DEV\_CHAR\_TABLE11\_LOC1)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-1 of Device11  
This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]2A0; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD62A0,FEDD\_[4:2]2A0]; FCHI3C=0000\_0000h

| Bits | Description                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------|
| 31:0 | <b>LSB_PROVISIONAL_ID.</b> Read-only, Volatile. Reset: 0000_0000h. The LSB 32-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]2A4 (FCH::I3C::DEV\_CHAR\_TABLE11\_LOC2)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-2 of Device11

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]2A4; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD62A4,FEDD\_[4:2]2A4]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                             |
| 15:0  | <b>MSB_PROVISIONAL_ID</b> . Read-only, Volatile. Reset: 0000h. The MSB 16-bit value of Provisional-ID |

**FCHI3Cx02DE[2...5]2A8 (FCH::I3C::DEV\_CHAR\_TABLE11\_LOC3)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-3 of Device11

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]2A8; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD62A8,FEDD\_[4:2]2A8]; FCHI3C=0000\_0000h

| Bits  | Description                                                               |
|-------|---------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                 |
| 15:8  | <b>BCR</b> . Read-only, Volatile. Reset: 00h. Bus Characteristic Value    |
| 7:0   | <b>DCR</b> . Read-only, Volatile. Reset: 00h. Device Characteristic Value |

**FCHI3Cx02DE[2...5]2AC (FCH::I3C::DEV\_CHAR\_TABLE11\_LOC4)**

Read-only, Volatile. Reset: 0000\_0000h.

Device Characteristic Table Location-4 of Device11

This register is used in Controller mode of operation.

1

\_link[3:0]\_aliasSMN; FCHI3Cx02DE\_[5:2]2AC; FCHI3C=0000\_0000h\_link[3:0]\_aliasHOSTLEGACY; FCHI3Cx[FEDD62AC,FEDD\_[4:2]2AC]; FCHI3C=0000\_0000h

| Bits | Description                                                                                 |
|------|---------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                   |
| 7:0  | <b>DEV_DYNAMIC_ADDR</b> . Read-only, Volatile. Reset: 00h. Device Dynamic Address assigned. |

**FCHI3Cx02DE[2...5]2C0 [Device Address Table Location of Device1] (FCH::I3C::DEV\_ADDR\_TABLE1\_LOC1)**

Read-write. Reset: 0000\_0000h.

Device Address Table Location of Device1

1

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]2C0; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD62C0,FEDD\_[4:2]2C0]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29 | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16 | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16])                                                                                                                                                                                                                |
| 15    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14    | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13    | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11    | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.<br>This bit controls whether PEC check should be performed for IBI data from device.<br>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.<br>- 0x0 - Packet Error Check disabled for IBI<br>- 0x1 - Packet Error Check enabled for IBI<br>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1. |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**FCHI3Cx02DE[2...5]2C4 [Device Address Table Location 2] (FCH::I3C::DEV\_ADDR\_TABLE2\_LOC1)**

Read-write. Reset: 0000\_0000h.

Device Address Table Location 2

1

[\\_link\[3:0\]\\_aliasSMN; FCHI3Cx02DE\\_\[5:2\]2C4; FCHI3C=0000\\_0000h](#)[\\_link\[3:0\]\\_aliasHOSTLEGACY; FCHI3Cx\[FEDD62C4,FEDD\\_\[4:2\]2C4\]; FCHI3C=0000\\_0000h](#)

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29 | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16 | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16])                                                                                                                                                                                                                |
| 15    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14    | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13    | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11    | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.<br>This bit controls whether PEC check should be performed for IBI data from device.<br>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.<br>- 0x0 - Packet Error Check disabled for IBI<br>- 0x1 - Packet Error Check enabled for IBI<br>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1. |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



**FCHI3Cx02DE[2...5]2C8 [Device Address Table Location 3] (FCH::I3C::DEV\_ADDR\_TABLE3\_LOC1)**

Read-write. Reset: 0000\_0000h.

Device Address Table Location 3

1

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]2C8; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD62C8,FEDD\_[4:2]2C8]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29 | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16 | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16])                                                                                                                                                                                                                |
| 15    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14    | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13    | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11    | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p><b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.</p> <p>This bit controls whether PEC check should be performed for IBI data from device.</p> <p>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.</p> <ul style="list-style-type: none"><li>- 0x0 - Packet Error Check disabled for IBI</li><li>- 0x1 - Packet Error Check enabled for IBI</li></ul> <p>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1.</p> |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**FCHI3Cx02DE[2...5]2CC [Device Address Table Location 4] (FCH::I3C::DEV\_ADDR\_TABLE4\_LOC1)**

Read-write. Reset: 0000\_0000h.

Device Address Table Location 4

1

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]2CC; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD62CC,FEDD\_[4:2]2CC]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29 | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16 | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16])                                                                                                                                                                                                                |
| 15    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14    | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13    | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11    | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p><b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.</p> <p>This bit controls whether PEC check should be performed for IBI data from device.</p> <p>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.</p> <ul style="list-style-type: none"><li>- 0x0 - Packet Error Check disabled for IBI</li><li>- 0x1 - Packet Error Check enabled for IBI</li></ul> <p>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1.</p> |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**FCHI3Cx02DE[2...5]2D0 [Device Address Table Location 5] (FCH::I3C::DEV\_ADDR\_TABLE5\_LOC1)**

Read-write. Reset: 0000\_0000h.

Device Address Table Location 5

1

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]2D0; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD62D0,FEDD\_[4:2]2D0]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29 | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16 | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16]))                                                                                                                                                                                                               |
| 15    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14    | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13    | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11    | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p><b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.</p> <p>This bit controls whether PEC check should be performed for IBI data from device.</p> <p>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.</p> <ul style="list-style-type: none"><li>- 0x0 - Packet Error Check disabled for IBI</li><li>- 0x1 - Packet Error Check enabled for IBI</li></ul> <p>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1.</p> |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**FCHI3Cx02DE[2...5]2D4 [Device Address Table Location 6] (FCH::I3C::DEV\_ADDR\_TABLE6\_LOC1)**

| Read-write. Reset: 0000_0000h.                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Address Table Location 6                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <a href="#">_link[3:0]_aliasSMN; FCHI3Cx02DE_[5:2]2D4; FCHI3C=0000_0000h</a>                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <a href="#">_link[3:0]_aliasHOSTLEGACY; FCHI3Cx[FEDD62D4,FEDD_[4:2]2D4]; FCHI3C=0000_0000h</a> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Bits                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 31                                                                                             | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29                                                                                          | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24                                                                                          | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16                                                                                          | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16])                                                                                                                                                                                                                |
| 15                                                                                             | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14                                                                                             | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13                                                                                             | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12                                                                                             | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11                                                                                             | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.<br>This bit controls whether PEC check should be performed for IBI data from device.<br>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.<br>- 0x0 - Packet Error Check disabled for IBI<br>- 0x1 - Packet Error Check enabled for IBI<br>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1. |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



**FCHI3Cx02DE[2...5]2D8 [Device Address Table Location 7] (FCH::I3C::DEV\_ADDR\_TABLE7\_LOC1)**

| Read-write. Reset: 0000_0000h.                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device Address Table Location 7                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <a href="#">_link[3:0]_aliasSMN; FCHI3Cx02DE_[5:2]2D8; FCHI3C=0000_0000h</a>                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <a href="#">_link[3:0]_aliasHOSTLEGACY; FCHI3Cx[FEDD62D8,FEDD_[4:2]2D8]; FCHI3C=0000_0000h</a> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Bits                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 31                                                                                             | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29                                                                                          | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24                                                                                          | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16                                                                                          | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16]))                                                                                                                                                                                                               |
| 15                                                                                             | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14                                                                                             | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13                                                                                             | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12                                                                                             | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11                                                                                             | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p><b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.</p> <p>This bit controls whether PEC check should be performed for IBI data from device.</p> <p>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.</p> <ul style="list-style-type: none"><li>- 0x0 - Packet Error Check disabled for IBI</li><li>- 0x1 - Packet Error Check enabled for IBI</li></ul> <p>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1.</p> |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**FCHI3Cx02DE[2...5]2DC [Device Address Table Location 8] (FCH::I3C::DEV\_ADDR\_TABLE8\_LOC1)**

Read-write. Reset: 0000\_0000h.

Device Address Table Location 8

1

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]2DC; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD62DC,FEDD\_[4:2]2DC]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29 | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16 | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16])                                                                                                                                                                                                                |
| 15    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14    | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13    | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11    | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.<br>This bit controls whether PEC check should be performed for IBI data from device.<br>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.<br>- 0x0 - Packet Error Check disabled for IBI<br>- 0x1 - Packet Error Check enabled for IBI<br>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1. |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**FCHI3Cx02DE[2...5]2E0 [Device Address Table Location 9] (FCH::I3C::DEV\_ADDR\_TABLE9\_LOC1)**

Read-write. Reset: 0000\_0000h.

Device Address Table Location 9

1

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]2E0; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD62E0,FEDD\_[4:2]2E0]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29 | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16 | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16]))                                                                                                                                                                                                               |
| 15    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14    | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13    | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11    | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <p><b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.</p> <p>This bit controls whether PEC check should be performed for IBI data from device.</p> <p>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.</p> <ul style="list-style-type: none"><li>- 0x0 - Packet Error Check disabled for IBI</li><li>- 0x1 - Packet Error Check enabled for IBI</li></ul> <p>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1.</p> |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

**FCHI3Cx02DE[2...5]2E4 [Device Address Table Location 10] (FCH::I3C::DEV\_ADDR\_TABLE10\_LOC1)**

Read-write. Reset: 0000\_0000h.

Device Address Table Location 10

1

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]2E4; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD62E4,FEDD\_[4:2]2E4]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29 | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16 | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16])                                                                                                                                                                                                                |
| 15    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14    | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13    | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11    | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.<br>This bit controls whether PEC check should be performed for IBI data from device.<br>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.<br>- 0x0 - Packet Error Check disabled for IBI<br>- 0x1 - Packet Error Check enabled for IBI<br>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1. |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |



**FCHI3Cx02DE[2...5]2E8 [Device Address Table Location 11] (FCH::I3C::DEV\_ADDR\_TABLE11\_LOC1)**

Read-write. Reset: 0000\_0000h.

Device Address Table Location 11

1

[\\_link\[3:0\]\\_aliasSMN](#); FCHI3Cx02DE\_[5:2]2E8; FCHI3C=0000\_0000h[\\_link\[3:0\]\\_aliasHOSTLEGACY](#); FCHI3Cx[FEDD62E8,FEDD\_[4:2]2E8]; FCHI3C=0000\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>DEVICE.</b> Read-write. Reset: 0.<br><b>Description:</b> Type of device<br>- 0: <a href="#">I3C</a><br>- 1: <a href="#">I2C</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 30:29 | <b>DEV_NACK_RETRY_CNT.</b> Read-write. Reset: 0h.<br><b>Description:</b> This field is used to set the Device NACK Retry count for the particular device.<br><br>If the Device NACK's for the device address, the controller automatically retries the same device until this count expires. If the Target does not ACK for the mentioned number of retries, then Controller generates an error response and move to the Halt state.<br><br>This feature is used for Retry Model for the following features mentioned in the <a href="#">I3C</a> Specification:<br>- Retry Model for Direct GET <a href="#">CCC</a> Commands.<br>- The incoming SIR-IBI matches with the Target address initiated by the Controller. |
| 28:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 23:16 | <b>DEV_DYNAMIC_ADDR.</b> Read-write. Reset: 00h.<br><b>Description:</b> Device Dynamic Address with parity.<br>This field consists of Dynamic address and Parity Bit.<br>The LSB bits [22:16] should consist of Dynamic Address field indicates the address to be assigned for the winning <a href="#">I3C</a> device when using <a href="#">ENTDAA</a> command.<br>The MSB[23] bit is the odd parity of the 7-bit Dynamic address used for ENTDAAs address assignment (~XOR(DEV_DYNAMIC_ADDR[22:16])                                                                                                                                                                                                                |
| 15    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 14    | <b>MR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Controller Request Reject field is used to control, per device, whether to accept Controller request from Devices.<br>- 0x0 - Accept: ACK the Controller Request<br>- 0x1 - Reject: NACK the Controller Request and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                   |
| 13    | <b>SIR_REJECT.</b> Read-write. Reset: 0.<br><b>Description:</b> In-Band Target Interrupt Request Reject field is used to control, per device, whether to accept Target Interrupt request from Devices.<br>- 0x0 - Accept: ACK the SIR<br>- 0x1 - Reject: NACK the SIR and send auto disable <a href="#">CCC</a> .                                                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>IBI_WITH_DATA.</b> Read-write. Reset: 0.<br><b>Description:</b> Mandatory one or more data bytes follow the accepted IBI from the device. Data byte continuation is indicated by T-Bit.<br>- 0x0 - IBI Without Mandatory Byte<br>- 0x1 - IBI with one or more Mandatory Bytes.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 11    | <b>IBI_PEC_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <b>Description:</b> Packet Error Check enabled for accepted IBI from the device. <a href="#">PEC</a> byte is appended at the end of IBI data from the device.<br>This bit controls whether PEC check should be performed for IBI data from device.<br>This bit also controls whether PEC byte has to be send for auto disable <a href="#">CCC</a> when controller NACKs the IBI.<br>- 0x0 - Packet Error Check disabled for IBI<br>- 0x1 - Packet Error Check enabled for IBI<br>This field is applicable only if configuration parameter 'IC_HAS_PEC' is set to 1. |
| 10:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6:0  | <b>STATIC_ADDRESS.</b> Read-write. Reset: 00h. Device Static Address.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

### 14.3.9 Watchdog Timer (WDT) Registers

Watchdog timer registers are accessed through two methods:

- Memory access to Watchdog Timer memory address range from FEB0\_0000h to FEB0\_000Fh. Program FCH::PM::DECODEEN[watchdogtmren]=1 to enable Watchdog Timer decoding.
- Memory mapped access through the AcpiMmio region. The Watchdog Timer registers start from FED8\_0000h+B00h. See FCH::PM::ISACONTROL[mmioen].

**WDTx00000000 (FCH::TMR::WDT::WATCHDOGCONTROL)**

Read-write.

\_aliasHOSTLEGACY; WDTx00000000; WDT=FED8\_0B00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7    | <b>watchdogtrigger_wo.</b> Read-write. Reset: 0.<br><b>Description:</b> Write only. Setting this bit triggers the watchdog to start a new count interval, counting down from the value that was last written to the Watchdog Count Register. This bit is always read as zero. Setting this bit has no effect if the watchdog is disabled or stopped.                                                                                                                                                                                                                                  |
| 6:5  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4    | <b>watchdogactionen.</b> Read-write. Reset: 1.<br><b>Description:</b> This bit is reset when enter POWER SAVING MODE. It can only be set after system enter POWER SAVING MODE and recover back. It is set by SMU.<br>When this bit is high, the actions in bit2 can be implemented.<br>When this bit is low, although WatchDogFired is set, the actions will still be suppressed.                                                                                                                                                                                                     |
| 3    | <b>watchdogdisable.</b> Read-write. Reset: 1.<br><b>Description:</b> This bit reflects the state of PMIO_Reg:00[7].<br>Writing to this bit has no effect.<br>0: Enable<br>1: Disable                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2    | <b>watchdogaction.</b> Read-write. Reset: 0.<br><b>Description:</b> This bit determines the action to be taken when the watchdog timer expires.<br>0: System reset<br>1: System power off<br>The bit is only valid when the watchdog is enabled.                                                                                                                                                                                                                                                                                                                                      |
| 1    | <b>watchdogfired.</b> Read-write. Reset: 0. A value of "1" indicates that the watchdog timer has expired and caused the current restart. The bit is cleared by writing a "1" to bit 1 in the Watchdog Control register. Writing a "0" has no effect. The bit is cleared by a power cycle or by the operating system and it must remain cleared for any restart that is not caused by the watchdog timer firing. The bit is only valid when the watchdog is enabled.                                                                                                                   |
| 0    | <b>watchdogrunstopb.</b> Read-write. Reset: 0.<br><b>Description:</b> This bit is used to control or indicate whether the watchdog is in the Running and Stopped states.<br>1: Watchdog is in the Running state<br>0: Watchdog is in the Stopped state<br>If the watchdog is in the Stopped state and a 1 is written to bit [0], the watchdog moves to the Running state, but a count interval is not started until a 1 is written to bit [7].<br>If the watchdog is in the Running state, writing a 1 to bit 0 has no effect.<br>The bit is only valid when the watchdog is enabled. |

**WDTx00000004 (FCH::TMR::WDT::WATCHDOGCOUNT)**

Read-write.

\_aliasHOSTLEGACY; WDTx00000004; WDT=FED8\_0B00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                           |
| 15:0  | <b>watchdogcount.</b> Read-write. Reset: XXXXXXXXXXXXXXXXb.<br><b>Description:</b> This defines the countdown time for the counter. The units are defined in the Units field in the Watchdog Resource Table (WDRT). The maximum value is defined in the Max Count field in the WDRT.<br>Reading this register returns in the current counter value. |

**14.3.10 Wake Alarm Device (AcDcTimer) Registers**

In Family 19h Models 10h-1Fh , signal iBatteryModeB is hardwired to '1', so it implements the AC only (power

independent) wake contain one programmable timer that can be configured to wake up the system regardless of the platform's power source when the timer expires. To simplify the programming interface the AC wake uses the AC timer portion of the AC/DC wake; writes to the DC timer when AC only wake is supported is ignored. The AC/DC timer registers are used to control the wake alarm device. They are accessed through the AcpiMmio region. The AC DC timer registers range from FED8\_0000h+1D00h to FED8\_0000h+1DFFh.

**ACDCx00000000 (FCH::TMR::ACDC::AC\_TIMER\_VALUE)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; ACDCx00000000; ACDC=FED8\_1D00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>ac_timer_value.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> Writing the register to a value other than FFFFFFFFh will start the AC timer:<br>Reading this register returns the current value of AC timer.<br>When AC or DC Timer generate wake up event, this register will be reset to FFFFFFFFh by hardware.<br>FFFFFFFFh: Disable AC timer.<br>FFFFFFFFh ~ 00000001h: The value indicates the number of seconds between the time when the AC timer is programmed and the time when it expires.<br>00000000h: The AC timer will wake up the system instantly. |

**ACDCx00000004 (FCH::TMR::ACDC::AC\_EXPIRED\_TIMER\_POLICY)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; ACDCx00000004; ACDC=FED8\_1D00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>ac_expired_timer_policy.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> If AC timer expired when current power source is DC, the wake signal won't be asserted. If the power source is switched back to AC when the AC timer is already expired, we will wait for the number of seconds defined in this register and then wake up the system.<br>When AC or DC Timer generate wake up event, this register will be reset to FFFFFFFFh by hardware.<br>FFFFFFFFh: Disable AC expired timer policy.<br>FFFFFFFFh ~ 00000001h: The value indicates the number of seconds between the time when the power is switched to AC and the time when it generates the wake-up event.<br>00000000h: The expired AC timer will wake up the system instantly once the power source is switched to AC. |

**ACDCx00000008 (FCH::TMR::ACDC::AC\_TIMER\_STATUS)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; ACDCx00000008; ACDC=FED8\_1D00h

| Bits | Description                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:2 | Reserved.                                                                                                                                                                              |
| 1    | <b>ac_timer_wakeup.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Wake-up was caused by AC timer expiration<br>0: Wake-up was not caused by AC timer expiration<br>Write 1 clear |
| 0    | <b>ac_timer_expired.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: AC timer expired<br>0: AC timer not expired<br>Write 1 clear                                                  |

**ACDCx00000010 (FCH::TMR::ACDC::DC\_TIMER\_VALUE)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; ACDCx00000010; ACDC=FED8\_1D00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>dc_timer_value.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> Writing the register to a value other than FFFFFFFFh will start the DC timer:<br>Reading this register returns the current value of DC timer.<br>When AC or DC Timer generate wake up event, this register will be reset to FFFFFFFFh by hardware.<br>FFFFFFFFh: Disable DC timer.<br>FFFFFFFEh ~ 00000001h: The value indicates the number of seconds between the time when the DC timer is programmed and the time when it expires.<br>00000000h: The DC timer will wake up the system instantly. |

**ACDCx00000014 (FCH::TMR::ACDC::DC\_EXPIRED\_TIMER\_POLICY)**

Read-write. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; ACDCx00000014; ACDC=FED8\_1D00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>dc_expired_timer_policy.</b> Read-write. Reset: FFFF_FFFFh.<br><b>Description:</b> If DC timer expired when current power source is AC, the wake signal won't be asserted. If the power source is switched back to DC when the DC timer is already expired, we will wait for the number of seconds defined in this register and then wake up the system.<br>When AC or DC Timer generate wake up event, this register will be reset to FFFFFFFFh by hardware.<br>FFFFFFFFh: Disable DC expired timer policy.<br>FFFFFFFEh ~ 00000001h: The value indicates the number of seconds between the time when the power is switched to DC and the time when it generates the wake-up event.<br>00000000h: The expired DC timer will wake up the system instantly once the power source is switched to DC. |

**ACDCx00000018 (FCH::TMR::ACDC::DC\_TIMER\_STATUS)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; ACDCx00000018; ACDC=FED8\_1D00h

| Bits | Description                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:2 | Reserved.                                                                                                                                                                              |
| 1    | <b>dc_timer_wakeup.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Wake-up was caused by DC timer expiration<br>0: Wake-up was not caused by DC timer expiration<br>Write 1 clear |
| 0    | <b>dc_timer_expired.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: DC timer expired<br>0: DC timer not expired<br>Write 1 clear                                                  |

**ACDCx00000020 (FCH::TMR::ACDC::ACDCTIMER\_CTRL)**

Read-write. Reset: 0000\_1000h.

\_aliasHOSTLEGACY; ACDCx00000020; ACDC=FED8\_1D00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:13 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 12    | <b>no_event_in_g0</b> . Read-write. Reset: 1. Debug purpose. Please don't change it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 11:10 | <b>sel_wake_rst_1_0</b> . Read-write. Reset: 0h. Debug purpose. Please don't change it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 9     | <b>dc_timer_event_en</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Disable AC Timer to wake up system<br>0: Disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8     | <b>ac_timer_event_en</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Enable AC Timer to wake up system<br>0: Disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7:1   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0     | <b>busy</b> . Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>Right after AC_TIMER_VALUE or DC_TIMER_VALUE is programmed, the hardware will set Busy bit to 1. The hardware will clear Busy bit once the programming is done and the corresponding timer is started properly. Before the software write the AC_TIMER_VALUE or DC_TIMER_VALUE registers, it has to read the Busy bit and make sure it is 0, otherwise, the hardware will just ignore the programming action from software. For the registers other than AC_TIMER_VALUE and DC_TIMER_VALUE, there is no such limitation. |

**14.3.11 Always On Always Connected (AOAC) Registers**

Table 252: FCH Device D3 control/status mapping

| Controller Block        | FCH Power Group | D3 Control Register | D3 Status Register | ClkOk (From block to ACPI) | AOACtrl (From ACPI to block)  |
|-------------------------|-----------------|---------------------|--------------------|----------------------------|-------------------------------|
| Clk Gen                 | PG1             | DevCtrl 0           | DevSts 0           | Tied high                  | Connected PllRstB and PllLock |
| AB                      | PG1             | DevCtrl 1           | DevSts 1           | Tied high                  | Connected                     |
| ACPI S0 (fch_acpismbus) | PG1             | DevCtrl 2           | DevSts 2           | Tied high                  | Connected                     |
| ACPI S5 (fch_acpi_s5)   | S5              | DevCtrl 3           | DevSts 3           | Tied high                  | Not used                      |
| I2C0                    | PG1a            | DevCtrl 5           | DevSts 5           | Tied high                  | Connected                     |
| I2C1                    | PG2             | DevCtrl 6           | DevSts 6           | Tied high                  | Connected                     |
| I2C2                    | PG2             | DevCtrl 7           | DevSts 7           | Tied high                  | Connected                     |
| I2C3                    | PG2             | DevCtrl 8           | DevSts 8           | Tied high                  | Connected                     |
| I2C4                    | PG2             | DevCtrl 9           | DevSts 9           | Tied high                  | Connected                     |
| I2C5                    | PG2             | DevCtrl 10          | DevSts 10          | Tied high                  | Connected                     |
| UART0                   | PG2             | DevCtrl 11          | DevSts 11          | Tied high                  | Connected                     |
| UART1                   | PG2             | DevCtrl 12          | DevSts 12          | Tied high                  | Connected                     |
| I3C1                    | PG2             | DevCtrl 13          | DevSts 13          | Tied high                  | Connected                     |
| I3C2                    | PG2             | DevCtrl 14          | DevSts 14          | Tied high                  | Connected                     |
| I3C3                    | PG2             | DevCtrl 15          | DevSts 15          | Tied high                  | Connected                     |
| UART2                   | PG2             | DevCtrl 16          | DevSts 16          | Tied high                  | Connected                     |
| AMBA                    | PG1             | DevCtrl 17          | DevSts 17          | Tied high                  | Connected                     |
| UART4                   | PG2             | DevCtrl 20          | DevSts 20          | Tied high                  | Connected                     |

|       |      |            |           |           |                                                     |
|-------|------|------------|-----------|-----------|-----------------------------------------------------|
| I3C0  | PG1a | DevCtrl 21 | DevSts 21 | Tied high | Connected                                           |
| eSPI1 | PG1  | DevCtrl 22 | DevSts 22 | Tied high | Connected                                           |
| UART3 | PG2  | DevCtrl 26 | DevSts 26 | Tied high | Connected                                           |
| eSPI0 | PG1  | DevCtrl 27 | DevSts 27 | Tied high | Connected                                           |
| CPU   | PG1  | DevCtrl 31 | DevSts 31 | Connected | FCH pin name<br>FCH_SOC_RESETn and<br>FCH_SOC_PWROK |

**AOACx00000000 (FCH::AOAC::PERFMON\_CONTROL)**

Read-write. Reset: 0000\_0000h.

**NOTE 1:**

Each block's busy signal has a traffic counter. The traffic counter keeps track of the amount of traffic in a specified timer interval. Each time interval consists of 65280 time slots. We can change the length of the time slot and time interval by changing the MonPeriodSel[1:0] register. If the busy signal from a block has asserted during a time slot, the counter will increase by 1 no matter how long the busy signal was during that time slot. At the end of the time interval, the counter value will be scaled by a weight and updated to the output of the traffic counter.

Each block has a traffic counter and software can read the weighted counter values of each block (TrafficCount\_00 ~ 07). Hardware also provides a sum of those weighted counter values through register TrafficCount\_All.

\_aliasHOSTLEGACY; AOACx00000000; AOAC=FED8\_1E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                                                                 |
| 29    | <b>trafficsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Write-1-clear<br>1: There are less (TrafficLess=1) or more (TrafficLess=0) traffic than the threshold defined in BusyTimeThreshold.<br>0: There are more (TrafficLess=1) or less (TrafficLess=0) traffic than the threshold defined in BusyTimeThreshold. |
| 28    | <b>intrsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Write-1-clear<br>1: There are less (IntrLess=1) or more (IntrLess=0) traffic than the threshold defined in IntrTimeThreshold.<br>0: There are more (IntrLess=1) or less (IntrLess=0) traffic than the threshold defined in IntrTimeThreshold.                |
| 27    | <b>cnt1source.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Assign traffic_cnt01 to Sata traffic.<br>0: Assign traffic_cnt01 to Sata Port 1 active (not in partial/slumber).                                                                                                                                       |
| 26    | <b>cnt0source.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Assign traffic_cnt00 to "AnyBusy".<br>0: Assign traffic_cnt00 to Sata Port 0 active (not in partial/slumber).<br>(See NOTE 2)                                                                                                                          |
| 25    | <b>checkinterrupt.</b> Read-write. Reset: 0. Enable interrupt counter (IntrCount)                                                                                                                                                                                                                                         |
| 24    | <b>checkc3.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Check C3 state when counting "AnyBusy" in traffic_cnt00.<br>0: Don't check C3.<br>(See NOTE 2)                                                                                                                                                            |
| 23:8  | Reserved.                                                                                                                                                                                                                                                                                                                 |
| 7:6   | <b>monperiodsel_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Time slot = 15 ns Time interval = 979.2 us<br>01: Time slot = 240 ns Time interval = 15.667 ms<br>10: Time slot = 1.92 us Time interval = 125.34 ms<br>11: Time slot = 15.36 us Time interval = 1 s<br>(See NOTE 1)                            |
| 5     | Reserved.                                                                                                                                                                                                                                                                                                                 |
| 4     | <b>intrless.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Generate interrupt status (IntrSts) when there is less interrupt than the specified threshold (IntrTimeThreshold)<br>0: Generate interrupt status (IntrSts) when there is more traffic than the specified threshold (IntrTimeThreshold)                  |
| 3     | <b>trafficless.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Generate traffic status (TrafficSts) when there is less traffic than the specified threshold (BusyTimeThreshold)<br>0: Generate traffic status (TrafficSts) when there is more traffic than the specified threshold (BusyTimeThreshold)               |



|   |                                                                                                                                            |
|---|--------------------------------------------------------------------------------------------------------------------------------------------|
| 2 | <b>interruptscien.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Generate SCI when there is interrupt status (IntrSts)<br>0: Disable |
| 1 | <b>trafficscien.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Generate SCI when there is traffic status (TrafficSts)<br>0: Disable  |
| 0 | <b>perfmoneable.</b> Read-write. Reset: 0. Global enable of Performance Monitor                                                            |

**AOACx00000004 (FCH::AOAC::PERFMON\_TIME\_LIMIT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx00000004; AOAC=FED8\_1E00h

| Bits  | Description                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>intrtimethreshold.</b> Read-write. Reset: 0000h. Specify the counter threshold for generating Interrupt SCI. |
| 15:0  | <b>busytimethreshold.</b> Read-write. Reset: 0000h. Specify the counter threshold for generating Traffic SCI.   |

**AOACx00000008 (FCH::AOAC::PERFMON\_WEIGHT\_3\_2\_1\_0)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx00000008; AOAC=FED8\_1E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**AOACx0000000C (FCH::AOAC::PERFMON\_WEIGHT\_7\_6\_5\_4)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx0000000C; AOAC=FED8\_1E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**AOACx00000010 (FCH::AOAC::PERFMON\_TRAF\_CNT\_1\_0)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx00000010; AOAC=FED8\_1E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**AOACx00000014 (FCH::AOAC::PERFMON\_TRAF\_CNT\_3\_2)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx00000014; AOAC=FED8\_1E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**AOACx00000018 (FCH::AOAC::PERFMON\_TRAF\_CNT\_5\_4)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx00000018; AOAC=FED8\_1E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**AOACx0000001C (FCH::AOAC::PERFMON\_TRAF\_CNT\_7\_6)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx0000001C; AOAC=FED8\_1E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**AOACx00000020 (FCH::AOAC::PERFMON\_TRAF\_CNT\_ALL)**

Read-only. Reset: 0000\_0000h.

TrafficCount\_All[26:0] = Internal-Weighted-Count00 [23:0] + + Internal-Weighted-Count07 [23:0]

\_aliasHOSTLEGACY; AOACx00000020; AOAC=FED8\_1E00h

| Bits  | Description                                                                                                        |
|-------|--------------------------------------------------------------------------------------------------------------------|
| 31:27 | Reserved.                                                                                                          |
| 26:0  | <b>trafficcount_all_26_0</b> . Read-only. Reset: 000_0000h. Sum of weighted counts from each traffic_cntXX blocks. |

**AOACx00000024 (FCH::AOAC::PERFMON\_INTR\_CNT)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx00000024; AOAC=FED8\_1E00h

| Bits  | Description                                      |
|-------|--------------------------------------------------|
| 31:24 | <b>swbusy</b> . Read-only. Reset: 00h.           |
| 23:16 | <b>swbusyen</b> . Read-only. Reset: 00h.         |
| 15:0  | <b>intrcount_15_0</b> . Read-only. Reset: 0000h. |

**AOACx00000030 (FCH::AOAC::ALTHPET\_TIMER\_L)**

Read-only.

Register 30h and 34hare read-only. It shows the current time with higher resolution than traditional RTC timer. AltMmTimer field shows the time with 64kHz resolution. It will wrap around at every second.

\_aliasHOSTLEGACY; AOACx00000030; AOAC=FED8\_1E00h

| Bits  | Description                                                                                     |
|-------|-------------------------------------------------------------------------------------------------|
| 31:24 | <b>minute_7_0</b> . Read-only. Reset: XXXXXXXXXb. Minute field of the ALTHPET_TIMER             |
| 23:16 | <b>second_7_0</b> . Read-only. Reset: XXXXXXXXXb. Second field of the ALTHPET_TIMER             |
| 15:0  | <b>altmmtimer</b> . Read-only. Reset: XXXXXXXXXXXXXXXXXb. AltMmTimer field of the ALTHPET_TIMER |

**AOACx00000034 (FCH::AOAC::ALTHPET\_TIMER\_H)**

Read-only.

\_aliasHOSTLEGACY; AOACx00000034; AOAC=FED8\_1E00h

| Bits  | Description                                                                       |
|-------|-----------------------------------------------------------------------------------|
| 31:24 | <b>year_7_0</b> . Read-only. Reset: XXXXXXXXXb. Year field of the ALTHPET_TIMER   |
| 23:16 | <b>month_7_0</b> . Read-only. Reset: XXXXXXXXXb. Month field of the ALTHPET_TIMER |
| 15:8  | <b>day_7_0</b> . Read-only. Reset: XXXXXXXXXb. Day field of the ALTHPET_TIMER     |
| 7:0   | <b>hour_7_0</b> . Read-only. Reset: XXXXXXXXXb. Hour field of the ALTHPET_TIMER   |

**AOACx00000038 (FCH::AOAC::ALTHPET\_ALARM\_L)**

Read-write. Reset: 0000\_0000h.

AltMmTimer can be programmed to trigger [SMI](#) (Event 43).

Whenever ALTHPET\_TIMER (L and H) matches the value defined in ALTHPET\_ALARM (L and H), we will generate a pulse on AltMmTimerSts (Event 43).

\_aliasHOSTLEGACY; AOACx00000038; AOAC=FED8\_1E00h

| Bits  | Description                                                                               |
|-------|-------------------------------------------------------------------------------------------|
| 31    | Reserved.                                                                                 |
| 30:24 | <b>minute_alarm_6_0</b> . Read-write. Reset: 00h. Minute field of the ALTHPET_ALARM       |
| 23    | Reserved.                                                                                 |
| 22:16 | <b>second_alarm_6_0</b> . Read-write. Reset: 00h. Second field of the ALTHPET_ALARM       |
| 15:0  | <b>altmmtimer_alarm</b> . Read-write. Reset: 0000h. AltMmTimer field of the ALTHPET_ALARM |

**AOACx0000003C (FCH::AOAC::ALTHPET\_ALARM\_H)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx0000003C; AOAC=FED8\_1E00h

| Bits  | Description                                                                       |
|-------|-----------------------------------------------------------------------------------|
| 31:24 | <b>year_alarm_7_0</b> . Read-write. Reset: 00h. Year field of the ALTHPET_ALARM   |
| 23:21 | Reserved.                                                                         |
| 20:16 | <b>month_alarm_4_0</b> . Read-write. Reset: 00h. Month field of the ALTHPET_ALARM |
| 15:14 | Reserved.                                                                         |
| 13:8  | <b>day_alarm_5_0</b> . Read-write. Reset: 00h. Day field of the ALTHPET_ALARM     |
| 7:6   | Reserved.                                                                         |
| 5:0   | <b>hour_alarm_5_0</b> . Read-write. Reset: 00h. Hour field of the ALTHPET_ALARM   |

**AOACx00000040 (FCH::AOAC::DEVCTRL\_0)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000040; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwronrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                        |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000041 (FCH::AOAC::DEVSTS\_0)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000041; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000042 (FCH::AOAC::DEVCTRL\_1)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000042; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000043 (FCH::AOAC::DEVSTS\_1)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000043; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000044 (FCH::AOAC::DEVCTRL\_2)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000044; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                             |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000045 (FCH::AOAC::DEVSTS\_2)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000045; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwr rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.          |

**AOACx00000046 (FCH::AOAC::DEVCTRL\_3)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000046; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwr onrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 3    | <b>pwr ondev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000047 (FCH::AOAC::DEVSTS\_3)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000047; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000048 (FCH::AOAC::DEVCTRL\_4)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000048; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                             |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000049 (FCH::AOAC::DEVSTS\_4)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000049; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx0000004A (FCH::AOAC::DEVCTRL\_5)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx0000004A; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |



**AOACx0000004B (FCH::AOAC::DEVSTS\_5)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000004B; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx0000004C (FCH::AOAC::DEVCTRL\_6)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx0000004C; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                             |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx0000004D (FCH::AOAC::DEVSTS\_6)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000004D; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwr rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.          |

**AOACx0000004E (FCH::AOAC::DEVCTRL\_7)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx0000004E; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwr onrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 3    | <b>pwr ondev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx0000004F (FCH::AOAC::DEVSTS\_7)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000004F; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000050 (FCH::AOAC::DEVCTRL\_8)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000050; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000051 (FCH::AOAC::DEVSTS\_8)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000051; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000052 (FCH::AOAC::DEVCTRL\_9)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000052; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondv</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000053 (FCH::AOAC::DEVSTS\_9)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000053; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000054 (FCH::AOAC::DEVCTRL\_10)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000054; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondv</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000055 (FCH::AOAC::DEVSTS\_10)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000055; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000056 (FCH::AOAC::DEVCTRL\_11)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000056; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                             |
| 3    | <b>pwrondv</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000057 (FCH::AOAC::DEVSTS\_11)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000057; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000058 (FCH::AOAC::DEVCTRL\_12)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000058; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000059 (FCH::AOAC::DEVSTS\_12)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000059; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx0000005A (FCH::AOAC::DEVCTRL\_13)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx0000005A; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                             |
| 3    | <b>pwronddev</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |



**AOACx0000005B (FCH::AOAC::DEVSTS\_13)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000005B; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwr rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.          |

**AOACx0000005C (FCH::AOAC::DEVCTRL\_14)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx0000005C; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwr onrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 3    | <b>pwr ondev</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx0000005D (FCH::AOAC::DEVSTS\_14)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000005D; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwr rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.          |

**AOACx0000005E (FCH::AOAC::DEVCTRL\_15)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx0000005E; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwr onrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 3    | <b>pwr ondev</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx0000005F (FCH::AOAC::DEVSTS\_15)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000005F; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000060 (FCH::AOAC::DEVCTRL\_16)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000060; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondv</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000061 (FCH::AOAC::DEVSTS\_16)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000061; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000062 (FCH::AOAC::DEVCTRL\_17)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000062; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000063 (FCH::AOAC::DEVSTS\_17)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000063; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000064 (FCH::AOAC::DEVCTRL\_18)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx00000064; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondv</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000065 (FCH::AOAC::DEVSTS\_18)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000065; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000066 (FCH::AOAC::DEVCTRL\_19)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx00000066; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                             |
| 3    | <b>pwrondev</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                        |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000067 (FCH::AOAC::DEVSTS\_19)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000067; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000068 (FCH::AOAC::DEVCTRL\_20)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx00000068; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondv</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000069 (FCH::AOAC::DEVSTS\_20)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000069; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwr rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.          |

**AOACx0000006A (FCH::AOAC::DEVCTRL\_21)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx0000006A; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwr onrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 3    | <b>pwr ondev</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |



**AOACx0000006B (FCH::AOAC::DEVSTS\_21)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000006B; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx0000006C (FCH::AOAC::DEVCTRL\_22)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx0000006C; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondv</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx0000006D (FCH::AOAC::DEVSTS\_22)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000006D; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx0000006E (FCH::AOAC::DEVCTRL\_23)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx0000006E; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondv</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx0000006F (FCH::AOAC::DEVSTS\_23)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000006F; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000070 (FCH::AOAC::DEVCTRL\_24)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx00000070; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondv</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000071 (FCH::AOAC::DEVSTS\_24)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000071; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwr rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.          |

**AOACx00000072 (FCH::AOAC::DEVCTRL\_25)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000072; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwr onrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 3    | <b>pwr ondev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000073 (FCH::AOAC::DEVSTS\_25)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000073; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000074 (FCH::AOAC::DEVCTRL\_26)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000074; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000075 (FCH::AOAC::DEVSTS\_26)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000075; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwr rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.          |

**AOACx00000076 (FCH::AOAC::DEVCTRL\_27)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000076; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwr onrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 3    | <b>pwr ondev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000077 (FCH::AOAC::DEVSTS\_27)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000077; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000078 (FCH::AOAC::DEVCTRL\_28)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx00000078; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                             |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx00000079 (FCH::AOAC::DEVSTS\_28)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx00000079; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx0000007A (FCH::AOAC::DEVCTRL\_29)**

Read-write. Reset: 74h.

\_aliasHOSTLEGACY; AOACx0000007A; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwrondv</b> . Read-write. Reset: 0.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                         |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 0h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |



**AOACx0000007B (FCH::AOAC::DEVSTS\_29)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000007B; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx0000007C (FCH::AOAC::DEVCTRL\_30)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx0000007C; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx0000007D (FCH::AOAC::DEVSTS\_30)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000007D; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx0000007E (FCH::AOAC::DEVCTRL\_31)**

Read-write. Reset: 7Dh.

\_aliasHOSTLEGACY; AOACx0000007E; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>isswcontrol</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Sw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>0: Hw controls control signals (PwrRstB, RefClkOk, RstB) to the device.<br>To avoid glitch on the reset signals, SW must set IsSwControl bit first and then program SwPwrOnRstB, SwRefClkOk and SwRstB. |
| 6    | <b>swrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RstB is asserted to the device if IsSwControl is set to 1.<br>1: RstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                                       |
| 5    | <b>swrefclkok</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: RefClkOk is de-asserted to the device if IsSwControl is set to 1.<br>1: RefClkOk is asserted to the device if IsSwControl is set to 1.                                                                                                                                           |
| 4    | <b>swpwrnrrstb</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRstB is asserted to the device if IsSwControl is set to 1.<br>1: PwrRstB is de-asserted to the device if IsSwControl is set to 1.                                                                                                                                            |
| 3    | <b>pwronddev</b> . Read-write. Reset: 1.<br><b>Description:</b> If SwControl is set to 0, SW can write this bit to trigger a HW controlled reset sequence to the device.<br>0: put device in reset state.<br>1: put device out of reset state.                                                                                                       |
| 2    | <b>devicestate</b> . Read-write. Reset: 1.<br><b>Description:</b> SW records the device power state in this field.<br>0: device power is removed<br>1: device power is applied                                                                                                                                                                       |
| 1:0  | <b>targeteddevicestate</b> . Read-write. Reset: 1h.<br><b>Description:</b> SW records D-State of the device in this field.<br>00: D0 UnInitialized<br>01: D0 Initialized<br>10: D1/D2/D3 Hot<br>11: D3 Cold                                                                                                                                          |

**AOACx0000007F (FCH::AOAC::DEVSTS\_31)**

Read-write. Reset: 27h.

\_aliasHOSTLEGACY; AOACx0000007F; AOAC=FED8\_1E00h

| Bits | Description                                                                                         |
|------|-----------------------------------------------------------------------------------------------------|
| 7    | <b>stat1</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 6    | <b>stat0</b> . Read-write. Reset: 0. Read-only. State of device.                                    |
| 5    | <b>clkok_state</b> . Read-write. Reset: 1. Read-only. State of device's ClkOk signal.               |
| 4    | <b>d3cold</b> . Read-write. Reset: 0. Read-only. State of device's D3Cold signal.                   |
| 3    | <b>devoffgating_state</b> . Read-write. Reset: 0. Read-only. State of device's DevOffGating signal. |
| 2    | <b>rstb_state</b> . Read-write. Reset: 1. Read-only. State of device's RstB signal.                 |
| 1    | <b>refclkok_state</b> . Read-write. Reset: 1. Read-only. State of device's RefClkOk signal.         |
| 0    | <b>pwrrstb_state</b> . Read-write. Reset: 1. Read-only. State of device's PwrRstB signal.           |

**AOACx00000080 (FCH::AOAC::SHADOW\_REG\_REQUEST)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx00000080; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>shadow_rst</b> . Read-write. Reset: 0.<br><b>Description:</b> SW can write this bit to reset shadow register control logics. (Write 1 to reset, and then write 0 to release the reset)<br>Note: SHADOW_hang means somehow the current request is not completed. SW should do a SHADOW_rst and request again. The Shadow Reg Status bit should remain previous state from last successful save/restore of the corresponding IP. So we don't want to reset Shadow Reg Status bit by SHADOW_rst.                                                                                                                                          |
| 30   | <b>request_sel</b> . Read-write. Reset: 0.<br><b>Description:</b> SW write this bit to select Request bank.<br>0: Select StoreRequest<br>1: Select RestoreRequest                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 29:0 | <b>request</b> . Read-write. Reset: 0000_0000h.<br><b>Description:</b> Request[29:0] are actually shared by two register banks: StoreRequest[29:0] and RestoreRequest[29:0]. SW write Request_Sel bit to choose which bank to access.<br>SW write 1 to these bits to request store or restore. The request type depends on which bank the bit is located (StoreRequest or RestoreRequest). Each bit is corresponding to a device being stored/restored. Write 0 has no effect to these registers.<br>Once store/restore is done for a device, the corresponding bit will be cleared to 0 by HW. Only HW will be able to clear these bits. |

**AOACx00000084 (FCH::AOAC::SHADOW\_REG\_STATUS)**

Read-write. Reset: 3FFF\_FFFFh.

\_aliasHOSTLEGACY; AOACx00000084; AOAC=FED8\_1E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                               |
| 29:0  | <b>status</b> . Read-write. Reset: 3FFF_FFFFh.<br><b>Description:</b> Reading these bits returns the status of the store/restore action. Each bit represent the status of a device. 0 means shadow register logic has finished store action for a device. 1 means shadow register has finished restore action for a device.<br>SW can also write this register to 1 or 0. This provide SW a way to override the status. |

**AOACx00000088 (FCH::AOAC::SHADOW\_REG\_SRAM\_ADDR)**

Read-write. Reset: 0000\_0000h.

SW has indirect access to the SRAM in shadow register block via ShadowSRAM\_Addr and ShadowSRAM\_Data. SW can write ShadowSRAM\_Addr to an address it wants to access, and then read or write ShadowSRAM\_Data.

\_aliasHOSTLEGACY; AOACx00000088; AOAC=FED8\_1E00h

| Bits  | Description                                                                       |
|-------|-----------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                         |
| 15:0  | <b>shadowsram_addr</b> . Read-write. Reset: 0000h. Shadow SRAM address (see NOTE) |

**AOACx0000008C (FCH::AOAC::SHADOW\_REG\_SRAM\_DATA)**

Read-write. Reset: 0000\_0000h.

SW has indirect access to the SRAM in shadow register block via ShadowSRAM\_Addr and ShadowSRAM\_Data. SW can write ShadowSRAM\_Addr to an address it wants to access, and then read or write ShadowSRAM\_Data.

\_aliasHOSTLEGACY; AOACx0000008C; AOAC=FED8\_1E00h

| Bits | Description                                                                         |
|------|-------------------------------------------------------------------------------------|
| 31:0 | <b>shadowsram_data</b> . Read-write. Reset: 0000_0000h. Shadow SRAM data (see NOTE) |

**AOACx00000090 (FCH::AOAC::SHADOW\_REG\_HW\_INIT\_EN)**

Read-write.

\_aliasHOSTLEGACY; AOACx00000090; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>shadowsram_inds</b> . Read-write. Reset: X.<br><b>Description:</b> Deep Sleep status of Shadow SRAM. Read-only.<br>0: Shadow SRAM is in normal function mode<br>1: Shadow SRAM is in Deep Sleep mode                                                                                                                                                   |
| 30   | <b>forcemasterack</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Shadow Master dynamically turn on/off its internal clock base on the requests.<br>1: Shadow Master internal clock always enabled.                                                                                                                                                 |
| 29:0 | <b>hwinit_en</b> . Read-write. Reset: 0800_0016h. These bits configure the capability of "hardware initiated store/restore" for each device. When a device's HwInit_En bit is enabled, hardware will automatically initiate a store for that device before going into Power saving mode state, and a restore when waking up from Power saving mode state. |

**AOACx00000094 (FCH::AOAC::POWERSAVINGMODECTRL)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx00000094; AOAC=FED8\_1E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:19 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 18    | <b>clock_slow_down_mode_req.</b> Read-write. Reset: 0.<br><b>Description:</b> Write-only. Read always returns 0.<br>SMU can write 1 to this bit to request <a href="#">FCH</a> to Slow Down clock (CLOCK SLOW DOWN MODE) when PM_regx50[0] =1                                                                                                                                                                         |
| 17    | <b>apuplloffreq.</b> Read-write. Reset: 0.<br><b>Description:</b> Write-only. Read always returns 0.<br>SMU can write 1 to this bit to request <a href="#">FCH</a> to shutdown CGPLL (CLOCK STOP MODE) when PM_regx50[0] =1                                                                                                                                                                                           |
| 16    | <b>intrblocked.</b> Read-write. Reset: 0.<br><b>Description:</b> Read-only. If <a href="#">FCH</a> is sending an interrupt message when SW or SMU write 1 to InterruptDis, the interrupt will be blocked "gracefully". We will wait until the interrupt is sent and then block the interrupt. This bit indicates the interrupt blocking status.<br>0: FCH interrupts are not blocked<br>1: FCH interrupts are blocked |
| 15    | <b>interruptdis.</b> Read-write. Reset: 0.<br><b>Description:</b> SW is able to set and clear this bit. HW can only set this bit.<br>1: interrupt from <a href="#">FCH</a> is disabled.<br>0: interrupt from FCH is enabled.                                                                                                                                                                                          |
| 14    | <b>arbiterdis.</b> Read-write. Reset: 0.<br><b>Description:</b> SW is able to set and clear this bit. HW can only set this bit.<br>1: <a href="#">FCH</a> upstream arbiter is disabled.<br>0: FCH upstream arbiter is enabled.                                                                                                                                                                                        |
| 13    | <b>maskioapic.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: IOAPIC mask bits function normally<br>1: Mask all IOAPIC interrupts<br>This bit is OR'd with all IOAPIC interrupt mask bits.                                                                                                                                                                                                                       |
| 12    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 11    | <b>anyupwakepowersavingmodeen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Power saving mode state can be wake up by GPIO controller and Shadow Timer.<br>1: Power saving mode state can be wake up by GPIO controller, Shadow Timer and any pending upstream message in ACPI.                                                                                                                                |
| 10    | <b>shadowtimerwakests.</b> Read-write. Reset: 0.<br><b>Description:</b> Read-only. HW will clear this bit during Power saving mode entry.<br>1: Shadow Timer fired and caused wake-up from Power saving mode state.<br>0: Shadow Timer did not fire.                                                                                                                                                                  |
| 9     | <b>powersavingmoderesume.</b> Read-write. Reset: 0. Set to 1 by Power saving mode exit to indicate the system has resumed from Power saving mode state. The bit is cleared by Power saving mode entry or write-1-cleared by Sw.                                                                                                                                                                                       |
| 8     | <b>powersavingmodeenter.</b> Read-write. Reset: 0.<br><b>Description:</b> Read-only<br>1: indicate the system is in Power saving mode state.<br>0: indicate the system is in S0 state.                                                                                                                                                                                                                                |
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 4     | <b>powersavingmodetrigger.</b> Read-write. Reset: 0.<br><b>Description:</b> Write-only, read always return 0.<br>Setting the bit to 1 to trigger Power saving mode power down sequence.                                                                                                                                                                                                                               |

|   |                                                                                                                                                                                        |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3 | Reserved.                                                                                                                                                                              |
| 2 | <b>powersavingmodeready2.</b> Read-write. Reset: 0.<br><b>Description:</b> Programmed by Driver.<br>0: Power saving mode can't be entered<br>1: Power saving mode can be entered       |
| 1 | <b>powersavingmodeonslps3b.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SLP_S3# doesn't assert in Power saving mode state<br>1: SLP_S3# will assert in Power saving mode state |
| 0 | <b>powersavingmodeready.</b> Read-write. Reset: 0.<br><b>Description:</b> Programmed by BIOS.<br>0: Power saving mode can't be entered<br>1: Power saving mode can be entered          |

**AOACx00000098 (FCH::AOAC::POWER\_SAVING\_MODE\_DEBUG)**

| Read-write.                                      |                                                                                                                                                                                                                                     |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| _aliasHOSTLEGACY; AOACx00000098; AOAC=FED8_1E00h |                                                                                                                                                                                                                                     |
| Bits                                             | Description                                                                                                                                                                                                                         |
| 31:14                                            | Reserved.                                                                                                                                                                                                                           |
| 13                                               | <b>power_saving_mode_dontwaitstpgnt.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Power saving mode state machine will wait for STPGNT after it sens STPCLK<br>1: Don't wait                                                 |
| 12                                               | <b>power_saving_mode_dontwaitsdb.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Wait for SDB (in PG1) to re-initialize before restore during Power saving mode wake-up<br>1: Don't wait                                       |
| 11                                               | <b>power_saving_mode_dontblockpmiorst.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Block PciRstB to PMIO register block during Power saving mode<br>1: Do not block PciRstB to PMIO register block during Power saving mode |
| 10                                               | <b>power_saving_mode_force_clk.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Power saving mode state machine dynamically gate its local clock<br>1: Power saving mode state machine local clock always on                    |
| 9                                                | <b>power_saving_mode_bypassshdwreg.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: During Power saving mode entry and exit, use ShadowReg to store and restore context of PG1 devices.<br>1: Bypass ShadowReg                  |
| 8                                                | <b>power_saving_mode_bypassshdwtmr.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: During Power saving mode entry and exit, use ShadowTimer to store and restore HPET and AcpiTimer.<br>1: Bypass ShadowTimer                  |
| 7:0                                              | <b>power_saving_mode_xtal_settle_time.</b> Read-write. Reset: XXXXXXXXb.<br><b>Description:</b> Debug purpose.<br>SW should leave it at default value.                                                                              |

**AOACx0000009C (FCH::AOAC::SHADOW\_TIMER\_CTRL)**

Read-write.

\_aliasHOSTLEGACY; AOACx0000009C; AOAC=FED8\_1E00h

| Bits  | Description                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>restoredoffset64.</b> Read-write. Reset: 00h. A programmable offset to be added to restored HPET 64bit timer this is to accounted for restore time uncertainty.              |
| 23:16 | <b>restoredoffset.</b> Read-write. Reset: 00h. A programmable offset to be added to restored HPET 32bit timer and ACPI timer this is to accounted for restore time uncertainty. |
| 15:8  | <b>earlycount.</b> Read-write. Reset: XXXXXXXXb. A programmable offset to be subtracted from the alarm value                                                                    |
| 7:3   | Reserved.                                                                                                                                                                       |
| 2     | <b>earlycountunit.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: The unit of EarlyCount is 1 RTC clock period.<br>1: The unit of EarlyCount is 16 RTC clock period        |
| 1     | <b>shadowacpitimeren.</b> Read-write. Reset: 0. Enable control to perform store/restore operation for ACPI timer                                                                |
| 0     | <b>shadowhpeten.</b> Read-write. Reset: 0. Enable control to perform store/restore operation from HPET timer                                                                    |

**AOACx000000A0 (FCH::AOAC::PWRGOOD\_CTRL)**

Read-write. Reset: EC00\_A03Bh.

NOTE1:

NOTE2:

\_aliasHOSTLEGACY; AOACx000000A0; AOAC=FED8\_1E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>swxhc0s5rstb.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: XHC0 PHY is powered down.<br>1: XHC0 PHY is powered.                                                                                                                                                                                                                                                                                                                             |
| 30    | <b>swu2phys5rstb.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: USB2 PHY is powered down.<br>1: USB2 PHY is powered.<br>(See NOTE2 for the relationship between USB2 PHY and XHC controller)                                                                                                                                                                                                                                                    |
| 29    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 28    | <b>swsatalockphyif.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SATA PHY interface is not locked<br>1: SATA PHY interface is locked                                                                                                                                                                                                                                                                                                           |
| 27    | <b>swusbpllrstb.</b> Read-write. Reset: 1.<br><b>Description:</b> SW can program this bit to toggle these USB related reset signals:<br>oUsbPllRstB<br>oUsb3PllRstB<br>oUsbPllLockB<br>oUsbDllRstB<br>oGoodClkForUsb<br>If SwUsbPllRstB is cleared to 0, the above signals will be driven to 0. After SW set SwUsbPllRstB to 1, HW will generate a de-assertion sequence for the above signals.                                                       |
| 26    | <b>swu3phys5rstb.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: USB3 PHY is powered down.<br>1: USB3 PHY is powered.<br>(See NOTE2 for the relationship between USB2 PHY and XHC controller)                                                                                                                                                                                                                                                    |
| 25    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 24    | <b>spi_pad_disable.</b> Read-write. Reset: 0. SW set this bit to 1 to disable the OE and PU of the SPI pads.                                                                                                                                                                                                                                                                                                                                          |
| 23    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 22:16 | <b>swblockusbrst.</b> Read-write. Reset: 00h.<br><b>Description:</b> SW can set these bits to 1 to individually block the reset signals to USB2 and USB3. Combined with SwUsbPllRstB bit, SW can control the way it assert reset for USB.<br>Bit 0: Block oUsbPllRstB<br>Bit 1: Block oUsb3PllRstB<br>Bit 2: Block oUsbPllLockB<br>Bit 3: Block oUsbDllRstB<br>Bit 4: Block oGoodClkForUsb<br>Bit 5: Block oSoftKB_2PciRst<br>Bit 6: Block oS3PciRstB |
| 15    | <b>powerallpwrland.</b> Read-write. Reset: 1.<br><b>Description:</b> If PowerAllPwrIsland is 1, we will sequentially power up all the power groups to avoid current surge. If the platform support AOAC, SW has to set this bit to 0 during boot-up. Otherwise, SW can leave it at 1. (See diagram in the NOTE1 below)                                                                                                                                |
| 14    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 13    | <b>swxhc1s5rstb.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: XHC1 PHY is powered down.<br>1: XHC1 PHY is powered.                                                                                                                                                                                                                                                                                                                             |



|      |                                                                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12   | <b>swotgs5rstb.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Change OtgS5RstB signal to 0<br>1: Change OtgS5RstB signal to 1                       |
| 11:9 | Reserved.                                                                                                                                                 |
| 8    | <b>swsatablockoob_aoac.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SATA <a href="#">OOB</a> is not blocked<br>1: SATA OOB is blocked             |
| 7:6  | Reserved.                                                                                                                                                 |
| 5    | <b>gbepwren.</b> Read-write. Reset: 1.<br><b>Description:</b> Control the power of GBE power island<br>0: Gbe is powered down<br>1: Gbe is powered up     |
| 4    | <b>otgpwren.</b> Read-write. Reset: 1.<br><b>Description:</b> Control the power of OTG power island<br>0: Otg is powered down<br>1: Otg is powered up     |
| 3    | <b>xhcpwren.</b> Read-write. Reset: 1.<br><b>Description:</b> Control the power of SSIC power island<br>0: Ssic is powered down<br>1: Ssic is powered up  |
| 2    | <b>imcpwren.</b> Read-write. Reset: 0.<br><b>Description:</b> Control the power of IMC<br>0: IMC is powered down<br>1: IMC is powered up                  |
| 1    | <b>pg2pwren.</b> Read-write. Reset: 1.<br><b>Description:</b> Control the power of PG2 power island<br>0: PG2 is powered down<br>1: PG2 is powered up     |
| 0    | <b>pg1apwren.</b> Read-write. Reset: 1.<br><b>Description:</b> Control the power of PG1a power island<br>0: PG1a is powered down<br>1: PG1a is powered up |

**AOACx000000F0 (FCH::AOAC::SHDWREG\_DEBUG)**

Read-write.

\_aliasHOSTLEGACY; AOACx000000F0; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 30   | <b>shadow_hang.</b> Read-write. Reset: 0.<br><b>Description:</b> Read-only. We have a timer to keep track of the save/restore process. When save/restore is on-going, the timer will be counting up. When there is no save/restore, the timer is reset to 0. SHADOW_hang bit goes high when the timer timeout and save/restore is not done yet. SW can toggle SHADOW_rst bit to reset the shadow register control logic if the hang condition is detected.<br>0: Save/restore has been on-going for less than 8~12 ms<br>1: Save/restore has been on-going for at least 8~12 ms<br>Note: SHADOW_hang means somehow the current request is not completed.SW should do a SHADOW_rst and request again.The Shadow Reg Status bit should remain previous state from last successful save/restore of the corresponding IP.So we don't want to reset Shadow Reg Status bit by SHADOW_rst.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 29:0 | <b>shadow_err.</b> Read-write. Reset: XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXb.<br><b>Description:</b> Write-1-clear. Write 0 has no effect. This register report errors encountered during each device's save/restore. The following error conditions will be recorded:<br>AHB bus response error<br>PCI bus master abort<br>PCI bus slave abort<br>Bit29: Error condition encountered during device 29 save/restore<br>Bit28: Error condition encountered during device 28 save/restore<br>Bit27: Error condition encountered during device 27 save/restore<br>Bit26: Error condition encountered during device 26 save/restore<br>Bit25: Error condition encountered during device 25 save/restore<br>Bit24: Error condition encountered during device 24 save/restore<br>Bit23: Error condition encountered during device 23 save/restore<br>Bit22: Error condition encountered during device 22 save/restore<br>Bit21: Error condition encountered during device 21 save/restore<br>Bit20: Error condition encountered during device 20 save/restore<br>Bit19: Error condition encountered during device 19 save/restore<br>Bit18: Error condition encountered during device 18 save/restore<br>Bit17: Error condition encountered during device 17 save/restore<br>Bit16: Error condition encountered during device 16 save/restore<br>Bit15: Error condition encountered during device 15 save/restore<br>Bit14: Error condition encountered during device 14 save/restore<br>Bit13: Error condition encountered during device 13 save/restore<br>Bit12: Error condition encountered during device 12 save/restore<br>Bit11: Error condition encountered during device 11 save/restore<br>Bit10: Error condition encountered during device 10 save/restore<br>Bit9 : Error condition encountered during device 9 save/restore<br>Bit8 : Error condition encountered during device 8 save/restore<br>Bit7 : Error condition encountered during device 7 save/restore<br>Bit6 : Error condition encountered during device 6 save/restore<br>Bit5 : Error condition encountered during device 5 save/restore<br>Bit4 : Error condition encountered during device 4 save/restore<br>Bit3 : Error condition encountered during device 3 save/restore<br>Bit2 : Error condition encountered during device 2 save/restore<br>Bit1 : Error condition encountered during device 1 save/restore<br>Bit0 : Error condition encountered during device 0 save/restore |

**AOACx000000F4 (FCH::AOAC::PWRRST\_DEBUG)**

Read-write.

NOTE1: When SW read 1 at oPG1a\_PwrGood, oPG2\_PwrGood, oGbe\_PwrGood, oOtg\_PwrGood or oSsic\_PwrGood, that means power-on sequence has completed for power group PG1a, PG2, GBE, OTG or SSIC, respectively.

NOTE2: When SW read 1 at PG1a\_CtrlSts, PG2\_CtrlSts, Gbe\_CtrlSts, Otg\_CtrlSts or Ssic\_CtrlSts, that means power-down sequence has completed for power group PG1a, PG2, GBE, OTG or SSIC, respectively.

\_aliasHOSTLEGACY; AOACx000000F4; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <p><b>pwrrstdebugbus.</b> Read-write. Reset: XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXb.</p> <p><b>Description:</b> Read-only. Debug signals from PwrRst block:</p> <p>Bit31: Gbe_CtrlSts (see NOTE2)</p> <p>Bit30: PG1a_CtrlSts (see NOTE2)</p> <p>Bit29: PG2_CtrlSts (see NOTE2)</p> <p>Bit28: Ssic_CtrlSts (see NOTE2)</p> <p>Bit27: Otg_CtrlSts (see NOTE2)</p> <p>Bit26: oDebouncedRsmRstB</p> <p>Bit25: oPciRstB</p> <p>Bit24: oCpuPwrGood</p> <p>Bit23: oCpuRstB</p> <p>Bit22: oU2PHY5RstB</p> <p>Bit21: oUsb3S5RstB</p> <p>Bit20: oPllRstB</p> <p>Bit19: oPllLock</p> <p>Bit18: oGbe_PwrGood (see NOTE1)</p> <p>Bit17: oPG1a_PwrGood (see NOTE1)</p> <p>Bit16: oPG2_PwrGood (see NOTE1)</p> <p>Bit15: oSsic_PwrGood (see NOTE1)</p> <p>Bit14: oOtg_PwrGood (see NOTE1)</p> <p>Bit13: oPG1_ToAllEn</p> <p>Bit12: oPG1a_ToAllEn</p> <p>Bit11: oPG2_ToAllEn</p> <p>Bit10: oSsic_ToAllEn</p> <p>Bit9 : oSlpRstAsrtDone</p> <p>Bit8 : PowerSavingModeRstDsrtDone</p> <p>Bit7 : S0ResetB_Osc</p> <p>Bit6 : S5ResetB_Osc</p> <p>Bit5 : oDebouncedUsrRstB</p> <p>Bit4 : DevPwrRstB[iDebugSelDev][4]</p> <p>Bit3 : DevPwrRstB[iDebugSelDev][3]</p> <p>Bit2 : DevPwrRstB[iDebugSelDev][2]</p> <p>Bit1 : DevPwrRstB[iDebugSelDev][1]</p> <p>Bit0 : DevPwrRstB[iDebugSelDev][0]</p> |

**AOACx000000F8 (FCH::AOAC::SW\_SEMOPHORE)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; AOACx000000F8; AOAC=FED8\_1E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>drvrownaoacreg.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>Driver write Set_DrvrOwnAoacReg and Clr_DrvrOwnAoacReg bits to set or clear DrvrOwnAoacReg. When DrvrOwnAoacReg is 1, BiosOwnAoacReg will stay at 0 and cannot be written. Driver should always make sure DrvrOwnAoacReg is 1 before it writes AOAC Registers, and clear DrvrOwnAoacReg after it finishes the programming. |
| 30   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                    |
| 29   | <b>clr_drvrownaoacreg.</b> Read-write. Reset: 0. Write only. Driver write 1 to this bit to clear DrvrOwnAoacReg.                                                                                                                                                                                                                                                                                             |
| 28   | <b>set_drvrownaoacreg.</b> Read-write. Reset: 0. Write only. Driver write 1 to this bit to set DrvrOwnAoacReg.                                                                                                                                                                                                                                                                                               |
| 27   | <b>biosownaoacreg.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>BIOS write Set_BiosOwnAoacReg and Clr_BiosOwnAoacReg bits to set or clear BiosOwnAoacReg. When BiosOwnAoacReg is 1, DrvrOwnAoacReg will stay at 0 and cannot be written. BIOS should always make sure BiosOwnAoacReg is 1 before it writes AOAC Registers, and clear BiosOwnAoacReg after it finishes the programming.     |
| 26   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                    |
| 25   | <b>clr_biosownaoacreg.</b> Read-write. Reset: 0. Write only. BIOS write 1 to this bit to clear BiosOwnAoacReg.                                                                                                                                                                                                                                                                                               |
| 24   | <b>set_biosownaoacreg.</b> Read-write. Reset: 0. Write only. BIOS write 1 to this bit to set BiosOwnAoacReg.                                                                                                                                                                                                                                                                                                 |
| 23:0 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                    |

**AOACx000000FC (FCH::AOAC::AOAC\_DEBUG)**

Read-write. Reset: 0080\_0000h.

\_aliasHOSTLEGACY; AOACx000000FC; AOAC=FED8\_1E00h

| Bits  | Description                                                             |
|-------|-------------------------------------------------------------------------|
| 31:26 | Reserved.                                                               |
| 25    | <b>shdwm_aclk_enable.</b> Read-write. Reset: 0. Read only               |
| 24    | <b>shdwm_any_request.</b> Read-write. Reset: 0. Read only               |
| 23    | <b>shdwm_sram_awake.</b> Read-write. Reset: 1. Read only                |
| 22    | <b>shdwm_shadowclkon_s.</b> Read-write. Reset: 0. Read only             |
| 21    | <b>shdwm_shadowinprogress_s.</b> Read-write. Reset: 0. Read only        |
| 20    | <b>shdwm_ahbm32_aclk_enbf.</b> Read-write. Reset: 0. Read only          |
| 19    | <b>shdwm_ahbm32_aclk_done.</b> Read-write. Reset: 0. Read only          |
| 18    | <b>shdwm_ahbm32_aclk_req.</b> Read-write. Reset: 0. Read only           |
| 17:13 | <b>shdwm_ctrlrid_4_0.</b> Read-write. Reset: 00h. Read only             |
| 12:8  | <b>shdwm_state_4_0.</b> Read-write. Reset: 00h. Read only               |
| 7     | <b>power_saving_mode_xtalpciclk_en.</b> Read-write. Reset: 0. Read only |
| 6:4   | <b>power_saving_mode_pd_st_2_0.</b> Read-write. Reset: 0h. Read only    |
| 3:0   | <b>power_saving_mode_state_3_0.</b> Read-write. Reset: 0h. Read only    |

## 14.3.12 SMBUS Host Controller

## 14.3.12.1 Device 14h Function 0 (SMBus) Configuration Registers

**SMBUSPCIx00000000 (FCH::SMBUSPCI::VENDORID)**

Read-only. Reset: 1022h.

\_aliasHOSTLEGACY; SMBUSPCIx00000000; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>vendorid.</b> Read-only. Reset: 1022h. Vendor Identifier. The vendor ID is 0x1022. This is the former ATI vendor ID which is now owned by AMD. AMD officially has two vendor IDs. |

**SMBUSPCIx00000002 (FCH::SMBUSPCI::DEVICEID)**

Read-only. Reset: 790Bh.

\_aliasHOSTLEGACY; SMBUSPCIx00000002; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>deviceid.</b> Read-only. Reset: 790Bh. Device Identifier. This 16-bit field is assigned by the device manufacturer and identifies the type of device. The device ID is selected by internal e-fuses. |

**SMBUSPCIx00000004 (FCH::SMBUSPCI::COMMAND)**

Read-write. Reset: 0403h.

\_aliasHOSTLEGACY; SMBUSPCIx00000004; SMBUSPCI=FED8\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:10 | Reserved.                                                                                                                                                                                                                                                                                                             |
| 9     | <b>fast_back_to_back_enable.</b> Read-write. Reset: 0. This bit indicates whether device is fast back-to-back capable. ACPI/Smbus does not support this function and so this bit is always 0. [Read-only]                                                                                                             |
| 8     | <b>serr_enable.</b> Read-write. Reset: 0. This bit is an enable bit for SERR# driver. A value of 0 disables the SERR# and a value of 1 enables it.                                                                                                                                                                    |
| 7     | <b>wait_cycle_control.</b> Read-write. Reset: 0. This bit is used to control whether or not a device does address/data stepping. This module does not use address stepping and so the value is always 0. [Read-only]                                                                                                  |
| 6     | <b>parity_error_response.</b> Read-write. Reset: 0. This bit controls the device's response to parity errors. When the bit is set, the device must take its normal action when a parity error is detected. When the bit is 0, the device must ignore any parity errors that it detects and continue normal operation. |
| 5     | <b>vga_palette_snoop.</b> Read-write. Reset: 0. This bit controls how VGA compatible and graphics devices handle accesses to VGA palette registers. This does not apply to this module and so the bit is always 0. [Read-only]                                                                                        |
| 4     | <b>memory_write_and_invalidate_enable.</b> Read-write. Reset: 0. This bit is an enable bit for using the Memory Write and Invalidate command. This module will not generate this command and so the bit is always 0. [Read-only]                                                                                      |
| 3     | <b>special_cycle.</b> Read-write. Reset: 0. A value of 0 causes the devices to ignore all special cycle operations. A value of 1 allows the device to monitor Special Cycle operations. This module does not respond to special cycle and so the bit is hardcoded to 0                                                |
| 2     | <b>bus_master.</b> Read-write. Reset: 0. A value of 0 disables the device from generating PCI accesses. A value of 1 allows it to behave as a bus master. ACPI/SMBus does not have PCI master and so the bit is always 0. [Read-only]                                                                                 |
| 1     | <b>memory_space.</b> Read-write. Reset: 1. This bit controls a device's response to memory space accesses. A value of 1 enables it and a value of 0 disables it. Since this module does claim certain memory cycles if BIOS is strapped to the PCI bus, this bit's default value 1.                                   |
| 0     | <b>i_o_space.</b> Read-write. Reset: 1. This bit controls a device's response to I/O space accesses. A value of 1 enables it and a value of 0 disables it. Since this module does claim certain legacy IO cycles, this bit's default value is 1.                                                                      |

**SMBUSPCIx00000006 (FCH::SMBUSPCI::STATUS)**

Read-write. Reset: 0220h.

\_aliasHOSTLEGACY; SMBUSPCIx00000006; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | <b>detected_parity_error.</b> Read-write. Reset: 0. This bit is set by the device whenever it detects a parity error, even if parity error handling is disabled.                                                                           |
| 14   | <b>signaled_system_error.</b> Read-write. Reset: 0. This bit is set by the device whenever the device asserts SERR#.                                                                                                                       |
| 13   | <b>received_master_abort.</b> Read-write. Reset: 0. This bit is set by a slave device whenever it terminates its transaction with Master-Abort.                                                                                            |
| 12   | <b>received_target_abort.</b> Read-write. Reset: 0. This bit is set by a master device whenever its transaction is terminated with a Target-Abort.                                                                                         |
| 11   | <b>signaled_target_abort.</b> Read-write. Reset: 0. This bit is set by a slave device whenever it terminates a cycle with a Target-Abort.                                                                                                  |
| 10:9 | <b>devsel_timing.</b> Read-write. Reset: 1h. These bits encode the timing of DEVSEL#. This module will always respond in medium timing and so these bits are always 11.                                                                    |
| 8    | <b>data_parity_error_detected.</b> Read-write. Reset: 0. Set to 1 if the Parity Error Response bit is set and the module has detected PERR# asserted while acting as a PCI master (regardless of whether PERR# was driven by this module). |
| 7    | <b>fast_back_to_back_capable.</b> Read-write. Reset: 0. This bit indicates whether the device is capable of fast back-to-back cycles. This module does not support this feature and so the bit is always 0. [Read-only]                    |
| 6    | <b>udf_supported.</b> Read-write. Reset: 0. This bit indicates whether the device supports user-definable feature. This module does not support this feature and so the bit is always 0. [Read-only]                                       |
| 5    | <b>capable_66mhz.</b> Read-write. Reset: 1. This bit indicates whether the device can support 66 MHz. This device is 66 MHz capable. [Read-only]                                                                                           |
| 4    | <b>msi_mapping_capability.</b> Read-write. Reset: 0. [Read-only] This bit indicates whether the device can support MSI mapping.                                                                                                            |
| 3:0  | Reserved.                                                                                                                                                                                                                                  |

**SMBUSPCIx00000008 (FCH::SMBUSPCI::REVISION\_ID\_CLASS\_CODE)**

Read-only. Reset: 0C05\_0071h.

\_aliasHOSTLEGACY; SMBUSPCIx00000008; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                        |
|------|------------------------------------------------------------------------------------|
| 31:8 | <b>class_code.</b> Read-only. Reset: 0C_0500h. 0C0500h denotes a SMBUS controller. |
| 7:0  | <b>revisionid.</b> Read-only. Reset: 71h. Revision ID                              |

**SMBUSPCIx0000000C (FCH::SMBUSPCI::CACHE\_LINE\_SIZE)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSPCIx0000000C; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>cache_line_size.</b> Read-only. Reset: 00h. This register specifies the system cacheline size. This module does not use Memory Write and Invalidate command and so this register is not applicable. It is hardcoded to 0. |

**SMBUSPCIx0000000D (FCH::SMBUSPCI::LATENCY\_TIMER)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSPCIx0000000D; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>latency_timer.</b> Read-only. Reset: 00h. This register specifies the value of the Latency Timer. This is not used in this module and so it is always 0. |

**SMBUSPCIx0000000E (FCH::SMBUSPCI::HEADER\_TYPE)**

Read-only. Reset: 80h.

\_aliasHOSTLEGACY; SMBUSPCIx0000000E; SMBUSPCI=FED8\_0000h

**Bits Description**7:0 **header\_type**. Read-only. Reset: 80h. This device is a multifunction device.**SMBUSPCIx0000000F (FCH::SMBUSPCI::BIST)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSPCIx0000000F; SMBUSPCI=FED8\_0000h

**Bits Description**7:0 **bist**. Read-only. Reset: 00h. The module has no built-in self-test and so this is always 0.**SMBUSPCIx00000010 (FCH::SMBUSPCI::BASE\_ADDRESS\_0)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMBUSPCIx00000010; SMBUSPCI=FED8\_0000h

**Bits Description**31:0 **base\_address\_0**. Read-only. Reset: 0000\_0000h. Not used and is hardcoded to 0.**SMBUSPCIx00000014 (FCH::SMBUSPCI::BASE\_ADDRESS\_1)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMBUSPCIx00000014; SMBUSPCI=FED8\_0000h

**Bits Description**31:0 **base\_address\_1**. Read-only. Reset: 0000\_0000h. Not used and is hardcoded to 0.**SMBUSPCIx00000018 (FCH::SMBUSPCI::BASE\_ADDRESS\_2)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMBUSPCIx00000018; SMBUSPCI=FED8\_0000h

**Bits Description**31:0 **base\_address\_2**. Read-only. Reset: 0000\_0000h. Not used and is hardcoded to 0.**SMBUSPCIx0000001C (FCH::SMBUSPCI::BASE\_ADDRESS\_3)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMBUSPCIx0000001C; SMBUSPCI=FED8\_0000h

**Bits Description**31:0 **base\_address\_3**. Read-only. Reset: 0000\_0000h. Not used and is hardcoded to 0.**SMBUSPCIx00000020 (FCH::SMBUSPCI::BASE\_ADDRESS\_4)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMBUSPCIx00000020; SMBUSPCI=FED8\_0000h

**Bits Description**31:0 **base\_address\_4**. Read-only. Reset: 0000\_0000h. Not used and is hardcoded to 0.**SMBUSPCIx00000024 (FCH::SMBUSPCI::BASE\_ADDRESS\_5)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMBUSPCIx00000024; SMBUSPCI=FED8\_0000h

**Bits Description**31:0 **base\_address\_5**. Read-only. Reset: 0000\_0000h. Not used and is hardcoded to 0.**SMBUSPCIx00000028 (FCH::SMBUSPCI::CARDBUS\_CIS\_POINTER)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMBUSPCIx00000028; SMBUSPCI=FED8\_0000h

**Bits Description**31:0 **cardbus\_cis\_pointer**. Read-only. Reset: 0000\_0000h. Not used and is hardcoded to 0.

**SMBUSPCIx0000002C (FCH::SMBUSPCI::SUBSYSTEM\_VENDOR\_ID)**

Write-only. Reset: 1022h.

\_aliasHOSTLEGACY; SMBUSPCIx0000002C; SMBUSPCI=FED8\_0000h

| Bits | Description                                                        |
|------|--------------------------------------------------------------------|
| 15:0 | <b>subsystem_vendor_id</b> . Write-only. Reset: 1022h. Write once. |

**SMBUSPCIx0000002E (FCH::SMBUSPCI::SUBSYSTEM\_ID)**

Write-only. Reset: 790Bh.

\_aliasHOSTLEGACY; SMBUSPCIx0000002E; SMBUSPCI=FED8\_0000h

| Bits | Description                                                 |
|------|-------------------------------------------------------------|
| 15:0 | <b>subsystem_id</b> . Write-only. Reset: 790Bh. Write once. |

**SMBUSPCIx00000030 (FCH::SMBUSPCI::EXPANSION\_ROM\_BASE\_ADDRESS)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSPCIx00000030; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                |
|------|--------------------------------------------------------------------------------------------|
| 7:0  | <b>expansion_rom_base_address</b> . Read-only. Reset: 00h. Not used and is hardcoded to 0. |

**SMBUSPCIx00000034 (FCH::SMBUSPCI::CAPABILITY\_POINTER)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSPCIx00000034; SMBUSPCI=FED8\_0000h

| Bits | Description                                                             |
|------|-------------------------------------------------------------------------|
| 7:0  | <b>capability_pointer</b> . Read-only. Reset: 00h. Default value is 00h |

**SMBUSPCIx0000003C (FCH::SMBUSPCI::INTERRUPT\_LINE)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSPCIx0000003C; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>interrupt_line</b> . Read-only. Reset: 00h. This module does not generate interrupt. This register is hardcoded to 0. |

**SMBUSPCIx0000003D (FCH::SMBUSPCI::INTERRUPT\_PIN)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSPCIx0000003D; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>interrupt_pin</b> . Read-only. Reset: 00h. This register specifies which interrupt pin the device issues. This module does not generate interrupt but contains the actual interrupt controller. This register is hardcoded to 0. |

**SMBUSPCIx0000003E (FCH::SMBUSPCI::MIN\_GNT)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSPCIx0000003E; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>min_gnt</b> . Read-only. Reset: 00h. This register specifies the desired settings for Latency Timer values. Value of 0 indicates that the device has no major requirements for the setting. This register is hardcoded to 0. |

**SMBUSPCIx0000003F (FCH::SMBUSPCI::MAX\_LAT)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSPCIx0000003F; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>max_lat</b> . Read-only. Reset: 00h. This register specifies the desired settings for Latency Timer values. Value of 0 indicates that the device has no major requirements for the setting. This register is hardcoded to 0. |



**SMBUSPCIx00000040 (FCH::SMBUSPCI::HCECONTROL)**

Read-write.

\_aliasHOSTLEGACY; SMBUSPCIx00000040; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:9 | Reserved.                                                                                                                                                                                                                                                                                                                                        |
| 8    | <b>a20state.</b> Read-write. Reset: 0.<br><b>Description:</b> Indicates current state of Gate A20 on keyboard controller. Used to compare against value written to 60h when GateA20Sequenceis active.                                                                                                                                            |
| 7    | <b>irq12active.</b> Read-write. Reset: 0.<br><b>Description:</b> Indicates that a positive transition on IRQ12 from the keyboard controller has occurred. Software may write a 1 to this bit to clear it (set it to 0). Software write of a 0 to this bit has no effect.                                                                         |
| 6    | <b>irq1active.</b> Read-write. Reset: 0.<br><b>Description:</b> Indicates that a positive transition on IRQ1 from the keyboard controller has occurred. Software may write a 1 to this bit to clear it (set it to 0). Software write of a 0 to this bit has no effect.                                                                           |
| 5    | <b>gatea20sequence.</b> Read-write. Reset: 0.<br><b>Description:</b> Set by HC when a data value of D1h is written to I/O port 64h. Cleared by HC on write to I/O port 64h of any value other than D1h.                                                                                                                                          |
| 4    | <b>externalirqen.</b> Read-write. Reset: 0.<br><b>Description:</b> When set to 1, IRQ1 and IRQ12 from the keyboard controller causes an emulation interrupt. The function controlled by this bit is independent of the setting of the EmulationEnablebit in this register.                                                                       |
| 3    | <b>irqen.</b> Read-write. Reset: 0.<br><b>Description:</b> When set, the HC generates IRQ1 or IRQ12 as long as the OutputFullbit in HceStatusis set to 1. If the AuxOutputFullbit of HceStatusis 0, then IRQ1 is generated if it is 1, then an IRQ12 is generated.                                                                               |
| 2    | <b>characterpending.</b> Read-write. Reset: 0.<br><b>Description:</b> When set, an emulation interrupt is generated when the OutputFullbit of the HceStatusregister is set to 0.                                                                                                                                                                 |
| 1    | <b>emulationinterrupt.</b> Read-write. Reset: X.<br><b>Description:</b> Read-only. This bit is a static decode of the emulation interrupt condition.                                                                                                                                                                                             |
| 0    | <b>emulationenable.</b> Read-write. Reset: 0.<br><b>Description:</b> When set to 1, the HC is enabled for legacy emulation. The HC decodes accesses to I/O registers 60h and 64h and generates IRQ1 and/or IRQ12 when appropriate. Additionally, the HC generate s an emulation interrupt at appropriate times to invoke the emulation software. |

**SMBUSPCIx00000044 (FCH::SMBUSPCI::HCEINPUT)**

Read-write.

\_aliasHOSTLEGACY; SMBUSPCIx00000044; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                                                              |
| 7:0  | <b>inputdata.</b> Read-write. Reset: 00h. This register holds data that is written to I/O ports 60h and 64h when emulation is enabled. |

**SMBUSPCIx00000048 (FCH::SMBUSPCI::HCEOUTPUT)**

Read-write.

\_aliasHOSTLEGACY; SMBUSPCIx00000048; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                                                                                                                                                                       |
| 7:0  | <b>outputdata</b> . Read-write. Reset: 00h. The data placed in this register by the emulation software is returned when I/O port 60h is read and emulation is enabled. On a read of this location, the OutputFull bit in HceStatus is set to 0. |

**SMBUSPCIx0000004C (FCH::SMBUSPCI::HCESTATUS)**

Read-write.

\_aliasHOSTLEGACY; SMBUSPCIx0000004C; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 7    | <b>parity</b> . Read-write. Reset: 0. Indicates parity error on keyboard/mouse data.                                                                                                                                                                                                                                                                                                                                             |
| 6    | <b>time_out</b> . Read-write. Reset: 0. Used to indicate a time-out                                                                                                                                                                                                                                                                                                                                                              |
| 5    | <b>auxoutputfull</b> . Read-write. Reset: 0. IRQ12 is asserted whenever this bit is set to 1 and OutputFull is set to 1 and the IRQEn bit is set.                                                                                                                                                                                                                                                                                |
| 4    | <b>inhibit_switch</b> . Read-write. Reset: 0. This bit reflects the state of the keyboard inhibit switch and is set if the keyboard is NOT inhibited.                                                                                                                                                                                                                                                                            |
| 3    | <b>cmddata</b> . Read-write. Reset: 0. The HC sets this bit to 0 on an I/O write to port 60h and to 1 on an I/O write to port 64h.                                                                                                                                                                                                                                                                                               |
| 2    | <b>flag</b> . Read-write. Reset: 0. Nominally used as a system flag by software to indicate a warm or cold boot.                                                                                                                                                                                                                                                                                                                 |
| 1    | <b>inputfull</b> . Read-write. Reset: 0. Except for the case of a Gate A20 sequence, this bit is set to 1 on an I/O write to address 60h or 64h. While this bit is set to 1 and emulation is enabled, an emulation interrupt condition exists.                                                                                                                                                                                   |
| 0    | <b>outputfull</b> . Read-write. Reset: 0. The HC sets this bit to 0 on a read of I/O port 60h. If IRQEn is set and AuxOutputFull is set to 0, then an IRQ1 is generated as long as this bit is set to 1. If IRQEn is set and AuxOutputFull is set to 1, then an IRQ12 is generated as long as this bit is set to 1. While this bit is 0 and CharacterPending in HceControl is set to 1, an emulation interrupt condition exists. |

**SMBUSPCIx00000050 (FCH::SMBUSPCI::HCEINTREN)**

Read-write.

\_aliasHOSTLEGACY; SMBUSPCIx00000050; SMBUSPCI=FED8\_0000h

| Bits  | Description                                                                                                                                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:11 | Reserved.                                                                                                                                                                                                     |
| 10    | <b>a20state_inv</b> . Read-write. Reset: 0. Set to 1 to use inverted version of A20State                                                                                                                      |
| 9:8   | <b>sofsrc</b> . Read-write. Reset: 0h.<br><b>Description:</b> Indicate which HCI_MSofN source used in USB Legacy.<br>2'b00: HCI_MSofN[0]<br>2'b01: HCI_MSofN[1]<br>2'b10: HCI_MSofN[2]<br>2'b11: HCI_MSofN[3] |
| 7:5   | <b>sofcounter</b> . Read-write. Reset: 5h. The amount of SOF number that emulation interrupt has to wait before <a href="#">SMI</a> is sent.                                                                  |
| 4     | <b>emulationsmien</b> . Read-write. Reset: 0. Set to 1 to generate <a href="#">SMI</a> if emulation interrupt happens                                                                                         |
| 3     | <b>blockirq12</b> . Read-write. Reset: 0. Set to 1 to block IRQ12                                                                                                                                             |
| 2     | <b>blockirq1</b> . Read-write. Reset: 0. Set to 1 to block IRQ1                                                                                                                                               |
| 1     | <b>enirq12</b> . Read-write. Reset: 1. Set to 1 to enable IRQ12Active                                                                                                                                         |
| 0     | <b>enirq1</b> . Read-write. Reset: 1. Set to 1 to enable IRQ1Active                                                                                                                                           |

**SMBUSPCIx000000FC (FCH::SMBUSPCI::UARTCONTROLREG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SMBUSPCIx000000FC; SMBUSPCI=FED8\_0000h

| Bits | Description                                                                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>uart_1p843mclk_en3.</b> Read-write. Reset: 0.<br><b>Description:</b> UART-3 Baud Rate control,<br>1: enable UART legacy mode to use 1.843MHz as baud rate clock<br>0: use 48MHz as UART baud rate clock     |
| 30   | <b>uart_1p843mclk_en2.</b> Read-write. Reset: 0.<br><b>Description:</b> UART-2 Baud Rate control,<br>1: enable UART legacy mode to use 1.843MHz as baud rate clock<br>0: use 48MHz as UART baud rate clock     |
| 29   | <b>uart_1p843mclk_en1.</b> Read-write. Reset: 0.<br><b>Description:</b> UART-1 Baud Rate control<br>1: enable UART legacy mode to use 1.843MHz as baud rate clock<br>0: use 48MHz as UART baud rate clock      |
| 28   | <b>uart_1p843mclk_en0.</b> Read-write. Reset: 0.<br><b>Description:</b> UART-0 Baud Rate control,<br>1: enable UART legacy mode to use 1.843MHz as baud rate clock<br>0: use 48MHz as UART baud rate clock     |
| 27   | <b>uart_legacy_pw_fcr_dis3.</b> Read-write. Reset: 0.<br><b>Description:</b> UART-3 FCR register write control,<br>1: enable UART legacy mode, disable write to FCR register<br>0: FCR register can be written |
| 26   | <b>uart_legacy_pw_fcr_dis2.</b> Read-write. Reset: 0.<br><b>Description:</b> UART-2 FCR register write control,<br>1: enable UART legacy mode, disable write to FCR register<br>0: FCR register can be written |
| 25   | <b>uart_legacy_pw_fcr_dis1.</b> Read-write. Reset: 0.<br><b>Description:</b> UART-1 FCR register write control,<br>1: enable UART legacy mode, disable write to FCR register<br>0: FCR register can be written |
| 24   | <b>uart_legacy_pw_fcr_dis0.</b> Read-write. Reset: 0.<br><b>Description:</b> UART-0 FCR register write control,<br>1: enable UART legacy mode, disable write to FCR register<br>0: FCR register can be written |
| 23:0 | Reserved.                                                                                                                                                                                                      |

**14.3.12.2 ASF (Alert Standard Format) Registers**

The ASF registers range from FED8\_0000h+900h to FED8\_0000h+9FFh. See FCH::PM::ISACONTROL[mmioen].

**ASFx00000000 (FCH::ASF::HOSTSTATUS)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000000; ASF=FED8\_0900h

| Bits | Description                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>lastbyte.</b> Read-only. Reset: 0.<br><b>Description:</b> 0: Last byte has not received<br>1: Last byte has received<br>This bit is "write one clear".                                                                                 |
| 6:5  | Reserved.                                                                                                                                                                                                                                 |
| 4    | <b>pecerror.</b> Read-only. Reset: 0.<br><b>Description:</b> 0: No CRC error<br>1: CRC error happens<br>This bit is "write one clear".                                                                                                    |
| 3    | <b>buscollision.</b> Read-only. Reset: 0.<br><b>Description:</b> 0: No bus collision<br>1: Bus collision<br>This bit is "write one clear".                                                                                                |
| 2    | <b>deverror.</b> Read-only. Reset: 0.<br><b>Description:</b> 0: Slave device behave correctly<br>1: No ACK or Slave device responses incorrectly<br>This bit is "write one clear".                                                        |
| 1    | <b>intr.</b> Read-only. Reset: 0.<br><b>Description:</b> Whenever HostBusy is cleared, this bit will be set.<br>This bit is "write one clear".                                                                                            |
| 0    | <b>hostbusy.</b> Read-only. Reset: 0.<br><b>Description:</b> 0: SM bus host is idle<br>1: SM bus host is busy<br>This register is set and cleared by hardware only to report the status of the ASF Host. Software can only read this bit. |

**ASFx00000002 (FCH::ASF::HOSTCONTROL)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000002; ASF=FED8\_0900h

| Bits | Description                                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>pecenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: <a href="#">PEC</a> disable<br>1: PEC enable, enable CRC checking when ASF HC presents as SM master and SM slave.                       |
| 6    | <b>start.</b> Read-write. Reset: 0.<br><b>Description:</b> WO:<br>0: Always read 0 on reads<br>1: Writing 1 to initiate the command                                                                       |
| 5    | <b>pecappend.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: No <a href="#">PEC</a> append<br>1: Automatic PEC append. ASF HC calculates CRC code and appends to the tail of the data packets.       |
| 4:2  | <b>protocol.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Quick<br>001: Byte<br>010: Byte Data<br>011: Word Data<br>100: Process call<br>101: Block<br>110: Block write-Block read-process call |
| 1    | <b>killhost.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Enable SM master<br>1: Reset SM master                                                                                                   |
| 0    | Reserved.                                                                                                                                                                                                 |

**ASFx00000003 (FCH::ASF::HOSTCOMMAND)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000003; ASF=FED8\_0900h

| Bits | Description                                                                     |
|------|---------------------------------------------------------------------------------|
| 7:0  | <b>hostcommand.</b> Read-write. Reset: 00h. Command to be transmitted by master |

**ASFx00000004 (FCH::ASF::SLAVEADDRESS)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000004; ASF=FED8\_0900h

| Bits | Description                                                                 |
|------|-----------------------------------------------------------------------------|
| 7:1  | <b>address.</b> Read-write. Reset: 00h. Provide the SM address of Slave     |
| 0    | <b>rw.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Write<br>1: Read |

**ASFx00000005 (FCH::ASF::DATA0)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000005; ASF=FED8\_0900h

| Bits | Description                                                                        |
|------|------------------------------------------------------------------------------------|
| 7:0  | <b>data0.</b> Read-write. Reset: 00h. Contains count or DATA0 field of transaction |

**ASFx00000006 (FCH::ASF::DATA1)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000006; ASF=FED8\_0900h

| Bits | Description                                                                |
|------|----------------------------------------------------------------------------|
| 7:0  | <b>data1</b> . Read-write. Reset: 00h. Contains DATA1 field of transaction |

**ASFx00000007 (FCH::ASF::DATAINDEX)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000007; ASF=FED8\_0900h

| Bits | Description                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>dataindex</b> . Read-write. Reset: 00h. It is mapped to 72 data registers in data buffer for block write/read and BlockWrite-BlockRead process call. |

**ASFx00000008 (FCH::ASF::PEC)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000008; ASF=FED8\_0900h

| Bits | Description                                                                        |
|------|------------------------------------------------------------------------------------|
| 7:0  | <b>pec</b> . Read-write. Reset: 00h. <a href="#">PEC</a> byte to be sent to slave. |

**ASFx00000009 (FCH::ASF::LISTENADR)**

Read-write. Reset: 40h.

\_aliasHOSTLEGACY; ASFx00000009; ASF=FED8\_0900h

| Bits | Description                                                                                                                                                                                                     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | <b>listenadr</b> . Read-write. Reset: 20h. The slave address which ASF slave response as ListenMode.                                                                                                            |
| 0    | <b>listenadren</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Enable ListenMode when the slave address equals to ListenAdr[7:1]<br>0: Disable ListenMode when the slave address equals to ListenAdr[7:1] |

**ASFx0000000A (FCH::ASF::ASFSTATUS)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx0000000A; ASF=FED8\_0900h

| Bits | Description                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>slavebusy</b> . Read-write. Reset: 0. Indicate if ASF slave is receiving data                                                                                                     |
| 6    | <b>slaveintr</b> . Read-write. Reset: 0. ASF Slave interrupt Status. Can be cleared to 0 by writing 1.                                                                               |
| 5:4  | Reserved.                                                                                                                                                                            |
| 3    | <b>remotepowercycle</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Power cycle has been triggered by ASF.<br>0: No power cycle ASF event<br>Can be cleared to 0 by writing 1. |
| 2    | <b>remotepowerup</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Power up has been triggered by ASF.<br>0: No Power up ASF event<br>Can be cleared to 0 by writing 1.          |
| 1    | <b>remotepowerdown</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Power down has been triggered by ASF<br>0: no Power down ASF event<br>Can be cleared to 0 by writing 1.     |
| 0    | <b>remoterest</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Reset has been triggered by ASF<br>0: no Reset cycle ASF event<br>Can be cleared to 0 by writing 1.              |

**ASFx0000000B (FCH::ASF::STATUSMASK0)**

Read-write. Reset: FFh.

\_aliasHOSTLEGACY; ASFx0000000B; ASF=FED8\_0900h

| Bits | Description                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>fanspeed2statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report Fan2 Speed Status to ASF<br>0: No report |
| 6    | <b>fanspeed1statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report Fan1 Speed Status to ASF<br>0: No report |
| 5    | <b>fanspeed0statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report Fan0 Speed Status to ASF<br>0: No report |
| 4    | <b>amdsistatusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report AMDSI status to ASF<br>0: No report          |
| 3    | <b>temp3statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report Temp3 status to ASF<br>0: No report          |
| 2    | <b>temp2statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report Temp2 status to ASF<br>0: No report          |
| 1    | <b>temp1statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report Temp1 status to ASF<br>0: No report.         |
| 0    | <b>temp0statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report Temp0 status to ASF<br>0: No report          |

**ASFx0000000C (FCH::ASF::STATUSMASK1)**

Read-write. Reset: FFh.

\_aliasHOSTLEGACY; ASFx0000000C; ASF=FED8\_0900h

| Bits | Description                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>analogio7statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report AnalogIo7 status to ASF<br>0: No report  |
| 6    | <b>analogio6statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report AnalogIo6 status to ASF<br>0: No report  |
| 5    | <b>analogio5statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report AnalogIo5 status to ASF<br>0: No report  |
| 4    | <b>analogio4statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report AnalogIo4 status to ASF<br>0: No report. |
| 3    | <b>analogio3statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report AnalogIo3 status to ASF<br>0: No report  |
| 2    | <b>analogio2statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report AnalogIo2 status to ASF<br>0: No report  |
| 1    | <b>analogio1statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report AnalogIo1 status to ASF<br>0: No report. |
| 0    | <b>analogio0statusenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Report AnalogIo0 status to ASF<br>0: No report  |



**ASFx0000000D (FCH::ASF::SLAVESTATUS)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx0000000D; ASF=FED8\_0900h

| Bits | Description                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | Reserved.                                                                                                                                                           |
| 3    | <b>wrongsp.</b> Read-write. Reset: 0.<br><b>Description:</b> RW<br>0: No SP error<br>1: No SP symbol is detected when bus turns to read<br>Writing 1 to clear to 0. |
| 2    | <b>slavedeverror.</b> Read-write. Reset: 0.<br><b>Description:</b> RW<br>0: Expected response<br>1: Unexpected response<br>Writing 1 to clear to 0.                 |
| 1    | <b>slavebuscollision.</b> Read-write. Reset: 0.<br><b>Description:</b> RW<br>0: No BusCollision<br>1: BusCollision happens<br>Writing 1 to clear to 0.              |
| 0    | <b>slavepecerror.</b> Read-write. Reset: 0.<br><b>Description:</b> RW<br>0: No <a href="#">PEC</a> error<br>1: PEC error<br>Writing 1 to clear to 0.                |

**ASFx0000000E (FCH::ASF::REMOTECTRLADR)**

Read-write. Reset: A8h.

\_aliasHOSTLEGACY; ASFx0000000E; ASF=FED8\_0900h

| Bits | Description                                                                       |
|------|-----------------------------------------------------------------------------------|
| 7:1  | <b>remotectladr.</b> Read-write. Reset: 54h. SM address of Remote Control device. |
| 0    | Reserved.                                                                         |

**ASFx0000000F (FCH::ASF::SENSORADR)**

Read-write. Reset: AAh.

\_aliasHOSTLEGACY; ASFx0000000F; ASF=FED8\_0900h

| Bits | Description                                                     |
|------|-----------------------------------------------------------------|
| 7:1  | <b>sensoradr.</b> Read-write. Reset: 55h. SM address of Sensor. |
| 0    | Reserved.                                                       |

**ASFx00000010 (FCH::ASF::DATAREADPOINTER)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000010; ASF=FED8\_0900h

| Bits | Description                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>datareadpointer.</b> Read-only. Reset: 00h. Current read pointer to the value specified in this register |

**ASFx00000011 (FCH::ASF::DATAWRITEPOINTER)**

Read-only. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000011; ASF=FED8\_0900h

| Bits | Description                                                                                                        |
|------|--------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>datawritepointer.</b> Read-only. Reset: 00h. Show current write pointer to the value specified in this register |

**ASFx00000012 (FCH::ASF::SETDATAREADPOINTER)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000012; ASF=FED8\_0900h

| Bits | Description                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>setdatareadpointer</b> . Read-write. Reset: 00h. Force the current write pointer to the value specified in this register |

**ASFx00000013 (FCH::ASF::DATABANKSEL)**

Read-write. Reset: 81h.

\_aliasHOSTLEGACY; ASFx00000013; ASF=FED8\_0900h

| Bits | Description                                                                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>setreadhostdatabank</b> . Read-write. Reset: 1.<br><b>Description:</b> 1: Select to read data from Host Data Bank.<br>0: Select to read data from Data Bank 0 or Data Bank 1 decided By SetReadRevDataBank bits.                                     |
| 6    | Reserved.                                                                                                                                                                                                                                               |
| 5:4  | <b>setreadrevdatabank</b> . Read-write. Reset: 0h.<br><b>Description:</b> 00: Select to read data from Data Bank 0.<br>01: Select to read data from Data Bank 1<br>10: Select to read data from Data Bank 1<br>11: Select to read data from Data Bank 1 |
| 3    | <b>databank1_full</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Data Bank 1 is free.<br>1: Data Bank 1 is full.<br>Writing 1 clears status.                                                                                                     |
| 2    | <b>databank0_full</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Data Bank 0 is free.<br>1: Data Bank 0 is full.<br>Writing 1 clears status.                                                                                                     |
| 1:0  | <b>databank</b> . Read-write. Reset: 1h.<br><b>Description:</b> Bit 0:<br>0: Data Bank 0 is the latest touched data bank.<br>1: Data Bank 1 is the latest touched data bank.<br>Bit 1:<br>0: Data Bank still has space.<br>1: Data Bank is now full.    |

**ASFx00000014 (FCH::ASF::SEMAPHORE)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000014; ASF=FED8\_0900h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 3    | <b>clrecsemaphore</b> . Read-write. Reset: 0. Write 1 to clear EcSemaphore bit. Writing 0 has no effect and read always returns 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2    | <b>ecsemaphore</b> . Read-write. Reset: 0. Write 1 to set this bit. This bit can only be set when HostSemaphore is clear. Writing 0 has no effect. Read returns the value of this bit. Software should never write both bit[2]=1 and bit[0]=1 at same time.                                                                                                                                                                                                                                                                                                                                                                                         |
| 1    | <b>clrhostsemaphore</b> . Read-write. Reset: 0. Write 1 to clear HostSemaphore bit. Write 0 has no effect and read always returns 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0    | <b>hostsemaphore</b> . Read-write. Reset: 0.<br><b>Description:</b> Bits [0] and [2] are meant to be used as software semaphore by 2 different agents, and they are named Host and Ec. When both host and <a href="#">EC</a> want to use the same resource, they can write to these semaphore bits first, then followed by a read. If the read returns a 1 in the semaphore bit, it means it has established the semaphore first.<br>Write 1 to set this bit. This bit can only be set when EcSemaphore is clear. Writing 0 has no effect. Read returns the value of this bit. Software should never write both bit[2]=1 and bit[0]=1 at same time. |

**ASFx00000015 (FCH::ASF::SLAVEEN)**

Read-write. Reset: 41h.

\_aliasHOSTLEGACY; ASFx00000015; ASF=FED8\_0900h

| Bits | Description                                                                                                                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>fairarben.</b> Read-write. Reset: 0.<br><b>Description:</b> RW<br>0: Disable Fair arbiter logic<br>1: Enable Fair Arbiter logic, which forces ASF master to give up <a href="#">SMBus</a> for a certain time, specified in register at offset 16h. |
| 6    | <b>tmrouten.</b> Read-write. Reset: 1.<br><b>Description:</b> RW<br>0: Disable timer out function<br>1: Enable timer out function                                                                                                                     |
| 5    | <b>legacysensoren.</b> Read-write. Reset: 0.<br><b>Description:</b> RW<br>0: Disable Legacy Sensor<br>1: Enable Legacy Sensor                                                                                                                         |
| 4    | <b>killslave.</b> Read-write. Reset: 0.<br><b>Description:</b> RW<br>Write 1 to reset Slave ASF Slave state machine                                                                                                                                   |
| 3    | <b>suspendslave.</b> Read-write. Reset: 0.<br><b>Description:</b> RW<br>Write 1 to Suspend (stop) ASF Slave state machine                                                                                                                             |
| 2    | <b>intruderalerttsen.</b> Read-write. Reset: 0.<br><b>Description:</b> RW:<br>0: FanSpeed2Status is returned<br>1: IntruderAlertSts is returned                                                                                                       |
| 1    | <b>slaveintrlistenen.</b> Read-write. Reset: 0. Set to 1 to allow ASF slave to generate slave interrupt when the address of received packet is the same as the one specified in ListenAdr register [ASF_IO:09h].                                      |
| 0    | <b>io_smb1_rxsel.</b> Read-write. Reset: 1.<br><b>Description:</b> For debug purpose only.<br>SW should always keep it at 1.                                                                                                                          |

**ASFx00000016 (FCH::ASF::DELAYMASTERTIMER)**

Read-write. Reset: 04h.

\_aliasHOSTLEGACY; ASFx00000016; ASF=FED8\_0900h

| Bits | Description                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fairarbtimer.</b> Read-write. Reset: 04h. Specify how long ASF master has to wait before submitting next packet.<br>Wait time = the value * 2us. |

**ASFx00000017 (FCH::ASF::DEBUG)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; ASFx00000017; ASF=FED8\_0900h

| Bits | Description                                                                                                                                                                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>asfdebouncecnten.</b> Read-write. Reset: 0.<br><b>Description:</b> This bit enable configurable debounce counter for ASF slave clock and data.<br>1: Use configurable debounce counter<br>0: Use simple 3-flop debounce logic                                                                                       |
| 6:4  | <b>asfdebouncecnt_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> This register is only used when AsfDebounceCntEn is 1. It configures the debounce length for ASF slave clock and data.<br>000: 60 ns<br>001: 120 ns<br>010: 180 ns<br>011: 240 ns<br>100: 300 ns<br>101: 360 ns<br>110: 420 ns<br>111: 480 ns |
| 3:2  | Reserved.                                                                                                                                                                                                                                                                                                              |
| 1    | <b>debouncebeforestretch.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: The ASF clock stretching logic checks the SMBUS clock from pad directly.<br>1: The ASF clock stretching logic checks the debounced version of SMBUS clock.                                                                               |
| 0    | <b>slavealwayschkpec.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                        |

**14.3.12.3 SMBUS Register**

The [SMBus](#) register space is accessed through two methods:

- IO access through SMBus IO base address:
  - Program the base address of SMBus IO space through FCH::PM::DECODEEN[smbusasfiobase].
  - Enable SMBus IO decoding through FCH::PM::DECODEEN[smbusasfioen].
- Direct memory mapped access through the AcpiMmio region.

The SMBus registers range from FED8\_0000h+A00h to FED8\_0000h+AFFh. See FCH::PM::ISACONTROL[mmioen].

The SMBus controller can operate up to 4 separate SMBus channels which are mux'd via a control in FCH::PM::DECODEEN[smbus0sel]. Software must set this mux prior to initiate bus commands.

Note: "Port 1" is missing in the list because it is the ASF bus which has its own control space.

The control register space for the SMBus control is enabled in FCH::PM::DECODEEN[smbusasfiobase], FCH::PM::DECODEEN[smbusasfioen] and FCH::PM::ISACONTROL[mmioen].

Software must set the value in FCH::PM::DECODEEN[smbusasfiobase], then set FCH::PM::DECODEEN[smbusasfioen] to gain access to the control IP spaces. Access can also be enabled via the ACPI [MMIO](#) space by setting FCH::PM::ISACONTROL[mmioen]

**SMBUSx00000000 (FCH::SMBUS::SMBUSSTATUS)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000000; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:5  | Reserved.                                                                                                                                                                                                                              |
| 4    | <b>failed.</b> Read-write. Reset: 0. This bit is set by hardware to indicate a failed bus transaction, set when SMBusControl. Kill bit is set. This bit is cleared by writing a 1 to it.                                               |
| 3    | <b>buscollision.</b> Read-write. Reset: 0. This bit is set by hardware to indicate an <a href="#">SMBus</a> transaction collision this bit can be cleared by writing a 1 to it.                                                        |
| 2    | <b>deviceerr.</b> Read-write. Reset: 0. This bit is set by hardware to indicate an error of one of the following: 1) illegal command field, 2) unclaimed cycle, 3) host device time-out. This bit can be cleared by writing a 1 to it. |
| 1    | <b>smbusinterrupt.</b> Read-write. Reset: 0. This bit is set by hardware to indicate the completion of the last host command. This bit can be cleared by writing a 1 to it.                                                            |
| 0    | <b>hostbusy.</b> Read-write. Reset: 0. This bit indicates the <a href="#">SMBus</a> controller is in the process of completing a command. When this bit is set, software should not access any other SMBus registers. [Read-only]      |

**SMBUSx00000001 (FCH::SMBUS::SMBUSSLAVESTATUS)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000001; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:6  | Reserved.                                                                                                                                                                                                                                         |
| 5    | <b>alertstatus.</b> Read-write. Reset: 0. This bit is set by hardware to indicate an SMBALERT_ signal. This function is not supported. [Read-only]                                                                                                |
| 4    | <b>shadow2status.</b> Read-write. Reset: 0. This bit is set by hardware to indicate a slave cycle address match of the SMB_Shadow2 port. This bit can be cleared by writing a 1 to it.                                                            |
| 3    | <b>shadow1status.</b> Read-write. Reset: 0. This bit is set by hardware to indicate a slave cycle address match of the SMB_Shadow1 port. This bit can be cleared by writing a 1 to it.                                                            |
| 2    | <b>slavestatus.</b> Read-write. Reset: 0. This bit is set by hardware to indicate a slave cycle event match of the <a href="#">SMBus</a> slave command and SMBus Slave Event match. This bit can be cleared by writing a 1 to it.                 |
| 1    | <b>slaveinit.</b> Read-write. Reset: 0. Writing a 1 to this bit will initialize the slave. It is unnecessary to write it back to 0. A read from it will always return a 0.                                                                        |
| 0    | <b>slavebusy.</b> Read-write. Reset: 0. This bit indicates the <a href="#">SMBus</a> controller slave interface is in the process of receiving data. Software should not try to access any other SMBus register when this bit is set. [Read-only] |

**SMBUSx00000002 (FCH::SMBUS::SMBUSCONTROL)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000002; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>reset_smbus.</b> Read-write. Reset: 0.<br><b>Description:</b> Write only.<br>Set the bit to 1 to stop smbus transaction and reset smbus controller state machine.                                                                                                          |
| 6    | <b>start.</b> Read-write. Reset: 0. Writing a 1 in this field initiates <a href="#">SMBus</a> controller host interface to execute the command programmed in the SMBusProtocol field.                                                                                         |
| 5    | Reserved.                                                                                                                                                                                                                                                                     |
| 4:2  | <b>smbusprotocol.</b> Read-write. Reset: 0h.<br><b>Description:</b> 000: Quick Command<br>001: Send Byte or Receive Byte<br>010: Write Byte or Read Byte<br>011: Write Word or Read Word<br>100: Reserved<br>101: Block Write or Block Read<br>110: Reserved<br>111: Reserved |
| 1    | <b>kill.</b> Read-write. Reset: 0. Stop the current host transaction in process.                                                                                                                                                                                              |
| 0    | <b>interruptenable.</b> Read-write. Reset: 0. Enable the generation of interrupts on the completion of current host transaction.                                                                                                                                              |

**SMBUSx00000003 (FCH::SMBUS::SMBUSHOSTCMD)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000003; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>smbushostcmd.</b> Read-write. Reset: 00h. This field contains the data transmitted in the command field of <a href="#">SMBus</a> host transaction. |

**SMBUSx00000004 (FCH::SMBUS::SMBUSADDRESS)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000004; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                           |
|------|-----------------------------------------------------------------------------------------------------------------------|
| 7:1  | <b>smbusaddr.</b> Read-write. Reset: 00h. This field contains the 7-bit address of the target slave device.           |
| 0    | <b>smbusrdwr.</b> Read-write. Reset: 0.<br><b>Description:</b> 0:Execute a Write command<br>1: Execute a Read command |

**SMBUSx00000005 (FCH::SMBUS::SMBUSDATA0)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000005; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>smbusdata0.</b> Read-write. Reset: 00h. This register should be programmed with a value to be transmitted in the data 0 field of an <a href="#">SMBus</a> host interface transaction. For Block Write commands, the count of the memory should be stored in this field. The value of this register is loaded into the block transfer count field. A valid value for block command count is between 1 and 32. For block reads, count received from SMBus device is stored here. |

**SMBUSx00000006 (FCH::SMBUS::SMBUSDATA1)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000006; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>smbusdata1.</b> Read-write. Reset: 00h.<br><b>Description:</b> This register should be programmed with a value to be transmitted in the data 1 field of an <a href="#">SMBus</a> host interface transaction.<br>When SMBusPoll2Byte is set, and SMBusProtocol = Byte Read/Write or Byte Data Read/Write, this register is used as the data field in the second transaction. |

**SMBUSx00000007 (FCH::SMBUS::SMBUSBLOCKDATA)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000007; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                        |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>smbusblockdata.</b> Read-write. Reset: 00h. This register is used to transfer data into or out of the block data storage array. |

**SMBUSx00000008 (FCH::SMBUS::SMBUSSLAVECONTROL)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000008; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>clrecsemaphore.</b> Read-write. Reset: 0. Write 1 to clear EcSemaphore bit. Writing 0 has no effect and reading always returns 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6    | <b>ecsemaphore.</b> Read-write. Reset: 0. Write 1 to set this bit. This bit can only be set when HostSemaphore is clear. Writing 0 has no effect. Reading returns the value of this bit. Software should never write both bit[6]=1 and bit[4]=1 at same time.                                                                                                                                                                                                                                                                                                                                                                             |
| 5    | <b>clrhostsemaphore.</b> Read-write. Reset: 0. Write 1 to clear HostSemaphore bit. Writing 0 has no effect and reading always returns 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 4    | <b>hostsemaphore.</b> Read-write. Reset: 0.<br><b>Description:</b> Bits 4 and 6 are meant to be used as software semaphore by 2 different agents, and they are named Host and Ec. When both host and <a href="#">EC</a> want to use the same resource, they can write to these semaphore bits first, followed by a read. If the read returns a 1 in the semaphore bit, it means it has established the semaphore first. Write 1 to set this bit. This bit can only be set when EcSemaphore is clear. Writing 0 has no effect. Reading returns the value of this bit. Software should never write both bit[6]=1 and bit[4]=1 at same time. |
| 3    | <b>smbusalertenable.</b> Read-write. Reset: 0. Enable the generation of an interrupt or resume event on the assertion of AMBALERT_ signal. (This function is not supported). [Read-only]                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 2    | <b>smbusshadow2en.</b> Read-write. Reset: 0. Enable the generation of an interrupt or resume event upon an external <a href="#">SMBus</a> master generating a transaction with an address that matches the SMBus Shadow 2 register.                                                                                                                                                                                                                                                                                                                                                                                                       |
| 1    | <b>smbusshadow1en.</b> Read-write. Reset: 0. Enable the generation of an interrupt or resume event upon an external <a href="#">SMBus</a> master generating a transaction with an address that matches the SMBus Shadow 1 register.                                                                                                                                                                                                                                                                                                                                                                                                       |
| 0    | <b>slaveenable.</b> Read-write. Reset: 0. Enable the generation of an interrupt or resume event upon an external <a href="#">SMBus</a> master generating a transaction with an address that matches the host controller slave port of 10h, a command field that matches the SMBus slave control register, and a match of corresponding enabled events.                                                                                                                                                                                                                                                                                    |

**SMBUSx00000009 (FCH::SMBUS::SMBUSSHADOWCMD)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000009; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>smbusshadowcmd.</b> Read-write. Reset: 00h. This field contains the command value that was received during an external <a href="#">SMBus</a> master access whose address field matched the host slave address (10h) or one of the slave shadow ports. |



**SMBUSx0000000A (FCH::SMBUS::SMBUSSLAVEEVENT)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; SMBUSx0000000A; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>smbusslaveevent.</b> Read-write. Reset: 0000h. This field contains data bits used to compare against incoming data to the <a href="#">SMBus</a> Slave Data register. When a bit in this register is 1 and a corresponding bit in SMBus Slave register is set, then an interrupt or resume event is generated if the command value matches the value in the SMBus slave control register and the access was to SMBus host address 10h. |

**SMBUSx0000000C (FCH::SMBUS::SLAVEDATA)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; SMBUSx0000000C; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>slavedata.</b> Read-write. Reset: 0000h. This field contains the data value which was transmitted during an external <a href="#">SMBus</a> master access whose address field matched one of the slave shadow port addresses or the SMBus host controller slave port address of 10h. |

**SMBUSx0000000E (FCH::SMBUS::SMBUSTIMING)**

Read-write. Reset: B0h.

\_aliasHOSTLEGACY; SMBUSx0000000E; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>smbustiming.</b> Read-write. Reset: B0h.<br><b>Description:</b> This register controls the frequency on the SMBUS. The formula to calculate the frequency is:<br>Frequency = 66MHz/(SmBusTiming * 4) |

**SMBUSx00000010 (FCH::SMBUS::I2CBUSCONFIG)**

Read-write. Reset: 00h.

I2CbusConfig register: Registers D2-D5 control the interface when this chip is the [I2C](#) slave.

\_aliasHOSTLEGACY; SMBUSx00000010; SMBUS=FED8\_0A00h

| Bits | Description                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------|
| 7:4  | <b>i2crevision.</b> Read-write. Reset: 0h. Sm bus controller revision                                     |
| 3:1  | Reserved.                                                                                                 |
| 0    | <b>i2cbusinterrupt.</b> Read-write. Reset: 0.<br><b>Description:</b> 0 : <a href="#">SMI</a> #<br>1 : IRQ |

**SMBUSx00000011 (FCH::SMBUS::I2CCOMMAND)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000011; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>i2ccommand.</b> Read-write. Reset: 00h. <a href="#">I2C</a> Host Slave Command this value specifies the command value to be matched for I2C master accesses to the I2Ccontroller host slave interface. |

**SMBUSx00000012 (FCH::SMBUS::I2CSHADOW1)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000012; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | <b>i2cslaveaddr1.</b> Read-write. Reset: 00h.<br><b>Description:</b> <a href="#">SMBus</a> Slave Address for shadow port 1<br>This value specifies the address used to match against incoming <a href="#">I2C</a> addresses for Shadow port 1. |
| 0    | <b>read_write_shadowport1.</b> Read-write. Reset: 0.<br><b>Description:</b> Read/Write for Shadow Port 1<br>This bit must be programmed to 0 because <a href="#">I2C</a> slave controller only responds to Word Write Transaction.             |

**SMBUSx00000013 (FCH::SMBUS::I2CSHADOW2)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000013; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | <b>i2cslaveaddr2.</b> Read-write. Reset: 00h.<br><b>Description:</b> <a href="#">SMBus</a> Slave Address for shadow port 2<br>This value specifies the address used to match against incoming <a href="#">I2C</a> addresses for Shadow port 2. |
| 0    | <b>read_write_shadowport2.</b> Read-write. Reset: 0.<br><b>Description:</b> Read/Write for Shadow Port 2<br>This bit must be programmed to 0 because <a href="#">I2C</a> slave controller only responds to Word Write Transaction.             |

**SMBUSx00000014 (FCH::SMBUS::SMBUSAUTOPOLL)**

Read-write. Reset: 84h.

\_aliasHOSTLEGACY; SMBUSx00000014; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>smbuspoll2byte.</b> Read-write. Reset: 1.<br><b>Description:</b> When set, if SMBusProtocol = Byte Read/Write or Byte Data Read/Write, the last command will be performed twice at each polling interval.<br>SMBusData1 contains data transmitted and retrieved from the second transaction.<br>SMBusHostCmd2 contains data transmitted in the command field in the second transaction. |
| 6:4  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                  |
| 3:1  | <b>smbuspollperiod.</b> Read-write. Reset: 2h.<br><b>Description:</b> 000b: poll every 1/16 second<br>001b: poll every 1/8 second<br>010b: poll every 1/ 4 second<br>011b: poll every 1/ 2 second<br>1xxb: poll every 1 second                                                                                                                                                             |
| 0    | <b>smbusautopollen.</b> Read-write. Reset: 0. When set, SMBUS will periodically execute the last command whenever SMBusCounter expires. The polling interval is defined by SMBusPollPeriod. The purpose of this function is to use the SMBUS to read CPU temperature via <a href="#">TSI</a> . Values in offset 0x06 and 0x05 are to be used for controlling the fan speed                 |

**SMBUSx00000015 (FCH::SMBUS::SMBUSCOUNTER)**

Read-only.

\_aliasHOSTLEGACY; SMBUSx00000015; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>smbuscounter.</b> Read-only. Reset: XXXXXXXXb. This counter shows the remaining time until the next read. The interval is based on SMBusPollPeriod. For example, if SMBusPollPeriod is 000b, then each tick in this counter represents 1/16 / 256 = 244us. |

**SMBUSx00000016 (FCH::SMBUS::SMBUSSTOP)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000016; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 0    | <b>smbuspausepoll.</b> Read-write. Reset: 0. This is to be used as a semaphore mechanism by software to stop the next polling. If SW wants to use the SMBUS for other purpose but the hardware has already been enabled (SMBusAutoPollEn=1) to perform auto polling, SW can halt the next polling by setting this bit to 1. If the halt is successful, this bit will return 1. If the HW has already started the polling operation at the same time SW tries to set this bit, this bit will return 0. When the pending polling operation is finally complete, this bit will then get set. When this bit is set, polling counter will temporarily stop (not reset). It is now safe for SW to access the SMBUS. Upon completion, SW must restore the previous command in the appropriate register prior to setting this bit. SW should also clear this bit so the counter can resume counting. |

**SMBUSx00000017 (FCH::SMBUS::SMBUSHOSTCMD2)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SMBUSx00000017; SMBUS=FED8\_0A00h

| Bits | Description                                                                                                                                                                                                        |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>smbushostcmd2.</b> Read-write. Reset: 00h. This field contains the data transmitted in the command field in the second command when SMBusPoll2Byte is set and SMBusProtocol = Byte Data Read or Write (3'b010). |

**14.3.13 ISA Bridge****14.3.13.1 TPM Block****14.3.13.1.1 TPM Control**

When the tpm\_amd bit (FCH::LPCPCICFG::TPM\_TRUSTED\_PLATFORM\_MODULE[tpm\_amd]) is 0, the processor only supports normal TPM cycles. Below are the mapping from system-software memory addresses to TPM address.

Table 253: System/Software Memory to TPM Address Mapping when tpm\_amd is 0

| System/software Memory Address | TPM Address |
|--------------------------------|-------------|
| FED4_0xxxh                     | 0xxxh       |
| FED4_1xxxh                     | 1xxxh       |
| FED4_2xxxh                     | 2xxxh       |
| FED4_3xxxh                     | 3xxxh       |
| FED4_4xxxh                     | 4xxxh       |

When the tpm\_amd bit (FCH::LPCPCICFG::TPM\_TRUSTED\_PLATFORM\_MODULE[tpm\_amd]) is 1, it only support following TPM cycles. Below are the mappings from system-software memory addresses to TPM address.

Table 254: System/Software Memory to TPM Address Mapping when tpm\_amd is 1

| System/software Memory Address | TPM Address |
|--------------------------------|-------------|
| FED4_0xxxh                     | 0xxxh       |
| FED4_1xxxh                     | 1xxxh       |
| FED4_2xxxh                     | 2xxxh       |

|                             |             |
|-----------------------------|-------------|
| FED4_3xxxh                  | 3xxxh       |
| FD_F923_FFFFh-FD_F920_0000h | 4028h       |
| FD_F928_0003h-FD_F928_0000h | 4020h       |
| FD_F928_0007h-FD_F928_0004h | 4027h-4024h |

#### 14.3.13.2 Device 14h Function 3 (SPI/eSPI Bridge) Configuration Registers

##### D14F3x000 (FCH::LPCPCICFG::VENDOR\_ID)

Read-only. Reset: 1022h.

\_aliasHOST; D14F3x000

| Bits | Description                                                                                                                                                                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>vendor_id</b> . Read-only. Reset: 1022h. Vendor Identifier. The vendor ID is 0x1022. This is the former ATI vendor ID which is now owned by AMD. AMD officially has two vendor IDs |

##### D14F3x002 (FCH::LPCPCICFG::DEVICE\_ID)

Read-only. Reset: 790Eh.

\_aliasHOST; D14F3x002

| Bits | Description                                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>device_id</b> . Read-only. Reset: 790Eh. Device Identifier. This 16-bit field is assigned by the device manufacturer and identifies the type of device. The device ID is selected by internal e-fuses. |

**D14F3x004 (FCH::LPCPCICFG::COMMAND)**

Read-write. Reset: 000Fh.

Command Register: The PCI specification defines this register to control a PCI device's ability to generate and respond to PCI cycles. Writes to this register, except bit 6, have no effect. Bits[3:0]=0fh and are read-only.

\_aliasHOST; D14F3x004

| Bits  | Description                                                                                                                                                                                                                              |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:10 | Reserved.                                                                                                                                                                                                                                |
| 9     | <b>fast_back_to_back_enable</b> . Read-write. Reset: 0. Fast Back-to-back Enable. The SB only acts as a master to a single device, so this functionality is not needed. This bit is always 0.                                            |
| 8     | <b>serr_enable</b> . Read-write. Reset: 0. SERR# Enable. If set to 1, the SB asserts SERR# when it detects an address parity error. SERR# is not asserted if this bit is 0.                                                              |
| 7     | <b>stepping_control</b> . Read-write. Reset: 0. Wait Cycle Enable. The SB does not need to insert a wait state between the address and data on the AD lines. This bit is always 0.                                                       |
| 6     | <b>parity_error_response</b> . Read-write. Reset: 0. PERR# (Response) Detection Enable bit. If set to 1, The SB asserts PERR# when it is the agent receiving data AND it detects a parity error. PERR# is not asserted if this bit is 0. |
| 5     | <b>vga_palette_snoop</b> . Read-write. Reset: 0. VGA Palette Snoop Enable. The SB (Southbridge) does not need to snoop VGA palette cycles. This bit is always 0.                                                                         |
| 4     | <b>memory_write_and_invalidate_enable</b> . Read-write. Reset: 0. Memory Write and Invalidate Enable. Not implemented. This bit is always 0.                                                                                             |
| 3     | <b>special_cycles</b> . Read-write. Reset: 1. Special Cycle Recognition Enable. This feature is not implemented and this bit is always 1.                                                                                                |
| 2     | <b>bus_master</b> . Read-write. Reset: 1.<br><b>Description:</b> Bus Master Enable.<br>1: Enable<br>0: Disable.                                                                                                                          |
| 1     | <b>memory_space</b> . Read-write. Reset: 1. Memory Access Enable. This function is not implemented. This bit is always 1.                                                                                                                |
| 0     | <b>io_space</b> . Read-write. Reset: 1. I/O Access Enable. This bit controls access to the I/O space registers. When this bit is 1, it enables access to the legacy IDE ports, and PCI bus master IDE I/O registers are enabled.         |

**D14F3x006 (FCH::LPCPCICFG::STATUS)**

Read-write. Reset: 0220h.

Status Register: The PCI specification defines this register to record status information for PCI related events. This is a read/write register. However, writes can only reset bits. A bit is reset when the register is written and the data in the corresponding bit location is a 1.

\_aliasHOST; D14F3x006

| Bits | Description                                                                                                                                                                                                                                            |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | <b>detected_parity_error</b> . Read-write. Reset: 0. Detected Parity Error. This bit is set to 1 when the SB detects a parity error.                                                                                                                   |
| 14   | <b>signaled_system_error</b> . Read-write. Reset: 0. SERR# status. This bit is set to 1 when the SB detects a PCI address parity error.                                                                                                                |
| 13   | <b>received_master_abort</b> . Read-write. Reset: 0. Received Master Abort Status. Set to 1 when the SB acts as a PCI master and aborts a PCI bus memory cycle. Cleared by writing a 1 to this bit.                                                    |
| 12   | <b>received_target_abort</b> . Read-write. Reset: 0. Received Target Abort. Set to 1 when an SB generated PCI cycle (the SB is the PCI master) is aborted by a PCI target. Cleared by writing a 1 to it.                                               |
| 11   | <b>signaled_target_abort</b> . Read-write. Reset: 0. Signaled Target Abort. Read-only, Set to 1 when the SB signals Target Abort.                                                                                                                      |
| 10:9 | <b>device_select_timing</b> . Read-write. Reset: 1h.<br><b>Description:</b> DEVSEL# Timing. Read-only. Indicates DEVSEL# timing when performing a positive decode. Since DEVSEL# is asserted to meet the medium timing, these bits are encoded as 01b. |
| 8    | <b>master_data_parity_error</b> . Read-write. Reset: 0. Data Parity Reported. Set to 1 if the SB detects PERR# asserted while acting as PCI master (whether PERR# was driven by the SB or not.)                                                        |
| 7:5  | Reserved.                                                                                                                                                                                                                                              |
| 4    | <b>capabilities_list</b> . Read-write. Reset: 0. Read-only. When reg0x78[1] (Msi On) is 1, this bit reads 1 when reg0x78[1] (Msi On) is 0, this bit reads 0.                                                                                           |
| 3:0  | Reserved.                                                                                                                                                                                                                                              |

**D14F3x008 (FCH::LPCPCICFG::REVISION\_ID\_CLASS\_CODE)**

Read-only. Reset: 0601\_0051h.

Revision ID/Class Code Register: This read only register contains the device's revision information and generic function. Since SB is an ISA bridge, its assigned class code is 060100h.

\_aliasHOST; D14F3x008

| Bits | Description                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | <b>class_code</b> . Read-only. Reset: 06_0100h. Class Code.                                                                    |
| 7:0  | <b>revision_id</b> . Read-only. Reset: 51h. These bits are hardwired to 11h to indicate the revision level of the chip design. |

**D14F3x00C (FCH::LPCPCICFG::CACHE\_LINE\_SIZE)**

Read-only. Reset: 00h.

Cache Line Size Register: This register specifies the system cache line size. This register is not implemented.

\_aliasHOST; D14F3x00C

| Bits | Description                                                      |
|------|------------------------------------------------------------------|
| 7:0  | <b>cache_line_size</b> . Read-only. Reset: 00h. Cache Line Size. |

**D14F3x00D (FCH::LPCPCICFG::LATENCY\_TIMER)**

Read-only. Reset: 00h.

Latency Timer Register: This register specifies the value of the Latency Timer in units of PCICLKs.

\_aliasHOST; D14F3x00D

| Bits | Description                                                  |
|------|--------------------------------------------------------------|
| 7:0  | <b>latency_timer</b> . Read-only. Reset: 00h. Latency Timer. |

**D14F3x00E (FCH::LPCPCICFG::HEADER\_TYPE)**

Read-only. Reset: 80h.

Header Type Register: This register identifies the type of the predefined header in the configuration space. Since THE SB is a multifunction device, the most significant bit is set.

\_aliasHOST; D14F3x00E**Bits Description**7:0 **header\_type**. Read-only. Reset: 80h. Header Type.**D14F3x00F (FCH::LPCPCICFG::BIST)**

Read-only. Reset: 00h.

Built-in Self Test Register: This register is used for control and status for Built-in Self Test. LPC has no [BIST](#) modes.\_aliasHOST; D14F3x00F**Bits Description**7:0 **bist**. Read-only. Reset: 00h. [BIST](#).**D14F3x040 (FCH::LPCPCICFG::PCI\_CONTROL)**

Read-write. Reset: 18h.

\_aliasHOST; D14F3x040**Bits Description**

7 Reserved.

6 **ecsemaphore**. Read-write. Reset: 0.**Description:** Read only.1: SPI rom is locked by External [EC](#).

0: SPI rom is not locked by External EC.

5 **biossemaphore**. Read-write. Reset: 0.**Description:** This bit is writeable by Host to block [EC](#) Rom access.

1: SPI rom is locked by Host.

0: SPI rom is not locked by Host.

Whenever the Host want to lock down Rom access then The host should keep I the bit to 1 untill read back to 1.

4 **extromsharingen**. Read-write. Reset: 1.**Description:** 1: support Rom sharing in External mode.

0: disable Rom sharing in External mode.

3 **vwromsharingen**. Read-write. Reset: 1.**Description:** 1: support Rom sharing in Virtual write mode.

0: disable Rom sharing in Virtual write mode.

2:0 Reserved.

**D14F3x044 (FCH::LPCPCICFG::IO\_PORT\_DECODE\_ENABLE)**

Reset: 0000\_0000h.

This register controls the decoding of parallel, serial, audio, MID, and MSS, FDC, game, [KBC](#), ACPI micro-controller, & Ad-lib ports. Writing 1 to a bit enables the corresponding I/O range.\_aliasHOST; D14F3x044**Bits Description**

31:0 Reserved.

**D14F3x048 (FCH::LPCPCICFG::IO\_MEM\_PORT\_DECODE\_ENABLE)**

Reset: 0000\_0000h.

This register controls the decoding of Super I/O configuration, alternate Super I/O configuration, wide generic ports, ROM1 &amp; ROM2 ports, and memory port. Writing a 1 to a bit enables the corresponding IO/ROM/Memory range.

\_aliasHOST; D14F3x048**Bits Description**

31:0 Reserved.

**D14F3x04C (FCH::LPCPCICFG::MEMORYRANGE)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x04C

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x050 (FCH::LPCPCICFG::ROM\_PROTECT\_0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x050

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | <b>rombase.</b> Read-write. Reset: 0_0000h.<br><b>Description:</b> Protected range is defined below:<br>{RomBase, 12'h0} ~ {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1)<br>Note: To support 64MB ROM, RomBase has been changed to use a Unified Mapped Address for all three ROMAddressRange (LPC bridge, PCI 0x68:6B, 0x6C:6F, 0xA8:AF) definition as shown below. Hardware will then further convert the Unified Mapped Address into physical address at the flash ROM<br>ROM1 mapping to Unified Address:<br>000F_FFFF: 000E_8000 FFFF_FFFF:FFFE_8000<br>ROM2 mapping to Unified Address:<br>FFFF_FFFF:FF00_0000 FFFF_FFFF:FF00_0000<br>ROM3 mapping to Unified Address:<br>FD_03FF_FFFF:FD_0000_0000 FFFF_FFFF:FC00_0000<br>Examples of :<br>1: To set write and read protection for 16K space at the unified address mapping of FFF8_3FFF: FFF8_0000<br>Set this register to 0xFFF8_0603:<br>2: To set write and read protection for 512K space at the unified address mapping of FC07_FFFF: FC00_0000<br>Set this register to 0xFC00_0707: |
| 11    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 10    | <b>write_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is write-protected. Writing to the range has no effect.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 9     | <b>read_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is read-protected. Reading any location in the range returns FFFF_FFFFh.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 8     | <b>rangeunit.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: 4K<br>1: 64K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 7:0   | <b>range.</b> Read-write. Reset: 00h.<br><b>Description:</b> Specify the protected range in either 4KB or 64KB unit depending on setting on bit 8.<br>Note: the protection definition is remapped to be within a 4GB range. BIOS should make sure resultant {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1) is within a valid range and will not cross FFFF_FFFF.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |



**D14F3x054 (FCH::LPCPCICFG::ROM\_PROTECT\_1)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x054

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | <b>rombase.</b> Read-write. Reset: 0_0000h.<br><b>Description:</b> Protected range is defined below:<br>{RomBase, 12'h0} ~ {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1)<br>Note: please refer to offsetx50 bit[31:12] for more information                                                                                                                     |
| 11    | Reserved.                                                                                                                                                                                                                                                                                                                                                                 |
| 10    | <b>write_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is write-protected. Writing to the range has no effect.                                                                                                                                                                                                       |
| 9     | <b>read_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is read-protected. Reading any location in the range returns FFFF_FFFFh.                                                                                                                                                                                       |
| 8     | <b>rangeunit.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: 4K<br>1: 64K                                                                                                                                                                                                                                                                                            |
| 7:0   | <b>range.</b> Read-write. Reset: 00h.<br><b>Description:</b> Specify the protected range in either 4KB or 64KB unit depending on setting on bit 8.<br>Note: the protection definition is remapped to be within a 4GB range. BIOS should make sure resultant {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1) is within a valid range and will not cross FFFF_FFFF. |

**D14F3x058 (FCH::LPCPCICFG::ROM\_PROTECT\_2)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x058

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | <b>rombase.</b> Read-write. Reset: 0_0000h.<br><b>Description:</b> Protected range is defined below:<br>{RomBase, 12'h0} ~ {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1)<br>Note: please refer to offsetx50 bit[31:12] for more information                                                                                                                     |
| 11    | Reserved.                                                                                                                                                                                                                                                                                                                                                                 |
| 10    | <b>write_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is write-protected. Writing to the range has no effect.                                                                                                                                                                                                       |
| 9     | <b>read_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is read-protected. Reading any location in the range returns FFFF_FFFFh.                                                                                                                                                                                       |
| 8     | <b>rangeunit.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: 4K<br>1: 64K                                                                                                                                                                                                                                                                                            |
| 7:0   | <b>range.</b> Read-write. Reset: 00h.<br><b>Description:</b> Specify the protected range in either 4KB or 64KB unit depending on setting on bit 8.<br>Note: the protection definition is remapped to be within a 4GB range. BIOS should make sure resultant {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1) is within a valid range and will not cross FFFF_FFFF. |

**D14F3x05C (FCH::LPCPCICFG::ROM\_PROTECT\_3)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x05C

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | <b>rombase.</b> Read-write. Reset: 0_0000h.<br><b>Description:</b> Protected range is defined below:<br>{RomBase, 12'h0} ~ {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1)<br>Note: please refer to offsetx50 bit[31:12] for more information                                                                                                                     |
| 11    | Reserved.                                                                                                                                                                                                                                                                                                                                                                 |
| 10    | <b>write_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is write-protected. Writing to the range has no effect.                                                                                                                                                                                                       |
| 9     | <b>read_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is read-protected. Reading any location in the range returns FFFF_FFFFh.                                                                                                                                                                                       |
| 8     | <b>rangeunit.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: 4K<br>1: 64K                                                                                                                                                                                                                                                                                            |
| 7:0   | <b>range.</b> Read-write. Reset: 00h.<br><b>Description:</b> Specify the protected range in either 4KB or 64KB unit depending on setting on bit 8.<br>Note: the protection definition is remapped to be within a 4GB range. BIOS should make sure resultant {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1) is within a valid range and will not cross FFFF_FFFF. |

**D14F3x060 (FCH::LPCPCICFG::PCI\_MEMORY\_START\_ADDRESS\_FOR\_LPC\_TARGET\_CYCLES)**

Reset: 0000h.

\_aliasHOST; D14F3x060

| Bits | Description |
|------|-------------|
| 15:0 | Reserved.   |

**D14F3x062 (FCH::LPCPCICFG::PCI\_MEMORY\_END\_ADDRESS\_FOR\_LPC\_TARGET\_CYCLES)**

Reset: 0000h.

\_aliasHOST; D14F3x062

| Bits | Description |
|------|-------------|
| 15:0 | Reserved.   |

**D14F3x064 (FCH::LPCPCICFG::PCI\_IO\_BASE\_ADDRESS\_FOR\_WIDEGENERICPORT)**

Reset: 0000\_0000h.

This register contains two 16-bits of I/O base address for LPC I/O (wide generic port) target range. The limit address is found by adding 512 to the base address.

\_aliasHOST; D14F3x064

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x068 (FCH::LPCPCICFG::ROMADDRESSRANGE\_1\_START\_ADDRESS)**

Read-write. Reset: 0000h.

\_aliasHOST; D14F3x068

| Bits | Description                                                                                                                                                                                                                                                                                                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>rom_start_address_1.</b> Read-write. Reset: 0000h. 16-bit starting address of the ROM (memory) address range 1. Default is set to 1M below 1M. This register contains the upper 16 bits of the starting address of the ROM address range 1. The lower 16 bits of the starting address are considered 0's. |

**D14F3x06A (FCH::LPCPCICFG::ROMADDRESSRANGE\_1\_END\_ADDRESS)**

Read-write. Reset: 000Fh.

\_aliasHOST; D14F3x06A

| Bits | Description                                                                                                                                                                                                                                                     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>rom_end_address_1.</b> Read-write. Reset: 000Fh. 16-bit END address of the ROM (memory) address range 1. This register contains the upper 16 bits of the ending address of the ROM address range 1. The lower 16 bits of the end address are considered 1's. |

**D14F3x06C (FCH::LPCPCICFG::ROMADDRESSRANGE\_2\_START\_ADDRESS)**

Read-write. Reset: FF00h.

\_aliasHOST; D14F3x06C

| Bits | Description                                                                                                                                                                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>rom_start_address_2.</b> Read-write. Reset: FF00h. 16-bit starting address of the ROM (memory) address range 2. Default is set to 16M below 4GB. This register contains the upper 16 bits of the starting address of the ROM address range 2. The lower 16 bits of the starting address are considered 0's. |

**D14F3x06E (FCH::LPCPCICFG::ROMADDRESSRANGE\_2\_END\_ADDRESS)**

Read-write. Reset: FFFFh.

\_aliasHOST; D14F3x06E

| Bits | Description                                                                                                                                                                                                                                                     |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>rom_end_address_2.</b> Read-write. Reset: FFFFh. 16-bit END address of the ROM (memory) address range 2. This register contains the upper 16 bits of the ending address of the ROM address range 2. The lower 16 bits of the end address are considered 1's. |

**D14F3x070 (FCH::LPCPCICFG::FIRMWARE\_HUB\_SELECT)**

Read-write. Reset: 0023\_4567h.

\_aliasHOST; D14F3x070

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x074 (FCH::LPCPCICFG::ALTERNATIVE\_WIDE\_IO\_RANGE\_ENABLE)**

Reset: 00h.

\_aliasHOST; D14F3x074

| Bits | Description |
|------|-------------|
| 7:0  | Reserved.   |

**D14F3x078 (FCH::LPCPCICFG::MISCELLANEOUS\_CONTROL\_BITS)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x078

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x07C (FCH::LPCPCICFG::TPM\_TRUSTED\_PLATFORM\_MODULE)**

Read-write. Reset: 03h.

any tpm cycle above is decoded only when the cycle is started by AlinkBridge. Access from bus master devices is not allowed.

\_aliasHOST; D14F3x07C

| Bits | Description                                                                                                                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>widertpmen.</b> Read-write. Reset: 0. Set to 1 to force logic to decode FED4xxxx as TPM cycles instead of FED4_0xxx, FED4_1xxx, FED4_2xxx, FED4_3xxx, and FED4_4xxx.                                                                                                                          |
| 6:2  | Reserved.                                                                                                                                                                                                                                                                                        |
| 1    | <b>tpm_amd.</b> Read-write. Reset: 1.<br><b>Description:</b> When set to 1, the following cycles covert into TPM cycle.<br>0xFFFF_FFFD_F920_0000~0xFFFF_FFFD_F923_FFFF 0x4028<br>0xFFFF_FFFD_F928_0000~0xFFFF_FFFD_F928_0003 0x4020<br>0xFFFF_FFFD_F928_0004~0xFFFF_FFFD_F928_0007 0x4024~0x4027 |
| 0    | <b>tpm12_en.</b> Read-write. Reset: 1.<br><b>Description:</b> When set to 1 then the follow memory cycles pass to TPM device.<br>0xFED4_0xxx<br>0xFED4_1xxx<br>0xFED4_2xxx<br>0xFED4_3xxx<br>0xFED4_4xxx                                                                                         |

**D14F3x07D (FCH::LPCPCICFG::LPCCLKCNTL)**

Read-write. Reset: 00h.

\_aliasHOST; D14F3x07D

| Bits | Description                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------------------|
| 7:6  | Reserved.                                                                                                                |
| 5    | <b>tpmbufferen.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Enable TPM buffer.<br>0: Disable TPM buffer.         |
| 4    | <b>tpmpfetchen.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Enable TPM burst read.<br>0: Disable TPM burst read. |
| 3:0  | Reserved.                                                                                                                |

**D14F3x084 (FCH::LPCPCICFG::TMKBC\_BASEADDRLOW)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x084

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x088 (FCH::LPCPCICFG::TMKBC\_BASEADDRHIGH)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x088

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x08C (FCH::LPCPCICFG::TMKBC\_REMAP)**

Reset: 0000h.

\_aliasHOST; D14F3x08C

| Bits | Description |
|------|-------------|
| 15:0 | Reserved.   |

**D14F3x090 (FCH::LPCPCICFG::WIDE\_IO\_2)**

Reset: 0000h.

\_aliasHOST; D14F3x090

**Bits Description**

15:0 Reserved.

**D14F3x098 (FCH::LPCPCICFG::EC\_LPC\_CNTRL)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x098

**Bits Description**

31:0 Reserved.

**D14F3x09C (FCH::LPCPCICFG::GEC\_SHADOWROM\_ADDRESS)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x09C

**Bits Description**

31:0 Reserved.

**D14F3x0A0 (FCH::LPCPCICFG::SPI\_BASE\_ADDR)**

Read-write. Reset: FEC1\_0012h.

\_aliasHOST; D14F3x0A0

**Bits Description**31:8 **spi\_espi\_baseaddr**. Read-write. Reset: FE\_C100h.**Description:** This register defines the [MMIO](#) base address for the SPI ROM controller and eSPI host controller:SPI [BAR](#) = {SPI\_BaseAddr[31:8],8'b0}

HFP BAR = {SPI\_BaseAddr[31:8],8'b0} + 0x0000\_1000

eSPI BAR = {SPI\_BaseAddr[31:8],8'b0} + 0x0001\_0000

These bits will be write protected when PSP\_RegxFC[8] (Security\_Mem\_RegxFC[8]) = 1

7:5 Reserved.

4 **pspsimmiosel**. Read-write. Reset: 1.**Description:** 0: Spi Mmio register space map to Host spi\_reg1: Spi Mmio register space map to [PSP](#) psp\_reg3 **routetpm2spi**. Read-write. Reset: 0. When set, TPM cycles are routed to SPI bus with TPM\_SPI\_CS# asserted (note TCG has not finalized the SPI TPM specification yet)2 **abortenable**. Read-write. Reset: 0. LPC Abort enable1 **spiromenable**. Read-write. Reset: 1. When this bit is set and chip is strapped to SPI Rom, SPI Rom is enabled, otherwise SPI Rom is disabled.0 **altspicsenable**. Read-write. Reset: 0. Alternative SPI CS enable**D14F3x0A4 (FCH::LPCPCICFG::EC\_PORTADDRESS)**

Read-write. Reset: 012Eh.

\_aliasHOST; D14F3x0A4

**Bits Description**

15:0 Reserved.

**D14F3x0A8 (FCH::LPCPCICFG::PCI\_BAR\_64MB\_ROM3\_LOW)**

Read-only. Reset: 0000\_0000h.

\_aliasHOST; D14F3x0A8

**Bits Description**31:26 **bar\_64mb\_31\_26**. Read-only. Reset: 00h. Base address [31:26] of 64MB BIOS ROM space.

25:0 Reserved.

**D14F3x0AC (FCH::LPCPCICFG::PCI\_BAR\_64MB\_ROM3\_HIGH)**

Read-only. Reset: 0000\_00FDh.

\_aliasHOST; D14F3x0AC

| Bits | Description                                                                                       |
|------|---------------------------------------------------------------------------------------------------|
| 31:0 | <b>bar_64mb_63_32.</b> Read-only. Reset: 0000_00FDh. Base address [63:32] of 64MB BIOS ROM space. |

**D14F3x0B0 (FCH::LPCPCICFG::ROMDMASRCADDR)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x0B0

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x0B4 (FCH::LPCPCICFG::ROMDMADSTADDR)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x0B4

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x0B8 (FCH::LPCPCICFG::ROMDMACONTROL)**

Read-write. Reset: 0000h.

\_aliasHOST; D14F3x0B8

| Bits | Description |
|------|-------------|
| 15:0 | Reserved.   |

**D14F3x0BA (FCH::LPCPCICFG::ECCONTROL)**

Reset: 00h.

\_aliasHOST; D14F3x0BA

| Bits | Description |
|------|-------------|
| 7:0  | Reserved.   |

**D14F3x0BB (FCH::LPCPCICFG::HOSTCONTROL)**

Read-write. Reset: 01h.

\_aliasHOST; D14F3x0BB

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | Reserved.                                                                                                                                                                    |
| 0    | <b>prefetchenspifromhost.</b> Read-write. Reset: 1. This is for performance enhancement purpose. When set, SPI controller will prefetch from the flash on behalf of the host |

**D14F3x0C0 (FCH::LPCPCICFG::ECROMWROFFSET)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x0C0

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x0C4 (FCH::LPCPCICFG::ECROMRDOFFSET)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x0C4

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**D14F3x0C8 (FCH::LPCPCICFG::CLIENTROMPROTECT)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x0C8

**Bits Description**

31:0 Reserved.

**D14F3x0CC (FCH::LPCPCICFG::AUTOROMCFG)**

Reset: 0000\_0000h.

\_aliasHOST; D14F3x0CC

**Bits Description**

31:0 Reserved.

**D14F3x0D0 (FCH::LPCPCICFG::CLKGATECNTRL)**

Read-write. Reset: 04h.

\_aliasHOST; D14F3x0D0

**Bits Description**

7:3 Reserved.

2 **spionclkrun.** Read-write. Reset: 1.**Description:** 0: Spi request can assert ClkRun#

1: Spi request doesn't assert ClkRun#

1:0 Reserved.

**D14F3x0D1 (FCH::LPCPCICFG::CLKPINCNTRL)**

Reset: 00h.

\_aliasHOST; D14F3x0D1

**Bits Description**

7:0 Reserved.

**D14F3x0D2 (FCH::LPCPCICFG::CLKRUNOVRID)**

Reset: 00h.

\_aliasHOST; D14F3x0D2

**Bits Description**

7:0 Reserved.

**D14F3x0D3 (FCH::LPCPCICFG::CLKRUNEN)**

Reset: 00h.

\_aliasHOST; D14F3x0D3

**Bits Description**

7:0 Reserved.

**D14F3x0D4 (FCH::LPCPCICFG::CLKRUNOPTION)**

Reset: 00h.

\_aliasHOST; D14F3x0D4

**Bits Description**

7:0 Reserved.

**D14F3x0DC (FCH::LPCPCICFG::PCI\_MISCCNTRL)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x0DC

**Bits Description**

31:0 Reserved.

**D14F3x0E0 (FCH::LPCPCICFG::ROM\_PROTECT\_4)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x0E0

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | <b>rombase.</b> Read-write. Reset: 0_0000h.<br><b>Description:</b> Protected range is defined below:<br>{RomBase, 12'h0} ~ {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1)<br>Note: please refer to offsetx50 bit[31:12] for more information                                                                                                                     |
| 11    | Reserved.                                                                                                                                                                                                                                                                                                                                                                 |
| 10    | <b>write_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is write-protected. Writing to the range has no effect.                                                                                                                                                                                                       |
| 9     | <b>read_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is read-protected. Reading any location in the range returns FFFF_FFFFh.                                                                                                                                                                                       |
| 8     | <b>rangeunit.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: 4K<br>1: 64K                                                                                                                                                                                                                                                                                            |
| 7:0   | <b>range.</b> Read-write. Reset: 00h.<br><b>Description:</b> Specify the protected range in either 4KB or 64KB unit depending on setting on bit 8.<br>Note: the protection definition is remapped to be within a 4GB range. BIOS should make sure resultant {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1) is within a valid range and will not cross FFFF_FFFF. |

**D14F3x0E4 (FCH::LPCPCICFG::ROM\_PROTECT\_5)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x0E4

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | <b>rombase.</b> Read-write. Reset: 0_0000h.<br><b>Description:</b> Protected range is defined below:<br>{RomBase, 12'h0} ~ {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1)<br>Note: please refer to offsetx50 bit[31:12] for more information                                                                                                                     |
| 11    | Reserved.                                                                                                                                                                                                                                                                                                                                                                 |
| 10    | <b>write_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is write-protected. Writing to the range has no effect.                                                                                                                                                                                                       |
| 9     | <b>read_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is read-protected. Reading any location in the range returns FFFF_FFFFh.                                                                                                                                                                                       |
| 8     | <b>rangeunit.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: 4K<br>1: 64K                                                                                                                                                                                                                                                                                            |
| 7:0   | <b>range.</b> Read-write. Reset: 00h.<br><b>Description:</b> Specify the protected range in either 4KB or 64KB unit depending on setting on bit 8.<br>Note: the protection definition is remapped to be within a 4GB range. BIOS should make sure resultant {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1) is within a valid range and will not cross FFFF_FFFF. |



**D14F3x0E8 (FCH::LPCPCICFG::ROM\_PROTECT\_6)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x0E8

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | <b>rombase.</b> Read-write. Reset: 0_0000h.<br><b>Description:</b> Protected range is defined below:<br>{RomBase, 12'h0} ~ {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1)<br>Note: please refer to offsetx50 bit[31:12] for more information                                                                                                                     |
| 11    | Reserved.                                                                                                                                                                                                                                                                                                                                                                 |
| 10    | <b>write_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is write-protected. Writing to the range has no effect.                                                                                                                                                                                                       |
| 9     | <b>read_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is read-protected. Reading any location in the range returns FFFF_FFFFh.                                                                                                                                                                                       |
| 8     | <b>rangeunit.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: 4K<br>1: 64K                                                                                                                                                                                                                                                                                            |
| 7:0   | <b>range.</b> Read-write. Reset: 00h.<br><b>Description:</b> Specify the protected range in either 4KB or 64KB unit depending on setting on bit 8.<br>Note: the protection definition is remapped to be within a 4GB range. BIOS should make sure resultant {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1) is within a valid range and will not cross FFFF_FFFF. |

**D14F3x0EC (FCH::LPCPCICFG::ROM\_PROTECT\_7)**

Read-write. Reset: 0000\_0000h.

\_aliasHOST; D14F3x0EC

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | <b>rombase.</b> Read-write. Reset: 0_0000h.<br><b>Description:</b> Protected range is defined below:<br>{RomBase, 12'h0} ~ {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1)<br>Note: please refer to offsetx50 bit[31:12] for more information                                                                                                                     |
| 11    | Reserved.                                                                                                                                                                                                                                                                                                                                                                 |
| 10    | <b>write_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is write-protected. Writing to the range has no effect.                                                                                                                                                                                                       |
| 9     | <b>read_protect.</b> Read-write. Reset: 0. When this bit is set, the memory range defined by this register is read-protected. Reading any location in the range returns FFFF_FFFFh.                                                                                                                                                                                       |
| 8     | <b>rangeunit.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: 4K<br>1: 64K                                                                                                                                                                                                                                                                                            |
| 7:0   | <b>range.</b> Read-write. Reset: 00h.<br><b>Description:</b> Specify the protected range in either 4KB or 64KB unit depending on setting on bit 8.<br>Note: the protection definition is remapped to be within a 4GB range. BIOS should make sure resultant {RomBase, 12'h0} + (Range+1) *4K (or 64K if bit 8 is 1) is within a valid range and will not cross FFFF_FFFF. |

**14.3.13.3 SPI Registers**[FCH](#) SPI design is mode 0 only.

Register FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER, field SpiReadMode[2:0] specifies the SPI read mode:

Table 255: *SpiReadMode[2:0] Definitions*

| Bits | Definition              |
|------|-------------------------|
| 000b | Normal read (up to 33M) |
| 001b | Reserved                |
| 010b | Dual IO (1-1-2)         |
| 011b | Quad IO (1-1-4)         |
| 100b | Dual IO (1-2-2)         |
| 101b | Quad IO (1-4-4)         |
| 110b | Normal read (up to 66M) |
| 111b | Fast Read               |

The SPI configuration registers are accessed through SPI base address specified by FCH::LPCPCICFG::SPI\_BASE\_ADDR. Software can communicate with the SPI ROM through the default memory or an alternate programming (a.k.a "Indexed mode") method:

- Memory access to the BIOS ROM address space is automatically handled by the hardware. The SPI ROM controller translates the memory address onto the SPI bus and accesses the SPI ROM data. Any other commands besides memory read or memory write to the SPI ROM need to go through the alternate programming (a.k.a "Indexed mode") method. SPI ROM access through the memory address is limited to a 24-bit address (16MB addressable space).
- With the alternate programming (a.k.a "Indexed mode") method, software needs to program the SPI\_OPCODE, SpiAddress, TXBYTE\_COUNT, RXBYTE\_COUNT, put the data into the transmit FIFO, and then execute the command. The hardware communicates with the SPI ROM using these parameters. This alternate programming (a.k.a "Indexed mode") method basically allows software to issue any flash vendor specific commands such as ERASE and STATUS. The alternate programming method can generate up to 32-bit addresses, but transfers are limited to "normal" (1-bit) mode.

The programming sequence is described as follows.

Change Rom Read mode/Speed:

1. Rom read is handled by old Spi engine first which running 16.7M without prefetch enabled.
2. Program FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER to set up Read mode.
3. Program dummy cycle in FCH::LPCHOSTSPIREG::SPI100\_DUMMY\_CYCLE\_CONFIG\_REGISTER to match the dummy cycle number of SPI Rom.
4. Program speed for all operations in FCH::LPCHOSTSPIREG::SPI100ENABLE\_REGISTER.
5. Program UseSpi100 to 1 in FCH::LPCHOSTSPIREG::SPI100ENABLE\_REGISTER.
6. Program PrefetchEnSPIFromHost in FCH::LPCPCICFG::HOSTCONTROL to enable read prefetch.

Command Mode:

1. Set the number of the bytes to be written in FCH::LPCHOSTSPIREG::TXBYTECOUNT[txbytecount] including the command code in the first byte or the bytes to be read in FCH::LPCHOSTSPIREG::RXBYTECOUNT[rxbytecount].
2. Set the command code in SPIx00 FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER.
3. Put the write data byte into FCH::LPCHOSTSPIREG::FIFO0~FCH::LPCHOSTSPIREG::FIFO71.
4. Issue the read status command to SPI device to make sure the SPI device is ready to take commands.
5. Wait for FCH::LPCHOSTSPIREG::SPISTATUS[spibusy] == 0 then check bit[7:0] to make sure that the write data and read data are done correctly.
6. Read FCH::LPCHOSTSPIREG::FIFO0[fifodata]~FCH::LPCHOSTSPIREG::FIFO71[fifodata] to retrieve the read data.

**14.3.13.3.1 Programming for ROM Protection register****14.3.13.3.1.1 SPI ROM Physical Address**

Using a 64MB SPI ROM as an example, since only address[25:0] are needed for 64MB, SPI ROM will simply ignore value of address[31:26]. To access SPI ROM, the 64MB SPI ROM are addressed using following physical address:

- 16MB bank-0: 0xFC00\_0000 – 0xFCFF\_FFFF
- 16MB bank-1: 0xFD00\_0000 – 0xFDFF\_FFFF
- 16MB bank-2: 0xFE00\_0000 – 0xFEFF\_FFFF
- 16MB bank-3: 0xFF00\_0000 – 0xFFFF\_FFFF

**14.3.13.3.1.2 Program ROM Protection Register**

Once physical address of protection region is determined, the ROM protection register can be programmed accordingly. Bits[11:0] are straight forward, bits[31:12]: RomBase need to match start physical address of protection region.

**14.3.13.3.1.2.1 Case-1, 0xFFFF0\_0000 – 0xFFFF0\_7FFF**

If 32K SPI ROM space at physical address 0xFFFF0\_0000 – 0xFFFF0\_7FFF need to have both Write and Read protection, ROM protection register can be programmed to 0xFFFF0\_0607.

[7:0]: Range = 0x7, with bit[8] = 0 (unit = 4K), range = 32K

[9] = 1, Read protect

[10] = 1, Write protect

[31:12] = 0xFFFF0\_0, starting address is 0xFFFF0\_0000, with range = 32K, the protection region is 0xFFFF0\_0000 – 0xFFFF0\_7FFF

**14.3.13.3.1.2.2 Case-2, 0xFD46\_0000 - 0xFD4D\_FFFF**

If 512K SPI ROM space at physical address 0xFD46\_0000 – 0xFD4D\_FFFF needs to have Write protection only, the ROM protection register can be programmed to 0xFD46\_0507.

[7:0]: Range = 0x7, with bit[8] = 1 (unit = 64K), range = 512K

[9] = 0, no Read protect

[10] = 1, Write protect

[31:12] = 0xFD46\_0, starting address is 0xFD46\_0000, with range = 512K, the protection region is 0xFD46\_0000 – 0xFD4D\_FFFF

**14.3.13.3.1.3 Index Mode (Indirect) Access**

Opcode used in Index Mode determines if it is using 24-bit address mode or 32-bit address mode. For example:

- OpCode = 0x03, 24-bit address Read command
- OpCode = 0x13, 32-bit address Read command
- OpCode = 0x02, 24-bit address byte program command
- OpCode = 0x12, 32-bit address byte program command

**14.3.13.3.1.3.1 24-bit Address Index Mode**

For Index mode, only address[25:0] are used for ROM protection address comparison. The address mapping of 24-bit address Index mode is:

Address[31:24]: 0x00, address[25:24] = 00b, point to bank-0 of 64 MB ROM (0xFC00\_0000 – 0xFCFF\_FFFF)

Address[23:16]: SPI\_regx80

Address[15:8]: SPI\_regx81

Address[7:0]: SPI\_regx82

#### 14.3.13.3.1.3.2 32-bit Address Index Mode

For Index mode, only address[25:0] are used for ROM protection address comparison. The address mapping of 32-bit address Index mode is:

Address[31:24]: SPI\_regx80, only address[25:24] used.

Address[23:16]: SPI\_regx81

Address[15:8]: SPI\_regx82

Address[7:0]: SPI\_regx83

#### 14.3.13.3.1.4 Exception

Register 'ROM Address Override' has been added to change [Logical address](#) to [Physical address](#) mapping, since Protection checking is using Address before 'ROM Address Override' mapping, caution need be paid when 'ROM Address Override[4:0]' not equal to 0x0.

Function of "ROM Address Override"

1. Bit[3:0] only apply to ROM2 (16MB) space, it is used to change ROM2 logical address to physical address mapping, it has no effect on ROM1 and ROM3
  1. Bit[3:0] = 0x0 or 0xF, ROM2 (0xFF00\_0000 – 0xFFFF\_FFFF) map to bank-3 (0xFF00\_0000 – 0xFFFF\_FFFF)
  2. Bit[3:0] = 0xC, ROM2 (0xFF00\_0000 – 0xFFFF\_FFFF) map to bank-0 (0xFC00\_0000 – 0xFCFF\_FFFF)
  3. Bit[3:0] = 0xD, ROM2 (0xFF00\_0000 – 0xFFFF\_FFFF) map to bank-1 (0xFD00\_0000 – 0xFDFD\_FFFF)
  4. Bit[3:0] = 0xE, ROM2 (0xFF00\_0000 – 0xFFFF\_FFFF) map to bank-2 (0xFE00\_0000 – 0xFEFF\_FFFF)
2. Bit[4] apply to all ROM1/ROM2/ROM3 space. However, since ROM1 and ROM2 are map to 0xFFxx\_xxxx by default, setting bit[4] = 1 will not affect ROM1/ROM2, only ROM3 (64MB) will be map to 0xFFxx\_xxxx (16MB).
  1. Bit[4] = 1, ROM3 (0xfd\_0000\_0000 – 0xfd\_00FF\_FFFF) map to bank-3 (0xFF00\_0000 – 0xFFFF\_FFFF)
    1. ROM3 (0xfd\_0100\_0000 – 0xfd\_01FF\_FFFF) map to bank-3 (0xFF00\_0000 – 0xFFFF\_FFFF)
    2. ROM3 (0xfd\_0200\_0000 – 0xfd\_02FF\_FFFF) map to bank-3 (0xFF00\_0000 – 0xFFFF\_FFFF)
    3. ROM3 (0xfd\_0300\_0000 – 0xfd\_03FF\_FFFF) map to bank-3 (0xFF00\_0000 – 0xFFFF\_FFFF)

#### 14.3.13.3.1.4.1 Case-1 (0xFFF0\_0000 – 0xFFF0\_7FFF)

When "ROM Address Override" register bit[4:0] != 0x00, there will be following side effects:

1. ROM1: no side effect.
2. ROM2:
  1. Bit[3:0] = 0x0 or 0xF, no side effect.
  2. Bit[3:0] = 0xC, host address 0xFFF0\_0000 – 0xFFF0\_7FFF is blocked.
    1. Side effect: SPI ROM 0xFCF0 – 0xFCF0\_7FFF is protected.
  3. Bit[3:0] = 0xD, host address 0xFFF0\_0000 – 0xFFF0\_7FFF is blocked.
    1. Side effect: SPI ROM 0xFDF0 – 0xFDF0\_7FFF is protected.
  4. Bit[3:0] = 0xE, host address 0xFFF0\_0000 – 0xFFF0\_7FFF is blocked.
    1. Side effect: SPI ROM 0xFE00 – 0xFE00\_7FFF is protected.
3. ROM3:
  1. Bit[4] = 0, no side effect.
  2. Bit[4] = 1, since all accesses through ROM3 are mapped to 0xFF00\_0000 – 0xFFFF\_FFFF. If accessed through 0xFD\_0000\_0000 – 0xFD\_02FF\_FFFF, it will access the ROM at 0xFF00\_0000 – 0xFFFF\_FFFF, which should be protected, but ROM protection comparison will not block it as address

before "ROM Address Override" is used.

1. Side effect: Need 3 more protection registers to block "0xFCF0\_0000 – 0xFCF0\_7FFF" and "0xFDF0\_0000 – 0xFDF0\_7FFF" and "0xFE00\_0000 – 0xFE00\_7FFF".

#### 14.3.13.3.1.4.2 Case-2 (0xFD46\_0000 – 0xFD4D\_FFFF)

When "ROM Address Override" register bit[4:0] != 0x00, there will be following side effects:

1. ROM1: no side effect.
2. ROM2:
  1. Bit[3:0] = 0x0 or 0xF, no side effect.
  2. Bit[3:0] = 0xC, no side effect, as ROM2 is mapped to 0xFC00\_0000 – 0xFCFF\_FFFF.
  3. Bit[3:0] = 0xD, since ROM2 is mapped to 0xFD00\_0000 – 0xFDFF\_FFFF, access to ROM2 address 0xFF46\_0000 – 0xFF4D\_FFFF needs to be blocked, which needs to program another ROM protection register with 0xFF46\_0407.
    1. Side effect: This will make SPI ROM 0xFF46\_0000 – 0xFF4D\_FFFF be protected when accessed through ROM1/ROM3.
  4. Bit[3:0] = 0xE, no side effect as ROM2 is mapped to 0xFE00\_0000 – 0xFEFF\_FFF.
3. ROM3:
  1. Bit[4]=0, no side effect
  2. Bit[4]=1, for ROM3 (0xFD\_0100\_0000 – 0xFD\_01FF\_FFFF) map to bank-3 (0xFF00\_0000 – 0xFFFF\_FFFF) case, ROM3 range 0xFD\_0146\_0000 – 0xFD\_014D\_FFFF will be blocked,
    1. Side effect: If bit[4]=1, all ROM access (ROM1/ROM2/ROM3) are map to 0xFF00\_0000 – 0xFFFF\_FFFF, which set protection for 0xFD46\_0000 – 0xFD4D\_FFFF should not exist.

**14.3.13.3.2 Serial Peripheral Interface (SPI) Registers**

**SPIx00000000 (FCH::LPCHOSTSPIREG::SPI\_CNTRL0\_REGISTER)**

Reset: 0FC0\_0000h.

\_aliasHOSTLEGACY; SPIx00000000; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>spibusy.</b> Read-write. Reset: 0.<br><b>Description:</b> Read-only.<br>0: SPI bus is idle<br>1: SPI bus is busy<br>Note: When SpiBusy=1, ROM access is going on, SpiReadMode[2:0] should not be changed and ExecuteOpCode should not be set to 1 to start new operation.                                                                                                                                                                                                                                                                                                                                                                                     |
| 30:29 | <b>spireadmode_2_1.</b> Read-write. Reset: 0h.<br><b>Description:</b> These two bits and bit 18 in the same register<br>Specify the Spi read mode:<br>000: Normal read (up to 33M)<br>001: Reserved<br>010: Dual-io (1-1-2)<br>011: Quad-io (1-1-4)<br>100: Dual-io (1-2-2)<br>101: Quad-io (1-4-4)<br>110: Normal read (up to 66M)<br>111: Fast read<br>Note: before change SpiReadMode[2:0], software should read bit[31]: SpiBusy=0 to make sure no SPI operation is going on.                                                                                                                                                                                |
| 28    | <b>spiclkgate.</b> Read-write. Reset: 0. Set to 1 to skip the 8th spiclk at the end data when doing read.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 27:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 23    | <b>spihostaccessromen.</b> Read-write. Reset: 1.<br><b>Description:</b> This is a clear-once protection bit once it is cleared to 0 it cannot be set back to 1. Once cleared to 0, some SPI registers cannot be written.<br>Note: bit[23] and bit[22] were used for other function also, but those function were removed, so they are used for write-protection for certain registers only.<br>Following registers are write-protect when bit[23:22] != 2'b11,<br>0x04[31:0]<br>0x08[31:0]<br>0x0C[31:24]<br>0x14[31:0]<br>0x18[31:0]<br>0x1C[1:0]<br>0x1D[1:0]<br>0x30[4:0]<br>0x40[31:0]<br>0x50[31:8]<br>0x54[31:8]<br>0x5C[7:0]<br>0x60[31:26]<br>0x64[31:0] |
| 22    | <b>spiaccessmacromen.</b> Read-write. Reset: 1. This is a clear-once protection bit once it is cleared to 0 it cannot be set back to 1. Once cleared to 0, some SPI registers cannot be written.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 21    | <b>illegalaccess.</b> Read-only. Reset: 0.<br><b>Description:</b> Read Only.<br>0. Legal Index mode Access<br>1. Illegal Index mode Access                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 20:19 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18    | <b>spireadmode_0.</b> Read-write. Reset: 0.<br><b>Description:</b> It is bit 0 of SpiReadMode to specify the spi read mode. Please see the definition of SpiReadMode in bit [30:29] below.<br>Note: before change SpiReadMode[2:0], software should read bit[31]: SpiBusy=0 to make sure no SPI operation is going on.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 17:12 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 11:10 | <b>prot_compare_range.</b> Read-write. Reset: 0h.<br><b>Description:</b> Specify what address range used for ROM protection range comparison. This bit will be write protected when x00[23:22] not equal 11b.<br>00: address [25:12] are used<br>01: address [24:12] are used<br>10: address [23:12] are used<br>11: reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 9     | <b>index_cacheline_stop.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Index transfer stop at TxByteCnt or RxByteCnt<br>1: Index transfer stop at smaller value of Cacheline-boundary-ByteCnt or TxByteCnt/RxByteCnt. Cacheline-boundary-ByteCnt = 0x40 Address[5:0]<br>Only when OpCode (x45[7:0]) match following command will Index_cacheline_stop =1 work,<br>24-bit address command:<br>ByteCmd (x0C[31:24], default = 0x02)<br>PAGEWR (0x18[23:16], default = 0x0A)<br>DPWCmd (0x14[7:0], default = 0xB2)<br>QPWCmd (0x14[15:8], default = 0xE2)<br>ReadCmd (0x18[7:0], default = 0x03)<br>FReadCmd (0x18[15:8], default = 0x0B)<br>DPRCmd (0x40[23:16], default = 0xBB)<br>QPRCmd (0x40[31:24], default = 0xEB)<br>32-bit address command:<br>ByteCmd32 (0x50[15:8], default = 0x12)<br>ReadCmd32(0x50[23:16],default= 0x13)<br>FReadCmd32(0x50[31:24],default= 0x0C)<br>DPRCmd32(0x54[23:16],default= 0xBC)<br>QPRCmd32(0x54[31:24],default= 0xEC)<br>This bit will be write protected when x00[23:22] not equal 11b.                                                                                                            |
| 8     | <b>dis_index_retry.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: PCI cycle write to x47[7]=1 to start Index mode transfer will be kept retried till Index mode transfer complete. This is backward compatible to previous design.<br>1: Once Index mode transfer is going on after write x47[7]=1, all register write PCI cycles to the same register block will be completed normally but write will be blocked thus register content is not changed (quietly dropped) till Index mode transfer complete thus software should not do any register write to the same register block when Index mode transfer is going on, all register read cycles to the same register block will be completed normally with read data returned, write/read to other register block will not be affected. For example, if SPIx47[7] is written =1 to start Index mode transfer, all register write to SPI register block will be quietly dropped while all register read to SPI register block will be completed normally, write/read to other register block will not be affected.<br>This bit will be write protected when x00[23:22] not equal 11b. |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |



**SPIx00000004 (FCH::LPCHOSTSPIREG::SPI\_RESTRICTEDCMD\_REGISTER)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx00000004; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>restrictedcmd3</b> . Read-write. Reset: 00h. Same as RestrictedCmd0                                                                                                                                                                                                                                                                                                                                                                                      |
| 23:16 | <b>restrictedcmd2</b> . Read-write. Reset: 00h. Same as RestrictedCmd0                                                                                                                                                                                                                                                                                                                                                                                      |
| 15:8  | <b>restrictedcmd1</b> . Read-write. Reset: 00h. Same as RestrictedCmd0                                                                                                                                                                                                                                                                                                                                                                                      |
| 7:0   | <b>restrictedcmd0</b> . Read-write. Reset: 00h.<br><b>Description:</b> This defines a restricted command code. If SPI_Opcode matches this register and SpiAccessRomEn and/or SpiHostAccessRomEn bit are cleared , 'Execute' bit of Index mode cannot be written, and the Index mode transfer will not occur.<br>Note when either SpiAccessRomEn and/or SpiHostAccessRomEn bit are cleared, these registers become read-only and cannot be changed any more. |

**SPIx00000008 (FCH::LPCHOSTSPIREG::SPI\_RESTRICTEDCMD2\_REGISTER)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx00000008; SPI=FEC1\_0000h

| Bits  | Description                                                            |
|-------|------------------------------------------------------------------------|
| 31:24 | <b>restrictedcmd7</b> . Read-write. Reset: 00h. Same as RestrictedCmd0 |
| 23:16 | <b>restrictedcmd6</b> . Read-write. Reset: 00h. Same as RestrictedCmd0 |
| 15:8  | <b>restrictedcmd5</b> . Read-write. Reset: 00h. Same as RestrictedCmd0 |
| 7:0   | <b>restrictedcmd4</b> . Read-write. Reset: 00h. Same as RestrictedCmd0 |

**SPIx0000000C (FCH::LPCHOSTSPIREG::SPI\_CNTRL1\_REGISTER)**

Read-write. Reset: 0200\_0000h.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[31:24] of Host\_Mem\_Reg:0Ch become read-only and cannot be changed any more.

\_aliasHOSTLEGACY; SPIx0000000C; SPI=FEC1\_0000h

| Bits  | Description                                                                        |
|-------|------------------------------------------------------------------------------------|
| 31:24 | <b>bytecommand</b> . Read-write. Reset: 02h. Specify the Byte Programmand Op Code. |
| 23:0  | Reserved.                                                                          |

**SPIx00000010 (FCH::LPCHOSTSPIREG::ESPI\_CNTRL\_REGISTER)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; SPIx00000010; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | <b>lock_all_use_spi1.</b> Read-write. Reset: 0. when this bit set to =1, bit [15] and bit[13:8] are locked and cannot be written, it can only be written again when PSPx2E[1]: override_iSMMWrite =1, or LPC_RstB could reset it back to 0.                                                                                                                                       |
| 14   | Reserved.                                                                                                                                                                                                                                                                                                                                                                         |
| 13   | <b>romcp_use_spi1.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: default, Host RomCopy mux on SPI0 Data/Clk,<br>1: Host RomCopy mux on SPI1 Data/Clk                                                                                                                                                                                                                        |
| 12   | <b>hfp_index_use_spi1.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: default, HFP Index mux on SPI0 Data/Clk,<br>1: HFP Index mux on SPI1 Data/Clk                                                                                                                                                                                                                          |
| 11   | <b>psp_index_use_spi1.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: default, <a href="#">PSP</a> Index mux on SPI0 Data/Clk,<br>1: PSP Index mux on SPI1 Data/Clk                                                                                                                                                                                                          |
| 10   | <b>host_index_use_spi1.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: default, Host Index mode mux on SPI0 Data/Clk,<br>1: Host Index mode mux on SPI1 Data/Clk                                                                                                                                                                                                             |
| 9    | <b>tpm_psp_use_spi1.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: default, TPM and <a href="#">PSP</a> mux on SPI0 Data/Clk,<br>1: TPM and PSP mux on SPI1 Data/Clk                                                                                                                                                                                                        |
| 8    | <b>espi0_use_spi1.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: default, eSPI0 mux on SPI0 Data/Clk,<br>1: eSPI0 mux on SPI1 Data/Clk.                                                                                                                                                                                                                                     |
| 7:2  | Reserved.                                                                                                                                                                                                                                                                                                                                                                         |
| 1    | <b>rom_addr_wr_prot.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: ROM1/ROM2/ROM3 address registers can be written<br>1: ROM1/ROM2/ROM3 address registers cannot be written<br>This bit will be write-protected once it is written 1, it can be enable for write again in <a href="#">SMI</a> , or <a href="#">PSP</a> can set PSPx2E[1]=1 to write to this register.       |
| 0    | <b>dis_espi_masctl_reg_wr.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: eSPI master control registers can be programmed<br>1: eSPI master control registers cannot be programmed<br>This bit will be write-protected once it is written 1, it can be enable for write again in <a href="#">SMI</a> , or <a href="#">PSP</a> can set PSPx2E[1]=1 to write to this register. |

**SPIx00000014 (FCH::LPCHOSTSPIREG::SPI\_CMDVALUE1\_REGISTER)**

Read-write. Reset: 0000\_E2B2h.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[15:0] of Host\_Mem\_Reg:14h become read-only and cannot be changed any more.

\_aliasHOSTLEGACY; SPIx00000014; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                               |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                 |
| 15:8  | <b>qpw_cmd.</b> Read-write. Reset: E2h. Command code of QPW (1-4-4 write), used to compare against the opcode sent out by the Index mode. |
| 7:0   | <b>dpw_cmd.</b> Read-write. Reset: B2h. Command code of DPW (1-2-2 write), used to compare against the opcode sent out by the Index mode. |

**SPIx00000018 (FCH::LPCHOSTSPIREG::SPI\_CMDVALUE2\_REGISTER)**

Read-write. Reset: 020A\_0B03h.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[31:0] of Host\_Mem\_Reg:18h become read-only and cannot be changed any more.

\_aliasHOSTLEGACY; SPIx00000018; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                             |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>bytewr.</b> Read-write. Reset: 02h. This is used to compare against the opcode sent out by the Index mode. This is a predefined value to decode for the BYTEWR (byte write) command. |
| 23:16 | <b>pagewr.</b> Read-write. Reset: 0Ah. This is used to compare against the opcode sent out by the Index mode. This is a predefined value to decode for the PAGEWR (page write) command. |
| 15:8  | <b>fread.</b> Read-write. Reset: 0Bh. This is used to compare against the opcode sent out by the Index mode. This is a predefined value to decode for the Fread (fast read) command.    |
| 7:0   | <b>read_cmd.</b> Read-write. Reset: 03h. This is used to compare against the opcode sent out by the Index mode. This is a predefined value to decode for the Read (Read byte) command.  |

**SPIx0000001C (FCH::LPCHOSTSPIREG::ROMCP\_CS\_REGISTER)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000001C; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                      |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:3  | Reserved.                                                                                                                                                                                                                                                                                                        |
| 2    | <b>lock_romcp_cs.</b> Read-write. Reset: 0. When this bit is set to 1, SPIx1C[1:0] will be write protected. Once this bit is set to 1, this bit is write protected, only when PSPx2E[1] =1b can this bit be written again, or LPC_RstB could reset it back to 0.                                                 |
| 1:0  | <b>romcps.</b> Read-write. Reset: 0h.<br><b>Description:</b> These two bits select the alternate SPI_CS# for Romcp, these 2 bits will be write protected when SPIx00[23:22] != 11b or SPIx1C[2] =1b<br>00b Romcp select SPI_CS1_L<br>01b Romcp select SPI_CS2_L<br>10b Romcp select SPI_CS3_L<br>11b -- reserved |

**SPIx0000001D (FCH::LPCHOSTSPIREG::ALT\_SPI\_CS\_REGISTER)**

Read-write. Reset: 08h.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[1:0] of Host\_Mem\_Reg:1Dh become read-only and cannot be changed any more

\_aliasHOSTLEGACY; SPIx0000001D; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | Reserved.                                                                                                                                                                                                                                                                                                                        |
| 6    | <b>lock_spi_cs.</b> Read-write. Reset: 0. When this bit is set to 1, SPIx1D[1:0] will be write protected. Once this bit is set to 1, this bit is write protected, only when PSPx2E[1] =1b can this bit be written again, or LPC_RstB could reset it back to 0.                                                                   |
| 5    | <b>spiprotectlock.</b> Read-write. Reset: 0. Once set, bits 3, 4, and 5 are no longer writeable.                                                                                                                                                                                                                                 |
| 4    | Reserved.                                                                                                                                                                                                                                                                                                                        |
| 3    | <b>spiprotecten0.</b> Read-write. Reset: 1. Enable SPI read/write protection base on PCI reg 0x50, 0x54, 0x58, 0x5C.                                                                                                                                                                                                             |
| 2    | Reserved.                                                                                                                                                                                                                                                                                                                        |
| 1:0  | <b>altspicsen.</b> Read-write. Reset: 0h.<br><b>Description:</b> These two bits select the alternate SPI_CS# for BIOS_ROM, these 2 bits will be write protected when SPIx00[23:22] != 11b or SPIx1D[6] =1b<br>00b BIOS ROM select SPI_CS1_L<br>01b BIOS ROM select SPI_CS2_L<br>10b BIOS ROM select SPI_CS3_L<br>11b -- reserved |

**SPIx00000020 (FCH::LPCHOSTSPIREG::SPI100ENABLE\_REGISTER)**

Read-write. Reset: 3133\_0700h.

\_aliasHOSTLEGACY; SPIx00000020; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:28 | <b>normspeed.</b> Read-write. Reset: 3h.<br><b>Description:</b> Configure the SPI bus normal speed in new SPI100 engine. If the command is not using TpmSpeed and FastSpeed, it will use NormSpeed.<br>The speed definition:<br>000: 66.66MHz<br>001: 33.33MHz<br>010: 22.22MHz<br>011: 16.66MHz<br>100: 100MHz<br>101: 800kHz<br>110: spi_spd6, defined in SPIx6C[5:0], default = 50MHz<br>111: spi_spd7, defined in SPIx6C[13:8], default = 4MHz                                         |
| 27:24 | <b>fastspeednew.</b> Read-write. Reset: 1h.<br><b>Description:</b> Configure the SPI bus speed for the following command in new SPI100 engine:<br>FAST READ<br>DDR READ (1-1-2)<br>QDR READ (1-1-4)<br>DPR READ (1-2-2)<br>QPR READ (1-4-4)<br>The speed definition:<br>000: 66.66MHz<br>001: 33.33MHz<br>010: 22.22MHz<br>011: 16.66MHz<br>100: 100MHz<br>101: 800kHz<br>110: spi_spd6, defined in SPIx6C[5:0], default = 50MHz<br>111: spi_spd7, defined in SPIx6C[13:8], default = 4MHz |
| 23:20 | <b>altspeednew.</b> Read-write. Reset: 3h.<br><b>Description:</b> Configure the SPI bus speed for the AltOpCode mode in new SPI100 engine<br>The speed definition:<br>000: 66.66MHz<br>001: 33.33MHz<br>010: 22.22MHz<br>011: 16.66MHz<br>100: 100MHz<br>101: 800kHz<br>110: spi_spd6, defined in SPIx6C[5:0], default = 50MHz<br>111: spi_spd7, defined in SPIx6C[13:8], default = 4MHz                                                                                                   |
| 19:16 | <b>tpmspeed.</b> Read-write. Reset: 3h.                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <b>Description:</b> Configure the SPI bus speed for TPM read and write in new SPI100 engine.<br>The speed definition:<br>000: 66.66MHz<br>001: 33.33MHz<br>010: 22.22MHz<br>011: 16.66MHz<br>100: 100MHz<br>101: 800kHz<br>110: spi_spd6, defined in SPIx6C[5:0], default = 50MHz<br>111: spi_spd7, defined in SPIx6C[13:8], default = 4MHz                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 15:12 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 11:8  | <b>spicsdlysel.</b> Read-write. Reset: 7h.<br><b>Description:</b> 0000: 0 SPI clock cycles of SpiCs# de-assertion time.<br>0001: 1 SPI clock cycles of SpiCs# de-assertion time.<br>0010: 2 SPI clock cycles of SpiCs# de-assertion time.<br>0011: 3 SPI clock cycles of SpiCs# de-assertion time.<br>0100: 4 SPI clock cycles of SpiCs# de-assertion time.<br>0101: 5 SPI clock cycles of SpiCs# de-assertion time.<br>0110: 6 SPI clock cycles of SpiCs# de-assertion time.<br>0111: 7 SPI clock cycles of SpiCs# de-assertion time.<br>1000: 8 SPI clock cycles of SpiCs# de-assertion time.<br>1001: 9 SPI clock cycles of SpiCs# de-assertion time.<br>1010: 10 SPI clock cycles of SpiCs# de-assertion time.<br>1011: 11 SPI clock cycles of SpiCs# de-assertion time.<br>1100: 12 SPI clock cycles of SpiCs# de-assertion time.<br>1101: 13 SPI clock cycles of SpiCs# de-assertion time.<br>1110: 14 SPI clock cycles of SpiCs# de-assertion time.<br>1111: 15 SPI clock cycles of SpiCs# de-assertion time. |
| 7:1   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 0     | <b>usespi100.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Use old SPI100 engine (verified in SB900)<br>1: Use new SPI100 engine (new since ERIE, support 100MHz)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

**SPIx00000024 (FCH::LPCHOSTSPIREG::SPI100PRECYC0\_REGISTER)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx00000024; SPI=FEC1\_0000h

**Bits Description**

31:0 Reserved.

**SPIx00000028 (FCH::LPCHOSTSPIREG::SPI100PRECYC1\_CONFIG\_REGISTER)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx00000028; SPI=FEC1\_0000h

**Bits Description**

31:0 Reserved.

**SPIx0000002C (FCH::LPCHOSTSPIREG::SPI100\_HOST\_PREFETCH\_CONFIG\_REGISTER)**

Read-write. Reset: 8000h.

**Prefetch Mechanism:**

Whenever there is read from Host, we will start to prefetch ROM data into the Host Prefetch Buffer.

The prefetch will start from the first address requested by Host, and finish when any of the following happens:

We have reached the maximum prefetch size defined in HostPrefetchSize and HostPrefOn64ByteBoundary register.

When Host requests an address that is not already prefetched and not going to be prefetched shortly, we stop current prefetch action and re-start a new prefetch with the first address being the current address requested by Host.

When there is a ROM-Write or AltOpCode request from Host, the on-going prefetch will be terminated, and the prefetch buffer will be flushed.

When there is a TPM-Write, TPM-Read, USB-Read or [EC](#)-Read request, the on-going prefetch will be halted. The contents of prefetch buffer will be preserved so that Host can access them later.

**Prefetch Prediction:**

We have logics to predict how soon the data in the current requested address will be fetched into the buffer. There are two algorithms in place. And depends on the setting of HostWillHitEn and HostHitSoonEn, we use either one of them:

"Will Hit" algorithm

We say the current requested address will be fetched "shortly" if this equation is true:

Current Requested Address  $\leq$  (First Prefetched Address + HostPrefetchSize)

"Hit Soon" algorithm

We say the current requested address will be fetched "shortly" if this equation is true:

Current Requested Address  $\leq$  (Last Prefetched Address + HostHitRange)

Of course, the current requested address also has to be equal to or larger than the first prefetched address.

If current requested address meet the criteria, the on-going prefetch will continue and Host will wait until the prefetch buffer received the requested data.

With the register default values, the prefetch behavior should be,

Start prefetch only when host request on 64 byte boundary. Otherwise, get 1~4 bytes from ROM depends on byte-enable.

When there is a ROM-write or a miss, flush the prefetch buffer immediately (including on-going prefetch action).

Once a 64 byte prefetch begins, we don't stop it until all 64 bytes are fetched, unless a ROM-write or a miss happens.

"Miss" means the requested address is not in the range of the 64 bytes being prefetched.

\_aliasHOSTLEGACY; SPIx0000002C; SPI=FEC1\_0000h

| Bits | Description                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------|
| 15   | <b>rd4dw_en_host.</b> Read-write. Reset: 1. Enable host burst to 4 Dword. This bit need be set to 1 for SPI DMA Copy to work properly. |
| 14:0 | Reserved.                                                                                                                              |

**SPIx0000002E (FCH::LPCHOSTSPIREG::TPM\_SPI\_DI\_TIMEOUT\_REGISTER)**

Read-write. Reset: 8000h.

\_aliasHOSTLEGACY; SPIx0000002E; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                    |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:11 | <b>tpm_di_to_cnt.</b> Read-write. Reset: 10h. SPI_DI_TO_counter is using 125ms Rtc8HzClk to increment, and these bits specify the timeout value. Due to synchronization, the actual delay is $N \times 125\text{ms}$ to $(N+1) \times 125\text{ms}$ . Default=16 for 2 seconds to 2.125 seconds. In spec, TIMEOUT_B is specified at 2 seconds. |
| 10    | <b>tpm_di_to_status.</b> Read-write. Reset: 0. When bit[9]=1 and TPM_DI_TO_counter reaches timeout, this bit is set to '1'. Software can write '1' to clear it., write '0' has no effect.                                                                                                                                                      |
| 9     | <b>tpm_di_to_enable.</b> Read-write. Reset: 0. When set to 1, if SPI_DI=0 during TPM access wait window, internal TPM_DI_TO_counter will start counting, once it reaches timeout value, internal SPI_DI will be forced to '1' to terminate TPM access, read cycle will return all '1' data, write cycle will be discarded.                     |
| 8:0   | Reserved.                                                                                                                                                                                                                                                                                                                                      |

**SPIx00000030 (FCH::LPCHOSTSPIREG::ROM2\_ADDR\_OVERRIDE\_REGISTER)**

Read-write. Reset: 14C0h.

NOTE1: When either bit[22] and/or bit[23] of offset 00h are cleared, bit[4:0] of Host\_Mem\_Reg:30h become read-only and cannot be changed any more.

NOTE2: 'ROM Address' used in this register description will be used as LPC ROM Address when LPC ROM is used, LPC ROM address does not use XOR. If SPI ROM is used, SPI ROM address [31:24] will be after XOR with offset x5C[7:0], SPIROM\_page[31:24]

\_aliasHOSTLEGACY; SPIx00000030; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:4 | Reserved.                                                                                                                                                                                                                                                         |
| 3    | <b>r2msk25.</b> Read-write. Reset: 0. When set to '1', R2VAL25 (bit[1]) will replace ROM Address[25] value when ROM2 range (defined by LPC PCIX6C[31:0]) is used to access ROM.                                                                                   |
| 2    | <b>r2msk24.</b> Read-write. Reset: 0. When set to '1', R2VAL24 (bit[0]) will replace ROM Address[24] value when ROM2 range (defined by LPC PCIX6C[31:0]) is used to access ROM.                                                                                   |
| 1    | <b>r2val25.</b> Read-write. Reset: 0.<br><b>Description:</b> When bit[3]=1, this bit will replace ROM Address[25] value when ROM2 range (defined by LPC PCIX6C[31:0]) is used to access ROM.<br>When bit[3]=0, ROM Address[25] will be derived from Host Address. |
| 0    | <b>r2val24.</b> Read-write. Reset: 0.<br><b>Description:</b> When bit[2]=1, this bit will replace ROM Address[24] value when ROM2 range (defined by LPC PCIX6C[31:0]) is used to access ROM.<br>When bit[2]=0, ROM Address[24] will be derived from Host Address. |

**SPIx00000032 (FCH::LPCHOSTSPIREG::SPI100\_DUMMY\_CYCLE\_CONFIG\_REGISTER)**

Read-write. Reset: 4608h.

The cycle count for mode bits shall be included in the Dummy Cycle Config Register.

For example, if ROM vendor requires 4 cycles of dummy bits and 2 cycles of mode bits in QPR Read, we should put 6 in QPR\_DummyCyc[3:0] register. As a result, [FCH](#) will send 1 byte of mode bits and 2 bytes of dummy bits through Dout, Din, WP# and HOLD# pins.

NOTE:

For IndexMode, the dummy byte should be programmed into the IndexMode FIFO. The SPI100 Dummy Cycle Config Registers are not for IndexMode dummy bytes.

\_aliasHOSTLEGACY; SPIx00000032; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                        |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:12 | <b>dpr_dummycyc.</b> Read-write. Reset: 4h. Configure dummy cycle count for DPR READ (1-2-2) command. The cycle count shall include the cycle counts for both mode bits and dummy bits. (See NOTE) |
| 11:8  | <b>qpr_dummycyc.</b> Read-write. Reset: 6h. Configure dummy cycle count for QPR READ (1-4-4) command. The cycle count shall include the cycle counts for both mode bits and dummy bits. (See NOTE) |
| 7:5   | Reserved.                                                                                                                                                                                          |
| 4:0   | <b>fastread_dummycyc.</b> Read-write. Reset: 08h. Configure dummy cycle count for FAST READ command.                                                                                               |



**SPIx00000034 (FCH::LPCHOSTSPIREG::SPI100\_RX\_TIMING\_CONFIG0\_REGISTER)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx00000034; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:7 | Reserved.                                                                                                                                                              |
| 6    | <b>dis_index_flush_rbuf.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Index access will flush prefetch buffer<br>1: Index access will not flush prefetch buffer |
| 5    | Reserved.                                                                                                                                                              |
| 4    | <b>flushhostprefetch.</b> Read-write. Reset: 0. If SW wants to flush the SPI100 Host prefetch buffer, SW can set this bit and then clear it.                           |
| 3:0  | Reserved.                                                                                                                                                              |

**SPIx00000038 (FCH::LPCHOSTSPIREG::SPI100\_RX\_TIMING\_CONFIG1\_REGISTER)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx00000038; SPI=FEC1\_0000h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**SPIx0000003C (FCH::LPCHOSTSPIREG::SPI100\_RX\_TIMING\_CONFIG2\_REGISTER)**

Read-write. Reset: 8800h.

\_aliasHOSTLEGACY; SPIx0000003C; SPI=FEC1\_0000h

| Bits | Description |
|------|-------------|
| 15:0 | Reserved.   |

**SPIx00000040 (FCH::LPCHOSTSPIREG::DDRCMDCODE)**

Read-write. Reset: 3Bh.

\_aliasHOSTLEGACY; SPIx00000040; SPI=FEC1\_0000h

| Bits | Description                                                              |
|------|--------------------------------------------------------------------------|
| 7:0  | <b>ddr_cmd.</b> Read-write. Reset: 3Bh. Command code of DDR (1-1-2 read) |

**SPIx00000041 (FCH::LPCHOSTSPIREG::QDRCMDCODE)**

Read-write. Reset: 6Bh.

\_aliasHOSTLEGACY; SPIx00000041; SPI=FEC1\_0000h

| Bits | Description                                                              |
|------|--------------------------------------------------------------------------|
| 7:0  | <b>qdr_cmd.</b> Read-write. Reset: 6Bh. Command code of QDR (1-1-4 read) |

**SPIx00000042 (FCH::LPCHOSTSPIREG::DPRCMDCODE)**

Read-write. Reset: BBh.

\_aliasHOSTLEGACY; SPIx00000042; SPI=FEC1\_0000h

| Bits | Description                                                              |
|------|--------------------------------------------------------------------------|
| 7:0  | <b>dpr_cmd.</b> Read-write. Reset: BBh. Command code of DPR (1-2-2 read) |

**SPIx00000043 (FCH::LPCHOSTSPIREG::QPRCMDCODE)**

Read-write. Reset: EBh.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[31:0] of Host\_Mem\_Reg:40h become read-only and cannot be changed any more.

\_aliasHOSTLEGACY; SPIx00000043; SPI=FEC1\_0000h

| Bits | Description                                                              |
|------|--------------------------------------------------------------------------|
| 7:0  | <b>qpr_cmd.</b> Read-write. Reset: EBh. Command code of QPR (1-4-4 read) |

**SPIx00000044 (FCH::LPCHOSTSPIREG::MODEBYTE)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000044; SPI=FEC1\_0000h

| Bits | Description                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>modebyte.</b> Read-write. Reset: 00h. Whenever DPR_CMD or QPR_CMD is used, ModeByte is also sent out onto the SPI stream |

**SPIx00000045 (FCH::LPCHOSTSPIREG::CMDCODE)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000045; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>spi_opcode.</b> Read-write. Reset: 00h. When software uses the alternate program method to communicate with the SPI ROM, this register contains the OPCODE (command code). |

**SPIx00000047 (FCH::LPCHOSTSPIREG::CMDTRIGGER)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000047; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>execute.</b> Read-write. Reset: 0.<br><b>Description:</b> Write 1 to execute the transaction in the alternate program registers. Writing 0 has no effect. When the transaction is complete, this bit will return 0. If the command is an illegal command, the bit cannot be set and thereby cannot execute.<br>Note: before set Execute=1 software should read Host_Mem_Regx4C[31]: SpiBusy=0 to make sure no SPI operation is going on. |
| 6:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**SPIx00000048 (FCH::LPCHOSTSPIREG::TXBYTECOUNT)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000048; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>txbytecount.</b> Read-write. Reset: 00h. Number of bytes to be sent to SPI ROM in Index mode. This number does not include SPI_OpCode specified in offset 0x45 |

**SPIx0000004B (FCH::LPCHOSTSPIREG::RXBYTECOUNT)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000004B; SPI=FEC1\_0000h

| Bits | Description                                                                                                |
|------|------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>rxbytecount.</b> Read-write. Reset: 00h. Number of bytes to be received from the SPI ROM in Index mode. |

**SPIx0000004C (FCH::LPCHOSTSPIREG::SPISTATUS)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx0000004C; SPI=FEC1\_0000h

| Bits | Description                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>spibusy.</b> Read-only. Reset: 0.<br><b>Description:</b> 1: Spi bus is busy<br>0: Spi busy is idle                      |
| 30:8 | Reserved.                                                                                                                  |
| 7:0  | <b>donebytecount.</b> Read-only. Reset: 00h. Indicate how many bytes has been received or sent in previous spi transaction |

**SPIx00000050 (FCH::LPCHOSTSPIREG::ADDR32CTRL0)**

Read-write. Reset: 0C13\_1200h.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[31:1] of Host\_Mem\_Reg:50h become read-only and cannot be changed any more.

\_aliasHOSTLEGACY; SPIx00000050; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>freadcmd_addr_32.</b> Read-write. Reset: 0Ch. Fast Read Command Code of 32-bit address mode                                                                                                                                                                                                                        |
| 23:16 | <b>readcmd_addr_32.</b> Read-write. Reset: 13h. Read Command Code of 32-bit address mode                                                                                                                                                                                                                              |
| 15:8  | <b>bytewrcmd_addr_32.</b> Read-write. Reset: 12h. Byte Program Command Code of 32-bit address mode                                                                                                                                                                                                                    |
| 7:1   | Reserved.                                                                                                                                                                                                                                                                                                             |
| 0     | <b>spi_rom_addr_32.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: 24-bit address SPI ROM<br>1: 32-bit address SPI ROM<br>This bit can only be changed when Host_Mem_Regx00[31]: SpiBusy =0, since this bit will affect Host Address to SPI ROM address mapping.<br>This bit is write-protect when PSPxFC[14] =1 |

**SPIx00000054 (FCH::LPCHOSTSPIREG::ADDR32CTRL1)**

Read-write. Reset: ECBC\_6C3Ch.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[31:0] of Host\_Mem\_Reg:54h become read-only and cannot be changed any more.

\_aliasHOSTLEGACY; SPIx00000054; SPI=FEC1\_0000h

| Bits  | Description                                                                                     |
|-------|-------------------------------------------------------------------------------------------------|
| 31:24 | <b>qpr_cmd_addr_32.</b> Read-write. Reset: ECh. QPR (1-4-4) Command Code of 32-bit address mode |
| 23:16 | <b>dpr_cmd_addr_32.</b> Read-write. Reset: BCh. DPR (1-2-2) Command Code of 32-bit address mode |
| 15:8  | <b>qdr_cmd_addr_32.</b> Read-write. Reset: 6Ch. QDR (1-1-4) Command Code of 32-bit address mode |
| 7:0   | <b>ddr_cmd_addr_32.</b> Read-write. Reset: 3Ch. DDR (1-1-2) Command Code of 32-bit address mode |

**SPIx00000058 (FCH::LPCHOSTSPIREG::ADDR32CTRL2)**

Read-write. Reset: 0000\_4608h.

\_aliasHOSTLEGACY; SPIx00000058; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                 |
| 15:12 | <b>dpr_dummycyc_addr_32.</b> Read-write. Reset: 4h. Configure dummy cycle count for DPR READ (1-2-2) command in 32-bit address mode (24-bit address mode at 0x32[15:12]). |
| 11:8  | <b>qpr_dummycyc_addr_32.</b> Read-write. Reset: 6h. Configure dummy cycle count for QPR READ (1-4-4) command in 32-bit address mode (24-bit address mode at 0x32[11:8]).  |
| 7:5   | Reserved.                                                                                                                                                                 |
| 4:0   | <b>fastread_dummycyc_addr_32.</b> Read-write. Reset: 08h. Configure dummy cycle count for FAST READ command in 32-bit address mode (24-bit address mode at 0x32[4:0])     |

**SPIx0000005C (FCH::LPCHOSTSPIREG::ADDR32CTRL3)**

Read-write. Reset: 0000\_0000h.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[7:0] of Host\_Mem\_Reg:5Ch become read-only and cannot be changed any more.

\_aliasHOSTLEGACY; SPIx0000005C; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:8 | Reserved.                                                                                                                                                                                                                                                                                                                                                   |
| 7:0  | <b>spirom_page_31_24</b> . Read-write. Reset: 00h.<br><b>Description:</b> Used for 32-bit address mode, produce SPIROMAddr[31:24] use XOR.<br>SPIROMAddr[31:24] =<br>SPI_ROM_page[31:24] ^ HostMemAddr[31:24]<br>These bits can only be changed when Host_Mem_Regx00[31]: SpiBusy =0, since these bits will affect Host Address to SPI ROM address mapping. |

**SPIx00000060 (FCH::LPCHOSTSPIREG::BAR\_64MB\_ROM3\_LOW)**

Read-write. Reset: 0000\_0000h.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[31:26] of Host\_Mem\_Reg:60h become read-only and cannot be changed any more.

\_aliasHOSTLEGACY; SPIx00000060; SPI=FEC1\_0000h

| Bits  | Description                                                                                          |
|-------|------------------------------------------------------------------------------------------------------|
| 31:26 | <b>bar_64mb_31_26</b> . Read-write. Reset: 00h. Specify Base address [31:26] of 64MB BIOS ROM space. |
| 25:0  | Reserved.                                                                                            |

**SPIx00000064 (FCH::LPCHOSTSPIREG::BAR\_64MB\_ROM3\_HIGH)**

Read-write. Reset: 0000\_00FDh.

When either bit[22] and/or bit[23] of offset 00h are cleared, bit[31:0] of Host\_Mem\_Reg:64h become read-only and cannot be changed any more.

\_aliasHOSTLEGACY; SPIx00000064; SPI=FEC1\_0000h

| Bits | Description                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>bar_64mb_63_32</b> . Read-write. Reset: 0000_00FDh. Specify Base address [63:32] of 64MB BIOS ROM space. |

**SPIx00000068 (FCH::LPCHOSTSPIREG::SPI\_RESTRICTEDCMD3\_REGISTER)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx00000068; SPI=FEC1\_0000h

| Bits  | Description                                                             |
|-------|-------------------------------------------------------------------------|
| 31:24 | <b>restrictedcmd11</b> . Read-write. Reset: 00h. Same as RestrictedCmd0 |
| 23:16 | <b>restrictedcmd10</b> . Read-write. Reset: 00h. Same as RestrictedCmd0 |
| 15:8  | <b>restrictedcmd9</b> . Read-write. Reset: 00h. Same as RestrictedCmd0  |
| 7:0   | <b>restrictedcmd8</b> . Read-write. Reset: 00h. Same as RestrictedCmd0  |

**SPIx0000006C (FCH::LPCHOSTSPIREG::SPISPEED)**

Read-write. Reset: 3204h.

\_aliasHOSTLEGACY; SPIx0000006C; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 13:8  | <b>spi_spd7.</b> Read-write. Reset: 32h.<br><b>Description:</b> This register define SPI clock frequency when speed = 0x7 is specified in NormSpeed(offset 0x20[31:28]), FastSpeedNew(offset 0x20[27:24]), AltSpeedNew(offset 0x20[23:20]), TpmSpeed(offset 0x20[19:16]).<br>For register value definition, please refer to bit[5:0] of this register                                                                                                                                                                                       |
| 7:6   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 5:0   | <b>spi_spd6.</b> Read-write. Reset: 04h.<br><b>Description:</b> This register define SPI clock frequency when speed = 0x6 is specified in NormSpeed(offset 0x20[31:28]), FastSpeedNew(offset 0x20[27:24]), AltSpeedNew(offset 0x20[23:20]), TpmSpeed(offset 0x20[19:16]).<br>With value = N in this register, SPI clock = 200MHz/N<br>Value 0 3 are not allowed to write to this register, hardware will write 0x4 if software is writing 0 3 to this register<br>0x4: 50MHz, 200MHz/4<br>0x32: 4MHz, 200MHz/50<br>0x3f: 3.17MHz, 200MHz/63 |

**SPIx00000070 (FCH::LPCHOSTSPIREG::HOST\_INT)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx00000070; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                                                                                                              |
| 23:16 | <b>host_int_enable.</b> Read-write. Reset: 00h.<br><b>Description:</b> 0: event occurrence will not set Host_Int_status<br>1: event occurrence will set Host_Int_status                                                                                                |
| 15:8  | <b>host_int_mask.</b> Read-write. Reset: 00h.<br><b>Description:</b> 0: associated Host_Int_status =1 will generate LPC_SPI_INT<br>1: mask interrupt when Host_Int_Status =1                                                                                           |
| 7:0   | <b>host_int_status.</b> Read-write. Reset: 00h.<br><b>Description:</b> Each bit will set to '1' when event occur and corresponding Host_Int_enable =1, software need write '1' to clear it.<br>[0]: Host Index transfer done<br>[1]: DMA Romcp done<br>[7:2]: reserved |

**SPIx00000074 (FCH::LPCHOSTSPIREG::LPC\_SPI\_INT)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; SPIx00000074; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 15:8  | <b>hid_int_status</b> . Read-only. Reset: 00h.<br><br><b>Description:</b> HID interrupt normally route to MP2 (HIDx150[15] =0), and bit[15:8] will always read back 0x00. When HID interrupt route to Host (HIDx150[15] =1), bit[15:8] are HID Interrupt status, software can read these bits to find Interrupt source. To clear those HID interrupt status, software need write to HIDx154<br>[8]: HID Index transfer done<br>[9]: HID DMA Copy done<br>[10]: HID-basic-write done<br>[11]: HID-basic-read done<br>[12]: HID-acc-write done<br>[13]: HID-acc-read done<br>[14]: HID Int_do_DMA_acc_rd done<br>[15]: HID device IRQ input |
| 7:4   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 3     | <b>psp_index_int</b> . Read-only. Reset: 0. If read '1', PSP_Index_Int is active, to clear it, write PSPx70[0] =1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2     | <b>hpf_index_int</b> . Read-only. Reset: 0. If read '1', HFP_Index_Int is active, to clear it, write HFPx70[0] =1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1     | <b>romcp_int</b> . Read-only. Reset: 0. If read '1', Romcp_Int is active, to clear it, write SPIx70[1] =1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0     | <b>host_index_int</b> . Read-only. Reset: 0. If read '1', Host_Index_Int is active, to clear it, write SPIx70[0] =1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

**SPIx00000080 (FCH::LPCHOSTSPIREG::FIFO0)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000080; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000081 (FCH::LPCHOSTSPIREG::FIFO1)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000081; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000082 (FCH::LPCHOSTSPIREG::FIFO2)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000082; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000083 (FCH::LPCHOSTSPIREG::FIFO3)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000083; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000084 (FCH::LPCHOSTSPIREG::FIFO4)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000084; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000085 (FCH::LPCHOSTSPIREG::FIFO5)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000085; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000086 (FCH::LPCHOSTSPIREG::FIFO6)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000086; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000087 (FCH::LPCHOSTSPIREG::FIFO7)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000087; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000088 (FCH::LPCHOSTSPIREG::FIFO8)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000088; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000089 (FCH::LPCHOSTSPIREG::FIFO9)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000089; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000008A (FCH::LPCHOSTSPIREG::FIFO10)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000008A; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000008B (FCH::LPCHOSTSPIREG::FIFO11)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000008B; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000008C (FCH::LPCHOSTSPIREG::FIFO12)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000008C; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000008D (FCH::LPCHOSTSPIREG::FIFO13)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000008D; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000008E (FCH::LPCHOSTSPIREG::FIFO14)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000008E; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000008F (FCH::LPCHOSTSPIREG::FIFO15)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000008F; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000090 (FCH::LPCHOSTSPIREG::FIFO16)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000090; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000091 (FCH::LPCHOSTSPIREG::FIFO17)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000091; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |



**SPIx00000092 (FCH::LPCHOSTSPIREG::FIFO18)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000092; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000093 (FCH::LPCHOSTSPIREG::FIFO19)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000093; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000094 (FCH::LPCHOSTSPIREG::FIFO20)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000094; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000095 (FCH::LPCHOSTSPIREG::FIFO21)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000095; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000096 (FCH::LPCHOSTSPIREG::FIFO22)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000096; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000097 (FCH::LPCHOSTSPIREG::FIFO23)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000097; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000098 (FCH::LPCHOSTSPIREG::FIFO24)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000098; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx00000099 (FCH::LPCHOSTSPIREG::FIFO25)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx00000099; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000009A (FCH::LPCHOSTSPIREG::FIFO26)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000009A; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000009B (FCH::LPCHOSTSPIREG::FIFO27)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000009B; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000009C (FCH::LPCHOSTSPIREG::FIFO28)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000009C; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000009D (FCH::LPCHOSTSPIREG::FIFO29)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000009D; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000009E (FCH::LPCHOSTSPIREG::FIFO30)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000009E; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx0000009F (FCH::LPCHOSTSPIREG::FIFO31)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx0000009F; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A0 (FCH::LPCHOSTSPIREG::FIFO32)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A0; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A1 (FCH::LPCHOSTSPIREG::FIFO33)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A1; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A2 (FCH::LPCHOSTSPIREG::FIFO34)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A2; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A3 (FCH::LPCHOSTSPIREG::FIFO35)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A3; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A4 (FCH::LPCHOSTSPIREG::FIFO36)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A4; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A5 (FCH::LPCHOSTSPIREG::FIFO37)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A5; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A6 (FCH::LPCHOSTSPIREG::FIFO38)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A6; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A7 (FCH::LPCHOSTSPIREG::FIFO39)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A7; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A8 (FCH::LPCHOSTSPIREG::FIFO40)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A8; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000A9 (FCH::LPCHOSTSPIREG::FIFO41)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000A9; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000AA (FCH::LPCHOSTSPIREG::FIFO42)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000AA; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000AB (FCH::LPCHOSTSPIREG::FIFO43)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000AB; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000AC (FCH::LPCHOSTSPIREG::FIFO44)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000AC; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000AD (FCH::LPCHOSTSPIREG::FIFO45)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000AD; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000AE (FCH::LPCHOSTSPIREG::FIFO46)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000AE; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000AF (FCH::LPCHOSTSPIREG::FIFO47)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000AF; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B0 (FCH::LPCHOSTSPIREG::FIFO48)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B0; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B1 (FCH::LPCHOSTSPIREG::FIFO49)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B1; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B2 (FCH::LPCHOSTSPIREG::FIFO50)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B2; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B3 (FCH::LPCHOSTSPIREG::FIFO51)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B3; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B4 (FCH::LPCHOSTSPIREG::FIFO52)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B4; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifo</b> data. Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B5 (FCH::LPCHOSTSPIREG::FIFO53)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B5; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B6 (FCH::LPCHOSTSPIREG::FIFO54)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B6; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B7 (FCH::LPCHOSTSPIREG::FIFO55)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B7; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B8 (FCH::LPCHOSTSPIREG::FIFO56)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B8; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000B9 (FCH::LPCHOSTSPIREG::FIFO57)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000B9; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000BA (FCH::LPCHOSTSPIREG::FIFO58)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000BA; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000BB (FCH::LPCHOSTSPIREG::FIFO59)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000BB; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000BC (FCH::LPCHOSTSPIREG::FIFO60)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000BC; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000BD (FCH::LPCHOSTSPIREG::FIFO61)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000BD; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000BE (FCH::LPCHOSTSPIREG::FIFO62)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000BE; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000BF (FCH::LPCHOSTSPIREG::FIFO63)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000BF; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000C0 (FCH::LPCHOSTSPIREG::FIFO64)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000C0; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000C1 (FCH::LPCHOSTSPIREG::FIFO65)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000C1; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000C2 (FCH::LPCHOSTSPIREG::FIFO66)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000C2; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000C3 (FCH::LPCHOSTSPIREG::FIFO67)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000C3; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000C4 (FCH::LPCHOSTSPIREG::FIFO68)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000C4; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000C5 (FCH::LPCHOSTSPIREG::FIFO69)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000C5; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000C6 (FCH::LPCHOSTSPIREG::FIFO70)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000C6; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |

**SPIx000000C7 (FCH::LPCHOSTSPIREG::FIFO71)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; SPIx000000C7; SPI=FEC1\_0000h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>fifodata</b> . Read-write. Reset: 00h. Data FiFo byte N which used in command mode to send or receive data. These FIFO can be written/read in Double-word, Word, or Byte. |



**SPIx000000FC (FCH::LPCHOSTSPIREG::SPI\_MISCCNTRL)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; SPIx000000FC; SPI=FEC1\_0000h

| Bits  | Description                                                                                                                                                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                                                                                                                                                                                                                        |
| 10    | <b>psp_own_spi</b> . Read-write. Reset: 0. Read Only. Value of PSPxFC[5]                                                                                                                                                                                                                         |
| 9     | <b>hfp_own_spi</b> . Read-write. Reset: 0. Read Only. Value of HFPxFC[5]                                                                                                                                                                                                                         |
| 8     | Reserved.                                                                                                                                                                                                                                                                                        |
| 7     | <b>en_map_spi_bx_to_cfg</b> . Read-write. Reset: 0. When set to 1, write/read to SPIxB0 xBF will be map to write/read LPC PCICFGxB0 xBF. This will allow processor to do SPI DMA Copy (controlled by LPC PCICFGxB0 xB9) through SPI Register space.                                              |
| 6     | <b>romcp_new_scheme</b> . Read-write. Reset: 0.<br><b>Description:</b> Set to '1' to allow direct or Index mode memory read to SPI ROM to be 'inserted' between SPI DMA Copy cache line transfer<br>If this bit is '0', software should not do any SPI Memory read during SPI DMA Copy operation |
| 5     | <b>hbios_spimutex</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Host-BIOS doesn't own SpiMutex<br>1: Host-BIOS owns SpiMutex<br>Host-BIOS can't write the bit to 1 if the SpiMutex is owned by <a href="#">PSP</a> (PSPxFC[5]=1) or Host-FP (HFPxFC[5]=1)                                |
| 4:0   | Reserved.                                                                                                                                                                                                                                                                                        |

**14.3.13.4 eSPI Registers**

The [MMIO](#) base address for accessing eSPI registers is defined in SPI BASE ADDR register (FCH::LPCPCICFG::SPI\_BASE\_ADDR).

**ESPI1SMNx00000000...ESPISMNx00000000 (FCH::ITF::ESPI::DN\_TXHDR\_0th)**

Reset: 0020\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000000; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000000; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000000; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000000; ESPI=FEC20000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>DNCMD_HDATA2.</b> Read-write. Reset: 00h.<br><b>Description:</b> The definition for this field is depended on SW_CMD_TYPE<br>Independent channel command selected:<br>Reserved, Always be 00h<br>Peripheral selected : Length[7:0]<br>VW selected: Reserved, Always be 00h<br>OOB selected: Length[7:0]<br>FLASH selected: Length[7:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 23:16 | <b>DNCMD_HDATA1.</b> Read-write. Reset: 20h.<br><b>Description:</b> The definition for this field is depended on SW_CMD_TYPE<br>Independent channel command selected:<br>Addres [7:0] of SET_CONFIGURATION/GET_CONFIGURATION<br>Bit[1:0] needs to be 00.<br>Note: In-Band command. These bits are ignored.<br>Peripheral selected<br>[23:20]: Tag<br>[19:16]: Length[11:8]<br>VW selected: Reserved, Always be 00h<br>OOB selected:<br>[23:20]: Tag<br>[19:16]: Length[11:8]<br>FLASH selected:<br>[23:20]: Tag<br>[19:16]: Length[11:8]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 15:8  | <b>DNCMD_HDATA0.</b> Read-write. Reset: 00h.<br><b>Description:</b> The definition for this field is depended on SW_CMD_TYPE.<br>Independent command selected:<br>Address [15:8] of SET_CONFIGURATION/GET_CONFIGURATION.<br>[15:12]:0h<br>[11:8]: address[11:8]<br>Note: In-Band command. These bits are ignored.<br>Peripheral selected:<br>a) SW programs this byte to be Message cycle type(0001xxxy) to instruct the eSPI controller send down peripheral message with data(8 bytes+data byte N) or without data(total 8 bytes)<br>b) SW programs this byte to be 'nsuccessful completion to instruct the eSPI controller send down unsuccessful completion<br>VW selected: It indicates the Virtual Wire Count will be send down. Bit[5:0] represends how many Virtual Wire groups to be commnunicated in the same packets.<br>NOTE: In the current design, it is limited to 16 groups(bit[5:4]=00) to save the registers needed. Only 8 indexes are defined In current eSPI spec v0.7.<br>OOB selected: SW programs this byte to be 0x21 to instruct the eSPI controller send send down Tunneled SMBUS message to slave<br>SW prgrames this byte to be CycleType for Flash Completion, including Cpl/Unsuccessful Cpl/CplID |
| 7:4   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3     | <b>DNCMD_STATUS.</b> Read,Write-0-only. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> RW0C.<br>Set: The bit needs to be set last by Software after all eSPI specific registers are all programmed to inform the protocol layer to send down command or packet.<br>Clear: Hardware will automatically clear this bit after the packet is sent down.                                                                                                                                                                                                |
| 2:0 | <b>DNCMD_TYPE.</b> Read-write. Reset: 0h.<br><b>Description:</b> TX Command Type:<br>000: Set Configuration (Independent command)<br>001: Get Configuration (Independent command)<br>010: In-band RESET command (Independent command)<br>011: Peripheral Unsuccessful Cpl down stream<br>100: Peripheral Channel message down stream<br>101: VW Channel down stream<br>110: <a href="#">OOB</a> Channel down stream<br>111: Flash Channel Cpl/CplD/Unsuccessful Cpl down stream |

#### ESPI1SMNx00000004...ESPISMNx00000004 (FCH::ITF::ESPI::DN\_TXHDR\_1)

| Read-write. Reset: 0000_0001h.                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">_link1_aliasSMN</a> ; ESPI1SMNx00000004; ESPI1SMN=02DCA000h  |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">_link0_aliasSMN</a> ; ESPISMNx00000004; ESPISMN=02DC5000h    |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">_link1_aliasHOSTLEGACY</a> ; ESPI1x00000004; ESPI1=FEC30000h |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <a href="#">_link0_aliasHOSTLEGACY</a> ; ESPIx00000004; ESPI=FEC20000h   |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Bits                                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 31:24                                                                    | <b>DNCMD_HDATA6.</b> Read-write. Reset: 00h.<br><b>Description:</b> The definition for this field is depended on SW_CMD_TYPE<br>Independent channel command selected: data[31:24]<br>Peripheral selected : Message specific byte 2<br>VW selected: Reserved, Always be 00h<br><a href="#">OOB</a> selected: Reserved, Always be 00h<br>FLASH selected: Reserved, Always be 00h                                                        |
| 23:16                                                                    | <b>DNCMD_HDATA5.</b> Read-write. Reset: 00h.<br><b>Description:</b> The definition for this field is depended on SW_CMD_TYPE<br>Independent channel command selected: data[23:16]<br>Peripheral selected : Message specific byte 1<br>VW selected: Reserved, Always be 00h<br><a href="#">OOB</a> selected:<br><a href="#">SMBus</a> Byte Count. Need to program not greater than 128bytes<br>FLASH selected: Reserved, Always be 00h |
| 15:8                                                                     | <b>DNCMD_HDATA4.</b> Read-write. Reset: 00h.<br><b>Description:</b> The definition for this field is depended on SW_CMD_TYPE<br>Independent channel command selected:<br>data[15:8]<br>Peripheral selected : Message specific byte 0<br>VW selected: Reserved, Always be 00h<br><a href="#">OOB</a> selected: <a href="#">SMBus</a> Command Op Code.<br>FLASH selected: Reserved, Always be 00h                                       |
| 7:0                                                                      | <b>DNCMD_HDATA3.</b> Read-write. Reset: 01h.<br><b>Description:</b> The definition for this field is depended on SW_CMD_TYPE<br>Independent channel command selected: data[7:0]<br>Peripheral selected : Message code [7:0]<br>VW selected: Reserved, Always be 00h<br><a href="#">OOB</a> selected:<br><a href="#">SMBus</a> Slave Address. Bit[0] needs to program to 1<br>FLASH selected: Reserved, Always be 00h                  |

**ESPI1SMNx00000008...ESPISMNx00000008 (FCH::ITF::ESPI::DN\_TXHDR\_2)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000008; ESPI1SMN=02DCA000h\_link0\_aliasSMN; ESPISMNx00000008; ESPISMN=02DC5000h\_link1\_aliasHOSTLEGACY; ESPI1x00000008; ESPI1=FEC30000h\_link0\_aliasHOSTLEGACY; ESPIx00000008; ESPI=FEC20000h**Bits Description**

31:8 Reserved.

7:0 **DNCMD\_HDATA7**. Read-write. Reset: 00h.**Description:** The definition for this field is depended on SW\_CMD\_TYPE

Independent channel command selected: Reserved, Always be 00h

Peripheral selected : Message specific byte 3

VW selected: Reserved, Always be 00h

[OOB](#) selected: Reserved, Always be 00h

FLASH selected: Reserved, Always be 00h

**ESPI1SMNx0000000C...ESPISMNx0000000C (FCH::ITF::ESPI::DN\_TXDATA\_PORT)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx0000000C; ESPI1SMN=02DCA000h\_link0\_aliasSMN; ESPISMNx0000000C; ESPISMN=02DC5000h\_link1\_aliasHOSTLEGACY; ESPI1x0000000C; ESPI1=FEC30000h\_link0\_aliasHOSTLEGACY; ESPIx0000000C; ESPI=FEC20000h**Bits Description**31:24 **DN\_TXDATA\_B3**. Read-write. Reset: 00h.**Description:** The definition for this field is depended on SW\_CMD\_TYPE

Independent channel command selected: Reserved, Always be 00h

Peripheral selected : Message Data DWn[31:24]

VW selected: VW Index Group 2n+1 data

[OOB](#) selected: OOB Message DWn[31:24]

FLASH selected: Flash Cpl Data DWn[31:24]

23:16 **DN\_TXDATA\_B2**. Read-write. Reset: 00h.**Description:** The definition for this field is depended on SW\_CMD\_TYPE

Independent channel command selected: Reserved, Always be 00h

Peripheral selected : Message Data DWn[23:16]

VW selected: VW Index Group 2n+1

[OOB](#) selected: OOB Message DWn[23:16]

FLASH selected: Flash Cpl Data DWn[23:16]

15:8 **DN\_TXDATA\_B1**. Read-write. Reset: 00h.**Description:** The definition for this field is depended on SW\_CMD\_TYPE

Independent channel command selected: Reserved, Always be 00h

Peripheral selected : Message Data DWn[15:8]

VW selected: VW Index Group 2n Data

[OOB](#) selected: OOB Message DWn[15:8]

FLASH selected: Flash Cpl Data DWn[15:8]

7:0 **DN\_TXDATA\_B0**. Read-write. Reset: 00h.**Description:** The definition for this field is depended on SW\_CMD\_TYPE

Independent channel command selected: Reserved, Always be 00h

Peripheral selected : Message Data DWn[7:0]

VW selected: VW Index Group 2n

[OOB](#) selected: OOB Message DWn[7:0]

FLASH selected: Flash Cpl Data DWn[7:0]

**ESPI1SMNx00000010...ESPISMNx00000010 (FCH::ITF::ESPI::UP\_RXHDR\_0)**

Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000010; ESPI1SMN=02DCA000h\_link0\_aliasSMN; ESPISMNx00000010; ESPISMN=02DC5000h\_link1\_aliasHOSTLEGACY; ESPI1x00000010; ESPI1=FEC30000h\_link0\_aliasHOSTLEGACY; ESPIx00000010; ESPI=FEC20000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                      |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>UPCMD_HDATA2.</b> Read-only. Reset: 00h.<br><b>Description:</b> RX_LOW_LEN:<br>This field store the Length[7:0] from GET_FLASH_NP/GET_OOB<br>[31:24] Length[7:0]                                                                                                                                                                                              |
| 23:16 | <b>UPCMD_HDATA1.</b> Read-only. Reset: 00h.<br><b>Description:</b> RX_TAG_LEN:<br>This field store the Tag and Length[11:8] which from eSPI packet received by GET_FLASH_NP/GET_OOB<br>[23:20], Tag<br>[19:16] Length[11:8]                                                                                                                                      |
| 15:8  | <b>UPCMD_HDATA0.</b> Read-only. Reset: 00h. Cycle Type This field store the cycle type from GET_FLASH_NP and GET_OOB                                                                                                                                                                                                                                             |
| 7:6   | Reserved.                                                                                                                                                                                                                                                                                                                                                        |
| 5:4   | <b>SLAVE_SEL.</b> Read-only. Reset: 0h.<br><b>Description:</b> Slave N Received selected<br>00: The upstream packet is from Slave0<br>Others: Reserved                                                                                                                                                                                                           |
| 3     | <b>UPCMD_STATUS.</b> Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> RW0C Valid bit Status:<br>This bit will be set after received <a href="#">OOB</a> message packet or Flash request packet, and eSPI will not send down another GET_OOB or GET_FLASH_NP before the Valid bit cleared by SW.<br>This bit can be cleared by SW writing 1 to this field. |
| 2:0   | <b>UPCMD_TYPE.</b> Read-only. Reset: 0h.<br><b>Description:</b> 000: Flash Channel Request (GET_FLASH_NP)<br>001: Upstream <a href="#">OOB</a> message (GET_OOB) Others: Reserved.                                                                                                                                                                               |

**ESPI1SMNx00000014...ESPISMNx00000014 (FCH::ITF::ESPI::UP\_RXHDR\_1)**

Read-only. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000014; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000014; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000014; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000014; ESPI=FEC20000h**Bits Description**31:24 **UPCMD\_HDATA6**. Read-only. Reset: 00h.**Description:** The definition for this field is depended on RX\_REQ\_TYPE[OOB](#) selected: Reserved

FLASH selected: Address[7:0]

23:16 **UPCMD\_HDATA5**. Read-only. Reset: 00h.**Description:** The definition for this field is depended on RX\_REQ\_TYPE[OOB](#) selected: [SMBus](#) Byte Count

FLASH selected: Address[15:8]

15:8 **UPCMD\_HDATA4**. Read-only. Reset: 00h.**Description:** The definition for this field is depended on RX\_REQ\_TYPE[OOB](#) selected: [SMBus](#) Command Opcode

FLASH selected: Address[23:16]

7:0 **UPCMD\_HDATA3**. Read-only. Reset: 00h.**Description:** The definition for this field is depended on RX\_REQ\_TYPE[OOB](#) selected [SMBus](#) Slave Address

FLASH selected Address[31:24]

**ESPI1SMNx00000018...ESPISMNx00000018 (FCH::ITF::ESPI::UP\_RXDATA\_PORT)**

Read-only. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000018; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000018; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000018; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000018; ESPI=FEC20000h**Bits Description**31:0 **UP\_RXDATA**. Read-only. Reset: 0000\_0000h.**ESPI1SMNx0000001C...ESPISMNx0000001C (FCH::ITF::ESPI::RESERVED\_REG0)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx0000001C; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx0000001C; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x0000001C; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx0000001C; ESPI=FEC20000h**Bits Description**

31:0 Reserved.

**ESPI1SMNx00000020...ESPISMNx00000020 (FCH::ITF::ESPI::RESERVED\_REG1)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000020; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000020; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000020; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000020; ESPI=FEC20000h**Bits Description**

31:0 Reserved.

**ESPI1SMNx00000024...ESPISMNx00000024 (FCH::ITF::ESPI::RESERVED\_REG2)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000024; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000024; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000024; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000024; ESPI=FEC20000h

**Bits Description**

31:0 Reserved.

**ESPI1SMNx0000002C...ESPISMNx0000002C (FCH::ITF::ESPI::MASTER\_CAP)**

Read-only. Reset: Fixed,E649\_E91Fh.

\_link1\_aliasSMN; ESPI1SMNx0000002C; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx0000002C; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x0000002C; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx0000002C; ESPI=FEC20000h

**Bits Description**31 **CRC\_CHECK\_SUPPORT**. Read-only. Reset: Fixed,1.30 **ALERT\_MODE\_SUPPORT**. Read-only. Reset: Fixed,1.29:28 **IO\_MODE\_SUPPORT**. Read-only. Reset: Fixed,2h. IO Mode support by Controller, Quad mode, Dual mode, single mode27:25 **CLK\_FREQ\_SUPPORT**. Read-only. Reset: Fixed,3h. Operating Support Frequency 16.7 MHz, 33 MHz, 66 MHz24:22 **SLAVE\_NUM**. Read-only. Reset: Fixed,1h.21:19 **PR\_MAX\_SIZE**. Read-only. Reset: Fixed,1h. 64 bytes address aligned max payload size.18:13 **VW\_MAX\_SIZE**. Read-only. Reset: Fixed,0Fh. Operating Maximum Virtual Wire Count support12:10 **OOB\_MAX\_SIZE**. Read-only. Reset: Fixed,2h. 128 bytes max payload size9:7 **FLASH\_MAX\_SIZE**. Read-only. Reset: Fixed,2h. 128 bytes max payload size6:4 **ESPI\_VERSION**. Read-only. Reset: Fixed,1h.3 **PR\_SUPPORT**. Read-only. Reset: Fixed,1.2 **VW\_SUPPORT**. Read-only. Reset: Fixed,1.1 **OOB\_SUPPORT**. Read-only. Reset: Fixed,1.0 **FLASH\_SUPPORT**. Read-only. Reset: Fixed,1.**ESPI1SMNx00000030...ESPISMNx00000030 (FCH::ITF::ESPI::GLOBAL\_CONTROL\_0)**

Read-write. Reset: 0000\_0008h.

\_link1\_aliasSMN; ESPI1SMNx00000030; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000030; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000030; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000030; ESPI=FEC20000h

**Bits Description**

31 Reserved.

30 **PR\_RST\_EN\_PLTRST**. Read-write. Reset: 0.29:24 **WAIT\_CNT**. Read-write. Reset: 00h.23:8 **WDG\_CNT**. Read-write. Reset: 0000h.7 **RG\_dbgclk\_gating\_en**. Read-write. Reset: 0.6:4 **AL\_IDLE\_TIMER**. Read-write. Reset: 0h.3 **AL\_STOP\_EN**. Read-write. Reset: 1.2 **PR\_CLKGAT\_EN**. Read-write. Reset: 0.1 **WAIT\_CHKEN**. Read-write. Reset: 0.0 **WDG\_EN**. Read-write. Reset: 0.

**ESPI1SMNx00000034...ESPISMNx00000034 (FCH::ITF::ESPI::GLOBAL\_CONTROL\_1)**

| Read-write.                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| _link1_aliasSMN; ESPI1SMNx00000034; ESPI1SMN=02DCA000h  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| _link0_aliasSMN; ESPISMNx00000034; ESPISMN=02DC5000h    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| _link1_aliasHOSTLEGACY; ESPI1x00000034; ESPI1=FEC30000h |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| _link0_aliasHOSTLEGACY; ESPIx00000034; ESPI=FEC20000h   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Bits                                                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 31:22                                                   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 21                                                      | <b>ESPI_REQ_NOTWITH_VW_REQ.</b> Read-write. when 0: eSPI_Req and eSPI_VW_Req go to the same bus. Then if eSPI_VW_Req gets granted, no eSPI_Req shall be asserted to the same SPI controller further; when set to 1: eSPI_Req and eSPI_VW_Req don't go to the same bus. Then even if eSPI_VW_Req gets granted, eSPI_Req still need be asserted to the related SPI controller further before issuing any transaction (like PUT_VM).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 20                                                      | <b>ALERT_ENABLE.</b> Read-write. Reset: 0. when 0: ALERT# to eSPI keep inactive(1); when set to 1: Alert# value from Alert# or Din[1] depending on Alert_mode setting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 19                                                      | <b>FL_REQ_VALID_DIS.</b> Read-write. Reset: 0. when 0: 1 index45 bit4/5 are treated as valid bits for FL_REQ/FL_REQ_ATOMIC. 2. index 43h[4], the valid bit will be 1 when send down FL_GNT.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 18                                                      | <b>ROM_SHARING_DIS.</b> Read-write. Reset: 0.<br><b>Description:</b> When this bit is 0b:<br>1. GET_VW will be sent down to eSPI bus as long as VW available status in GET_STATUS is set to 1 no matter VW channel is enabled or not.<br>2. If GET_VW without FL_Req bit set, this GET_VW should be discarded.<br>3. When receiving eSPI_VW_GNT indicating that SPI has granted this FL_Req and already tri-stated the bus, eSPI controller will unconditionally send down PUT_VW with the FL_ACK set without requesting SPI the bus usage (i.e., no eSPI_Req is asserted to SPI controller if bit [21] is 0b, otherwise, eSPI_Req is required to assert before issuing PUT_VW).<br>When this bit is 1b, disable ROM access:<br>1. Get_VW/Put_VW will follow the normal eSPI protocol, i.e., the VW channel need be enabled first.<br>2. No eSPI_VW_Req/eSPI_VW_Atomic will be sent to eSPI host controller.<br>3. No response to eSPI_VW_GNT from eSPI controller. |
| 17:13                                                   | <b>RGCMD_INT_MAP.</b> Read-write. Reset: 17h. Register CMD interrupt mapping, When Register command (Downstream/Upstream peripheral message, Downstream/Upstream <a href="#">OOB</a> , Downstream VW, Channel Independent command) has finished, eSPI controller generated interrupt will mapping to interrupt pin according to following register setting. Mapping refer to bit12:8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 12:8                                                    | <b>ERR_INT_MAP.</b> Read-write. Reset: 1Fh.<br><b>Description:</b> ERR interrupt mapping, When Slave transaction has error happened, and the error interrupt enable has been set. The error interrupt will map to interrupt pin according to following register setting.<br>Bits Mapping<br>00000: IRQ0<br>0001: IRQ1<br>....<br>10111: IRQ23<br>11111: <a href="#">SMI</a> #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 7                                                       | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 6                                                       | <b>MEMWR_LEN_DIS.</b> Read-write. Reset: 0. Control bit to fix the memwr length overridden issue when 2nd mem wr timeout. 1: Disable the fix 0: Enable the fix                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 5                                                       | <b>TAR_RM_DOUTEN_DIS.</b> Read-write. Reset: 0. Control bit to remove the DoutEn(let DoutEn=0 to tristate the DAT PADs) for the 2nd cycle of Turn around cycles. 1: Don't remove 0: Remove                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4                                                       | <b>ROM_INIT_ACCESS_EN.</b> Read-write. Reset: 1. Enable bit for ROM access before ESPI initialization.Rom access before ESPI initialization is enabled by default and must be disabled by writing it to 0 during ESPI initialization then design will ignore this bit. 1: Enable 0: Disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 3                                                       | <b>ROM_PREFETCH_EN.</b> Read-write. Reset: 0. Enable 16DW prefetch to save rom access time. 16DW data will be returned at one time and saved in FIFO. 1: Enable 0: Disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |



|   |                                             |
|---|---------------------------------------------|
| 2 | <b>SUB_DECODE_EN.</b> Read-write. Reset: 0. |
| 1 | <b>BUS_MASTER_EN.</b> Read-write. Reset: 0. |
| 0 | <b>SW_RST.</b> Read-write. Reset: 0.        |

**ESPI1SMNx0000003C...ESPISMNx0000003C (FCH::ITF::ESPI::MISC\_CONTROL\_1)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx0000003C; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx0000003C; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x0000003C; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx0000003C; ESPI=FEC20000h

**Bits Description**31:0 **MISC\_CONTROL\_1\_RESERVED.** Read-write. Reset: 0000\_0000h.**ESPI1SMNx00000040...ESPISMNx00000040 (FCH::ITF::ESPI::SLAVE0\_DECODE\_EN)**

Read-write. Reset: 0000\_0004h.

\_link1\_aliasSMN; ESPI1SMNx00000040; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000040; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000040; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000040; ESPI=FEC20000h

**Bits Description**

31:29 Reserved.

28 **MMIO\_RANGE4\_EN.** Read-write. Reset: 0.27 **IO\_RANGE15\_EN.** Read-write. Reset: 0.26 **IO\_RANGE14\_EN.** Read-write. Reset: 0.25 **IO\_RANGE13\_EN.** Read-write. Reset: 0.24 **IO\_RANGE12\_EN.** Read-write. Reset: 0.23 **IO\_RANGE11\_EN.** Read-write. Reset: 0.22 **IO\_RANGE10\_EN.** Read-write. Reset: 0.21 **IO\_RANGE9\_EN.** Read-write. Reset: 0.20 **IO\_RANGE8\_EN.** Read-write. Reset: 0.19 **IO\_RANGE7\_EN.** Read-write. Reset: 0.18 **IO\_RANGE6\_EN.** Read-write. Reset: 0.17 **IO\_RANGE5\_EN.** Read-write. Reset: 0.16 **IO\_RANGE4\_EN.** Read-write. Reset: 0.15 **MMIO\_RANGE3\_EN.** Read-write. Reset: 0.14 **MMIO\_RANGE2\_EN.** Read-write. Reset: 0.13 **MMIO\_RANGE1\_EN.** Read-write. Reset: 0.12 **MMIO\_RANGE0\_EN.** Read-write. Reset: 0.11 **IO\_RANGE3\_EN.** Read-write. Reset: 0.10 **IO\_RANGE2\_EN.** Read-write. Reset: 0.9 **IO\_RANGE1\_EN.** Read-write. Reset: 0.8 **IO\_RANGE0\_EN.** Read-write. Reset: 0.

7:3 Reserved.

2 **IO\_80\_EN.** Read-write. Reset: 1.1 **IO\_60\_AND\_64\_EN.** Read-write. Reset: 0.0 **IO\_2E\_TO\_2F\_EN.** Read-write. Reset: 0.

**ESPI1SMNx00000044...ESPISMNx00000044 (FCH::ITF::ESPI::SLAVE0\_IO\_BASE\_REG0)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000044; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000044; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000044; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000044; ESPI=FEC20000h**Bits Description**31:16 **RANGE1**. Read-write. Reset: 0000h. IO decode base address for Range 115:0 **RANGE0**. Read-write. Reset: 0000h. IO decode base address for Range 0**ESPI1SMNx00000048...ESPISMNx00000048 (FCH::ITF::ESPI::SLAVE0\_IO\_BASE\_REG1)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000048; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000048; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000048; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000048; ESPI=FEC20000h**Bits Description**31:16 **RANGE3**. Read-write. Reset: 0000h. IO decode base address for Range 315:0 **RANGE2**. Read-write. Reset: 0000h. IO decode base address for Range 2**ESPI1SMNx0000004C...ESPISMNx0000004C (FCH::ITF::ESPI::SLAVE0\_IO\_SIZE)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx0000004C; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx0000004C; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x0000004C; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx0000004C; ESPI=FEC20000h**Bits Description**31:24 **RANGE3**. Read-write. Reset: 00h. Programmable IO Range3 size23:16 **RANGE2**. Read-write. Reset: 00h. Programmable IO Range2 size15:8 **RANGE1**. Read-write. Reset: 00h. Programmable IO Range1 size7:0 **RANGE0**. Read-write. Reset: 00h. Programmable IO Range0 size**ESPI1SMNx00000050...ESPISMNx00000050 (FCH::ITF::ESPI::SLAVE0\_MMIO\_BASE\_REG0)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000050; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000050; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000050; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000050; ESPI=FEC20000h**Bits Description**31:0 **RANGE0**. Read-write. Reset: 0000\_0000h. [MMIO](#) decode base address for Range 0**ESPI1SMNx00000054...ESPISMNx00000054 (FCH::ITF::ESPI::SLAVE0\_MMIO\_BASE\_REG1)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000054; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000054; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000054; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000054; ESPI=FEC20000h**Bits Description**31:0 **RANGE1**. Read-write. Reset: 0000\_0000h. [MMIO](#) decode base address for Range 1

**ESPI1SMNx00000058...ESPISMNx00000058 (FCH::ITF::ESPI::SLAVE0\_MMIO\_BASE\_REG2)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000058; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000058; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000058; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000058; ESPI=FEC20000h**Bits Description**31:0 **RANGE2**. Read-write. Reset: 0000\_0000h. [MMIO](#) decode base address for Range 2**ESPI1SMNx0000005C...ESPISMNx0000005C (FCH::ITF::ESPI::SLAVE0\_MMIO\_BASE\_REG3)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx0000005C; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx0000005C; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x0000005C; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx0000005C; ESPI=FEC20000h**Bits Description**31:0 **RANGE3**. Read-write. Reset: 0000\_0000h. [MMIO](#) decode base address for Range 3**ESPI1SMNx00000060...ESPISMNx00000060 (FCH::ITF::ESPI::SLAVE0\_MMIO\_SIZE\_REG0)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000060; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000060; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000060; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000060; ESPI=FEC20000h**Bits Description**31:16 **RANGE1**. Read-write. Reset: 0000h. Programmable [MMIO](#) Range1 size.15:0 **RANGE0**. Read-write. Reset: 0000h. Programmable [MMIO](#) Range0 size.**ESPI1SMNx00000064...ESPISMNx00000064 (FCH::ITF::ESPI::SLAVE0\_MMIO\_SIZE\_REG1)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx00000064; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx00000064; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x00000064; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx00000064; ESPI=FEC20000h**Bits Description**31:16 **RANGE3**. Read-write. Reset: 0000h. Programmable [MMIO](#) Range3 size.15:0 **RANGE2**. Read-write. Reset: 0000h. Programmable [MMIO](#) Range2 size.

**ESPI1SMNx00000068...ESPISMNx00000068 (FCH::ITF::ESPI::SLAVE0\_CONFIG)**

Read-write. Reset: 0000\_000Ch.

\_link1\_aliasSMN; ESPI1SMNx00000068; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000068; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000068; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000068; ESPI=FEC20000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                    |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>CRC_CHECK_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                     |
| 30    | <b>ALERT_MODE_SEL.</b> Read-write. Reset: 0.<br><b>Description:</b> 0b: IO bit1 pin is used to signal the Alert event.<br>1b: A dedicated Alert# pin is used to signal the Alert event.<br>Note: This bit can only be 0 in a single master-single slave topology. For single master-multiple slave topology, this bit must be programmed to 1. |
| 29:28 | <b>IO_MODE_SEL.</b> Read-write. Reset: 0h. I/O Mode Select. 00 Single I/O 01 Dual I/O 10 Quad I/O 11 Reserved.                                                                                                                                                                                                                                 |
| 27:25 | <b>CLK_FREQ_SEL.</b> Read-write. Reset: 0h.<br><b>Description:</b> Operating Frequency: 000 16.6 MHz 001 33 MHz 010 66 MHz<br>Others Reserved.                                                                                                                                                                                                 |
| 24:4  | Reserved.                                                                                                                                                                                                                                                                                                                                      |
| 3     | <b>PR_EN.</b> Read-write. Reset: 1.                                                                                                                                                                                                                                                                                                            |
| 2     | <b>VW_EN.</b> Read-write. Reset: 1.                                                                                                                                                                                                                                                                                                            |
| 1     | <b>OOB_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                           |
| 0     | <b>FLASH_EN.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                         |

**ESPI1SMNx0000006C...ESPISMNx0000006C (FCH::ITF::ESPI::SLAVE0\_INT\_EN)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx0000006C; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx0000006C; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x0000006C; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx0000006C; ESPI=FEC20000h

| Bits  | Description                                           |
|-------|-------------------------------------------------------|
| 31    | <b>FLASH_REQ_INT_EN.</b> Read-write. Reset: 0.        |
| 30    | <b>RXOOB_INT_EN.</b> Read-write. Reset: 0.            |
| 29    | <b>RXMSG_INT_EN.</b> Read-write. Reset: 0.            |
| 28    | <b>DNCMD_INT_EN.</b> Read-write. Reset: 0.            |
| 27    | <b>RXVW_GRP3_INT_EN.</b> Read-write. Reset: 0.        |
| 26    | <b>RXVW_GRP2_INT_EN.</b> Read-write. Reset: 0.        |
| 25    | <b>RXVW_GRP1_INT_EN.</b> Read-write. Reset: 0.        |
| 24    | <b>RXVW_GRP0_INT_EN.</b> Read-write. Reset: 0.        |
| 23:20 | Reserved.                                             |
| 19    | <b>WDG_TIMEOUT_INT_EN.</b> Read-write. Reset: 0.      |
| 18    | <b>MST_ABORT_INT_EN.</b> Read-write. Reset: 0.        |
| 17:16 | Reserved.                                             |
| 15    | <b>PROTOCOL_ERR_INT_EN.</b> Read-write. Reset: 0.     |
| 14    | <b>RXFLASH_OVERFLOW_INT_EN.</b> Read-write. Reset: 0. |
| 13    | <b>RXMSG_OVERFLOW_INT_EN.</b> Read-write. Reset: 0.   |
| 12    | <b>RXOOB_OVERFLOW_INT_EN.</b> Read-write. Reset: 0.   |
| 11    | <b>ILLEGAL_LEN_INT_EN.</b> Read-write. Reset: 0.      |
| 10    | <b>ILLEGAL_TAG_INT_EN.</b> Read-write. Reset: 0.      |
| 9     | <b>UNSUCSS_CPL_INT_EN.</b> Read-write. Reset: 0.      |
| 8     | <b>INVALID_CT_INT_EN.</b> Read-write. Reset: 0.       |
| 7     | <b>INVALID_RSP_INT_EN.</b> Read-write. Reset: 0.      |
| 6     | <b>NON_FATAL_ERR_INT_EN.</b> Read-write. Reset: 0.    |
| 5     | <b>FATAL_ERR_INT_EN.</b> Read-write. Reset: 0.        |
| 4     | <b>NO_RSP_INT_EN.</b> Read-write. Reset: 0.           |
| 3     | Reserved.                                             |
| 2     | <b>CRC_ERR_INT_EN.</b> Read-write. Reset: 0.          |
| 1     | <b>WAIT_TIMEOUT_INT_EN.</b> Read-write. Reset: 0.     |
| 0     | <b>BUS_ERR_INT_EN.</b> Read-write. Reset: 0.          |

**ESPI1SMNx00000070...ESPISMNx00000070 (FCH::ITF::ESPI::SLAVE0\_INT\_STS)**

Read,Write-1-to-clear. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000070; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000070; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000070; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000070; ESPI=FEC20000h

| Bits  | Description                                                    |
|-------|----------------------------------------------------------------|
| 31    | <b>FLASH_REQ_INT</b> . Read,Write-1-to-clear. Reset: 0.        |
| 30    | <b>RXOOB_INT</b> . Read,Write-1-to-clear. Reset: 0.            |
| 29    | <b>RXMSG_INT</b> . Read,Write-1-to-clear. Reset: 0.            |
| 28    | <b>DNCMD_INT</b> . Read,Write-1-to-clear. Reset: 0.            |
| 27    | <b>RXVW_GRP3_INT</b> . Read,Write-1-to-clear. Reset: 0.        |
| 26    | <b>RXVW_GRP2_INT</b> . Read,Write-1-to-clear. Reset: 0.        |
| 25    | <b>RXVW_GRP1_INT</b> . Read,Write-1-to-clear. Reset: 0.        |
| 24    | <b>RXVW_GRP0_INT</b> . Read,Write-1-to-clear. Reset: 0.        |
| 23:20 | Reserved.                                                      |
| 19    | <b>WDG_TIMEOUT_INT</b> . Read,Write-1-to-clear. Reset: 0.      |
| 18    | <b>MST_ABORT_INT</b> . Read,Write-1-to-clear. Reset: 0.        |
| 17:16 | Reserved.                                                      |
| 15    | <b>PROTOCOL_ERR_INT</b> . Read,Write-1-to-clear. Reset: 0.     |
| 14    | <b>RXFLASH_OVERFLOW_INT</b> . Read,Write-1-to-clear. Reset: 0. |
| 13    | <b>RXMSG_OVERFLOW_INT</b> . Read,Write-1-to-clear. Reset: 0.   |
| 12    | <b>RXOOB_OVERFLOW_INT</b> . Read,Write-1-to-clear. Reset: 0.   |
| 11    | <b>ILLEGAL_LEN_INT</b> . Read,Write-1-to-clear. Reset: 0.      |
| 10    | <b>ILLEGAL_TAG_INT</b> . Read,Write-1-to-clear. Reset: 0.      |
| 9     | <b>UNSUCSS_CPL_INT</b> . Read,Write-1-to-clear. Reset: 0.      |
| 8     | <b>UNKNOWN_CT_INT</b> . Read,Write-1-to-clear. Reset: 0.       |
| 7     | <b>UNKNOWN_RSP_INT</b> . Read,Write-1-to-clear. Reset: 0.      |
| 6     | <b>NON_FATAL_ERR_INT</b> . Read,Write-1-to-clear. Reset: 0.    |
| 5     | <b>FATAL_ERR_INT</b> . Read,Write-1-to-clear. Reset: 0.        |
| 4     | <b>NO_RSP_INT</b> . Read,Write-1-to-clear. Reset: 0.           |
| 3     | Reserved.                                                      |
| 2     | <b>CRC_ERR_INT</b> . Read,Write-1-to-clear. Reset: 0.          |
| 1     | <b>WAIT_TIMEOUT_INT</b> . Read,Write-1-to-clear. Reset: 0.     |
| 0     | <b>BUS_ERR_INT</b> . Read,Write-1-to-clear. Reset: 0.          |

**ESPI1SMNx00000074...ESPISMNx00000074 (FCH::ITF::ESPI::SLAVE0\_RXMSG\_HDR0)**

Read-only. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000074; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000074; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000074; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000074; ESPI=FEC20000h

| Bits  | Description                                                                     |
|-------|---------------------------------------------------------------------------------|
| 31:24 | <b>BYTE3</b> . Read-only. Reset: 00h. Received Peripheral Message code          |
| 23:16 | <b>BYTE2</b> . Read-only. Reset: 00h. Received Peripheral Message Length[7:0]   |
| 15:8  | <b>BYTE1</b> . Read-only. Reset: 00h. [15:12]: Tag [11:8]: Length[11:8]         |
| 7:0   | <b>CYCLETYP</b> . Read-only. Reset: 00h. CycleType[7:0] for Peripheral Msg/MsgD |

**ESPI1SMNx00000078...ESPISMNx00000078 (FCH::ITF::ESPI::SLAVE0\_RXMSG\_HDR1)**

Read-only. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000078; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000078; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000078; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000078; ESPI=FEC20000h

**Bits Description**31:24 **SPECIFIC\_BYTE3**. Read-only. Reset: 00h. Peripheral Message Specific Byte323:16 **SPECIFIC\_BYTE2**. Read-only. Reset: 00h. Peripheral Message Specific Byte215:8 **SPECIFIC\_BYTE1**. Read-only. Reset: 00h. Peripheral Message Specific Byte17:0 **SPECIFIC\_BYTE0**. Read-only. Reset: 00h. Peripheral Message Specific Byte0**ESPI1SMNx0000007C...ESPISMNx0000007C (FCH::ITF::ESPI::SLAVE0\_RXMSG\_DATA\_PORT)**

Read-only. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx0000007C; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx0000007C; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x0000007C; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx0000007C; ESPI=FEC20000h

**Bits Description**31:0 **RXMSG\_DATA**. Read-only. Reset: 0000\_0000h.**ESPI1SMNx00000080...ESPISMNx00000080 (FCH::ITF::ESPI::SLAVE0\_IO\_BASE\_REG2)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000080; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000080; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000080; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000080; ESPI=FEC20000h

**Bits Description**31:16 **RANGE5**. Read-write. Reset: 0000h. IO decode base address for Range 515:0 **RANGE4**. Read-write. Reset: 0000h. IO decode base address for Range 4**ESPI1SMNx00000084...ESPISMNx00000084 (FCH::ITF::ESPI::SLAVE0\_IO\_BASE\_REG3)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000084; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000084; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000084; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000084; ESPI=FEC20000h

**Bits Description**31:16 **RANGE7**. Read-write. Reset: 0000h. IO decode base address for Range 715:0 **RANGE6**. Read-write. Reset: 0000h. IO decode base address for Range 6**ESPI1SMNx00000088...ESPISMNx00000088 (FCH::ITF::ESPI::SLAVE0\_IO\_SIZE1)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000088; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000088; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000088; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000088; ESPI=FEC20000h

**Bits Description**31:24 **RANGE7**. Read-write. Reset: 00h. Programmable IO Range7 size23:16 **RANGE6**. Read-write. Reset: 00h. Programmable IO Range6 size15:8 **RANGE5**. Read-write. Reset: 00h. Programmable IO Range5 size7:0 **RANGE4**. Read-write. Reset: 00h. Programmable IO Range4 size

**ESPI1SMNx0000008C...ESPISMNx0000008C (FCH::ITF::ESPI::SLAVE0\_IO\_BASE\_REG4)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx0000008C; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx0000008C; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x0000008C; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx0000008C; ESPI=FEC20000h

**Bits Description**31:16 **RANGE9.** Read-write. Reset: 0000h. IO decode base address for Range 915:0 **RANGE8.** Read-write. Reset: 0000h. IO decode base address for Range 8**ESPI1SMNx00000090...ESPISMNx00000090 (FCH::ITF::ESPI::SLAVE0\_IO\_BASE\_REG5)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000090; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000090; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000090; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000090; ESPI=FEC20000h

**Bits Description**31:16 **RANGE11.** Read-write. Reset: 0000h. IO decode base address for Range 1115:0 **RANGE10.** Read-write. Reset: 0000h. IO decode base address for Range 10**ESPI1SMNx00000094...ESPISMNx00000094 (FCH::ITF::ESPI::SLAVE0\_IO\_SIZE2)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000094; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000094; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000094; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000094; ESPI=FEC20000h

**Bits Description**31:24 **RANGE11.** Read-write. Reset: 00h. Programmable IO Range11 size23:16 **RANGE10.** Read-write. Reset: 00h. Programmable IO Range10 size15:8 **RANGE9.** Read-write. Reset: 00h. Programmable IO Range9 size7:0 **RANGE8.** Read-write. Reset: 00h. Programmable IO Range8 size**ESPI1SMNx00000098...ESPISMNx00000098 (FCH::ITF::ESPI::RESERVED\_RXMSG\_REG6)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx00000098; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx00000098; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x00000098; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx00000098; ESPI=FEC20000h

**Bits Description**

31:24 Reserved.

23:0 **VW\_IRQ\_Inactive\_Polarity.** Read-write. Reset: 00\_0000h. This field is used to change the polarity of default value of Rx VW IRQ status. The default value of Rx VW IRQ is 0h, and a transition of 0 to 1b with this field shall change the default value of Rx VW IRQ to 1b



**ESPI1SMNx0000009C...ESPISMNx0000009C (FCH::ITF::ESPI::SLAVE0\_RXVW)**

Reset: 0007\_0C00h.

\_link1\_aliasSMN; ESPI1SMNx0000009C; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx0000009C; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x0000009C; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx0000009C; ESPI=FEC20000h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>CPUTEMP_REQ.</b> Read,Write-1-to-clear. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 30    | <b>RTCTIME_REQ.</b> Read,Write-1-to-clear. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 29:20 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 19    | <b>HOST_RST_ACK.</b> Read,Write-1-to-clear. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 18    | <b>RCIN_B.</b> Read,Write-1-only. Reset: 1.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 17    | <b>SMI_B.</b> Read,Write-1-only. Reset: 1.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 16    | <b>SCI_B.</b> Read,Write-1-only. Reset: 1.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 15    | <b>SLAVE_BOOT_LOAD_STS.</b> Read,Write-1-to-clear. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 14    | <b>ERROR_NONFATAL.</b> Read,Write-1-to-clear. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 13    | <b>ERROR_FATAL.</b> Read,Write-1-to-clear. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 12    | <b>SLAVE_BOOT_LOAD_DONE.</b> Read,Write-1-to-clear. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 11    | <b>PME_B.</b> Read,Write-1-only. Reset: 1.                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 10    | <b>WAKE_B.</b> Read,Write-1-only. Reset: 1.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 9     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 8     | <b>OOB_RST_ACK.</b> Read,Write-1-to-clear. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 7:5   | <b>IRQ_STS.</b> Read-only. Reset: 0h.<br><b>Description:</b> IRQ Satus:<br>IRQ status which specified by IRQ selection:<br>Bits Status<br>000 IRQ keep 0 unchanged<br>001 IRQ keep 1 unchanged<br>010 IRQ changed from 1 to 0 (Clear)<br>011 IRQ changed from 0 to 1 (Set)<br>100 IRQ changed from 0->1->0(High pulse)<br>101 IRQ changed from 1->0->1(Low pulse)<br>110 IRQ changed from 1->1->0 or 1->0->0 or 0->0->0<br>111 IRQ changed from 0->0->1 or 0->1->1 or 1->1->1 |
| 4:0   | <b>IRQ_SEL.</b> Read-write. Reset: 00h.<br><b>Description:</b> This field determine Slave N Received Virtual Wires Register bit[7:5] output which IRQ status.<br>Bits Selection<br>00000: IRQ0<br>00001 IRQ1<br>....<br>10111:IRQ23<br>others:Reserved                                                                                                                                                                                                                        |

**ESPI1SMNx000000A0...ESPISMNx000000A0 (FCH::ITF::ESPI::SLAVE0\_RXVW\_DATA)**

Read-only. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000A0; ESPI1SMN=02DCA000h\_link0\_aliasSMN; ESPISMNx000000A0; ESPISMN=02DC5000h\_link1\_aliasHOSTLEGACY; ESPI1x000000A0; ESPI1=FEC30000h\_link0\_aliasHOSTLEGACY; ESPIx000000A0; ESPI=FEC20000h**Bits Description**31:24 **GRP3**. Read-only. Reset: 00h.**Description:** Group3 Virtual Wire Data Register:

When VW MISC CONTRL register bit3 set, eSPI master will check each received VW Index, if the received Index matches with Group3 Virtual Wire Index Selection Register, eSPI Master will update this field with the new received value.

23:16 **GRP2**. Read-only. Reset: 00h.**Description:** Group2 Virtual Wire Data Register:

When VW MISC CONTRL register bit2 set, eSPI master will check each received VW Index, if the received Index matches with Group2 Virtual Wire Index Selection Register, eSPI Master will update this field with the new received value.

15:8 **GRP1**. Read-only. Reset: 00h.**Description:** Group1 Virtual Wire Data Register:

When VW MISC CONTRL register bit1 set, eSPI master will check each received VW Index, if the received Index matches with Group1 Virtual Wire Index Selection Register, eSPI Master will update this field with the new received value.

7:0 **GRP0**. Read-only. Reset: 00h.**Description:** Group0 Virtual Wire Data Register:

When VW MISC CONTRL register bit0 set, eSPI master will check each received VW Index, if the received Index matches with Group0 Virtual Wire Index Selection Register, eSPI Master will update this field with the new received value.

**ESPI1SMNx000000A4...ESPISMNx000000A4 (FCH::ITF::ESPI::SLAVE0\_RXVW\_INDEX)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000A4; ESPI1SMN=02DCA000h\_link0\_aliasSMN; ESPISMNx000000A4; ESPISMN=02DC5000h\_link1\_aliasHOSTLEGACY; ESPI1x000000A4; ESPI1=FEC30000h\_link0\_aliasHOSTLEGACY; ESPIx000000A4; ESPI=FEC20000h**Bits Description**31:24 **GRP3**. Read-write. Reset: 00h. Group3 Virtual Wire Index Selection Register23:16 **GRP2**. Read-write. Reset: 00h. Group2 Virtual Wire Index Selection Register15:8 **GRP1**. Read-write. Reset: 00h. Group1 Virtual Wire Index Selection Register7:0 **GRP0**. Read-write. Reset: 00h. Group0 Virtual Wire Index Selection Register

**ESPI1SMNx000000A8...ESPISMNx000000A8 (FCH::ITF::ESPI::SLAVE0\_RXVW\_MISC\_CNTL)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000A8; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx000000A8; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x000000A8; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx000000A8; ESPI=FEC20000h

| Bits | Description                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>IRQ23_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 30   | <b>IRQ22_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 29   | <b>IRQ21_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 28   | <b>IRQ20_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 27   | <b>IRQ19_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 26   | <b>IRQ18_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 25   | <b>IRQ17_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 24   | <b>IRQ16_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 23   | <b>IRQ15_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 22   | <b>IRQ14_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 21   | <b>IRQ13_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 20   | <b>IRQ12_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 19   | <b>IRQ11_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 18   | <b>IRQ10_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                 |
| 17   | <b>IRQ9_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 16   | <b>IRQ8_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 15   | <b>IRQ7_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 14   | <b>IRQ6_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 13   | <b>IRQ5_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 12   | <b>IRQ4_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 11   | <b>IRQ3_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 10   | <b>IRQ2_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 9    | <b>IRQ1_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 8    | <b>IRQ0_MASK.</b> Read-write. Reset: 0.                                                                                                                                                                  |
| 7    | <b>CPUTEMP_RTCTIME_VW_INDEX_SEL.</b> Read-write. Reset: 0. CPU_TEMP/RTC TIME VW index selection. 0:index 53(decimal), 1: index 63(decimal)                                                               |
| 6    | <b>CPUTEMP_RTCTIME_VW_EN.</b> Read-write. Reset: 0. Enabled Hardware to Receive VW packet of index 53(63) from Slave to indicate the slave to request CPU_TEMP/RTC Time info. 1: Enable, 0: Disable.     |
| 5    | Reserved.                                                                                                                                                                                                |
| 4    | <b>SUS_STAT_VWEN.</b> Read-write. Reset: 0. Enabled Hardware to send VW packet when SUS_STAT# changes. 1: Enable, 0: Disable.                                                                            |
| 3    | <b>GRP3_EN.</b> Read-write. Reset: 0. GRP3 Enable : When Set, VW channel will check received Index, if Index is same as Group3 Index register setting; VW will store the Data into Group3 Data register. |
| 2    | <b>GRP2_EN.</b> Read-write. Reset: 0. GRP2 Enable : When Set, VW channel will check received Index, if Index is same as Group2 Index register setting; VW will store the Data into Group2 Data register. |
| 1    | <b>GRP1_EN.</b> Read-write. Reset: 0. GRP1 Enable : When Set, VW channel will check received Index, if Index is same as Group1 Index register setting; VW will store the Data into Group1 Data register. |
| 0    | <b>GRP0_EN.</b> Read-write. Reset: 0. GRP0 Enable : When Set, VW channel will check received Index, if Index is same as Group0 Index register setting; VW will store the Data into Group0 Data register. |

**ESPI1SMNx000000AC...ESPISMNx000000AC (FCH::ITF::ESPI::SLAVE0\_RXVW\_POLARITY)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000AC; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx000000AC; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x000000AC; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx000000AC; ESPI=FEC20000h

| Bits  | Description                                     |
|-------|-------------------------------------------------|
| 31:29 | Reserved.                                       |
| 28    | <b>OOB_RST_WARN_EN.</b> Read-write. Reset: 0.   |
| 27    | <b>NMIOUT_EN.</b> Read-write. Reset: 0.         |
| 26    | <b>SMIOUT_EN.</b> Read-write. Reset: 0.         |
| 25    | <b>HOST_RST_WARN_DIS.</b> Read-write. Reset: 0. |
| 24    | <b>PLTRSTB_DIS.</b> Read-write. Reset: 0.       |
| 23    | <b>IRQ23_POLARITY.</b> Read-write. Reset: 0.    |
| 22    | <b>IRQ22_POLARITY.</b> Read-write. Reset: 0.    |
| 21    | <b>IRQ21_POLARITY.</b> Read-write. Reset: 0.    |
| 20    | <b>IRQ20_POLARITY.</b> Read-write. Reset: 0.    |
| 19    | <b>IRQ19_POLARITY.</b> Read-write. Reset: 0.    |
| 18    | <b>IRQ18_POLARITY.</b> Read-write. Reset: 0.    |
| 17    | <b>IRQ17_POLARITY.</b> Read-write. Reset: 0.    |
| 16    | <b>IRQ16_POLARITY.</b> Read-write. Reset: 0.    |
| 15    | <b>IRQ15_POLARITY.</b> Read-write. Reset: 0.    |
| 14    | <b>IRQ14_POLARITY.</b> Read-write. Reset: 0.    |
| 13    | <b>IRQ13_POLARITY.</b> Read-write. Reset: 0.    |
| 12    | <b>IRQ12_POLARITY.</b> Read-write. Reset: 0.    |
| 11    | <b>IRQ11_POLARITY.</b> Read-write. Reset: 0.    |
| 10    | <b>IRQ10_POLARITY.</b> Read-write. Reset: 0.    |
| 9     | <b>IRQ9_POLARITY.</b> Read-write. Reset: 0.     |
| 8     | <b>IRQ8_POLARITY.</b> Read-write. Reset: 0.     |
| 7     | <b>IRQ7_POLARITY.</b> Read-write. Reset: 0.     |
| 6     | <b>IRQ6_POLARITY.</b> Read-write. Reset: 0.     |
| 5     | <b>IRQ5_POLARITY.</b> Read-write. Reset: 0.     |
| 4     | <b>IRQ4_POLARITY.</b> Read-write. Reset: 0.     |
| 3     | <b>IRQ3_POLARITY.</b> Read-write. Reset: 0.     |
| 2     | <b>IRQ2_POLARITY.</b> Read-write. Reset: 0.     |
| 1     | <b>IRQ1_POLARITY.</b> Read-write. Reset: 0.     |
| 0     | <b>IRQ0_POLARITY.</b> Read-write. Reset: 0.     |

**ESPI1SMNx000000B0...ESPISMNx000000B0 (FCH::ITF::ESPI::SLAVE0\_IO\_BASE\_REG6)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000B0; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx000000B0; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x000000B0; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx000000B0; ESPI=FEC20000h

| Bits  | Description                                                                   |
|-------|-------------------------------------------------------------------------------|
| 31:16 | <b>RANGE13.</b> Read-write. Reset: 0000h. IO decode base address for Range 13 |
| 15:0  | <b>RANGE12.</b> Read-write. Reset: 0000h. IO decode base address for Range 12 |

**ESPI1SMNx000000B4...ESPISMNx000000B4 (FCH::ITF::ESPI::SLAVE0\_IO\_BASE\_REG7)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000B4; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx000000B4; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x000000B4; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx000000B4; ESPI=FEC20000h

**Bits Description**31:16 **RANGE15.** Read-write. Reset: 0000h. IO decode base address for Range 1515:0 **RANGE14.** Read-write. Reset: 0000h. IO decode base address for Range 14**ESPI1SMNx000000B8...ESPISMNx000000B8 (FCH::ITF::ESPI::SLAVE0\_IO\_SIZE3)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000B8; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx000000B8; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x000000B8; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx000000B8; ESPI=FEC20000h

**Bits Description**31:24 **RANGE15.** Read-write. Reset: 00h. Programmable IO Range15 size23:16 **RANGE14.** Read-write. Reset: 00h. Programmable IO Range14 size15:8 **RANGE13.** Read-write. Reset: 00h. Programmable IO Range13 size7:0 **RANGE12.** Read-write. Reset: 00h. Programmable IO Range12 size**ESPI1SMNx000000BC...ESPISMNx000000BC (FCH::ITF::ESPI::SLAVE0\_MMIO\_BASE\_REG4)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000BC; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx000000BC; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x000000BC; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx000000BC; ESPI=FEC20000h

**Bits Description**31:0 **RANGE4.** Read-write. Reset: 0000\_0000h. [MMIO](#) decode base address for Range 4**ESPI1SMNx000000C0...ESPISMNx000000C0 (FCH::ITF::ESPI::SLAVE0\_MMIO\_SIZE\_REG2)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000C0; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx000000C0; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x000000C0; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx000000C0; ESPI=FEC20000h

**Bits Description**

31:25 Reserved.

24:0 **RANGE4.** Read-write. Reset: 000\_0000h. Programmable [MMIO](#) Range4 size, 1M - vA.6, 32M - vA.7**ESPI1SMNx000000C4...ESPISMNx000000C4 (FCH::ITF::ESPI::MMIO\_CPUTEMP)**

Read-write. Reset: 0000\_0000h.

\_link1\_aliasSMN; ESPI1SMNx000000C4; ESPI1SMN=02DCA000h

\_link0\_aliasSMN; ESPISMNx000000C4; ESPISMN=02DC5000h

\_link1\_aliasHOSTLEGACY; ESPI1x000000C4; ESPI1=FEC30000h

\_link0\_aliasHOSTLEGACY; ESPIx000000C4; ESPI=FEC20000h

**Bits Description**31:0 **MMIO\_CPUTEMP.** Read-write. Reset: 0000\_0000h. [MMIO](#) address to send down CPU temperture

**ESPI1SMNx000000C8...ESPISMNx000000C8 (FCH::ITF::ESPI::MMIO\_RTCTIME)**

Read-write. Reset: 0000\_0000h.

[\\_link1\\_aliasSMN](#); ESPI1SMNx000000C8; ESPI1SMN=02DCA000h[\\_link0\\_aliasSMN](#); ESPISMNx000000C8; ESPISMN=02DC5000h[\\_link1\\_aliasHOSTLEGACY](#); ESPI1x000000C8; ESPI1=FEC30000h[\\_link0\\_aliasHOSTLEGACY](#); ESPIx000000C8; ESPI=FEC20000h**Bits Description**31:0 **MMIO\_RTCTIME**. Read-write. Reset: 0000\_0000h. [MMIO](#) address to send down RTC TIME**14.3.14 MISC Registers****14.3.14.1 Miscellaneous (MISC) Registers**

**MISCx00000000 (FCH::MISC::GPPCLKCONTROL)**

Read-write. Reset: 0000\_CFFh.

\_aliasHOSTLEGACY; MISCx00000000; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 11:10 | <b>gpp_clk15_clock_request_mapping.</b> Read-write. Reset: 3h.<br><b>Description:</b> GPP15 PCIE clock pins (GPP_CLK15P/GPP_CLK15N) output control by CLKREQ15# (or CLKREQ1G#) pin<br>GPP_CLK15P/GPP_CLK15N pins can be controlled (turn OFF, ON, CLKREQ15# controlled) by the following options.<br>GPP15_CLKREQ_Mapping:<br>00 Off<br>01 CLKREQ15#<br>10 Off, reserved<br>11 On (default)                                                                        |
| 9:8   | <b>gpp_clk13_clock_request_mapping.</b> Read-write. Reset: 3h.<br><b>Description:</b> GPP13 PCIE clock pins (GPP_CLK13P/GPP_CLK13N) output control by CLKREQ13# pin<br>GPP_CLK13P/GPP_CLK13N pins can be controlled (turn OFF, ON, CLKREQ13# controlled) by the following options.<br>GPP13_CLKREQ_Mapping:<br>00 Off<br>01 CLKREQ13#<br>10 Off, reserved<br>11 On (default)<br>Note: MISCx00000004[24] is an override bit which is XOR with GPP13_CLKREQ_Mapping. |
| 7:6   | <b>gpp_clk12_clock_request_mapping.</b> Read-write. Reset: 3h.<br><b>Description:</b> GPP12 PCIE clock pins (GPP_CLK12P/GPP_CLK12N) output control by CLKREQ12# pin<br>GPP_CLK12P/GPP_CLK12N pins can be controlled (turn OFF, ON, CLKREQ12# controlled) by the following options.<br>GPP12_CLKREQ_Mapping:<br>00 Off<br>01 CLKREQ12#<br>10 Off, reserved<br>11 On (default)<br>Note: MISCx00000004[12] is an override bit which is XOR with GPP12_CLKREQ_Mapping. |
| 5:4   | <b>gpp_clk14_clock_request_mapping.</b> Read-write. Reset: 3h.<br><b>Description:</b> GPP14 PCIE clock pins (GPP_CLK14P/GPP_CLK14N) output control by CLKREQ14# pin<br>GPP_CLK14P/GPP_CLK14N pins can be controlled (turn OFF, ON, CLKREQ14# controlled) by the following options.<br>GPP14_CLKREQ_Mapping:<br>00 Off<br>01 CLKREQ14#<br>10 Off, reserved<br>11 On (default)<br>Note: MISCx00000004[29] is an override bit which is XOR with GPP14_CLKREQ_Mapping. |
| 3:2   | <b>gpp_clk11_clock_request_mapping.</b> Read-write. Reset: 3h.                                                                                                                                                                                                                                                                                                                                                                                                     |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <p><b>Description:</b> GPP11 PCIE clock pins (GPP_CLK11P/GPP_CLK11N) output control by CLKREQ11# pin. GPP_CLK11P/GPP_CLK11N pins can be controlled (turn OFF, ON, CLKREQ11# controlled) by the following options.</p> <p>GPP11_CLKREQ_Mapping:</p> <p>00 Off</p> <p>01 CLKREQ11#</p> <p>10 Off, reserved</p> <p>11 On (default)</p> <p>Note: MISCx00000004[9] is an override bit which is XOR with GPP11_CLKREQ_Mapping.</p>                                                                                                                                                                                                                                                                          |
| 1:0 | <p><b>gpp_clk10_clock_request_mapping.</b> Read-write. Reset: 3h.</p> <p><b>Description:</b> GPP10 PCIE clock pins (GPP_CLK10P/GPP_CLK10N) output control by CLKREQ10# pin. Socket0 GPP10 is default output 100MHz (CG1PLL 100MHz) for Socket1 CG1PLL as refclk thru Socket1 GPP10 input.</p> <p>Socket1 GPP10 is default with input mode.</p> <p>If the platform is one socket only, then socket0 GPP10 can be controlled (turn OFF, ON, CLKREQ10# controlled) by the following options.</p> <p>GPP10_CLKREQ_Mapping:</p> <p>00 Off</p> <p>01 CLKREQ10#</p> <p>10 Off, reserved</p> <p>11 On (default)</p> <p>Note: MISCx00000004[11] is an override bit which is XOR with GPP10_CLKREQ_Mapping.</p> |

#### MISCx00000004 (FCH::MISC::CLKOUTPUTCNTRL)

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000004; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |



**MISCx00000008 (FCH::MISC::CGPLLCONFIG1)**

Read-write. Reset: 6000\_0000h.

\_aliasHOSTLEGACY; MISCx00000008; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>xtal_refclk2x_clkenb.</b> Read-write. Reset: 0.<br><b>Description:</b> REFCLK2X clock enable/disable<br>0: Turn on<br>1: Turn off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 30   | <b>xtal_clkgen_s5_clken.</b> Read-write. Reset: 1.<br><b>Description:</b> CLKGEN_S5 RefClk Enable/disable<br>0: Turn off<br>1: Turn on                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 29   | <b>xtal_clkgen_s0_clken.</b> Read-write. Reset: 1.<br><b>Description:</b> CLKGEN_S0 RefClk Enable/disable<br>0: Turn off<br>1: Turn on                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 28:9 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 8    | <b>cg1_refdivsrc_override.</b> Read-write. Reset: 0.<br><b>Description:</b> CG1PLL Refclk Source Select Override<br>This bit is used to override CG1PLL refclk source select.<br>By default, Master_die CG1PLL refclk source is from CG_XTAL 48MHz and Slave_die is 100MHz from external clock chip<br>For Master_die:<br>0: CG1PLL refclk source is from CG_XTAL 48MHz<br>1: CG1PLL refclk source is 100MHz from external clock chip<br>For Slave_die:<br>0: CG1PLL refclk source is 100MHz from external clock chip<br>1: CG1PLL refclk source is from CG_XTAL 48MHz<br>Note: Need to apply a reset after set this bit |
| 7:5  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 4    | <b>refclk_source_switch_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> Refclk Source Switch Mode<br>For CGPLL refclk source, it can be from either CG_XTAL or external reference source.<br>0: Switch refclk source on-the-fly<br>1: Need to apply a Cold Reset for switching different refclk source                                                                                                                                                                                                                                                                                                            |
| 3:1  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 0    | <b>cg1_spread_spectrum_enable.</b> Read-write. Reset: 0.<br><b>Description:</b> CG1_PLL Spread Spectrum Enable<br>0: Disable Spread Spectrum (default)<br>1: Enable Spread Spectrum                                                                                                                                                                                                                                                                                                                                                                                                                                      |

**MISCx0000000C (FCH::MISC::CGPLLCONFIG2)**

Read-write. Reset: 0001\_0000h.

\_aliasHOSTLEGACY; MISCx0000000C; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISCx00000010 (FCH::MISC::CGPLLCONFIG3)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000010; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                 |
| 29    | <b>cg1pll_fracn_en_override</b> . Read-write. Reset: 0. Used with CG1PLL frac-N and <a href="#">SSC</a> clocking only.                                                    |
| 28:13 | Reserved.                                                                                                                                                                 |
| 12:4  | <b>cg1pll_fcw0_int_override</b> . Read-write. Reset: 000h. CG1PLL Override: Integer portion of Frequency Control Word0 (a.k.a. feedback divisor0).                        |
| 3:2   | Reserved.                                                                                                                                                                 |
| 1:0   | <b>cg1pll_refclk_div_override</b> . Read-write. Reset: 0h.<br><b>Description:</b> CG1PLL Override: Reference clock divisor.<br>Settings:<br>2'b00=1<br>2'b01=2<br>2'b1x=4 |

**MISCx00000014 (FCH::MISC::CGPLLCONFIG4)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000014; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                                                                                              |
| 23:8  | <b>cg1pll_fcw1_frac_override</b> . Read-write. Reset: 0000h. CG1PLL Override: Fractional portion of Frequency Control Word1 (a.k.a. feedback divisor1). Intended to be used with frequency ramping. Also used to step PLL frequency and phase for DFT. |
| 7:0   | Reserved.                                                                                                                                                                                                                                              |

**MISCx00000018 (FCH::MISC::CGPLLCONFIG5)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000018; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>cg1pll_fcw_slew_frac_override</b> . Read-write. Reset: 0000h.<br><b>Description:</b> CG1PLL Override: Sets <a href="#">SSC</a> freq ramp rate. Set fractional change in programmed frequency per refclk cycle.<br>e.g. 0.5% downspread SSC at 33.3kHz and fbdiv=80.<br>FCW_slewrates_frac =<br>$216 * 0.00485 * 80 / (15\mu s / 10ns) = 17$<br>Need 31.5kHz and -0.375% |
| 15:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                  |

**MISCx0000001C (FCH::MISC::CGPLLCONFIG6)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx0000001C; MISC=FED8\_0E00h

| Bits  | Description                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------|
| 31:28 | Reserved.                                                                                                                 |
| 27:26 | <b>cgpll_gi_coarse_mant_override</b> . Read-write. Reset: 0h. CG1PLL Coarse integral path fp mult mantissa                |
| 25:21 | Reserved.                                                                                                                 |
| 20:17 | <b>cg1pll_gp_coarse_exp_override</b> . Read-write. Reset: 0h. CG1PLL Override: Coarse proportional path fp mult exponent  |
| 16:13 | <b>cg1pll_gp_coarse_mant_override</b> . Read-write. Reset: 0h. CG1PLL Override: Coarse proportional path fp mult mantissa |
| 12:9  | <b>cg1pll_gi_coarse_exp_override</b> . Read-write. Reset: 0h. CG1PLL Override: Coarse integral path fp mult exponent      |
| 8:0   | Reserved.                                                                                                                 |

**MISCx00000020 (FCH::MISC::IMPCALIBRATION)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000020; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISCx00000024 (FCH::MISC::CLKDRVSTH1)**

Read-write. Reset: 0024\_0249h.

\_aliasHOSTLEGACY; MISCx00000024; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                                                                                                                                                                                               |
| 23:21 | <b>gpp5_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP5 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001          |
| 20:18 | <b>gpp4_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP4 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001          |
| 17:12 | Reserved.                                                                                                                                                                                                                                                                                                                                               |
| 11:9  | <b>gpp_clk3_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP_CLK_3 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001 |
| 8:6   | <b>gpp_clk2_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP_CLK_2 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001 |
| 5:3   | <b>gpp_clk1_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP_CLK_1 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001 |
| 2:0   | <b>gpp_clk0_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.                                                                                                                                                                                                                                                                           |

|  |                                                                                                                                                                                                                                                                        |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Description:</b> Drive Strength control for GPP_CLK_0 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001 |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**MISCx00000028 (FCH::MISC::CLKDRVSTH2)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000028; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:22 | Reserved.                                                                                                                                                                                                                                                                                                                                                    |
| 21    | <b>bp_x48m_output_override.</b> Read-write. Reset: 0.<br><b>Description:</b> BP_X48M clock output override, set this to "1" will override the default enable/disable of BP_X48M_1 clock output.<br>For BP_X48M output clocks, default enabled, this bit is to override default value.                                                                        |
| 20    | Reserved.                                                                                                                                                                                                                                                                                                                                                    |
| 19:12 | <b>debug_sig_sel_0.</b> Read-write. Reset: 00h.<br><b>Description:</b> Debug Signal Selection 0<br>These bits are used to select internal signal for debug observation thru uPAD_S5_X48MO pin with MISCx00000028[27]=1<br>Bit19 is used to select debug signal either from S0 or S5.<br>0: Select S5 Debug Signal Group0<br>1: Select S0 Debug Signal Group0 |
| 11:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                    |

**MISCx0000002C (FCH::MISC::CLKGATEDCNTL)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx0000002C; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:18 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 17    | <b>blinkclk_gateoffen.</b> Read-write. Reset: 0. Check: 1.<br><b>Description:</b> B-Link Clock Gate-Off Enable<br>Internal B-Link clock has two clock trees: one is a free running clock and the other is a gated clock.<br>When all controllers agree to stop the gated B-Link clock and this bit got set, clkgating logic will gate off the clock tree from clock root.<br>Note: Need to enable with PM04[16]=1(non-sticky) and rsmu_local_reg_1C[31]=1(non-sticky).<br>0: Disable B-Link Clock Gate-Off function. Default<br>1: Enable B-Link Clock Gate-Off function |
| 16    | <b>alinkclk_gateoffen.</b> Read-write. Reset: 0. Check: 1.<br><b>Description:</b> A-Link Clock Gate-Off Enable<br>Internal A-Link clock has two clock trees: one is a free-running clock and the other is a gated clock.<br>When all controllers agree to stop the gated A-Link clock and this bit got set, clkgating logic will gate off the clock tree from clock root.<br>Note: Need to enable with PM04[16]=1(non-sticky).<br>0: Disable A-Link Clock Gate-Off function. Default<br>1: Enable A-Link Clock Gate-Off function                                         |
| 15:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

**MISCx00000030 (FCH::MISC::CGPLL\_CONFIGURATION0)**

Read-write.

\_aliasHOSTLEGACY; MISCx00000030; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 16    | <b>uart3_14_77mhz_sclk_enable.</b> Read-write. Reset: 0.<br><b>Description:</b> UART3 14.77MHz SCLK Enable<br>0: Turn off UART0 14.77MHz SCLK<br>1: Turn on UART0 14.77MHz SCLK                                                                                                                                                                                                                                                                |
| 15    | <b>uart2_14_77mhz_sclk_enable.</b> Read-write. Reset: 0.<br><b>Description:</b> UART2 14.77MHz SCLK Enable<br>0: Turn off UART0 14.77MHz SCLK<br>1: Turn on UART0 14.77MHz SCLK                                                                                                                                                                                                                                                                |
| 14    | <b>uart1_14_77mhz_sclk_enable.</b> Read-write. Reset: 0.<br><b>Description:</b> UART1 14.77MHz SCLK Enable<br>0: Turn off UART0 14.77MHz SCLK<br>1: Turn on UART0 14.77MHz SCLK                                                                                                                                                                                                                                                                |
| 13    | <b>uart0_14_77mhz_sclk_enable.</b> Read-write. Reset: 0.<br><b>Description:</b> UART0 14.77MHz SCLK Enable<br>0: Turn off UART0 14.77MHz SCLK<br>1: Turn on UART0 14.77MHz SCLK                                                                                                                                                                                                                                                                |
| 12:4  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3     | <b>usb0_rtcclk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> USB0 RTCCLK (root_RTC_CLK_0) Enable/Disable<br>Note: USB0 RTCCLK clock root is in remote <a href="#">FCH</a> . This bit is from remote FCH Misc register.<br>0: Turn on<br>1: Turn off                                                                                                                                                                                    |
| 2     | <b>usb1_rtcclk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> USB1 RTCCLK (root_RTC_CLK_1) Enable/Disable<br>Note: USB1 RTCCLK clock root is in local <a href="#">FCH</a> . This bit is from local FCH Misc register.<br>0: Turn on<br>1: Turn off                                                                                                                                                                                      |
| 1:0   | <b>pcie_phy_refclk_selection_override.</b> Read-write. Reset: XXb.<br><b>Description:</b> PCIE PHY Refclk Selection Override (P-PHY)<br>CLKB_PCIE_PHY_P/N (fch_tile/CLKGEN)<br>00 = 100MHz CG1_PLL generated (default)<br>01 = 100MHz CG2_PLL generated<br>10 = EXT_GPP0_SRC<br>11 = EXT_GPP2_RX<br>Note: MISCx000000BC[12] to select EXT_GPP0_SRC either is from GPP0 external input or GPP0 external input with divided-by-2 or divided-by-4 |

**MISCx00000034 (FCH::MISC::CGPLL\_CONFIGURATION1)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000034; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:23 | <b>cg1pll_fcw1_int_override.</b> Read-write. Reset: 000h. CG1PLL Override: Integer portion of Frequency Control Word0 (a.k.a. feedback divisor1). Intended to be used with frequency ramping. Also used to step PLL frequency and phase for DFT. |
| 22:0  | Reserved.                                                                                                                                                                                                                                        |

**MISCx00000038 (FCH::MISC::CGPLL\_CONFIGURATION2)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000038; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:23 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 22:21 | <b>cg1pll_external_refclk_sel_override.</b> Read-write. Reset: 0h.<br><b>Description:</b> CG1PLL External Refclk Selection Override<br>This is the refclk mux source inside CLKGEN_S0.<br>00 = local GPP0_RX (external 100M)<br>01 = GPP2_RX<br>10 = CG1_100M<br>11 = remote GPP0_RX<br>The MISCx00000038[22:21] are override bits to this mux inside CLKGEN_S0.<br>The local CLKGEN_S0 default is "00" in iCLK mode and "01" in eCLK mode.<br>The remote CLKGEN_S0 default is "10" in iCLK mode and "01" in eCLK mode.<br>For example:<br>If we want to use external 100MHz as CG1PLL refclk source thru local GPP0_RX in eCLK mode, then program MISCx00000038[22:21]="01". (default "01" XOR program MISCx00000038[22:21]="01" = "00")<br>If we want to use external 100MHz as CG2PLL refclk source thru remote GPP2_RX in iCLK mode, then program MISCx00000038[22:21]="11". (default "10" XOR program MISCx00000038[22:21]="11" = "01") |
| 20    | <b>usb1_phy_cmlclk_static_off.</b> Read-write. Reset: 0.<br><b>Description:</b> USB1 PHY CML Clock Static Turn Off<br>0: USB1 PHY CML Clock will not be static turn off<br>1: USB1 PHY CML Clock will be static turn off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 19    | <b>usb0_phy_cmlclk_static_off.</b> Read-write. Reset: 0.<br><b>Description:</b> USB0 PHY CML Clock Static Turn Off<br>0: USB0 PHY CML Clock will not be static turn off<br>1: USB0 PHY CML Clock will be static turn off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 18    | <b>usb1_dbgclk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> USB1 DBGCLK Enable/Disable<br>0: Enable<br>1: Disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 17    | <b>usb0_dbgclk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> USB0 DBGCLK Enable/Disable<br>0: Enable<br>1: Disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 16    | <b>usb1_host_clk_selection_fw.</b> Read-write. Reset: 0.<br><b>Description:</b> USB1 Host Clock Selection_FW<br>USB1 Host Clock can be either 300M generated from CGPLL or 48M generated from XTAL_PAD when CGPLL 300M generated clock is not available.<br>MISCx00000038[15] is used to select the switching mechanism.<br>For MISCx00000038[16]:<br>1: 300MHz generated by CGPLL<br>0: 48MHz generated from XTAL_PAD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 15    | <b>usb1_host_clk_switch_mechanism.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|      | <b>Description:</b> USB1 Host Clock Switch_ Mechanism<br>USB1 Host Clock can be either 300M generated from CGPLL or 48M generated from XTAL_PAD, default is 300M.<br>Please refer USB's requirement. (pair-up with USB's sequence)<br>USB1 Host Clock can be either 300M generated from CGPLL or 48M generated from XTAL_PAD when CGPLL 300M generated clock is not available.<br>MISCx00000038[15] is used to select the switching mechanism.<br>MISCx00000038[15] =0, HW dynamic switching between 300M and 48M according whether CGPLL is available or not.<br>MISCx00000038[15]=1, FW switch thru MISCx00000038[16]                                                                 |
| 14   | <b>usb1_host_clk_staticoff.</b> Read-write. Reset: 0.<br><b>Description:</b> USB1 Host Clock Static Off<br>If USB1 is not used in a given platform, USB1_Host_Clk can be turn off by set this bit to "1".<br>0: Static Off function is not enabled<br>1: Static Off function is enabled                                                                                                                                                                                                                                                                                                                                                                                                 |
| 13   | <b>usb0_host_clk_selection_fw.</b> Read-write. Reset: 0.<br><b>Description:</b> USB0 Host Clock Selection_FW<br>USB0 Host Clock can be either 300M generated from CGPLL or 48M generated from XTAL_PAD when CGPLL 300M generated clock is not available.<br>MISCx00000038[12] is used to select the switching mechanism.<br>For MISCx00000038[13]:<br>1: 300MHz generated by CGPLL<br>0: 48MHz generated from XTAL_PAD                                                                                                                                                                                                                                                                  |
| 12   | <b>usb0_host_clk_switch_mechanism.</b> Read-write. Reset: 0.<br><b>Description:</b> USB0 Host Clock Switch_ Mechanism<br>USB0 Host Clock can be either 300M generated from CGPLL or 48M generated from XTAL_PAD, default is 300M.<br>Please refer USB's requirement. (pair-up with USB's sequence)<br>USB0 Host Clock can be either 300M generated from CGPLL or 48M generated from XTAL_PAD when CGPLL 300M generated clock is not available.<br>MISCx00000038[12] is used to select the switching mechanism.<br>MISCx00000038[12] =0, HW dynamic switching between 300M and 48M according whether CGPLL is available or not.<br>MISCx00000038[12]=1, FW switch thru MISCx00000038[13] |
| 11   | <b>usb0_host_clk_staticoff.</b> Read-write. Reset: 0.<br><b>Description:</b> USB0 Host Clock Static Off<br>If USB0 is not used in a given platform, USB0_Host_Clk can be turn off by set this bit to "1".<br>0: Static Off function is not enabled<br>1: Static Off function is enabled                                                                                                                                                                                                                                                                                                                                                                                                 |
| 10:0 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

**MISCx0000003C (FCH::MISC::CGPLL\_CONFIGURATION3)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx0000003C; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |



**MISCx00000040 (FCH::MISC::MISCCLKCNTRL0)**

Read-write. Reset: 0000\_4040h.

\_aliasHOSTLEGACY; MISCx00000040; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                     |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | Reserved.                                                                                                                                                                                                                                                       |
| 30    | <b>cg1_cfg_update_req.</b> Read-write. Reset: 0.<br><b>Description:</b> Set this bit will request CG1_PLL to load spread related value into CG1_PLL.<br>The bit will be clear "0" by hardware after request send to CG1_PLL.                                    |
| 29:28 | Reserved.                                                                                                                                                                                                                                                       |
| 27    | <b>blink266m_gate_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> Blink 288MHz Clock Gate_EnB<br>Program "0" to gated off this clock from free running                                                                                                    |
| 26    | Reserved.                                                                                                                                                                                                                                                       |
| 25    | <b>cg1_fbdiv_loaden.</b> Read-write. Reset: 0. Set "1" to enable loading CG1_PLL FB_DIV value form register                                                                                                                                                     |
| 24:18 | Reserved.                                                                                                                                                                                                                                                       |
| 17    | <b>usb0_refclk_drvier_pwdn_s5.</b> Read-write. Reset: 0.<br><b>Description:</b> USB0 (USB controller 0) Refclk Driver (differential) Power-Down in S5<br>When set this bit to "1", USB0 Refclk Driver will be turn off in S5 state.                             |
| 16:15 | Reserved.                                                                                                                                                                                                                                                       |
| 14    | <b>osclkswitchen.</b> Read-write. Reset: 1.<br><b>Description:</b> When this bit is set, the <a href="#">FCH</a> will use CG_XTAL 48M to generate the average 14MHz clock as OSC.<br>0: OSC is 12MHz (CG_XTAL 48M divided by 4)<br>1: OSC is average 14.318MHz. |
| 13:12 | Reserved.                                                                                                                                                                                                                                                       |
| 11    | <b>az_48mclk_pwdn.</b> Read-write. Reset: 0.<br><b>Description:</b> AZ 48MHz Clock Output Driver PWDN<br><a href="#">FCH</a> provided 48MHz clock for AZ.<br>0: AZ 48MHz Clock Output Driver is enable<br>1: AZ 48MHz Clock Output Driver is disable            |
| 10    | Reserved.                                                                                                                                                                                                                                                       |
| 9     | <b>xtal48m_s5_pwdn_en.</b> Read-write. Reset: 0.<br><b>Description:</b> CG 48MHz XTAL Pad Power Down Enable when in S5 state<br>Set this bit allow to power down CG XTAL pad in S5 state.                                                                       |
| 8:7   | Reserved.                                                                                                                                                                                                                                                       |
| 6     | <b>usb0_refclk_drvier_en_s3.</b> Read-write. Reset: 1.<br><b>Description:</b> USB0 (USB controller 0) Refclk Driver (differential) Enable in S3<br>When set this bit to "0", USB0 Refclk Driver will be turn off in S3 state.                                   |
| 5     | <b>usb1_refclk_drvier_pwdn_s3.</b> Read-write. Reset: 0.<br><b>Description:</b> USB1 (USB controller 1) Refclk Driver (differential) Power-Down in S3<br>When set this bit to "1", USB1 Refclk Driver will be turn off in S3 state.                             |
| 4     | Reserved.                                                                                                                                                                                                                                                       |
| 3     | <b>usb1_refclk_drvier_pwdn_s5.</b> Read-write. Reset: 0.<br><b>Description:</b> USB1 (USB controller 1) Refclk Driver (differential) Power-Down in S5<br>When set this bit to "1", USB1 Refclk Driver will be turn off in S5 state.                             |
| 2     | Reserved.                                                                                                                                                                                                                                                       |
| 1     | <b>corespeedmode.</b> Read-write. Reset: 0.<br><b>Description:</b> Slow down Internal Core Clock (B-Link clock) for power saving.<br>0: Full speed B-Link clock<br>1: Slow speed B-Link clock                                                                   |

|   |           |
|---|-----------|
| 0 | Reserved. |
|---|-----------|

**MISCx00000044 (FCH::MISC::MISCCLKCNTL1)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000044; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISCx00000048 (FCH::MISC::MISCCLKCNTL2)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000048; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISCx0000004C (FCH::MISC::MISCCLKCNTL3)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx0000004C; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>low_power_display_400mclk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> Low Power Display 400MHz Clock Output Enable/Disable<br>Non Sticky bit.<br>0: Enable<br>1: Disable                                                                                                                                       |
| 30    | <b>low_power_display_300mclk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> Low Power Display 300MHz Clock Output Enable/Disable<br>Non Sticky bit.<br>0: Enable<br>1: Disable                                                                                                                                       |
| 29    | <b>low_power_display_600mclk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> Low Power Display 600MHz Clock Output Enable/Disable<br>Non Sticky bit.<br>0: Enable<br>1: Disable                                                                                                                                       |
| 28:12 | Reserved.                                                                                                                                                                                                                                                                                                                   |
| 11    | <b>usb1_host_clk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> USB1 Host Clock EnableB/Disable<br>Non Sticky bit.<br>When resume, it will be enable (default setting "0") regardless what has been programmed and BIOS needs to program it again if this bit has been programmed to "1".<br>0: Enable<br>1: Disable |
| 10    | <b>usb0_host_clk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> USB0 Host Clock EnableB/Disable<br>Non Sticky bit.<br>When resume, it will be enable (default setting "0") regardless what has been programmed and BIOS needs to program it again if this bit has been programmed to "1".<br>0: Enable<br>1: Disable |
| 9     | <b>sata_sgpio_2_3_clk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> SATA SGPIO Clock Enable/Disable (for host controller 2 & 3)<br>Non Sticky bit.<br>0: Turn on (Enable)<br>1: Turn off (Disable)                                                                                                                  |
| 8     | <b>sata_sgpio_0_1_clk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> SATA SGPIO Clock Enable/Disable (for host controller 1 & 2)<br>Non Sticky bit.<br>0: Turn on (Enable)<br>1: Turn off (Disable)                                                                                                                  |
| 7     | <b>sata_100m_core_3_clk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> SATA 100M Core Clock Enable/Disable (for host controller 3)<br>Non Sticky bit.<br>0: Turn on (Enable)<br>1: Turn off (Disable)                                                                                                                |
| 6     | <b>sata_100m_core_2_clk_enb.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                      |

|     |                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> SATA 100M Core Clock Enable/Disable (for host controller 2)<br>Non Sticky bit.<br>0: Turn on (Enable)<br>1: Turn off (Disable)                                                                                                                                                                                                                                             |
| 5   | <b>sata_100m_core_1_clk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> SATA 100M Core Clock Enable/Disable (for host controller 1)<br>Non Sticky bit.<br>0: Turn on (Enable)<br>1: Turn off (Disable)                                                                                                                                                                                   |
| 4   | <b>sata_100m_core_0_clk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> SATA 100M Core Clock Enable/Disable (for host controller 0)<br>Non Sticky bit.<br>0: Turn on (Enable)<br>1: Turn off (Disable)                                                                                                                                                                                   |
| 3:2 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                      |
| 1   | <b>pcie_phy_p_refclk_static_pwdn_cntl.</b> Read-write. Reset: 0.<br><b>Description:</b> PCIE_P_Set_PHY Refclk Driver Static PWDN Control<br>0: Enable on-chip driver (default)<br>1: Disable on-chip driver (if it is unused)<br>When resume, it will be static shut-down with on-chip driver enable (default setting) regardless what has been programmed and BIOS needs to program it again. |
| 0   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                      |

**MISCx00000050 (FCH::MISC::JTAGCNTRL)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; MISCx00000050; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 7:0  | Reserved.   |

**MISCx00000058 (FCH::MISC::CLKCNTRL58)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000058; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISCx0000005C (FCH::MISC::CLKCNTRL5C)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx0000005C; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISCx00000060 (FCH::MISC::IDLECNTRL)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000060; MISC=FED8\_0E00h

| Bits  | Description                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------|
| 31:24 | <b>idlecount.</b> Read-write. Reset: 00h. This returns the idle count from the latest monitored period. |
| 23:0  | Reserved.                                                                                               |

**MISCx00000064 (FCH::MISC::DSMXTRIG\_ROUTING)**

Read-write. Reset: FFFFh.

\_aliasHOSTLEGACY; MISCx00000064; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | <b>dsmxtrig3_to_ec.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable CNB2ACPI_AL_trigB[3] low-pulse will trigger a 8051 ( <a href="#">EC</a> ) action<br>0: Disabled                          |
| 14   | <b>dsmxtrig2_to_ec.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable CNB2ACPI_AL_trigB[2] low-pulse will trigger a 8051 ( <a href="#">EC</a> ) action<br>0: Disabled                          |
| 13   | <b>dsmxtrig1_to_ec.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable CNB2ACPI_AL_trigB[1] low-pulse will trigger a 8051 ( <a href="#">EC</a> ) action<br>0: Disabled                          |
| 12   | <b>dsmxtrig0_to_ec.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable CNB2ACPI_AL_trigB[0] low-pulse will trigger a 8051 ( <a href="#">EC</a> ) action<br>0: Disabled                          |
| 11   | <b>dsmxtrig3_to_autotransaction.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable the falling of CNB2ACPI_AL_trigB[3] to trigger a pre-programmed transaction.<br>0: Disabled                 |
| 10   | <b>dsmxtrig2_to_autotransaction.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable the falling of CNB2ACPI_AL_trigB[2] to trigger a pre-programmed transaction.<br>0: Disabled                 |
| 9    | <b>dsmxtrig1_to_autotransaction.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable the falling of CNB2ACPI_AL_trigB[1] to trigger a pre-programmed transaction.<br>0: Disabled                 |
| 8    | <b>dsmxtrig0_to_autotransaction.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable the falling of CNB2ACPI_AL_trigB[0] to trigger a pre-programmed transaction.<br>0: Disabled                 |
| 7    | <b>dsmxtrig3_to_irq.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable CNB2ACPI_AL_trigB[3] being low to trigger a specific IRQ (depend on the setting in IO C00/C01 registers)<br>0: Disabled |
| 6    | <b>dsmxtrig2_to_irq.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable CNB2ACPI_AL_trigB[2] being low to trigger a specific IRQ (depend on the setting in IO C00/C01 registers)<br>0: Disabled |
| 5    | <b>dsmxtrig1_to_irq.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable CNB2ACPI_AL_trigB[1] being low to trigger a specific IRQ (depend on the setting in IO C00/C01 registers)<br>0: Disabled |
| 4    | <b>dsmxtrig0_to_irq.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable CNB2ACPI_AL_trigB[0] being low to trigger a specific IRQ (depend on the setting in IO C00/C01 registers)<br>0: Disabled |
| 3    | <b>dsmxtrig3_to_smi.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable the falling of CNB2ACPI_AL_trigB[3] to trigger an <a href="#">SMI</a><br>0: Disabled                                    |
| 2    | <b>dsmxtrig2_to_smi.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: Enable the falling of CNB2ACPI_AL_trigB[2] to trigger an <a href="#">SMI</a><br>0: Disabled                                    |

|   |                                                                                                                    |
|---|--------------------------------------------------------------------------------------------------------------------|
| 1 | <b>dsmxtrig1_to_smi.</b> Read-write. Reset: 1.                                                                     |
|   | <b>Description:</b> 1: Enable the falling of CNB2ACPI_AL_trigB[1] to trigger an <a href="#">SMI</a><br>0: Disabled |
| 0 | <b>dsmxtrig0_to_smi.</b> Read-write. Reset: 1.                                                                     |
|   | <b>Description:</b> 1: Enable the falling of CNB2ACPI_AL_trigB[0] to trigger an <a href="#">SMI</a><br>0: Disabled |

**MISCx00000068 (FCH::MISC::MEMPWRSVCNTRL)**

Read-write.

\_aliasHOSTLEGACY; MISCx00000068; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>ec_bypass_mem_dsd.</b> Read-write. Reset: 1. <a href="#">EC</a> memory BypassMemdsd control.                                                                                                                                                                                                                       |
| 30    | <b>bios_ram_bypass_mem_dsd.</b> Read-write. Reset: 1.<br><b>Description:</b> BIOS RAM BypassMemdsd control.<br>0: Enable memory DS features.<br>1: Disable memory DS.<br>Default is 1'b1, memory DS is disabled.                                                                                                      |
| 29    | <b>shdw_ram_bypass_mem_dsd.</b> Read-write. Reset: 1.<br><b>Description:</b> Shadow RAM BypassMemdsd control.<br>0: Enable memory DS features.<br>1: Disable memory DS.<br>Default is 1'b1, memory DS is disabled.                                                                                                    |
| 28    | <b>hub_mem_slp_dis.</b> Read-write. Reset: 0. USB Hub BypassMemdsd control                                                                                                                                                                                                                                            |
| 27    | <b>ehci_mem_slp_dis.</b> Read-write. Reset: 0. USB2 BypassMemdsd control                                                                                                                                                                                                                                              |
| 26:21 | Reserved.                                                                                                                                                                                                                                                                                                             |
| 20    | <b>gbe_bypass_mem_dsd.</b> Read-write. Reset: 1.<br><b>Description:</b> GBE Memory BypassMemdsd control<br>1: Disable Memory <a href="#">Shutdown</a> .<br>0: Enable Memory Shutdown.<br>Default is 1'b1, memory Shutdown is disabled.                                                                                |
| 19    | <b>ufs_bypass_mem_dsd.</b> Read-write. Reset: 1.<br><b>Description:</b> UFS Memory BypassMemdsd control<br>1: Disable Memory shutdown.<br>0: Enable Memory shutdown.<br>Default is 1'b1, memory <a href="#">Shutdown</a> is disabled.<br>Note: While PCI reset assert, memory will keep shutdown regardless this bit. |
| 18    | <b>sd_bypass_mem_dsd.</b> Read-write. Reset: 1. SD memory BypassMemdsd control.                                                                                                                                                                                                                                       |
| 17    | <b>sata_bypass_mem_dsd.</b> Read-write. Reset: 0. SATAMemory BypassMemdsd control.                                                                                                                                                                                                                                    |
| 16    | <b>cfga_xhc_ncpu_mem_slp_dis.</b> Read-write. Reset: 0. USB3 memory BypassMemdsd control.                                                                                                                                                                                                                             |
| 15    | <b>hid_bypass_mem_dsd.</b> Read-write. Reset: 1.<br><b>Description:</b> HID memory (2 of them, 1024x32 and 512x32) BypassMemdsd control.<br>0: Enable memory DS and SD features.<br>1: Disable memory DS and SD.<br>Default is 1'b1, memory DS/SD is disabled.                                                        |
| 14:6  | Reserved.                                                                                                                                                                                                                                                                                                             |
| 5     | <b>i3c_bypass_mem_dsd.</b> Read-write. Reset: 1. <a href="#">I3C</a> memory BypassMemdsd control.                                                                                                                                                                                                                     |
| 4     | <b>ila_bypass_mem_dsd.</b> Read-write. Reset: 1. ILA memory BypassMemdsd control.                                                                                                                                                                                                                                     |
| 3     | <b>amba_bypass_mem_dsd.</b> Read-write. Reset: 1.<br><b>Description:</b> AMBA memory BypassMemdsd control.<br>0: Enable memory DS and SD features.<br>1: Disable memory DS and SD.<br>Default is 1'b1, memory DS/SD is disabled.                                                                                      |
| 2     | <b>ab_bypass_mem_dsd.</b> Read-write. Reset: 1. AB memory BypassMemdsd control.                                                                                                                                                                                                                                       |
| 1     | <b>sbg_bypass_mem_dsd.</b> Read-write. Reset: 1.                                                                                                                                                                                                                                                                      |

|   |                                                                                                                                                                            |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <b>Description:</b> SBG memory BypassMemdsd control.<br>0: Enable memory DS and SD features.<br>1: Disable memory DS and SD.<br>Default is 1'b1, memory DS/SD is disabled. |
| 0 | Reserved.                                                                                                                                                                  |



**MISCx0000006C (FCH::MISC::ECOBIT2)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx0000006C; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>postcodewidthsel.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Reading MISC_Reg:78h returns a 8-bit PostCode<br>0: Reading MISC_Reg:78h returns a 32-bit PostCode                                                                                                                                         |
| 30   | <b>type2straponce_dis.</b> Read-write. Reset: 0. Set to 0 to capture TypeII straps only at the first time PwrGood goes up.                                                                                                                                                                                          |
| 29   | <b>fixrtcintratesel_dis.</b> Read-write. Reset: 0. Set to 0 to avoid interrupts when RTC RegisterA[3:0] (Rate Selection) is changed.                                                                                                                                                                                |
| 28   | <b>fixsmbusystuck_dis.</b> Read-write. Reset: 0. Reserved, set to 0                                                                                                                                                                                                                                                 |
| 27   | <b>xtalpciclk_notalwaysen.</b> Read-write. Reset: 0. Check: 1. XtalPciClk switch config                                                                                                                                                                                                                             |
| 26   | <b>xtalpciclk_simplesel_dis.</b> Read-write. Reset: 0. XtalPciClk switch config                                                                                                                                                                                                                                     |
| 25   | <b>xtalpciclk_undoasrtrstchg.</b> Read-write. Reset: 0. XtalPciClk switch config                                                                                                                                                                                                                                    |
| 24   | Reserved.                                                                                                                                                                                                                                                                                                           |
| 23   | <b>enlpcclk0leakfix.</b> Read-write. Reset: 0.<br><b>Description:</b> LPCCLK0 is randomly 1 or 0 during S3/S5. There will be leakage when LPCCLK0 is 1. This bit enable the fix for it.<br>1: Enable<br>0: Disable                                                                                                  |
| 22   | <b>undoacpicdcchange.</b> Read-write. Reset: 0. Undo ACPI CDC changes                                                                                                                                                                                                                                               |
| 21   | <b>sel_intr_disable_method.</b> Read-write. Reset: 0. Check: 1.<br><b>Description:</b> 0: Setting InterruptDis bit to 1 will keep K8_arb and MSI state machine at idle state, and therefore, block all upstream messages except STPCLK.<br>1: Setting InterruptDis bit to 1 will block the output of IOAPIC and PIC |
| 20   | <b>en_t62_5msstart_fix.</b> Read-write. Reset: 0. Enable the fix for T62_5msStart                                                                                                                                                                                                                                   |
| 19   | <b>en_kbrstfix.</b> Read-write. Reset: 0. Enable the fix for kb_reset_sync1                                                                                                                                                                                                                                         |
| 18   | <b>block_ila_rst.</b> Read-write. Reset: 0. Block ILA reset                                                                                                                                                                                                                                                         |
| 17   | <b>fatalerrfixecoen0.</b> Read-write. Reset: 0.<br><b>Description:</b> Gating S0Rst to toggle reset.<br>1: Enable<br>0: Disable                                                                                                                                                                                     |
| 16   | <b>fatalerrfixecoen1.</b> Read-write. Reset: 0.<br><b>Description:</b> Gating DevAllRstB to toggle reset.<br>1: Enable<br>0: Disable                                                                                                                                                                                |
| 15   | Reserved.                                                                                                                                                                                                                                                                                                           |
| 14   | <b>fullcf92devrsten_n.</b> Read-write. Reset: 0.<br><b>Description:</b> For DERVXRTL-1749<br>0: enable fix for CF9=0xE to 15ns wide oDevAllRstB<br>1: disable the fix                                                                                                                                               |
| 13   | <b>pwrbut4sshdn2devrsten_n.</b> Read-write. Reset: 0.<br><b>Description:</b> For DERVXRTL-1749<br>0: enable fix for 4s shutdown to 15ns wide oDevAllRstB<br>1: disable the fix                                                                                                                                      |
| 12:6 | Reserved.                                                                                                                                                                                                                                                                                                           |
| 5    | <b>posdecwillwaitisa.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Will wait the positive decode<br>1: Won't wait the positive decode                                                                                                                                                                        |

|   |                                                                                                                                                                                                                                                                                            |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4 | <b>io92willtrigisa</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: IO92 won't trigger IsaBridge<br>1: IO92 will trigger unnecessary IsaBridge                                                                                                                                        |
| 3 | Reserved.                                                                                                                                                                                                                                                                                  |
| 2 | <b>undoacpisupportabretry</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: ACPI PCI master support target retry from AB<br>1: ACPI PCI master doesn't support target retry from AB                                                                                                    |
| 1 | <b>pic2topicpic1directly</b> . Read-write. Reset: 0.<br><b>Description:</b> In the old code, PIC2 will send another INT rising edge to PIC1 when:<br>PIC2 INT is still high.<br>And<br>Software write any of PIC2 registers.<br>This bit enable the fix for it.<br>1: Enable<br>0: Disable |
| 0 | <b>usenew8254ctl</b> . Read-write. Reset: 0.<br><b>Description:</b> There are some CDC violations in 8254 timers. This bit enable the fix for it.<br>1: Enable 8254 timer CDC fix<br>0: Disable                                                                                            |

**MISCx00000070 (FCH::MISC::OSCFREQCOUNTER)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000070; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>countenable</b> . Read-write. Reset: 0. When set, it enables the internal counter to count the number of OSC clocks. When SW is not using this function, it should always set it back to 0 to conserve power.                            |
| 30    | <b>countisvalid</b> . Read-write. Reset: 0. When OscCountPerSec is valid, this bit is set. This bit is read only. SW should always wait for this bit to be set before it should read OscCountPerSec                                         |
| 29:28 | Reserved.                                                                                                                                                                                                                                   |
| 27:0  | <b>osccountpersec</b> . Read-write. Reset: 000_0000h. Number of OSC clocks per 1 second. Whenever bit 31 (CountEnable) is set, an internal counter will start counting the number of OSC clocks per second and record the count value here. |

**MISCx00000074 (FCH::MISC::HPETCLKPERIOD)**

Read-write. Reset: 0429\_B17Eh.

\_aliasHOSTLEGACY; MISCx00000074; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>hpetclkperiod</b> . Read-write. Reset: 0429_B17Eh. The register controls the value of clkperiod register in HPET <a href="#">MMIO</a> register space. |

**MISCx00000078 (FCH::MISC::POSTCODE)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000078; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>postcode_31_0</b> . Read-only. Reset: 0000_0000h.<br><b>Description:</b> IO-Wr 83h~80h can write an internal 32-bit PostCode Register (PostCode[31:0]).<br>IO-Rd 80h will return PostCode[7:0].<br>Reading Misc_Reg:7Bh~78h with PostCodeWidthSel=1 will return {24'b0, PostCode[7:0]}.<br>Reading Misc_Reg:7Bh~78h with PostCodeWidthSel=0 will return PostCode[31:0]. |

**MISCx0000007C (FCH::MISC::POSTCODESTACK)**

Read-only. Reset: FFFF\_FFFFh.

\_aliasHOSTLEGACY; MISCx0000007C; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                  |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>postcodestack_31_0.</b> Read-only. Reset: FFFF_FFFFh.<br><b>Description:</b> 8 deep post code STACK read out window.<br>Each time read will get 32 bits post code. Unused byte will return all 0s.<br>If write full, will lost the oldest data and fill in the new data.<br>When SW read out the STACK, it will read from new data to old data, and don't flush them. Extra read will get duplicate data. |

**MISCx00000080 (FCH::MISC::STRAPSTATUS)**

Read-only.

\_aliasHOSTLEGACY; MISCx00000080; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                          |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:28 | <b>packageid.</b> Read-only. Reset: XXXXb. Specify the package ID of the current Die.                                                                                                                                                                                                |
| 27:24 | <b>dieid.</b> Read-only. Reset: XXXXb. Specify the Die ID of the current Die.                                                                                                                                                                                                        |
| 23:18 | Reserved.                                                                                                                                                                                                                                                                            |
| 17    | <b>clkgenstrap.</b> Read-only. Reset: X.<br><b>Description:</b> 1: Select 48MHz Crystal clock<br>0: Select 100MHz <a href="#">PCIe</a> clock                                                                                                                                         |
| 16:6  | Reserved.                                                                                                                                                                                                                                                                            |
| 5     | <b>waf_train_disable.</b> Read-only. Reset: 0.<br><b>Description:</b> WAFL_TRAINING_DISABLE = 0 (default) means WAFL link will be trained.<br>WAFL_TRAINING_DISABLE = 1 means WAFL link will not be trained.                                                                         |
| 4     | Reserved.                                                                                                                                                                                                                                                                            |
| 3     | <b>romtype_1.</b> Read-only. Reset: X.<br><b>Description:</b> 1: Rom Strap is eSPI without SAFS support (in combination with MISCx00000080[romtype_0])<br>0: Rom strap is SPI (in combination with MISCx00000080[romtype_0]).                                                        |
| 2     | Reserved.                                                                                                                                                                                                                                                                            |
| 1     | <b>romtype_0.</b> Read-only. Reset: X.<br><b>Description:</b> Reserved, set to 1<br>1: Rom Strap is SPI                                                                                                                                                                              |
| 0     | <b>alternateimage.</b> Read-only. Reset: X.<br><b>Description:</b> 1: SPI ROM support Alternate Image, Software need to enable 32-bit address SPI mode to access ROM > 16MB<br>0: SPI ROM does not support Alternate Image, 24-bit address SPI mode can be used to access ROM = 16MB |

**MISCx00000084 (FCH::MISC::HSP\_SECURITY)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000084; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISCx00000088 (FCH::MISC::POSTCODE\_CONTROL)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000088; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:1 | Reserved.                                                                                                                                                                         |
| 0    | <b>post_rd_ptr_rst.</b> Read-write. Reset: 0. Software reset bit for read pointer. When new turn of read needed, set this bit and read pointer will go back for next turn's read. |

**MISCx00000090 (FCH::MISC::AUTOTRANSACTION)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; MISCx00000090; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | <b>transactiontype</b> . Read-write. Reset: 0h. PCI Command type used for this transaction. For example, if this is programmed with a value of 3h, the transaction issue will be an IO write.                                                           |
| 3:2  | <b>bytecount</b> . Read-write. Reset: 0h.<br><b>Description:</b> 00: 1 byte<br>01: 2 bytes<br>10: 4 bytes<br>11: 4 bytes                                                                                                                                |
| 1    | <b>dualaddr</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Use single address cycle<br>1: Use dual address cycle                                                                                                                                 |
| 0    | <b>autoexecute</b> . Read-write. Reset: 0. Writing this bit will cause the HW to execute the transaction defined by the definition below. Once it is written, this bit stays as 1 until the transaction is completed, in which case it will return to 0 |

**MISCx00000091 (FCH::MISC::ALLOWEC)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; MISCx00000091; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                                                                                                                                        |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:3  | Reserved.                                                                                                                                                                                                                                                                                          |
| 2    | <b>autotriggerfromcpuen</b> . Read-write. Reset: 0. If this bit is set, a falling edge on KSO15/XDB[1] /(&cnb_fch_dsm_xtrig[3:0]) will trigger this autotransaction logic                                                                                                                          |
| 1    | <b>disableauto</b> . Read-write. Reset: 0. If this bit is set, the entire AutoTransaction logic is disabled. Once this bit is set, it cannot be cleared except by system reset                                                                                                                     |
| 0    | <b>allowectoautotransacten</b> . Read-write. Reset: 0. When this bit is 0, <a href="#">EC</a> cannot write to any of registers relating to any of the registers in the Auto Transaction Generation logic. When this bit is 1, then EC can change any of these bits. Only BIOS can change this bit. |

**MISCx00000094 (FCH::MISC::AUTOADDRLOW)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000094; MISC=FED8\_0E00h

| Bits | Description                                                                                             |
|------|---------------------------------------------------------------------------------------------------------|
| 31:0 | <b>autoaddrlow</b> . Read-write. Reset: 0000_0000h. Low address to be used by the AutoExecute operation |

**MISCx00000098 (FCH::MISC::AUTOADDRHIGH)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx00000098; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>autoaddrhigh</b> . Read-write. Reset: 0000_0000h. High address to be used by the AutoExecute operation. This register is only applicable when DualAddr = 1. |

**MISCx0000009C (FCH::MISC::AUTODATA)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx0000009C; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>autodata</b> . Read-write. Reset: 0000_0000h. If the operation is read, this register will return the read data. If the TransactionType is a write command, this register will contain the write data. Note byte is aligned accordingly. |

**MISCx000000A0 (FCH::MISC::CGPLLCNTRL0)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000A0; MISC=FED8\_0E00h

**Bits Description**

31:0 Reserved.

**MISCx000000A4 (FCH::MISC::CGPLLCNTRL1)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000A4; MISC=FED8\_0E00h

**Bits Description**

31:0 Reserved.

**MISCx000000A8 (FCH::MISC::CGPLLCNTRL2)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000A8; MISC=FED8\_0E00h

**Bits Description**

31:0 Reserved.

**MISCx000000AC (FCH::MISC::CGPLLCNTRL3)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000AC; MISC=FED8\_0E00h

**Bits Description**

31:23 Reserved.

22 **cpu\_refclk\_driver\_static\_pwdn.** Read-write. Reset: 0.

**Description:** CPU Refclk (100MHz) Driver Static PWDN  
 Program this bit will static turn off or on CPU refclk.  
 0 = enable (turn on)  
 1 = disable (turn off)

21 **cpu\_100m\_refclk\_selection.** Read-write. Reset: 0.

**Description:** CPU\_100M Refclk Selection  
 0 = 100MHz CG1\_PLL generated (default)  
 1 = GPP1 external refclk  
 Note: MISCx000000BC[11] to select EXT\_GPP1\_SRC either is from GPP1 external input or GPP1 external input with divided-by-2 or divided-by-4

20 **smu\_100m\_refclk\_selection.** Read-write. Reset: 0.

**Description:** SMU\_100M Refclk Selection  
 0 = 100MHz CG1\_PLL generated (default)  
 1 = GPP0 external refclk

19:0 Reserved.

**MISCx000000B0 (FCH::MISC::CGPLLCNTRL4)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000B0; MISC=FED8\_0E00h

**Bits Description**

31:0 Reserved.

**MISCx000000B4 (FCH::MISC::CGPLLCNTRL5)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000B4; MISC=FED8\_0E00h

**Bits Description**

31:0 Reserved.

**MISCx000000B8 (FCH::MISC::CGPLLCNTRL6)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000B8; MISC=FED8\_0E00h

**Bits Description**

31:0 Reserved.

**MISCx000000BC (FCH::MISC::CGPLLCNTRL7)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000BC; MISC=FED8\_0E00h

**Bits Description**

31:28 Reserved.

27:26 **gpp5\_refclk\_selection\_override**. Read-write. Reset: 0h.**Description:** GPP5 Refclk selection Override

00 = CG\_PLL generated 100MHz

01 = CG2\_PLL generated 100MHz (not a valid option)

10 = Reserved

11 = Ext\_GPP0\_SRC

Note: MISCx000000BC[13] to select EXT\_GPP0\_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4

25:24 **gpp4\_refclk\_selection\_override**. Read-write. Reset: 0h.**Description:** GPP4 Refclk selection Override

00 = CG\_PLL generated 100MHz

01 = CG2\_PLL generated 100MHz (not a valid option)

10 = Reserved

11 = Ext\_GPP0\_SRC

Note: MISCx000000BC[13] to select EXT\_GPP0\_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4

23:22 **gpp3\_refclk\_selection\_override**. Read-write. Reset: 0h.**Description:** GPP3 Refclk selection Override

00 = CG\_PLL generated 100MHz

01 = CG2\_PLL generated 100MHz (not a valid option)

10 = Reserved

11 = Reserved

Note: MISCx000000BC[13] to select EXT\_GPP0\_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4

21:20 **gpp2\_refclk\_selection\_override**. Read-write. Reset: 0h.**Description:** GPP2 Refclk selection Override

00 = CG\_PLL generated 100MHz

01 = CG2\_PLL generated 100MHz (not a valid option)

10 = Reserved

11 = Reserved

Note: MISCx000000BC[13] to select EXT\_GPP0\_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4

19:18 **gpp1\_refclk\_selection\_override**. Read-write. Reset: 0h.**Description:** GPP1 Refclk selection Override

00 = CG\_PLL generated 100MHz

01 = CG2\_PLL generated 100MHz (not a valid option)

10 = Reserved

11 = Reserved

Note: MISCx000000BC[13] to select EXT\_GPP0\_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4

17:16 **gpp0\_refclk\_selection\_override**. Read-write. Reset: 0h.

|     |                                                                                                                                                                                                                                                                                                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> GPP0 Refclk selection Override<br>00 = CG1_PLL generated 100MHz<br>01 = CG2_PLL generated 100MHz (not a valid option)<br>10 = GPP2_RX (GPP2_BOT)<br>11 = Reserved<br>Note: MISCx000000BC[12] to select EXT_GPP2_SRC either is from GPP2 external input or GPP2 external input with divided-by-2/divided-by-4 |
| 15  | <b>ext_bypassclk_en.</b> Read-write. Reset: 0.<br><b>Description:</b> EXT_BYPASSCLK_EN<br>Enable/Disable REF_BYPASSCLK external input thru GPP3_RX<br>0 = disable<br>1 = enable                                                                                                                                                  |
| 14  | <b>clk_cgpll_ext_pwdn.</b> Read-write. Reset: 0.<br><b>Description:</b> CLK_CGPLL_EXT_PWDN<br>Enable/Disable CGPLL external refclk source thru GPP0 input.<br>0 = enable<br>1 = disable                                                                                                                                          |
| 13  | <b>ext_gpp0_refclk_sel.</b> Read-write. Reset: 0.<br><b>Description:</b> EXT_GPP0_REFCLK_SEL<br>0 = GPP0 external input<br>1 = GPP0 external input divided-by-2(div-2 or div-4 depend on MISCx0000003C[1])                                                                                                                       |
| 12  | Reserved.                                                                                                                                                                                                                                                                                                                        |
| 11  | <b>ext_gpp1_refclk_sel.</b> Read-write. Reset: 0.<br><b>Description:</b> EXT_GPP1_REFCLK_SEL<br>0 = GPP1 external input<br>1 = GPP1 external input divided-by-2(div-2 or div-4 depend on MISCx0000003C[3])                                                                                                                       |
| 10  | <b>smu_100m_refclk_driver_pwdn.</b> Read-write. Reset: 0.<br><b>Description:</b> SMU_100M Refclk Driver PWDN<br>0 = enable<br>1 = disable                                                                                                                                                                                        |
| 9:0 | Reserved.                                                                                                                                                                                                                                                                                                                        |

**MISCx000000C0 (FCH::MISC::IOTRAPPING0)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; MISCx000000C0; MISC=FED8\_0E00h

| Bits | Description                                                                                                        |
|------|--------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>iotrappingadr0.</b> Read-write. Reset: 0000h. Specify the I/O address 0 which causes <a href="#">SMI</a> event. |

**MISCx000000C2 (FCH::MISC::IOTRAPPING1)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; MISCx000000C2; MISC=FED8\_0E00h

| Bits | Description                                                                                                        |
|------|--------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>iotrappingadr1.</b> Read-write. Reset: 0000h. Specify the I/O address 1 which causes <a href="#">SMI</a> event. |

**MISCx000000C4 (FCH::MISC::IOTRAPPING2)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; MISCx000000C4; MISC=FED8\_0E00h

| Bits | Description                                                                                                        |
|------|--------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>iotrappingadr2.</b> Read-write. Reset: 0000h. Specify the I/O address 2 which causes <a href="#">SMI</a> event. |

**MISCx000000C6 (FCH::MISC::IOTRAPPING3)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; MISCx000000C6; MISC=FED8\_0E00h

| Bits | Description                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>iotrappingadr3</b> . Read-write. Reset: 0000h. Specify the I/O address 3 which causes <a href="#">SMI</a> event. |

**MISCx000000C8 (FCH::MISC::CFGTRAPPING0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000C8; MISC=FED8\_0E00h

| Bits | Description                                                                                                               |
|------|---------------------------------------------------------------------------------------------------------------------------|
| 31:2 | <b>cfgtrappingadr0</b> . Read-write. Reset: 0000_0000h. Specify the CFG address 0 which causes <a href="#">SMI</a> event. |
| 1:0  | Reserved.                                                                                                                 |



**MISCx000000CC (FCH::MISC::SMITRAPPINGRWRDOVR)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000CC; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:25 | Reserved.                                                                                                                                                                                                                                                                                                                                                                    |
| 24:17 | <b>i3cpadsmodeppodselect.</b> Read-write. Reset: 00h.<br><b>Description:</b> 0x00: <a href="#">I3C</a> pads work on Open-Drain mode.<br>0xFF: I3C pads work on Push-Pull mode.<br>Others: Reserved.<br>Note: This field should only be 0x00 or 0xFF, other values are not recommended. This field should be set to 0xFF if any of pads I2C0 ~ 3 is set to the GPIO function. |
| 16    | <b>cfgtrappingrw0.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Trap on CFG read access on the address specified in CfgTrappingAdr0<br>1: Trap on CFG write access on the address specified in CfgTrappingAdr0                                                                                                                                                        |
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                                                                                                    |
| 12    | <b>memtrappingrdovr0.</b> Read-write. Reset: 0. Set to 1 to force read data to be replaced by MemRdOvrData0.                                                                                                                                                                                                                                                                 |
| 11:9  | Reserved.                                                                                                                                                                                                                                                                                                                                                                    |
| 8     | <b>memtrappingrw0.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Trap on MEM read access on the address specified in MemTrappingAdr0<br>1: Trap on MEM write access on the address specified in MemTrappingAdr0                                                                                                                                                        |
| 7:4   | Reserved.                                                                                                                                                                                                                                                                                                                                                                    |
| 3     | <b>iotrappingrw3.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Trap on I/O read access on the address specified in IoTrappingAdr3<br>1: Trap on I/O write access on the address specified in IoTrappingAdr3                                                                                                                                                           |
| 2     | <b>iotrappingrw2.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Trap on I/O read access on the address specified in IoTrappingAdr2<br>1: Trap on I/O write access on the address specified in IoTrappingAdr2                                                                                                                                                           |
| 1     | <b>iotrappingrw1.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Trap on I/O read access on the address specified in IoTrappingAdr1<br>1: Trap on I/O write access on the address specified in IoTrappingAdr1                                                                                                                                                           |
| 0     | <b>iotrappingrw0.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Trap on I/O read access on the address specified in IoTrappingAdr0<br>1: Trap on I/O write access on the address specified in IoTrappingAdr0                                                                                                                                                           |

**MISCx000000D0 (FCH::MISC::MEMTRAPPING0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000D0; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:2 | <b>memtrappingadr0</b> . Read-write. Reset: 0000_0000h. Specify the 30-bit MEM address 0 which causes <a href="#">SMI</a> even, lowest 2 bits are ignored. |
| 1:0  | Reserved.                                                                                                                                                  |

**MISCx000000D4 (FCH::MISC::MEMRDOVRDATA0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000D4; MISC=FED8\_0E00h

| Bits | Description                                                                                                                                                                               |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>memtrappingrdata0</b> . Read-write. Reset: 0000_0000h. The 32 bit data is used as the return data when the memory read trapping is enabled in MemTrapping0. with MemTrappingRdOvr0 = 1 |

**MISCx000000D8 (FCH::MISC::I2C0\_PADCTRL)**

Read-write. Reset: 0000\_0030h.

This 32-bit register is used to control [I3C](#) and [I2C](#) pad.\_aliasHOSTLEGACY; MISCx000000D8; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>spikercsel_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Select <a href="#">RC</a> constant for <a href="#">I2C</a> spike suppression. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1 = 20ns spike suppression<br>0 = 50ns spike suppression                                                 |
| 29:28 | <b>mode_1p8_1p1_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Voltage Select pin. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1 = 1.8V operation<br>0 = 1.1V operation                                                                                                                         |
| 27:26 | <b>mode_i3c_i2c_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> <a href="#">I3C</a> or <a href="#">I2C</a> mode enable. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= I3C mode<br>0 = I2C mode                                                                                                  |
| 25:24 | <b>slewp_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Active Rise Slew Compensation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Faster rise slew<br>0= Disable                                                                                                                            |
| 23:22 | <b>resbiasen_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> When the bias circuit is enabled (BiasCrtEn=1), ResBiasEn determines the bias current type. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Constant-gm resistive current<br>0= Temperature compensated resistive current            |
| 21:20 | <b>compsel_1_0.</b> Read-write. Reset: 0h. Unused.                                                                                                                                                                                                                                                |
| 19:18 | <b>spare_1_0.</b> Read-write. Reset: 0h. Spare pins. Bit 1 for PAD1 and Bit 0 for PAD0.                                                                                                                                                                                                           |
| 17:16 | <b>biascrten_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Bias circuit should be ON only in <a href="#">I2C</a> mode. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Enable for I2C Fast/Fast plus mode slew spec<br>0= Disable for I2C Standard mode                                         |
| 15    | <b>rsel1p1.</b> Read-write. Reset: 0. When asserted decreases or increases resistance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                       |
| 14    | <b>rsel0p9.</b> Read-write. Reset: 0. When asserted decreases or increases resistance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                       |
| 13    | <b>csel1p1.</b> Read-write. Reset: 0. When asserted decreases or increases capacitance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                      |
| 12    | <b>csel0p9.</b> Read-write. Reset: 0. When asserted decreases or increases capacitance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                      |
| 11:10 | <b>spikercen_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Enables spike suppression in <a href="#">I2C</a> mode only. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Filter enable (must be used in conjunction with pin Mode_I3c_I2c = 0 only for 1.8V and 1.1V system)<br>0= Disable        |
| 9:8   | <b>fallslewsel_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> <a href="#">I2C</a> Mode select bits. Bit 1 for PAD1 and Bit 0 for PAD0. Must be used when Mode_I3c_I2c=0 pin.<br>1= I2C Fast Mode (Tx Freq=400kHz), Fast Plus Mode (Tx Freq=1MHz)<br>0= I2C Standard Mode, Tx Freq=100kHz. |
| 7:6   | <b>slewn_1_0.</b> Read-write. Reset: 0h.                                                                                                                                                                                                                                                          |

|     |                                                                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> Active Fall Slew Compensation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Faster fall slew<br>0= Disable                                                                                                                       |
| 5:4 | <b>rxsel_1_0.</b> Read-write. Reset: 3h.<br><b>Description:</b> Select RX of operation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= 1.8V or 1.1V RX (must be used in conjunction with pin Mode_1p8_1p1 = 1/0 for 1.8V and 1.1V system)<br>0= RX OFF |
| 3:0 | <b>od_rp_sw_3_0.</b> Read-write. Reset: 0h. Open-drain pull-up switch. Bit 3 and 2 for PAD1, and Bit 1 and 0 for PAD0.                                                                                                                           |

**MISCx000000DC (FCH::MISC::I2C1\_PADCTRL)**

Read-write. Reset: 0000\_0030h.

This 32-bit register is used to control [I3C](#) and [I2C](#) pad.

\_aliasHOSTLEGACY; MISCx000000DC; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>spikercsel_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Select <a href="#">RC</a> constant for <a href="#">I2C</a> spike suppression. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1 = 20ns spike suppression<br>0 = 50ns spike suppression                                                 |
| 29:28 | <b>mode_1p8_1p1_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Voltage Select pin. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1 = 1.8V operation<br>0 = 1.1V operation                                                                                                                         |
| 27:26 | <b>mode_i3c_i2c_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> <a href="#">I3C</a> or <a href="#">I2C</a> mode enable. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= I3C mode<br>0 = I2C mode                                                                                                  |
| 25:24 | <b>slewp_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Active Rise Slew Compensation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Faster rise slew<br>0= Disable                                                                                                                            |
| 23:22 | <b>resbiasen_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> When the bias circuit is enabled (BiasCrtEn=1), ResBiasEn determines the bias current type. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Constant-gm resistive current<br>0= Temperature compensated resistive current            |
| 21:20 | <b>compsel_1_0.</b> Read-write. Reset: 0h. Unused.                                                                                                                                                                                                                                                |
| 19:18 | <b>spare_1_0.</b> Read-write. Reset: 0h. Spare pins. Bit 1 for PAD1 and Bit 0 for PAD0.                                                                                                                                                                                                           |
| 17:16 | <b>biascrten_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Bias circuit should be ON only in <a href="#">I2C</a> mode. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Enable for I2C Fast/Fast plus mode slew spec<br>0= Disable for I2C Standard mode                                         |
| 15    | <b>rsel1p1.</b> Read-write. Reset: 0. When asserted decreases or increases resistance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                       |
| 14    | <b>rsel0p9.</b> Read-write. Reset: 0. When asserted decreases or increases resistance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                       |
| 13    | <b>csel1p1.</b> Read-write. Reset: 0. When asserted decreases or increases capacitance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                      |
| 12    | <b>csel0p9.</b> Read-write. Reset: 0. When asserted decreases or increases capacitance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                      |
| 11:10 | <b>spikercen_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Enables spike suppression in <a href="#">I2C</a> mode only. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Filter enable (must be used in conjunction with pin Mode_I3c_I2c = 0 only for 1.8V and 1.1V system)<br>0= Disable        |
| 9:8   | <b>fallslewsel_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> <a href="#">I2C</a> Mode select bits. Bit 1 for PAD1 and Bit 0 for PAD0. Must be used when Mode_I3c_I2c=0 pin.<br>1= I2C Fast Mode (Tx Freq=400kHz), Fast Plus Mode (Tx Freq=1MHz)<br>0= I2C Standard Mode, Tx Freq=100kHz. |
| 7:6   | <b>slewn_1_0.</b> Read-write. Reset: 0h.                                                                                                                                                                                                                                                          |

|     |                                                                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> Active Fall Slew Compensation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Faster fall slew<br>0= Disable                                                                                                                       |
| 5:4 | <b>rxsel_1_0.</b> Read-write. Reset: 3h.<br><b>Description:</b> Select RX of operation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= 1.8V or 1.1V RX (must be used in conjunction with pin Mode_1p8_1p1 = 1/0 for 1.8V and 1.1V system)<br>0= RX OFF |
| 3:0 | <b>od_rp_sw_3_0.</b> Read-write. Reset: 0h. Open-drain pull-up switch. Bit 3 and 2 for PAD1, and Bit 1 and 0 for PAD0.                                                                                                                           |

**MISCx000000E0 (FCH::MISC::I2C2\_PADCTRL)**

Read-write. Reset: 0000\_0030h.

This 32-bit register is used to control [I3C](#) and [I2C](#) pad.

\_aliasHOSTLEGACY; MISCx000000E0; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>spikercsel_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Select <a href="#">RC</a> constant for <a href="#">I2C</a> spike suppression. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1 = 20ns spike suppression<br>0 = 50ns spike suppression                                                 |
| 29:28 | <b>mode_1p8_1p1_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Voltage Select pin. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1 = 1.8V operation<br>0 = 1.1V operation                                                                                                                         |
| 27:26 | <b>mode_i3c_i2c_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> <a href="#">I3C</a> or <a href="#">I2C</a> mode enable. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= I3C mode<br>0 = I2C mode                                                                                                  |
| 25:24 | <b>slewp_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Active Rise Slew Compensation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Faster rise slew<br>0= Disable                                                                                                                            |
| 23:22 | <b>resbiasen_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> When the bias circuit is enabled (BiasCrtEn=1), ResBiasEn determines the bias current type. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Constant-gm resistive current<br>0= Temperature compensated resistive current            |
| 21:20 | <b>compsel_1_0.</b> Read-write. Reset: 0h. Unused.                                                                                                                                                                                                                                                |
| 19:18 | <b>spare_1_0.</b> Read-write. Reset: 0h. Spare pins. Bit 1 for PAD1 and Bit 0 for PAD0.                                                                                                                                                                                                           |
| 17:16 | <b>biascrten_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Bias circuit should be ON only in <a href="#">I2C</a> mode. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Enable for I2C Fast/Fast plus mode slew spec<br>0= Disable for I2C Standard mode                                         |
| 15    | <b>rsel1p1.</b> Read-write. Reset: 0. When asserted decreases or increases resistance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                       |
| 14    | <b>rsel0p9.</b> Read-write. Reset: 0. When asserted decreases or increases resistance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                       |
| 13    | <b>csel1p1.</b> Read-write. Reset: 0. When asserted decreases or increases capacitance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                      |
| 12    | <b>csel0p9.</b> Read-write. Reset: 0. When asserted decreases or increases capacitance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                      |
| 11:10 | <b>spikercen_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Enables spike suppression in <a href="#">I2C</a> mode only. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Filter enable (must be used in conjunction with pin Mode_I3c_I2c = 0 only for 1.8V and 1.1V system)<br>0= Disable        |
| 9:8   | <b>fallslewsel_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> <a href="#">I2C</a> Mode select bits. Bit 1 for PAD1 and Bit 0 for PAD0. Must be used when Mode_I3c_I2c=0 pin.<br>1= I2C Fast Mode (Tx Freq=400kHz), Fast Plus Mode (Tx Freq=1MHz)<br>0= I2C Standard Mode, Tx Freq=100kHz. |
| 7:6   | <b>slewn_1_0.</b> Read-write. Reset: 0h.                                                                                                                                                                                                                                                          |

|     |                                                                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> Active Fall Slew Compensation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Faster fall slew<br>0= Disable                                                                                                                       |
| 5:4 | <b>rxsel_1_0.</b> Read-write. Reset: 3h.<br><b>Description:</b> Select RX of operation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= 1.8V or 1.1V RX (must be used in conjunction with pin Mode_1p8_1p1 = 1/0 for 1.8V and 1.1V system)<br>0= RX OFF |
| 3:0 | <b>od_rp_sw_3_0.</b> Read-write. Reset: 0h. Open-drain pull-up switch. Bit 3 and 2 for PAD1, and Bit 1 and 0 for PAD0.                                                                                                                           |



**MISCx000000E4 (FCH::MISC::I2C3\_PADCTRL)**

Read-write. Reset: 0000\_0030h.

This 32-bit register is used to control [I3C](#) and [I2C](#) pad.\_aliasHOSTLEGACY; MISCx000000E4; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>spikercsel_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Select <a href="#">RC</a> constant for <a href="#">I2C</a> spike suppression. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1 = 20ns spike suppression<br>0 = 50ns spike suppression                                                 |
| 29:28 | <b>mode_1p8_1p1_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Voltage Select pin. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1 = 1.8V operation<br>0 = 1.1V operation                                                                                                                         |
| 27:26 | <b>mode_i3c_i2c_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> <a href="#">I3C</a> or <a href="#">I2C</a> mode enable. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= I3C mode<br>0 = I2C mode                                                                                                  |
| 25:24 | <b>slewp_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Active Rise Slew Compensation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Faster rise slew<br>0= Disable                                                                                                                            |
| 23:22 | <b>resbiasen_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> When the bias circuit is enabled (BiasCrtEn=1), ResBiasEn determines the bias current type. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Constant-gm resistive current<br>0= Temperature compensated resistive current            |
| 21:20 | <b>compsel_1_0.</b> Read-write. Reset: 0h. Unused.                                                                                                                                                                                                                                                |
| 19:18 | <b>spare_1_0.</b> Read-write. Reset: 0h. Spare pins. Bit 1 for PAD1 and Bit 0 for PAD0.                                                                                                                                                                                                           |
| 17:16 | <b>biascrten_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Bias circuit should be ON only in <a href="#">I2C</a> mode. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Enable for I2C Fast/Fast plus mode slew spec<br>0= Disable for I2C Standard mode                                         |
| 15    | <b>rsel1p1.</b> Read-write. Reset: 0. When asserted decreases or increases resistance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                       |
| 14    | <b>rsel0p9.</b> Read-write. Reset: 0. When asserted decreases or increases resistance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                       |
| 13    | <b>csel1p1.</b> Read-write. Reset: 0. When asserted decreases or increases capacitance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                      |
| 12    | <b>csel0p9.</b> Read-write. Reset: 0. When asserted decreases or increases capacitance by 10% for all <a href="#">RC</a> timers. Must be used in conjunction with pin Mode_I3c_I2c = 0 only.                                                                                                      |
| 11:10 | <b>spikercen_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> Enables spike suppression in <a href="#">I2C</a> mode only. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Filter enable (must be used in conjunction with pin Mode_I3c_I2c = 0 only for 1.8V and 1.1V system)<br>0= Disable        |
| 9:8   | <b>fallslewsel_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> <a href="#">I2C</a> Mode select bits. Bit 1 for PAD1 and Bit 0 for PAD0. Must be used when Mode_I3c_I2c=0 pin.<br>1= I2C Fast Mode (Tx Freq=400kHz), Fast Plus Mode (Tx Freq=1MHz)<br>0= I2C Standard Mode, Tx Freq=100kHz. |
| 7:6   | <b>slewn_1_0.</b> Read-write. Reset: 0h.                                                                                                                                                                                                                                                          |

|     |                                                                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> Active Fall Slew Compensation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= Faster fall slew<br>0= Disable                                                                                                                       |
| 5:4 | <b>rxsel_1_0.</b> Read-write. Reset: 3h.<br><b>Description:</b> Select RX of operation. Bit 1 for PAD1 and Bit 0 for PAD0.<br>1= 1.8V or 1.1V RX (must be used in conjunction with pin Mode_1p8_1p1 = 1/0 for 1.8V and 1.1V system)<br>0= RX OFF |
| 3:0 | <b>od_rp_sw_3_0.</b> Read-write. Reset: 0h. Open-drain pull-up switch. Bit 3 and 2 for PAD1, and Bit 1 and 0 for PAD0.                                                                                                                           |

**MISCx000000E8 (FCH::MISC::I2C4\_PADCTRL)**

Read-write. Reset: 0000\_003Ch.

\_aliasHOSTLEGACY; MISCx000000E8; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:19 | Reserved.                                                                                                                                                                                                                                                           |
| 18    | <b>spare1.</b> Read-write. Reset: 0. Spare Pin                                                                                                                                                                                                                      |
| 17    | <b>spare0.</b> Read-write. Reset: 0. Spare Pin                                                                                                                                                                                                                      |
| 16    | <b>biascrten.</b> Read-write. Reset: 0. Pbias should be on in Fast/Fast+ Mode internally overridden to turn on by FallSlew setting, to save power can be turned off in <a href="#">I2C</a> Standard Mode FallSlew=00                                                |
| 15    | <b>rsl1p1.</b> Read-write. Reset: 0. When asserted increases resistance by 10% for all <a href="#">RC</a> timers                                                                                                                                                    |
| 14    | <b>rsl0p9.</b> Read-write. Reset: 0. When asserted decreases resistance by 10% for all <a href="#">RC</a> timers                                                                                                                                                    |
| 13    | <b>csel1p1.</b> Read-write. Reset: 0. When asserted increases capacitance by 10% for all <a href="#">RC</a> timers                                                                                                                                                  |
| 12    | <b>csel0p9.</b> Read-write. Reset: 0. When asserted decreases capacitance by 10% for all <a href="#">RC</a> timers                                                                                                                                                  |
| 11    | <b>spikercsel.</b> Read-write. Reset: 0. Select <a href="#">RC</a> constant for <a href="#">I2C</a> spike suppression, 0=50ns, 1=20ns                                                                                                                               |
| 10    | <b>spikercen.</b> Read-write. Reset: 0. Enable Rx spike suppression, default=0                                                                                                                                                                                      |
| 9     | <b>slewn.</b> Read-write. Reset: 0. Enable Changing Strength of pre-drive for fall time compensation by 25%. Slewn=1 Enabled, 0=disabled.                                                                                                                           |
| 8:7   | <b>fallslewsel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00= Standard Mode Tx Freq=100kHz<br>01= low speed 12ns..120ns Frequency TX<1MHz (FM/FM+) 10= not used<br>11= not used                                                                             |
| 6     | <b>pden.</b> Read-write. Reset: 0. Pull-down enable for 1=Enable 0=Disable                                                                                                                                                                                          |
| 5:4   | <b>i2crxsel.</b> Read-write. Reset: 3h.<br><b>Description:</b> i2cRxSel<1:0>=0b01 Schmitt trigger for 3.3V input<br>i2cRxSel<1:0>=0b10 Schmitt trigger for 3.3V input<br>i2cRxSel<1:0>=0b11 Schmitt trigger for 1.8V input<br>i2cRxSel<1:0>=0b00 All receivers off. |
| 3:0   | <b>ng.</b> Read-write. Reset: Ch. N Strength Control                                                                                                                                                                                                                |

**MISCx000000EC (FCH::MISC::I2C5\_PADCTRL)**

Read-write. Reset: 0000\_003Ch.

\_aliasHOSTLEGACY; MISCx000000EC; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:19 | Reserved.                                                                                                                                                                                                                                                            |
| 18    | <b>spare1</b> . Read-write. Reset: 0. Spare Pin                                                                                                                                                                                                                      |
| 17    | <b>spare0</b> . Read-write. Reset: 0. Spare Pin                                                                                                                                                                                                                      |
| 16    | <b>biascrten</b> . Read-write. Reset: 0. Pbias should be on in Fast/Fast+ Mode internally overridden to turn on by FallSlew setting, to save power can be turned off in <a href="#">I2C</a> Standard Mode FallSlew=00                                                |
| 15    | <b>rsel1p1</b> . Read-write. Reset: 0. When asserted increases resistance by 10% for all <a href="#">RC</a> timers                                                                                                                                                   |
| 14    | <b>rsel0p9</b> . Read-write. Reset: 0. When asserted decreases resistance by 10% for all <a href="#">RC</a> timers                                                                                                                                                   |
| 13    | <b>csel1p1</b> . Read-write. Reset: 0. When asserted increases capacitance by 10% for all <a href="#">RC</a> timers                                                                                                                                                  |
| 12    | <b>csel0p9</b> . Read-write. Reset: 0. When asserted decreases capacitance by 10% for all <a href="#">RC</a> timers                                                                                                                                                  |
| 11    | <b>spikercsel</b> . Read-write. Reset: 0. Select <a href="#">RC</a> constant for <a href="#">I2C</a> spike suppression, 0=50ns, 1=20ns                                                                                                                               |
| 10    | <b>spikercen</b> . Read-write. Reset: 0. Enable Rx spike suppression, default=0                                                                                                                                                                                      |
| 9     | <b>slewn</b> . Read-write. Reset: 0. Enable Changing Strength of pre-drive for fall time compensation by 25%. Slewn=1 Enabled, 0=disabled.                                                                                                                           |
| 8:7   | <b>fallslewsel</b> . Read-write. Reset: 0h.<br><b>Description:</b> 00= Standard Mode Tx Freq=100kHz)<br>01= low speed 12ns..120ns Frequency TX<1MHz (FM/FM+) 10= not used<br>11= not used                                                                            |
| 6     | <b>pden</b> . Read-write. Reset: 0. Pull-down enable for 1=Enable 0=Disable                                                                                                                                                                                          |
| 5:4   | <b>i2crxsel</b> . Read-write. Reset: 3h.<br><b>Description:</b> i2cRxSel<1:0>=0b01 Schmitt trigger for 3.3V input<br>i2cRxSel<1:0>=0b10 Schmitt trigger for 3.3V input<br>i2cRxSel<1:0>=0b11 Schmitt trigger for 1.8V input<br>i2cRxSel<1:0>=0b00 All receivers off. |
| 3:0   | <b>ng</b> . Read-write. Reset: Ch. N Strength Control                                                                                                                                                                                                                |

**MISCx000000F0 (FCH::MISC::CLKCNTRLF0)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000F0; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISCx000000F4 (FCH::MISC::CLKCNTRLF4)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISCx000000F4; MISC=FED8\_0E00h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISCx000000F8 (FCH::MISC::TDR\_SECURITY\_STATUS)**

Read-only.

\_aliasHOSTLEGACY; MISCx000000F8; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:22 | Reserved.                                                                                                                                                                                                                                                                      |
| 21    | <b>tdr_i3c_control_updated.</b> Read-only. Reset: X. Tdr_I3C_CONTROL_accessed                                                                                                                                                                                                  |
| 20    | <b>tdr_ioconfig_updated.</b> Read-only. Reset: X. Tdr_S5IODIRCONFIG_accessed Tdr_S0IODIRCONFIG_accessed Tdr_S5IODIRAPPEND_accessed Tdr_S0IODIRAPPEND_accessed                                                                                                                  |
| 19    | <b>tdr_s5_clkconfig_updated.</b> Read-only. Reset: X. Tdr_S5_PLLDFTCONFIG_accessed Tdr_S5_PLLDFTSTS_accessed Tdr_S5_CLKDFTCONFIG_accessed                                                                                                                                      |
| 18    | <b>tdr_sdio_control_updated.</b> Read-only. Reset: X. Tdr_SDIO_CONTROL_accessed                                                                                                                                                                                                |
| 17    | <b>tdr_i2c_control_updated.</b> Read-only. Reset: X. Tdr_I2C_CONTROL_accessed                                                                                                                                                                                                  |
| 16    | <b>tdr_gpio_control_updated.</b> Read-only. Reset: X. Tdr_GPIO18_CONTROL_accessed Tdr_GPIO33_CONTROL_accessed                                                                                                                                                                  |
| 15    | <b>tdr_iodftcfg_updated.</b> Read-only. Reset: X. Tdr_IODFTCFG_accessed                                                                                                                                                                                                        |
| 14    | <b>tdr_spare_updated.</b> Read-only. Reset: X. Tdr_SPARE_accessed                                                                                                                                                                                                              |
| 13    | <b>classb_tdr_updated.</b> Read-only. Reset: X. A TDR that is only enabled after Cpl_aeb_valid has been updated.                                                                                                                                                               |
| 12    | <b>s0stateobserve_tdr_updated.</b> Read-only. Reset: X. Tdr_S0STATEOBSERVE_accessed                                                                                                                                                                                            |
| 11    | <b>ila_tdr_updated.</b> Read-only. Reset: X. Tdr_ILA_accessed                                                                                                                                                                                                                  |
| 10    | <b>idcode_tdr_updated.</b> Read-only. Reset: X. Tdr_IDCODE_accessed   Tdr_PKGID_accessed                                                                                                                                                                                       |
| 9     | <b>scan_tdr_updated.</b> Read-only. Reset: X. Tdr_SYSSCAN_accessed   Tdr_SCANCONFIG_accessed                                                                                                                                                                                   |
| 8     | <b>sysdebug_tdr_updated.</b> Read-only. Reset: X. Tdr_SYSDEBUG_accessed                                                                                                                                                                                                        |
| 7     | <b>sdb_tdr_updated.</b> Read-only. Reset: X. Tdr_SDB_CONFIG_accessed   Tdr_SDB_STATUS_accessed                                                                                                                                                                                 |
| 6     | <b>mbist_tdr_updated.</b> Read-only. Reset: X. Tdr_MBISTCONFIG_accessed   Tdr_MEMBISTSTS_accessed                                                                                                                                                                              |
| 5     | <b>iotstcntrl_tdr_updated.</b> Read-only. Reset: X. Tdr_IOTSTCNTRL_accessed                                                                                                                                                                                                    |
| 4     | <b>testclkdis_tdr_updated.</b> Read-only. Reset: X. Tdr_TESTCLKDIS_accessed                                                                                                                                                                                                    |
| 3     | <b>ateconfig_tdr_updated.</b> Read-only. Reset: X. Tdr_ATECONFIG_accessed                                                                                                                                                                                                      |
| 2     | <b>pll_clk_dft_tdr_updated.</b> Read-only. Reset: X. Tdr_PLLDFT_accessed   Tdr_CLKDFT_accessed                                                                                                                                                                                 |
| 1     | <b>acpi_tdr_updated.</b> Read-only. Reset: X. Tdr_ACPIJTAGCMD_accessed   Tdr_ACPIJTAGSTATUS_accessed                                                                                                                                                                           |
| 0     | <b>any_tdr_updated.</b> Read-only. Reset: X.<br><b>Description:</b> Read-only. Set if any of the <a href="#">FCH</a> TDR are Updated (Dfx Security Feature)<br>0: No FCH TDR has been updated since RsmrstB was asserted<br>1: One or More FCH TDR registers have been updated |

**MISCx000000FC (FCH::MISC::CLKGATING\_CNTRL)**

Read-write. Reset: FFFF\_FC6Fh.

\_aliasHOSTLEGACY; MISCx000000FC; MISC=FED8\_0E00h

| Bits  | Description                                                                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:27 | Reserved.                                                                                                                                                                                                                       |
| 26    | <b>hpet64_tmr_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of HPET64 Timer block can be stopped if not being Accessed<br>1: the clock of HPET64 Timer block can't be stopped.                    |
| 25    | <b>hpet32_tmr_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of HPET32 Timer block can be stopped if not being Accessed<br>1: the clock of HPET32 Timer block can't be stopped.                    |
| 24    | <b>boot_tmr_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Boot Timer block can be stopped if not being Accessed<br>1: the clock of Boot Timer block can't be stopped.                          |
| 23    | <b>long_tmr_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Long Timer block can be stopped if not being Accessed<br>1: the clock of Long Timer block can't be stopped.                          |
| 22    | <b>shrt_tmr_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Short Timer block can be stopped if not being Accessed<br>1: the clock of Short Timer block can't be stopped.                        |
| 21    | <b>asf_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of ASF block can be stopped if not being Accessed<br>1: the clock of ASF block can't be stopped.                                             |
| 20    | <b>asf_clkgen_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of ASF Clock Generator block can be stopped if not being Accessed<br>1: the clock of ASF Clock Generator block can't be stopped.      |
| 19    | <b>shdw_syscnt_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Shadow System Counter block can be stopped if not being Accessed<br>1: the clock of Shadow System Counter block can't be stopped. |
| 18    | <b>acpi_msi_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of ACPI MSI block can be stopped if not being Accessed.<br>1: the clock of ACPI MSI block can't be stopped.                             |
| 17    | <b>usb_legacy_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of USB Legacy block can be stopped if not being Accessed.<br>1: the clock of USB Legacy block can't be stopped.                       |
| 16    | <b>isa_bridge_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of ISA Bridge block can be stopped if not being Accessed.<br>1: the clock of ISA Bridge block can't be stopped.                       |
| 15    | <b>acpi_debug_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of ACPI debug block can be stopped if not being Accessed.<br>1: the clock of ACPI debug block can't be stopped.                       |
| 14    | <b>tmr_8254_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of 8254 timer block can be stopped if not being Accessed.<br>1: the clock of 8254 timer block can't be stopped.                         |
| 13    | <b>shdw_pcislave_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Shadow pcislave block can be stopped if not being Accessed.<br>1: the clock of Shadow pcislave block can't be stopped.          |
| 12    | <b>watchdogtimerblk_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of WatchDogTimer block can be stopped if not being Accessed.<br>1: the clock of WatchDogTimer block can't be stopped.           |

|     |                                                                                                                                                                                                                    |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11  | <b>acdctmrblk_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of AcDcTmr block can be stopped if not being Accessed.<br>1: the clock of AcDcTmr block can't be stopped.                |
| 10  | <b>biosramblk_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Bios Ram block can be stopped if not being Accessed.<br>1: the clock of Bios Ram block can't be stopped.              |
| 9:7 | Reserved.                                                                                                                                                                                                          |
| 6   | <b>obffblk_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of OBFF block can be stopped if not being Accessed.<br>1: the clock of OBFF block can't be stopped.                         |
| 5   | <b>smbus0_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Smbus0 block can be stopped if not being Accessed.<br>1: the clock of Smbus0 block can't be stopped.                      |
| 4   | Reserved.                                                                                                                                                                                                          |
| 3   | <b>aoacregblk_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Aoac register block can be stopped if not being Accessed.<br>1: the clock of Aoac register block can't be stopped.    |
| 2   | <b>pmio2regblk_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Pmio2 register block can be stopped if not being Accessed.<br>1: the clock of Pmio2 register block can't be stopped. |
| 1   | <b>pmioregblk_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of Pmio register block can be stopped if not being Accessed.<br>1: the clock of Pmio register block can't be stopped.    |
| 0   | <b>miscregblk_clkgatingdis.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: the clock of MISC register block can be stopped if not being Accessed.<br>1: the clock of MISC register block can't be stopped.    |

**14.3.14.2    Miscellaneous (MISC2) Registers**

**MISC2x00000000 (FCH::MISC2::RMT\_CLKCNTL\_0\_REG)**

Read-write. Reset: 1000\_CFFCh.

\_aliasHOSTLEGACY; MISC2x00000000; MISC2=FED8\_1300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:12 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 11:10 | <b>gpp_clk05_clock_request_mapping.</b> Read-write. Reset: 3h.<br><b>Description:</b> GPP05 PCIE clock pins (GPP_CLK05P/GPP_CLK05N) output control by CLKREQ05# (or CLKREQG#) pin<br>GPP_CLK05P/GPP_CLK05N pins can be controlled (turn OFF, ON, CLKREQ05# controlled) by the following options.<br>GPP05_CLKREQ_Mapping:<br>00 Off<br>01 CLKREQ05#<br>10 Off, reserved<br>11 On (default)                                                                                                                                                                                                                            |
| 9:8   | <b>gpp_clk03_clock_request_mapping.</b> Read-write. Reset: 3h.<br><b>Description:</b> GPP03 PCIE clock pins (GPP_CLK03P/GPP_CLK03N) output control by CLKREQ03# pin<br>GPP_CLK03P/GPP_CLK03N pins can be controlled (turn OFF, ON, CLKREQ03# controlled) by the following options.<br>GPP03_CLKREQ_Mapping:<br>00 Off<br>01 CLKREQ03#<br>10 Off, reserved<br>11 On (default)<br>Note: MISC2x00000004[24] is an override bit which is XOR with GPP03_CLKREQ_Mapping.                                                                                                                                                   |
| 7:6   | <b>gpp_clk02_clock_request_mapping.</b> Read-write. Reset: 3h.<br><b>Description:</b> GPP02 PCIE clock pins (GPP_CLK02P/GPP_CLK02N) output control by CLKREQ02# pin<br>eCLK Mode:<br>Socket0/1 GPP02 is default with input mode.<br>Cannot be controlled by these bits. But can be override by MISC2x00000004[12].<br>iCLK Mode:<br>Socket0/1 GPP02 can be controlled (turn OFF, ON, CLKREQ02# controlled) by the following options.<br>GPP02_CLKREQ_Mapping:<br>00 Off<br>01 CLKREQ02#<br>10 Off, reserved<br>11 On (default)<br>Note: MISC2x00000004[12] is an override bit which is XOR with GPP02_CLKREQ_Mapping. |
| 5:4   | <b>gpp_clk04_clock_request_mapping.</b> Read-write. Reset: 3h.<br><b>Description:</b> GPP04 PCIE clock pins (GPP_CLK04P/GPP_CLK04N) output control by CLKREQ04# pin<br>GPP_CLK04P/GPP_CLK04N pins can be controlled (turn OFF, ON, CLKREQ04# controlled) by the following options.<br>GPP04_CLKREQ_Mapping:<br>00 Off<br>01 CLKREQ04#<br>10 Off, reserved<br>11 On (default)<br>Note: MISC2x00000004[29] is an override bit which is XOR with GPP04_CLKREQ_Mapping.                                                                                                                                                   |
| 3:2   | <b>gpp_clk01_clock_request_mapping.</b> Read-write. Reset: 3h.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



|     |                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> GPP01 PCIE clock pins (GPP_CLK01P/GPP_CLK01N) output control by CLKREQ01# pin<br>GPP_CLK01P/GPP_CLK01N pins can be controlled (turn OFF, ON, CLKREQ01# controlled) by the following options.<br>GPP01_CLKREQ_Mapping:<br>00 Off<br>01 CLKREQ01#<br>10 Off, reserved<br>11 On (default)<br>Note: MISC2x00000004[9] is an override bit which is XOR with GPP01_CLKREQ_Mapping. |
| 1:0 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                        |

#### MISC2x00000004 (FCH::MISC2::RMT\_CLKCNTL\_1\_REG)

Read-write. Reset: 0000\_00F0h.

\_aliasHOSTLEGACY; MISC2x00000004; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

#### MISC2x00000008 (FCH::MISC2::RMT\_CLKCNTL\_2\_REG)

Read-write. Reset: 1800\_1E00h.

\_aliasHOSTLEGACY; MISC2x00000008; MISC2=FED8\_1300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:9 | Reserved.                                                                                                                                                                                                                                                                                                                                                                  |
| 8    | <b>cg2_refdivsrc_override.</b> Read-write. Reset: 0.<br><b>Description:</b> CG2PLL Refclk Source Select Override<br>This bit is used to override CG2PLL refclk source select.<br>By default, Master_die and Slave_die CG2PLL refclk source is from CG1PLL generated 100MHz.<br>This bit shall be kept to default value.<br>Note: Need to apply a reset after set this bit  |
| 7:5  | Reserved.                                                                                                                                                                                                                                                                                                                                                                  |
| 4    | <b>refclk_source_switch_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> Refclk Source Switch Mode<br>For CGPLL refclk source, it can be from either CG_XTAL or external reference source.<br>Note: for CG2PLL, refclk source has no CG_XTAL connected.<br>0: Switch refclk source on-the-fly<br>1: Need to apply a Cold Reset for switching different refclk source |
| 3:1  | Reserved.                                                                                                                                                                                                                                                                                                                                                                  |
| 0    | <b>cg2_spread_spectrum_enable.</b> Read-write. Reset: 0.<br><b>Description:</b> CG2_PLL Spread Spectrum Enable<br>0: Disable Spread Spectrum (default)<br>1: Enable Spread Spectrum                                                                                                                                                                                        |

#### MISC2x0000000C (FCH::MISC2::RMT\_CLKCNTL\_3\_REG)

Read-write. Reset: 0000\_5555h.

\_aliasHOSTLEGACY; MISC2x0000000C; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x00000010 (FCH::MISC2::RMT\_CLKCNTL\_4\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000010; MISC2=FED8\_1300h

| Bits  | Description                                                                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                 |
| 29    | <b>cg2pll_fracn_en_override</b> . Read-write. Reset: 0. Used with CG2PLL frac-N and <a href="#">SSC</a> clocking only.                                                    |
| 28:13 | Reserved.                                                                                                                                                                 |
| 12:4  | <b>cg2pll_fcw0_int_override</b> . Read-write. Reset: 000h. CG2PLL Override: Integer portion of Frequency Control Word0 (a.k.a. feedback divisor0).                        |
| 3:2   | Reserved.                                                                                                                                                                 |
| 1:0   | <b>cg2pll_refclk_div_override</b> . Read-write. Reset: 0h.<br><b>Description:</b> CG2PLL Override: Reference clock divisor.<br>Settings:<br>2'b00=1<br>2'b01=2<br>2'b1x=4 |

**MISC2x00000014 (FCH::MISC2::RMT\_CLKCNTL\_5\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000014; MISC2=FED8\_1300h

| Bits  | Description                                                                                                                                                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                                                                                              |
| 23:8  | <b>cg2pll_fcw1_frac_override</b> . Read-write. Reset: 0000h. CG2PLL Override: Fractional portion of Frequency Control Word1 (a.k.a. feedback divisor1). Intended to be used with frequency ramping. Also used to step PLL frequency and phase for DFT. |
| 7:0   | Reserved.                                                                                                                                                                                                                                              |

**MISC2x00000018 (FCH::MISC2::RMT\_CLKCNTL\_6\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000018; MISC2=FED8\_1300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>cg2pll_fcw_slew_frac_override</b> . Read-write. Reset: 0000h.<br><b>Description:</b> CG2PLL Override: Sets <a href="#">SSC</a> freq ramp rate. Set fractional change in programmed frequency per refclk cycle.<br>e.g. 0.5% downspread SSC at 33.3kHz and fbdiv=80.<br>$FCW\_slewr\_rate\_frac = 216 * 0.00485 * 80 / (15us/10ns) = 17$<br>Need 31.5kHz and -0.375% |
| 15:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                              |

**MISC2x0000001C (FCH::MISC2::RMT\_CLKCNTL\_7\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x0000001C; MISC2=FED8\_1300h

| Bits  | Description                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------|
| 31:28 | Reserved.                                                                                                                 |
| 27:26 | <b>cgpll_gi_coarse_mant_override</b> . Read-write. Reset: 0h. CG2PLL Coarse integral path fp mult mantissa                |
| 25:21 | Reserved.                                                                                                                 |
| 20:17 | <b>cg2pll_gp_coarse_exp_override</b> . Read-write. Reset: 0h. CG2PLL Override: Coarse proportional path fp mult exponent  |
| 16:13 | <b>cg2pll_gp_coarse_mant_override</b> . Read-write. Reset: 0h. CG2PLL Override: Coarse proportional path fp mult mantissa |
| 12:9  | <b>cg2pll_gi_coarse_exp_override</b> . Read-write. Reset: 0h. CG2PLL Override: Coarse integral path fp mult exponent      |
| 8:0   | Reserved.                                                                                                                 |

**MISC2x00000020 (FCH::MISC2::RMT\_CLKDRVSTH\_0\_REG)**

Read-write.

\_aliasHOSTLEGACY; MISC2x00000020; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x00000024 (FCH::MISC2::RMT\_CLKDRVSTH\_1\_REG)**

Read-write. Reset: 0024\_0249h.

\_aliasHOSTLEGACY; MISC2x00000024; MISC2=FED8\_1300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                                                                                                                                                                                                |
| 23:21 | <b>gpp05_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP05 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001         |
| 20:18 | <b>gpp04_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP04 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001         |
| 17:12 | Reserved.                                                                                                                                                                                                                                                                                                                                                |
| 11:9  | <b>gpp_clk03_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP_CLK03 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001 |
| 8:6   | <b>gpp_clk02_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP_CLK02 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001 |
| 5:3   | <b>gpp_clk01_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.<br><b>Description:</b> Drive Strength control for GPP_CLK01 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001 |
| 2:0   | <b>gpp_clk00_clock_buffer_driving_strength_control.</b> Read-write. Reset: 1h.                                                                                                                                                                                                                                                                           |

|  |                                                                                                                                                                                                                                                                        |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Description:</b> Drive Strength control for GPP_CLK00 differential Clock Buffers<br>Drive strength addition/subtraction relative to IMP_CTRL[4:0].<br>3'bx00 : ~ -10%<br>3'bx01 : no change from IMP_CTRL<br>3'bx10 : ~ +10%<br>3'bx11 : ~ + 20%<br>Default: 3'b001 |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**MISC2x00000028 (FCH::MISC2::RMT\_CLKDRVSTH\_2\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000028; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x0000002C (FCH::MISC2::RMT\_CLKDRVSTH\_3\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x0000002C; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x00000030 (FCH::MISC2::RMT\_PLLCNTL\_0\_REG)**

Read-write.

\_aliasHOSTLEGACY; MISC2x00000030; MISC2=FED8\_1300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:4 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3    | <b>usb0_rtcclk_enb.</b> Read-write. Reset: 0.<br><b>Description:</b> USB0 RTCCLK (root_RTC_CLK_0) Enable/Disable<br>0: Turn on<br>1: Turn off                                                                                                                                                                                                                                                                                  |
| 2    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 1:0  | <b>pcie_phy_refclk_selection_override.</b> Read-write. Reset: XXb.<br><b>Description:</b> PCIE PHY Refclk Selection Override ( G-PHY)<br>CLKB_PCIE_PHY_P/N<br>00 = 100MHz CG2_PLL generated (default)<br>01 = 100MHz CG2_PLL generated<br>10 = EXT_GPP0_SRC<br>11 = EXT_GPP2_RX<br>Note: MISC2x000000BC[12] to select EXT_GPP0_SRC either is from GPP0 external input or GPP0 external input with divided-by-2 or divided-by-4 |

**MISC2x00000034 (FCH::MISC2::RMT\_PLLCNTL\_1\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000034; MISC2=FED8\_1300h

| Bits  | Description                                                                                                                                                                                                                                      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:23 | <b>cg2pll_fcw1_int_override.</b> Read-write. Reset: 000h. CG2PLL Override: Integer portion of Frequency Control Word0 (a.k.a. feedback divisor1). Intended to be used with frequency ramping. Also used to step PLL frequency and phase for DFT. |
| 22:0  | Reserved.                                                                                                                                                                                                                                        |

**MISC2x00000038 (FCH::MISC2::RMT\_PLLCNTL\_2\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000038; MISC2=FED8\_1300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:23 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 22:21 | <b>cg2pll_external_refclk_sel_override.</b> Read-write. Reset: 0h.<br><b>Description:</b> CG2PLL External Refclk Selection Override<br>This is the refclk mux source inside CLKGEN_S0.<br>00 = local GPP0_RX (external 100M)<br>01 = GPP2_RX<br>10 = CG2_100M<br>11 = remote GPP0_RX<br>The MISC2x00000038[22:21] are override bits to this mux inside CLKGEN_S0.<br>The local CLKGEN_S0 default is "00" in iCLK mode and "01" in eCLK mode.<br>The remote CLKGEN_S0 default is "10" in iCLK mode and "01" in eCLK mode.<br>For example:<br>If we want to use external 100MHz as CG2PLL refclk source thru local GPP0_RX in eCLK mode, then program MISC2x00000038[22:21]="01". (default "01" XOR program MISC2x00000038[22:21]="01" = "00")<br>If we want to use external 100MHz as CG2PLL refclk source thru remote GPP2_RX in iCLK mode, then program MISC2x00000038[22:21]="11". (default "10" XOR program MISC2x00000038[22:21]="11" = "01") |
| 20:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

**MISC2x0000003C (FCH::MISC2::CGPLL\_CONFIGURATION)**

Read-write. Reset: 0000\_3310h.

\_aliasHOSTLEGACY; MISC2x0000003C; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x00000040 (FCH::MISC2::RMT\_MISCCNTL\_0\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000040; MISC2=FED8\_1300h

| Bits  | Description                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | Reserved.                                                                                                                                                                                                                    |
| 30    | <b>cg2_cfg_update_req.</b> Read-write. Reset: 0.<br><b>Description:</b> Set this bit will request CG2_PLL to load spread related value into CG2_PLL.<br>The bit will be clear "0" by hardware after request send to CG2_PLL. |
| 29:26 | Reserved.                                                                                                                                                                                                                    |
| 25    | <b>cg2_fbdiv_loaden.</b> Read-write. Reset: 0. Set "1" to enable loading CG2_PLL FB_DIV value form register                                                                                                                  |
| 24:0  | Reserved.                                                                                                                                                                                                                    |

**MISC2x00000044 (FCH::MISC2::RMT\_MISCCNTL\_1\_REG)**

Read-write. Reset: 0060\_0000h.

\_aliasHOSTLEGACY; MISC2x00000044; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x00000048 (FCH::MISC2::RMT\_MISCCNTL\_2\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000048; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x0000004C (FCH::MISC2::RMT\_MISCCNTL\_3\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x0000004C; MISC2=FED8\_1300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:4 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                      |
| 3    | <b>pcie_phy_g_refclk_static_pwdn_cntl.</b> Read-write. Reset: 0.<br><b>Description:</b> PCIE_G_Set_PHY Refclk Driver Static PWDN Control<br>0: Enable on-chip driver (default)<br>1: Disable on-chip driver (if it is unused)<br>When resume, it will be static shut-down with on-chip driver enable (default setting) regardless what has been programmed and BIOS needs to program it again. |
| 2:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                      |

**MISC2x00000058 (FCH::MISC2::RMT\_MISCCNTL58)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000058; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x0000005C (FCH::MISC2::RMT\_MISCCNTL5C)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x0000005C; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x00000060 (FCH::MISC2::IDLECNTRLCOPY)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000060; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x00000070 (FCH::MISC2::RMT\_CTRL0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x00000070; MISC2=FED8\_1300h

| Bits | Description                                                                                                                                                                                                                                                                                         |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:2 | Reserved.                                                                                                                                                                                                                                                                                           |
| 1    | <b>rst_usb_s5_rmt.</b> Read-write. Reset: 0.<br><b>Description:</b> Write '1' to make Cpl_VDDCR_S5_RESETn_rmt=0 to reset USB in remote tile.<br>Need to write '0' to deassert Cpl_VDDCR_S5_RESETn=1. This bit is not for normal operation, it is to be used for recovery when something went wrong. |
| 0    | <b>mask_usb_s5_rst_rmt.</b> Read-write. Reset: 0.<br><b>Description:</b> Set to '1' to mask oDevAllRstB to gen Cpl_VDDCR_S5_RESETn_rmt =0 in remote tile to reset USB, default=0                                                                                                                    |

**MISC2x000000A0 (FCH::MISC2::RMT\_CG1PLLCNTL\_0\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000A0; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x000000A4 (FCH::MISC2::RMT\_CG1PLLCNTL\_1\_REG)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000A4; MISC2=FED8\_1300h**Bits    Description**

31:0    Reserved.

**MISC2x000000A8 (FCH::MISC2::RMT\_CG1PLLCNTL\_2\_REG)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000A8; MISC2=FED8\_1300h**Bits    Description**

31:0    Reserved.

**MISC2x000000AC (FCH::MISC2::RMT\_CG1PLLCNTL\_3\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000AC; MISC2=FED8\_1300h**Bits    Description**

31:0    Reserved.

**MISC2x000000B0 (FCH::MISC2::RMT\_CG2PLLCNTL\_0\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000B0; MISC2=FED8\_1300h**Bits    Description**

31:0    Reserved.

**MISC2x000000B4 (FCH::MISC2::RMT\_CG2PLLCNTL\_1\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000B4; MISC2=FED8\_1300h**Bits    Description**

31:0    Reserved.

**MISC2x000000B8 (FCH::MISC2::RMT\_CG2PLLCNTL\_2\_REG)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000B8; MISC2=FED8\_1300h**Bits    Description**

31:0    Reserved.



**MISC2x000000BC (FCH::MISC2::RMT\_CG2PLLCNTL\_3\_REG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000BC; MISC2=FED8\_1300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:28 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                   |
| 27:26 | <b>gpp5_refclk_selection_override.</b> Read-write. Reset: 0h.<br><b>Description:</b> GPP5 Refclk selection Override<br>00 = CG_PLL generated 100MHz<br>01 = CG2_PLL generated 100MHz (not a valid option)<br>10 = Reserved<br>11 = Ext_GPP0_SRC<br>Note: MISC2x000000BC[13] to select EXT_GPP0_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4 |
| 25:24 | <b>gpp4_refclk_selection_override.</b> Read-write. Reset: 0h.<br><b>Description:</b> GPP4 Refclk selection Override<br>00 = CG_PLL generated 100MHz<br>01 = CG2_PLL generated 100MHz (not a valid option)<br>10 = Reserved<br>11 = Ext_GPP0_SRC<br>Note: MISC2x000000BC[13] to select EXT_GPP0_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4 |
| 23:22 | <b>gpp3_refclk_selection_override.</b> Read-write. Reset: 0h.<br><b>Description:</b> GPP3 Refclk selection Override<br>00 = CG_PLL generated 100MHz<br>01 = CG2_PLL generated 100MHz (not a valid option)<br>10 = Reserved<br>11 = Reserved<br>Note: MISC2x000000BC[13] to select EXT_GPP0_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4     |
| 21:20 | <b>gpp2_refclk_selection_override.</b> Read-write. Reset: 0h.<br><b>Description:</b> GPP2 Refclk selection Override<br>00 = CG_PLL generated 100MHz<br>01 = CG2_PLL generated 100MHz (not a valid option)<br>10 = Reserved<br>11 = Reserved<br>Note: MISC2x000000BC[13] to select EXT_GPP0_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4     |
| 19:18 | <b>gpp1_refclk_selection_override.</b> Read-write. Reset: 0h.<br><b>Description:</b> GPP1 Refclk selection Override<br>00 = CG_PLL generated 100MHz<br>01 = CG2_PLL generated 100MHz (not a valid option)<br>10 = Reserved<br>11 = Reserved<br>Note: MISC2x000000BC[13] to select EXT_GPP0_SRC either is from GPP0 external input or GPP0 external input with divided-by-2/divided-by-4     |
| 17:16 | <b>gpp0_refclk_selection_override.</b> Read-write. Reset: 0h.                                                                                                                                                                                                                                                                                                                               |

|     |                                                                                                                                                                                                                                                                                                                            |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> GPP0 Refclk selection Override<br>00 = CG2_PLL generated 100MHz<br>01 = CG2_PLL generated 100MHz (not a valid)<br>10 = GPP2_RX (GPP2_BOT)<br>11 = Reserved<br>Note: MISC2x000000BC[12] to select EXT_GPP2_SRC either is from GPP2 external input or GPP2 external input with divided-by-2/divided-by-4 |
| 15  | Reserved.                                                                                                                                                                                                                                                                                                                  |
| 14  | <b>clk_cgpll_ext_pwdn.</b> Read-write. Reset: 0.<br><b>Description:</b> CLK_CGPLL_EXT_PWDN<br>Enable/Disable CGPLL external refclk source thru GPP0 input.<br>0 = enable<br>1 = disable                                                                                                                                    |
| 13  | <b>ext_gpp0_refclk_sel.</b> Read-write. Reset: 0.<br><b>Description:</b> EXT_GPP0_REFCLK_SEL<br>0 = GPP0 external input<br>1 = GPP0 external input divided-by-2(div-2 or div-4 depend on MISC2x0000003C[1])                                                                                                                |
| 12  | Reserved.                                                                                                                                                                                                                                                                                                                  |
| 11  | <b>ext_gpp1_refclk_sel.</b> Read-write. Reset: 0.<br><b>Description:</b> EXT_GPP1_REFCLK_SEL<br>0 = GPP1 external input<br>1 = GPP1 external input divided-by-2(div-2 or div-4 depend on MISC2x0000003C[3])                                                                                                                |
| 10  | <b>smu_100m_refclk_driver_pwdn.</b> Read-write. Reset: 0.<br><b>Description:</b> SMU_100M Refclk Driver PWDN<br>0 = enable<br>1 = disable                                                                                                                                                                                  |
| 9:0 | Reserved.                                                                                                                                                                                                                                                                                                                  |

**MISC2x000000F0 (FCH::MISC2::RMT\_MISCCNTLF0)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000F0; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**MISC2x000000F4 (FCH::MISC2::RMT\_MISCCNTLF4)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; MISC2x000000F4; MISC2=FED8\_1300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**14.3.14.3 Power Management (PM) Registers and Standard ACPI Registers**

Table 256: ACPI MMIO Space Allocation

|             |                                                                                                                           |
|-------------|---------------------------------------------------------------------------------------------------------------------------|
| 00FFh-0000h | <a href="#">SMBus</a> PCI configuration registers, see 14.3.12.1 [Device 14h Function 0 (SMBus) Configuration Registers]. |
| 01FFh-0100h | Reserved.                                                                                                                 |
| 02FFh-0200h | <a href="#">SMI</a> , see 14.3.5 [SMI Registers].                                                                         |
| 03FFh-0300h | PMIO and ACPI, see 14.3.14.3 [Power Management (PM) Registers and Standard ACPI Registers].                               |
| 08FFh-0800h |                                                                                                                           |

|             |                                                                                   |
|-------------|-----------------------------------------------------------------------------------|
| 05FFh-0500h | BIOS RAM                                                                          |
| 06FFh-0600h | CMOS RAM                                                                          |
| 07FFh-0700h | CMOS                                                                              |
| 09FFh-0900h | ASF registers, see 14.3.12.2 [ASF (Alert Standard Format) Registers].             |
| 0AFFh-0A00h | SMBus registers, see 14.3.12.3 [SMBUS Register].                                  |
| 0BFFh-0B00h | Watchdog registers, see 14.3.9 [Watchdog Timer (WDT) Registers].                  |
| 0CFFh-0C00h | HPET, see 14.3.6 [High Precision Event Timer (HPET) Registers].                   |
| 0DFFh-0D00h | IOMux, see 14.3.15.1 [IOMUX Registers].                                           |
| 0EFFh-0E00h | Miscellaneous registers, see 14.3.14.1 [Miscellaneous (MISC) Registers].          |
| 10FFh-1000h | Reserved.                                                                         |
| 11FFh-1100h | Reserved.                                                                         |
| 12FFh-1200h | Remote Tile GPIO+IOMUX, see 14.3.15.3 [REMOTE GPIO and IOMUX Registers].          |
| 13FFh-1300h | Remote CGPLL-MISC2, see 14.3.14.2 [Miscellaneous (MISC2) Registers].              |
| 14FFh-1400h | DP-VGA                                                                            |
| 19FFh-1500h | GPIO Registers, see 14.3.15.2 [GPIO Registers].                                   |
| 1BFFh-1B00h | Reserved.                                                                         |
| 1CFFh-1C00h | Reserved.                                                                         |
| 1DFFh-1D00h | Wake Device (AC DC timer), see 14.3.10 [Wake Alarm Device (AcDcTimer) Registers]. |
| 1EFFh-1E00h | Reserved.                                                                         |
| 1FFFh-1F00h | Reserved.                                                                         |

The PM register space is accessed through two methods:

- Indirect IO access through IOCD6 [PM\_Index] and IOCD7 [PM\_Data]. Software first programs the offset into the index register IOCD6 and then reads from or writes to the data register IOCD7.
- Direct memory mapped access through the AcpiMmio region. The PM registers range from FED8\_0000h+300h to FED8\_0000h+3FFh. See PMx04[MmioEn] for details on the AcpiMmio region.

**PMx00000000 (FCH::PM::DECODEEN)**

Read-write. Reset: E302\_0B10h.

\_aliasHOSTLEGACY; PMx00000000; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                       |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | <b>ioapicconfig.</b> Read-write. Reset: 3h. Set to 11 to improve IoApic latency                                                                                                                                                                                                                                   |
| 29    | <b>hpet_msi_en.</b> Read-write. Reset: 1. Set to 1 to expose MSI cap in HPET Cap register..                                                                                                                                                                                                                       |
| 28    | <b>hpet_width_sel.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: HPET is 32-bit<br>1: HPET is 64-bit                                                                                                                                                                                                        |
| 27:26 | <b>watchdogoptions.</b> Read-write. Reset: 0h. Set to 00 for normal WatchDogTimer operation.                                                                                                                                                                                                                      |
| 25:24 | <b>watchdogfreq.</b> Read-write. Reset: 3h.<br><b>Description:</b> This registers define the clock frequency used by the WatchDogTimer<br>00: 32us<br>01: 10ms<br>10: 100ms<br>11: 1s                                                                                                                             |
| 23:21 | <b>asfclkssel.</b> Read-write. Reset: 0h.<br><b>Description:</b> The value controls the frequency of ASF master clock its definition is:<br>000: ~100kHz<br>001: ~200kHz<br>010: ~300kHz<br>011:~ 400kHz<br>100:~ 600kHz<br>101:~ 800kHz<br>110:~ 900kHz<br>111:~1MHz                                             |
| 20:19 | <b>smbus0sel.</b> Read-write. Reset: 0h.<br><b>Description:</b> SmBus port selection.<br>There is only one SMBUS engine controlling four SMBUS ports. This register routes the SMBUS engine to the desired port.<br>00: Port 0 (For SMBUS on the board)<br>01: Port 2 (Dedicated for <a href="#">TSI</a> polling) |
| 18    | <b>asfclkswitch.</b> Read-write. Reset: 0. Set to 1 to change ASF master clock from RTC(32k) to the clock defined in ASFClkSel of the same register.                                                                                                                                                              |
| 17    | <b>asfclkstretchen.</b> Read-write. Reset: 1. Set to 1 to enable clock stretching support.                                                                                                                                                                                                                        |
| 16    | <b>asfsmmasteren.</b> Read-write. Reset: 0. Set to 1 to enable ASF SMBUS master function.                                                                                                                                                                                                                         |
| 15:8  | <b>smbusasfiobase.</b> Read-write. Reset: 0Bh.<br><b>Description:</b> Smbus Io base = {Smbus0AsfIoBase[7:0], 0x00}<br>ASF Io base = {Smbus0AsfIoBase[7:0], 0x20}<br>By default Smbus Io base is 0xB00 and ASF Io base is 0xB20                                                                                    |
| 7     | <b>watchdogtmren.</b> Read-write. Reset: 0. Set to 1 to enable WatchDog Timer memory(FEB00000 ~ FEB00003) decoding, and enable WatchDog Timer operation.                                                                                                                                                          |
| 6     | <b>hpeten.</b> Read-write. Reset: 0. Set to 1 to enable HPET memory(FED00000 ~ FED001FF) decoding.                                                                                                                                                                                                                |
| 5     | <b>ioapicen.</b> Read-write. Reset: 0. Set to 1 to enable IoApic memory(FEC00000 ~ FEC0007F) decoding.                                                                                                                                                                                                            |
| 4     | <b>smbusasfioen.</b> Read-write. Reset: 1. Set to 1 to enable Smbus and Asf Io decoding. Smbus and Asf Io range are defined in Smbus0AsfIoBase.                                                                                                                                                                   |
| 3     | <b>dmaport80.</b> Read-write. Reset: 0. Set to 1 to pass Io port 0x80, 0x81, 0x82, 0x83 to legacy Dma Io range.                                                                                                                                                                                                   |
| 2     | <b>legacydmaioen.</b> Read-write. Reset: 0. Set to 1 to enable legacy Dma Io range.                                                                                                                                                                                                                               |
| 1     | <b>cf9ioen.</b> Read-write. Reset: 0. Set to 1 to enable CF9 Io port decoding                                                                                                                                                                                                                                     |
| 0     | <b>legacyioen.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                          |

|  |                                                                                                                                                                                        |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <b>Description:</b> Set to 1 to enable the following Io decoding:<br>0x20, 0x21, 0xA0, 0xA1 (PIC)<br>0x40, 0x41, 0x42, 0x43, 0x61 (8254 timer)<br>0x70, 0x71, 0x72, 0x73 (Rtc)<br>0x92 |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

#### PMx00000004 (FCH::PM::ISACONTROL)

Read-write.

PrtcControl registers are reset by RsmRstB and UsrRstB. RsmRstB is asserted during G3 state. UsrRstB is asserted when the system reset button is pressed (in S0 state). dPassEnable will be 0 during G3 state so that when S5 power is down, the invalid values of PrtcControl Registers will be isolated to 0 for PRTC.

One important requirement is:

Rtc32kHz clock's availability should not have any dependency with the PrtcControl Registers. This is because we use Rtc32kHz clock for various reset de-bounce logics including the reset for PrtcControl Registers.

\_aliasHOSTLEGACY; PMx00000004; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>rtc_bg_adj5.</b> Read-write. Reset: 0. PRTC band-gap control bit                                                                                                                                                                                       |
| 30    | <b>rtc_bg_adj4.</b> Read-write. Reset: 0. PRTC band-gap control bit                                                                                                                                                                                       |
| 29    | <b>rtc_bg_adj3.</b> Read-write. Reset: 0. PRTC band-gap control bit                                                                                                                                                                                       |
| 28    | <b>rtc_bg_adj2.</b> Read-write. Reset: 0. PRTC band-gap control bit                                                                                                                                                                                       |
| 27    | <b>rtc_bg_adj1.</b> Read-write. Reset: 0. PRTC band-gap control bit                                                                                                                                                                                       |
| 26    | <b>rtc_bg_adj0.</b> Read-write. Reset: 0. PRTC band-gap control bit                                                                                                                                                                                       |
| 25    | <b>rtc_bg_set_en.</b> Read-write. Reset: 0.<br><b>Description:</b> This is the enable for the PRTC 32kHz band-gap control bits (rtc_bg_adj0~5).<br>0: PRTC band-gap uses its own hard-wired configuration.<br>1: PRTC band-gap uses the rtc_bg_adj* bits. |
| 24    | <b>rtc_high_res.</b> Read-write. Reset: 0. Debug purpose register requested by PRTC designer.                                                                                                                                                             |
| 23:22 | Reserved.                                                                                                                                                                                                                                                 |
| 21    | <b>rtc_osc_op1.</b> Read-write. Reset: 0. PRTC oscillator control bit                                                                                                                                                                                     |
| 20    | <b>rtc_osc_op0.</b> Read-write. Reset: 0. PRTC oscillator control bit                                                                                                                                                                                     |
| 19    | <b>rtc_osc_set1.</b> Read-write. Reset: 0. PRTC oscillator control bit                                                                                                                                                                                    |
| 18    | <b>rtc_osc_set0.</b> Read-write. Reset: 0. PRTC oscillator control bit                                                                                                                                                                                    |
| 17    | <b>rtc_osc_set_en.</b> Read-write. Reset: 0. This is the latch enable for the PRTC 32kHz oscillator control bits (rtc_osc_set0~1, rtc_osc_op0~1).                                                                                                         |
| 16    | <b>abclkgateen.</b> Read-write. Reset: 0.<br><b>Description:</b> Master switch for Alink and Blink clock gating.<br>0: Disabled<br>1: Enabled                                                                                                             |
| 15:12 | Reserved.                                                                                                                                                                                                                                                 |
| 11    | <b>bm_req_en.</b> Read-write. Reset: 0. Legacy BM_REQ# function enable bit it is now for debug purpose only.                                                                                                                                              |
| 10    | Reserved.                                                                                                                                                                                                                                                 |
| 9     | <b>drqmasken.</b> Read-write. Reset: 0. Set to 1 to allow DMA DRQ input to block clock gating                                                                                                                                                             |
| 8     | <b>dmaenhanceen.</b> Read-write. Reset: 0. Set to 1 to enable enhancement of legacy Dma.                                                                                                                                                                  |
| 7:6   | Reserved.                                                                                                                                                                                                                                                 |
| 5     | <b>read_shadow.</b> Read-write. Reset: 0. Set to 1 to allow to read Pic ICWX, OCWX register through Io port 21h/A1h.                                                                                                                                      |
| 4:2   | Reserved.                                                                                                                                                                                                                                                 |
| 1     | <b>mmioen.</b> Read-write. Reset: 1. Set to 1 to enable Acpi Mmio range (FED8_0000 ~FED8_1FFF). The space is allocated as specified in Table shows the Address mapping of each ACPI block in front of register PM_regx00                                  |
| 0     | <b>biosramen.</b> Read-write. Reset: 0. Set to 1 to enable bios Ram whose base is FED1_0000 (256 bytes)                                                                                                                                                   |

**PMx00000008 (FCH::PM::PCICONTROL)**

Read-write.

SLP\_TYP is located in AcpiPm1CntBlk offset 00h, bits 10~12.

\_aliasHOSTLEGACY; PMx00000008; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:26 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 25    | <b>force_slpstate_retry.</b> Read-write. Reset: 0. Set to 1 to send out <a href="#">SMI</a> message before the completion response of IO write to SLP_TYP register. This is to be used in conjunction with SMI trapping on write to SLP_TYP register                                                                                                                                                                                                        |
| 24    | <b>force_stpclk_retry.</b> Read-write. Reset: 1.<br><b>Description:</b> Set to 1 to send out STPCLK message before the completion response of the following 3 types of request:<br>I/O write to Slp_typ register<br>I/O write Ldt_stp command<br>C1e cycle<br>Normally it is required to send out STPCLK before completion of the cycles listed above, except for the case of <a href="#">SMI</a> trapping. In that case, this bit should be left as 0      |
| 23    | <b>ab_stall_en.</b> Read-write. Reset: 0. Set to 1 to allow the legacy DMA engine to hold the internal bus before completing legacy DMA on the LPC bus. This is only needed for certain old LPC devices.                                                                                                                                                                                                                                                    |
| 22:21 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 20    | <b>shutdownoption.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Issue Init message upstream when receiving shutdown message.<br>1 : Generate Pci reset when receiving shutdown message.                                                                                                                                                                                                                                                              |
| 19    | <b>masternowait.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: ACPI PCI Master doesn't wait for Slave idle when it wants to request bus.<br>0: Old behavior. PCI Master will wait for Slave idle.<br>Note: Software need keep this bit to1. In rare case, if there is downstream cycle to ACPI register (from CPU/ <a href="#">SMN</a> /M2P) and there is upstream Interrupt message pending, state machine will lock up if this bit is not set to 1. |
| 18    | <b>echohostfix.</b> Read-write. Reset: 1. Not used.                                                                                                                                                                                                                                                                                                                                                                                                         |
| 17    | <b>changedma.</b> Read-write. Reset: 0. Not used.                                                                                                                                                                                                                                                                                                                                                                                                           |
| 16    | <b>gatedma.</b> Read-write. Reset: 0. Not used.                                                                                                                                                                                                                                                                                                                                                                                                             |
| 15    | <b>clock_slow_mask.</b> Read-write. Reset: 0. Not used.                                                                                                                                                                                                                                                                                                                                                                                                     |
| 14:12 | <b>ext_intr_time.</b> Read-write. Reset: 0h. Specify the extended interrupt time in 2 microsecond intervals. This is used for preventing APU from re-entering C state right away when it just breaks out from a C state                                                                                                                                                                                                                                     |
| 11    | <b>dly_en.</b> Read-write. Reset: 0. Not used.                                                                                                                                                                                                                                                                                                                                                                                                              |
| 10    | <b>ignr_usb_smi_req.</b> Read-write. Reset: 1. Not used.                                                                                                                                                                                                                                                                                                                                                                                                    |
| 9     | <b>block_acpi_s5_intr_st.</b> Read-write. Reset: 0. Set to 1, it will block the acpi_s5 interrupt status to send out.                                                                                                                                                                                                                                                                                                                                       |
| 8     | <b>pic_apic_arbiter.</b> Read-write. Reset: 1. Set to 1 to arbitrate between PIC request and IOAPIC request                                                                                                                                                                                                                                                                                                                                                 |
| 7     | <b>force_smaf_match.</b> Read-write. Reset: 0. Set to 1 to enable STPGNT message matching to the expected SMAF.                                                                                                                                                                                                                                                                                                                                             |
| 6     | <b>nmimsgsel.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Encode NMI request as legacy PIC NMI message type.<br>1: NMI request as NMI message type.                                                                                                                                                                                                                                                                                                 |
| 5     | <b>picmsgsel.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Encode PIC interrupt request as Legacy PIC ExtInt message type<br>1: encode PIC interrupt request as ExtInt message type                                                                                                                                                                                                                                                                  |
| 4     | <b>msg_intr_enable.</b> Read-write. Reset: 0. Set to 1 to deliver legacy PIC interrupt as message type.                                                                                                                                                                                                                                                                                                                                                     |
| 3     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2     | <b>fake_dma_en.</b> Read-write. Reset: 0. Not used                                                                                                                                                                                                                                                                                                                                                                                                          |
| 1     | <b>undo_wrtdt_done.</b> Read-write. Reset: 0. Not used                                                                                                                                                                                                                                                                                                                                                                                                      |

|   |                                                        |
|---|--------------------------------------------------------|
| 0 | <b>undo_dma_change.</b> Read-write. Reset: 0. Not used |
|---|--------------------------------------------------------|

**PMx0000000C (FCH::PM::STPCLKSMAF)**

Read-write. Reset: 5543\_2106h.

\_aliasHOSTLEGACY; PMx0000000C; PM=FED8\_0300h

| Bits  | Description                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------|
| 31    | Reserved.                                                                                                  |
| 30:28 | <b>ttsmaf.</b> Read-write. Reset: 5h. System management action field for Thermal Throttling STPCLK message |
| 27    | Reserved.                                                                                                  |
| 26:24 | <b>nssmaf.</b> Read-write. Reset: 5h. System management action field for Normal Throttling STPCLK message  |
| 23    | Reserved.                                                                                                  |
| 22:20 | <b>s3smaf.</b> Read-write. Reset: 4h. System management action field for S3 STPCLK message                 |
| 19    | Reserved.                                                                                                  |
| 18:16 | <b>s1smaf.</b> Read-write. Reset: 3h. System management action field for S1 STPCLK message                 |
| 15    | Reserved.                                                                                                  |
| 14:12 | <b>vfsmaf.</b> Read-write. Reset: 2h. System management action field for VFID STPCLK message               |
| 11    | Reserved.                                                                                                  |
| 10:8  | <b>c3smaf.</b> Read-write. Reset: 1h. System management action field for C3 STPCLK message                 |
| 7     | Reserved.                                                                                                  |
| 6:4   | <b>c2smaf.</b> Read-write. Reset: 0h. System management action field for C2 STPCLK message                 |
| 3     | Reserved.                                                                                                  |
| 2:0   | <b>s4s5smaf.</b> Read-write. Reset: 6h. System management action field for S4/5 STPCLK message             |

**PMx00000010 (FCH::PM::PWRRSTCFG)**

Read-write.

\_aliasHOSTLEGACY; PMx00000010; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:7 | Reserved.                                                                                                                                                                                                                                                                                                                                                                           |
| 6    | <b>fatalrsttos5rst.</b> Read-write. Reset: 1.                                                                                                                                                                                                                                                                                                                                       |
| 5    | <b>rtcclkchken_s5reg.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: RtcClkChk function is disabled<br>1: RtcClkChk function depends on the strap pin. If the strapped value is 1, RtcClkChk function is enabled. The strap name is RtcClkChkEn_strap.                                                                                                                         |
| 4    | <b>slps3waitrstasrt.</b> Read-write. Reset: 1. Debug purpose                                                                                                                                                                                                                                                                                                                        |
| 3:2  | <b>rstblkssel_1_0.</b> Read-write. Reset: 1h.<br><b>Description:</b> 00: Always use old logic for reset signals.<br>01: Use new logic for reset signals during Power saving mode entry/exit, otherwise, use old logic.<br>10: Always use new logic for reset signals.<br>11: Always use new logic for reset signals.<br>Note:<br>Old logic: fch_pwr_detect<br>New logic: fch_PwrRst |
| 1    | <b>toggleallpwrgoodoncf9.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: De-assert and then assert all PwrGood signals (for PG1, PG1a, PG2 and XHC) during CF9 reset.<br>0: During CF9 reset, PG1_PwrGood stay at high. PG1a_PwrGood, PG2_PwrGood and Xhc_PwrGood behavior depend on TogglePG1aPG2PGXHConCf9 register bit during CF9 reset.                                    |
| 0    | <b>togglepg1apg2pgxhconcf9.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: De-assert and then assert PG1a_PwrGood, PG2_PwrGood and Xhc_PwrGood during CF9 reset.<br>0: PG1a_PwrGood, PG2_PwrGood and Xhc_PwrGood stay at high during CF9 reset.                                                                                                                                |

**PMx00000012 (FCH::PM::FCHPWRGOODTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000012; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                                                                                    |
| 10:0  | <b>fchpwrgoodtmr</b> . Read-write. Reset: 150h. T1, Debouncetimer value for FCHPwrGood, and the delay timer value for PwrRstB deassertion of devices in PG1. |

**PMx00000014 (FCH::PM::PGPWRGOODTOALLENTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000014; PM=FED8\_0300h

| Bits  | Description                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                                |
| 10:0  | <b>pgpwrgoodtoallentmr</b> . Read-write. Reset: 016h. T18, Timer for PwrGood to AllEn for all the tiles. |

**PMx00000016 (FCH::PM::PGPWRGOODASSERTIONTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000016; PM=FED8\_0300h

| Bits  | Description                                                                                                         |
|-------|---------------------------------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                                           |
| 10:0  | <b>pgpwrgoodassertiontmr</b> . Read-write. Reset: 020h. T19, Timer for DebouncedPwrGood to PwrGood of all the tiles |

**PMx00000018 (FCH::PM::PWRGATETMR)**

Read-write. Reset: A864\_8642h.

\_aliasHOSTLEGACY; PMx00000018; PM=FED8\_0300h

| Bits  | Description                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------|
| 31:28 | <b>pwrlanddaughterassertiontmr</b> . Read-write. Reset: Ah. T24, Timer for PwrOn=0 to Daughter_SD=1 of PG1a          |
| 27:24 | <b>pwrlandmotherassertiontmr</b> . Read-write. Reset: 8h. T23, Timer for PwrOn=0 to Mother_SD=1 of PG1a              |
| 23:20 | <b>pwrlandmemdaughterassertiontmr</b> . Read-write. Reset: 6h. T22, Timer for PwrOn=0 to Mem_Daughter_SD=1 of PG1a   |
| 19:16 | <b>pwrlandmemmotherassertiontmr</b> . Read-write. Reset: 4h. T21, Timer for PwrOn=0 to Mem_Mother_SD=1 of PG1a       |
| 15:12 | <b>pwrlandmemmotherdeassertiontmr</b> . Read-write. Reset: 8h. T17, Timer for PwrOn=1 to Mem_Mother_SD=0 of PG1a     |
| 11:8  | <b>pwrlandmemdaughterdeassertiontmr</b> . Read-write. Reset: 6h. T16, Timer for PwrOn=1 to Mem_Daughter_SD=0 of PG1a |
| 7:4   | <b>pwrlandmotherdeassertiontmr</b> . Read-write. Reset: 4h. T15, Timer for PwrOn=1 to Mother_SD =0 of PG1a           |
| 3:0   | <b>pwrlanddaughterdeassertiontmr</b> . Read-write. Reset: 2h. T14, Timer for PwrOn=1 to Daughter_SD =0 of PG1a       |

**PMx0000001C (FCH::PM::PLLSTBTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx0000001C; PM=FED8\_0300h

| Bits  | Description                                                                             |
|-------|-----------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                               |
| 10:0  | <b>pllrstbtmr</b> . Read-write. Reset: 040h. Delay timer value for PLLRstB deassertion. |



**PMx0000001E (FCH::PM::PLLLOCKTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx0000001E; PM=FED8\_0300h

| Bits  | Description                                                                           |
|-------|---------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                             |
| 10:0  | <b>plllocktmr</b> . Read-write. Reset: 03Ch. Delay timer value for PllLock assertion. |

**PMx00000020 (FCH::PM::PCIRSTBTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000020; PM=FED8\_0300h

| Bits  | Description                                                                             |
|-------|-----------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                               |
| 10:0  | <b>pcirstbtmr</b> . Read-write. Reset: 120h. Delay timer value for PciRstB deassertion. |

**PMx00000022 (FCH::PM::CPURSTBTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000022; PM=FED8\_0300h

| Bits  | Description                                                                            |
|-------|----------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                              |
| 10:0  | <b>cpurstbtmr</b> . Read-write. Reset: 19Ch. Delaytimer value for CpuRstB deassertion. |

**PMx00000024 (FCH::PM::NBPWRGOODTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000024; PM=FED8\_0300h

| Bits  | Description                                                                              |
|-------|------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                |
| 10:0  | <b>nbpwrgoodtmr</b> . Read-write. Reset: 0A0h. Delay timer value for NBPwrGood assertion |

**PMx00000026 (FCH::PM::CPUPWRGOODTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000026; PM=FED8\_0300h

| Bits  | Description                                                                                |
|-------|--------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                  |
| 10:0  | <b>cpupwrgoodtmr</b> . Read-write. Reset: 0E0h. Delay timer value for CpuPwrGood assertion |

**PMx00000028 (FCH::PM::S0TOS5ENTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000028; PM=FED8\_0300h

| Bits  | Description                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                                                              |
| 10:0  | <b>s0tos5entmr</b> . Read-write. Reset: 055h. T12, timer for DebouncedPwrGood=1 to S5TilePwrMux=1, PwrGoodOsc=0 to xSLP_S3_/xSLP_S5_=0 |

**PMx0000002A (FCH::PM::S5TOS0ENTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx0000002A; PM=FED8\_0300h

| Bits  | Description                                                                                    |
|-------|------------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                      |
| 10:0  | <b>s5tos0entmr</b> . Read-write. Reset: 010h. T3, timer for DebouncedPwrGood=1 to S5_ToAllEn=1 |

**PMx0000002C (FCH::PM::SLPRSTTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx0000002C; PM=FED8\_0300h

| Bits  | Description                                                                              |
|-------|------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                |
| 10:0  | <b>slprsttmr</b> . Read-write. Reset: 080h. T13, Timer for S3/S5 Sleep Reset pulse width |

**PMx0000002E (FCH::PM::OVERRIDEDEVSTTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx0000002E; PM=FED8\_0300h

| Bits  | Description                                                                                       |
|-------|---------------------------------------------------------------------------------------------------|
| 15:12 | Reserved.                                                                                         |
| 11:8  | <b>overriderstbtmr</b> . Read-write. Reset: 0h. Override Default RstBTmr for device resets        |
| 7:4   | <b>overrideefclkoktmr</b> . Read-write. Reset: 0h. Override Default RefClkOkTmr for device resets |
| 3:0   | <b>overridepwrstbtmr</b> . Read-write. Reset: 0h. Override Default PwrRstBTmr for device resets   |

**PMx00000030 (FCH::PM::OVERRIDEPWRRSTBASRTTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000030; PM=FED8\_0300h

| Bits  | Description                                                                                               |
|-------|-----------------------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                                 |
| 10:0  | <b>overridepwrstbasrttmr</b> . Read-write. Reset: 000h. Override Default PwrRstBASrtTmr for device resets |

**PMx00000032 (FCH::PM::OVERRIDEDEFCLKOKDSRTTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx00000032; PM=FED8\_0300h

| Bits  | Description                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------|
| 15:11 | Reserved.                                                                                                  |
| 10:0  | <b>overrideefclkokdsrttmr</b> . Read-write. Reset: 000h. Override Default RefClkOkDsrTmr for device resets |

**PMx00000034 (FCH::PM::OVERRIDEENABLE)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx00000034; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>overrideenable</b> . Read-write. Reset: 0000_0000h. Enable DevRst block timing override. Each bit enables the override for a device's DevRst block. When the bit is 1, the corresponding device's DevRst block will use the timing defined in PM_Reg:2E~33h, instead of the original hardwired values. |

**PMx00000038 (FCH::PM::DEVST\_RSTBONLY)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx00000038; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>devrst_rstbonly</b> . Read-write. Reset: 0000_0000h. Each bit is corresponding to a device's DevRst block. This is for debug purpose. When a bit is set 1, the corresponding device's PwrRsB and RefClkOk will be identical with RstB. When the bit is 0, the device's resets have default behaviors. Please refer to FCH_AOAC_Design_Specification for detail reset sequence and timing. |

**PMx0000003C (FCH::PM::S0RESETTMR)**

Read-write.

\_aliasHOSTLEGACY; PMx0000003C; PM=FED8\_0300h

| Bits  | Description                                                           |
|-------|-----------------------------------------------------------------------|
| 15:12 | Reserved.                                                             |
| 11:0  | <b>s0resettmr</b> . Read-write. Reset: 800h. Timer value for S0ResetB |

**PMx0000003E (FCH::PM::PWRRSTCNFG)**

Read-write. Reset: E601h.

\_aliasHOSTLEGACY; PMx0000003E; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | <b>osc16usinterval</b> . Read-write. Reset: E6h. In decimal 230 (or 16us) by default. It is used to set the Osc16Us counter limit. It is the base timer tick to increment various reset counter. Whenever we should enter into Power saving mode, SMU FW can change Osc16UsInterval to a smaller value. Upon Power saving mode exit, it can change it back to the default value. |
| 7    | Reserved.                                                                                                                                                                                                                                                                                                                                                                        |
| 6    | <b>keep_warmrst_bf_mp1_ack</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: warm reset sequence does not wait for MP1_WarmResetAck when en_MP1_WarmResetAck (PMx70[0]) =1<br>1: warm reset sequence waits for MP1_WarmResetAck indefinitely when en_MP1_WarmResetAck (PMx70[0]) =1                                                                                          |
| 5:4  | Reserved.                                                                                                                                                                                                                                                                                                                                                                        |
| 3    | <b>en_coldrst_mp1ack</b> . Read-write. Reset: 0.<br><b>Description:</b> Before set this bit =1, PMx70[0]: en_MP1_WarmResetAck need be set =1 first. This bit is reset by RsmRstB.<br>0: Cold Reset is not blocked by MP1_WarmResetAck protocol<br>1: Cold Reset is blocked by MP1_WarmResetAck protocol                                                                          |
| 2    | <b>smerr_l_en</b> . Read-write. Reset: 0.<br><b>Description:</b> The output enable of SMERR_L.<br>1: SMERR_L pad is output mode,<br>0: SMERR_L pad is input mode.                                                                                                                                                                                                                |
| 1    | <b>en_mp1_warmresetack2</b> . Read-write. Reset: 0.<br><b>Description:</b> Chiken bit to select the gated iOsc16Us.<br>1: select the gated iOsc16Us by warm_rst_block_noncpu_rst which synced to osc clock domain.<br>0: select the original iOsc16Us.                                                                                                                           |
| 0    | <b>encpuwaitdev</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: PwrRst block doesn't wait for non-D3 devices RstB de-assertion before it de-asserts CpuRstB.<br>1: PwrRst block will wait until all non-D3 devices RstB are de-asserted, and then de-assert CpuRstB.<br>Note: "non-D3 devices" refer to those devices that are not currently in D3.                        |

**PMx00000040 (FCH::PM::ESPIINTRCTRL)**

Read-write. Reset: 00FF\_FFFFh.

\_aliasHOSTLEGACY; PMx00000040; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | Reserved.                                                                                                                                                                    |
| 23:0  | <b>espidvintrmask</b> . Read-write. Reset: FF_FFFFh.<br><b>Description:</b> SW can set these bits to mask of eSPI Device IRQ23~0.<br>1: Mask off the interrupt<br>0: No mask |

**PMx00000044 (FCH::PM::BOOTTIMEREN)**

Reset: 1800\_0000h.

\_aliasHOSTLEGACY; PMx00000044; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>boottmrdisable.</b> Read-write. Reset: 0. Set to 1 to stop boot timer. Once set it to 1, timer will reset back to 0. When this bit is set back to 0, timer starts counting from 0. This bit will clear itself on any reset (this bit is non-sticky).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 30    | <b>failbootrststs.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: Boot timer has not been fired.<br>1: Boot timer has been fired.<br>Write 1 to clearit to 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 29    | <b>expireboottmr.</b> Read-write. Reset: 0. Set to 1 to force boot timer to expire then, NB PwrGood can be asserted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 28    | <b>boottmrstopongalink.</b> Read-write. Reset: 1. Set to 1 to stop boot timer when <a href="#">FCH</a> observes the good boot after PCI reset.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 27    | <b>boottmrfuncen.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Disable boot timer function.<br>1: Enable boot timer function. The boot timer will start to count down and toggle NBPwrGood after 1.26s ( $2^{**24}/(66.6M/5)$ ) if SW has not set the bit 29 to 1 after PCI reset or resuming from S3/S4/S5. This bit is persistent through any reset or sleep (this bit is sticky).                                                                                                                                                                                                                                                                                                                                                                                                       |
| 26:25 | <b>boottmrdivsel.</b> Read-write. Reset: 0h.<br><b>Description:</b> Select the Fail Boot Timer limit.<br>00b: 1.26 Second, using Divide_by_5, default<br>01b: 2.26 Second, using Divide_by_9<br>10b: 3.26 Second, using Divide_by_13<br>11b: 4.26 Second, using Divide_by_17                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 24:0  | <b>failboottimer.</b> Read-write. Reset: 000_0000h.<br><b>Description:</b> Read Only. The counter of APU Boot timer (66.6MHz/5), which starts counting when all of the the following conditions are met:<br>Bit31 of this register =0<br>Bit27 of this register =1<br>PCI reset is not asserted.<br>Set bit31=1 will stop the timer and reset timer back to 0, it will count from 0 again if bit31 is set back to 0, bit31 is a non-sticky bit and will get clear to 0 on any reset.<br>Set bit27=0 will disable timer permanently because bit27 is a sticky bit. Bit27 will reset back to 1 if there is a S5 power loss or it can be written back to 1 by software.<br>The timer itself cannot be written directly. It will reset back to 0 whenever bit31=1 or bit27=0 or a reset has occurred. |

**PMx00000048 (FCH::PM::PGPWRENDLY)**

Read-write. Reset: 5935\_2B21h.

\_aliasHOSTLEGACY; PMx00000048; PM=FED8\_0300h

| Bits  | Description                                                                  |
|-------|------------------------------------------------------------------------------|
| 31:24 | <b>pg1pwrdownldlytmr.</b> Read-write. Reset: 59h. PG1 Power Down delay timer |
| 23:16 | <b>xhcpwrendlytmr.</b> Read-write. Reset: 35h. XhcPwrEn delay timer          |
| 15:8  | <b>pg2pwrendlytmr.</b> Read-write. Reset: 2Bh. PG2PwrEn delay timer          |
| 7:0   | <b>pg1apwrendlytmr.</b> Read-write. Reset: 21h. PG1aPwrEn delay timer        |

**PMx0000004C (FCH::PM::I2CINPUTTHRESHOLD)**

Read-write. Reset: 493F\_0000h.

\_aliasHOSTLEGACY; PMx0000004C; PM=FED8\_0300h

| Bits  | Description                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>gbepwrendlytmr.</b> Read-write. Reset: 49h. GbePwrEn delay timer                                                                               |
| 23:16 | <b>otgpwrendlytmr.</b> Read-write. Reset: 3Fh. OtgPwrEn delay timer                                                                               |
| 15:6  | Reserved.                                                                                                                                         |
| 5     | <b>i2c5inputthresholdhi.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure I2C5 input threshold<br>0: Low threshold<br>1: High threshold |
| 4     | <b>i2c4inputthresholdhi.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure I2C4 input threshold<br>0: Low threshold<br>1: High threshold |
| 3     | <b>i2c3inputthresholdhi.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure I2C3 input threshold<br>0: Low threshold<br>1: High threshold |
| 2     | <b>i2c2inputthresholdhi.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure I2C2 input threshold<br>0: Low threshold<br>1: High threshold |
| 1     | <b>i2c1inputthresholdhi.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure I2C1 input threshold<br>0: Low threshold<br>1: High threshold |
| 0     | <b>i2c0inputthresholdhi.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure I2C0 input threshold<br>0: Low threshold<br>1: High threshold |

**PMx00000050 (FCH::PM::APUPLLCTRL)**

Read-write. Reset: 3F3E\_0000h.

\_aliasHOSTLEGACY; PMx00000050; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>apupllrstbtmr.</b> Read-write. Reset: 3Fh. Configure the PllLock assertion time. Unit is 16usec.                                                                                                                                                                                                                                                                                                                     |
| 23:16 | <b>apupllpwrrstbtmr.</b> Read-write. Reset: 3Eh. Configure the PllRstB deassertion time. Unit is 16usec.                                                                                                                                                                                                                                                                                                                |
| 15:2  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                               |
| 1     | <b>vidchg_use_newscheme.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: old scheme, 8xPcick wide VIDCHG_req generated when wakeup from Clock slow down mode or Clock stop mode<br>1: new scheme, while in Clock slow down mode or Clock stop mode, VIDCHG_req assert =1 when receiving wakeup signal, VIDCHG_req deassert =0 after receiving VIDCHG_ack =1, VIDCHG_ack will deassert =0 after seeing VIDCHG_req =0 |
| 0     | <b>apupllctrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable<br>1: Eable APU to power down CGPLL(CLOCK STOP MODE) or slow down clock (CLOCK SLOW DOWN MODE)<br>When this bit is 1, ApuPllOffReq or CLOCK SLOW DOWN MODE_req bit in AOAC_Reg0x94 can be written by APU (or to be more precise, SMU) to power down the CGPLL or slow down clock.                                                          |

**PMx00000054 (FCH::PM::SERIALIRQCONFIG)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx00000054; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 14   | <b>pmx54bit7_nonsticky</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: bit[7] is non-Sticky<br>0: bit[7] is Sticky                                                                                                                                                                                                                                                                                                                                                  |
| 13   | <b>softwareirq12</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: IRQ12 input is high<br>0: IRQ12 input is low                                                                                                                                                                                                                                                                                                                                                       |
| 12   | <b>softwareirq1</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: IRQ1 input is high<br>0: IRQ1 input is low                                                                                                                                                                                                                                                                                                                                                          |
| 11   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 10   | <b>undo_ser_irq_change</b> . Read-write. Reset: 0. Not used.                                                                                                                                                                                                                                                                                                                                                                                                              |
| 9    | <b>legacy_dis</b> . Read-write. Reset: 0. Not used.                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 8    | <b>siirq_cntrl</b> . Read-write. Reset: 0. Not used.                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 7    | <b>serialirqenable</b> . Read-write. Reset: 0.<br><b>Description:</b> Setting this bit to 1 enable the serial IRQ function.<br>When bit[14]=0, this bit is stick, default.<br>When bit[14]=1, this bit is non-sticky                                                                                                                                                                                                                                                      |
| 6    | <b>serirqmode</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Continuous mode<br>1: Active (quiet) mode                                                                                                                                                                                                                                                                                                                                                             |
| 5:2  | <b>numserirqbits</b> . Read-write. Reset: 0h.<br><b>Description:</b> Total number of serial IRQ's = 17 + NumSerIrqbits<br>0: 17 serial IRQ's (15 IRQ, <a href="#">SMI</a> #, + IOCHK#)<br>1: 18 serial IRQ's (15 IRQ, SMI#, IOCHK#, INTA#)<br>...<br>15: 32 serial IRQ's<br>The serial IRQ can support 15 IRQ#, SMI#, IOCHK#, INTA#, INTB#, INTC#, and INTD#.<br>When serial SMI# is used, BIOS will need to check SIO (or device that generates serial SMI#) for status. |
| 1:0  | <b>numstartbits</b> . Read-write. Reset: 0h.<br><b>Description:</b> This field defines the number of clocks in the start frame.<br>Start Frame Width = 4 + 2 * NumStartBits                                                                                                                                                                                                                                                                                               |

**PMx00000056 (FCH::PM::RTCCONTROL)**

Read-write.

\_aliasHOSTLEGACY; PMx00000056; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | Reserved.                                                                                                                                                                                                                                                                                                                       |
| 14   | <b>extrartccmosen</b> . Read-write. Reset: 0. When it is 1, SW can access the extra 16 bytes of RTC CMOS RAM.                                                                                                                                                                                                                   |
| 13   | <b>altcmosmapen</b> . Read-write. Reset: 0. When enabled, bank 1 of CMOS RAM is changed. Index 00:0D will still return the time and alarm settings. Index 0E:7F will return the absolute offset 8E:FF.                                                                                                                          |
| 12   | <b>centuryen</b> . Read-write. Reset: 1. Enable RTC Century support.                                                                                                                                                                                                                                                            |
| 11   | <b>mask_rtc_clk_out</b> . Read-write. Reset: 0. Set to 1 to disable RtcClk output.                                                                                                                                                                                                                                              |
| 10   | <b>rtcclkdrive</b> . Read-write. Reset: 1.<br><b>Description:</b> 0: HIGHDRIVE tied low for RtcClkOut pad<br>1: HIGHDRIVE tied high for RtcClkOut pad                                                                                                                                                                           |
| 9:7  | Reserved.                                                                                                                                                                                                                                                                                                                       |
| 6    | <b>rtc_test_en</b> . Read-write. Reset: 0.<br><b>Description:</b> This bit is for simulation only. Please do not change its value in real silicon.<br>0: Normal speed for prescaler<br>1: We replace 8kHz by OSC clock in prescaler. As the result, the prescaler 8kHz clock and all the clock divided from it can be speed up. |
| 5    | Reserved.                                                                                                                                                                                                                                                                                                                       |
| 4    | <b>rtcprotectc0_cf</b> . Read-write. Reset: 0. When set, RTC RAM index C0:CFh will be locked from read/write. This bit can only be written once.                                                                                                                                                                                |
| 3    | <b>rtcprotectd0_df</b> . Read-write. Reset: 0. When set, RTC RAM index D0:DFh will be locked from read/write. This bit can only be written once.                                                                                                                                                                                |
| 2    | <b>rtcprotecte0_ef</b> . Read-write. Reset: 0. When set, RTC RAM index E0:EFh will be locked from read/write. This bit can only be written once.                                                                                                                                                                                |
| 1    | <b>rtcprotectf0_ff</b> . Read-write. Reset: 0. When set, RTC RAM index F0:FFh will be locked from read/write. This bit can only be written once.                                                                                                                                                                                |
| 0    | <b>rt_cprotect38_3f</b> . Read-write. Reset: 0. When set, RTC RAM index 38:3Fh will be locked from read/write. This bit can only be written once.                                                                                                                                                                               |

**PMx00000058 (FCH::PM::VRT\_T1)**

Read-write. Reset: 01h.

\_aliasHOSTLEGACY; PMx00000058; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>vrt_t1</b> . Read-write. Reset: 01h. To conserve power, the RTC battery is sampled periodically for checking its state of health. VRT_T1 and VRT_T2 make up the interval of the checking. When VRT_Enable is high, the battery is being sampled. When VRT_enable is low, the battery is not being sampled. This register defines the time of VRT enable being high for RTC battery monitor circuit, in milliseconds. |

**PMx00000059 (FCH::PM::VRT\_T2)**

Read-write. Reset: FFh.

\_aliasHOSTLEGACY; PMx00000059; PM=FED8\_0300h

| Bits | Description                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>vrt_t2</b> . Read-write. Reset: FFh. This register defines the time of VRT enable being low for the RTC battery monitor circuit, in 4 ms increments. |

**PMx0000005A (FCH::PM::INTRUDERCONTROL)**

Read-write. Reset: 11h.

\_aliasHOSTLEGACY; PMx0000005A; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | Reserved.                                                                                                                                                                                                                                                                                                                                                                       |
| 6    | <b>cmoserasests.</b> Read-write. Reset: 0. Indicate that a CMOS Erase has been occurred.                                                                                                                                                                                                                                                                                        |
| 5    | <b>cmoserasectl.</b> Read-write. Reset: 0. Write to 1 to clear CMOS Erase status.                                                                                                                                                                                                                                                                                               |
| 4    | <b>cmoserasedis.</b> Read-write. Reset: 1. Set to 1 to disable CMOS Erase.                                                                                                                                                                                                                                                                                                      |
| 3    | Reserved.                                                                                                                                                                                                                                                                                                                                                                       |
| 2    | <b>intruderalertsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read Only. The status bit will be set to 1 if an Intruder alter event (pad-IntruderAlert#=0) has occurred .<br>Software need to set bit[1] to clear this status bit.<br>set bit[0]=1, bit[1]=0<br>After Trigger:<br>read bit[2] for status<br>set bit[0]=1, bit[1]=1<br>wait a short time<br>set bit[1]=0 |
| 1    | <b>intruderalertclr.</b> Read-write. Reset: 0.<br><b>Description:</b> Write 1 to clear the IntruderAlert status bit (bit[2]: IntruderAlertSts).<br>Software need to write this bit 0 to enable pad-IntruderAlert#=0 to set bit[2]: IntruderAlertSts                                                                                                                             |
| 0    | <b>intruderalertpuen.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: enable internal pullup (>200K). Please noted that pullup is enabled when both bit[0]=1 and bit[2]=0, if pad-IntruderAlert# is '0' while bit[1]=0, bit[2] will be set, and internal pullup will be disabled to save power.<br>0: disable internal pullup, when bit[0]=0, internal pullup is disabled.  |

**PMx0000005B (FCH::PM::RTCShadow)**

Read-write.

RTC\_AIE is defined at RTC\_Reg:0Bh[bit5].

Note1: These four bits(bit[7:4]) don't have any default value. After power on, their values are undertermined. SW has to program PwrFailShadow to give them values.

\_aliasHOSTLEGACY; PMx0000005B; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>forcepwron.</b> Read-write. Reset: X.<br><b>Description:</b> 0: If RTC AIE = 1, will wakeup when RTC alarm fires after a power failure/resume. (See Note)<br>1: If RTC AIE =1, will force power on after power resumes regardless of Bit[5:4] setting. (See Note)                                                                                                         |
| 6    | <b>powerstate.</b> Read-write. Reset: X.<br><b>Description:</b> Power state indicator.<br>0: Off<br>1: On                                                                                                                                                                                                                                                                    |
| 5:4  | <b>pwrfailoption.</b> Read-write. Reset: 0h.<br><b>Description:</b> These two bits will determine how system should resume after a power failure.<br>00: Always offalways power off after power resumes<br>01: Always onalways power on after power resumes<br>10: Always offalways power off after power resumes<br>11: Use previousresume to same setting when power fails |
| 3:0  | <b>pwrfailshadow.</b> Read-write. Reset: 0h. Writing to these four bits will set the value onto bits [7:4]. Software should always set bit 2 = 1.                                                                                                                                                                                                                            |



**PMx0000005C (FCH::PM::LLBCNTRL)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx0000005C; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:3  | Reserved.                                                                                                                                                                                                  |
| 2    | <b>allowwakes3en.</b> Read-write. Reset: 0. Set to 1 to allow LLB# as wake event in S3.                                                                                                                    |
| 1    | <b>useaswakeen.</b> Read-write. Reset: 0. Set to 1 to treat LLB# as wake event.                                                                                                                            |
| 0    | <b>blockwakeen.</b> Read-write. Reset: 0. Set 1 to block wake event if LLB# is asserted. But if UseAsWakeEn and AllowWakeS3En are all set to 1, LLB# and other wake events can wake the system up from S3. |

**PMx0000005D (FCH::PM::RTCGPIO)**

Read-write.

\_aliasHOSTLEGACY; PMx0000005D; PM=FED8\_0300h

| Bits | Description                                                                                                                              |
|------|------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>ortclkpub.</b> Read-write. Reset: 0.<br><b>Description:</b> Pull up control for RTCCLK pad<br>0: Enable pull up<br>1: Disable pull up |
| 6:2  | Reserved.                                                                                                                                |
| 1    | <b>rtcirqgpioenb.</b> Read-write. Reset: 1. RtcIrq Gpio Output Enable                                                                    |
| 0    | <b>rtcirqgpioout.</b> Read-write. Reset: 0. RtcIrq Gpio Output                                                                           |

**PMx0000005E (FCH::PM::RTCEXTINDEX)**

Read-write.

\_aliasHOSTLEGACY; PMx0000005E; PM=FED8\_0300h

| Bits | Description                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>index.</b> Read-write. Reset: XXXXXXXXb. Specify the offset of RTC Extended Registers to be read/written from PM_REG:5Fh |

**PMx0000005F (FCH::PM::RTCEXTDATA)**

Read-write.

\_aliasHOSTLEGACY; PMx0000005F; PM=FED8\_0300h

| Bits | Description                                                                                  |
|------|----------------------------------------------------------------------------------------------|
| 7:0  | <b>data.</b> Read-write. Reset: XXXXXXXXb. Read data or write data of RTC Extended Registers |

**PMx00000060 (FCH::PM::ACPIPM1EVTBLK)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx00000060; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:2 | <b>acpipm1evtblk.</b> Read-write. Reset: 0000h. These bits define the least significant byte of the 16 bit I/O range base address of the ACPI power management Event Block. Bit 2 corresponds to Addr[2] and bit 7 corresponds to Addr[7]. |
| 1:0  | Reserved.                                                                                                                                                                                                                                  |

**PMx00000062 (FCH::PM::ACPIPM1CNTBLK)**

Read-write. Reset: 0400h.

\_aliasHOSTLEGACY; PMx00000062; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:1 | <b>acpipm1cntblk.</b> Read-write. Reset: 0200h. These bits define the least significant byte of the 16 bit I/O base address of the ACPI power management Control block. Bit 1 corresponds to Addr[1] and bit 7 corresponds to Addr[7]. |
| 0    | Reserved.                                                                                                                                                                                                                              |

**PMx00000064 (FCH::PM::ACPIPMTMRBLK)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx00000064; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                         |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:1 | <b>acpipmtmrblk.</b> Read-write. Reset: 0000h. These bits define the least significant byte of the 16 bit I/O base address of the ACPI power management Timer block. Bit 1 corresponds to Addr[1] and bit 7 corresponds to Addr[7]. |
| 0    | Reserved.                                                                                                                                                                                                                           |

**PMx00000066 (FCH::PM::P\_CNTBLK)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx00000066; PM=FED8\_0300h

| Bits | Description |
|------|-------------|
| 15:0 | Reserved.   |

**PMx00000068 (FCH::PM::ACPIGPE0BLK)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx00000068; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:2 | <b>acpigpe0blk.</b> Read-write. Reset: 0000h. These bits define the least significant byte of the 16 bit I/O base address of the ACPI power management General Purpose Event block. Bit 2 corresponds to Addr[2] and bit 7 corresponds to Addr[7]. Addr[1:0] are ignored because this register block is 4 byte long. |
| 1:0  | Reserved.                                                                                                                                                                                                                                                                                                            |

**PMx0000006A (FCH::PM::ACPISMICMD)**

Read-write. Reset: 00B0h.

\_aliasHOSTLEGACY; PMx0000006A; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>acpismicmd.</b> Read-write. Reset: 00B0h. These bits define the least significant byte of the 16 bit I/O base address of the ACPI <a href="#">SMI</a> Command block. Bit 0 corresponds to Addr[0] and bit 7 corresponds to Addr[7]. The address is required to be WORD-aligned (Addr[0]=0) |

**PMx0000006E (FCH::PM::ACPIPMACNTBLK)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx0000006E; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:0 | <b>acpipmacntblk.</b> Read-write. Reset: 0000h. These bits define the most significant byte of the 16 bit I/O base address. Bit 0 corresponds to Addr[8] and bit 7 corresponds to Addr[15]. |

**PMx00000070 (FCH::PM::MP1\_WARMRESETACK\_WLAN\_CLK\_REQ)**

Read-write. Reset: 3F3E\_0170h.

\_aliasHOSTLEGACY; PMx00000070; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>wlanpllrbtmtmr.</b> Read-write. Reset: 3Fh. Configure the PllLock assertion (0->1) time after PllRstB deasserts (0->1). Unit is 16usec.                                                                                                                                             |
| 23:16 | <b>wlanpllprstbtmtmr.</b> Read-write. Reset: 3Eh. Configure the PllRstB deassertion (0->1) time. Unit is 16usec.                                                                                                                                                                       |
| 15    | <b>wlanpllctrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable<br>1: Enable WLAN to Control CGPLL<br>When this bit is 1 and system in CLOCK STOP MODE/POWER SAVING MODE/S3/S4/S5, wlan_clk_req could control FCH_APU_PllRstB/FCH_APU_PllLock which go to CGPLL          |
| 14    | <b>wlan_pme_gate.</b> Read-write. Reset: 0. WLAN_FCH_WLAN_PME (= event[37]) and WLAN_FCH_BT_PME (= event[38]) gate. Set to 1 will block these PMEs to generate event.                                                                                                                  |
| 13:3  | <b>wdt_warmresetack_tmr.</b> Read-write. Reset: 02Eh.<br><b>Description:</b> Time out value of MP1_Watchdog timer, this time is using 16.2us Osc clock to do increment.<br>default=0x2E<br>After append 2'b11, the Timer is 187.<br>(16.2us*187=3.029ms),<br>max=16.2us*8191=132.694ms |
| 2     | <b>en_slv_cpurst.</b> Read-write. Reset: 0. when set to 1, Master <a href="#">FCH</a> will do reset when Slave FCH asserts CpuRstB=0. Slave FCH will do Warm Reset for sync_flood.                                                                                                     |
| 1     | <b>en_wdt_warmresetack.</b> Read-write. Reset: 0. when bit[0]=1, set to 1 to enable MP1_Watchdog Timer for WarmResetAck handshake, if MP1_WarmResetAck rising edge not detected before MP1_Watchdog Timer time out, <a href="#">Cold reset</a> will be generated.                      |
| 0     | <b>en_mp1_warmresetack.</b> Read-write. Reset: 0. set to 1 to enable MP1_WarmResetAck handshake, all <a href="#">FCH</a> controlled Devices Reset, except PLL and APU, will be blocked till MP1_WarmResetAck rising edge or MP1_Watchdog Timer time out.                               |

**PMx00000074 (FCH::PM::ACPICONFIG)**

Read-write. Reset: 1000\_0140h.

\_aliasHOSTLEGACY; PMx00000074; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>delay_gpp_off_time.</b> Read-write. Reset: 0.<br><b>Description:</b> When BP_PWR_GOOD is used as RESET button, once pressed, PCI GPP clock will be stopped right away while PciRST# will be active (=0) some time later, this bit is used to meet customer's request to have PCI GPP clock running after PciRST# active. This bit is reset by RsmRstB only.<br>1: PCI BPP clock keep running for some time after PciRST#=0<br>0: PCI BPP clock stopped right away when BP_PWR_GOOD=0 |
| 30   | <b>use_gatedalinkclk.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Some logics in LPC are using free-run AlinkClk, same as before.<br>1: Some logics in LPC are using gated AlinkClk to save power.                                                                                                                                                                                                                                                                              |
| 29   | <b>rtcwakealarm.</b> Read-write. Reset: 0. Set to 1 to only rtc alarm to wake up the system.                                                                                                                                                                                                                                                                                                                                                                                            |
| 28   | <b>pcieeventmap.</b> Read-write. Reset: 1. Set to 1 to route pme message from APU to gevent 24, Hotplug message from APU to gevent 7..                                                                                                                                                                                                                                                                                                                                                  |
| 27   | <b>wakepinasgevent.</b> Read-write. Reset: 0. Set to 1 to treat Wake# pin as Gevent input.                                                                                                                                                                                                                                                                                                                                                                                              |
| 26   | <b>pcie_wak_intr_dis.</b> Read-write. Reset: 0. Set to 1 to disable interrupt from Pcie_wak_sts                                                                                                                                                                                                                                                                                                                                                                                         |
| 25   | <b>pcie_wak_mask.</b> Read-write. Reset: 0. Set to 1 to disable PCIE_WAK_STS and PCIE_WAK_DIS function. This is used for supporting ACPI 3.0 specification. If ACPI 3.0 is not supported, this bit should be left as 1                                                                                                                                                                                                                                                                  |
| 24   | <b>pcienative.</b> Read-write. Reset: 0. Setting to 1 will block <a href="#">PCIe</a> GPP PME message and HotPlug message from generating SCI. This is used for supporting ACPI 3.0 specification. If ACPI 3.0 is not supported, this bit should be left as 0                                                                                                                                                                                                                           |
| 23   | <b>rst_usb_s5.</b> Read-write. Reset: 0. Write '1' to make Cpl_VDDCR_S5_RESETEn=0 to reset USB,MP2, etc. Need to write '0' to deassert Cpl_VDDCR_S5_RESETEn=1. This bit is not for normal operation, it is to be used for recovery when something went wrong.                                                                                                                                                                                                                           |
| 22   | <b>usersmureset.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: fch_rsmu_hard_resestb is not used to reset fch_rsmu, SDP, AB.<br>1: fch_rsmu_hard_resestb is used to reset fch_rsmu, SDP, AB.                                                                                                                                                                                                                                                                                      |
| 21   | <b>mask_usb_s5_rst.</b> Read-write. Reset: 0. Set to '1' to mask oDevAllRstB to gen Cpl_VDDCR_S5_RESETEn=0 to reset USB/MP2/..., default=0                                                                                                                                                                                                                                                                                                                                              |
| 20   | <b>en_df_intrwake.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: disable FCH_DF_IntrWake in s5_misc.vpp<br>1: enable FCH_DF_IntrWake in s5_misc.vpp                                                                                                                                                                                                                                                                                                                               |
| 19   | <b>force_spiusepin_0.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SPI shared pad cannot be used as GPIO<br>1: SPI shared pad can be used as GPIO if selected by IOMUX                                                                                                                                                                                                                                                                                                           |
| 18   | <b>en_sync_flood.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: disable SYNC_FLOOD<br>1: enable SYNC_FLOOD                                                                                                                                                                                                                                                                                                                                                                        |
| 17   | <b>en_shutdown_msg.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: disable SHUTDOWN message<br>1: enable SHUTDOWN message                                                                                                                                                                                                                                                                                                                                                          |
| 16   | <b>sw_s5pwrmux.</b> Read-write. Reset: 0.<br><b>Description:</b> When PM_regx74[15]=0, this bit control oS5PwrMux to USB. This bit will be reset to 0 by PllLock=0, thus it will be reset to 0 on any S5/S3 entry.<br>0: USB is powered by S5-domain VDDCR_SOC_S5 (bypass to S5 rail mode)<br>1: USB is powered by CLDO Vout (regulation mode, to be used in S0)                                                                                                                        |
| 15   | <b>sw_s5pwrmux_override_n.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                    |

|    |                                                                                                                                                                                                                                                                                                                              |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Description:</b> 0: use PM_regx74[16] to control oS5PwrMux to USB.<br>1: use iPlLlLock to control oS5PwrMux to USB.<br>When iPlLlLock is unlock (=0), USB is powered by S5-domain VDDCR_SOC_S5 (bypass to S5 rail mode)<br>When iPlLlLock is locked (=1), USB is powered by CLDO Vout (regulation mode, to be used in S0) |
| 14 | <b>sel_smu_thermtrip.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: use synced SMU_THERmTrip to drive pad BP_THERMTRIP_L<br>1: use SMU_THERmTrip directly to drive pad BP_THERMTRIP_L                                                                                                                                  |
| 13 | <b>sel_pwrgd_pad.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SPI_FBCLK_mux will use debounced PwrGood PAD input (~10ms) as ResetB to avoid glitch.<br>1: SPI_FBCLK_mux will use PwrGood PAD input as ResetB                                                                                                         |
| 12 | <b>lpc_rst_dis.</b> Read-write. Reset: 0.<br><b>Description:</b> assign IO_LPC_RST_L_IN_iA = LPC_RESETB & !LPC_RST_Dis<br>It is used to gate LPC_RST_L. default eanbled.                                                                                                                                                     |
| 11 | <b>s5_lpcpinmode.</b> Read-write. Reset: 0. Co-work with above bit in the above logic.                                                                                                                                                                                                                                       |
| 10 | <b>s5_lpcpinmodesel.</b> Read-write. Reset: 0. Not used                                                                                                                                                                                                                                                                      |
| 9  | <b>acpireducedhwen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Acpi Fixed register interface is enabled<br>1: the decoding of Acpi Fixed registers and Sci are disabled. In addition wake function from pm1a is disabled as well.                                                                                   |
| 8  | <b>pwnbtnen.</b> Read-write. Reset: 1. Set to 1 to enable power button support in pm1a block                                                                                                                                                                                                                                 |
| 7  | <b>bios_rls.</b> Read-write. Reset: 0. Set to1 to generate SCI. Read always return 0                                                                                                                                                                                                                                         |
| 6  | <b>maskarbdis.</b> Read-write. Reset: 1. Set to1 to disable ArbDis function in the ACPI register. ArbDis is not really used, but it still needs to be accessible by OS.                                                                                                                                                      |
| 5  | Reserved.                                                                                                                                                                                                                                                                                                                    |
| 4  | <b>tmr_en_en.</b> Read-write. Reset: 0. Set to 1 enable TMR_EN function in the standard ACPI register.                                                                                                                                                                                                                       |
| 3  | <b>slpbtn_en_en.</b> Read-write. Reset: 0. Set to1 to enable SLPBTN_EN function in the standard ACPI register.                                                                                                                                                                                                               |
| 2  | <b>rtc_en_en.</b> Read-write. Reset: 0. Set to1 to enable RTC_EN function in the standard ACPI register.                                                                                                                                                                                                                     |
| 1  | <b>gbl_en_en.</b> Read-write. Reset: 0. Set to 1 to enable GBL function in the standard ACPI PmControl register.                                                                                                                                                                                                             |
| 0  | <b>decen_acpi.</b> Read-write. Reset: 0. Set to 1 to enable decoding of the standard ACPI registers                                                                                                                                                                                                                          |

**PMx00000078 (FCH::PM::WAKEIOADDR)**

Read-write. Reset: 0000\_FFFFh.

\_aliasHOSTLEGACY; PMx00000078; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:17 | Reserved.                                                                                                                                                                                                                   |
| 16    | <b>mp1_wdt_cnt_clr.</b> Read-write. Reset: 0.<br><b>Description:</b> Write one clear.<br>Write this bit to one, it will clear the MP1_WDT_cnt in PwrRst.                                                                    |
| 15:0  | <b>wakeiobaseaddress.</b> Read-write. Reset: FFFFh. The register specifies the wake I/O address. Any I/O write to the I/O address can cause APU to wake from C state. This is an obsolete function that is not used anymore |

**PMx0000007E (FCH::PM::CSTATEEN)**

Read-write. Reset: 0060h.

\_aliasHOSTLEGACY; PMx0000007E; PM=FED8\_0300h

| Bits | Description                                                                               |
|------|-------------------------------------------------------------------------------------------|
| 15:6 | Reserved.                                                                                 |
| 5    | <b>k8c1etoc3en.</b> Read-write. Reset: 1. Set to 1 to put APU into C3 state in C1e state. |
| 4    | <b>k8c1etoc2en.</b> Read-write. Reset: 0. Set to 1 to put APU into C2 state in C1e state. |
| 3:0  | Reserved.                                                                                 |

**PMx00000080 (FCH::PM::BREAKEVENT)**

Read-write. Reset: 0026\_0000h.

\_aliasHOSTLEGACY; PMx00000080; PM=FED8\_0300h

| Bits  | Description                                                             |
|-------|-------------------------------------------------------------------------|
| 31:24 | Reserved.                                                               |
| 23:16 | <b>scratchbit16_23</b> . Read-write. Reset: 26h.                        |
| 15:5  | Reserved.                                                               |
| 4     | <b>scratchbit4</b> . Read-write. Reset: 0. Scratch bit used by Software |
| 3     | <b>scratchbit3</b> . Read-write. Reset: 0. Scratch bit used by Software |
| 2     | <b>scratchbit2</b> . Read-write. Reset: 0. Scratch bit used by Software |
| 1     | Reserved.                                                               |
| 0     | <b>scratchbit0</b> . Read-write. Reset: 0. Scratch bit used by Software |

**PMx00000088 (FCH::PM::CSTATECONTROL)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx00000088; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:11 | Reserved.                                                                                                                                                                                                                                                                                                  |
| 10    | <b>sw_sync_flood</b> . Read-write. Reset: 0. When PMx8c[27]=1, write this bit =1 will generate SW_sync_flood. Since this register is non-sticky, this bit will be cleared by PciRstB, thus software no need to write 0 after write 1 should this SW_sync_flood is enabled to generate Reset (PMx74[18]=1). |
| 9:6   | Reserved.                                                                                                                                                                                                                                                                                                  |
| 5     | <b>slp_en</b> . Read-write. Reset: 0. Enable LDTSTOP# as an output                                                                                                                                                                                                                                         |
| 4:3   | Reserved.                                                                                                                                                                                                                                                                                                  |
| 2     | <b>dlyslpen</b> . Read-write. Reset: 0. Set to 1 to delay recognition of STPGNT# until there is no pending read in AB                                                                                                                                                                                      |
| 1:0   | Reserved.                                                                                                                                                                                                                                                                                                  |

**PMx0000008C (FCH::PM::BTTNSCICONTROL)**

Read-write. Reset: 00FF\_FFFFh.

\_aliasHOSTLEGACY; PMx0000008C; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>btn_diswake_pressins0.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: no wakeup when button pressed in S0, and software put system in S3 while button still pressed and then released in less than 4s<br>0: default, when button is pressed and released in less than 4s, if system is in S3/S5, system will always do wakeup. |
| 30   | <b>sci_btn_fall_edge.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: set PWRBTN_STS_reg to gen SCI at button falling edge (after 16ms debounce)<br>0: default, no SCI at button falling edge                                                                                                                                      |
| 29   | <b>wlan_fp6_iso_en.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: WLAN ISO off, S5 WLAN <a href="#">PMA</a> to S5 WLAN digital signals are disconnected.<br>1: WLAN ISO on, S5 WLAN PMA to S5 WLAN digital signals are connected.                                                                                                |
| 28   | <b>smireg_use_s5rst.</b> Read-write. Reset: 0.<br><b>Description:</b> This bit is reset by RsmRstB only.<br>0: some registers inside <a href="#">SMI</a> (acpi_event) module are reset by PciRstB<br>1: some registers inside SMI (acpi_event) module are reset by S5RstB                                                              |
| 27   | <b>sw_sync_flood_enable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: PMx88[10] cannot generate SW_sync_flood.<br>1: PMx88[10] can generate SW_sync_flood.                                                                                                                                                                      |
| 26   | <b>dis_sci_wakeup_power_saving_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: sci_wakeup can wake Power saving mode<br>1: sci_wakeup cannot wake Power saving mode                                                                                                                                                          |
| 25   | <b>dis_pic_irq_8_power_saving_mode.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: pic_irq_8 can wakeup Power saving mode<br>1: pic_irq_8 cannot wakeup Power saving mode                                                                                                                                                         |
| 24   | <b>acpialarm_nchk_tmren.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: ACPIAlarm will check TMR_EN =1<br>1: ACPIAlarm will not check TMR_EN                                                                                                                                                                                      |
| 23:0 | <b>espi1devintrmask.</b> Read-write. Reset: FF_FFFFh.<br><b>Description:</b> SW can set these bits to mask of eSPI_1 Device IRQ23~0.<br>1: Mask off the interrupt<br>0: No mask                                                                                                                                                        |

**PMx00000090 (FCH::PM::SPIPADPUPD)**

Read-write. Reset: 88A4\_0959h.

\_aliasHOSTLEGACY; PMx00000090; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>boot_timer_cold_rst.</b> Read-write. Reset: 1. When bit[31]=0, Boot Timer Reset is doing warm reset (toggle Cpu_ResetB only). By setting bit[31]=1, Boot Timer Reset will do cold reset (toggle both Cpu_ResetB and Cpu_PwrGood).                                                                                                                                                                                                                                                                                                                                                              |
| 30    | <b>dis_spi_pupd_sw.</b> Read-write. Reset: 0.<br><b>Description:</b> Set to 1 to disable all SPI PAD pullup/pulldown defined in this register. When we are in S3/S5, external S0VDD will be power-off, though SPI PADs are in S5, depends on board design, S0VDD may be connected to SPI PAD. To enable SPI ROM sharing with external <a href="#">EC</a> , all pullup/pulldown need be disabled so external EC can drive SPI signals to access SPI ROM. Software can set bit[30]=1 without change bit [11:0] value before getting into S3/S5, after exit S3/S5, software need to reset bit[30]=0. |
| 29    | <b>dis_spi_pupd_hw.</b> Read-write. Reset: 0. If set to 1, when system is in Power saving mode or S3 or S5, all SPI PAD pullup/pulldown defined in this register will be disabled.                                                                                                                                                                                                                                                                                                                                                                                                                |
| 28    | <b>spi2_cs1_l_pulldown.</b> Read-write. Reset: 0. control pad SPI2_CS1 pulldown, 1: Enable 0: disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 27    | <b>spi2_cs1_l_pullup.</b> Read-write. Reset: 1. control pad SPI2_CS1 pullup, 1: Enable 0: disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 26:24 | <b>st_sx.</b> Read-write. Reset: 0h.<br><b>Description:</b> Read Only, indicate the current S-state value, for debug purpose only, as in clock stop mode/Power saving mode/S3/S4/S5, Host cannot read register value,<br>0x0: S0 state<br>0x1: clock slow down state<br>0x2: clock stop state<br>0x3: S3 state<br>0x4: Power saving mode state<br>0x5: S4/S5 state<br>0x6, 0x7: reserved                                                                                                                                                                                                          |
| 23    | <b>spi1_hold_l_pulldown.</b> Read-write. Reset: 1. control pad SPI1_HOLD_L pulldown, 1: Enable 0: disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 22    | <b>spi1_hold_l_pullup.</b> Read-write. Reset: 0. control pad SPI1_HOLD_L pullup, 1: Enable 0: disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 21    | <b>spi1_wp_l_pulldown.</b> Read-write. Reset: 1. control pad SPI1_WP_L pulldown, 1: Enable 0: disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 20    | <b>spi1_wp_l_pullup.</b> Read-write. Reset: 0. control pad SPI1_WP_L pullup, 1: Enable 0: disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 19    | <b>spi1_cs1_l_pulldown.</b> Read-write. Reset: 0. control pad SPI1_CS1 pulldown, 1: Enable 0: disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 18    | <b>spi1_cs1_l_pullup.</b> Read-write. Reset: 1. control pad SPI1_CS1 pullup, 1: Enable 0: disable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 17    | <b>bt_gpio_1p8v_receive.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: GPIO7/GPIO8 are 3.3V receiver<br>1: GPIO7/GPIO8 are 1.8V receiver, when they used as BT receiver                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 16    | <b>espics_use_data2.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: eSPI device is using SPI_DATA1<br>1: eSPI device is using SPI_DATA2                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 15    | <b>tpmcs_use_data2.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SPI device use TPM_CS_L is using SPI_DATA1<br>1: SPI device use TPM_CS_L is using SPI_DATA2                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 14    | <b>spics3_use_data2.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SPI device use SPI_CS3_L (spi_regx1D[1:0]=10b) is using SPI_DATA1<br>1: SPI device use SPI_CS3_L (spi_regx1D[1:0]=10b) is using SPI_DATA2                                                                                                                                                                                                                                                                                                                                                                                |
| 13    | <b>spics2_use_data2.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SPI device use SPI_CS2_L (spi_regx1D[1:0]=01b) is using SPI_DATA1<br>1: SPI device use SPI_CS2_L (spi_regx1D[1:0]=01b) is using SPI_DATA2                                                                                                                                                                                                                                                                                                                                                                                |
| 12    | <b>spics1_use_data2.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



|    |                                                                                                                                                                  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | <b>Description:</b> 0: SPI device use SPI_CS1_L (spi_regx1D[1:0]=00b) is using SPI_DATA1<br>1: SPI device use SPI_CS1_L (spi_regx1D[1:0]=00b) is using SPI_DATA2 |
| 11 | <b>spi_clk_pulldown.</b> Read-write. Reset: 1. control pad SPI_CLK pulldown, 1: Enable 0: disable                                                                |
| 10 | <b>spi_clk_pullup.</b> Read-write. Reset: 0. control pad SPI_CLK pullup, 1: Enable 0: disable                                                                    |
| 9  | <b>spi_cs1_pulldown.</b> Read-write. Reset: 0. control pad SPI_CS1 pulldown, 1: Enable 0: disable                                                                |
| 8  | <b>spi_cs1_pullup.</b> Read-write. Reset: 1. control pad SPI_CS1 pullup, 1: Enable 0: disable                                                                    |
| 7  | <b>spi_hold_l_pulldown.</b> Read-write. Reset: 0. control pad SPI_HOLD_L pulldown, 1: Enable 0: disable                                                          |
| 6  | <b>spi_hold_l_pullup.</b> Read-write. Reset: 1. control pad SPI_HOLD_L pullup, 1: Enable 0: disable                                                              |
| 5  | <b>spi_wp_l_pulldown.</b> Read-write. Reset: 0. control pad SPI_WP_L pulldown, 1: Enable 0: disable                                                              |
| 4  | <b>spi_wp_l_pullup.</b> Read-write. Reset: 1. control pad SPI_WP_L pullup, 1: Enable 0: disable                                                                  |
| 3  | <b>spi_do_pulldown.</b> Read-write. Reset: 1. control pad SPI_DO pulldown, 1: Enable 0: disable                                                                  |
| 2  | <b>spi_do_pullup.</b> Read-write. Reset: 0. control pad SPI_DO pullup, 1: Enable 0: disable                                                                      |
| 1  | <b>spi_di_pulldown.</b> Read-write. Reset: 0. control pad SPI_DI pulldown, 1: Enable 0: disable                                                                  |
| 0  | <b>spi_di_pullup.</b> Read-write. Reset: 1. control pad SPI_DI pullup, 1: Enable 0: disable                                                                      |

**PMx00000094 (FCH::PM::CSTATETIMING0)**

Read-write. Reset: 0010\_0000h.

\_aliasHOSTLEGACY; PMx00000094; PM=FED8\_0300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**PMx000000A0 (FCH::PM::MESSAGECSTATE)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx000000A0; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>checkcoreiddis.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Ignore Core Id check when MultiCoreEnbit set 1.<br>0: Enable Core Id check when MultiCoreEnbit set 1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 30:21 | <b>pmxa4reserved.</b> Read-write. Reset: 000h.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 20:16 | <b>tmrseloverride.</b> Read-write. Reset: 00h.<br><b>Description:</b> This is to be used with bit TimerTickChgMsgEn. In case <a href="#">FCH</a> auto-timer detection logic is not functioning properly, one can use these bits to override the logic and force the logic to monitor the specific timer.<br>Bit 0 When set, use HPET<br>Bit 1 When set, use RTC<br>Bit 2 if HPET is selected, setting bit 2 will force the logic to monitor HPET timer 0.<br>Bit 3 if HPET is selected, setting bit 3 will force the logic to monitor HPET timer 1.<br>Bit 4 if HPET is selected, setting bit 4 will force the logic to monitor HPET timer2. |
| 15:11 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 10    | <b>clkintrtagen.</b> Read-write. Reset: 0. When enabled, <a href="#">FCH</a> will mark the periodic timer interrupt.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 9:2   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 1     | <b>timertickchgmsgen.</b> Read-write. Reset: 0. When enabled, <a href="#">FCH</a> will send a message to CPU indicating the latest periodic timer interval. FCH will automatically determine which timer (PIT, RTC, or HPET) is being used.                                                                                                                                                                                                                                                                                                                                                                                                  |
| 0     | <b>obattmodechgmsgen.</b> Read-write. Reset: 0. When enabled, <a href="#">FCH</a> will automatically send a message to CPU indicating the power mode (AC vs battery). In addition, every time it is changed, FCH will generate a message to indicate the update.                                                                                                                                                                                                                                                                                                                                                                             |

**PMx000000A4 (FCH::PM::MISCDEBUG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx000000A4; PM=FED8\_0300h

| Bits | Description                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------|
| 31:3 | Reserved.                                                                                                                       |
| 2    | <b>blockeventsmi</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Block GEVENT <a href="#">SMI</a><br>0: Unblock GEVEN SMI |
| 1    | <b>blocksci</b> . Read-write. Reset: 0.<br><b>Description:</b> 1: Block SCI<br>0: Unblock SCI                                   |
| 0    | Reserved.                                                                                                                       |

**PMx000000A8 (FCH::PM::VIRTUALWIRE)**

Read-write. Reset: 0023\_0085h.

\_aliasHOSTLEGACY; PMx000000A8; PM=FED8\_0300h

| Bits  | Description                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------|
| 31:27 | Reserved.                                                                                                  |
| 26:16 | <b>otherpwrrsttmr</b> . Read-write. Reset: 023h. Parametised the reset timing wait time for other devices. |
| 15:11 | Reserved.                                                                                                  |
| 10:0  | <b>pg1pwrrsttmr</b> . Read-write. Reset: 085h. Parametised the reset timing wait time for devices in PG1.  |

**PMx000000B0 (FCH::PM::DEFERTIMETICK\_OBFFENABLE)**

Read-write.

\_aliasHOSTLEGACY; PMx000000B0; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>sw_obff_override.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: CpuActive/MemActive from Data Fabric are used for OBFF.<br>1: bit[30:29] are used for OBFF.                                                                                                                                                                                                                                                         |
| 30    | <b>sw_obff_cpuactive.</b> Read-write. Reset: 0. When bit[31]=Sw_obff_override=1, this bit will override CpuActive from Data Fabric to control OBFF operation when bit[11]=OBFFEn=1                                                                                                                                                                                                                                           |
| 29    | <b>sw_obff_memactive.</b> Read-write. Reset: 0. When bit[31]=Sw_obff_override=1, this bit will override MemActive from Data Fabric to control OBFF operation when bit[11]=OBFFEn=1                                                                                                                                                                                                                                           |
| 28    | <b>long10usdelay.</b> Read-write. Reset: 0.<br><b>Description:</b> Spec requires ACTIVE/OBFF -> IDLE to wait at least 10us, for other transition ACTIVE/IDLE -> OBFF or OBFF/IDLE -> ACTIVE, the delay is controlled by this bit.<br>0: use 3x400ns (400ns controlled by bit[6:2])<br>1: use 10us (10us controlled by bit[21:12])                                                                                            |
| 27    | <b>cpuactive.</b> Read-write. Reset: X.<br><b>Description:</b> Read Only.<br>State of Internal signal CpuActive that from Data Fabric                                                                                                                                                                                                                                                                                        |
| 26    | <b>memactive.</b> Read-write. Reset: X.<br><b>Description:</b> Read Only.<br>State of Internal signal MemActive that from Data Fabric.                                                                                                                                                                                                                                                                                       |
| 25:24 | <b>obff_state.</b> Read-write. Reset: 0h.<br><b>Description:</b> Read Only.<br>Internal OBFF state machine states.<br>00: IDLE (when bit[11]=OBFFEn=0, OBFF_state is IDLE)<br>01: OBFF<br>11: ACTIVE                                                                                                                                                                                                                         |
| 23    | <b>obff_transition_done.</b> Read-write. Reset: 1.<br><b>Description:</b> Read Only.<br>1: OBFF transition is done<br>0: OBFF transition is going on.                                                                                                                                                                                                                                                                        |
| 22    | <b>en_obff_blk_wake.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: input of pad_WAKE_L will not be blocked from generating event when OBFF state machine output pattern.<br>1: input of pad_WAKE_L will be blocked from generating event when OBFF state machine output pattern of "IDLE/ACTIVE -> OBFF" or "IDLE/OBFF -> ACTIVE", and will be enabled when OBFF state machine output pattern of "OBFF/ACTIVE -> IDLE) |
| 21:12 | <b>cnt10usper48m.</b> Read-write. Reset: 1E0h.<br><b>Description:</b> Number of 48MHz clock for 10us delay, used for spec requirement "when platform enter CPU Active or OBFF, platform should not return to Idle in less than 10us". Software can change this value to adjust the delay.<br>Default=0x1e0=480*(1sec/48M)=10us                                                                                               |
| 11    | <b>obffen.</b> Read-write. Reset: 0. Set to 1 to enable OBFF function which toggling WAKE#                                                                                                                                                                                                                                                                                                                                   |
| 10:8  | <b>defertimertickvalue.</b> Read-write. Reset: 0h.                                                                                                                                                                                                                                                                                                                                                                           |

|     |                                                                                                                                                                                                                                                                                           |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | <b>Description:</b> 000: No skipping<br>001: Skip 1 timer tick<br>010: Skip 2 timer ticks<br>011: Skip 3 timer ticks<br>100: Skip 4 timer ticks<br>101: Skip 5 timer ticks<br>110: Skip 6 timer ticks<br>111: Skip 7 timer ticks                                                          |
| 7   | <b>force_obff_blk_wake.</b> Read-write. Reset: 0. When set to 1, input of pad_WAKE_L will always be blocked from generating event.                                                                                                                                                        |
| 6:2 | <b>cnt400nsper48m.</b> Read-write. Reset: 14h.<br><b>Description:</b> Number of 48MHz (20.83ns) clock for 400ns delay, used for pulse width (both high and low pulse) of WAKE# protocol, software can change this value to adjust the WAKE# pulse width.<br>Default=0x14=20*20.83 =417ns. |
| 1   | <b>forcetmrticken.</b> Read-write. Reset: 0. If bit 0 is set along with this bit and <a href="#">FCH</a> has skipped a timer tick interrupt, FCH will immediately generate the timer tick interrupt upon C state exit                                                                     |
| 0   | <b>defertimerticken.</b> Read-write. Reset: 0. When set, <a href="#">FCH</a> will skip a number of timer tick interrupts based on the defined value in DeterTimeTickValue when CPU is in C state. When CPU is not in C state, FCH will not skip any timer tick interrupts.                |

**PMx000000B4 (FCH::PM::ACPIMISCDEBUG)**

Read-write. Reset: 0580\_0000h.

\_aliasHOSTLEGACY; PMx000000B4; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 29:24 | <b>tpreset1b.</b> Read-write. Reset: 05h.<br><b>Description:</b> Timing parameter used for S* -> S0 state transition. This determines the delay between CPU_STP# de-assertion and LPC_PD# de-assertion, in 8s increment with 8s uncertainty.                                                                                                                                                                                                                                           |
| 23    | <b>intrtoggleonldtstp.</b> Read-write. Reset: 1.<br><b>Description:</b> Configure the behavior of "DMA/Interrupt indicator":<br>When LdtStpB behaves as interrupt indicator, this bit configure the way it indicates interrupt:<br>0: The interrupt is signaled as a level signal.<br>1: The interrupt is signaled as a clock signal with 2us or 4us period depends on LDTSTPBTrSel.                                                                                                   |
| 22    | <b>ldtstpoutputdis.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure the behavior of "DMA/Interrupt indicator":<br>Disable LdtStpB output:<br>0: LdtStpB output enabled<br>1: LdtStpB output disabled                                                                                                                                                                                                                                                                        |
| 21    | <b>wakecstateinslp.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure the behavior of "DMA/Interrupt indicator":<br>0: Do not wake C State before entering S State.<br>1: Wake C State before entering S State.                                                                                                                                                                                                                                                               |
| 20    | <b>mergeallowldtstpwithldtstp.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure the behavior of "DMA/Interrupt indicator":<br>Change the definition of pin DMAACTIVE_L (ALLOWLDTSTP).<br>0: Drive DMAACTIVE_L to low when there is pending upstream request.<br>1: Output 2us/4us toggling on DMAACTIVE_L (ALLOWLDTSTP) when there is pending interrupt request and not in C0 state to DMAACTIVE_L otherwise drive DMAACTIVE_L to low when there is pending upstream request |
| 19    | <b>intronldtstpben.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure the behavior of "DMA/Interrupt indicator":<br>Change the definition of LdtStpB pin to "interrupt indicator".<br>0: LdtStpB behaves as LdtStpB<br>1: LdtStpB behaves as interrupt indicator                                                                                                                                                                                                              |
| 18    | <b>ldtstpbtmrsl.</b> Read-write. Reset: 0.<br><b>Description:</b> Configure the behavior of "DMA/Interrupt indicator":<br>Select 2us or 4us toggling on DMAACTIVE_L (ALLOWLDTSTP) or LDTSTPB_L according to Merge_Interrupt_Dma_Reg value.<br>0: 2us<br>1: 4us                                                                                                                                                                                                                         |
| 17    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 16    | <b>oallowldtstpasout.</b> Read-write. Reset: 0.<br><b>Description:</b> Control the input/output direction of pin AllowLdtStp.<br>0: (default) AllowLdtStp is input to <a href="#">FCH</a> driven by APU.<br>1: AllowLdtStp as output pin to APU to indicate FCH traffic activities.                                                                                                                                                                                                    |
| 15:1  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0     | <b>fid_protect_en.</b> Read-write. Reset: 0. Set to 1 to skip C-state transition when <a href="#">FID/VID</a> message is received concurrently.                                                                                                                                                                                                                                                                                                                                        |

**PMx000000B8 (FCH::PM::TPRESET2)**

Read-write. Reset: 88h.

\_aliasHOSTLEGACY; PMx000000B8; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:6  | <b>clkgatecntrl.</b> Read-write. Reset: 2h.<br><b>Description:</b> These two bits control whether SMBUS module will allow clock gating to the internal 66MHz core clock<br>00: Disable the clock gating function<br>01: Wait 16 clocks before allowing clock gating to the SMBUS module<br>10: Wait 64 clocks before allowing clock gating to the SMBUS module<br>11: Wait 256 clocks before allowing clock gating to the SMBUS module |
| 5:0  | <b>tpreset2.</b> Read-write. Reset: 08h. Timing parameter used for S* -> S0 state transitions. This register determines the LDTSTOP# deassertion delay in 8s increment with 8s uncertainty.                                                                                                                                                                                                                                            |

**PMx000000B9 (FCH::PM::LPCMISCDEBUG)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000B9; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                        |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | Reserved.                                                                                                                                                                                                                                                                                                                                          |
| 6    | <b>test_pwr_btn.</b> Read-write. Reset: 0.<br><b>Description:</b> This bit is used to speed up simulation<br>1'b1: debounce 16ms counter is 0x12<br>1'b0: debounce 16ms counter is 0x1ff                                                                                                                                                           |
| 5:4  | Reserved.                                                                                                                                                                                                                                                                                                                                          |
| 3    | <b>clkrundisable.</b> Read-write. Reset: 0. Legacy DMA and serial IRQ logic reside in this module and they are running on the 33MHz LPCCLK. Setting this bit will disable this module's ability to support CLKRUN# function from PCIBridge. In other words, when this bit is set, this module will prevent PCIBridge from stopping the 33MHz clock |
| 2    | <b>test_pm_tmr.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: normal PM timer function<br>1: PM timer function run in test mode                                                                                                                                                                                                              |
| 1    | <b>test_1ms.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: normal delay in S state timing<br>1: Speed up S state delay by 7 times                                                                                                                                                                                                            |
| 0    | <b>t32_64.</b> Read-write. Reset: 0.<br><b>Description:</b> Delay timer for S2:<br>0: 32 pci clk<br>1: 64 pci clk                                                                                                                                                                                                                                  |

**PMx000000BA (FCH::PM::S\_STATECONTROL)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx000000BA; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15   | <b>maskpmsgen.</b> Read-write. Reset: 0. When set (along with PmeMsgEn=1), PmeAck message coming from <a href="#">PCIe</a> device will be ignored and ACPI S state logic will solely use the timeout mechanism to sequence through the S3 state. This bit is used as an option to guard against multiple PmeAck messages coming from CNB and internal <a href="#">FCH</a> PCIe bridge so FCH S state logic will not sequence into S3 state prematurely. |
| 14   | <b>wakepinenable.</b> Read-write. Reset: 0. Set to 1 to enable wakeup from WAKE# pin.                                                                                                                                                                                                                                                                                                                                                                   |
| 13   | <b>agptimeadj.</b> Read-write. Reset: 0. If set to 1, S* -> S0 state transitions will use 1ms clock for timing sequence otherwise, 8s clock will be used.                                                                                                                                                                                                                                                                                               |
| 12:6 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 5    | <b>overrideshorttimemode.</b> Read-write. Reset: 0.<br><b>Description:</b> It is only available when ShortTimeMode is enabled.<br>When this bit is set high, ShortTimeMode will be disabled for Osc16Us_new.                                                                                                                                                                                                                                            |
| 4    | <b>pmmsgstrig.</b> Read-write. Reset: 0.<br><b>Description:</b> SW write this bit to trigger a PmeTurnOff sequence to NB.<br>Reading this bit returns the status of the PmeTurnOff sequence (1 means not done 0 means done)                                                                                                                                                                                                                             |
| 3    | <b>pmmsgen.</b> Read-write. Reset: 0. Set to 1 to enable PmeTurnOff/PmeMsgAck handshake.                                                                                                                                                                                                                                                                                                                                                                |
| 2    | <b>stpcklen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: disable STPCLK/STPGNT handshake in S state transition<br>1: Enable STPCLK/STPGNT handshake in S state transition                                                                                                                                                                                                                                                                       |
| 1    | <b>allowoffset.</b> Read-write. Reset: 0. Set to 1 to add extra delay for STPCLK. Only valid if AgpTimeAdj is set.                                                                                                                                                                                                                                                                                                                                      |
| 0    | <b>longslps3.</b> Read-write. Reset: 0. Set to 1 to extend SLP_S3# assertion to 1s minimum.                                                                                                                                                                                                                                                                                                                                                             |

**PMx000000BC (FCH::PM::THROTTLINGCONTROL)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000BC; PM=FED8\_0300h

| Bits | Description |
|------|-------------|
| 7:0  | Reserved.   |

**PMx000000BD (FCH::PM::RCINBRSTENABLE)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000BD; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                       |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>rcinbrstenable.</b> Read-write. Reset: 0.                                                                                                                                                                                      |
| 6    | <b>hostrstwarndisable.</b> Read-write. Reset: 0.<br><b>Description:</b> Disable early indicator of PLTRST_B<br>eSPI0 need to work in boot, so enable the two resets by default.<br>Set this bit to 1 will disable the two resets. |
| 5    | <b>rcinbrstenable1.</b> Read-write. Reset: 0.                                                                                                                                                                                     |
| 4    | <b>hostrstwarnenable_1.</b> Read-write. Reset: 0. Enable early indicator of PLTRST_B                                                                                                                                              |
| 3:0  | Reserved.                                                                                                                                                                                                                         |

**PMx000000BE (FCH::PM::RESETCONTROL1)**

Read-write. Reset: 72h.

\_aliasHOSTLEGACY; PMx000000BE; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                      |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>rsttocpupwrgden.</b> Read-write. Reset: 0. If set to 1, <a href="#">FCH</a> toggles CPUPG on every reset.                                                                                                                                                                     |
| 6    | <b>hwm_resetooption.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Hwm function(Pmio2 register block) is reset by RsmRst.<br>1: Hwm function(Pmio2 register block) is reset by PciRst.                                                                                     |
| 5    | <b>slp_typen_control.</b> Read-write. Reset: 1.<br><b>Description:</b> Set to 1 to enable the function of SLP_TYPEn bit in PmControl register [AcpiPm1CntBlk:00h]. The SLP_TYPEn bit in PmControl register has no effect if this bit is clear.                                   |
| 4    | <b>kbrsten.</b> Read-write. Reset: 1. Set to 1 to enable KB_RST# pin to trigger KB Reset.                                                                                                                                                                                        |
| 3    | <b>kb_trig_type.</b> Read-write. Reset: 0.<br><b>Description:</b> If kb_trig_type == 0, we will use the level trigger as kb_reset_src.<br>If kb_sel_edge == 1, we will use edge trigger as kb_reset_src.<br>This will give customers the choice how to trigger a keyboard reset. |
| 2    | <b>kb_sel_edge.</b> Read-write. Reset: 0.<br><b>Description:</b> This bit is applicable when kb_trig_type is set high.<br>If kb_sel_edge == 0, negedge will be used as the kb_reset_src.<br>If kb_sel_edge == 1, posedge will be used as the kb_reset_src.                       |
| 1    | <b>kb_pcirst_en.</b> Read-write. Reset: 1. Set to 1 to make PCI reset during KB Reset, which can be triggered by KB_RST# pin or <a href="#">EC</a> .                                                                                                                             |
| 0    | <b>softreseten.</b> Read-write. Reset: 0. Set to 1 to block any reset request until the system is not C state.                                                                                                                                                                   |

**PMx000000BF (FCH::PM::RESETCONTROL2)**

Read-write. Reset: C0h.

\_aliasHOSTLEGACY; PMx000000BF; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                             |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>pwrgoodenb.</b> Read-write. Reset: 1. Output enable for PwrGood pin (active low)                                                                                                                     |
| 6    | <b>pwrgoodout.</b> Read-write. Reset: 1. Output data for PwrGood pin                                                                                                                                    |
| 5    | <b>bypass_pwr_good.</b> Read-write. Reset: 0. If asserted, Southbridge will not wait for deassertion of PWRGOOD to monitor wakeup events.                                                               |
| 4    | <b>resetpcie2.</b> Read-write. Reset: 0. This bit is to be used with GEVENT4 mux select bits. If GEVENT4 IO mux select bits are set to 10b, GEVENT4# pin will become PCIE_RST2#                         |
| 3    | <b>shutdownpinen.</b> Read-write. Reset: 0. <a href="#">Shutdown</a> system if seeing a negative edge on pinSHUTDOWN#                                                                                   |
| 2    | <b>overridewarmrst.</b> Read-write. Reset: 0. Add a S5 sticky bit "OverrideWarmRst" to override DevAllRstB so S5 GPIO can be persistent and not being reset by warm reset.                              |
| 1    | <b>selnew16us.</b> Read-write. Reset: 0.<br><b>Description:</b> When this bit is set high, Osc16Us_new will be used in PwrRst except MP1_WDT_cnt. When this bit is low, Osc16Us will be used in PwrRst. |
| 0    | Reserved.                                                                                                                                                                                               |



**PMx000000C0 (FCH::PM::S5\_RESET\_STATUS)**

Reset: 0000\_0800h.

This register shows the source of previous reset.

\_aliasHOSTLEGACY; PMx000000C0; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31   | <b>sw_sync_flood_flag.</b> Read-write,Read,Write-1-to-clear. Reset: 0. PMxC0[31] will be set if sw_sync_flood (PMx88[10]) trigger reset. Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                                            |
| 30   | <b>sdp_parity_err.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> When there is 'parity error', Sync_Flood reset will occur and PMxC0[27] will be set if enabled, in order to distinguish 'parity error' and 'CPU sync flood', PMxC0[30] will be set when there is SDP parity Error, thus software can distinguish 'parity error' and 'CPU sync flood'.<br>This bit will not be cleared by other reset event, software need write 1 to clear. SDP parity error will not clear any status bit in this register. |
| 29   | <b>mp1_wdtout.</b> Read-write,Read,Write-1-to-clear. Reset: 0. This bit will be set to 1 when MP1_Watchdog timer time out (this indicates there was a failed warm reset handshake between SMU and <a href="#">FCH</a> ). This bit will not be cleared by other reset event, software need write 1 to clear. MP1_Watchdog timer time out will not clear any status bit in this register.                                                                                                                                                |
| 28   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 27   | <b>sync_flood.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system reset was caused by a SYNC_FLOOD event which was due to an UE error( when PMx74[18]=1). Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                       |
| 26   | <b>remotereseetfromasf.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system reset was caused by a remote RESET command from ASF. Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                                                 |
| 25   | <b>watchdogissuereset.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system reset was caused by MICROSOFT WatchDog Timer. Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                                                         |
| 24   | <b>failbootrst.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system reset was caused by AMD Fail boot timer. Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                                                                     |
| 23   | <b>shutdown_msg.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system reset was caused by a SHUTDOWN command from CPU (when PMx08[20]=1 and PMx74[17]=1). Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                         |
| 22   | <b>kb_reset.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system reset was caused by assertion of KB_RST_L.. Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                                                                     |
| 21   | <b>sleepreset.</b> Read-write,Read,Write-1-to-clear. Reset: 0. Reset status from Sleep state (Power saving mode, S3, 4, or 5) transition. Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                                           |
| 20   | <b>do_k8_full_reset.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> system reset was caused by CF9 = 0x0E. Write 1 to clear.<br>[Note] Write CF9=0xE will set this bit=1, but write CF9=0xE will generate SLpRst later which will set bit[21]=SleepReset. In order to keep this bit =1, this bit will not be cleared by hardware, software need to write 1 to clear this bit .<br>Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.             |
| 19   | <b>do_k8_reset.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system reset was caused by CF9 = 0x06. Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                                                                              |
| 18   | <b>do_k8_init.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system reset was caused by CF9 = 0x04. Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                                                                                                                                                                                                                                                                                               |

|       |                                                                                                                                                                                                                                                                   |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17    | <b>soft_pcirst.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system reset was caused by writing to PMIO 0xC4[0] (PciReset). Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set. |
| 16    | <b>usrrstb.</b> Read-write,Read,Write-1-to-clear. Reset: 0. Last reset was caused by BP_SYS_RST_L assertion. Write 1 to clear. Bit[31] and Bit[28:16] except bit[20] will be cleared by Last reset event except the associated bit will be set.                   |
| 15:14 | <b>pmeturnofftime.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: 1ms<br>01: 2ms<br>10: 4ms<br>11: 8ms                                                                                                                                                     |
| 13:10 | Reserved.                                                                                                                                                                                                                                                         |
| 9     | <b>intthermaltrip.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system was shut down due to an internal ThermalTrip event. Write 1 to clear                                                                                                                    |
| 8:5   | Reserved.                                                                                                                                                                                                                                                         |
| 4     | <b>remotepowerdownfromasf.</b> Read-write,Read,Write-1-to-clear. Reset: 0. SOC has received a remote Power Off command from ASF. Write 1 to clear.                                                                                                                |
| 3     | Reserved.                                                                                                                                                                                                                                                         |
| 2     | <b>shutdown.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system was shut down due to ShutDown event (SHUTDOWN# pin). Write 1 to clear.                                                                                                                        |
| 1     | <b>pwrbtn4second.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system was shut down due to 4s PwrButton event. Write 1 to clear.                                                                                                                               |
| 0     | <b>thermaltrip.</b> Read-write,Read,Write-1-to-clear. Reset: 0. system was shut down due to BP_THERMTRIP_L assertion. Write 1 to clear.                                                                                                                           |

**PMx000000C4 (FCH::PM::RESETCOMMAND)**

Read-write. Reset: 28h.

\_aliasHOSTLEGACY; PMx000000C4; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>reseten.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Not allow to write bit 0<br>1: Allow to write bit 0.                                                                                                |
| 6    | <b>resetallacpi.</b> Read-write. Reset: 0. Writing 1 to emulate a Reset Button event.                                                                                                                               |
| 5    | <b>resetbuttonen.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: reset from reset button (SYS_RESET_L) will generate S5Reset.<br>0: Disable SYS_RESET_L to generate S5Reset, it can be used as Gevent19/GPIO1. |
| 4    | <b>resetpcie.</b> Read-write. Reset: 0. Set to 1 to reset Gpp port.                                                                                                                                                 |
| 3    | <b>usrrstb2pll.</b> Read-write. Reset: 1. Set to 1 to stop Pll when reset button is pressed.                                                                                                                        |
| 2    | <b>selectdebug.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Select the PM_Reg C0 to be S5/Reset Status register.<br>1: Select the PM_Reg C0 to be a debug status register.                                  |
| 1    | <b>memrstdisable.</b> Read-write. Reset: 0. When set, the memory reset function at DDR_RST# pin will be disabled.                                                                                                   |
| 0    | <b>pcireset.</b> Read-write. Reset: 0. Writing 1 to do a PCI reset                                                                                                                                                  |

**PMx000000C5 (FCH::PM::CF9SHADOW)**

Read-write.

\_aliasHOSTLEGACY; PMx000000C5; PM=FED8\_0300h

| Bits | Description                                                                                                                                           |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | Reserved.                                                                                                                                             |
| 3    | <b>fullrst.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Assert reset signals only<br>1: Place system in S5 state for 3 to 5 seconds           |
| 2    | <b>rstcmd.</b> Read-write. Reset: 0.<br><b>Description:</b> Write with 1 to generate reset as specified by bit[3,1].<br>Write only. Always read as 0. |
| 1    | <b>sysrst.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Send INIT HT message<br>1: Reset as specified by bit3                                  |
| 0    | Reserved.                                                                                                                                             |

**PMx000000C6 (FCH::PM::HTCONTROL)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx000000C6; PM=FED8\_0300h

| Bits | Description |
|------|-------------|
| 15:0 | Reserved.   |

**PMx000000C8 (FCH::PM::MISC\_PMIO)**

Read-write. Reset: 0028\_800Ch.

\_aliasHOSTLEGACY; PMx000000C8; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                              |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:24 | <b>clkintrvectorord.</b> Read-write. Reset: 00h. Specify the value used to indentify the clock interrupt.                                                                                                                                |
| 23    | Reserved.                                                                                                                                                                                                                                |
| 22    | <b>clkintrvectororden.</b> Read-write. Reset: 0. When set, the system timer interrupt in the IOAPIC will be tagged with a value defined by ClkIntrVectorOrd                                                                              |
| 21    | <b>align_s3s5.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Old scheme, BP_SLP_S5_L and BP_SLP_S3_L may not be aligned, this is default.<br>1: new scheme, BP_SLP_S5_L and BP_SLP_S3_L are aligned.                               |
| 20    | Reserved.                                                                                                                                                                                                                                |
| 19    | <b>usecpurst.</b> Read-write. Reset: 1. If this bit is not set, system reset will cause INIT# instead of CPURST#.                                                                                                                        |
| 18    | <b>usebypassrom.</b> Read-write. Reset: 0. When this bit is set, it will override the ROM straps and use bits 17:16 of this register to set the strap. This is for BIOS debugging purpose or for system having multiple BIOSes on board. |
| 17    | <b>romtype_strap_bypass.</b> Read-write. Reset: 0. This bit will override the strap for ROMTYPE_strap when bit 18 of this register set.                                                                                                  |
| 16    | <b>spi32bit_strap_bypass.</b> Read-write. Reset: 0. This bit will override the strap for Spi32Bit_strap when bit 18 of this register set.                                                                                                |
| 15    | <b>rsmu_cpl_gate_dis.</b> Read-write. Reset: 1.<br><b>Description:</b> Chicken bit to disable the gating for Cpl signals between RSMU and SMU.<br>If it is low, gating enabled.<br>If it is high, gating disabled.                       |
| 14    | <b>ocpupwrgood_sel.</b> Read-write. Reset: 0.<br><b>Description:</b> Chicken bit to disable the gating for signals between RSMU and SMU.<br>If it is low, gating enabled.<br>If it is high, gating disabled.                             |
| 13    | <b>id_change_en.</b> Read-write. Reset: 0. Setting this bit will allow the software to change the DeviceID and RevisionID.                                                                                                               |
| 12    | <b>s5resetoverride.</b> Read-write. Reset: 0. Set to 1 to mask off internet PCI reset used in ACPI.                                                                                                                                      |
| 11    | <b>writebackenable.</b> Read-write. Reset: 0. HD audio/modem write back enable. If set, the WakeOnRing status bit will be written back to HD Audio controller upon system power up.                                                      |
| 10    | <b>llb_en.</b> Read-write. Reset: 0. If set, LLB function is enabled, and system won't wakeup from ACPI S state until LLB# is de-asserted.                                                                                               |
| 9:8   | <b>temp_polarity.</b> Read-write. Reset: 0h.<br><b>Description:</b> Temperature polarity control for THRMTRIP and TALERT respectively.<br>0: Active low<br>1: Active high                                                                |
| 7     | <b>disablepcirom.</b> Read-write. Reset: 0. Set to 1 to disable PCI from strap.                                                                                                                                                          |
| 6     | <b>twarnen.</b> Read-write. Reset: 0. If set, it enables TALERT# pin                                                                                                                                                                     |
| 5     | <b>eprom_efuseindex_select.</b> Read-write. Reset: 0. If this bit is 0, PMIO D8h and D9h are for accessing the EPROM strap bits. When this bit is set to 1, PMIO D8h and D9h is for accessing the Efuse bits.                            |
| 4     | <b>ouseacpistraps.</b> Read-write. Reset: 0. When set, it will use the config bits from index D8h and D9h to override the EpromStraps.                                                                                                   |
| 3     | <b>tdeaden.</b> Read-write. Reset: 1. When set, GEVENT2 takes up the THRMTRIP function. When THRMTRIP pin is low and TFATAL_EN(bit2 of the same register) is set, hardware will switch the system to S5 automatically.                   |
| 2     | <b>tfatal_en.</b> Read-write. Reset: 1. This bit enables both the soft PCIRST and the THRMTRIP function.                                                                                                                                 |
| 1     | <b>instatnoffenable.</b> Read-write. Reset: 0. Enable fast shutdown upon THERMTRIP# event                                                                                                                                                |
| 0     | <b>cpu_io_pulldowndrvstrength.</b> Read-write. Reset: 0. When set, the integrated pull-down drive strength of all CPU Ios are increased by 50%.                                                                                          |

**PMx000000CC (FCH::PM::IODRVSTH)**

Read-write. Reset: DB6D\_B6DBh.

\_aliasHOSTLEGACY; PMx000000CC; PM=FED8\_0300h

**Bits Description**

31:0 Reserved.

**PMx000000D2 (FCH::PM::PMIODEBUG)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000D2; PM=FED8\_0300h

**Bits Description**

7 Reserved.

6 **cf9rstdisable**. Read-write. Reset: 0. When set, write to CF9 will not generate a reset. The purpose of this bit is to allow BIOS to trap CF9

5:0 Reserved.

**PMx000000D4 (FCH::PM::PWRRSTDEBSEL)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000D4; PM=FED8\_0300h

**Bits Description**

7:5 Reserved.

4:0 **debugseldev**. Read-write. Reset: 00h. Debug Bus Select for PwrRst signals**PMx000000D5 (FCH::PM::ALTMIOEN)**

Read-write.

\_aliasHOSTLEGACY; PMx000000D5; PM=FED8\_0300h

**Bits Description**

7:2 Reserved.

1 **alt\_addr\_width\_sel**. Read-write. Reset: 0.**Description:** 1h : alternate address is 64bit width,  
0h : alternate address is 32bit width.0 **alt\_addr\_en**. Read-write. Reset: 0. Alternate address enable bit.**PMx000000D6 (FCH::PM::ALTMIOBASE)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx000000D6; PM=FED8\_0300h

**Bits Description**15:0 **lower\_addr\_alt**. Read-write. Reset: 0000h. Lower bits of base address. Its value only take effect when alternate address is enabled. For example, when it is set as 0xABCD and alternate address width is 32bit, base address will be 0xABCD\_0000. If it is 64 bit then the base address is 0xFFFF\_FFFF\_ABCD\_0000.**PMx000000D8 (FCH::PM::EPROM\_EFUSEINDEX)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000D8; PM=FED8\_0300h

**Bits Description**7:0 **eprom\_efuseindex**. Read-write. Reset: 00h.**Description:** Index register to access Eprom setting (PM\_regxC8[5]=0) or Efuse bits (PM\_regxC8[5]=1). Write to this port sets the initial value of the index. Writing to the EpromStrapData port will auto-increment this index. Programming through the index/data port will not take effect until the next reset. This register is used as the index register to read the value of efuse when PM\_regxC8[5] is set to 1.

**PMx000000D9 (FCH::PM::EPROM\_EFUSEDATA)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000D9; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>eprom_efusedata.</b> Read-write. Reset: 00h.<br><b>Description:</b> Data register to access Eprom bits (PM_regxC8[5] =0) or Efuse bits (PM_regxC8[5]=1). Writing to the EpromStrapData port will auto-increment the index at PMIO_CCh. Programming through the index/data port will not take effect until the next reset.<br>This register is used as the data register to read the value of efuse when PM_regxC8[5] is set to 1. |

**PMx000000DA (FCH::PM::SATACONFIG)**

Read-write. Reset: 0011h.

\_aliasHOSTLEGACY; PMx000000DA; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:8 | <b>pmio_xda_sataconfig.</b> Read-write. Reset: 00h. Not used.                                                                                                                                                                                                                                                                                                                |
| 7:6  | <b>ref_div_sel.</b> Read-write. Reset: 0h.<br><b>Description:</b> This is CP_PLL_CLKR, the reference clock divider setting.<br>00: Divide by 1 (25MHz reference clock)<br>01: Divide by 2<br>10: Divide by 4 (100MHz reference clock)<br>11: Same as 10                                                                                                                      |
| 5:4  | <b>ref_clk_sel.</b> Read-write. Reset: 1h.<br><b>Description:</b> This is CP_PLL_REFCLK_SEL, the reference clock source selection for SATA PLL.<br>00/10: Reference clock from crystal oscillator via PAD_XTALI and PAD_XTALO<br>01: Reference clock from internal clock through CP_PLL_REFCLK_P and CP_PLL_REFCLK_N via RDL<br>11: same as 01                               |
| 3    | <b>hiddenide.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: IDE controller is exposed and Combined Mode is enabled. SATA controller has control over Port0 through Port3, IDE controller has control over Port4 and Port5<br>1: IDE controller is hidden and Combined Mode is disabled, SATA controller has full control of all 6 Ports when operating in non-IDE mode |
| 2    | <b>setmaxgen2.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SATA controller operates in maximum Gen3 (3.0Gbps) speed<br>1: SATA controller operates in maximum Gen2 (3.0Gbps) speed and saves more power on PLL.                                                                                                                                                      |
| 1    | <b>channel_sel.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: SATA Port4 and Port5 utilizing Primary IDE channel<br>1: SATA Port4 and Port5 utilizing Secondary IDE channel                                                                                                                                                                                            |
| 0    | <b>sataenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: SATA controller is disabled<br>1: SATA controller is enabled                                                                                                                                                                                                                                              |

**PMx000000DC (FCH::PM::PMIO\_XDC\_SATACONFIG)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx000000DC; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>pmio_xdc_sataconfig.</b> Read-write. Reset: 0000_0000h.<br><b>Description:</b> Used by Sata DEVSLP.<br>This register only reset by RsmRstB or UsmRstB, BIOS need to clear this register to 0x0 after WarmReset so that device can response with SATA controller's <a href="#">OOB</a> . |

**PMx000000E0 (FCH::PM::ABREGBAR)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx000000E0; PM=FED8\_0300h

| Bits  | Description                                                                                                               |
|-------|---------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                 |
| 15:3  | <b>abregbar</b> . Read-write. Reset: 0000h. The 13 bits are mapped into AB IO base[15:3]. AB IO Base [2:0] are tied to 0. |
| 2:0   | Reserved.                                                                                                                 |

**PMx000000E4 (FCH::PM::ABDEBUG)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx000000E4; PM=FED8\_0300h

| Bits | Description                                                                     |
|------|---------------------------------------------------------------------------------|
| 15   | <b>ab_jtagmuxhost_en</b> . Read-write. Reset: 0. oAB_JtagMuxHost_en             |
| 14:7 | Reserved.                                                                       |
| 6    | <b>sdpdebugmode</b> . Read-write. Reset: 0. Select the SDP Debug mode.          |
| 5    | <b>sdpdebugen</b> . Read-write. Reset: 0. Enable the SDP Debug mode.            |
| 4    | <b>reqfilteron</b> . Read-write. Reset: 0. Not used.                            |
| 3:2  | <b>arbcontrol</b> . Read-write. Reset: 0h. Not used.                            |
| 1:0  | <b>blinkcontrol</b> . Read-write. Reset: 0h. Control the behavior on Blink pad. |

**PMx000000E6 (FCH::PM::DACCNTL)**

Read-write. Reset: 0001h.

\_aliasHOSTLEGACY; PMx000000E6; PM=FED8\_0300h

| Bits | Description                                             |
|------|---------------------------------------------------------|
| 15:0 | <b>dacntrl</b> . Read-write. Reset: 0001h. For VGA tile |

**PMx000000EA (FCH::PM::UFSCNTL)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000EA; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                            |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:4  | Reserved.                                                                                                                                                                                                              |
| 3    | <b>mphy_pwrokvddc</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: No deassertion or don't force PwrOkVddc going to MPHY to low<br>1: Deassert or force PwrOkVddc going to MPHY to low                            |
| 2    | <b>mphy_cpbybypassrefclk_en</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Pad is output or Enable 26MHz output or Disable BypRefClk input<br>1: Pad is input or Disable 26MHz output or Enable BypRefClk input |
| 1    | <b>byp_mphy_debugbus</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Don't Bypass MPHY DSM Debug Bus<br>1: Bypass MPHY DSM Debug Bus                                                                             |
| 0    | <b>dis26mhzpadgt</b> . Read-write. Reset: 0.<br><b>Description:</b> 0: Enable 26MHz pad gating as per design logic<br>1: Disable 26MHz pad gating and always enable 26 MHz output                                      |

**PMx000000EB (FCH::PM::AZEN)**

Read-write. Reset: 01h.

\_aliasHOSTLEGACY; PMx000000EB; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                                                                                  |
| 1    | <b>aznosnoopenable.</b> Read-write. Reset: 0.<br><b>Description:</b> When set, HD AUDIO data transfer will not cause the BM_STS bit to be set and to wake up the CPU from C3 state.<br>Under current C1e implementation, there is no need to set this bit. |
| 0    | <b>azenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Disable HD audio controller<br>1: Enable HD audio controller                                                                                                                              |

**PMx000000EC (FCH::PM::LPCGATING)**

Read-write. Reset: 11h.

\_aliasHOSTLEGACY; PMx000000EC; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:6  | Reserved.                                                                                                                                                                                                                                         |
| 5    | <b>iss5rom.</b> Read-write. Reset: 0.<br><b>Description:</b> 1: Rom_Rst output is deasserted from G3 to S0.<br>0: Rom_Rst output is toggled along with Arst#.                                                                                     |
| 4    | <b>dislpctoacpi.</b> Read-write. Reset: 1.<br><b>Description:</b> 1: The ReqB from ACPI won't be blocked when there is cycle to LPC.<br>0: The ReqB from ACPI will be blocked when there is cycle to LPC.<br>(This bit is also related to bit 3.) |
| 3    | Reserved.                                                                                                                                                                                                                                         |
| 2    | <b>abnobypassen.</b> Read-write. Reset: 0. Set to 0 to not allow AB bypass any LPC cycles.                                                                                                                                                        |
| 1    | <b>lpc_a20en.</b> Read-write. Reset: 0. Set to 1 to enable A20# input.                                                                                                                                                                            |
| 0    | <b>lpc_enable.</b> Read-write. Reset: 1. Set to 1 to enable LPC bridge                                                                                                                                                                            |

**PMx000000ED (FCH::PM::USBGATING)**

Read-write. Reset: 04h.

\_aliasHOSTLEGACY; PMx000000ED; PM=FED8\_0300h

| Bits | Description                                                                                   |
|------|-----------------------------------------------------------------------------------------------|
| 7    | Reserved.                                                                                     |
| 6    | <b>usb_smiact_en.</b> Read-write. Reset: 0. Not used.                                         |
| 5    | <b>undo_smi_change.</b> Read-write. Reset: 0. Not used.                                       |
| 4    | <b>usb_smi_en.</b> Read-write. Reset: 0. Set to 1 to enable USB <a href="#">SMI</a> #.        |
| 3    | <b>usb_pme_gate.</b> Read-write. Reset: 0. XHC0/1 PME gate. Set to 1 will block the PME.      |
| 2    | <b>pmio_ohci_arb_req_q_vld_en.</b> Read-write. Reset: 1. Not used.                            |
| 1    | <b>usb_irq_en.</b> Read-write. Reset: 0. Set to 1 to route IRQ1/IRQ12 from usb to PIC/IOAPIC. |
| 0    | <b>usb_a20_en.</b> Read-write. Reset: 0. Set to 1 to enable USB A20#.                         |



**PMx000000EE (FCH::PM::USB3CNTRL)**

Read-write.

\_aliasHOSTLEGACY; PMx000000EE; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                     |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:6  | Reserved.                                                                                                                                                                                                                                                                                       |
| 5:4  | <b>xhc1powersel.</b> Read-write. Reset: 0h.                                                                                                                                                                                                                                                     |
|      | <b>Description:</b> 00: oXhc1S5RstB assert when CF9, KbRst, RsmRst, UsrRstB, S5, S3, Power saving mode.<br>01: oXhc1S5RstB assert when CF9, KbRst, RsmRst, UsrRstB, S5.<br>10: oXhc1S5RstB assert when CF9, KbRst, RsmRst, UsrRstB.<br>11: oXhc1S5RstB assert when CF9, KbRst, RsmRst, UsrRstB. |
| 3    | <b>usb3efusstat.</b> Read-write. Reset: X. Read-Only. This bit means "USB3 is Efuse-enabled"                                                                                                                                                                                                    |
| 2    | Reserved.                                                                                                                                                                                                                                                                                       |
| 1:0  | <b>xhc0powersel.</b> Read-write. Reset: 0h.                                                                                                                                                                                                                                                     |
|      | <b>Description:</b> 00: oXhc0S5RstB assert when CF9, KbRst, RsmRst, UsrRstB, S5, S3, Power saving mode.<br>01: oXhc0S5RstB assert when CF9, KbRst, RsmRst, UsrRstB, S5.<br>10: oXhc0S5RstB assert when CF9, KbRst, RsmRst, UsrRstB.<br>11: oXhc0S5RstB assert when CF9, KbRst, RsmRst, UsrRstB. |

**PMx000000EF (FCH::PM::USBDEBUG)**

Read-write. Reset: 7Fh.

\_aliasHOSTLEGACY; PMx000000EF; PM=FED8\_0300h

| Bits | Description                                           |
|------|-------------------------------------------------------|
| 7:0  | <b>s3_usbenable.</b> Read-write. Reset: 7Fh. Not used |

**PMx000000F0 (FCH::PM::USBCONTROL)**

Read-write. Reset: 739Ch.

\_aliasHOSTLEGACY; PMx000000F0; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:13 | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 12    | <b>usb2blglobalclkgateen.</b> Read-write. Reset: 1. Set to 1 to enable USB2 B-Link Global Clock Gating                                                                                                                                                                                                                                                                                                                                                                                                        |
| 11    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 10:8  | <b>usbsleepctrl.</b> Read-write. Reset: 3h.<br><b>Description:</b> Control on USB advanced async sleep function. Setting of 000b:100b are for the advanced async sleep.<br>000b: Standard 10s sleep<br>001b: Advanced sleep up to 2 uframes<br>010b: Advanced sleep up to 4 uframes<br>011b: Advanced sleep up to 6 uframes (default)<br>100b: Advanced sleep up to next uframe 0<br>101b, 110b: Reserved<br>111b: EHCI will stop fetching descriptor once it has completed the list while CPU is in C state. |
| 7     | <b>usbresumeenable.</b> Read-write. Reset: 1. Set to 1 to enable S3 wakeup on USB device resume.                                                                                                                                                                                                                                                                                                                                                                                                              |
| 6     | <b>usbs3wakeresumeonlydisable.</b> Read-write. Reset: 0. Default (0) is to support USB Wake-Up event on Resume only. When set to 1, USB can wakeup on all wake events, connection, disconnect, over-current, and/or resume detect.                                                                                                                                                                                                                                                                            |
| 5     | <b>pmio_ohci_mem_slp_dis.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: enable OHCI memory sleep mode<br>1: disable OHCI memory sleep mode                                                                                                                                                                                                                                                                                                                                                              |
| 4     | <b>usb11pdresistorenable.</b> Read-write. Reset: 1. Set to 0 to disconnect pull-down resistors on stand-alone USB1.1 pads.                                                                                                                                                                                                                                                                                                                                                                                    |
| 3     | <b>usbs5resetenable.</b> Read-write. Reset: 1. Set to 1 to enable USB reset on S4/S5 resume detection.                                                                                                                                                                                                                                                                                                                                                                                                        |
| 2     | <b>usbkbresetenable.</b> Read-write. Reset: 1. Set to 1 to enable resetting USB on KB reset.                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 1     | <b>usbforceregenable.</b> Read-write. Reset: 0. This function is not implemented.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 0     | <b>usbphys5pwrdownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> Set to 1 to disable S4/S5 USB PHY power down support and to enable S4 USB wakeup support. The bit has to be set to 1 to support S4 USB wakeup.                                                                                                                                                                                                                                                                                     |

**PMx000000F2 (FCH::PM::USBCONTROL\_1)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000F2; PM=FED8\_0300h

| Bits | Description |
|------|-------------|
| 7:0  | Reserved.   |

**PMx000000F3 (FCH::PM::USBDEBUG\_2)**

Read-write. Reset: 00h.

\_aliasHOSTLEGACY; PMx000000F3; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7    | <b>forcephyportreset.</b> Read-write. Reset: 0. Forces USB PHY port reset.                                                                                                                 |
| 6    | <b>forcephyearlyreset.</b> Read-write. Reset: 0. Forces USB PHY early reset.                                                                                                               |
| 5    | <b>forcephydllreset.</b> Read-write. Reset: 0. Forces USB PHY DLL reset.                                                                                                                   |
| 4    | <b>forcephypllreset.</b> Read-write. Reset: 0. Forces USB PHY PLL reset.                                                                                                                   |
| 3    | <b>forcephyprdown.</b> Read-write. Reset: 0. Forces USB PHY into power down mode.                                                                                                          |
| 2:0  | <b>forcereset2usb.</b> Read-write. Reset: 0h. These are software control bits that can be used to force resetting of USB host controllers. Each bit corresponds to one USB major function. |

**PMx000000F4 (FCH::PM::USBCONTROL\_2)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx000000F4; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                 |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:3 | Reserved.                                                                                                                                                                                                                                                                                   |
| 2    | <b>usb_s3_ohci_discon_fix</b> . Read-write. Reset: 0. Not used.                                                                                                                                                                                                                             |
| 1    | <b>ohci_hidden_enable</b> . Read-write. Reset: 0.<br><b>Description:</b> This bit is used to hide OHCI controllers inside USB2.0.<br>0: USB2.0 contains OHCI and EHCIs.<br>1: All OHCI controllers inside USB2.0 are hidden and only EHCI controller exist as single function.<br>Not used. |
| 0    | <b>usb_s3_dis_con_wo_wakeen</b> . Read-write. Reset: 0.<br><b>Description:</b> Set to open EHCI/OHCI S3 disconnect and connect fix.<br>Not used.                                                                                                                                            |

**PMx000000F6 (FCH::PM::OTGCFG)**

Read-write. Reset: 0000h.

\_aliasHOSTLEGACY; PMx000000F6; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:2 | Reserved.                                                                                                                                                                                                                                                                                                                                      |
| 1:0  | <b>otgpowersel</b> . Read-write. Reset: 0h.<br><b>Description:</b> 00: oOtgS5RstB assert when CF9, KbRst, RsmRst, UsrcRstB, S5, S3, Power saving mode.<br>01: oOtgS5RstB assert when CF9, KbRst, RsmRst, UsrcRstB, S5.<br>10: oOtgS5RstB assert when CF9, KbRst, RsmRst, UsrcRstB.<br>11: oOtgS5RstB assert when CF9, KbRst, RsmRst, UsrcRstB. |

**PMx000000F8 (FCH::PM::SPBBASE)**

Read-write. Reset: FEF0\_003Fh.

\_aliasHOSTLEGACY; PMx000000F8; PM=FED8\_0300h

| Bits | Description                                                  |
|------|--------------------------------------------------------------|
| 31:0 | <b>spbs5cntrl</b> . Read-write. Reset: FEF0_003Fh. Not used. |

**PMx000000FC (FCH::PM::TRACEMEMORYEN)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx000000FC; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                 |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:20 | <b>tracememorybaseaddr</b> . Read-write. Reset: 000h. The base address of trace memory in ILA (Internal Logic Analyzer) connect to ILA.ILA_Mem_Addr. It is 1M memory space. |
| 19:1  | Reserved.                                                                                                                                                                   |
| 0     | <b>tracememoryen</b> . Read-write. Reset: 0. Set to 1 to enable trace memory decoding in ILA (Internal Logic Analyzer), connect to ILA.Acpi2ILA_Enable                      |

**PMx00000500 (FCH::PM::PM1STATUS)**

Read-write.

This register is located at the base address defined by AcpiPm1EvtBlk.

\_aliasHOSTLEGACY; PMx00000500; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | <b>wakestatus.</b> Read-write. Reset: 0. This bit is set when the system is in the sleep state and a wake-up event occurs.                                                                                                                                                                                                       |
| 14    | <b>pciexpwakestatus.</b> Read-write. Reset: 0. This bit is set by hardware to indicate that the system wake is due to a PCI Express® wakeup event.                                                                                                                                                                               |
| 13:11 | Reserved.                                                                                                                                                                                                                                                                                                                        |
| 10    | <b>rtcstatus.</b> Read-write. Reset: 0. This bit is set when RTC generates an alarm.                                                                                                                                                                                                                                             |
| 9     | Reserved.                                                                                                                                                                                                                                                                                                                        |
| 8     | <b>pwrbtnstatus.</b> Read-write. Reset: 0. Power button status bit                                                                                                                                                                                                                                                               |
| 7:6   | Reserved.                                                                                                                                                                                                                                                                                                                        |
| 5     | <b>gblstatus.</b> Read-write. Reset: 0. This bit is set when an SCI is generated due to the BIOS wanting the attention of the SCI handler. This is set by writing 1 to PM_Reg: 74h bit [7].                                                                                                                                      |
| 4     | <b>bmstatus.</b> Read-write. Reset: 0. Bus master status bit. This bit is set any time a system bus master requests the system bus, and can only be cleared by writing an one to this bit position. Note: this bit is no longer used except for server CPU. For server CPU, this will be used in conjunction with IDLE_EXIT# pin |
| 3:1   | Reserved.                                                                                                                                                                                                                                                                                                                        |
| 0     | <b>tmrstatus.</b> Read-write. Reset: 0. Timer carry status bit. This bit gets set anytime the 31st bit of 32 bit counter changes (whenever the MSB changes from low to high or high to low. While TmrEn and TmrStatus are set, an interrupt event is raised). [Read-only]                                                        |

**PMx00000502 (FCH::PM::PM1ENABLE)**

Read-write.

This register is located at the base address defined by AcpiPm1EvtBlk.

\_aliasHOSTLEGACY; PMx00000502; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                              |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15    | Reserved.                                                                                                                                                                                                                                                |
| 14    | <b>pciexpwakedis.</b> Read-write. Reset: 1. This bit disables the inputs to the PciExpWakeStatus from waking the system.                                                                                                                                 |
| 13:11 | Reserved.                                                                                                                                                                                                                                                |
| 10    | <b>rtcen.</b> Read-write. Reset: 0. RTC enable. If this bit is set, SCI is generated whenever RtcStatus is true.                                                                                                                                         |
| 9     | Reserved.                                                                                                                                                                                                                                                |
| 8     | <b>pwrbtnen.</b> Read-write. Reset: 0. If this bit is set, SCI is generated whenever PwrBtnStatus is true.                                                                                                                                               |
| 7:6   | Reserved.                                                                                                                                                                                                                                                |
| 5     | <b>gblen.</b> Read-write. Reset: 0. If this bit is set, SCI is raised whenever both GblEn and GblStatus are true.                                                                                                                                        |
| 4:1   | Reserved.                                                                                                                                                                                                                                                |
| 0     | <b>tmren.</b> Read-write. Reset: 0. This is the timer carry interrupt enable bit. When this bit is set then an SCI event is generated anytime the TmrStatus is set. When this bit is reset then no interrupt is generated when the TmrStatus bit is set. |

**PMx00000504 (FCH::PM::PMCONTROL)**

Read-write.

This register is located at the base address defined by AcpiPm1CntBlk (PM\_Reg:62h).

\_aliasHOSTLEGACY; PMx00000504; PM=FED8\_0300h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                  |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15:14 | Reserved.                                                                                                                                                                                                                                                                                                                                    |
| 13    | <b>slp_en.</b> Read-write. Reset: 0.<br><b>Description:</b> This is a write-only bit and reads from it always return zero.<br>If PM_Reg:Beh bit5 (SLP_TYP_EN) is 1 (default), setting this bit will cause the system to sequence into the sleeping state associated with the SLP_TYP fields programmed. Writing 0 to this bit has no effect. |
| 12:10 | <b>slp_typ.</b> Read-write. Reset: 0h. Defines the sleep state the system enters when the SLP_TYPEn is set to one. This design currently implements 5 states: S0, S1, S3, S4, and S5.                                                                                                                                                        |
| 9:3   | Reserved.                                                                                                                                                                                                                                                                                                                                    |
| 2     | <b>gbl_rls.</b> Read-write. Reset: 0. If SMI_Reg:B0h[17:16] is set to 01b, writing 1 to this bit will generate <a href="#">SMI#</a> and set SMI_Reg:88h bit[8]. Reading, this bit will always return 0.                                                                                                                                      |
| 1     | <b>bmrlld.</b> Read-write. Reset: 0. If this bit is set, any bus master activity will cause the C state logic to break out from C3. This is no longer needed for current C state implementation                                                                                                                                              |
| 0     | <b>sci_en.</b> Read-write. Reset: 0. Selects the power management event to be either an SCI or <a href="#">SMI#</a> interrupt for the following events. When this bit is set, then PM events will generate an SCI interrupt otherwise, it will be SMI#.                                                                                      |

**PMx00000508 (FCH::PM::TMRVALUE\_ETMRVALUE)**

Read-only.

AcpiPmTmrBlk is defined in PM\_Reg:64h

\_aliasHOSTLEGACY; PMx00000508; PM=FED8\_0300h

| Bits | Description                                                                                                                                                      |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>tmrvalue.</b> Read-only. Reset: XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXb. This read-only field returns the running count of the power management timer (ACPI timer). |

**PMx0000050C (FCH::PM::CLKVALUE)**

Read-write. Reset: 0000\_0000h.

CpuControl Io base is defined in PM\_Reg 0x66

\_aliasHOSTLEGACY; PMx0000050C; PM=FED8\_0300h

| Bits | Description |
|------|-------------|
| 31:0 | Reserved.   |

**PMx00000510 (FCH::PM::PLVL2)**

Read-only. Reset: 00h.

CpuControl Io base is defined in PM\_Reg 0x66

\_aliasHOSTLEGACY; PMx00000510; PM=FED8\_0300h

| Bits | Description                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>plvl2.</b> Read-only. Reset: 00h. Reads to this register return all zeros and generate a "enter C2" sequence to APU writes to this register have no effect. |

**PMx00000511 (FCH::PM::PLVL3)**

Read-only. Reset: 00h.

CpuControl Io base is defined in PM\_Reg 0x66

\_aliasHOSTLEGACY; PMx00000511; PM=FED8\_0300h

| Bits | Description                                                                                                                                                    |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>plvl3.</b> Read-only. Reset: 00h. Reads to this register return all zeros and generate a "enter C3" sequence to APU writes to this register have no effect. |

**PMx00000514 (FCH::PM::EVENT\_STATUS\_STDACPI)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx00000514; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                                                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>eventstatus.</b> Read-write. Reset: 0000_0000h. Each bit represents an ACPI event status. Writing 1 to any bit clears it. Each event status is set when the selected event input equals to the corresponding value in SciTrig. Note: configuration for EVENTS are located at SMI_Reg: 08h through 70h. The status bits are also mirrored in SMI_Reg: 00h |

**PMx00000518 (FCH::PM::EVENT\_ENABLE\_STDACPI)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; PMx00000518; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                  |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:0 | <b>eventenable.</b> Read-write. Reset: 0000_0000h. Each bit controls whether ACP should generate wakeup and Sci interrupt. The enable bits are also mirrored in SMI_Reg: 04h |

**PMx0000051C (FCH::PM::SMICMDPORT)**

Read-write. Reset: 00h.

This register is located at the base address defined by AcpiSmiCmd [PM\_Reg:6Ah] + offset 0. When **SMI** command port is enabled, write to this port will generate SMI# (only IOW can generate SMI#, MEMW will not generate SMI#). A read of this address will return the previously written value but will not generate SMI. The SMI command port has to be located at an even address (ie, 0, 2, 4, 6, 8, A, C, or E). SmiCmdStatus is always located one byte immediately after the SmiCmdPort. The implementation actually contains four (4) bytes of address space. If SMI command port is assigned to byte 0, then byte 1 is SmiCmdStatus. Byte 2 and 3 can be used as scratch. If SmiCmdPort is assigned to byte 2, then bytes 0 and 1 are not available.

\_aliasHOSTLEGACY; PMx0000051C; PM=FED8\_0300h

| Bits | Description                                                                   |
|------|-------------------------------------------------------------------------------|
| 7:0  | <b>smicmdport.</b> Read-write. Reset: 00h. Writing the Port can generate Smi. |

**PMx0000051D (FCH::PM::SMICMDSTATUS)**

Read-write. Reset: 00h.

This register is located at the base address defined by AcpiSmiCmd [PM\_Reg:6Ah] + offset 1.

\_aliasHOSTLEGACY; PMx0000051D; PM=FED8\_0300h

| Bits | Description                                                      |
|------|------------------------------------------------------------------|
| 7:0  | <b>smicmdstatus.</b> Read-write. Reset: 00h. Used by BIOS and OS |

**PMx00000524 (FCH::PM::PMACONTROL)**

Read-write.

AcpiPmaCntBlk is defined in PM\_Reg:6Eh

\_aliasHOSTLEGACY; PMx00000524; PM=FED8\_0300h

| Bits | Description                                                                                                                                                                                          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | Reserved.                                                                                                                                                                                            |
| 0    | <b>arb_dis.</b> Read-write. Reset: 0. System arbiter is disabled when this bit is set. Note: under this current AMD C state implementation, this is no longer used and should not be reported to OS. |

**PMx00000528 (FCH::PM::PMACONTROL\_1)**

Read-write.

AcpiPmaCntBlk is defined in PM\_Reg:6Eh

\_aliasHOSTLEGACY; PMx00000528; PM=FED8\_0300h

| Bits | Description                                                                                                                                                             |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | Reserved.                                                                                                                                                               |
| 0    | <b>arb_dis2.</b> Read-write. Reset: 1. System arbiter is disabled when this bit is set if ARB_DIS is also set. This bit is hidden from OS and meant to be used by BIOS. |

## 14.3.14.4 RTC External Registers

**PMx5F\_x00 [RTCEXT DltSavEn] (FCH::PM::RTCEXT::RTCEXTDltSavEn)**

Read-write.

\_aliasHOSTLEGACY; PMx5F\_x00; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits | Description                                                                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:1  | Reserved.                                                                                                                                                                       |
| 0    | <b>DltSavEnable</b> . Read-write. Reset: X. 1=Enable RTC daylight saving feature. The value of this register is undefined/non-deterministic when powered up for the first time. |

**PMx5F\_x01 [RTCEXT SprFwdCtrl] (FCH::PM::RTCEXT::RTCEXTSprFwdCtrl)**

Read-write.

\_aliasHOSTLEGACY; PMx5F\_x01; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |             |     |      |     |      |     |      |         |           |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-----|------|-----|------|-----|------|---------|-----------|
| 7       | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |             |     |      |     |      |     |      |         |           |
| 6       | <b>SprFwdWeek</b> . Read-write. Reset: X. This specifies which Sunday morning to do the "spring forward". Spring forward is usually at the 1st Sunday of April in United States and last Sunday of March in Europe. 0=The 1st Sunday of the month. 1=The last Sunday of the month. The value of this register is undefined/non-deterministic when powered up for the first time.                                                                                                                                                                                                                                                                                                                                         |       |             |     |      |     |      |     |      |         |           |
| 5:0     | <b>SprFwdHour</b> . Read-write. Reset: XXXXXXb.<br><b>Description:</b> This Binary-Coded Decimal ( <a href="#">BCD</a> ) value determines which hour (24 hour mode) to do the "spring forward". Setting of 02h means 2am. Default is 00h which also denotes 2am. Spring forward is usually 2am in United States and 1am in Europe.<br>Note: the value of this register is undefined/indeterministic when power up first time.<br><b>ValidValues:</b> <table border="1"> <thead> <tr> <th>Value</th><th>Description</th></tr> </thead> <tbody> <tr> <td>00h</td><td>2AM.</td></tr> <tr> <td>01h</td><td>1AM.</td></tr> <tr> <td>02h</td><td>2AM.</td></tr> <tr> <td>3Fh-03h</td><td>Reserved.</td></tr> </tbody> </table> | Value | Description | 00h | 2AM. | 01h | 1AM. | 02h | 2AM. | 3Fh-03h | Reserved. |
| Value   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |             |     |      |     |      |     |      |         |           |
| 00h     | 2AM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |     |      |     |      |     |      |         |           |
| 01h     | 1AM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |     |      |     |      |     |      |         |           |
| 02h     | 2AM.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |     |      |     |      |     |      |         |           |
| 3Fh-03h | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |             |     |      |     |      |     |      |         |           |

**PMx5F\_x02 [RTCEXT SprFwdMonth] (FCH::PM::RTCEXT::RTCEXTSprFwdMonth)**

Read-write.

\_aliasHOSTLEGACY; PMx5F\_x02; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |             |   |       |   |       |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|---|-------|---|-------|
| 7:5   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |             |   |       |   |       |
| 4:0   | <b>SprFwdMonth</b> . Read-write. Reset: XXXXXb. This Binary-Coded Decimal ( <a href="#">BCD</a> ) value determines which month to do the "spring forward". The value of this register is undefined/non-deterministic when powered up for the first time.<br><b>ValidValues:</b> <table border="1"> <thead> <tr> <th>Value</th><th>Description</th></tr> </thead> <tbody> <tr> <td>0</td><td>April</td></tr> <tr> <td>1</td><td>March</td></tr> </tbody> </table> | Value | Description | 0 | April | 1 | March |
| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |             |   |       |   |       |
| 0     | April                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |   |       |   |       |
| 1     | March                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |   |       |   |       |

**PMx5F\_x03 [RTCEXT FallBackCtrl] (FCH::PM::RTCEXT::RTCEXTFallBackCtrl)**

Read-write.

\_aliasHOSTLEGACY; PMx5F\_x03; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|---------|-----------|-----|-------|-----|-------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|-----|------|---------|-----------|-----|------|-----|------|-----|-------|-----|-------|-----|----------|---------|-----------|
| 7       | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 6       | <b>FallBackWeek.</b> Read-write. Reset: X. 0=The last week of the month. 1=The first week of the month. This value specifies which Sunday morning to do the "fall back". The value of this register is undefined/non-deterministic when powered up for the first time.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 5:0     | <p><b>FallBackHour.</b> Read-write. Reset: XXXXXXb. This Binary-Coded Decimal (<a href="#">BCD</a>) value specifies which hour (24 hour mode) to do the "fall back". Fall back is usually 2 AM in United States and 1 AM in Europe. 02h=2 AM. The value of this register is undefined/non-deterministic when powered up for the first time.</p> <p><b>ValidValues:</b></p> <table> <tr> <th>Value</th><th>Description</th></tr> <tr><td>00h</td><td>2 am</td></tr> <tr><td>01h</td><td>1 am</td></tr> <tr><td>02h</td><td>2 am</td></tr> <tr><td>03h</td><td>3 am</td></tr> <tr><td>04h</td><td>4 am</td></tr> <tr><td>05h</td><td>5 am</td></tr> <tr><td>06h</td><td>6 am</td></tr> <tr><td>07h</td><td>7 am</td></tr> <tr><td>08h</td><td>8 am</td></tr> <tr><td>09h</td><td>9 am</td></tr> <tr><td>0Fh-0Ah</td><td>Reserved.</td></tr> <tr><td>10h</td><td>10 am</td></tr> <tr><td>11h</td><td>11 am</td></tr> <tr><td>12h</td><td>Noon</td></tr> <tr><td>13h</td><td>1 pm</td></tr> <tr><td>14h</td><td>2 pm</td></tr> <tr><td>15h</td><td>3 pm</td></tr> <tr><td>16h</td><td>4 pm</td></tr> <tr><td>17h</td><td>5 pm</td></tr> <tr><td>18h</td><td>6 pm</td></tr> <tr><td>19h</td><td>7 pm</td></tr> <tr><td>1Fh-1Ah</td><td>Reserved.</td></tr> <tr><td>20h</td><td>8 pm</td></tr> <tr><td>21h</td><td>9 pm</td></tr> <tr><td>22h</td><td>10 pm</td></tr> <tr><td>23h</td><td>11 pm</td></tr> <tr><td>24h</td><td>Midnight</td></tr> <tr><td>3Fh-25h</td><td>Reserved.</td></tr> </table> | Value | Description | 00h | 2 am | 01h | 1 am | 02h | 2 am | 03h | 3 am | 04h | 4 am | 05h | 5 am | 06h | 6 am | 07h | 7 am | 08h | 8 am | 09h | 9 am | 0Fh-0Ah | Reserved. | 10h | 10 am | 11h | 11 am | 12h | Noon | 13h | 1 pm | 14h | 2 pm | 15h | 3 pm | 16h | 4 pm | 17h | 5 pm | 18h | 6 pm | 19h | 7 pm | 1Fh-1Ah | Reserved. | 20h | 8 pm | 21h | 9 pm | 22h | 10 pm | 23h | 11 pm | 24h | Midnight | 3Fh-25h | Reserved. |
| Value   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 00h     | 2 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 01h     | 1 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 02h     | 2 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 03h     | 3 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 04h     | 4 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 05h     | 5 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 06h     | 6 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 07h     | 7 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 08h     | 8 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 09h     | 9 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 0Fh-0Ah | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 10h     | 10 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 11h     | 11 am                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 12h     | Noon                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 13h     | 1 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 14h     | 2 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 15h     | 3 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 16h     | 4 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 17h     | 5 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 18h     | 6 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 19h     | 7 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 1Fh-1Ah | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 20h     | 8 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 21h     | 9 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 22h     | 10 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 23h     | 11 pm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 24h     | Midnight                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |
| 3Fh-25h | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |       |     |       |     |      |     |      |     |      |     |      |     |      |     |      |     |      |     |      |         |           |     |      |     |      |     |       |     |       |     |          |         |           |



**PMx5F\_x04 [RTCEXT FallBackMonth] (FCH::PM::RTCEXT::RTCEXTFallBackMonth)**

Read-write.

\_aliasHOSTLEGACY; PMx5F\_x04; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|-----|---------|-----|---------|-----|----------|-----|-------|-----|-------|-----|-----|-----|------|-----|------|-----|--------|-----|-----------|---------|-----------|-----|---------|-----|----------|-----|----------|---------|-----------|
| 7:5     | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 4:0     | <b>FallBackMonth.</b> Read-write. Reset: XXXXXb. This Binary-Coded Decimal ( <b>BCD</b> ) value specifies which month to "fall back". The value of this register is undefined/non-deterministic when powered up for the first time. <p><b>ValidValues:</b></p> <table> <tr> <th>Value</th><th>Description</th></tr> <tr><td>00h</td><td>October</td></tr> <tr><td>01h</td><td>January</td></tr> <tr><td>02h</td><td>February</td></tr> <tr><td>03h</td><td>March</td></tr> <tr><td>04h</td><td>April</td></tr> <tr><td>05h</td><td>May</td></tr> <tr><td>06h</td><td>June</td></tr> <tr><td>07h</td><td>July</td></tr> <tr><td>08h</td><td>August</td></tr> <tr><td>09h</td><td>September</td></tr> <tr><td>0Fh-0Ah</td><td>Reserved.</td></tr> <tr><td>10h</td><td>October</td></tr> <tr><td>11h</td><td>November</td></tr> <tr><td>12h</td><td>December</td></tr> <tr><td>1Fh-13h</td><td>Reserved.</td></tr> </table> | Value | Description | 00h | October | 01h | January | 02h | February | 03h | March | 04h | April | 05h | May | 06h | June | 07h | July | 08h | August | 09h | September | 0Fh-0Ah | Reserved. | 10h | October | 11h | November | 12h | December | 1Fh-13h | Reserved. |
| Value   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 00h     | October                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 01h     | January                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 02h     | February                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 03h     | March                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 04h     | April                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 05h     | May                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 06h     | June                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 07h     | July                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 08h     | August                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 09h     | September                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 0Fh-0Ah | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 10h     | October                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 11h     | November                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 12h     | December                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |
| 1Fh-13h | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |             |     |         |     |         |     |          |     |       |     |       |     |     |     |      |     |      |     |        |     |           |         |           |     |         |     |          |     |          |         |           |

**PMx5F\_x10 [RTCEXT WeekTimerControl] (FCH::PM::RTCEXT::RTCEXTWeekTimerCtl)**

Read-write.

The 16-bit Week Timer is a battery-powered down counter timer that supports 1ms, 1 second, and 1minute resolution and auto reloads when the timer reaches 0. The WEEK\_ALRM interrupt is asserted when the timer reaches 0 and stays asserted until the timer is disabled.

\_aliasHOSTLEGACY; PMx5F\_x10; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |             |    |         |    |         |    |     |    |           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------|----|---------|----|---------|----|-----|----|-----------|
| 7:3   | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |    |         |    |         |    |     |    |           |
| 2:1   | <b>Resolution.</b> Read-write. Reset: XXb. This field specifies the resolution of the Week Timer counter. Before programing this bit, software should program Enable to 0 to disable the Week Timer. The value of this register is undefined/non-deterministic when powered up for the first time. <p><b>ValidValues:</b></p> <table> <tr> <th>Value</th><th>Description</th></tr> <tr><td>0h</td><td>1minute</td></tr> <tr><td>1h</td><td>1second</td></tr> <tr><td>2h</td><td>1ms</td></tr> <tr><td>3h</td><td>Reserved.</td></tr> </table> | Value | Description | 0h | 1minute | 1h | 1second | 2h | 1ms | 3h | Reserved. |
| Value | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |             |    |         |    |         |    |     |    |           |
| 0h    | 1minute                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |    |         |    |         |    |     |    |           |
| 1h    | 1second                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |       |             |    |         |    |         |    |     |    |           |
| 2h    | 1ms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |       |             |    |         |    |         |    |     |    |           |
| 3h    | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |             |    |         |    |         |    |     |    |           |
| 0     | <b>Enable.</b> Read-write. Reset: X. 0=Disable Week Timer. 1=Enable Week Timer. The value of this register is undefined/non-deterministic when powered up for the first time.                                                                                                                                                                                                                                                                                                                                                                 |       |             |    |         |    |         |    |     |    |           |

**PMx5F\_x11 [RTCEXT WeekTimerReloadLow] (FCH::PM::RTCEXT::RTCEXTWeekTimerReloadLo)**

Read-write.

\_aliasHOSTLEGACY; PMx5F\_x11; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>WeekTimerReloadLow.</b> Read-write. Reset: XXh. This field is used to program the lower 8 bits of the 16-bit WeekTimerReload register. Writing the WeekTimerReloadLow register causes the 16-bit WeekTimerReload to be written into the Week Timer. Software should program FCH::PM::RTCEXT::RTCEXTWeekTimerCtl[Enable] = 0 before writing to this register. The value of this register is undefined/non-deterministic when powered up for the first time. |

**PMx5F\_x12 [RTCEXT WeekTimerReloadHigh] (FCH::PM::RTCEXT::RTCEXTWeekTimerReloadHi)**

Read-write.

\_aliasHOSTLEGACY; PMx5F\_x12; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>WeekTimerReloadHigh.</b> Read-write. Reset: XXh. This field is used to program the upper 8 bits of the 16-bit WeekTimerReload register. This field should be programmed before programming FCH::PM::RTCEXT::RTCEXTWeekTimerReloadLo. Software should program FCH::PM::RTCEXT::RTCEXTWeekTimerCtl[Enable] = 0 before writing to this register. The value of this register is undefined/non-deterministic when powered up for the first time. |

**PMx5F\_x13 [RTCEXT WeekTimerDataLow] (FCH::PM::RTCEXT::RTCEXTWeekTimerDataLo)**

Read-only.

\_aliasHOSTLEGACY; PMx5F\_x13; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                              |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>WeekTimerDataLow.</b> Read-only. Reset: XXh. This field is used to read the current state of the 16-bit Week Timer. Reading from the WeekTimerDataLow register returns the lower 8 bits of the 16-bit Week Timer and causes the upper 8 bits to be latched into FCH::PM::RTCEXT::RTCEXTWeekTimerDataHi. The value of this register is undefined/non-deterministic when powered up for the first time. |

**PMx5F\_x14 [RTCEXT WeekTimerDataHigh] (FCH::PM::RTCEXT::RTCEXTWeekTimerDataHi)**

Read-only.

\_aliasHOSTLEGACY; PMx5F\_x14; PM=FED8\_0300h; DataPortWrite=FCH::PM::RTCEXTINDEX

| Bits | Description                                                                                                                                                                                                                                                                                                                                                                                   |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:0  | <b>WeekTimerDataHigh.</b> Read-only. Reset: XXh. This field is used to read the current state of the 16-bit Week Timer. Reading from the WeekTimerDataHigh register returns the upper 8 bits of the 16-bit Week Timer latched by a previous read from FCH::PM::RTCEXT::RTCEXTWeekTimerReloadLo. The value of this register is undefined/non-deterministic when powered up for the first time. |

**14.3.15 GPIO Pin control registers****14.3.15.1 IOMUX Registers**

Note: In function 0 (IOMUX==00) ~ function 3 (IOMUX==03) of the IOMUXx3 in the Table 257 [IOMUX Function Table], SPD\_HOST\_CTRL\_L is a SW related function, it is controlled by platform BIOS and [BMC](#), not a HW native function.

Table 257: IOMUX Function Table

| IOMUX | Pin Name | Do mai | GP IO | GEV ENT | Over ride_ | Over ride_ | IOMUX==00 | IOMUX==01 | IOMUX==02 | IOMUX==03 | Def ault | IOMUX |
|-------|----------|--------|-------|---------|------------|------------|-----------|-----------|-----------|-----------|----------|-------|
|-------|----------|--------|-------|---------|------------|------------|-----------|-----------|-----------|-----------|----------|-------|

|              |                                                     | n  | #  | #  | 0 | 1             |                               |                                   |                                   |                                   |     | value<br>at<br>reset |
|--------------|-----------------------------------------------------|----|----|----|---|---------------|-------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----|----------------------|
| IOMU<br>Xx0  | PWR_BTN_<br>L_AGPI<br>O0                            | S5 | 0  | 21 |   |               | PWR_BTN_<br>L                 | GPIO0_<br>pu1pd0                  | GPIO0                             | GPIO0                             | PU  | 0                    |
| IOMU<br>Xx1  | SYS_RESE<br>T_L_AGPI<br>O1                          | S5 | 1  | 19 |   | RST_<br>strap | SYS_RESE<br>T_L               | GPIO1                             | GPIO1                             | GPIO1                             | PU  | 0                    |
| IOMU<br>Xx2  | WAKE_L_<br>AGPIO2                                   | S5 | 2  | 8  |   |               | WAKE_L                        | GPIO2                             | GPIO2                             | GPIO2                             | PU  | 0                    |
| IOMU<br>Xx3  | AGPIO3_S<br>PD_HOST<br>_CTRL_L                      | S5 | 3  | 2  |   |               | GPIO3/<br>SPD_HOST<br>_CTRL_L | GPIO3/<br>SPD_H<br>OST_C<br>TRL_L | GPIO3/<br>SPD_H<br>OST_C<br>TRL_L | GPIO3/<br>SPD_H<br>OST_C<br>TRL_L | PU  | 0                    |
| IOMU<br>Xx4  | AGPIO4_S<br>ATA_ACT<br>_L                           | S5 | 4  | 4  |   |               | GPIO4                         | SATA_<br>ACT_L                    | GPIO4                             | GPIO4                             | PU  | 0                    |
| IOMU<br>Xx5  | AGPIO5_D<br>EVSLP0_S<br>ATA_ZP0_<br>L               | S5 | 5  | 7  |   |               | GPIO5                         | DEVSL<br>P0                       | GPIO5                             | GPIO5                             | PD  | 0                    |
| IOMU<br>Xx6  | AGPIO6_D<br>EVSLP1_S<br>ATA_ZP1_<br>L               | S5 | 6  | 10 |   |               | GPIO6                         | DEVSL<br>P1                       | GPIO6                             | GPIO6                             | PD  | 0                    |
| IOMU<br>Xx7  | AGPIO7                                              | S5 | 7  |    |   |               | GPIO7                         | GPIO7                             | GPIO7                             | GPIO7                             | PD  | 0                    |
| IOMU<br>XxC  | PWRGD_<br>OUT_AGPI<br>O12                           | S5 | 12 |    |   |               | PWRGD_O<br>UT                 | GPIO12                            | GPIO12                            | GPIO12                            | N/A | 0                    |
| IOMU<br>XxD  | I2C4_SCL<br>_HP_SCL_<br>UBM_SCL_<br>_AGPIO13        | S5 | 13 |    |   |               | I2C4_SCL                      | CLK_48<br>_24_0                   | GPIO13                            | GPIO13                            | N/A | 0                    |
| IOMU<br>XxE  | I2C4_SDA<br>_HP_SDA_<br>UBM_SDA_<br>_AGPIO14        | S5 | 14 |    |   |               | I2C4_SDA                      | PowerSa<br>vingMo<br>de           | GPIO14                            | GPIO14                            | N/A | 0                    |
| IOMU<br>Xx10 | USB10_O<br>C_L_AGPI<br>O16                          | S5 | 16 | 12 |   |               | USB10_OC<br>_L                | GPIO16                            | GPIO16                            | GPIO16                            | PU  | 0                    |
| IOMU<br>Xx11 | USB11_O<br>C_L_AGPI<br>O17                          | S5 | 17 | 13 |   |               | USB11_OC<br>_L                | GPIO17                            | GPIO17                            | GPIO17                            | PU  | 0                    |
| IOMU<br>Xx13 | I2C5_SCL<br>_BMC_SC<br>L_SMBUS<br>1_SCL_AG<br>PIO19 | S5 | 19 |    |   |               | I2C5_SCL                      | SMBUS<br>1_SCL                    | GPIO19                            | GPIO19                            | N/A | 0                    |
| IOMU<br>Xx14 | I2C5_SDA<br>_BMC_SD                                 | S5 | 20 |    |   |               | I2C5_SDA                      | SMBUS<br>1_SDA                    | GPIO20                            | GPIO20                            | N/A | 0                    |

|              |                                             |    |     |    |                     |                           |                      |                      |        |             |     |   |
|--------------|---------------------------------------------|----|-----|----|---------------------|---------------------------|----------------------|----------------------|--------|-------------|-----|---|
|              | A_SMBUS<br>1_SDA_A<br>GPIO20                |    |     |    |                     |                           |                      |                      |        |             |     |   |
| IOMU<br>Xx15 | AGPIO21                                     | S5 | 21  | 5  |                     |                           | GPIO21               | GPIO21               | GPIO21 |             | PD  | 0 |
| IOMU<br>Xx16 | AGPIO22                                     | S5 | 22  | 3  |                     |                           | GPIO22               |                      | GPIO22 | GPIO22      | PD  | 0 |
| IOMU<br>Xx17 | ESPI_RST<br>OUT_L_A<br>GPIO23               | S5 | 23  | 16 |                     |                           | ESPI_RSTO<br>UT_L    | GPIO23               |        | GPIO23      | N/A | 0 |
| IOMU<br>Xx18 | SMERR_L<br>_AGPIO24                         | S5 | 24  | 14 |                     |                           | SMERR_L              | GPIO24               | GPIO24 | GPIO24      | PU  | 0 |
| IOMU<br>Xx1A | PCIE_RST<br>1_L_AGPI<br>O26                 | S5 | 26  | 15 |                     |                           | PCIE_RST1<br>_L      | GPIO26               | GPIO26 | GPIO26      | N/A | 0 |
| IOMU<br>Xx4A | ESPI_CLK<br>2_AGPIO7<br>4                   | S0 | 74  |    |                     |                           | ESPI_CLK2            | GPIO74               | GPIO74 | GPIO74      | PD  | 0 |
| IOMU<br>Xx4B | ESPI_CLK<br>1_SPI_CL<br>K1_AGPI<br>O75      | S0 | 75  |    |                     | SPI3<br>2Bit_<br>strap    | ESPI_CLK1            | GPIO75               | GPIO75 |             | PD  | 0 |
| IOMU<br>Xx4C | AGPIO76_<br>SPI_TPM_<br>CS_L                | S5 | 76  | 11 |                     |                           | GPIO76               | SPI_TP<br>M_CS_<br>L | GPIO76 | GPIO76      | PD  | 0 |
| IOMU<br>Xx56 | NMI_SYN<br>C_FLOOD<br>_L                    | S0 | 86  | 9  |                     |                           | NMI_SYNC<br>_FLOOD_L |                      |        |             | PD  | 0 |
| IOMU<br>Xx57 | AGPIO87                                     | S0 | 87  |    |                     |                           | GPIO87               |                      | GPIO87 | GPIO87      | PD  | 0 |
| IOMU<br>Xx58 | AGPIO88                                     | S0 | 88  |    |                     |                           | GPIO88               |                      | GPIO88 | GPIO88      | PD  | 0 |
| IOMU<br>Xx59 | GENINT_<br>L_PM_INT<br>R_L_AGPI<br>O89      | S0 | 89  | 0  |                     |                           | GENINT_L             | PM_IN<br>TR_L        | GPIO89 | GPIO89      | PU  | 0 |
| IOMU<br>Xx68 | AGPIO104                                    | S0 | 104 | 20 |                     |                           | GPIO104              | GPIO10<br>4          |        | GPIO10<br>4 | PD  | 0 |
| IOMU<br>Xx69 | AGPIO105                                    | S0 | 105 | 22 |                     |                           | GPIO105              | GPIO10<br>5          |        | GPIO10<br>5 | PD  | 0 |
| IOMU<br>Xx6A | AGPIO106                                    | S0 | 106 | 23 |                     |                           | GPIO106              | GPIO10<br>6          |        | GPIO10<br>6 | PD  | 0 |
| IOMU<br>Xx6B | AGPIO107                                    | S0 | 107 |    |                     |                           | GPIO107              | GPIO10<br>7          |        | GPIO10<br>7 | PD  | 0 |
| IOMU<br>Xx6C | ESPI0_AL<br>ERT_L_ES<br>PI_IO1_A<br>GPIO108 | S0 | 108 |    |                     | ESPI<br>0_AL<br>ERT_<br>L | ESPI0_ALE<br>RT_D1   | GPIO10<br>8          |        | GPIO10<br>8 | PU  | 0 |
| IOMU<br>Xx6D | AGPIO109                                    | S0 | 109 |    | ROM<br>TYP<br>E_str |                           | GPIO109              | GPIO10<br>9          |        | GPIO10<br>9 | PU  | 0 |

|              |                                           |    |     |    |                          |                           |                              |                 |             |             |    |   |
|--------------|-------------------------------------------|----|-----|----|--------------------------|---------------------------|------------------------------|-----------------|-------------|-------------|----|---|
|              |                                           |    |     |    | ap                       |                           |                              |                 |             |             |    |   |
| IOMU<br>Xx6E | ESPI1_AL<br>ERT_L_A<br>GPIO110            | S0 | 110 |    |                          | ESPI<br>1_AL<br>ERT_<br>L | ESPI1_ALE<br>RT_D1           | GPIO11<br>0     | GPIO11<br>0 | GPIO11<br>0 | PU | 0 |
| IOMU<br>Xx73 | AGPIO115<br>_CLK_RE<br>Q11_L              | S0 | 115 | 1  |                          |                           | GPIO115                      | CLK_R<br>EQ11_L | GPIO11<br>5 | GPIO11<br>5 | PU | 0 |
| IOMU<br>Xx74 | AGPIO116<br>_CLK_RE<br>Q12_L              | S0 | 116 | 6  |                          |                           | GPIO116                      | CLK_R<br>EQ12_L | GPIO11<br>6 | GPIO11<br>6 | PU | 0 |
| IOMU<br>Xx75 | ESPI_CLK<br>0_SPI_CL<br>K0_AGPIO<br>117   | S0 | 117 |    | CLK<br>GEN<br>_stra<br>p |                           | ESPI_CLK0                    | GPIO11<br>7     | GPIO11<br>7 | GPIO11<br>7 | PD | 0 |
| IOMU<br>Xx76 | SPI_CS0_L<br>_AGPIO11<br>8                | S0 | 118 |    |                          |                           | SPI_CS0_L                    | GPIO11<br>8     | GPIO11<br>8 | GPIO11<br>8 | PU | 0 |
| IOMU<br>Xx77 | SPI_CS1_L<br>_AGPIO11<br>9                | S0 | 119 |    |                          |                           | SPI_CS1_L                    | GPIO11<br>9     | GPIO11<br>9 | GPIO11<br>9 | PU | 0 |
| IOMU<br>Xx78 | ESPI0_DA<br>T0_SPI0_D<br>AT0_AGP<br>IO120 | S0 | 120 |    |                          |                           | ESPI0_DAT<br>0/<br>SPI0_DAT0 | GPIO12<br>0     | GPIO12<br>0 | GPIO12<br>0 | PD | 0 |
| IOMU<br>Xx79 | ESPI0_DA<br>T1_SPI0_D<br>AT1_AGP<br>IO121 | S0 | 121 |    |                          |                           | ESPI0_DAT<br>1/<br>SPI0_DAT1 | GPIO12<br>1     | GPIO12<br>1 | GPIO12<br>1 | PU | 0 |
| IOMU<br>Xx7A | ESPI0_DA<br>T2_SPI0_D<br>AT2_AGP<br>IO122 | S0 | 122 |    |                          |                           | ESPI0_DAT<br>2/<br>SPI0_DAT2 | GPIO12<br>2     | GPIO12<br>2 | GPIO12<br>2 | PU | 0 |
| IOMU<br>Xx7B | ESPI0_DA<br>T3_SPI0_D<br>AT3_AGP<br>IO123 | S0 | 123 |    |                          |                           | ESPI0_DAT<br>3/<br>SPI0_DAT3 | GPIO12<br>3     | GPIO12<br>3 | GPIO12<br>3 | PU | 0 |
| IOMU<br>Xx7C | ESPI_CS0_<br>L_AGPIO1<br>24               | S0 | 124 |    |                          |                           | ESPI_CS0_<br>L               | GPIO12<br>4     | GPIO12<br>4 | GPIO12<br>4 | PU | 0 |
| IOMU<br>Xx7D | ESPI_CS1_<br>L_AGPIO1<br>25               | S0 | 125 |    |                          |                           | ESPI_CS1_<br>L               | GPIO12<br>5     | GPIO12<br>5 | GPIO12<br>5 | PU | 0 |
| IOMU<br>Xx7E | SPI_CS2_L<br>_AGPIO12<br>6                | S0 | 126 |    |                          |                           | SPI_CS2_L                    | GPIO12<br>6     | GPIO12<br>6 | GPIO12<br>6 | PU | 0 |
| IOMU<br>Xx81 | ESPI_RST<br>N_L_KBR<br>ST_L_AGP<br>IO129  | S0 | 129 | 17 |                          |                           | ESPI_RST<br>N_L              | KBRST<br>_L     | GPIO12<br>9 | GPIO12<br>9 | PU | 0 |
| IOMU<br>Xx83 | ESPI1_DA<br>T0_SPI1_D                     | S0 | 131 |    |                          | WAF<br>L_TI               | ESPI1_DAT<br>0/              | GPIO13<br>1     | GPIO13<br>1 | GPIO13<br>1 | PU | 0 |

|              |                                                                    |    |     |  |  |                            |                              |                               |                |             |     |   |
|--------------|--------------------------------------------------------------------|----|-----|--|--|----------------------------|------------------------------|-------------------------------|----------------|-------------|-----|---|
|              | AT0_AGPI<br>O131                                                   |    |     |  |  | MEO<br>UT_s<br>trap        | SPI1_DAT0                    |                               |                |             |     |   |
| IOMU<br>Xx84 | ESPI1_DA<br>T1_SPI1_D<br>AT1_AGPI<br>O132                          | S0 | 132 |  |  |                            | ESPI1_DAT<br>1/<br>SPI1_DAT1 | GPIO13<br>2                   | GPIO13<br>2    | GPIO13<br>2 | PU  | 0 |
| IOMU<br>Xx85 | ESPI1_DA<br>T2_SPI1_D<br>AT2_AGPI<br>O133                          | S0 | 133 |  |  |                            | ESPI1_DAT<br>2/<br>SPI1_DAT2 | GPIO13<br>3                   | GPIO13<br>3    | GPIO13<br>3 | PU  | 0 |
| IOMU<br>Xx86 | ESPI1_DA<br>T3_SPI1_D<br>AT3_AGPI<br>O134                          | S0 | 134 |  |  |                            | ESPI1_DAT<br>3/<br>SPI1_DAT3 | GPIO13<br>4                   | GPIO13<br>4    | GPIO13<br>4 | PU  | 0 |
| IOMU<br>Xx87 | UART0_C<br>TS_L_UA<br>RT2_TXD_<br>AGPIO135                         | S0 | 135 |  |  |                            | UART0_CT<br>S_L              | UART2<br>_TXD                 | GPIO13<br>5    | GPIO13<br>5 | PD  | 0 |
| IOMU<br>Xx88 | UART0_R<br>XD_AGPI<br>O136                                         | S0 | 136 |  |  |                            | UART0_RX<br>D                | GPIO13<br>6                   | GPIO13<br>6    | GPIO13<br>6 | PD  | 0 |
| IOMU<br>Xx89 | UART0_R<br>TS_L_UA<br>RT2_RXD_<br>_AGPIO13<br>7                    | S0 | 137 |  |  |                            | UART0_RT<br>S_L              | UART2<br>_RXD                 | GPIO13<br>7    | GPIO13<br>7 | PU  | 0 |
| IOMU<br>Xx8A | UART0_T<br>XD_AGPI<br>O138                                         | S0 | 138 |  |  |                            | UART0_TX<br>D                | GPIO13<br>8                   | GPIO13<br>8    | GPIO13<br>8 | PU  | 0 |
| IOMU<br>Xx8B | UART0_IN<br>TR_AGPI<br>O139                                        | S0 | 139 |  |  |                            | UART0_IN<br>TR               | GPIO13<br>9                   | GPIO13<br>9    | GPIO13<br>9 | PD  | 0 |
| IOMU<br>Xx8D | UART1_R<br>XD_AGPI<br>O141                                         | S0 | 141 |  |  |                            | UART1_RX<br>D                | GPIO14<br>1                   | GPIO14<br>1    | GPIO14<br>1 | PD  | 0 |
| IOMU<br>Xx8E | UART1_T<br>XD_AGPI<br>O142                                         | S0 | 142 |  |  | ROM<br>TYP<br>E1_st<br>rap | UART1_TX<br>D                | GPIO14<br>2                   | GPIO14<br>2    | GPIO14<br>2 | PU  | 0 |
| IOMU<br>Xx91 | I3C0_SCL<br>_I2C0_SC<br>L_SPD0_S<br>CL_SMBU<br>S0_SCL_A<br>GPIO145 | S0 | 145 |  |  |                            | I3C0_SCL/<br>SPD0_SCL        | I2C0_S<br>CL/<br>SPD0_S<br>CL | SMBUS<br>0_SCL | GPIO14<br>5 | n/a | 0 |
| IOMU<br>Xx92 | I3C0_SDA<br>_I2C0_SD<br>A_SPD0_S<br>DA_SMBU<br>S0_SDA_A<br>GPIO146 | S0 | 146 |  |  |                            | I3C0_SDA/<br>SPD0_SDA        | I2C0_S<br>DA/<br>SPD0_S<br>DA | SMBUS<br>0_SDA | GPIO14<br>6 | n/a | 0 |

|              |                                                     |    |     |  |  |  |                       |                               |             |             |     |   |
|--------------|-----------------------------------------------------|----|-----|--|--|--|-----------------------|-------------------------------|-------------|-------------|-----|---|
| IOMU<br>Xx93 | I3C1_SCL<br>_I2C1_SC<br>L_SPD1_S<br>CL_AGPI<br>O147 | S0 | 147 |  |  |  | I3C1_SCL/<br>SPD1_SCL | I2C1_S<br>CL/<br>SPD1_S<br>CL | GPIO14<br>7 | GPIO14<br>7 | n/a | 0 |
| IOMU<br>Xx94 | I3C1_SDA<br>_I2C1_SD<br>A_SPD1_S<br>DA_AGPI<br>O148 | S0 | 148 |  |  |  | I3C1_SDA/<br>SPD1_SDA | I2C1_S<br>DA/<br>SPD1_S<br>DA | GPIO14<br>8 | GPIO14<br>8 | n/a | 0 |
| IOMU<br>Xx95 | I3C2_SCL<br>_I2C2_SC<br>L_SPD2_S<br>CL_AGPI<br>O149 | S0 | 149 |  |  |  | I3C2_SCL/<br>SPD2_SCL | I2C2_S<br>CL/<br>SPD2_S<br>CL | GPIO14<br>9 | GPIO14<br>9 | n/a | 0 |
| IOMU<br>Xx96 | I3C2_SDA<br>_I2C2_SD<br>A_SPD2_S<br>DA_AGPI<br>O150 | S0 | 150 |  |  |  | I3C2_SDA/<br>SPD2_SDA | I2C2_S<br>DA/<br>SPD2_S<br>DA | GPIO15<br>0 | GPIO15<br>0 | n/a | 0 |
| IOMU<br>Xx97 | I3C3_SCL<br>_I2C3_SC<br>L_SPD3_S<br>CL_AGPI<br>O151 | S0 | 151 |  |  |  | I3C3_SCL/<br>SPD3_SCL | I2C3_S<br>CL/<br>SPD3_S<br>CL | GPIO15<br>1 | GPIO15<br>1 | n/a | 0 |
| IOMU<br>Xx98 | I3C3_SDA<br>_I2C3_SD<br>A_SPD3_S<br>DA_AGPI<br>O152 | S0 | 152 |  |  |  | I3C3_SDA/<br>SPD3_SDA | I2C3_S<br>DA/<br>SPD3_S<br>DA | GPIO15<br>2 | GPIO15<br>2 | n/a | 0 |

The IOMUX register space is accessed through the AcpiMmio region. The registers range from FED8\_0000h+D00h to FED8\_0000h+DFFh. See FCH::PM::ISACONTROL[mmioen].

If the platform supports Hardware-reduced ACPI where FCH::PM::ACPICONFIG[acpireducedhwen]==1, then the GPIO/GEvent pins do not generate SCI (As required per ACPI 5.0 specification). The GPIO pins identified as AGPIO are capable of generating the interrupt as GPIO-signaled ACPI Events.

AGPIO: Advanced GPIO - this pin can be used for interrupt, wake, or GPIO.

If the platform does not support Hardware-reduced ACPI where FCH::PM::ACPICONFIG[acpireducedhwen]==0, then the GPIO/Gevents function as supported previously.

Override\_0/1 columns in the IOMUX Function Table identify pin function in certain circumstances when the IOMUX selection is ignored. At power-on reset, any '\*\_strap' entry is used to provide advice input for the named function. In SPI mode, selected when FCH::PM::ACPICONFIG[force\_spiusepin\_0]==0, any '\*SPI\*' entry becomes the function of the pin.

For IOMUX registers, example FCH::IOMUX::IOMUX0\_GPIO, there are multi-function IO pin function select (function-0/function-1/function-2/function-3), it is general description in the table, more detailed description for these function-0/function-1/function-2/function-3 refer to Table 257 [IOMUX Function Table].

**IOMUXx00000000 (FCH::IOMUX::IOMUX0\_GPIO)**

Read-write. Reset: 00h.

**\*Note:**

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000000; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000001 (FCH::IOMUX::IOMUX1\_GPIO)**

Read-write. Reset: 00h.

**\*Note:**

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000001; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000002 (FCH::IOMUX::IOMUX2\_GPIO)**

Read-write. Reset: 00h.

**\*Note:**

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000002; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |



**IOMUXx00000003 (FCH::IOMUX::IOMUX3\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000003; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000004 (FCH::IOMUX::IOMUX4\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000004; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000005 (FCH::IOMUX::IOMUX5\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000005; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000006 (FCH::IOMUX::IOMUX6\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000006; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000007 (FCH::IOMUX::IOMUX7\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000007; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000008 (FCH::IOMUX::IOMUX8\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000008; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000009 (FCH::IOMUX::IOMUX9\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000009; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000000A (FCH::IOMUX::IOMUX10\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000000A; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000000B (FCH::IOMUX::IOMUX11\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000000B; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000000C (FCH::IOMUX::IOMUX12\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000000C; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000000D (FCH::IOMUX::IOMUX13\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000000D; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000000E (FCH::IOMUX::IOMUX14\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000000E; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000000F (FCH::IOMUX::IOMUX15\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000000F; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000010 (FCH::IOMUX::IOMUX16\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000010; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000011 (FCH::IOMUX::IOMUX17\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000011; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000012 (FCH::IOMUX::IOMUX18\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000012; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000013 (FCH::IOMUX::IOMUX19\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000013; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000014 (FCH::IOMUX::IOMUX20\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000014; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000015 (FCH::IOMUX::IOMUX21\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000015; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000016 (FCH::IOMUX::IOMUX22\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000016; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000017 (FCH::IOMUX::IOMUX23\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000017; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000018 (FCH::IOMUX::IOMUX24\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000018; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000019 (FCH::IOMUX::IOMUX25\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000019; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000001A (FCH::IOMUX::IOMUX26\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000001A; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |



**IOMUXx0000001B (FCH::IOMUX::IOMUX27\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000001B; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000001C (FCH::IOMUX::IOMUX28\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000001C; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000001D (FCH::IOMUX::IOMUX29\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000001D; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000001E (FCH::IOMUX::IOMUX30\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000001E; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000001F (FCH::IOMUX::IOMUX31\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000001F; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000020 (FCH::IOMUX::IOMUX32\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000020; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000021 (FCH::IOMUX::IOMUX33\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000021; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000022 (FCH::IOMUX::IOMUX34\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000022; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000023 (FCH::IOMUX::IOMUX35\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000023; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000024 (FCH::IOMUX::IOMUX36\_GPIO)**

Read-write. Reset: 00h.

**\*Note:**

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000024; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000025 (FCH::IOMUX::IOMUX37\_GPIO)**

Read-write. Reset: 00h.

**\*Note:**

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000025; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000026 (FCH::IOMUX::IOMUX38\_GPIO)**

Read-write. Reset: 00h.

**\*Note:**

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000026; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000027 (FCH::IOMUX::IOMUX39\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000027; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000028 (FCH::IOMUX::IOMUX40\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000028; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000029 (FCH::IOMUX::IOMUX41\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000029; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000002A (FCH::IOMUX::IOMUX42\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000002A; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000002B (FCH::IOMUX::IOMUX43\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000002B; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000002C (FCH::IOMUX::IOMUX44\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000002C; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000002D (FCH::IOMUX::IOMUX45\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000002D; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000002E (FCH::IOMUX::IOMUX46\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000002E; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000002F (FCH::IOMUX::IOMUX47\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000002F; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000030 (FCH::IOMUX::IOMUX48\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000030; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000031 (FCH::IOMUX::IOMUX49\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000031; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000032 (FCH::IOMUX::IOMUX50\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000032; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |



**IOMUXx00000033 (FCH::IOMUX::IOMUX51\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000033; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000034 (FCH::IOMUX::IOMUX52\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000034; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000035 (FCH::IOMUX::IOMUX53\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000035; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000036 (FCH::IOMUX::IOMUX54\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000036; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000037 (FCH::IOMUX::IOMUX55\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000037; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000038 (FCH::IOMUX::IOMUX56\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000038; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000039 (FCH::IOMUX::IOMUX57\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000039; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000003A (FCH::IOMUX::IOMUX58\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000003A; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000003B (FCH::IOMUX::IOMUX59\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000003B; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000003C (FCH::IOMUX::IOMUX60\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000003C; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000003D (FCH::IOMUX::IOMUX61\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000003D; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000003E (FCH::IOMUX::IOMUX62\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000003E; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000003F (FCH::IOMUX::IOMUX63\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000003F; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000040 (FCH::IOMUX::IOMUX64\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000040; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000041 (FCH::IOMUX::IOMUX65\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000041; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000042 (FCH::IOMUX::IOMUX66\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000042; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000043 (FCH::IOMUX::IOMUX67\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000043; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000044 (FCH::IOMUX::IOMUX68\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000044; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000045 (FCH::IOMUX::IOMUX69\_GPIO)**

Read-write. Reset: 00h.

**\*Note:**

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000045; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000046 (FCH::IOMUX::IOMUX70\_GPIO)**

Read-write. Reset: 00h.

**\*Note:**

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000046; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000047 (FCH::IOMUX::IOMUX71\_GPIO)**

Read-write. Reset: 00h.

**\*Note:**

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000047; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000048 (FCH::IOMUX::IOMUX72\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000048; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000049 (FCH::IOMUX::IOMUX73\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000049; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000004A (FCH::IOMUX::IOMUX74\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000004A; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |



**IOMUXx0000004B (FCH::IOMUX::IOMUX75\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000004B; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000004C (FCH::IOMUX::IOMUX76\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000004C; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000004D (FCH::IOMUX::IOMUX77\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000004D; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000004E (FCH::IOMUX::IOMUX78\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000004E; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000004F (FCH::IOMUX::IOMUX79\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000004F; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000050 (FCH::IOMUX::IOMUX80\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000050; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000051 (FCH::IOMUX::IOMUX81\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000051; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000052 (FCH::IOMUX::IOMUX82\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000052; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000053 (FCH::IOMUX::IOMUX83\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000053; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000054 (FCH::IOMUX::IOMUX84\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000054; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000055 (FCH::IOMUX::IOMUX85\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000055; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000056 (FCH::IOMUX::IOMUX86\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000056; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000057 (FCH::IOMUX::IOMUX87\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000057; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000058 (FCH::IOMUX::IOMUX88\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000058; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000059 (FCH::IOMUX::IOMUX89\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000059; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000005A (FCH::IOMUX::IOMUX90\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000005A; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000005B (FCH::IOMUX::IOMUX91\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000005B; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000005C (FCH::IOMUX::IOMUX92\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000005C; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000005D (FCH::IOMUX::IOMUX93\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000005D; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000005E (FCH::IOMUX::IOMUX94\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000005E; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000005F (FCH::IOMUX::IOMUX95\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000005F; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000060 (FCH::IOMUX::IOMUX96\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000060; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000061 (FCH::IOMUX::IOMUX97\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000061; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000062 (FCH::IOMUX::IOMUX98\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000062; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |



**IOMUXx00000063 (FCH::IOMUX::IOMUX99\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000063; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000064 (FCH::IOMUX::IOMUX100\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000064; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000065 (FCH::IOMUX::IOMUX101\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000065; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000066 (FCH::IOMUX::IOMUX102\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000066; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000067 (FCH::IOMUX::IOMUX103\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000067; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000068 (FCH::IOMUX::IOMUX104\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000068; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000069 (FCH::IOMUX::IOMUX105\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000069; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000006A (FCH::IOMUX::IOMUX106\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000006A; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000006B (FCH::IOMUX::IOMUX107\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000006B; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000006C (FCH::IOMUX::IOMUX108\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000006C; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000006D (FCH::IOMUX::IOMUX109\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000006D; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000006E (FCH::IOMUX::IOMUX110\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000006E; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000006F (FCH::IOMUX::IOMUX111\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000006F; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000070 (FCH::IOMUX::IOMUX112\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000070; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000071 (FCH::IOMUX::IOMUX113\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000071; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000072 (FCH::IOMUX::IOMUX114\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000072; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000073 (FCH::IOMUX::IOMUX115\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000073; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000074 (FCH::IOMUX::IOMUX116\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000074; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000075 (FCH::IOMUX::IOMUX117\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000075; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000076 (FCH::IOMUX::IOMUX118\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000076; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000077 (FCH::IOMUX::IOMUX119\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000077; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000078 (FCH::IOMUX::IOMUX120\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000078; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000079 (FCH::IOMUX::IOMUX121\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000079; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000007A (FCH::IOMUX::IOMUX122\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000007A; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |



**IOMUXx0000007B (FCH::IOMUX::IOMUX123\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000007B; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000007C (FCH::IOMUX::IOMUX124\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000007C; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000007D (FCH::IOMUX::IOMUX125\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000007D; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000007E (FCH::IOMUX::IOMUX126\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000007E; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000007F (FCH::IOMUX::IOMUX127\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000007F; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000080 (FCH::IOMUX::IOMUX128\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000080; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000081 (FCH::IOMUX::IOMUX129\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000081; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000082 (FCH::IOMUX::IOMUX130\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000082; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000083 (FCH::IOMUX::IOMUX131\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000083; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000084 (FCH::IOMUX::IOMUX132\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000084; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000085 (FCH::IOMUX::IOMUX133\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000085; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000086 (FCH::IOMUX::IOMUX134\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000086; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000087 (FCH::IOMUX::IOMUX135\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000087; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000088 (FCH::IOMUX::IOMUX136\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000088; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000089 (FCH::IOMUX::IOMUX137\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000089; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000008A (FCH::IOMUX::IOMUX138\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000008A; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000008B (FCH::IOMUX::IOMUX139\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000008B; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000008C (FCH::IOMUX::IOMUX140\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000008C; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000008D (FCH::IOMUX::IOMUX141\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000008D; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000008E (FCH::IOMUX::IOMUX142\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000008E; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx0000008F (FCH::IOMUX::IOMUX143\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx0000008F; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000090 (FCH::IOMUX::IOMUX144\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000090; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000091 (FCH::IOMUX::IOMUX145\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000091; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000092 (FCH::IOMUX::IOMUX146\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000092; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |



**IOMUXx00000093 (FCH::IOMUX::IOMUX147\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000093; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000094 (FCH::IOMUX::IOMUX148\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000094; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000095 (FCH::IOMUX::IOMUX149\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000095; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000096 (FCH::IOMUX::IOMUX150\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000096; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000097 (FCH::IOMUX::IOMUX151\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000097; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**IOMUXx00000098 (FCH::IOMUX::IOMUX152\_GPIO)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number in hexadecimal: 00h ~ 98h.

&lt;X&gt; denotes number in decimal: 0 ~ 152.

\_aliasHOSTLEGACY; IOMUXx00000098; IOMUX=FED8\_0D00h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**14.3.15.2 GPIO Registers**

The GPIO pins are controlled by a combination of device enables and by their specific GPIO and IOMUX register pair.

GPIO registers are accessed through memory-mapped ACPIMMIO region. The offset is relative to FED8\_0000h+1500h. GPIO bank 0 registers range from FED8\_0000h+1500h to FED8\_0000h+15FFh. GPIO Bank 1 registers range from

FED8\_0000h+1600h to FED8\_0000h+16FFh. GPIO Bank 2 registers range from FED8\_0000h+1700h to FED8\_0000h+17FFh. GPIO Bank 3 registers range from FED8\_0000h+1800h to FED8\_0000h+18FFh.

Table 258: I2C/I3C Pad Configuration Method

|           | PAD name                                           | GPIO register(bank, offset)<br>(No use) | Misc_Reg(Control I2C<br>PAD) |
|-----------|----------------------------------------------------|-----------------------------------------|------------------------------|
| I2C0/I3C0 | I3C0_SCL_I2C0_SCL_SPD0_SCL_<br>SMBUS0_SCL_AGPI0145 | 0x244                                   | 0xD8                         |
|           | I3C0_SDA_I2C0_SDA_SPD0_SDA_<br>SMBUS0_SDA_AGPI0146 | 0x248                                   |                              |
| I2C1/I3C1 | I3C1_SCL_I2C1_SCL_SPD1_SCL_<br>AGPI0147            | 0x24C                                   | 0xDC                         |
|           | I3C1_SDA_I2C1_SDA_SPD1_SDA_<br>AGPI0148            | 0x250                                   |                              |
| I2C2/I3C3 | I3C2_SCL_I2C2_SCL_SPD2_SCL_<br>AGPI0149            | 0x254                                   | 0xE0                         |
|           | I3C2_SDA_I2C2_SDA_SPD2_SDA_<br>AGPI0150            | 0x258                                   |                              |
| I2C3/I3C3 | I3C3_SCL_I2C3_SCL_SPD3_SCL_<br>AGPI0151            | 0x25C                                   | 0xE4                         |
|           | I3C3_SDA_I2C3_SDA_SPD3_SDA_<br>AGPI0152            | 0x260                                   |                              |
| I2C4      | I2C4_SCL_HP_SCL_UBM_SCL_A<br>GPI013                | 0x044                                   | 0xE8                         |
|           | I2C4_SDA_HP_SDA_UBM_SDA_<br>AGPI014                | 0x048                                   |                              |
| I2C5      | I2C5_SCL_BMC_SCL_SMBUS1_S<br>CL_AGPI019            | 0x04C                                   | 0xEC                         |
|           | I2C5_SDA_BMC_SDA_SMBUS1_S<br>DA_AGPI020            | 0x050                                   |                              |

Table 259: GPIO BANK0 Register Default Value

| Register   | Reset Value | Index | Function                           |
|------------|-------------|-------|------------------------------------|
| GPIOx00000 | 0014_0000h  | 0     | BP_PWR_BTN_L_AGPI00                |
| GPIOx00004 | 0014_0000h  | 1     | BP_SYS_RESET_L_AGPI01              |
| GPIOx00008 | 0014_0000h  | 2     | BP_WAKE_L_AGPI02                   |
| GPIOx0000C | 0014_0000h  | 3     | BP_AGPI03_SPD_HOST_CTRL_L          |
| GPIOx00010 | 0014_0000h  | 4     | BP_AGPI04_SATA_ACT_L               |
| GPIOx00014 | 0024_0000h  | 5     | BP_AGPI05_DEVSLP0_SATA_ZP0_L       |
| GPIOx00018 | 0024_0000h  | 6     | BP_AGPI06_DEVSLP1_SATA_ZP1_L       |
| GPIOx0001C | 0024_0000h  | 7     | BP_AGPI07                          |
| GPIOx00020 | 0024_0000h  | 8     | Reserved                           |
| GPIOx00024 | 0024_0000h  | 9     | Reserved                           |
| GPIOx00028 | 0014_0000h  | 10    | Reserved                           |
| GPIOx0002C | 0014_0000h  | 11    | Reserved                           |
| GPIOx00030 | 0004_0000h  | 12    | BP_PWRGD_OUT_AGPI012               |
| GPIOx00034 | 0004_0000h  | 13    | BP_I2C4_SCL_HP_SCL_UBM_SCL_AGPI013 |
| GPIOx00038 | 0004_0000h  | 14    | BP_I2C4_SDA_HP_SDA_UBM_SDA_AGPI014 |
| GPIOx0003C | 0000_0000h  | 15    | Reserved                           |

|            |            |    |                                                      |
|------------|------------|----|------------------------------------------------------|
| GPIOx00040 | 0014_0000h | 16 | BP_USB10_OC_L_AGPIO16                                |
| GPIOx00044 | 0014_0000h | 17 | BP_USB11_OC_L_AGPIO17                                |
| GPIOx00048 | 0014_0000h | 18 | Reserved                                             |
| GPIOx0004C | 0004_0000h | 19 | BP_I2C5_SCL_BMC_SCL_SMBUS1_SCL_AGPIO19               |
| GPIOx00050 | 0004_0000h | 20 | BP_I2C5_SDA_BMC_SDA_SMBUS1_SDA_AGPIO20               |
| GPIOx00054 | 0024_0000h | 21 | BP_AGPIO21                                           |
| GPIOx00058 | 0024_0000h | 22 | BP_AGPIO22                                           |
| GPIOx0005C | 0014_0000h | 23 | BP_ESPI_RSTOUT_L_AGPIO23                             |
| GPIOx00060 | 0014_0000h | 24 | BP_SMERR_L_AGPIO24                                   |
| GPIOx00064 | 0000_0000h | 25 | Reserved                                             |
| GPIOx00068 | 0004_0000h | 26 | BP_PCIE_RST1_L_AGPIO26                               |
| GPIOx0006C | 0024_0000h | 27 | Reserved                                             |
| GPIOx00070 | 0024_0000h | 28 | Reserved                                             |
| GPIOx00074 | 0014_0000h | 29 | Reserved                                             |
| GPIOx00078 | 0014_0000h | 30 | Reserved                                             |
| GPIOx0007C | 0014_0000h | 31 | Reserved                                             |
| GPIOx00080 | 0024_0000h | 32 | Reserved                                             |
| GPIOx00084 | 0000_0000h | 33 | Reserved                                             |
| GPIOx00088 | 0000_0000h | 34 | Reserved                                             |
| GPIOx0008C | 0000_0000h | 35 | Reserved                                             |
| GPIOx00090 | 0000_0000h | 36 | Reserved                                             |
| GPIOx00094 | 0000_0000h | 37 | Reserved                                             |
| GPIOx00098 | 0000_0000h | 38 | Reserved                                             |
| GPIOx0009C | 0000_0000h | 39 | Reserved                                             |
| GPIOx000A0 | 0024_0000h | 40 | Reserved                                             |
| GPIOx000A4 | 0000_0000h | 41 | Reserved                                             |
| GPIOx000A8 | 0014_0000h | 42 | Reserved                                             |
| GPIOx000AC | 0000_0000h | 43 | Int_FakeSts0                                         |
| GPIOx000B0 | 0000_0000h | 44 | Int_ShdwSysAlarmFire                                 |
| GPIOx000B4 | 0000_0000h | 45 | Int_pwr_btn<br>(NOTE: This Event needs to debounce.) |
| GPIOx000B8 | 0000_0000h | 46 | Int_FakeSts0                                         |
| GPIOx000BC | 0000_0000h | 47 | Int_Wake_up_WDTO                                     |
| GPIOx000C0 | 0000_0000h | 48 | Int_DhcpLeaseTimerExpire                             |
| GPIOx000C4 | 0000_0000h | 49 | Int_ASFSlaveIntr                                     |
| GPIOx000C8 | 0000_0000h | 50 | Int_sm_irq_                                          |
| GPIOx000CC | 0000_0000h | 51 | Int_WakeFromLLB                                      |
| GPIOx000D0 | 0000_0000h | 52 | Int_AcDcTimerEvent                                   |
| GPIOx000D4 | 0000_0000h | 53 | Int_ALTHPET_TimerSts                                 |
| GPIOx000D8 | 0000_0000h | 54 | Int_iSocEvent[0] (MP2 wakeup event)                  |
| GPIOx000DC | 0000_0000h | 55 | Int_iSocEvent[1] (MP2 GPIO[0])                       |
| GPIOx000E0 | 0000_0000h | 56 | Int_iSocEvent[2] (MP2 GPIO[1])                       |
| GPIOx000E4 | 0000_0000h | 57 | Reserved                                             |
| GPIOx000E8 | 0000_0000h | 58 | Int_usb_xhc_0_acpi_pme                               |
| GPIOx000EC | 0000_0000h | 59 | Int_usb_xhc_1_acpi_pme                               |
| GPIOx000F0 | 0000_0000h | 60 | Int_RTC_STS_reg                                      |
| GPIOx000F4 | 0000_0000h | 61 | Int_ACP_FCH_AZ_Wake                                  |
| GPIOx000F8 | 0000_0000h | 62 | Int_ACP_FCH_I2S_Wake                                 |

Table 260: GPIO BANK1 Register Default Value

| Register   | Reset Value | Index | Function                           |
|------------|-------------|-------|------------------------------------|
| GPIOx00100 | 0000_0000h  | 64    | Reserved                           |
| GPIOx00104 | 0000_0000h  | 65    | Reserved                           |
| GPIOx00108 | 0000_0000h  | 66    | Reserved                           |
| GPIOx0010C | 0000_0000h  | 67    | Reserved                           |
| GPIOx00110 | 0000_0000h  | 68    | Reserved                           |
| GPIOx00114 | 0000_0000h  | 69    | Reserved                           |
| GPIOx00118 | 0000_0000h  | 70    | Reserved                           |
| GPIOx0011C | 0000_0000h  | 71    | Reserved                           |
| GPIOx00120 | 0000_0000h  | 72    | Reserved                           |
| GPIOx00124 | 0000_0000h  | 73    | Reserved                           |
| GPIOx00128 | 0024_0000h  | 74    | BP_ESPI_CLK2_AGPI074               |
| GPIOx0012C | 0024_0000h  | 75    | BP_ESPI_CLK1_SPI_CLK1_AGPI075      |
| GPIOx00130 | 0014_0000h  | 76    | BP_AGPI076_SPI_TPM_CS_L            |
| GPIOx00134 | 0000_0000h  | 77    | Reserved                           |
| GPIOx00138 | 0000_0000h  | 78    | Reserved                           |
| GPIOx0013C | 0000_0000h  | 79    | Reserved                           |
| GPIOx00140 | 0000_0000h  | 80    | Reserved                           |
| GPIOx00144 | 0000_0000h  | 81    | Reserved                           |
| GPIOx00148 | 0000_0000h  | 82    | Reserved                           |
| GPIOx0014C | 0000_0000h  | 83    | Reserved                           |
| GPIOx00150 | 0000_0000h  | 84    | Reserved                           |
| GPIOx00154 | 0000_0000h  | 85    | Reserved                           |
| GPIOx00158 | 0024_0000h  | 86    | BP_NMI_SYNC_FLOOD_L                |
| GPIOx0015C | 0024_0000h  | 87    | BP_AGPI087                         |
| GPIOx00160 | 0024_0000h  | 88    | BP_AGPI088                         |
| GPIOx00164 | 0014_0000h  | 89    | BP_GENINT_L_PM_INTR_L_AGPI089      |
| GPIOx00168 | 0000_0001h  | 90    | Reserved                           |
| GPIOx0016C | 0000_0000h  | 91    | Reserved                           |
| GPIOx00170 | 0000_0000h  | 92    | Reserved                           |
| GPIOx00174 | 0000_0000h  | 93    | Reserved                           |
| GPIOx00178 | 0000_0000h  | 94    | Reserved                           |
| GPIOx0017C | 0000_0000h  | 95    | Reserved                           |
| GPIOx00180 | 0000_0000h  | 96    | Reserved                           |
| GPIOx00184 | 0000_0000h  | 97    | Reserved                           |
| GPIOx00188 | 0000_0000h  | 98    | Reserved                           |
| GPIOx0018C | 0000_0000h  | 99    | Reserved                           |
| GPIOx00190 | 0000_0000h  | 100   | Reserved                           |
| GPIOx00194 | 0000_0000h  | 101   | Reserved                           |
| GPIOx00198 | 0000_0000h  | 102   | Reserved                           |
| GPIOx0019C | 0000_0000h  | 103   | Reserved                           |
| GPIOx001A0 | 0024_0000h  | 104   | BP_AGPI0104                        |
| GPIOx001A4 | 0024_0000h  | 105   | BP_AGPI0105                        |
| GPIOx001A8 | 0024_0000h  | 106   | BP_AGPI0106                        |
| GPIOx001AC | 0024_0000h  | 107   | BP_AGPI0107                        |
| GPIOx001B0 | 0014_0000h  | 108   | BP_ESPI0_ALERT_L_ESPI_IO1_AGPI0108 |
| GPIOx001B4 | 0024_0000h  | 109   | BP_AGPI0109                        |

|            |            |     |                                  |
|------------|------------|-----|----------------------------------|
| GPIOx001B8 | 0000_0000h | 110 | BP_ESPI1_ALERT_L_AGPI0110        |
| GPIOx001BC | 0000_0000h | 111 | Reserved                         |
| GPIOx001C0 | 0000_0000h | 112 | Reserved                         |
| GPIOx001C4 | 0000_0000h | 113 | Reserved                         |
| GPIOx001C8 | 0000_0000h | 114 | Reserved                         |
| GPIOx001CC | 0014_0000h | 115 | BP_AGPI0115_CLK_REQ11_L          |
| GPIOx001D0 | 0014_0000h | 116 | BP_AGPI0116_CLK_REQ12_L          |
| GPIOx001D4 | 0024_0000h | 117 | BP_ESPI_CLK0_SPI_CLK0_AGPI0117   |
| GPIOx001D8 | 0014_0000h | 118 | BP_SPI_CS0_L_AGPI0118            |
| GPIOx001DC | 0014_0000h | 119 | BP_SPI_CS1_L_AGPI0119            |
| GPIOx001E0 | 0014_0000h | 120 | BP_ESPI0_DAT0_SPI0_DAT0_AGPI0120 |
| GPIOx001E4 | 0014_0000h | 121 | BP_ESPI0_DAT1_SPI0_DAT1_AGPI0121 |
| GPIOx001E8 | 0014_0000h | 122 | BP_ESPI0_DAT2_SPI0_DAT2_AGPI0122 |
| GPIOx001EC | 0014_0000h | 123 | BP_ESPI0_DAT3_SPI0_DAT3_AGPI0123 |
| GPIOx001F0 | 0014_0000h | 124 | BP_ESPI_CS0_L_AGPI0124           |
| GPIOx001F4 | 0014_0000h | 125 | BP_ESPI_CS1_L_AGPI0125           |
| GPIOx001F8 | 0014_0000h | 126 | BP_SPI_CS2_L_AGPI0126            |

Table 261: GPIO BANK2 Register Default Value

| Register   | Reset Value | Index | Function                                          |
|------------|-------------|-------|---------------------------------------------------|
| GPIOx00200 | 0000_0000h  | 128   | Reserved                                          |
| GPIOx00204 | 0014_0000h  | 129   | BP_ESPI_RSTIN_L_KBRST_L_AGPI0129                  |
| GPIOx00208 | 0000_0000h  | 130   | Reserved                                          |
| GPIOx0020C | 0014_0000h  | 131   | BP_ESPI1_DAT0_SPI1_DAT0_AGPI0131                  |
| GPIOx00210 | 0014_0000h  | 132   | BP_ESPI1_DAT1_SPI1_DAT1_AGPI0132                  |
| GPIOx00214 | 0014_0000h  | 133   | BP_ESPI1_DAT2_SPI1_DAT2_AGPI0133                  |
| GPIOx00218 | 0014_0000h  | 134   | BP_ESPI1_DAT3_SPI1_DAT3_AGPI0134                  |
| GPIOx0021C | 0024_0000h  | 135   | BP_UART0_CTS_L_UART2_TXD_AGPI0135                 |
| GPIOx00220 | 0024_0000h  | 136   | BP_UART0_RXD_AGPI0136                             |
| GPIOx00224 | 0014_0000h  | 137   | BP_UART0_RTS_L_UART2_RXD_AGPI0137                 |
| GPIOx00228 | 0014_0000h  | 138   | BP_UART0_TXD_AGPI0138                             |
| GPIOx0022C | 0024_0000h  | 139   | BP_UART0_INTR_AGPI0139                            |
| GPIOx00230 | 0000_0000h  | 140   | Reserved                                          |
| GPIOx00234 | 0024_0000h  | 141   | BP_UART1_RXD_AGPI0141                             |
| GPIOx00238 | 0014_0000h  | 142   | BP_UART1_TXD_AGPI0142                             |
| GPIOx0023C | 0000_0000h  | 143   | Reserved                                          |
| GPIOx00240 | 0000_0000h  | 144   | Reserved                                          |
| GPIOx00244 | 0004_0000h  | 145   | BP_I3C0_SCL_I2C0_SCL_SPD0_SCL_SMBUS0_SCL_AGPI0145 |
| GPIOx00248 | 0004_0000h  | 146   | BP_I3C0_SDA_I2C0_SDA_SPD0_SDA_SMBUS0_SDA_AGPI0146 |
| GPIOx0024C | 0004_0000h  | 147   | BP_I3C1_SCL_I2C1_SCL_SPD1_SCL_AGPI0147            |
| GPIOx00250 | 0004_0000h  | 148   | BP_I3C1_SDA_I2C1_SDA_SPD1_SDA_AGPI0148            |
| GPIOx00254 | 0004_0000h  | 149   | BP_I3C2_SCL_I2C2_SCL_SPD2_SCL_AGPI0149            |
| GPIOx00258 | 0004_0000h  | 150   | BP_I3C2_SDA_I2C2_SDA_SPD2_SDA_AGPI0150            |
| GPIOx0025C | 0004_0000h  | 151   | BP_I3C3_SCL_I2C3_SCL_SPD3_SCL_AGPI0151            |
| GPIOx00260 | 0004_0000h  | 152   | BP_I3C3_SDA_I2C3_SDA_SPD3_SDA_AGPI0152            |
| GPIOx00264 | 0000_0000h  | 153   | Reserved                                          |

|            |            |     |                                          |
|------------|------------|-----|------------------------------------------|
| GPIOx00268 | 0000_0000h | 154 | Reserved                                 |
| GPIOx0026C | 0000_0000h | 155 | Reserved                                 |
| GPIOx00270 | 0000_0000h | 156 | Reserved                                 |
| GPIOx00274 | 0000_0000h | 157 | Reserved                                 |
| GPIOx00278 | 0000_0000h | 158 | Reserved                                 |
| GPIOx0027C | 0000_0000h | 159 | Reserved                                 |
| GPIOx00280 | 0000_0000h | 160 | Reserved                                 |
| GPIOx00284 | 0000_0000h | 161 | Reserved                                 |
| GPIOx00288 | 0000_0000h | 162 | Reserved                                 |
| GPIOx0028C | 0000_0000h | 163 | Reserved                                 |
| GPIOx00290 | 0000_0000h | 164 | Reserved                                 |
| GPIOx00294 | 0000_0000h | 165 | Reserved                                 |
| GPIOx00298 | 0000_0000h | 166 | Reserved                                 |
| GPIOx0029C | 0000_0000h | 167 | Reserved                                 |
| GPIOx002A0 | 0000_0000h | 168 | Reserved                                 |
| GPIOx002A4 | 0000_0000h | 169 | Reserved                                 |
| GPIOx002A8 | 0000_0000h | 170 | Reserved                                 |
| GPIOx002AC | 0000_0000h | 171 | Reserved                                 |
| GPIOx002B0 | 0000_0000h | 172 | NBGppPmePulse                            |
| GPIOx002B4 | 0000_0000h | 173 | NBGppHpPulse                             |
| GPIOx002B8 | 0000_0000h | 174 | AcpiPerfIntr                             |
| GPIOx002BC | 0000_0000h | 175 | sata_sci_ & sata_sci2_                   |
| GPIOx002C0 | 0000_0000h | 176 | FanThermal_SCIOut                        |
| GPIOx002C4 | 0000_0000h | 177 | ASFMasterIntr                            |
| GPIOx002C8 | 0000_0000h | 178 | Ras_event                                |
| GPIOx002CC | 0000_0000h | 179 | GBL_RLS                                  |
| GPIOx002D0 | 0000_0000h | 180 | ShortTimerEvent   LongTimerEvent         |
| GPIOx002D4 | 0000_0000h | 181 | NBHwAssertion_r[3]   NBSciAssertion_r[3] |
| GPIOx002D8 | 0000_0000h | 182 | eSPI_WAKE_PME_B                          |
| GPIOx002DC | 0000_0000h | 183 | eSPI_SYS_EVT_B                           |
| GPIOx002E0 | 0000_0000h | 184 | eSPI1_WAKE_PME_B                         |
| GPIOx002E4 | 0000_0000h | 185 | eSPI1_SYS_EVT_B                          |
| GPIOx002E8 | 0000_0000h | 186 | Reserved                                 |
| GPIOx002EC | 0000_0000h | 187 | Reserved                                 |
| GPIOx002F0 | 0000_0000h | 188 | Reserved                                 |
| GPIOx002F4 | 0000_0000h | 189 | Reserved                                 |
| GPIOx002F8 | 0000_0000h | 190 | Reserved                                 |

Table 262: GPIO BANK3 Register Default Value

| Register   | Reset Value | Index | Function |
|------------|-------------|-------|----------|
| GPIOx00300 | 0000_0000h  | 192   |          |
| GPIOx00304 | 0000_0000h  | 193   |          |
| GPIOx00308 | 0000_0000h  | 194   |          |
| GPIOx0030C | 0000_0000h  | 195   |          |
| GPIOx00310 | 0000_0000h  | 196   |          |
| GPIOx00314 | 0000_0000h  | 197   |          |
| GPIOx00318 | 0000_0000h  | 198   |          |
| GPIOx0031C | 0000_0000h  | 199   |          |

|            |            |     |  |
|------------|------------|-----|--|
| GPIOx00320 | 0000_0000h | 200 |  |
| GPIOx00324 | 0000_0000h | 201 |  |
| GPIOx00328 | 0000_0000h | 202 |  |
| GPIOx0032C | 0000_0000h | 203 |  |
| GPIOx00330 | 0000_0000h | 204 |  |
| GPIOx00334 | 0000_0000h | 205 |  |
| GPIOx00338 | 0000_0000h | 206 |  |
| GPIOx0033C | 0000_0000h | 207 |  |
| GPIOx00340 | 0000_0000h | 208 |  |
| GPIOx00344 | 0000_0000h | 209 |  |
| GPIOx00348 | 0000_0000h | 210 |  |
| GPIOx0034C | 0000_0000h | 211 |  |
| GPIOx00350 | 0000_0000h | 212 |  |
| GPIOx00354 | 0000_0000h | 213 |  |
| GPIOx00358 | 0000_0000h | 214 |  |
| GPIOx0035C | 0000_0000h | 215 |  |
| GPIOx00360 | 0000_0000h | 216 |  |
| GPIOx00364 | 0000_0000h | 217 |  |
| GPIOx00368 | 0000_0000h | 218 |  |
| GPIOx0036C | 0000_0000h | 219 |  |
| GPIOx00370 | 0000_0000h | 220 |  |
| GPIOx00374 | 0000_0000h | 221 |  |
| GPIOx00378 | 0000_0000h | 222 |  |
| GPIOx0037C | 0000_0000h | 223 |  |
| GPIOx00380 | 0000_0000h | 224 |  |
| GPIOx00384 | 0000_0000h | 225 |  |
| GPIOx00388 | 0000_0000h | 226 |  |
| GPIOx0038C | 0000_0000h | 227 |  |
| GPIOx00390 | 0000_0000h | 228 |  |
| GPIOx00394 | 0000_0000h | 229 |  |
| GPIOx00398 | 0000_0000h | 230 |  |
| GPIOx0039C | 0000_0000h | 231 |  |
| GPIOx003A0 | 0000_0000h | 232 |  |
| GPIOx003A4 | 0000_0000h | 233 |  |
| GPIOx003A8 | 0000_0000h | 234 |  |
| GPIOx003AC | 0000_0000h | 235 |  |
| GPIOx003B0 | 0000_0000h | 236 |  |
| GPIOx003B4 | 0000_0000h | 237 |  |
| GPIOx003B8 | 0000_0000h | 238 |  |
| GPIOx003BC | 0000_0000h | 239 |  |
| GPIOx003C0 | 0000_0000h | 240 |  |
| GPIOx003C4 | 0000_0000h | 241 |  |
| GPIOx003C8 | 0000_0000h | 242 |  |
| GPIOx003CC | 0000_0000h | 243 |  |
| GPIOx003D0 | 0000_0000h | 244 |  |
| GPIOx003D4 | 0000_0000h | 245 |  |
| GPIOx003D8 | 0000_0000h | 246 |  |
| GPIOx003DC | 0000_0000h | 247 |  |



|            |            |     |  |
|------------|------------|-----|--|
| GPIOx003E0 | 0000_0000h | 248 |  |
| GPIOx003E4 | 0000_0000h | 249 |  |
| GPIOx003E8 | 0000_0000h | 250 |  |
| GPIOx003EC | 0000_0000h | 251 |  |
| GPIOx003F0 | 0000_0000h | 252 |  |
| GPIOx003F4 | 0000_0000h | 253 |  |
| GPIOx003F8 | 0000_0000h | 254 |  |
| GPIOx003FC | 0000_0000h | 255 |  |

*Table 263: Debounce Timer Definition*

| DebounceTmrLarge | DebounceTmrOutUnit | Timer Unit               | Max Debounce Time |
|------------------|--------------------|--------------------------|-------------------|
| 0                | 0                  | 61 usec (2 RtcClk)       | 915 usec          |
| 0                | 1                  | 183 usec (6 RtcClk)      | 2.75 msec         |
| 1                | 0                  | 15.56 msec (510 RtcClk)  | 233 msec          |
| 1                | 1                  | 62.44 msec (2046 RtcClk) | 936 msec          |

**GPIOx00000000 (FCH::GPIO::PWR\_BTN\_L\_AGPI00)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000000; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>less10secsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read-only. This bit is only valid for GPIO0. For other GPIO, this bit is Reserved.<br>When power button is pressed for less than 10 second in S0 state, this bit will become 1. This bit can be cleared by writing 1 to InterruptSts bit.                                                            |
| 30    | <b>less2secsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read-only. This bit is only valid for GPIO0. For other GPIO, this bit is Reserved.<br>When power button is pressed for less than 2 second in S0 state, this bit will become 1. When Less2secSts becomes 1, Less10secSts will also become 1. This bit can be cleared by writing 1 to InterruptSts bit. |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                                                                                                                     |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                                                                                                               |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                                                                                                              |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                                                                                                                         |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                                                                                                                      |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                                                                                                                 |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                                                                                                                   |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                                                                                                                        |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                                                                                                              |
| 19    | Reserved.                                                                                                                                                                                                                                                                                                                                                              |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                                                                                                             |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.                                                                                                                                                                                                                                                                                                                                   |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                                                                                                                                                                                                                                                            |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State)                                                                                                                                                                                               |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                      |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000004 (FCH::GPIO::SYS\_RESET\_L\_AGPI01)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000004; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000008 (FCH::GPIO::WAKE\_L\_AGPIO2)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000008; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000000C (FCH::GPIO::AGPIO3\_SPD\_HOST\_CTRL\_L)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx0000000C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000010 (FCH::GPIO::AGPIO4\_SATA\_ACT\_L)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000010; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000014 (FCH::GPIO::AGPIO5\_DEVSLP0\_SATA\_ZP0\_L)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000014; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000018 (FCH::GPIO::AGPIO6\_DEVSLP1\_SATA\_ZP1\_L)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000018; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000001C (FCH::GPIO::AGPIO7)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx0000001C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                              |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                                        |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                                  |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                              |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                                     |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec Debounce |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                                |

**GPIOx00000030 (FCH::GPIO::PWRGD\_OUT\_AGPI012)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000030; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000034 (FCH::GPIO::I2C4\_SCL\_HP\_SCL\_UBM\_SCL\_AGPI013)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000034; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000038 (FCH::GPIO::I2C4\_SDA\_HP\_SDA\_UBM\_SDA\_AGPI014)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000038; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000040 (FCH::GPIO::USB10\_OC\_L\_AGPI016)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000040; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000044 (FCH::GPIO::USB11\_OC\_L\_AGPI017)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000044; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000004C (FCH::GPIO::I2C5\_SCL\_BMC\_SCL\_SMBUS1\_SCL\_AGPI019)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx0000004C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000050 (FCH::GPIO::I2C5\_SDA\_BMC\_SDA\_SMBUS1\_SDA\_AGPI020)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000050; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000054 (FCH::GPIO::AGPIO21\_LPC\_PD\_L)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000054; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000058 (FCH::GPIO::AGPIO22\_LPC\_PME\_L)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000058; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000005C (FCH::GPIO::ESPI\_RSTOUT\_L\_AGPI023\_LPC\_RST\_L)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx0000005C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000060 (FCH::GPIO::SMERR\_L\_AGPI024)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000060; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000068 (FCH::GPIO::PCIE\_RST1\_L\_AGPI026)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000068; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000070 (FCH::GPIO::X48M\_OUT\_AGPI028)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000070; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000000AC (FCH::GPIO::INT\_FAKESTS0)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000AC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000B0 (FCH::GPIO::INT\_SHDWSYSALARMFIRE)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000B0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000B4 (FCH::GPIO::INT\_PWR\_BTTN)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000B4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000B8 (FCH::GPIO::INT\_FAKESTS0\_2ND)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000B8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000BC (FCH::GPIO::INT\_WAKE\_UP\_WDTO)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000BC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |



**GPIOx000000C0 (FCH::GPIO::INT\_DHCPLEASETIMEREXPIRE)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000C0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000C4 (FCH::GPIO::INT\_ASFSLAVEINTR)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000C4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000C8 (FCH::GPIO::INT\_SM\_IRQ)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000C8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000CC (FCH::GPIO::INT\_WAKEFROMLLB)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000CC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000D0 (FCH::GPIO::INT\_ACDCTIMEREVENT)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000D0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000D4 (FCH::GPIO::INT\_ALTHPET\_TIMERSTS)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000D4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000D8 (FCH::GPIO::INT\_ISOCEVENT0)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000D8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000DC (FCH::GPIO::INT\_ISOCEVENT1)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000DC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |



**GPIOx000000E0 (FCH::GPIO::INT\_ISOCEVENT2)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000E0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000E8 (FCH::GPIO::INT\_USB\_XHC\_0\_ACPI\_PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000E8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000EC (FCH::GPIO::INT\_USB\_XHC\_1\_ACPI\_PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000EC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000F0 (FCH::GPIO::INT\_RTC\_STS\_REG)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000F0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000F4 (FCH::GPIO::INT\_ACP\_FCH\_AZ\_WAKE)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000F4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000F8 (FCH::GPIO::INT\_ACP\_FCH\_I2S\_WAKE)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000000F8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000000FC (FCH::GPIO::GPIO\_WAKE\_INTERRUPT\_MASTER\_SWITCH)**

Read-write. Reset: FF00\_0000h.

\_aliasHOSTLEGACY; GPIOx000000FC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31    | <b>gpiowakeen.</b> Read-write. Reset: 1. When this bit is 0, all GPIO wake are blocked.                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 30    | <b>gpiointerrupten.</b> Read-write. Reset: 1. When this bit is 0, all GPIO interrupts are blocked.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 29    | <b>eoi.</b> Read-write. Reset: 1.<br><b>Description:</b> This bit is set to 1 by SW when it is allowed to send GPIO interrupt. HW clear this bit when interrupt occurs. When this bit is 0, the GPIO interrupt is blocked.<br>(This SW/HW handshake mechanism is the same with EOS of <a href="#">SMI</a> .)                                                                                                                                                                                                         |
| 28    | <b>mask_sts_en.</b> Read-write. Reset: 1. This bit enables HW to block all wake/intr status generation when SW writes any Debounce* registers. The length of blocking depends on mask_sts_length[3:0].                                                                                                                                                                                                                                                                                                               |
| 27:24 | <b>mask_sts_length_3_0.</b> Read-write. Reset: Fh.<br><b>Description:</b> See mask_sts_en as well.<br>The length of blocking = {mask_sts_length[11:0], 14'h3FFF}                                                                                                                                                                                                                                                                                                                                                     |
| 23:16 | <b>mask_sts_length_11_4.</b> Read-write. Reset: 00h.<br><b>Description:</b> See mask_sts_en as well.<br>The length of blocking = {mask_sts_length[11:0], 14'h3FFF}                                                                                                                                                                                                                                                                                                                                                   |
| 15    | <b>enwinbluebtn.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: GPIO0 detect debounced power button. Power button override is 4 sec.<br>1: GPIO0 detect debounced power button in S3/S5/Power saving mode, and detect "pressed less than 2sec" and "pressed 2~10sec" in S0. Power button override is 10 sec.                                                                                                                                                                                                    |
| 14    | <b>introutactivehi.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: GPIO controller interrupt output is low active<br>1: GPIO controller interrupt output is high active                                                                                                                                                                                                                                                                                                                                         |
| 13    | <b>selgpio0src.</b> Read-write. Reset: 0.<br><b>Description:</b> Select the source for GPIO0 detection.<br>0: Power button goes to a processing logic first and then goes to GPIO0 detection logic<br>1: Power button goes to GPIO0 debounce and then goes to GPIO0 detection logic.<br>Note: The "processing logic" includes 16ms debounce counter and a logic to detect how long the button has been pressed to generate press_less2s_sts and press_less4s_sts. "GPIO0 debounce block" only has debounce function. |
| 12    | <b>introutpulse.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: GPIO controller interrupt output is a level signal<br>1: GPIO controller interrupt output is pulse signal<br>Note: The polarity is defined by IntrOutActiveHi register bit.                                                                                                                                                                                                                                                                     |
| 11:0  | Reserved.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

**GPIOx00000128 (FCH::GPIO::ESPI\_CLK2\_AGPIO74\_LPCCLK0)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000128; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000012C (FCH::GPIO::ESPI\_CLK1\_SPI\_CLK1\_AGPIOW75\_LPCCLK1)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx0000012C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000130 (FCH::GPIO::AGPIO76\_SPI\_TPM\_CS\_L)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000130; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000158 (FCH::GPIO::NMI\_SYNC\_FLOOD\_L)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000158; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000015C (FCH::GPIO::AGPIO87\_SERIRQ)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx0000015C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000160 (FCH::GPIO::AGPIO88\_LPC\_CLKRUN\_L)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000160; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000164 (FCH::GPIO::GENINT\_L\_PM\_INTR\_L\_AGPI089)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000164; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                      |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001A0 (FCH::GPIO::AGPIO104\_LAD0)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx000001A0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001A4 (FCH::GPIO::AGPIO105\_LAD1)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx000001A4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001A8 (FCH::GPIO::AGPIO106\_LAD2)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx000001A8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001AC (FCH::GPIO::AGPIO107\_LAD3)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx000001AC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001B0 (FCH::GPIO::ESPI0\_ALERT\_L\_ESPI\_IO1\_AGPI0108\_LDRQ\_L)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001B0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001B4 (FCH::GPIO::AGPIO109\_LFRAME\_L)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx000001B4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001B8 (FCH::GPIO::ESPI1\_ALERT\_L\_AGPI0110)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001B8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001CC (FCH::GPIO::AGPIO115\_CLK\_REQ11\_L)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001CC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001D0 (FCH::GPIO::AGPIO116\_CLK\_REQ12\_L)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001D0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001D4 (FCH::GPIO::ESPI\_CLK0\_SPI\_CLK0\_AGPIO117)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx000001D4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001D8 (FCH::GPIO::SPI\_CS0\_L\_AGPI0118)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001D8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001DC (FCH::GPIO::SPI\_CS1\_L\_AGPI0119)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001DC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001E0 (FCH::GPIO::ESPI0\_DAT0\_SPI0\_DAT0\_AGPI0120)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001E0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001E4 (FCH::GPIO::ESPI0\_DAT1\_SPI0\_DAT1\_AGPI0121)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001E4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001E8 (FCH::GPIO::ESPI0\_DAT2\_SPI0\_DAT2\_AGPI0122)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001E8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001EC (FCH::GPIO::ESPI0\_DAT3\_SPI0\_DAT3\_AGPIO123)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001EC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001F0 (FCH::GPIO::ESPI\_CS0\_L\_AGPI0124)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001F0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001F4 (FCH::GPIO::ESPI\_CS1\_L\_AGPI0125)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001F4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000001F8 (FCH::GPIO::SPI\_CS2\_L\_AGPI0126)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx000001F8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000204 (FCH::GPIO::ESPI\_RSTIN\_L\_KBRST\_L\_AGPI0129)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000204; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000020C (FCH::GPIO::ESPI1\_DAT0\_SPI1\_DAT0\_AGPIO131)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx0000020C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000210 (FCH::GPIO::ESPI1\_DAT1\_SPI1\_DAT1\_AGPI0132)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000210; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                      |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000214 (FCH::GPIO::ESPI1\_DAT2\_SPI1\_DAT2\_AGPI0133)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000214; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000218 (FCH::GPIO::ESPI1\_DAT3\_SPI1\_DAT3\_AGPI0134)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000218; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000021C (FCH::GPIO::UART0\_CTS\_L\_UART2\_TXD\_AGPI0135)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx0000021C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000220 (FCH::GPIO::UART0\_RXD\_AGPI0136)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000220; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000224 (FCH::GPIO::UART0\_RTS\_L\_UART2\_RXD\_AGPI0137)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000224; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000228 (FCH::GPIO::UART0\_TXD\_AGPI0138)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; GPIOx00000228; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000022C (FCH::GPIO::UART0\_INTR\_AGPIO139)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx0000022C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000234 (FCH::GPIO::UART1\_RXD\_AGPI0141)**

Reset: 0044\_0000h.

\_aliasHOSTLEGACY; GPIOx00000234; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000238 (FCH::GPIO::UART1\_TXD\_AGPI0142)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; GPIOx00000238; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000244 (FCH::GPIO::I3C0\_SCL\_I2C0\_SCL\_SPD0\_SCL\_SMBUS0\_SCL\_AGPIO145)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000244; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000248 (FCH::GPIO::I3C0\_SDA\_I2C0\_SDA\_SPD0\_SDA\_SMBUS0\_SDA\_AGPIO146)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000248; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000024C (FCH::GPIO::I3C1\_SCL\_I2C1\_SCL\_SPD1\_SCL\_AGPI0147)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx0000024C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000250 (FCH::GPIO::I3C1\_SDA\_I2C1\_SDA\_SPD1\_SDA\_AGPI0148)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000250; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000254 (FCH::GPIO::I3C2\_SCL\_I2C2\_SCL\_SPD2\_SCL\_AGPI0149)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000254; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000258 (FCH::GPIO::I3C2\_SDA\_I2C2\_SDA\_SPD2\_SDA\_AGPI0150)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000258; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx0000025C (FCH::GPIO::I3C3\_SCL\_I2C3\_SCL\_SPD3\_SCL\_AGPI0151)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx0000025C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |



|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx00000260 (FCH::GPIO::I3C3\_SDA\_I2C3\_SDA\_SPD3\_SDA\_AGPI0152)**

Reset: 0004\_0000h.

\_aliasHOSTLEGACY; GPIOx00000260; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**GPIOx000002AC (FCH::GPIO::LPC\_SPI\_INT)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002AC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002B0 (FCH::GPIO::INT\_NBGPPMEPULSE)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002B0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002B4 (FCH::GPIO::INT\_NBGPPHPULSE)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002B4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002B8 (FCH::GPIO::INT ACPIPERFINTR)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002B8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002BC (FCH::GPIO::INT\_SATA\_SCI\_SCI2)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002BC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |



**GPIOx000002C0 (FCH::GPIO::INT\_FANTHERMAL\_SCIOUT)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002C0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002C4 (FCH::GPIO::INT\_ASFMASTERINTR)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002C4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002C8 (FCH::GPIO::INT\_RAS\_EVENT)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002C8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002CC (FCH::GPIO::INT\_GBL\_RLS)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002CC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002D0 (FCH::GPIO::INT\_SHORTTIMEREVENT\_LONGTIMEREVENT)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002D0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002D4 (FCH::GPIO::INT\_NBASSERTION)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002D4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002D8 (FCH::GPIO::INT\_ESPIPME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002D8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)<br>(NOTE: This Event is low enable)                                                                  |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002DC (FCH::GPIO::INT\_ESPISYSEVT)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002DC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)<br>(NOTE: This Event is low enable)                                                                  |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |



**GPIOx000002E0 (FCH::GPIO::INT\_ESIPME1)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002E0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)<br>(NOTE: This Event is low enable)                                                                  |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002E4 (FCH::GPIO::INT\_ESPISYSEVT1)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002E4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)<br>(NOTE: This Event is low enable)                                                                  |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx000002F0 (FCH::GPIO::GPIO\_WAKE\_STATUS\_INDEX\_0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002F0; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                           |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>wake_status_index_31_16.</b> Read-write. Reset: 0000h. Read-only. When this bit is 1, it means at least one of the wake status of GPIO N*4 ~ N*4+3 is 1. N = 16~31 |
| 15    | <b>wake_status_index_15.</b> Read-write. Reset: 0. Read-only. When this bit is 1, it means at least one of the wake status of GPIO 60~62 is 1.                        |
| 14:0  | <b>wake_status_index_14_0.</b> Read-write. Reset: 0000h. Read-only. When this bit is 1, it means at least one of the wake status of GPIO N*4 ~ N*4+3 is 1. N = 0~14   |

**GPIOx000002F4 (FCH::GPIO::GPIO\_WAKE\_STATUS\_INDEX\_1)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002F4; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                                                                                                                  |
| 15    | <b>nbgpppmewake.</b> Read-write. Reset: 0.<br><b>Description:</b> Read-only.<br>1: NBGpp has sent PME to wake the system. (one of wake status is set in Gpio Bank3 register)<br>0: no NBGpp Pme wake event |
| 14    | Reserved.                                                                                                                                                                                                  |
| 13:0  | <b>wake_status_index_45_32.</b> Read-write. Reset: 0000h. Read-only. When this bit is 1, it means at least one of the wake status of GPIO N*4 ~ N*4+3 is 1. N = 32~45                                      |

**GPIOx000002F8 (FCH::GPIO::GPIO\_INTERRUPT\_STATUS\_INDEX\_0)**

Read-write. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx000002F8; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                     |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:16 | <b>interrupt_status_index_31_16.</b> Read-write. Reset: 0000h. Read-only. When this bit is 1, it means at least one of the interrupt status of GPIO N*4 ~ N*4+3 is 1. N = 16~31 |
| 15    | <b>interrupt_status_index_15.</b> Read-write. Reset: 0. Read-only. When this bit is 1, it means at least one of the interrupt status of GPIO 60~62 is 1.                        |
| 14:0  | <b>interrupt_status_index_14_0.</b> Read-write. Reset: 0000h. Read-only. When this bit is 1, it means at least one of the interrupt status of GPIO N*4 ~ N*4+3 is 1. N = 0~14   |

**GPIOx000002FC (FCH::GPIO::GPIO\_INTERRUPT\_STATUS\_INDEX\_1)**

Read-write. Reset: 1F00\_0000h.

\_aliasHOSTLEGACY; GPIOx000002FC; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                              |
| 28    | <b>mask_sts_en.</b> Read-write. Reset: 1. This bit enables HW to block all wake/intr status generation when SW writes any Debounce* registers. The length of blocking depends on mask_sts_length[3:0]. |
| 27:24 | <b>mask_sts_length_3_0.</b> Read-write. Reset: Fh.<br><b>Description:</b> See mask_sts_en as well.<br>The length of blocking = {mask_sts_length[11:0], 14'h3FFF}                                       |
| 23:16 | <b>mask_sts_length_11_4.</b> Read-write. Reset: 00h.<br><b>Description:</b> See mask_sts_en as well.<br>The length of blocking = {mask_sts_length[11:0], 14'h3FFF}                                     |
| 15    | <b>nbgpppmeintr.</b> Read-write. Reset: 0.<br><b>Description:</b> Read-only.<br>1: NB GPP has sent PME (one of Intr status is set in Gpio Bank3 register)<br>0: no NB GPP Pme event                    |
| 14    | Reserved.                                                                                                                                                                                              |
| 13:0  | <b>interrupt_status_index_45_32.</b> Read-write. Reset: 0000h. Read-only. When this bit is 1, it means at least one of the interrupt status of GPIO N*4 ~ N*4+3 is 1. N = 32~45                        |

**GPIOx00000300 (FCH::GPIO::INT\_NBGPPORTDEV0PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000300; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000304 (FCH::GPIO::INT\_NBGPPORTDEV1PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000304; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000308 (FCH::GPIO::INT\_NBGPPORTDEV2PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000308; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx0000030C (FCH::GPIO::INT\_NBGPPPORTDEV3PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx0000030C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |



**GPIOx00000310 (FCH::GPIO::INT\_NBGPPORTDEV4PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000310; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000314 (FCH::GPIO::INT\_NBGPPORTDEV5PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000314; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000318 (FCH::GPIO::INT\_NBGPPORTDEV6PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000318; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx0000031C (FCH::GPIO::INT\_NBGPPPORTDEV7PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx0000031C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000320 (FCH::GPIO::INT\_NBGPPORTDEV8PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000320; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000324 (FCH::GPIO::INT\_NBGPPORTDEV9PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000324; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000328 (FCH::GPIO::INT\_NBGPPORTDEV10PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000328; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx0000032C (FCH::GPIO::INT\_NBGPPORTDEV11PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx0000032C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |



**GPIOx00000330 (FCH::GPIO::INT\_NBGPPORTDEV12PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000330; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000334 (FCH::GPIO::INT\_NBGPPORTDEV13PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000334; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000338 (FCH::GPIO::INT\_NBGPPORTDEV14PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000338; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx0000033C (FCH::GPIO::INT\_NBGPPORTDEV15PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx0000033C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000340 (FCH::GPIO::INT\_NBGPPORTDEV16PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000340; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000344 (FCH::GPIO::INT\_NBGPPORTDEV17PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000344; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000348 (FCH::GPIO::INT\_NBGPPORTDEV18PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000348; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx0000034C (FCH::GPIO::INT\_NBGPPPORTDEV19PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx0000034C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |



**GPIOx00000350 (FCH::GPIO::INT\_NBGPPORTDEV20PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000350; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000354 (FCH::GPIO::INT\_NBGPPORTDEV21PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000354; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000358 (FCH::GPIO::INT\_NBGPPORTDEV22PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000358; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx0000035C (FCH::GPIO::INT\_NBGPPORTDEV23PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx0000035C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000360 (FCH::GPIO::INT\_NBGPPORTDEV24PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000360; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000364 (FCH::GPIO::INT\_NBGPPORTDEV25PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000364; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000368 (FCH::GPIO::INT\_NBGPPORTDEV26PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000368; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx0000036C (FCH::GPIO::INT\_NBGPPPORTDEV27PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx0000036C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |



**GPIOx00000370 (FCH::GPIO::INT\_NBGPPORTDEV28PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000370; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000374 (FCH::GPIO::INT\_NBGPPORTDEV29PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000374; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx00000378 (FCH::GPIO::INT\_NBGPPORTDEV30PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx00000378; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**GPIOx0000037C (FCH::GPIO::INT\_NBGPPPORTDEV31PME)**

Reset: 0000\_0000h.

\_aliasHOSTLEGACY; GPIOx0000037C; GPIO=FED8\_1500h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23:17 | Reserved.                                                                                                                                                                                                                                                                    |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                       |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                 |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                           |
| 7:0   | Reserved.                                                                                                                                                                                                                                                                    |

**14.3.15.3 REMOTE GPIO and IOMUX Registers**

Remote GPIO register and Remote IOMUX register (GPIO256 ~ GPIO266) are in the same bank which is different with other GPIO (GPIO0 ~ GPIO255). Each Remote GPIO PAD requires 32-bit GPIO register and 8-bit IOMUX register.

Table 264: Remote GPIO Register Offset and Default Value

| Register offset | Reset      | Function            |
|-----------------|------------|---------------------|
| 0x00            | 0014_0000h | AGPIO256_SGPIO0_CLK |

|                                    |            |                                 |
|------------------------------------|------------|---------------------------------|
| 0x04                               | 0014_0000h | AGPIO257_SGPIO1_CLK_CLK_REQ01_L |
| 0x08                               | 0014_0000h | AGPIO258_SGPIO2_CLK_CLK_REQ02_L |
| 0x0C                               | 0014_0000h | AGPIO259_SGPIO3_CLK             |
| 0x10                               | 0014_0000h | SGPIO_DATAOUT_AGPIO260          |
| 0x14                               | 0014_0000h | SGPIO_LOAD_AGPIO261             |
| 0x20                               | 0014_0000h | USB00_OC_L_AGPIO264             |
| 0x24                               | 0014_0000h | USB01_OC_L_AGPIO265             |
| 0x28                               | 0004_0000h | PCIE_RST0_L_AGPIO266            |
| 0x2C                               | 0000_0000h | Reserved                        |
| 0x30                               | 0000_0000h | Reserved                        |
| 0x34                               | 0000_0000h | Reserved                        |
| 0x38                               | 0024_0000h | Reserved                        |
| 0x3C                               | 0024_0000h | Reserved                        |
| ** Offset 0x40 – 0xBF are reserved |            |                                 |

Table 265: Remote IOMUX Register Offset and Default Value

| Register offset                    | Reset | Function      |
|------------------------------------|-------|---------------|
| 0xC0                               | 00b   | GPIO256       |
| 0xC1                               | 00b   | GPIO257       |
| 0xC2                               | 00b   | GPIO258       |
| 0xC3                               | 00b   | GPIO259       |
| 0xC4                               | 00b   | SGPIO_DATAOUT |
| 0xC5                               | 00b   | SGPIO_LOAD    |
| 0xC8                               | 00b   | USB00_OC_L    |
| 0xC9                               | 00b   | USB01_OC_L    |
| 0xCA                               | 00b   | PCIE_RST0_L   |
| 0xCB                               | 00b   | Reserved      |
| 0xCC                               | 00b   | Reserved      |
| 0xCD                               | 00b   | Reserved      |
| 0xCE                               | 00b   | Reserved      |
| 0xCF                               | 00b   | Reserved      |
| ** Offset 0xD0 – 0xEF are reserved |       |               |

Table 266: Remote IOMUX Function Table

| IOMUX         | Pin Name                        | Do<br>mai<br>n | GP<br>IO | G<br>EV<br>EN<br>T | Ov<br>err<br>ide<br>_0 | Ov<br>err<br>ide<br>_1 | IOMUX<br>=0 | IOMUX<br>=1 | IO<br>MU<br>X=2         | IO<br>MU<br>X=3 | Defau<br>lt | Reset<br>value |
|---------------|---------------------------------|----------------|----------|--------------------|------------------------|------------------------|-------------|-------------|-------------------------|-----------------|-------------|----------------|
| IOMUXx<br>100 | AGPIO256_SGPIO0_CLK             | S5             | 256      |                    |                        |                        | GPIO256     | SGPIO0_CLK  | GPI<br>O256             | GPI<br>O256     | PU          | 00             |
| IOMUXx<br>101 | AGPIO257_SGPIO1_CLK_CLK_REQ01_L | S5             | 257      |                    |                        |                        | GPIO257     | SGPIO1_CLK  | CL<br>K_<br>RE<br>G01_L | GPI<br>O257     | PU          | 00             |
| IOMUXx<br>102 | AGPIO258_SGPIO2_CLK_CLK_REQ02_L | S5             | 258      |                    |                        |                        | GPIO258     | SGPIO2_CLK  | CL<br>K_<br>O25         | GPI<br>O25      | PU          | 00             |

|               |                            |    |     |  |  |  |                       |                |                 |                 |     |    |
|---------------|----------------------------|----|-----|--|--|--|-----------------------|----------------|-----------------|-----------------|-----|----|
|               |                            |    |     |  |  |  |                       |                | RE<br>Q02<br>_L | 8               |     |    |
| IOMUXx<br>103 | AGPIO259_SGPIO3_C<br>LK    | S5 | 259 |  |  |  | GPIO25<br>9           | SGPIO3<br>_CLK | GPI<br>O25<br>9 | GPI<br>O25<br>9 | PU  | 00 |
| IOMUXx<br>104 | SGPIO_DATAOUT_A<br>GPIO260 | S5 | 260 |  |  |  | SGPIO_<br>DATAO<br>UT | GPIO26<br>0    | GPI<br>O26<br>0 | GPI<br>O26<br>0 | PU  | 00 |
| IOMUXx<br>105 | SGPIO_LOAD_AGPIO<br>261    | S5 | 261 |  |  |  | SGPIO_<br>LOAD        | GPIO26<br>1    | GPI<br>O26<br>1 | GPI<br>O26<br>1 | PU  | 00 |
| IOMUXx<br>108 | USB00_OC_L_AGPIO<br>264    | S5 | 264 |  |  |  | USB00_<br>OC_L        | GPIO26<br>4    | GPI<br>O26<br>4 | GPI<br>O26<br>4 | PU  | 00 |
| IOMUXx<br>109 | USB01_OC_L_AGPIO<br>265    | S5 | 265 |  |  |  | USB01_<br>OC_L        | GPIO26<br>5    | GPI<br>O26<br>5 | GPI<br>O26<br>5 | PU  | 00 |
| IOMUXx<br>10A | PCIE_RST0_L_AGPIO<br>266   | S5 | 266 |  |  |  | PCIE_R<br>ST0_L       | GPIO26<br>6    | GPI<br>O26<br>6 | GPI<br>O26<br>6 | N/A | 00 |

**REMOTEGPIOx00000000 (FCH::RMTGPIO::AGPIO256\_SGPIO0\_CLK)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx00000000; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |



**REMOTEGPIOx00000004 (FCH::RMTGPIO::AGPIO257\_SGPIO1\_CLK\_CLK\_REQ01\_L)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx00000004; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**REMOTEGPIOx00000008 (FCH::RMTGPIO::AGPIO258\_SGPIO2\_CLK\_CLK\_REQ02\_L)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx00000008; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**REMOTEGPIOx0000000C (FCH::RMTGPIO::AGPIO259\_SGPIO3\_CLK)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx0000000C; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**REMOTEGPIOx00000010 (FCH::RMTGPIO::SGPIO\_DATAOUT\_AGPIO260)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx00000010; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |



**REMOTEGPIOx00000014 (FCH::RMTGPIO::SGPIO\_LOAD\_AGPIO261)**

Reset: 0014\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx00000014; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**REMOTEGPIOx00000020 (FCH::RMTGPIO::USB00\_OC\_L\_AGPI0264)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx00000020; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**REMOTEGPIOx00000024 (FCH::RMTGPIO::USB01\_OC\_L\_AGPI0265)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx00000024; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**REMOTEGPIOx00000028 (FCH::RMTGPIO::PCIE\_RST0\_L\_AGPIO266)**

Reset: 0024\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx00000028; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                                  |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:30 | Reserved.                                                                                                                                                                                                                                                                    |
| 29    | <b>wakests.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't generate wake event.<br>1: the pin is one of wake source<br>Writing 1 to clear the bit to 0.                                                                           |
| 28    | <b>interruptsts.</b> Read-write,Read,Write-1-to-clear. Reset: 0.<br><b>Description:</b> 0: the pin doesn't cause interrupt.<br>1: the pin is one of interrupt source<br>Writing 1 to clear the bit to 0.                                                                     |
| 27:26 | Reserved.                                                                                                                                                                                                                                                                    |
| 25    | <b>swcntrlen.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Disable software controlled GPIO in<br>1: Enable software controlled GPIO in                                                                                                                               |
| 24    | <b>swcntrlin.</b> Read-write. Reset: 0. When SwCntrlEn is 1, this bit can be used by software to control the GPIO input to the GPIO controller. This might be useful during debug or work-around.                                                                            |
| 23    | <b>outputenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Output is disabled on the pin<br>1: Output is enabled on the pin.                                                                                                                                       |
| 22    | <b>outputvalue.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: low.<br>1: high.                                                                                                                                                                                         |
| 21    | <b>pulldownenable.</b> Read-write. Reset: 1.<br><b>Description:</b> 0: Pull-down is disabled on the pin.<br>1: Pull-down is enabled on the pin.                                                                                                                              |
| 20    | <b>pullupenable.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Pull-up is disabled on the pin.<br>1: Pull-up is enabled on the pin.                                                                                                                                    |
| 19    | Reserved.                                                                                                                                                                                                                                                                    |
| 18:17 | <b>drvstrengthsel.</b> Read-write. Reset: 2h.<br><b>Description:</b> 3.3V PAD:<br>x0: Z= 40 ohms<br>x1: Z= 80 ohms<br>1.8V PAD:<br>00: not supported<br>01: Z= 60 ohms<br>10: Z= 40 ohms<br>11: Z= 80 ohms                                                                   |
| 16    | <b>pinsts.</b> Read-write. Reset: 0.<br><b>Description:</b> Read only.<br>0: the pin is low<br>1: the pin is high<br>(NOTE: the bit doesn't affected by debounce logic)                                                                                                      |
| 15:13 | <b>wakecntrl_2_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> WakeCntrl[0] enables wake in Power saving mode state<br>WakeCntrl[1] enables wake in S3 state (SLP_TYP = 3, and !G0_State)<br>WakeCntrl[2] enables wake in S4/S5 state (SLP_TYP = 4 or 5, and !G0_State) |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12:11 | <b>interruptenable_1_0.</b> Read-write. Reset: 0h.<br><b>Description:</b> InterruptEnable[0] enables interrupt status<br>InterruptEnable[1] enables interrupt delivery                                                                                                                                                                                                                                                                                                     |
| 10:9  | <b>activelevel.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: Active high<br>01: Active Low.<br>10: Active on both of edges if LevelTrig is set to 0<br>11: reserved                                                                                                                                                                                                                                                                                               |
| 8     | <b>leveltrig.</b> Read-write. Reset: 0.<br><b>Description:</b> 0: Edge trigger<br>1: Level trigger                                                                                                                                                                                                                                                                                                                                                                         |
| 7     | <b>debouncetmrmlarge.</b> Read-write. Reset: 0. Combined with DebounceTmrOutUnit, this bit change the unit and max debounce time for the debounce timer. Please refer to "DebounceTmrOutUnit" description for details.                                                                                                                                                                                                                                                     |
| 6:5   | <b>debouncecntrl.</b> Read-write. Reset: 0h.<br><b>Description:</b> 00: No debounce<br>01: Preserve low glitch<br>10: Preserve high glitch<br>11: Remove glitch                                                                                                                                                                                                                                                                                                            |
| 4     | <b>debouncetmroutunit.</b> Read-write. Reset: 0.<br><b>Description:</b> {DebounceTmrLarge, DebounceTmrOutUnit} defines the unit and max debounce time for the debounce timer,<br>00: Timer Unit = 61 usec (2 RtcClk), Max Debounce Time = 976 usec<br>01: Timer Unit = 244 usec (8 RtcClk), Max Debounce Time = 3.9 msec<br>10: Timer Unit = 15.6 msec (512 RtcClk), Max Debounce Time = 250 msec<br>11: Timer Unit = 62.5 msec (2048 RtcClk), Max Debounce Time = 1.0 sec |
| 3:0   | <b>debouncetmrout.</b> Read-write. Reset: 0h. Debounce timer out number ( 0 means that debouncing logic is disabled)                                                                                                                                                                                                                                                                                                                                                       |

**REMOTEGPIOx000000C0 (FCH::RMTGPIO::IOMUX0\_GPIO256)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000C0; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |



**REMOTEGPIOx000000C1 (FCH::RMTGPIO::IOMUX1\_GPIO257)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000C1; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000C2 (FCH::RMTGPIO::IOMUX2\_GPIO258)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000C2; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000C3 (FCH::RMTGPIO::IOMUX3\_GPIO259)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000C3; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000C4 (FCH::RMTGPIO::IOMUX4\_GPIO260)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000C4; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000C5 (FCH::RMTGPIO::IOMUX5\_GPIO261)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000C5; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000C8 (FCH::RMTGPIO::IOMUX8\_GPIO264)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000C8; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000C9 (FCH::RMTGPIO::IOMUX9\_GPIO265)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000C9; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000CA (FCH::RMTGPIO::IOMUX10\_GPIO266)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000CA; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000CB (FCH::RMTGPIO::IOMUX11\_GPIO267)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000CB; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000CC (FCH::RMTGPIO::IOMUX12\_GPIO268)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000CC; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000CD (FCH::RMTGPIO::IOMUX13\_GPIO269)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000CD; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000CE (FCH::RMTGPIO::IOMUX14\_GPIO270)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000CE; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                        |
| 1:0  | <b>iomux_gpio_x.</b> Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000CF (FCH::RMTGPIO::IOMUX15\_GPIO271)**

Read-write. Reset: 00h.

\*Note:

&lt;N&gt; denotes number of IoMux, from 0 to 15, IoMux0 for GPIO256, , IoMux register at C0h ~ CFh.

&lt;X&gt; denotes GPIO number in decimal: 256 ~ 271.

\_aliasHOSTLEGACY; REMOTEGPIOx000000CF; REMOTEGPIO=FED8\_1200h

| Bits | Description                                                                                                                                                                                       |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7:2  | Reserved.                                                                                                                                                                                         |
| 1:0  | <b>iomux_gpio_x</b> . Read-write. Reset: 0h.<br><b>Description:</b> Multi-function IO pin function select for pin GPIO<X><br>00: function-0<br>01: function-1<br>10: function-2<br>11: function-3 |

**REMOTEGPIOx000000F0 (FCH::RMTGPIO::RMT\_GPIO\_WAKE\_STATUS)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx000000F0; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                               |
| 15:0  | <b>gpioxxx_wake_status</b> . Read-only. Reset: 0000h. Wake status of GPIO256 (bit[0]) GPIO271 (bit[15]) |

**REMOTEGPIOx000000F4 (FCH::RMTGPIO::RMT\_GPIO\_INTERRUPT\_STATUS)**

Read-only. Reset: 0000\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx000000F4; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                  |
|-------|--------------------------------------------------------------------------------------------------------------|
| 31:16 | Reserved.                                                                                                    |
| 15:0  | <b>gpioxxx_intr_status</b> . Read-only. Reset: 0000h. Interrupt status of GPIO256 (bit[0]) GPIO271 (bit[15]) |

**REMOTEGPIOx000000FC (FCH::RMTGPIO::RMT\_GPIO\_MASTER\_SWITCH)**

Read-write. Reset: 1F00\_0000h.

\_aliasHOSTLEGACY; REMOTEGPIOx000000FC; REMOTEGPIO=FED8\_1200h

| Bits  | Description                                                                                                                                                                                                                                                    |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31:29 | Reserved.                                                                                                                                                                                                                                                      |
| 28    | <b>mask_sts_en</b> . Read-write. Reset: 1. This bit enables HW to block all wake/intr status generation when SW writes any Debounce* registers. The length of blocking depends on mask_sts_length[11:0] using pclk to count, default = 15.15ns * 0x40000 = 4ms |
| 27:24 | <b>mask_sts_length_3_0</b> . Read-write. Reset: Fh.<br><b>Description:</b> See mask_sts_en as well.<br>The length of blocking = {mask_sts_length[11:0], 14'h3FFF}                                                                                              |
| 23:16 | <b>mask_sts_length_11_4</b> . Read-write. Reset: 00h.<br><b>Description:</b> See mask_sts_en as well.<br>The length of blocking = {mask_sts_length[11:0], 14'h3FFF}                                                                                            |
| 15:0  | Reserved.                                                                                                                                                                                                                                                      |